

Stochastic Processes, Theory for Applications

Solutions to Exercises
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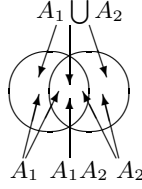
If any errors or typos are discovered, please notify Robert Gallager at gallager@mit.edu so that they can be corrected.

The author gratefully acknowledges the help of Shan-Yuan Ho, who has edited many of these solutions, and of a number of teaching assistants, particularly Natasha Blitvic and Mina Karzand, who wrote earlier drafts of some solutions. Thanks are also due to Kluwer Academic Press for permission to use a number of exercises that also appeared in Gallager, 'Discrete Stochastic Processes,' Kluwer, 1995. The original solutions to those exercises were prepared by Shan-Yuan Ho, but changed substantially here by the author both to be consistent with the new text and occasionally for added clarity.

A.1 Solutions for Chapter 1

Exercise 1.1: Let A_1 and A_2 be arbitrary events and show that $\Pr\{A_1 \cup A_2\} + \Pr\{A_1 A_2\} = \Pr\{A_1\} + \Pr\{A_2\}$. Explain which parts of the sample space are being double-counted on both sides of this equation and which parts are being counted once.

Solution: As shown in the figure below, $A_1 A_2$ is part of $A_1 \cup A_2$ and is thus being double counted on the left side of the equation. It is also being double counted on the right (and is in fact the meaning of $A_1 A_2$ as those sample points that are both in A_1 and in A_2).



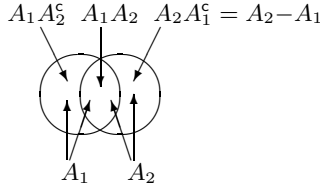
Exercise 1.2: This exercise derives the probability of an arbitrary (non-disjoint) union of events, derives the union bound, and derives some useful limit expressions.

a) For 2 arbitrary events A_1 and A_2 , show that

$$A_1 \cup A_2 = A_1 \cup (A_2 - A_1), \quad (\text{A.1})$$

where $A_2 - A_1 = A_2 A_1^c$. Show that A_1 and $A_2 - A_1$ are disjoint. Hint: This is what Venn diagrams were invented for.

Solution: Note that each sample point ω is in A_1 or A_1^c , but not both. Thus each ω is in exactly one of A_1 , $A_1^c A_2$ or $A_1^c A_2^c$. In the first two cases, ω is in both sides of (A.1) and in the last case it is in neither. Thus the two sides of (A.1) are identical. Also, as pointed out above, A_1 and $A_2 - A_1$ are disjoint. These results are intuitively obvious from the Venn diagram,



b) For any $n \geq 2$ and arbitrary events $A_1, \dots, A_n$, define $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$. Show that $B_1, B_2, \dots$ are disjoint events and show that for each $n \geq 2$, $\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n B_i$. Hint: Use induction.

Solution: Let $B_1 = A_1$. From (a) B_1 and B_2 are disjoint and (from (A.1)), $A_1 \cup A_2 = B_1 \cup B_2$. Let $C_n = \bigcup_{i=1}^n A_i$. We use induction to prove that $C_n = \bigcup_{i=1}^n B_i$ and that the B_n are disjoint. We have seen that $C_2 = B_1 \cup B_2$, which forms the basis for the induction. We assume that $C_{n-1} = \bigcup_{i=1}^{n-1} B_i$ and prove that $C_n = \bigcup_{i=1}^n B_i$.

$$\begin{aligned} C_n &= C_{n-1} \cup A_n = C_{n-1} \cup A_n C_{n-1}^c \\ &= C_{n-1} \cup B_n = \bigcup_{i=1}^n B_i. \end{aligned}$$

In the second equality, we used (A.1), letting C_{n-1} play the role of A_1 and A_n play the role

of A_2 . From this same application of (A.1), we also see that C_{n-1} and $B_n = A_n - C_{n-1}$ are disjoint. Since $C_{n-1} = \bigcup_{i=1}^{n-1} B_i$, this also shows that B_n is disjoint from $B_1, \dots, B_{n-1}$.

c) Show that

$$\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} = \Pr\left\{\bigcup_{n=1}^{\infty} B_n\right\} = \sum_{n=1}^{\infty} \Pr\{B_n\}.$$

Solution: If $\omega \in \bigcup_{n=1}^{\infty} A_n$, then it is in A_n for some $n \geq 1$. Thus $\omega \in \bigcup_{i=1}^n B_i$, and thus $\omega \in \bigcup_{n=1}^{\infty} B_n$. The same argument works the other way, so $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. This establishes the first equality above, and the second is the third axiom of probability.

d) Show that for each n , $\Pr\{B_n\} \leq \Pr\{A_n\}$. Use this to show that

$$\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} \leq \sum_{n=1}^{\infty} \Pr\{A_n\}.$$

Solution: Since $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$, we see that $\omega \in B_n$ implies that $\omega \in A_n$, i.e., that $B_n \subseteq A_n$. From (1.5), this implies that $\Pr\{B_n\} \leq \Pr\{A_n\}$ for each n . Thus

$$\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} = \sum_{n=1}^{\infty} \Pr\{B_n\} \leq \sum_{n=1}^{\infty} \Pr\{A_n\}.$$

e) Show that $\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} = \lim_{n \rightarrow \infty} \Pr\left\{\bigcup_{i=1}^n A_i\right\}$. Hint: Combine (c) and (b). Note that this says that the probability of a limit is equal to the limit of the probabilities. This might well appear to be obvious without a proof, but you will see situations later where similar appearing interchanges cannot be made.

Solution: From (c),

$$\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} = \sum_{n=1}^{\infty} \Pr\{B_n\} = \lim_{k \rightarrow \infty} \sum_{n=1}^k \Pr\{B_n\}.$$

From (b), however,

$$\sum_{n=1}^k \Pr\{B_n\} = \Pr\left\{\bigcup_{n=1}^k B_n\right\} = \Pr\left\{\bigcup_{n=1}^k A_n\right\}.$$

Combining the first equation with the limit in k of the second yields the desired result.

f) Show that $\Pr\left\{\bigcap_{n=1}^{\infty} A_n\right\} = \lim_{n \rightarrow \infty} \Pr\left\{\bigcap_{i=1}^n A_i\right\}$. Hint: Remember De Morgan's equalities.

Solution: Using De Morgan's equalities,

$$\begin{aligned} \Pr\left\{\bigcap_{n=1}^{\infty} A_n\right\} &= 1 - \Pr\left\{\bigcup_{n=1}^{\infty} A_n^c\right\} = 1 - \lim_{k \rightarrow \infty} \Pr\left\{\bigcup_{n=1}^k A_n^c\right\} \\ &= \lim_{k \rightarrow \infty} \Pr\left\{\bigcap_{n=1}^k A_n\right\}. \end{aligned}$$

Exercise 1.3: Find the probability that a five card poker hand, chosen randomly from a 52 card deck, contains 4 aces. That is, if all 52! arrangements of a deck of cards are equally likely, what is the probability that all 4 aces are in the first 5 cards of the deck.

Solution: The ace of spades can be in any of the first 5 positions, the ace of hearts in any of the 4 remaining positions out of the first 5, and so forth for the other two aces. The

remaining 48 cards can be in any of the remaining 48 positions. Thus there are $(5 \cdot 4 \cdot 3 \cdot 2)48!$ permutations of the 52 cards for which the first 5 cards contain 4 aces. Thus

$$\Pr\{4 \text{ aces}\} = \frac{5!48!}{52!} = \frac{5 \cdot 4 \cdot 3 \cdot 2}{52 \cdot 51 \cdot 50 \cdot 49} = 1.847 \times 10^{-5}.$$

Exercise 1.4: Consider a sample space of 8 equiprobable sample points and let A_1, A_2, A_3 be three events each of probability $1/2$ such that $\Pr\{A_1 A_2 A_3\} = \Pr\{A_1\} \Pr\{A_2\} \Pr\{A_3\}$.

a) Create an example where $\Pr\{A_1 A_2\} = \Pr\{A_1 A_3\} = \frac{1}{4}$ but $\Pr\{A_2 A_3\} = \frac{1}{8}$. Hint: Make a table with a row for each sample point and a column for each of the above 3 events and try different ways of assigning sample points to events (the answer is not unique).

Solution: Note that exactly one sample point must be in A_1, A_2 , and A_3 in order to make $\Pr\{A_1 A_2 A_3\} = 1/8$. In order to make $\Pr\{A_1 A_2\} = 1/4$, there must be one additional sample point that contains A_1 and A_2 but not A_3 . Similarly, there must be one sample point that contains A_1 and A_3 but not A_2 . These points give rise to the first three rows in the table below. There can be no additional sample point containing A_2 and A_3 since $\Pr\{A_2 A_3\} = 1/8$. Thus each remaining sample point can be in at most 1 of the events A_1, A_2 , and A_3 . Since $\Pr\{A_i\} = 1/2$ for $1 \leq i \leq 3$ two sample points must contain A_2 alone, two must contain A_3 alone, and a single sample point must contain A_1 alone. This uniquely specifies the table below except for which sample point lies in each event.

Sample points	A_1	A_2	A_3
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	0
6	0	1	0
7	0	0	1
8	0	0	1

b) Show that, for your example, A_2 and A_3 are not independent. Note that the definition of statistical independence would be very strange if it allowed A_1, A_2, A_3 to be independent while A_2 and A_3 are dependent. This illustrates why the definition of independence requires (1.14) rather than just (1.15).

Solution: Note that $\Pr\{A_2 A_3\} = 1/8 \neq \Pr\{A_2\} \Pr\{A_3\}$, so A_2 and A_3 are dependent. We also note that $\Pr\{A_1^c A_2^c A_3^c\} = 0 \neq \Pr\{A_1^c\} \Pr\{A_2^c\} \Pr\{A_3^c\}$, further reinforcing the conclusion that A_1, A_2, A_3 are not statistically independent. Although the definition in (1.14) of statistical independence of more than 2 events looks strange, it is clear from this example that (1.15) is insufficient in the sense that it only specifies part of the above table.

Exercise 1.5: This exercise shows that for all rv's X , $F_X(x)$ is continuous from the right.

a) For any given rv X , any real number x , and each integer $n \geq 1$, let $A_n = \{\omega : X > x + 1/n\}$, and show that $A_1 \subseteq A_2 \subseteq \dots$. Use this and the corollaries to the axioms of probability to show that $\Pr\left\{\bigcup_{n \geq 1} A_n\right\} = \lim_{n \rightarrow \infty} \Pr\{A_n\}$.

Solution: If $X(\omega) > x + \frac{1}{n}$, then (since $\frac{1}{n} > \frac{1}{n+1}$), we also have $X(\omega) > x + \frac{1}{n+1}$. Thus $A_n \subseteq A_{n+1}$ for all $n \geq 1$. Thus from (1.9), $\Pr\left\{\bigcup_{n \geq 1} A_n\right\} = \lim_{n \rightarrow \infty} \Pr\{A_n\}$.

b) Show that $\Pr\left\{\bigcup_{n \geq 1} A_n\right\} = \Pr\{X > x\}$ and that $\Pr\{X > x\} = \lim_{n \rightarrow \infty} \Pr\{X > x + 1/n\}$.

Solution: If $X(\omega) > x$, then there must be an n sufficiently large that $X(\omega) > x + 1/n$. Thus $\{\omega : X > x\} \subseteq \bigcup_{n \geq 1} A_n$. The subset inequality goes the other way also since $X(\omega) > x + 1/n$ for any $n \geq 1$ implies that $X(\omega) > x$. Since these represent the same events, they have the same probability and $\Pr\left\{\bigcup_{n \geq 1} A_n\right\} = \Pr\{X > x\}$. Then from (a) we also have

$$\Pr\{X > x\} = \lim_{n \rightarrow \infty} \Pr\{A_n\} = \lim_{n \rightarrow \infty} \Pr\{X > x + 1/n\}.$$

c) Show that for $\epsilon > 0$, $\lim_{\epsilon \rightarrow 0} \Pr\{X \leq x + \epsilon\} = \Pr\{X \leq x\}$.

Solution: Taking the complement of both sides of the above equation, $\Pr\{X \leq x\} = \lim_{n \rightarrow \infty} \Pr\{X \leq x + 1/n\}$. Since $\Pr\{X \leq x + \epsilon\}$ is non-decreasing in ϵ , it also follows that for $\epsilon > 0$, $\Pr\{X \leq x\} = \lim_{\epsilon \rightarrow 0} \Pr\{X \leq x + \epsilon\}$.

Exercise 1.6: Show that for a continuous nonnegative rv X ,

$$\int_0^\infty \Pr\{X > x\} dx = \int_0^\infty x f_X(x) dx. \quad (\text{A.2})$$

Hint 1: First rewrite $\Pr\{X > x\}$ on the left side of (A.2) as $\int_x^\infty f_X(y) dy$. Then think through, to your level of comfort, how and why the order of integration can be interchanged in the resulting expression.

Solution: We have $\Pr\{X > x\} = \int_x^\infty f_X(y) dy$ from the definition of a continuous rv. We look at $\mathbb{E}[X] = \int_0^\infty \Pr\{X > x\} dx$ as $\lim_{a \rightarrow \infty} \int_0^a F^c(x) dx$ since the limiting operation $a \rightarrow \infty$ is where the interesting issue is.

$$\begin{aligned} \int_0^a F^c(x) dx &= \int_0^a \int_x^\infty f_X(y) dy dx \\ &= \int_0^a \int_x^a f_X(y) dy dx + \int_0^a \int_a^\infty f_X(y) dy dx \\ &= \int_0^a \int_0^y f_X(y) dx dy + a F_X^c(a). \end{aligned}$$

We first broke the integral on the right into two parts, one for $y < x$ and the other for $y \geq x$. Since the limits of integration on the first part were finite, they could be interchanged. The inner integral of the first part is $y f_X(y)$, so

$$\lim_{a \rightarrow \infty} \int_0^a F_X^c(x) dx = \lim_{a \rightarrow \infty} \int_0^a y f_X(y) dy + \lim_{a \rightarrow \infty} a F_X^c(a).$$

Assuming that $\mathbb{E}[X]$ exists, the integral on the left is nondecreasing in A and has the finite limit $\overline{X}$. The first integral on the right is also nondecreasing and upper bounded by the first integral, so it also has a limit. This means that $\lim_{a \rightarrow \infty} a F_X^c(a)$ must also have a limit, say β . Now if $\beta > 0$, then for any $\epsilon \in (0, a)$, $a F_X(a) > \beta - \epsilon$ for all sufficiently large a . For

all such a , then $F_X^c(a) > (\beta - \epsilon)/a$. This would imply that $\overline{X} = \int_0^\infty F_X^c(x) dx = \infty$, which is a contradiction. Thus $\beta = 0$, i.e., $\lim_{a \rightarrow \infty} aF_X^c(a) = 0$, establishing (A.2) for the case where $E[X]$ is finite. The case where $E[X]$ is infinite is a minor perturbation.

The result that $\lim_{a \rightarrow \infty} aF_X^c(a) = 0$ is also important and can be seen intuitively from Figure 1.3.

Hint 2: As an alternate approach, derive (A.2) using integration by parts.

Solution: Using integration by parts and being less careful,

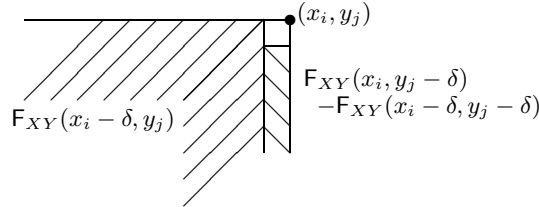
$$\int_0^\infty d(xF_X^c(x)) = - \int_0^\infty x f_X(x) dx + \int_0^\infty F_X^c(x) dx.$$

The left side is $\lim_{a \rightarrow \infty} aF_X^c(a) - 0F_X(0)$ so this shows the same thing, again requiring the fact that $\lim_{a \rightarrow \infty} aF_X^c(a) = 0$ when $E[X]$ exists.

Exercise 1.7: Suppose X and Y are discrete rv's with the PMF $p_{XY}(x_i, y_j)$. Show (a picture will help) that this is related to the joint CDF by

$$p_{XY}(x_i, y_j) = \lim_{\delta > 0, \delta \rightarrow 0} [F(x_i, y_j) - F(x_i - \delta, y_j) - F(x_i, y_j - \delta) + F(x_i - \delta, y_j - \delta)].$$

Solution: The picture below makes this equation obvious. Note that $F(x_i, y_j)$ is the probability of the quadrant of joint sample values (x, y) that satisfy $(x, y) \leq (x_i, y_j)$. The term $F_{XY}(x_i - \delta, y_j)$ is the probability of the hatched region on the left and $F_{XY}(x_i, y_j - \delta) - F_{XY}(x_i - \delta, y_j - \delta)$ is the probability of the hatched region on the right. Thus the expression on the right of the equation is the probability of the δ by δ square to the left and below (x_i, y_j) . When this square becomes too small to include any other sample point, the probability of the square is $p_{XY}(x_i, y_j)$.



Exercise 1.8: A variation of Example 1.5.1 is to let M be a random variable that takes on both positive and negative values with the PMF

$$p_M(m) = \frac{1}{2|m|(|m| + 1)}.$$

In other words, M is symmetric around 0 and $|M|$ has the same PMF as the nonnegative rv N of Example 1.5.1.

a) Show that $\sum_{m \geq 0} m p_M(m) = \infty$ and $\sum_{m < 0} m p_M(m) = -\infty$. (Thus show that the expectation of M not only does not exist but is undefined even in the extended real number system.)

Solution:

$$\begin{aligned}\sum_{m \geq 0} m p_M(m) &= \sum_{m \geq 0} \frac{1}{2(|m|+1)} = \infty \\ \sum_{m < 0} m p_M(m) &= \sum_{m < 0} \frac{-1}{2(|m|+1)} = -\infty.\end{aligned}$$

b) Suppose that the terms in $\sum_{m=-\infty}^{\infty} m p_M(m)$ are summed in the order of 2 positive terms for each negative term (i.e., in the order $1, 2, -1, 3, 4, -2, 5, \dots$). Find the limiting value of the partial sums in this series. Hint: You may find it helpful to know that

$$\lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{1}{i} - \int_1^n \frac{1}{x} dx \right] = \gamma,$$

where γ is the Euler-Mascheroni constant, $\gamma = 0.57721 \dots$.

Solution: The sum after $3n$ terms is

$$\sum_{m=1}^{2n} \frac{1}{2(m+1)} - \sum_{m=1}^n \frac{1}{2(m+1)} = \frac{1}{2} \sum_{i=n+2}^{2n+1} \frac{1}{i}.$$

Taking the limit as $n \rightarrow \infty$, the Euler-Mascheroni constant cancels out and

$$\lim_{n \rightarrow \infty} \frac{1}{2} \sum_{i=n+2}^{2n+1} \frac{1}{i} = \lim_{n \rightarrow \infty} \frac{1}{2} \int_{n+2}^{2n+1} \frac{1}{x} dx = \frac{1}{2} \ln 2.$$

c) Repeat (b) where, for any given integer $k > 0$, the order of summation is k positive terms for each negative term.

Solution: This is done the same way, and the answer is $\frac{1}{2} \ln k$. What the exercise essentially shows is that in a sum for which both the positive terms sum to infinity and the negative terms sum to $-\infty$, one can get any desired limit by summing terms in an appropriate order. In fact, to reach any desired limit, one alternates between positive terms until exceeding the desired limit, then negative terms until falling below the desired limit, then positive terms again, etc.

Exercise 1.9: (Proof of Theorem 1.4.1) The bounds on the binomial in this theorem are based on the *Stirling bounds*. These say that for all $n \geq 1$, $n!$ is upper and lower bounded by

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/12n}. \quad (\text{A.3})$$

The ratio, $\sqrt{2\pi n}(n/e)^n/n!$, of the first two terms is monotonically increasing with n toward the limit 1, and the ratio $\sqrt{2\pi n}(n/e)^n \exp(1/12n)/n!$ is monotonically decreasing toward 1. The upper bound is more accurate, but the lower bound is simpler and known as the Stirling approximation. See [8] for proofs and further discussion of the above facts.

a) Show from (A.3) and from the above monotone property that

$$\binom{n}{k} < \sqrt{\frac{n}{2\pi k(n-k)}} \frac{n^n}{k^k (n-k)^{n-k}}.$$

Hint: First show that $n!/k! < \sqrt{n/k} n^n k^{-k} e^{-n+k}$ for $k < n$.

Solution: Since the ratio of the first two terms of (A.3) is increasing in n , we have

$$\sqrt{2\pi k}(k/e)^k/k! < \sqrt{2\pi n}(n/e)^n/n!.$$

Rearranging terms, we have the result in the hint. Applying the first inequality of (A.3) to $n - k$ and combining this with the result on $n!/k!$ yields the desired result.

b) Use the result of (a) to upper bound $p_{S_n}(k)$ by

$$p_{S_n}(k) < \sqrt{\frac{n}{2\pi k(n-k)}} \frac{p^k(1-p)^{n-k}n^n}{k^k(n-k)^{n-k}}.$$

Show that this is equivalent to the upper bound in Theorem 1.4.1.

Solution: Using the binomial equation and then (a),

$$p_{S_n}(k) = \binom{n}{k} p^k (1-p)^{n-k} < \sqrt{\frac{n}{2\pi k(n-k)}} \frac{n^n}{k^k(n-k)^{n-k}} p^k (1-p)^{n-k}.$$

This is the the desired bound on $p_{S_n}(k)$. Letting $\tilde{p} = k/n$, this becomes

$$\begin{aligned} p_{S_n}(\tilde{p}n) &< \sqrt{\frac{1}{2\pi n\tilde{p}(1-\tilde{p})}} \frac{p^{\tilde{p}n}(1-p)^{n(1-\tilde{p})}}{\tilde{p}^{\tilde{p}n}(1-\tilde{p})^{n(1-\tilde{p})}} \\ &= \sqrt{\frac{1}{2\pi n\tilde{p}(1-\tilde{p})}} \exp\left(n\left[\tilde{p}\ln\frac{p}{\tilde{p}} + \tilde{p}\ln\frac{1-p}{1-\tilde{p}}\right]\right), \end{aligned}$$

which is the same as the upper bound in Theorem 1.4.1.

c) Show that

$$\binom{n}{k} > \sqrt{\frac{n}{2\pi k(n-k)}} \frac{n^n}{k^k(n-k)^{n-k}} \left[1 - \frac{n}{12k(n-k)}\right].$$

Solution: Use the factorial lower bound on $n!$ and the upper bound on k and $(n-k)!$. This yields

$$\begin{aligned} \binom{n}{k} &> \sqrt{\frac{n}{2\pi k(n-k)}} \frac{n^n}{k^k(n-k)^{n-k}} \exp\left(-\frac{1}{12k} - \frac{1}{12(n-k)}\right) \\ &> \sqrt{\frac{n}{2\pi k(n-k)}} \frac{n^n}{k^k(n-k)^{n-k}} \left[1 - \frac{n}{12k(n-k)}\right], \end{aligned}$$

where the latter equation comes from combining the two terms in the exponent and then using the bound $e^{-x} > 1 - x$.

d) Derive the lower bound in Theorem 1.4.1.

Solution: This follows by substituting $\tilde{p}n$ for k in the solution to c) and substituting this in the binomial formula.

e) Show that $\phi(p, \tilde{p}) = \tilde{p}\ln(\frac{\tilde{p}}{p}) + (1-\tilde{p})\ln(\frac{1-\tilde{p}}{1-p})$ is 0 at $\tilde{p} = p$ and nonnegative elsewhere.

Solution: It is obvious that $\phi(p, \tilde{p}) = 0$ for $\tilde{p} = p$. Taking the first two derivatives of $\phi(p, \tilde{p})$ with respect to $\tilde{p}$,

$$\frac{\partial\phi(p, \tilde{p})}{\partial\tilde{p}} = -\ln\left(\frac{p(1-\tilde{p})}{\tilde{p}(1-p)}\right) \quad \frac{\partial^2\phi(p, \tilde{p})}{\partial\tilde{p}^2} = \frac{1}{\tilde{p}(1-\tilde{p})}.$$

Since the second derivative is positive for $0 < \tilde{p} < 1$, the minimum of $\phi(p, \tilde{p})$ with respect to $\tilde{p}$ is 0, is achieved where the first derivative is 0, *i.e.*, at $\tilde{p} = p$. Thus $\phi(p, \tilde{p}) > 0$ for $\tilde{p} \neq p$. Furthermore, $\phi(p, \tilde{p})$ increases as $\tilde{p}$ moves in either direction away from p .

Exercise 1.10: Let X be a ternary rv taking on the 3 values 0, 1, 2 with probabilities p_0, p_1, p_2 respectively. Find the median of X for each of the cases below.

a) $p_0 = 0.2, p_1 = 0.4, p_2 = 0.4$.

b) $p_0 = 0.2, p_1 = 0.2, p_2 = 0.6$.

c) $p_0 = 0.2, p_1 = 0.3, p_2 = 0.5$.

Note 1: The median is not unique in (c). Find the interval of values that are medians. Note 2: Some people force the median to be distinct by defining it as the midpoint of the interval satisfying the definition given here.

Solution: The median of X is 1 for (a), 2 for (b), and the interval $[1, 2]$ for (c).

d) Now suppose that X is nonnegative and continuous with the density $f_X(x) = 1$ for $0 \leq x \leq 0.5$ and $f_X(x) = 0$ for $0.5 < x \leq 1$. We know that $f_X(x)$ is positive for all $x > 1$, but it is otherwise unknown. Find the median or interval of medians.

The median is sometimes (incorrectly) defined as that α for which $\Pr\{X > \alpha\} = \Pr\{X < \alpha\}$. Show that it is possible for no such α to exist. Hint: Look at the examples above.

Solution: The interval of medians is $[0.5, 1]$. In particular, $\Pr\{X \leq x\} = 1/2$ for all x in this interval and $\Pr\{X \geq x\} = 1/2$ in this interval.

For each of the first 3 examples, there is no α for which $\Pr\{X < \alpha\} = \Pr\{X > \alpha\}$. One should then ask why there must always be an x such that $\Pr\{X \geq x\} \geq 1/2$ and $\Pr\{X \leq x\} \geq 1/2$. To see this, let $x_o = \inf\{x : F_X(x) \geq 1/2\}$. We must have $F_X(x_o) \geq 1/2$ since F_X is continuous on the right. Because of the infimum, we must have $F_X(x_o - \epsilon) < 1/2$ for all $\epsilon > 0$, and therefore $\Pr\{X \geq x_o - \epsilon\} \geq 1/2$. But $\Pr\{X \geq x\}$ is continuous on the left for the same reason that $F_X(x)$ is continuous on the right, and thus x_o is a median of X . This is the kind of argument that makes many people hate analysis.

Exercise 1.11: a) For any given rv Y , express $E[|Y|]$ in terms of $\int_{y<0} F_Y(y) dy$ and $\int_{y\geq 0} F_Y^c(y) dy$. Hint: Review the argument in Figure 1.4.

Solution: We have seen in (1.34) that

$$E[Y] = - \int_{y<0} F_Y(y) dy + \int_{y\geq 0} F_Y^c(y) dy.$$

Since all negative values of Y become positive in $|Y|$,

$$E[|Y|] = + \int_{y<0} F_Y(y) dy + \int_{y\geq 0} F_Y^c(y) dy.$$

To spell this out in greater detail, let $Y = Y^+ + Y^-$ where $Y^+ = \max\{0, Y\}$ and $Y^- = \min\{Y, 0\}$. Then $Y = Y^+ + Y^-$ and $|Y| = Y^+ - Y^- = Y^+ + |Y^-|$. Since $E[Y^+] = \int_{y\geq 0} F_Y^c(y) dy$ and $E[Y^-] = - \int_{y<0} F_Y(y) dy$, the above results follow.

b) For some given rv X with $E[|X|] < \infty$, let $Y = X - \alpha$. Using (a), show that

$$E[|X - \alpha|] = \int_{-\infty}^{\alpha} F_X(x) dx + \int_{\alpha}^{\infty} F_X^c(x) dx.$$

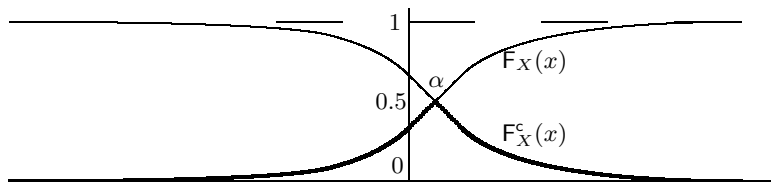
Solution: This follows by changing the variable of integration in (a). That is,

$$\begin{aligned} E[|X - \alpha|] &= E[|Y|] = \int_{y < 0} F_Y(y) dy + \int_{y \geq 0} F_Y^c(y) dy \\ &= \int_{-\infty}^{\alpha} F_X(x) dx + \int_{\alpha}^{\infty} F_X^c(x) dx, \end{aligned}$$

where in the last step, we have changed the variable of integration from y to $x - \alpha$.

c) Show that $E[|X - \alpha|]$ is minimized over α by choosing α to be a median of X . Hint: Both the easy way and the most instructive way to do this is to use a graphical argument illustrating the above two integrals. Be careful to show that when the median is an interval, all points in this interval achieve the minimum.

Solution: As illustrated in the picture, we are minimizing an integral for which the integrand changes from $F_X(x)$ to $F_X^c(x)$ at $x = \alpha$. If $F_X(x)$ is strictly increasing in x , then $F_X^c = 1 - F_X$ is strictly decreasing. We then minimize the integrand over all x by choosing α to be the point where the curves cross, *i.e.*, where $F_X(x) = .5$. Since the integrand has been minimized at each point, the integral must also be minimized.



If F_X is continuous but not strictly increasing, then there might be an interval over which $F_X(x) = .5$; all points on this interval are medians and also minimize the integral; Exercise 1.10 (c) gives an example where $F_X(x) = 0.5$ over the interval $[1, 2)$. Finally, if $F_X(\alpha) \geq 0.5$ and $F_X(\alpha - \epsilon) < 0.5$ for some α and all $\epsilon > 0$ (as in parts (a) and (b) of Exercise 1.10), then the integral is minimized at that α and that α is also the median.

Exercise 1.12: Let X be a rv with CDF $F_X(x)$. Find the CDF of the following rv's.

a) The maximum of n IID rv's, each with CDF $F_X(x)$.

Solution: Let M_+ be the maximum of the n rv's $X_1, \dots, X_n$. Note that for any real x , M_+ is less than or equal to x if and only if $X_j \leq x$ for each j , $1 \leq j \leq n$. Thus

$$\Pr\{M_+ \leq x\} = \Pr\{X_1 \leq x, X_2 \leq x, \dots, X_n \leq x\} = \prod_{j=1}^n \Pr\{X_j \leq x\},$$

where we have used the independence of the X_j 's. Finally, since $\Pr\{X_j \leq x\} = F_X(x)$ for each j , we have $F_{M_+}(x) = \Pr\{M_+ \leq x\} = (F_X(x))^n$.

b) The minimum of n IID rv's, each with CDF $F_X(x)$.

Solution: Let M_- be the minimum of $X_1, \dots, X_n$. Then, in the same way as in ((a), $M_- > y$ if and only if $X_j > y$ for $1 \leq j \leq n$ and for all choice of y . We could make the same statement using greater than or equal in place of strictly greater than, but the strict inequality is what is needed for the CDF. Thus,

$$\Pr\{M_- > y\} = \Pr\{X_1 > y, X_2 > y, \dots, X_n > y\} = \prod_{j=1}^n \Pr\{X_j > y\}.$$

It follows that $1 - F_{M_-}(y) = (1 - F_X(y))^n$.

c) The difference of the rv's defined in a) and b); assume X has a density $f_X(x)$.

Solution: There are many difficult ways to do this, but also a simple way, based on first conditioning on the event that $X_1 = x$. Then $X_1 = M_+$ if and only if $X_j \leq x$ for $2 \leq j \leq n$. Also, given $X_1 = M_+ = x$, we have $R = M_+ - M_- \leq r$ if and only if $X_j > x - r$ for $2 \leq j \leq n$. Thus, since the X_j are IID,

$$\begin{aligned} \Pr\{M_+ = X_1, R \leq r \mid X_1 = x\} &= \prod_{j=2}^n \Pr\{x-r < X_j \leq x\} \\ &= [\Pr\{x-r < X \leq x\}]^{n-1} = [F_X(x) - F_X(x-r)]^{n-1}. \end{aligned}$$

We can now remove the conditioning by averaging over $X_1 = x$. Assuming that X has the density $f_X(x)$,

$$\Pr\{X_1 = M_+, R \leq r\} = \int_{-\infty}^{\infty} f_X(x) [F_X(x) - F_X(x-r)]^{n-1} dx.$$

Finally, we note that the probability that two of the X_j are the same is 0 so the events $X_j = M_+$ are disjoint except with zero probability. Also we could condition on $X_j = x$ instead of X_1 with the same argument (*i.e.*, by using symmetry), so $\Pr\{X_j = M_+, R \leq r\} = \Pr\{X_1 = M_+, R \leq r\}$. It follows that

$$\Pr\{R \leq r\} = \int_{-\infty}^{\infty} n f_X(x) [F_X(x) - F_X(x-r)]^{n-1} dx.$$

The only place we really needed the assumption that X has a PDF was in asserting that the probability that two or more of the X_j 's are jointly equal to the maximum is 0. The formula can be extended to arbitrary CDF's by being careful about this possibility.

These expressions have a simple form if X is exponential with the PDF $\lambda e^{-\lambda x}$ for $x \geq 0$. Then

$$\Pr\{M_- \geq y\} = e^{-n\lambda y}; \quad \Pr\{M_+ \leq y\} = (1 - e^{-\lambda y})^n; \quad \Pr\{R \leq y\} = (1 - e^{-\lambda y})^{n-1}.$$

We will see how to derive the above expression for $\Pr\{R \leq y\}$ in Chapter 2.

Exercise 1.13: Let X and Y be rv's in some sample space Ω and let $Z = X + Y$, *i.e.*, for each $\omega \in \Omega$, $Z(\omega) = X(\omega) + Y(\omega)$. The purpose of this exercise is to show that Z is a rv. This is a mathematical fine point that many readers may prefer to simply accept without proof.

a) Show that the set of ω for which $Z(\omega)$ is infinite or undefined has probability 0.

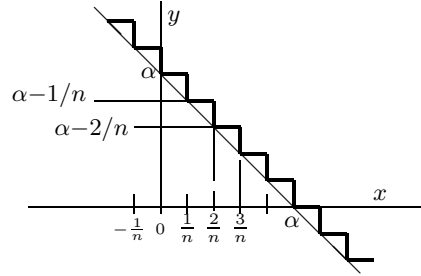
Solution: Note that Z can be infinite (either $\pm\infty$) or undefined only when either X or Y are infinite or undefined. Since these are events of zero probability, Z can be infinite or undefined only with probability 0.

b) We must show that $\{\omega \in \Omega : Z(\omega) \leq \alpha\}$ is an event for each real α , and we start by approximating that event. To show that $Z = X + Y$ is a rv, we must show that for each real number α , the set $\{\omega \in \Omega : X(\omega) + Y(\omega) \leq \alpha\}$ is an event. Let $B(n, k) = \{\omega : X(\omega) \leq k/n\} \cap \{Y(\omega) \leq \alpha + (1-k)/n\}$ for integer $k > 0$. Let $D(n) = \bigcup_k B(n, k)$, and show that $D(n)$ is an event.

Solution: We are trying to show that $\{Z \leq \alpha\}$ is an event for arbitrary α and doing this by first quantizing X and Y into intervals of size $1/n$ where k is used to number these quantized elements. Part (c) will make sense of how this is related to $\{Z \leq \alpha\}$, but for now we simply treat the sets as defined. Each set $B(n, k)$ is an intersection of two events, namely the event $\{\omega : X(\omega) \leq k/n\}$ and the event $\{\omega : Y(\omega) \leq \alpha + (1-k)/n\}$; these must be events since X and Y are rv's. For each n , $D(n)$ is a countable union (over k) of the sets $B(n, k)$, and thus $D(n)$ is an event for each n and each α .

c) On a 2 dimensional sketch for a given α , show the values of $X(\omega)$ and $Y(\omega)$ for which $\omega \in D(n)$. Hint: This set of values should be bounded by a staircase function.

Solution:



The region $D(n)$ is sketched for $\alpha n = 5$; it is the region below the staircase function above. The k th step of the staircase, extended horizontally to the left and vertically down is the set $B(n, k)$. Thus we see that $D(n)$ is an upper bound to the set $\{Z \leq \alpha\}$, which is the straight line of slope -1 below the staircase.

d) Show that

$$\{\omega : X(\omega) + Y(\omega) \leq \alpha\} = \bigcap_{n \geq 1} D(n). \quad (\text{A.4})$$

Explain why this shows that $Z = X + Y$ is a rv.

Solution: The region $\{\omega : X(\omega) + Y(\omega) \leq \alpha\}$ is the region below the diagonal line of slope -1 that passes through the point $(0, \alpha)$. This region is thus contained in $D(n)$ for each $n \geq 1$ and is thus contained in $\bigcap_{n \geq 1} D(n)$. On the other hand, each point ω for which $X(\omega) + Y(\omega) > \alpha$ is not contained in $D(n)$ for sufficiently large n . This verifies (A.4). Since $D(n)$ is an event, the countable intersection is also an event, so $\{\omega : X(\omega) + Y(\omega) \leq \alpha\}$ is an event. This applies for all α . This, in conjunction with (a), shows that Z is a rv.

e) Explain why this implies that if $X_1, X_2, \dots, X_n$ are rv's, then $Y = X_1 + X_2 + \dots + X_n$ is a rv. Hint:

Only one or two lines of explanation are needed.

Solution: We have shown that $X_1 + X_2$ is a rv, so $(X_1 + X_2) + X_3$ is a rv, etc.

Exercise 1.14: a) Let $X_1, X_2, \dots, X_n$ be rv's with expected values $\bar{X}_1, \dots, \bar{X}_n$. Show that $E[X_1 + \dots + X_n] = \bar{X}_1 + \dots + \bar{X}_n$. You may assume that the rv's have a joint density function, but do not assume that the rv's are independent.

Solution: We assume that the rv's have a joint density, and we ignore all mathematical fine points here. Then

$$\begin{aligned} E[X_1 + \dots + X_n] &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_1 + \dots + x_n) f_{X_1 \dots X_n}(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= \sum_{j=1}^n \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_j f_{X_1 \dots X_n}(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= \sum_{j=1}^n \int_{-\infty}^{\infty} x_j f_{X_j}(x_j) dx_j = \sum_{j=1}^n E[X_j]. \end{aligned}$$

Note that the separation into a sum of integrals simply used the properties of integration and that no assumption of statistical independence was made.

b) Now assume that $X_1, \dots, X_n$ are statistically independent and show that the expected value of the product is equal to the product of the expected values.

Solution: From the independence, $f_{X_1 \dots X_n}(x_1, \dots, x_n) = \prod_{j=1}^n f_{X_j}(x_j)$. Thus

$$\begin{aligned} E[X_1 X_2 \dots X_n] &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{j=1}^n x_j \prod_{j=1}^n f_{X_j}(x_j) dx_1 \dots dx_n \\ &= \prod_{j=1}^n \int_{-\infty}^{\infty} x_j f_{X_j}(x_j) dx_j = \prod_{j=1}^n E[X_j]. \end{aligned}$$

c) Again assuming that $X_1, \dots, X_n$ are statistically independent, show that the variance of the sum is equal to the sum of the variances.

Solution: Since (a) shows that $E\left[\sum_j X_j\right] = \sum_j \bar{X}_j$, we have

$$\begin{aligned} \text{VAR} \left[\sum_{j=1}^n X_j \right] &= E \left[\left(\sum_{j=1}^n X_j - \sum_{j=1}^n \bar{X}_j \right)^2 \right] \\ &= E \left[\sum_{j=1}^n \sum_{i=1}^n (X_j - \bar{X}_j)(X_i - \bar{X}_i) \right] \\ &= \sum_{j=1}^n \sum_{i=1}^n E[(X_j - \bar{X}_j)(X_i - \bar{X}_i)], \end{aligned} \tag{A.5}$$

where we have again used (a). Now from (b) (which used the independence of the X_j), $E[(X_j - \bar{X}_j)(X_i - \bar{X}_i)] = 0$ for $i \neq j$. Thus (A.5) simplifies to

$$\text{VAR} \left[\sum_{j=1}^n X_j \right] = \sum_{j=1}^n E[(X_j - \bar{X}_j)^2] = \sum_{j=1}^n \text{VAR}[X_j].$$

Exercise 1.15: (Stieltjes integration) **a)** Let $h(x) = u(x)$ and $F_X(x) = u(x)$ where $u(x)$ is the unit step, i.e., $u(x) = 0$ for $-\infty < x < 0$ and $u(x) = 1$ for $x \geq 0$. Using the definition of the Stieltjes integral in Footnote 19, show that $\int_{-1}^1 h(x) dF_X(x)$ does not exist. Hint: Look at the term in the Riemann sum including $x = 0$ and look at the range of choices for $h(x)$ in that interval. Intuitively, it might help initially to view $dF_X(x)$ as a unit impulse at $x = 0$.

Solution: The Riemann sum for this Stieltjes integral is $\sum_n h(x_n)[F(y_n) - F(y_{n-1})]$ where $y_{n-1} < x_n \leq y_n$. For any partition $\{y_n; n \geq 1\}$, consider the k such that $y_{k-1} < 0 \leq y_k$ and consider choosing either $x_n < 0$ or $x_n \geq 0$. In the first case $h(x_n)[F(y_n) - F(y_{n-1})] = 0$ and in the second $h(x_n)[F(y_n) - F(y_{n-1})] = 1$. All other terms are 0 and this can be done for all partitions as $\delta \rightarrow 0$, so the integral is undefined.

b) Let $h(x) = u(x - a)$ and $F_X(x) = u(x - b)$ where a and b are in $(-1, +1)$. Show that $\int_{-1}^1 h(x) dF_X(x)$ exists if and only if $a \neq b$. Show that the integral has the value 1 for $a < b$ and the value 0 for $a > b$. Argue that this result is still valid in the limit of integration over $(-\infty, \infty)$.

Solution: Using the same argument as in (a) for any given partition $\{y_n; n \geq 1\}$, consider the k such that $y_{k-1} < b \leq y_k$. If $a = b$, x_k can be chosen to make $h(x_k)$ either 0 or 1, causing the integral to be undefined as in (a). If $a < b$, then for a sufficiently fine partition, $h(x_k) = 1$ for all x_k such that $y_{k-1} < x_k \leq y_k$. Thus that term in the Riemann sum is 1. For all other n , $F_X(y_n) - F_X(y_{n-1}) = 0$, so the Riemann sum is 1. For $a > b$ and k as before, $h(x_k) = 0$ for a sufficiently fine partition, and the integral is 0. The argument does not involve the finite limits of integration, so the integral remains the same for infinite limits.

c) Let X and Y be independent discrete rv's, each with a finite set of possible values. Show that $\int_{-\infty}^{\infty} F_X(z - y) dF_Y(y)$, defined as a Stieltjes integral, is equal to the distribution of $Z = X + Y$ at each z other than the possible sample values of Z , and is undefined at each sample value of Z . Hint: Express F_X and F_Y as sums of unit steps. Note: This failure of Stieltjes integration is not a serious problem; $F_Z(z)$ is a step function, and the integral is undefined at its points of discontinuity. We automatically define $F_Z(z)$ at those step values so that F_Z is a CDF (i.e., is continuous from the right). This problem does not arise if either X or Y is continuous.

Solution: Let X have the PMF $\{p(x_1), \dots, p(x_K)\}$ and Y have the PMF $\{p_Y(y_1), \dots, p_Y(y_J)\}$. Then $F_X(x) = \sum_{k=1}^K p(x_k)u(x - x_k)$ and $F_Y(y) = \sum_{j=1}^J q(y_j)u(y - y_j)$. Then

$$\int_{-\infty}^{\infty} F_X(z - y) dF_Y(y) = \sum_{k=1}^K \sum_{j=1}^J \int_{-\infty}^{\infty} p(x_k)q(y_j)u(z - y_j - x_k)du(y - y_j).$$

From (b), the integral above for a given k, j exists unless $z = x_k + y_j$. In other words, the Stieltjes integral gives the CDF of $X + Y$ except at those z equal to $x_k + y_j$ for some k, j ,

i.e., equal to the values of Z at which $F_Z(z)$ (as found by discrete convolution) has step discontinuities.

To give a more intuitive explanation, $F_X(x) = \Pr\{X \leq x\}$ for any discrete rv X has jumps at the sample values of X and the value of $F_X(x_k)$ at any such x_k includes $p(x_k)$, *i.e.*, F_X is continuous to the right. The Riemann sum used to define the Stieltjes integral is not sensitive enough to ‘see’ this step discontinuity at the step itself. Thus, the stipulation that Z be continuous on the right must be used in addition to the Stieltjes integral to define F_Z at its jumps.

Exercise 1.16: Let $X_1, X_2, \dots, X_n, \dots$ be a sequence of IID continuous rv’s with the common probability density function $f_X(x)$; note that $\Pr\{X=\alpha\} = 0$ for all α and that $\Pr\{X_i=X_j\} = 0$ for all $i \neq j$. For $n \geq 2$, define X_n as a *record-to-date* of the sequence if $X_n > X_i$ for all $i < n$.

a) Find the probability that X_2 is a record-to-date. Use symmetry to obtain a numerical answer without computation. A one or two line explanation should be adequate).

Solution: X_2 is a record-to-date with probability $1/2$. The reason is that X_1 and X_2 are IID, so either one is larger with probability $1/2$; this uses the fact that they are equal with probability 0 since they have a density.

b) Find the probability that X_n is a record-to-date, as a function of $n \geq 1$. Again use symmetry.

Solution: By the same symmetry argument, each X_i , $1 \leq i \leq n$ is equally likely to be the largest, so that each is largest with probability $1/n$. Since X_n is a record-to-date if and only if it is the largest of $X_1, \dots, X_n$, it is a record-to-date with probability $1/n$.

c) Find a simple expression for the expected number of records-to-date that occur over the first m trials for any given integer m . Hint: Use indicator functions. Show that this expected number is infinite in the limit $m \rightarrow \infty$.

Solution: Let $\mathbb{I}_n$ be 1 if X_n is a record-to-date and be 0 otherwise. Thus $E[\mathbb{I}_i]$ is the expected value of the ‘number’ of records-to-date (either 1 or 0) on trial i . That is

$$E[\mathbb{I}_n] = \Pr\{\mathbb{I}_n = 1\} = \Pr\{X_n \text{ is a record-to-date}\} = 1/n.$$

Thus

$$E[\text{records-to-date up to } m] = \sum_{n=1}^m E[\mathbb{I}_n] = \sum_{n=1}^m \frac{1}{n}.$$

This is the harmonic series, which goes to ∞ in the limit $m \rightarrow \infty$. If you are unfamiliar with this, note that $\sum_{n=1}^{\infty} 1/n \geq \int_1^{\infty} \frac{1}{x} dx = \infty$.

Exercise 1.17: (Continuation of Exercise 1.16) a) Let N_1 be the index of the *first* record-to-date in the sequence. Find $\Pr\{N_1 > n\}$ for each $n \geq 2$. Hint: There is a far simpler way to do this than working from (b) in Exercise 1.16.

Solution: The event $\{N_1 > n\}$ is the event that no record-to-date occurs in the first n trials, which means that X_1 is the largest of $\{X_1, X_2, \dots, X_n\}$. By symmetry, this event has probability $1/n$. Thus $\Pr\{N_1 > n\} = 1/n$.

b) Show that N_1 is a rv (i.e., that N_1 is not defective).

Solution: Every sample sequence for $X_1, X_2, \dots$, maps into either a positive integer or infinity for N_1 . The probability that N_1 is infinite is $\lim_{n \rightarrow \infty} \Pr\{N_1 > n\} = \lim_{n \rightarrow \infty} (1/n)$, which is 0. Thus N_1 is finite with probability 1 and is thus a rv.

c) Show that $E[N_1] = \infty$.

Solution: Since N_1 is a nonnegative rv,

$$E[N_1] = \int_0^\infty \Pr\{N_1 > x\} dx = 1 + \sum_{n=1}^\infty \frac{1}{n} = \infty.$$

d) Let N_2 be the index of the *second* record-to-date in the sequence. Show that N_2 is a rv. You need not find the CDF of N_2 here.

Solution: For any given n_1 and n_2 , $2 \leq n_1 \leq n_2$, we start by finding $\Pr\{N_1 = n_1, N_2 > n_2\}$, which we show can be expressed as

$$\{N_1 = n_1, N_2 > n_2\} = \{X_{n_1} = \max(X_1, \dots, X_{n_2})\} \cap \{X_1 = \max(X_1, \dots, X_{n_1-1})\}. \quad (\text{A.6})$$

The first of these events, $\{X_{n_1} = \max(X_1, \dots, X_{n_2})\}$, means that X_{n_1} must be a record-to-date and also must be the last record to date up to and including n_2 . The second event, $\{X_1 = \max(X_1, \dots, X_{n_1-1})\}$ ensures that X_{n_1} is the first record-to-date. The first event has probability $1/n_2$. The second event has to do only with the ordering of the first n_1-1 terms, and thus is independent of the first event and has probability $1/n_1$. Thus,

$$\Pr\{\{N_1 = n_1, N_2 > n_2\}\} = \frac{1}{n_2(n_1-1)}.$$

The marginal probability $\Pr\{N_2 > n_2\}$ can now be found by summing over n_1 and including the event $\{N_1 > n_2, N_2 > n_2\}$, which has probability $1/n_2$.

$$\Pr\{N_2 > n_2\} = \frac{1}{n_2} + \sum_{n_1=2}^{n_2} \frac{1}{n_2(n_1-1)}$$

This approaches 0 with increasing n_2 , so N_2 is a rv. More precisely, we see that for large n_2 , $\Pr\{N_2 > n_2\} \approx (\ln n_2)/n_2$, so that this approaches 0 somewhat more slowly than $\Pr\{N_1 > n_2\}$, which is not surprising. These results do not depend on the CDF of X beyond the assumption that X is continuous since they are purely ordering results.

e) Contrast your result in (c) to the result from (c) of Exercise 1.16 saying that the expected number of records-to-date is infinite over an infinite number of trials. Note: this might be a shock to your intuition — there is an infinite expected wait for the first of an infinite sequence of occurrences.

Solution: Even though the expected wait for the first record-to-date is infinite, it is still a random variable, and thus the first record-to-date must eventually occur. We have also shown that the second record-to-date eventually occurs, and it can be shown that the n th

eventually occurs for all n . This makes the result in Exercise 1.16 unsurprising once it is understood.

Exercise 1.18: (Another direction from Exercise 1.16) a) For any given $n \geq 2$, find the probability that X_n and X_{n+1} are both records-to-date. Hint: The idea in Exercise 1.16 (b) is helpful here, but the result is not.

Solution: For both X_{n+1} and X_n to be records-to-date it is necessary and sufficient for X_{n+1} to be larger than all the earlier X_i (including X_n) and for X_n to be larger than all of the X_i earlier than it. Since any one of the first $n+1$ X_i is equally likely to be the largest, the probability that X_{n+1} is the largest is $1/(n+1)$. For X_n to also be a record-to-date, it must be the second largest of the $n+1$. Since all the first n terms are equally likely to be the second largest (given that X_{n+1} is the largest), the conditional probability that X_n is the second largest is $1/n$. Thus,

$$\Pr\{X_{n+1} \text{ and } X_n \text{ are records-to-date}\} = \frac{1}{n(n+1)}.$$

Note that there might be earlier records-to-date before n ; we have simply calculated the probability that X_n and X_{n+1} are records-to-date.

b) Is the event that X_n is a record-to-date statistically independent of the event that X_{n+1} is a record-to-date?

Solution: Yes, we have found the joint probability that X_n and X_{n+1} are records, and it is the product of the events that each are records individually.

c) Find the expected number of adjacent pairs of records-to-date over the sequence $X_1, X_2, \dots$. Hint: A helpful fact here is that $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.

Solution: Let $\mathbb{I}_n$ be the indicator function of the event that X_n and X_{n+1} are records-to-date. Then $E[\mathbb{I}_n] = \Pr\{\mathbb{I}_n = 1\} = \frac{1}{n(n+1)}$. The expected number of pairs of records (where, for example, records at 2, 3, and 4 are counted as two pairs of records), the expected number of pairs over the sequence is

$$\begin{aligned} E[\text{Number of pairs}] &= \sum_{n=2}^{\infty} \frac{1}{n(n+1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \sum_{n=2}^{\infty} \frac{1}{n} - \sum_{n=3}^{\infty} \frac{1}{n} = \frac{1}{2}, \end{aligned}$$

where we have used the hint and then summed the terms separately. This hint is often useful in analyzing stochastic processes.

The intuition here is that records-to-date tend to become more rare with increasing n (X_n is a record with probability $1/n$). As we have seen, the expected number of records from 2 to m is on the order of $\ln m$, which grows very slowly with m . The probability of an adjacent pair of records, as we have seen, decreases as $1/n(n+1)$ with n , which means that if one does not occur for small n , it will probably not occur at all. It can be seen from this that the time until the first pair of records is a defective random variable.

Exercise 1.19: a) Assume that X is a nonnegative discrete rv taking on values $a_1, a_2, \dots$, and let $Y = h(X)$ for some nonnegative function h . Let $b_i = h(a_i)$, $i \geq 1$ be the i^{th} value taken on by Y . Show that $E[Y] = \sum_i b_i p_Y(b_i) = \sum_i h(a_i) p_X(a_i)$. Find an example where $E[X]$ exists but $E[Y] = \infty$.

Solution: If we make the added assumption that $b_i \neq b_j$ for all $i \neq j$, then Y has the sample value b_i if and only if X has the sample value a_i ; thus $p_Y(b_i) = p_X(a_i)$ for each i . It then follows that $\sum_i b_i p_Y(b_i) = \sum_i h(a_i) p_X(a_i)$. This must be $E[Y]$ (which might be finite or infinite). The idea is the same without the assumption that $b_i \neq b_j$ for $i \neq j$, but now the more complicated notation $\Pr\{b\} = \sum_{i:h(a_i)=b} \Pr\{a_i\}$ must be used for each sample value b of Y .

A simple example where $E[X]$ is finite and $E[Y] = \infty$ is to choose $a_1, a_2, \dots$, to be 1, 2, $\dots$ and choose $p_X(i) = 2^{-i}$. Then $E[X] = 2$. Choosing $h(i) = 2^i$, we have $b_i = 2^i$ and $E[Y] = \sum_i 2^i \cdot 2^{-i} = \infty$. Without the assumption that $b_i \neq b_j$, the set of sample points of Y is the set of distinct values of b_i .

b) Let X be a nonnegative continuous rv with density $f_X(x)$ and let $h(x)$ be differentiable, nonnegative, and nondecreasing in x . Let $A(\delta) = \sum_{n \geq 1} h(n\delta)[F(n\delta) - F(n\delta - \delta)]$, i.e., $A(\delta)$ is a δ th order approximation to the Stieltjes integral $\int h(x) dF(x)$. Show that if $A(1) < \infty$, then $A(2^{-k}) \leq A(2^{-(k-1)}) < \infty$ for $k \geq 1$. Show from this that $\int h(x) dF(x)$ converges to a finite value. Note: this is a very special case, but it can be extended to many cases of interest. It seems better to consider these convergence questions as required rather than consider them in general.

Solution: Let $\delta = 2^{-k}$ for $k \geq 1$. We take the expression for $A(2\delta)$ and break each interval of size 2δ into two intervals each of size δ ; we use this to relate $A(2\delta)$ to $A(\delta)$.

$$\begin{aligned} A(2\delta) &= \sum_{n \geq 1} h(2n\delta) [F(2n\delta) - F(2n\delta - 2\delta)] \\ &= \sum_{n \geq 1} h(2n\delta) [F(2n\delta) - F(2n\delta - \delta)] + \sum_{n \geq 1} h(2n\delta) [F(2n\delta - \delta) - F(2n\delta - 2\delta)] \\ &\geq \sum_{n \geq 1} h(2n\delta) [F(2n\delta) - F(2n\delta - \delta)] + \sum_{n \geq 1} h(2n\delta - \delta) [F(2n\delta - \delta) - F(2n\delta - 2\delta)] \\ &= \sum_{m \geq 1} h(m\delta) [F(m\delta) - F(m\delta - \delta)] = A(\delta), \end{aligned}$$

where to get the final line, we substituted m for $2n$ in the first term of the preceding line and m for $2n-1$ in the second term, resulting in $A(\delta)$. Thus $A(2^{-k})$ is nonnegative and nonincreasing in k . Since $A(1)$ is finite by definition, $\lim_{k \rightarrow \infty} A(2^{-k})$ has a limit, which must be the value of the Stieltjes integral.

To be a little more precise about the Stieltjes integral, we see that $A(\delta)$ as defined above uses the largest value, $h(n\delta)$, of $h(x)$ over the interval $x \in [n\delta - \delta, n\delta]$. By replacing $h(n\delta)$ by $h(n\delta - \delta)$, we get the smallest value in each interval, and then the sequence is nonincreasing with the same limit. For an arbitrary partition of the real line, rather than the equi-spaced partition here, the argument here would have to be further extended.

Exercise 1.20: a) Consider a positive, integer-valued rv whose CDF is given at integer values by

$$F_Y(y) = 1 - \frac{2}{(y+1)(y+2)} \quad \text{for integer } y \geq 0.$$

Use (1.31) to show that $E[Y] = 2$. Hint: Note that $1/[(y+1)(y+2)] = 1/(y+1) - 1/(y+2)$.

Solution: Combining (1.31) with the hint, we have

$$\begin{aligned} E[Y] &= \sum_{y \geq 0} F_Y^c(y) = \sum_{y \geq 0} \frac{2}{y+1} - \sum_{y \geq 0} \frac{2}{y+2} \\ &= \sum_{y \geq 0} \frac{2}{y+1} - \sum_{y \geq 1} \frac{2}{y+1} = 2, \end{aligned}$$

where the second sum in the second line eliminates all but the first term of the first sum.

b) Find the PMF of Y and use it to check the value of $E[Y]$.

Solution: For $y = 0$, $p_Y(0) = F_Y(0) = 0$. For integer $y \geq 1$, $p_Y(y) = F_Y(y) - F_Y(y-1)$. Thus for $y \geq 1$,

$$p_Y(y) = \frac{2}{y(y+1)} - \frac{2}{(y+1)(y+2)} = \frac{4}{y(y+1)(y+2)}.$$

Finding $E[Y]$ from the PMF, we have

$$\begin{aligned} E[Y] &= \sum_{y=1}^{\infty} y p_Y(y) = \sum_{y=1}^{\infty} \frac{4}{(y+1)(y+2)} \\ &= \sum_{y=1}^{\infty} \frac{4}{y+1} - \sum_{y=2}^{\infty} \frac{4}{y+1} = 2. \end{aligned}$$

c) Let X be another positive, integer-valued rv. Assume its conditional PMF is given by

$$p_{X|Y}(x|y) = \frac{1}{y} \quad \text{for } 1 \leq x \leq y.$$

Find $E[X | Y = y]$ and use it to show that $E[X] = 3/2$. Explore finding $p_X(x)$ until you are convinced that using the conditional expectation to calculate $E[X]$ is considerably easier than using $p_X(x)$.

Solution: Conditioned on $Y = y$, X is uniform over $\{1, 2, \dots, y\}$ and thus has the conditional mean $(y+1)/2$. If you are unfamiliar with this fact, think of adding $\{1 + 2 + \dots + y\}$ to $\{y + (y-1) + \dots + 1\}$, getting $y(y+1)$. Thus $\{1 + 2 + \dots + y\} = y(y+1)/2$. It follows that

$$E[X] = E[E[X|Y]] = E\left[\frac{Y+1}{2}\right] = 3/2.$$

Calculating this expectation in the conventional way would require first calculating $p_X(x)$ and then calculating the expectation. Calculating $p_X(x)$,

$$p_X(x) = \sum_{y=x}^{\infty} p_Y(y) p_{X|Y}(x|y) = \sum_{y=x}^{\infty} \frac{4}{y(y+1)(y+2)} \frac{1}{y}.$$

It might be possible to calculate this in closed form, but it certainly does not look attractive. Only a dedicated algebraic masochist would pursue this further given the other approach.

d) Let Z be another integer-valued rv with the conditional PMF

$$p_{Z|Y}(z|y) = \frac{1}{y^2} \quad \text{for } 1 \leq z \leq y^2.$$

Find $E[Z | Y = y]$ for each integer $y \geq 1$ and find $E[Z]$.

Solution: As in (c), $E[Z|Y] = \frac{Y^2+1}{2}$. Since $p_Y(y)$ approaches 0 as y^{-3} , we see that $E[Y^2]$ is infinite and thus $E[Z] = \infty$.

Exercise 1.21: a) Show that, for uncorrelated rv's, the expected value of the product is equal to the product of the expected values (by definition, X and Y are uncorrelated if $E[(X - \bar{X})(Y - \bar{Y})] = 0$).

Solution: This results from a straightforward computation.

$$\begin{aligned} 0 &= E[(X - \bar{X})(Y - \bar{Y})] = E[XY] - E[X\bar{Y}] - E[\bar{X}Y] + E[\bar{X}\bar{Y}] \\ &= E[XY] - \bar{X}\bar{Y}. \end{aligned}$$

b) Show that if X and Y are uncorrelated, then the variance of $X + Y$ is equal to the variance of X plus the variance of Y .

Solution: This is also straightforward, but a bit more tedious.

$$\begin{aligned} \text{VAR}[X + Y] &= E[(X + Y)^2] - (E[X + Y])^2 \\ &= E[X^2] + 2E[XY] + E[Y^2] - \bar{X}^2 - 2\bar{X}\bar{Y} - \bar{Y}^2 \\ &= \text{VAR}[X] + \text{VAR}[Y] + 2E[XY] - 2\bar{X}\bar{Y} = \text{VAR}[X] + \text{VAR}[Y], \end{aligned}$$

where we used (a) in the last step.

c) Show that if $X_1, \dots, X_n$ are uncorrelated, then the variance of the sum is equal to the sum of the variances.

Solution: If $X_1, \dots, X_n$ are uncorrelated, then $X_1 + \dots, X_{n-1}$ is uncorrelated from X_n . Thus, from (b),

$$\text{VAR}[X_1 + \dots + X_n] = \text{VAR}[X_1 + \dots + X_{n-1}] + \text{VAR}[X_n].$$

Using this equation for $n = 2$ as the basis for induction, we see that $\text{VAR}[X_1 + \dots + X_n] = \sum_{i=1}^n \text{VAR}[X_i]$.

d) Show that independent rv's are uncorrelated.

Solution: If X and Y are independent, then $E[(X - \bar{X})(Y - \bar{Y})] = 0$, so X and Y are uncorrelated.

e) Let X, Y be identically distributed ternary valued random variables with the PMF $p_X(-1) = p_X(1) = 1/4$; $p_X(0) = 1/2$. Find a simple joint probability assignment such that X and Y are uncorrelated but dependent.

Solution: This becomes easier to understand if we express X as $|X| \cdot X_s$ where $|X|$ is the magnitude of X and X_s is ± 1 with equal probability. From the given PMF, we see that $|X|$ and X_s are independent. Similarly, let $Y = |Y| \cdot Y_s$. For reasons soon to be apparent, we

construct a joint PMF by taking $|Y| = |X|$ and by taking, X_s, Y_s , and $|X|$ to be statistically independent.

Since X and Y are zero mean, they are uncorrelated if $E[XY] = 0$. For the given joint PMF, we have

$$E[XY] = E[|X| \cdot X_s \cdot |Y| \cdot Y_s] = E[|X|^2] E[X_s Y_s] = 0,$$

where we have used the fact that X_s and Y_s are zero mean and independent. Thus X and Y are uncorrelated. On the other hand, X and Y are certainly not independent since $|X|$ and $|Y|$ are dependent and in fact identical. This should not be surprising since the correlation of two rv's is a single number, whereas many numbers are needed to specify the joint distribution. This example was constructed through the realization that the type of symmetry here between X and Y is sufficient to guarantee uncorrelatedness, but not enough to cause independence.

f) You have seen that the moment generating function of a sum of independent rv's is equal to the product of the individual moment generating functions. Give an example where this is false if the variables are uncorrelated but dependent.

Solution: We know (although we have not proven) that the MGF of a rv specifies the distribution, and thus the MGF of a dependent sum cannot be the same as the MGF of an independent sum with the same marginals. Thus any will work, but the one in (e) is quite simple.

$$\begin{aligned} g_{X+Y}(r) &= \frac{1}{8}e^{-2r} + \frac{3}{4} + \frac{1}{8}e^{2r} \\ g_X(r)g_Y(r) &= \left(\frac{1}{4}e^{-r} + \frac{1}{2} + \frac{1}{4}e^r\right)^2 = \frac{1}{16}e^{-2r} + \frac{1}{4}e^{-r} + \frac{3}{8} + \frac{1}{4}e^r + \frac{1}{16}e^{2r}. \end{aligned}$$

Exercise 1.22: Suppose X has the Poisson PMF, $p_X(n) = \lambda^n \exp(-\lambda)/n!$ for $n \geq 0$ and Y has the Poisson PMF, $p_Y(n) = \mu^n \exp(-\mu)/n!$ for $n \geq 0$. Assume that X and Y are independent. Find the distribution of $Z = X + Y$ and find the conditional distribution of Y conditional on $Z = n$.

Solution: The seemingly straightforward approach is to take the discrete convolution of X and Y (*i.e.*, the sum of the joint PMF's of X and Y for which $X + Y$ has a given value $Z = n$). Thus

$$\begin{aligned} p_Z(n) &= \sum_{k=0}^n p_X(k)p_Y(n-k) = \sum_{k=0}^n \frac{\lambda^k e^{-\lambda}}{k!} \times \frac{\mu^{n-k} e^{-\mu}}{(n-k)!} \\ &= e^{-(\lambda+\mu)} \sum_{k=0}^n \frac{\lambda^k \mu^{n-k}}{k!(n-k)!}. \end{aligned}$$

At this point, one needs some added knowledge or luck. One might hypothesize (correctly) that Z is also a Poisson rv with parameter $\lambda + \mu$; one might recognize the sum above, or one might look at an old solution. We multiply and divide the right hand expression above

by $(\lambda + \mu)^n/n!$.

$$\begin{aligned} p_Z(n) &= \frac{(\lambda + \mu)^n e^{-(\lambda + \mu)}}{n!} \sum_{k=0}^n \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{\lambda + \mu}\right)^k \left(\frac{\mu}{\lambda + \mu}\right)^{n-k} \\ &= \frac{(\lambda + \mu)^n e^{-(\lambda + \mu)}}{n!}, \end{aligned}$$

where we have recognized the sum on the right as a binomial sum.

Another approach that is actually more straightforward uses the fact (see (1.52)) that the MGF of the sum of independent rv's is the product of the MGF's of those rv's. From Table 1.2 (or a simple derivation), $g_X(r) = \exp[\lambda(e^r - 1)]$. Similarly, $g_Y(r) = \exp[\mu(e^r - 1)]$, so $g_Z(r) = \exp[(\lambda + \mu)(e^r - 1)]$. Since the MGF specifies the PMF, Z is a Poisson rv with parameter $\lambda + \mu$.

Finally, we must find $p_{Y|Z}(i | n)$. As a prelude to using Bayes' law, note that

$$p_{Z|Y}(n | i) = \Pr\{X + Y = n | Y = i\} = \Pr\{X = n - i\}.$$

Thus

$$\begin{aligned} p_{Y|Z}(i | n) &= \frac{p_Y(i)p_X(n-i)}{p_Z(n)} = \frac{\lambda^{n-i}e^{-\lambda}}{(n-i)!} \times \frac{\mu^i e^{-\mu}}{i!} \times \frac{n!}{(\mu + \lambda)^n e^{-(\lambda + \mu)}} \\ &= \binom{n}{i} \left(\frac{\lambda}{\lambda + \mu}\right)^{n-i} \left(\frac{\mu}{\lambda + \mu}\right)^i. \end{aligned}$$

Why this turns out to be a binomial PMF will be clarified when we study Poisson processes.

Exercise 1.23: a) Suppose X , Y and Z are binary rv's, each taking on the value 0 with probability 1/2 and the value 1 with probability 1/2. Find a simple example in which X , Y , Z are statistically *dependent* but are *pairwise* statistically *independent* (i.e., X , Y are statistically independent, X , Z are statistically independent, and Y , Z are statistically independent). Give $p_{XYZ}(x, y, z)$ for your example. Hint: In the simplest example, there are four joint values for x, y, z that have probability 1/4 each.

Solution: The simplest solution is also a very common relationship between 3 binary rv's. The relationship is that X and Y are IID and $Z = X \oplus Y$ where $\oplus$ is modulo two addition, i.e., addition with the table $0 \oplus 0 = 1 \oplus 1 = 0$ and $0 \oplus 1 = 1 \oplus 0 = 1$. Since Z is a function of X and Y , there are only 4 sample values, each of probability 1/4. The 4 possible sample values for (XYZ) are then (000), (011), (101) and (110). It is seen from this that all pairs of X, Y, Z are statistically independent

b) Is pairwise statistical independence enough to ensure that

$$\mathbb{E} \left[\prod_{i=1}^n X_i \right] = \prod_{i=1}^n \mathbb{E} [X_i]$$

for a set of rv's $X_1, \dots, X_n$?

Solution: No, (a) gives an example, i.e., $\mathbb{E} [XYZ] = 0$ and $\mathbb{E} [X] \mathbb{E} [Y] \mathbb{E} [Z] = 1/8$.

Exercise 1.24: Show that $\mathbb{E} [X]$ is the value of α that minimizes $\mathbb{E} [(X - \alpha)^2]$.

Solution: $E[(X - \alpha)^2] = E[X^2] - 2\alpha E[X] + \alpha^2$. This is clearly minimized over α by $\alpha = E[X]$.

Exercise 1.25: For each of the following random variables, find the endpoints r_- and r_+ of the interval for which the moment generating function $g(r)$ exists. Determine in each case whether $g(r)$ exists at r_- and r_+ . For parts a) and b) you should also find and sketch $g(r)$. For parts c) and d), $g(r)$ has no closed form.

a) Let λ, θ , be positive numbers and let X have the density.

$$f_X(x) = \frac{1}{2}\lambda \exp(-\lambda x); x \geq 0; \quad f_X(x) = \frac{1}{2}\theta \exp(\theta x); x < 0.$$

Solution: Integrating to find $g_X(r)$ as a function of λ and θ , we get

$$g_X(r) = \int_{-\infty}^0 \frac{1}{2}\theta e^{\theta x + rx} dx + \int_0^{\infty} \frac{1}{2}\lambda e^{-\lambda x + rx} dx = \frac{\theta}{2(\theta + r)} + \frac{\lambda}{2(\lambda - r)}.$$

The first integral above converges for $r > -\theta$ and the second for $r < \lambda$. Thus $r_- = -\theta$ and $r_+ = \lambda$. The MGF does not exist at either end point.

b) Let Y be a Gaussian random variable with mean m and variance σ^2 .

Solution: Calculating the MGF by completing the square in the exponent,

$$\begin{aligned} g_Y(r) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2} + ry\right) dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m-r\sigma^2)^2}{2\sigma^2} + rm + \frac{r^2\sigma^2}{2}\right) dy \\ &= \exp\left(rm + \frac{r^2\sigma^2}{2}\right), \end{aligned}$$

where the final equality arises from realizing that the other terms in the equation above represent a Gaussian density and thus have unit integral. Note that this is the same as the result in Table 1.1. This MGF is finite for all finite r so $r_- = -\infty$ and $r_+ = \infty$. Also $g_Y(r)$ is infinite at each endpoint.

c) Let Z be a nonnegative random variable with density

$$f_Z(z) = k(1+z)^{-2} \exp(-\lambda z); \quad z \geq 0,$$

where $\lambda > 0$ and $k = [\int_{z \geq 0} (1+z)^{-2} \exp(-\lambda z) dz]^{-1}$. Hint: Do not try to evaluate $g_Z(r)$. Instead, investigate values of r for which the integral is finite and infinite.

Solution: Writing out the formula for $g_Z(r)$, we have

$$g_Z(r) = \int_0^{\infty} k(1+z)^{-2} \exp((r-\lambda)z) dz.$$

This integral is clearly infinite for $r > \lambda$ and clearly finite for $r < \lambda$. For $r = \lambda$, the exponential term disappears, and we note that $(1+z)^{-2}$ is bounded for $z \leq 1$ and goes to 0 as z^{-2} as $z \rightarrow \infty$, so the integral is finite. Thus r_+ belongs to the region where $g_Z(r)$ is finite.

The whole point of this is that the random variables for which $r_+ = \lambda$ are those for which the density or PMF go to 0 with increasing z as $e^{-\lambda z}$. Whether or not $g_Z(\lambda)$ is finite depends on the coefficient of $e^{-\lambda z}$.

d) For the Z of (c), find the limit of $\gamma'(r)$ as r approaches λ from below. Then replace $(1+z)^2$ with $|1+z|^3$ in the definition of $f_Z(z)$ and K and show whether the above limit is then finite or not. Hint: no integration is required.

Solution: Differentiating $g_Z(r)$ with respect to r ,

$$g'_Z(r) = \int_0^\infty kz(1+z)^{-2} \exp((r-\lambda)z) dz.$$

For $r = \lambda$, the above integrand approaches 0 as $1/z$ and thus the integral does not converge. In other words, although $g_Z(\lambda)$ is finite, the slope of $g_Z(r)$ is unbounded as $r \rightarrow \lambda$ from below. If $(1+z)^{-2}$ is replaced with $(1+z)^{-3}$ (with k modified to maintain a probability density), we see that as $z \rightarrow \infty$, $z(1+z)^{-3}$ goes to 0 as $1/z^2$, so the integral converges. Thus in this case the slope of $g_Z(r)$ remains bounded for $r < \lambda$.

Exercise 1.26: **a)** Assume that the random variable X has a moment generating function $g_X(r)$ that is finite in the interval (r_-, r_+) , $r_- < 0 < r_+$, and assume $r_- < r < r_+$ throughout. For any finite constant c , express the moment generating function of $X - c$, i.e., $g_{(X-c)}(r)$ in terms of the moment generating function of X . Show that $g''_{(X-c)}(r) \geq 0$.

Solution: Note that $g_{(X-c)}(r) = E[\exp(r(X-c))] = g_X(r)e^{-cr}$. Thus r_+ and r_- are the same for X and $X - c$. Thus (see Footnote 24), the derivatives of $g_{(X-c)}(r)$ with respect to r are finite. The first two derivatives are then given by

$$g'_{(X-c)}(r) = E[(X-c)\exp(r(X-c))],$$

$$g''_{(X-c)}(r) = E[(X-c)^2 \exp(r(X-c))] \geq 0,$$

since $(X-c)^2 \exp(r(X-c)) \geq 0$ for all x .

b) Show that $g''_{(X-c)}(r) = [g''_X(r) - 2cg'_X(r) + c^2g_X(r)]e^{-rc}$.

Solution: Writing $(X-c)^2$ as $X^2 - 2cX + c^2$, we get

$$\begin{aligned} g''_{(X-c)}(r) &= E[X^2 \exp(r(X-c))] - 2cE[X \exp(r(X-c))] + c^2E[\exp(r(X-c))] \\ &= [E[X^2 \exp(rX)] - 2cE[X \exp(rX)] + c^2E[\exp(rX)]] \exp(-rc) \\ &= [g''_X(r) - 2cg'_X(r) + c^2g_X(r)] \exp(-rc). \end{aligned}$$

c) Use **a)** and **b)** to show that $g''_X(r)g_X(r) - [g'_X(r)]^2 \geq 0$, and that $\gamma''_X(r) \geq 0$. Hint: Let $c = g'_X(r)/g_X(r)$.

Solution: With the suggested choice for c ,

$$\begin{aligned} g''_{(X-c)}(r) &= \left[g''_X(r) - 2\frac{(g'_X(r))^2}{g_X(r)} + \frac{(g'_X(r))^2}{g_X(r)} \right] \exp(-rc) \\ &= \left[\frac{g_X(r)g''_X(r) - [g'_X(r)]^2}{g_X(r)} \right] \exp(-cr). \end{aligned}$$

Since this is nonnegative from (a), we see that

$$\gamma_X''(r) = g_X(r)g_X''(r) - [g_X'(r)]^2 \geq 0.$$

d) Assume that X is non-atomic, *i.e.*, that there is no value of c such that $\Pr\{X = c\} = 1$. Show that the inequality sign “ $\geq$ ” may be replaced by “ $>$ ” everywhere in a), b) and c).

Solution: Since X is non-atomic, $(X - c)$ must be non-zero with positive probability, and thus from (a), $g_{(X-c)}''(r) > 0$. Thus the inequalities in parts b) and c) are strict also.

Exercise 1.27: A computer system has n users, each with a unique name and password. Due to a software error, the n passwords are randomly permuted internally (*i.e.* each of the $n!$ possible permutations are equally likely. Only those users lucky enough to have had their passwords unchanged in the permutation are able to continue using the system.)

a) What is the probability that a particular user, say user 1, is able to continue using the system?

Solution: The obvious (and perfectly correct) solution is that user 1 is equally likely to be assigned any one of the n passwords in the permutation, and thus has probability $1/n$ of being able to continue using the system.

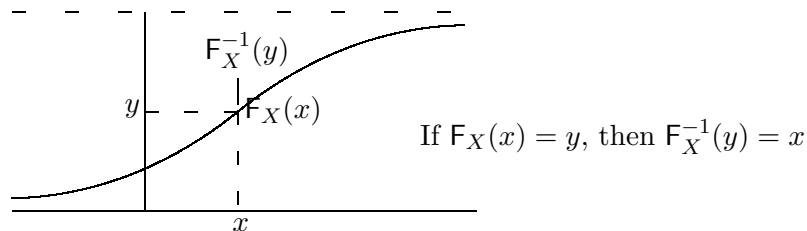
This might be somewhat unsatisfying since it does not reason directly from the property that all $n!$ permutations are equally likely. Thus a more direct argument is that the number of permutations starting with 1 is the number of permutations of the remaining $(n - 1)$ users, *i.e.*, $(n - 1)!$. These are equally likely, each with probability $1/n!$, so the probability that user 1 (or any given user i) can continue to use the system is $1/n$.

b) What is the expected number of users able to continue using the system? Hint: Let X_i be a rv with the value 1 if user i can use the system and 0 otherwise.

Solution: We have just seen that $\Pr\{X_i = 1\} = 1/n$, so $\Pr\{X_i = 0\} = 1 - 1/n$. Thus $E[X_i] = 1/n$. The number of users who can continue to use the system is $S_n = \sum_{i=1}^n X_i$, so $E[S_n] = \sum_i E[X_i] = n/n = 1$. It is important to understand here that the X_i are not independent, and that the expected value of a finite sum of rv's is equal to the sum of the expected values whether or not the rv's are independent. For example, with two users, either both can continue or neither.

Exercise 1.28: Suppose the rv X is continuous and has the CDF $F_X(x)$. Consider another rv $Y = F_X(X)$. That is, for each sample point ω such that $X(\omega) = x$, we have $Y(\omega) = F_X(x)$. Show that Y is uniformly distributed in the interval 0 to 1.

Solution: For simplicity, first assume that $F_X(x)$ is strictly increasing in x , thus having the following appearance:



Since $F_X(x)$ is continuous in x and strictly increasing from 0 to 1, there must be an inverse function F_X^{-1} such that for each $y \in (0, 1)$, $F_X^{-1}(y) = x$ for that x such that $F_X(x) = y$. For this y , then, the event $\{F_X(X) \leq y\}$ is the same as the event $\{X \leq F_X^{-1}(y)\}$. This is illustrated in the figure above. Using this equality for the given y ,

$$\begin{aligned} \Pr\{Y \leq y\} &= \Pr\{F_X(X) \leq y\} = \Pr\{X \leq F_X^{-1}(y)\} \\ &= F_X(F_X^{-1}(y)) = y. \end{aligned}$$

where in the final equation, we have used the fact that F_X^{-1} is the inverse function of F_X . This relation, for all $y \in (0, 1)$, shows that Y is uniformly distributed between 0 and 1.

If F_X is not strictly increasing, *i.e.*, if there is any interval over which $F_X(x)$ has a constant value y , then we can define $F_X^{-1}(y)$ to have any given value within that interval. The above argument then still holds, although F_X^{-1} is no longer the inverse of F_X .

If there is any discrete point, say z at which $\Pr\{X = z\} > 0$, then $F_X(x)$ cannot take on values in the open interval between $F_X(z) - a$ and $F_X(z)$ where $a = \Pr\{X = z\}$. Thus F_X is uniformly distributed only for continuous rv's.

Exercise 1.29: Let Z be an integer-valued rv with the PMF $p_Z(n) = 1/k$ for $0 \leq n \leq k-1$. Find the mean, variance, and moment generating function of Z . Hint: An elegant way to do this is to let U be a uniformly distributed continuous rv over $(0, 1]$ that is independent of Z . Then $U + Z$ is uniform over $(0, k]$. Use the known results about U and $U + Z$ to find the mean, variance, and MGF for Z .

Solution: It is not particularly hard to find the mean, variance, and MGF of Z directly, but the technique here is often useful in relating continuous rv's to their quantized versions. We have

$$\begin{aligned} E[Z] &= E[U + Z] - E[U] = k/2 - 1/2 = (k-1)/2 \\ \sigma_Z^2 &= \sigma_{(U+Z)}^2 - \sigma_U^2 = k^2/12 - 1/12 = (k^2-1)/12 \\ g_Z(r) &= g_{(Z+U)}(r)/g_U(r) = \frac{\exp(rk) - 1}{rk} \times \frac{r}{\exp(r) - 1} \\ &= \frac{\exp(kr) - 1}{k[\exp(r) - 1]}. \end{aligned}$$

Exercise 1.30: (Alternate approach 1 to the Markov inequality) a) Let Y be a nonnegative rv and $y > 0$ be some fixed number. Let A be the event that $Y \geq y$. Show that $y\mathbb{I}_A \leq Y$ (*i.e.*, that this inequality is satisfied for every $\omega \in \Omega$).

Solution If $Y \geq y$, then $\mathbb{I}_A = 1$ and $Y \geq y\mathbb{I}_A$. Alternatively, if $Y < y$, then $\mathbb{I}_A = 0$ and $Y \geq Y\mathbb{I}_A$ again since Y is nonnegative. Thus in all cases, $Y \geq y\mathbb{I}_A$.

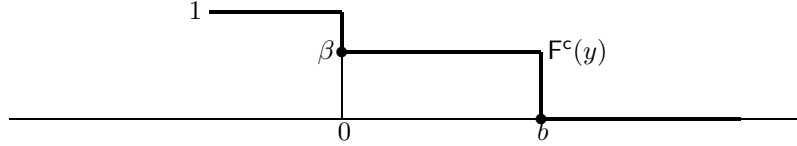
b) Use your result in (a) to prove the Markov inequality.

Solution Taking the expected value of both sides of (a), $E[Y] \geq yE[\mathbb{I}_A]$. Since $E[\mathbb{I}_A] = \Pr\{A \geq y\}$, the Markov inequality follows.

Exercise 1.31: (Alternate approach 2 to the Markov inequality) a) Minimize $E[Y]$ over all

nonnegative rv's such that $\Pr\{Y \geq b\} = \beta$ for some given $b > 0$ and $0 < \beta < 1$. Hint: Use a graphical argument similar to that in Figure 1.7. What is the rv that achieves the minimum? Hint: It is binary.

Solution: Since $E[Y] = \int_0^\infty F^c(y) dy$, we can try to minimize $E[Y]$ by minimizing $F^c(y)$ at each y subject to the constraints on $F^c(y)$. One constraint is that $F^c(y)$ is nonincreasing in y and the other is that $\Pr\{Y \geq b\} = \beta$. Choosing $F^c(y) = \beta$ for $0 \leq y < b$ and $F^c(y) = 0$ for $y \geq b$ satisfies these constraints and $F^c(y)$ cannot be further reduced anywhere. The minimizing complementary distribution function is then sketched below:



The rv Y is then recognized as binary with the sample values 0 and b and the probabilities $p_Y(0) = 1 - \beta$ and $p_Y(b) = \beta$. The minimizing $E[Y]$ is then βb .

b) Use (a) to prove the Markov inequality and also point out the distribution that meets the inequality with equality.

Solution: We have seen that $E[Y] \geq \beta b = b\Pr\{Y \geq b\}$, which, on dividing by b , is the Markov inequality. Thus, for any $b > 0$ and any $\Pr\{Y \geq b\}$ there is a binary rv that meets the Markov bound with equality.

Exercise 1.32: (The one-sided Chebyshev inequality) This inequality states that if a zero-mean rv X has a variance σ^2 , then it satisfies the inequality

$$\Pr\{X \geq b\} \leq \frac{\sigma^2}{\sigma^2 + b^2} \quad \text{for every } b > 0, \quad (\text{A.7})$$

with equality for some b only if X is binary and $\Pr\{X = b\} = \sigma^2/(\sigma^2 + b^2)$. We prove this here using the same approach as in Exercise 1.31. Let X be a zero-mean rv that satisfies $\Pr\{X \geq b\} = \beta$ for some given $b > 0$ and $0 < \beta < 1$. The variance σ^2 of X can be expressed as

$$\sigma^2 = \int_{-\infty}^{b^-} x^2 f_X(x) dx + \int_b^\infty x^2 f_X(x) dx. \quad (\text{A.8})$$

We will first minimize σ^2 over all zero-mean X satisfying $\Pr\{X \geq b\} = \beta$.

a) Show that the second integral in (A.8) satisfies $\int_b^\infty x^2 f_X(x) dx \geq b^2 \beta$.

Solution: We first should explain why we started trying to establish an upper bound to $\Pr\{X \geq b\}$ and then switched to minimizing the variance such that $\bar{X} = 0$ and $\Pr\{X \geq b\} = \beta$ for some given β . We will find that $\sigma_{\min}^2 = b^2 \beta / (1 - \beta)$, or, equivalently, $\beta = \sigma_{\min}^2 / (b^2 + \sigma_{\min}^2)$. Thus for each zero-mean X satisfying $\Pr\{X \geq b\} = \beta$, we have (A.8), i.e.,

$$\Pr\{X \geq b\} = \beta = \frac{\sigma_{\min}^2}{\sigma_{\min}^2 + b^2} \leq \frac{\sigma_X^2}{\sigma_X^2 + b^2}.$$

Solving part (a),

$$\int_b^\infty x^2 f_X(x) dx \geq \int_b^\infty b^2 f_X(x) dx = b^2 F_X^c(b) = b^2 \beta.$$

b) Show that the first integral in (A.8) is constrained by

$$\int_{-\infty}^{b^-} f_X(x) dx = 1 - \beta \quad \text{and} \quad \int_{-\infty}^{b^-} x f_X(x) dx \leq -b\beta.$$

Solution: $\int_{-\infty}^{b^-} f_X(x) dx = 1 - \int_b^{\infty} f_X(x) dx = 1 - \beta$. Similarly, since $\bar{X} = 0$,

$$\int_{-\infty}^{b^-} x f_X(x) dx = 0 - \int_b^{\infty} x f_X(x) dx \leq - \int_b^{\infty} b f_X(x) dx = -b\beta.$$

c) Minimize the first integral in (A.8) subject to the constraints in (b). Hint: If you scale $f_X(x)$ up by $1/(1 - \beta)$, it integrates to 1 over $(-\infty, b)$ and the second constraint becomes an expectation. You can then minimize the first integral in (A.8) by inspection.

Solution: Let $h(x) = f_X(x)/(1 - \beta)$ over $x \leq b$ and $h(x) = 0$ elsewhere. Then $h(x)$ is a PDF and the corresponding mean is $-b\beta/(1 - \beta)$. The corresponding second moment is lower bounded by the square of that mean, so

$$\int_{-\infty}^{b^-} x^2 h(x) dx \geq \frac{b^2 \beta^2}{(1 - \beta)^2}.$$

Since $h(x) = f_X(x)/(1 - \beta)$, we have $\int_{-\infty}^{b^-} x^2 f_X(x) dx \geq b^2 \beta^2 / (1 - \beta)$.

d) Combine the results in a) and c) to show that $\sigma^2 \geq b^2 \beta / (1 - \beta)$. Find the minimizing distribution. Hint: It is binary.

Solution: Substituting the results of (a) and (c) into (A.8)

$$\sigma^2 \geq \frac{b^2 \beta^2}{1 - \beta} + b^2 \beta = \frac{b^2 \beta}{1 - \beta}. \quad (\text{A.9})$$

This is met with equality when the inequalities in (a) and (c) are met with equality. Thus $X = b$ whenever $X \geq b$ and $X = -b\beta/(1 - \beta)$ when $X < b$. Since β is the probability that $X \geq b$, we see that $p_X(b) = \beta$ and $p_X(-b\beta/(1 - \beta)) = 1 - \beta$.

e) Use (d) to establish (A.7). Also show (trivially) that if Y has a mean $\bar{Y}$ and variance σ^2 , then $\Pr\{Y - \bar{Y} \geq b\} \leq \sigma^2 / (\sigma^2 + b^2)$.

Solution: From (A.9), $\sigma^2(1 - \beta) \geq b^2 \beta$, from which $\beta \leq \sigma^2 / (b^2 + \sigma^2)$, i.e., $\Pr\{X \geq b\} \leq \sigma^2 / (b^2 + \sigma^2)$. The conditions for equality are clearly the same as in (d). The final result follows by letting X be the fluctuation of Y and applying (A.7) to X .

Exercise 1.33: (Proof of (1.48)) Here we show that if X is a zero-mean rv with a variance σ^2 , then the median α satisfies $|\alpha| \leq \sigma$.

a) First show that $|\alpha| \leq \sigma$ for the special case where X is binary with equiprobable values at $\pm\sigma$.

Solution: In this special case, $F_X(x) = 1/2$ for $-\sigma \leq x \leq \sigma$, and any value in this range satisfies the condition to be a median.

b) For all zero-mean rv's X with variance σ^2 other than the special case in a), show that

$$\Pr\{X \geq \sigma\} < 0.5.$$

Hint: Use the one-sided Chebyshev inequality of Exercise 1.32.

Solution: The one-sided Chebyshev inequality for zero-mean X says that $\Pr\{X \geq b\} \leq \sigma^2/(b^2 + \sigma^2)$. Letting $b = \sigma$, this says that $\Pr\{X \geq \sigma\} \leq 0.5$. The condition for equality here (as shown in Exercise 1.32) is that X is binary with values $\pm\sigma$. Thus, outside of that special case, $\Pr\{X \geq \sigma\} < 0.5$.

c) Show that $\Pr\{X \geq \alpha\} \geq 0.5$. Other than the special case in a), show that this implies that $\alpha < \sigma$.

Solution: By definition of a median, $\Pr\{X \geq \alpha\} \geq 0.5$ and $\Pr\{X \leq \alpha\} \geq 0.5$. Since (b) (outside of the special case of (a)) shows that $\Pr\{X \geq \sigma\} < 0.5$, the median must be smaller than σ .

d) Other than the special case in a), show that $|\alpha| < \sigma$. Hint: repeat b) and c) for the rv $-X$. You have then shown that $|\alpha| \leq \sigma$ with equality only for the binary case with values $\pm\sigma$. For rv's Y with a non-zero mean, this shows that $|\alpha - \bar{Y}| \leq \sigma$.

Solution: Note that $-X$ is also zero-mean with variance σ^2 , so (outside of the case in (a)), $\Pr\{-X \geq \sigma\} = \Pr\{X \leq -\sigma\} < 0.5$. This implies that the median is greater than $-\sigma$. Thus $|\alpha| \leq \sigma$ with equality only in the binary equiprobable case. For Y with a non-zero mean, we simply apply this result to the fluctuation of Y .

Exercise 1.34: We stressed the importance of the mean of a rv X in terms of its association with the sample average via the WLLN. Here we show that there is a form of WLLN for the median and for the entire CDF, say $F_X(x)$ of X via sufficiently many independent sample values of X .

a) For any given x , let $\mathbb{I}_j(x)$ be the indicator function of the event $\{X_j \leq x\}$ where $X_1, X_2, \dots, X_j, \dots$ are IID rv's with the CDF $F_X(x)$. State the WLLN for the IID rv's $\{\mathbb{I}_1(x), \mathbb{I}_2(x), \dots\}$.

Solution: The mean value of $\mathbb{I}_j(x)$ is $F_X(x)$ and the variance (after a short calculation) is $F_X(x)F_X^c(x)$. This is finite (and in fact at most $1/4$), so Theorem 1.7.1 applies and

$$\lim_{n \rightarrow \infty} \Pr \left\{ \left| \frac{1}{n} \sum_{j=1}^n \mathbb{I}_j(x) - F_X(x) \right| > \epsilon \right\} = 0 \quad \text{for all } x \text{ and } \epsilon > 0. \quad (\text{A.10})$$

This says that if we take n samples of X and use $(1/n) \sum_{j=1}^n \mathbb{I}_j(x)$ to approximate the CDF $F_X(x)$ at each x , then the probability that the approximation error exceeds ϵ at any given x approaches 0 with increasing n .

b) Does the answer to (a) require X to have a mean or variance?

Solution: No. As pointed out in a), $\mathbb{I}_j(x)$ has a mean and variance whether or not X does, so Theorem 1.7.1 applies.

c) Suggest a procedure for evaluating the median of X from the sample values of $X_1, X_2, \dots$. Assume that X is a continuous rv and that its PDF is positive in an open interval around the median. You need not be precise, but try to think the issue through carefully.

What you have seen here, without stating it precisely or proving it is that the median has a law of large numbers associated with it, saying that the sample median of n IID samples of a rv is close to the true median with high probability.

Solution: Note that $(1/n) \sum_{j=1}^n \mathbb{I}_j(y)$ is a rv for each y . Any sample function $x_1, \dots, x_n$ of $X_1, \dots, X_n$ maps into a sample value of $(1/n) \sum_{j=1}^n \mathbb{I}_j(y)$ for each y . We can view this

collection of sample values as a function of y . Any such sample function is non-decreasing in y , and as seen in (a) is an approximation to $F_X(y)$ at each y . This function of y has all the characteristics of a CDF itself, so we can let $\hat{\alpha}_n$ be the median of $(1/n) \sum_{j=1}^n \mathbb{I}_j(y)$ as a function of y . Let α be the true median of X and let $\delta > 0$ be arbitrary. Note that if $(1/n) \sum_{j=1}^n \mathbb{I}_j(\alpha - \delta) < .5$, then $\hat{\alpha}_n > \alpha - \delta$. Similarly, if $(1/n) \sum_{j=1}^n \mathbb{I}_j(\alpha + \delta) > .5$, then $\hat{\alpha}_n < \alpha + \delta$. Thus,

$$\Pr\{|\hat{\alpha}_n - \alpha| \geq \delta\} \leq \Pr\left\{\frac{1}{n} \sum_{j=1}^n \mathbb{I}_j(\alpha - \delta) \geq .5\right\} + \Pr\left\{\frac{1}{n} \sum_{j=1}^n \mathbb{I}_j(\alpha + \delta) \leq .5\right\}.$$

Because of the assumption of a nonzero density, there is some $\epsilon_1 > 0$ such that $F_X(\alpha - \delta) < .5 - \epsilon_1$ and some $\epsilon_2 > 0$ such that $F_X(\alpha + \delta) > .5 + \epsilon_2$. Thus,

$$\begin{aligned} \Pr\{|\hat{\alpha}_n - \alpha| \geq \delta\} &\leq \Pr\left\{\left|\frac{1}{n} \sum_{j=1}^n \mathbb{I}_j(\alpha - \delta) - F_X(\alpha - \delta)\right| > \epsilon_1\right\} \\ &+ \Pr\left\{\left|\frac{1}{n} \sum_{j=1}^n \mathbb{I}_j(\alpha + \delta) - F_X(\alpha + \delta)\right| > \epsilon_2\right\}. \end{aligned}$$

From (A.10), the limit of this as $n \rightarrow \infty$ is 0, which is a WLLN for the median. With a great deal more fussing, the same result holds true without the assumption of a positive density if we allow α above to be any median in cases where the median is nonunique.

Exercise 1.35 a) Show that for any integers $0 < k < n$,

$$\binom{n}{k+1} \leq \binom{n}{k} \frac{n-k}{k}.$$

Solution:

$$\begin{aligned} \binom{n}{k+1} &= \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!}{k!(k+1)(n-k)!/(n-k)} \\ &= \binom{n}{k} \frac{n-k}{k+1} \leq \binom{n}{k} \frac{n-k}{k}. \end{aligned} \tag{A.11}$$

b) Extend (a) to show that, for all $\ell \leq n-k$,

$$\binom{n}{k+\ell} \leq \binom{n}{k} \left[\frac{n-k}{k}\right]^\ell. \tag{A.12}$$

Solution: Using $k+\ell$ in place of $k+1$ in (A.11),

$$\binom{n}{k+\ell} \leq \binom{n}{k+\ell-1} \left[\frac{n-k-(\ell-1)}{k+\ell-1}\right] \leq \binom{n}{k+\ell-1} \left[\frac{n-k}{k}\right].$$

Applying recursion on ℓ , we get (A.12).

c) Let $\tilde{p} = k/n$ and $\tilde{q} = 1 - \tilde{p}$. Let S_n be the sum of n binary IID rv's with $\mathbf{p}_X(0) = q$ and $\mathbf{p}_X(1) = p$. Show that for all $\ell \leq n - k$,

$$\mathbf{p}_{S_n}(k + \ell) \leq \mathbf{p}_{S_n}(k) \left(\frac{\tilde{q}p}{\tilde{p}q} \right)^\ell. \quad (\text{A.13})$$

Solution: Using (b),

$$\mathbf{p}_{S_n}(k + \ell) = \binom{n}{k + \ell} p^{k + \ell} q^{n - k - \ell} \leq \binom{n}{k} \left[\frac{n - k}{k} \right]^\ell \left[\frac{p}{q} \right]^\ell p^k q^{n - k} = \mathbf{p}_{S_n}(k) \left[\frac{\tilde{q}p}{\tilde{p}q} \right]^\ell.$$

d) For $k/n > p$, show that $\Pr\{S_n \geq k\} \leq \frac{\tilde{p}q}{\tilde{p} - p} \mathbf{p}_{S_n}(k)$.

Solution: Using the bound in (A.13), we get

$$\begin{aligned} \Pr\{S_n \geq k\} &= \sum_{\ell=0}^{n-k} \mathbf{p}_{S_n}(k + \ell) \leq \sum_{\ell=0}^{n-k} \mathbf{p}_{S_n}(k) \left(\frac{\tilde{q}p}{\tilde{p}q} \right)^\ell \\ &\leq \mathbf{p}_{S_n}(k) \frac{1}{1 - \tilde{q}p/\tilde{p}q} = \mathbf{p}_{S_n}(k) \frac{\tilde{p}q}{\tilde{p}q - \tilde{q}p} \\ &= \mathbf{p}_{S_n}(k) \frac{\tilde{p}q}{\tilde{p}(1 - p) - (1 - \tilde{p})p} = \mathbf{p}_{S_n}(k) \frac{\tilde{p}q}{\tilde{p} - p}. \end{aligned} \quad (\text{A.14})$$

e) Now let ℓ be fixed and $k = \lceil n\tilde{p} \rceil$ for fixed $\tilde{p}$ such that $1 > \tilde{p} > p$. Argue that as $n \rightarrow \infty$,

$$\mathbf{p}_{S_n}(k + \ell) \sim \mathbf{p}_{S_n}(k) \left(\frac{\tilde{q}p}{\tilde{p}q} \right)^\ell \quad \text{and} \quad \Pr\{S_n \geq k\} \sim \frac{\tilde{p}q}{\tilde{p} - p} \mathbf{p}_{S_n}(k),$$

where $a(n) \sim b(n)$ means that $\lim_{n \rightarrow \infty} a(n)/b(n) = 1$.

Solution: Note that (A.12) provides an upper bound to $\binom{n}{k + \ell}$ and a slight modification of the same argument provides the lower bound $\binom{n}{k + \ell} \geq \binom{n}{k} \left(\frac{n - k - \ell}{k + \ell} \right)^\ell$. Taking the ratio of the upper to lower bound for fixed ℓ and $\tilde{p}$ as $n \rightarrow \infty$, we see that

$$\begin{aligned} \binom{n}{k + \ell} &\sim \binom{n}{k} \left[\frac{n - k}{k} \right]^\ell \quad \text{so that} \\ \mathbf{p}_{S_n}(k + \ell) &\sim \mathbf{p}_{S_n}(k) (\tilde{q}p/\tilde{p}q)^\ell \end{aligned} \quad (\text{A.15})$$

follows. Replacing the upper bound in (A.14) with the asymptotic equality in (A.15), and letting ℓ grow very slowly with n , we get $\Pr\{S_n \geq k\} \sim \frac{\tilde{p}q}{\tilde{p} - p} \mathbf{p}_{S_n}(k)$.

Exercise 1.36 A sequence $\{a_n; n \geq 1\}$ of real numbers has the limit 0 if for all $\epsilon > 0$, there is an $m(\epsilon)$ such that $|a_n| \leq \epsilon$ for all $n \geq m(\epsilon)$. Show that the sequences in parts a) and b) below satisfy $\lim_{n \rightarrow \infty} a_n = 0$ but the sequence in (c) does not have a limit.

a) $a_n = \frac{1}{\ln(\ln(n+1))}$

Solution: This is very tedious, but most of us who have never done any analysis must carry out this kind of detail a few times before seeing automatically what is going on. For $a_n \leq \epsilon$, we have $\ln(\ln(n+1)) \geq 1/\epsilon$. Thus we must have $\ln(n+1) \geq e^{1/\epsilon}$, or, finally

$$n \geq 1 + \exp(\exp(1/\epsilon)).$$

Taking $m(\epsilon)$ to be the right-hand side (which is then finite for all $\epsilon > 0$, we see that $a_n \leq \epsilon$ for all $n \geq m(\epsilon)$.

b) $a_n = n^{10} \exp(-n)$

Solution: The technique in (a) doesn't work easily here, but the following criterion is often useful and applies here: If $b_n \geq a_n$ for all n greater than some n_0 , and if $\lim_{n \rightarrow \infty} b_n = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$ also. For the sequence here, let $b_n = \exp(-n/2)$. For large enough n , $n^{10} = \exp(10 \ln n) \leq \exp(n/2)$, and this implies that $a_n \leq b_n$ for all large enough n . Thus $\lim_{n \rightarrow \infty} a_n = 0$.

c) $a_n = 1$ for $n = 10^\ell$ for each positive integer ℓ and $a_n = 0$ otherwise.

Solution: Here, for any $\epsilon < 1$, $a_n > \epsilon$ for all n of the form $n = 10^\ell$ for ℓ integer. Since $a_n = 0$ for all other n , $\lim_{n \rightarrow \infty} a_n$ does not exist.

d) Show that the definition can be changed (with no change in meaning) by replacing ϵ with either $1/k$ or 2^{-k} for every positive integer k .

Solution: Suppose the original definition holds. Then for any k , we can let $\epsilon = 1/k$, and know that there is an $m(1/k)$ such that $a_n \leq 1/k$ for all $n \geq m(1/k)$. Conversely, suppose that for every k , there is an m_k such that $a_n \leq 1/k$ for all $n \geq m_k$. Then for every $\epsilon > 0$, we can choose a k such that $\epsilon > 1/k$. For this k , $a_n \leq \epsilon$ for all $n \geq m_k$. The same argument holds when ϵ is replaced by 2^{-k} .

Exercise 1.37: Represent the MGF of a rv X by

$$g_X(r) = \int_{-\infty}^0 e^{rx} dF(x) + \int_0^{\infty} e^{rx} dF(x).$$

In each of the following parts, you are welcome to restrict X to be either discrete or continuous.

a) Show that the first integral always exists (*i.e.*, is finite) for $r \geq 0$ and that the second integral always exists for $r \leq 0$.

Solution: For $r \geq 0$ and $x \leq 0$, we have $0 \leq e^{rx} \leq 1$. Thus the first integral lies between 0 and 1 and is thus finite. Similarly, for $r \leq 0$ and $x \geq 0$, we have $0 \leq e^{rx} \leq 1$, so the second integral is finite in this range.

b) Show that if the second integral exists for a given $r_1 > 0$, then it also exists for all r in the range $0 \leq r \leq r_1$.

Solution: For $x \geq 0$ (*i.e.*, in the domain of the first integral), we have $e^{rx} \leq e^{r_1 x}$, so we also have $0 \leq \int_0^{\infty} e^{rx} dF(x) \leq \int_0^{\infty} e^{r_1 x} dF(x)$. Thus the second integral exists for all $r \in [0, r_1]$.

c) Show that if the first integral exists for a given $r_2 < 0$, then it also exists for all r in the range $r_2 \leq r \leq 0$.

Solution: This follows in the same way as (b).

d) Show that the domain of r over which $g_X(r)$ exists is an interval from some $r_- \leq 0$ to some $r_+ \geq 0$ (the interval might or might not include each endpoint, and either or both end points might be 0, ∞ , or any point between).

Solution: Combining parts a) and b), we see that if the second integral exists for some $r_1 > 0$, then it exists for all $r \leq r_1$. Defining r_+ as the supremum of values of r for which

the second integral exists, we see that the second integral exists for all $r < r_+$. Since the second integral exists for $r = 0$, we see that $r_+ \geq 0$.

Similarly, the first integral exists for all $r > r_-$ where $r_- \leq 0$ is the infimum of r such that the first integral exists. From (a), both integrals then exist for $r \in (r_-, r_+)$.

e) Find an example where $r_+ = 1$ and the MGF does not exist for $r = 1$. Find another example where $r_+ = 1$ and the MGF does exist for $r = 1$. Hint: Consider $f_X(x) = e^{-x}$ for $x \geq 0$ and figure out how to modify it to $f_Y(y)$ so that $\int_0^\infty e^y f_Y(y) dy < \infty$ but $\int_0^\infty e^{y+\epsilon y} f_Y(y) dy = \infty$ for all $\epsilon > 0$.

Solution: Note: The hint should not have been necessary here. Anyone trying to imagine why $\int_0^\infty e^{rx} f(x) dx$ should converge for some r and not for others who doesn't consider exponentials was up too late the previous night. We observe that the second integral, $\int_0^\infty \exp(rx - x) dx$ converges for $r < 1$ and diverges for $r \geq 1$, so r_+ is not in the domain of convergence.

How do we modify e^{-x} for $x \geq 0$ to make $\int_0^\infty f(x) \exp(xr) dx$ converge for $r = 1$ but to diverge for all $r > 1$? We need an $f(x)$ that goes to 0 with increasing x just a little faster than e^{-x} , but it has to go to zero more slowly than $e^{-(1+\epsilon)x}$ for any $\epsilon > 0$. There are many choices, such as $\exp(-x - \sqrt{x})$ or $(1 + x^2)^{-1} \exp(-x)$. Each of these requires a normalizing constant to make the function a probability density, but there is no need to calculate the constant.

Exercise 1.38: Let $\{X_n; n \geq 1\}$ be a sequence of independent but not identically distributed rv's. We say that the weak law of large numbers (WLLN) holds for this sequence if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \Pr \left\{ \left| \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n} \right| \geq \epsilon \right\} = 0 \quad \text{where } S_n = X_1 + X_2 + \cdots + X_n.$$

a) Show that the WLLN holds if there is some constant A such that $\sigma_{X_n}^2 \leq A$ for all n .

Solution: From (1.41), the variance of S_n is given by $\sigma_{S_n}^2 = \sum_{i=1}^n \sigma_{X_i}^2$. Since $\sigma_{X_i}^2 \leq A$, we have $\sigma_{S_n}^2 \leq nA$ and thus $\sigma_{S_n/n}^2 \leq A/n$. Thus, from the Chebyshev inequality,

$$\Pr \left\{ \left| \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n} \right| \geq \epsilon \right\} \leq \frac{A}{n\epsilon^2}.$$

Then, exactly as in the proof of the WLLN for IID rv's, the quantity on the right approaches 0 as $n \rightarrow \infty$ for all $\epsilon > 0$, completing the demonstration.

b) Suppose that $\sigma_{X_n}^2 \leq A n^{1-\alpha}$ for some α , $0 < \alpha < 1$ and for all $n \geq 1$. Show that the WLLN holds in this case.

Solution: This is similar to (a), but it is considerably stronger, since it says that the variance of the X_i can be slowly increasing in i , and if this increase is bounded as above, the WLLN still holds. Since $An^{1-\alpha}$ is increasing in n , $\sigma_{X_i}^2 \leq An^{1-\alpha}$ for $i \leq n$. Thus,

$$\begin{aligned} \sigma_{S_n}^2 &= \sum_{i=1}^n \sigma_{X_i}^2 \leq nAn^{1-\alpha} \leq An^{2-\alpha} \\ \sigma_{S_n/n}^2 &\leq An^{-\alpha}. \end{aligned}$$

From the Chebyshev inequality,

$$\Pr\left\{\left|\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\right| \geq \epsilon\right\} \leq \frac{An^{-\alpha}}{\epsilon^2},$$

which for any $\epsilon > 0$ approaches 0 as $n \rightarrow \infty$.

Exercise 1.39: Let $\{X_i; i \geq 1\}$ be IID binary rv's. Let $\Pr\{X_i = 1\} = \delta$, $\Pr\{X_i = 0\} = 1 - \delta$. Let $S_n = X_1 + \dots + X_n$. Let m be an arbitrary but fixed positive integer. Think! then evaluate the following and explain your answers:

a) $\lim_{n \rightarrow \infty} \sum_{i: n\delta - m \leq i \leq n\delta + m} \Pr\{S_n = i\}.$

Solution: It is easier to reason about the problem if we restate the sum in the following way:

$$\begin{aligned} \sum_{i: n\delta - m \leq i \leq n\delta + m} \Pr\{S_n = i\} &= \Pr\{n\delta - m \leq S_n \leq n\delta + m\} \\ &= \Pr\{-m \leq S_n - n\bar{X} \leq m\} \\ &= \Pr\left\{\frac{-m}{\sigma\sqrt{n}} \leq \frac{S_n - n\bar{X}}{\sigma\sqrt{n}} \leq \frac{m}{\sigma\sqrt{n}}\right\}, \end{aligned}$$

where σ is the standard deviation of X . Now in the limit $n \rightarrow \infty$, $(S_n - n\bar{X})/\sigma\sqrt{n}$ approaches a normalized Gaussian rv in distribution, *i.e.*,

$$\lim_{n \rightarrow \infty} \Pr\left\{\frac{-m}{\sigma\sqrt{n}} \leq \frac{S_n - n\bar{X}}{\sigma\sqrt{n}} \leq \frac{m}{\sigma\sqrt{n}}\right\} = \lim_{n \rightarrow \infty} [\Phi(\frac{m}{\sigma\sqrt{n}}) - \Phi(\frac{-m}{\sigma\sqrt{n}})] = 0.$$

This can also be seen immediately from the binomial distribution as it approaches a discrete Gaussian distribution. We are looking only at essentially the central $2m$ terms of the binomial, and each of those terms goes to 0 as $1/\sqrt{n}$ with increasing n .

b) $\lim_{n \rightarrow \infty} \sum_{i: 0 \leq i \leq n\delta + m} \Pr\{S_n = i\}.$

Solution: Here all terms on lower side of the distribution are included and the upper side is bounded as in (a). Arguing in the same way as in (a), we see that

$$\sum_{i: 0 \leq i \leq n\delta + m} \Pr\{S_n = i\} = \Pr\left\{\frac{S_n - n\bar{X}}{\sigma\sqrt{n}} \leq \frac{m}{\sigma\sqrt{n}}\right\}.$$

In the limit, this is $\Phi(0) = 1/2$.

c) $\lim_{n \rightarrow \infty} \sum_{i: n(\delta - 1/m) \leq i \leq n(\delta + 1/m)} \Pr\{S_n = i\}.$

Solution: Here the number of terms included in the sum is increasing linearly with n , and the appropriate mechanism is the WLLN.

$$\sum_{i: n(\delta - 1/m) \leq i \leq n(\delta + 1/m)} \Pr\{S_n = i\} = \Pr\left\{-\frac{1}{m} \leq \frac{S_n - n\bar{X}}{n} \leq \frac{1}{m}\right\}.$$

In the limit $n \rightarrow \infty$, this is 1 by the WLLN. The essence of this exercise has been to scale the random variables properly to go the limit. We have used the CLT and the WLLN, but

one could guess the answers immediately by recognizing what part of the distribution is being looked at.

Exercise 1.40: The WLLN is said to hold for a zero-mean sequence $\{S_n; n \geq 1\}$ if

$$\lim_{n \rightarrow \infty} \Pr \left\{ \left| \frac{S_n}{n} \right| > \epsilon \right\} = 0 \quad \text{for every } \epsilon > 0.$$

The CLT is said to hold for $\{S_n; n \geq 1\}$ if for some $\sigma > 0$ and all $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \Pr \left\{ \frac{S_n}{\sigma\sqrt{n}} \leq z \right\} = \Phi(z),$$

where $\Phi(z)$ is the normalized Gaussian CDF. Show that if the CLT holds, then the WLLN holds also. Note 1: If you hate ϵ, δ arguments, you will hate this. Note 2: It will probably ease the pain if you convert the WLLN statement to: For every $\epsilon > 0, \delta > 0$ there exists an $n(\epsilon, \delta)$ such that for every $n \geq n(\epsilon, \delta)$,

$$\Pr \left\{ \frac{S_n}{n} < -\epsilon \right\} \leq \delta \quad \text{and} \quad \Pr \left\{ \frac{S_n}{n} > \epsilon \right\} \leq \delta. \quad (\text{A.16})$$

Solution: The alternate form above simply splits the two tails of the distribution and uses the definition of a limit. Note that if this statement is satisfied for one value of $\delta > 0$, it is satisfied for all larger δ . Thus we restrict attention to $0 < \delta < 1$. For any such δ and any $\epsilon > 0$, we now derive this alternate form from the CLT. To do this, choose $z < 0$ to satisfy $\delta = 2\Phi(z)$. Note that this also satisfies $\delta = 2(1 - \Phi(|z|))$ (it might help to look at Figure 1.12 while following this).

From the CLT, using this particular z , we see that since the limit is $\Phi(z)$, there must be an n_1 large enough that

$$\Pr \left\{ \frac{S_n}{\sigma\sqrt{n}} < z \right\} < 2\Phi(z) = \delta \quad \text{for } n \geq n_1.$$

Rewriting this,

$$\Pr \left\{ \frac{S_n}{n} < \frac{z\sigma}{\sqrt{n}} \right\} < \delta \quad \text{for } n \geq n_1.$$

Let $n(\epsilon, \delta)$ be any integer both greater than n_1 and such that $z\sigma/\sqrt{n} < \epsilon$. Then the first part of (A.16) is satisfied. The second part is a minor variation.

Exercise 1.41: (Details in the proof of Theorem 1.7.4) a) Show that if $X_1, X_2, \dots$ are IID, then the truncated versions $\check{X}_1, \check{X}_2, \dots$ are also IID.

Solution: Readers with an engineering orientation can be forgiven for saying this is obvious. For an actual proof, note that

$$F_{\check{X}_1 \dots \check{X}_n}(y_1, \dots, y_n) = \Pr \left\{ \bigcap_{j=1}^n \{\check{X}_j \leq y_j\} \right\}.$$

The rv $\check{X}_j$ is a function of the rv X_j and for each real number y_j , we have

$$\{\check{X}_j \leq y_j\} = \begin{cases} \emptyset & \text{for } y_j < \overline{X} - b, \\ \{X_j \leq y_j\} & \text{for } \overline{X} - b \leq y_j < \overline{X} + b, \\ \Omega & \text{for } y_j \geq \overline{X} + b. \end{cases}$$

If all the y_j are in the range $[\bar{X}-b, \bar{X}+b)$, then each event $\{\check{X}_j \leq y_j\}$ is the same as $\{X_j \leq y_j\}$, so it is clear that these n events are statistically independent. If some of the y_j are outside of this range, then the events are still independent since $\emptyset$ and Ω are independent of all other events, including themselves. Thus

$$F_{\check{X}_1 \dots \check{X}_n}(y_1, \dots, y_n) = \prod_{j=1}^n \Pr\{\check{X}_j \leq y_j\} = \prod_{j=1}^n F_{\check{X}_j}(y_j).$$

Finally, since the X_j are identically distributed, the $\check{X}_j$ are also.

b) Show that each $\check{X}_i$ has a finite mean $\mathbb{E}[\check{X}]$ and finite variance $\sigma_{\check{X}}^2$. Show that the variance is upper bounded by the second moment around the original mean $\bar{X}$, i.e., show that $\sigma_{\check{X}}^2 \leq \mathbb{E}[|\check{X} - \mathbb{E}[X]|^2]$.

Solution: Since $\bar{X} - b \leq \check{X} \leq \bar{X} + b$, we have

$$\bar{X} - b \leq \mathbb{E}[\check{X}] \leq \bar{X} + b,$$

so $\mathbb{E}[\check{X}]$ is finite (and in the same way, $\mathbb{E}[|\check{X}|]$ is finite).

Also, since $(\check{X})^2 \leq (|\bar{X}| + b)^2$, we have $\mathbb{E}[\check{X}^2] \leq (|\bar{X}| + b)^2$ so the variance $\sigma_{\check{X}}^2$ is also finite.

Finally, for any rv Y with a variance, and any real α , $\mathbb{E}[(Y - \alpha)^2] = \mathbb{E}[Y^2] - 2\alpha\bar{Y} + \alpha^2$, which is minimized by $\alpha = \bar{Y}$, i.e., $\mathbb{E}[(Y - \alpha)^2] \geq \sigma_Y^2$. Letting Y be $\check{X}$ and α be $\bar{X}$ shows that $\sigma_{\check{X}}^2 \leq \mathbb{E}[|\check{X} - \mathbb{E}[X]|^2]$.

c) Assume that $\check{X}_i$ is X_i truncated to $\bar{X} \pm b$. Show that $|\check{X} - \bar{X}| \leq b$ and that $|\check{X} - \bar{X}| \leq |X - \bar{X}|$. Use this to show that $\sigma_{\check{X}}^2 \leq b\mathbb{E}[|\check{X} - \bar{X}|] \leq 2b\mathbb{E}[|X|]$.

Solution: Since $\bar{X} - b \leq \check{X} \leq \bar{X} + b$, we have $-b \leq \check{X} - \bar{X} \leq b$, i.e., $|\check{X} - \bar{X}| \leq b$. To show that $|\check{X} - \bar{X}| \leq |X - \bar{X}|$, we look at the two cases, $|X - \bar{X}| \leq b$ and $|X - \bar{X}| > b$. In the first case, $\check{X} = X$, so $\check{X} - \bar{X} = X - \bar{X}$. In the second case, $|\check{X} - \bar{X}| = b$ so $|\check{X} - \bar{X}| < |X - \bar{X}|$.

$$|\check{X} - \bar{X}|^2 \leq b|X - \bar{X}|,$$

where we have used the upper bound b on the first $|\check{X} - \bar{X}|$ above and the bound $|X - \bar{X}|$ on the other. Combining this with (b),

$$\sigma_{\check{X}}^2 \leq \mathbb{E}[|\check{X} - \bar{X}|^2] \leq b\mathbb{E}[|X| + |\bar{X}|] \leq 2b\mathbb{E}[|X|]. \quad (\text{A.17})$$

d) Let $\check{S}_n = \check{X}_1 + \dots + \check{X}_n$ and show that for any $\epsilon > 0$,

$$\Pr\left\{\left|\frac{\check{S}_n}{n} - \mathbb{E}[\check{X}]\right| \geq \frac{\epsilon}{2}\right\} \leq \frac{8b\mathbb{E}[|X|]}{n\epsilon^2}. \quad (\text{A.18})$$

Solution: Note that $\check{S}_n/n$ has variance $\sigma_{\check{X}}^2/n$, so the Chebyshev inequality says that

$$\Pr\left\{\left|\frac{\check{S}_n}{n} - \mathbb{E}[\check{X}]\right| \geq \frac{\epsilon}{2}\right\} \leq \frac{4\sigma_{\check{X}}^2}{n\epsilon^2}.$$

Substituting (A.17) in this gives us the desired bound.

e) Use Figure 1.13, along with (1.34), to show that for all sufficiently large b , $|\mathbb{E}[X - \check{X}]| \leq \epsilon/2$. Use this to show that

$$\Pr\left\{\left|\frac{\check{S}_n}{n} - \mathbb{E}[X]\right| \geq \epsilon\right\} \leq \frac{8b\mathbb{E}[|X|]}{n\epsilon^2} \quad \text{for all large enough } b. \quad (\text{A.19})$$

Solution: Note that $\check{X}$ and X have the same CDF from $\bar{X} - b$ to $\bar{X} + b$. From (1.34), then,

$$\mathbb{E}[X - \check{X}] = - \int_{-\infty}^{\bar{X}-b} F_X(x) dx + \int_{\bar{X}+b}^{\infty} F_X^c(x) dx.$$

Since X has a finite mean, $\int_0^\infty F_X^c(x) dx$ is finite, so $\lim_{b \rightarrow \infty} \int_{\bar{X}+b}^\infty F_X^c(x) dx = 0$. This shows that $\mathbb{E}[X - \check{X}] \leq \epsilon/2$ for all sufficiently large b . In the same way $\mathbb{E}[X - \check{X}] \geq -\epsilon/2$ for all sufficiently large b and thus $|\mathbb{E}[X - \check{X}]| \leq \epsilon/2$. Next, recall that for all numbers a, b , we have $|a - b| \leq |a| + |b|$. Thus,

$$\begin{aligned} \left|\frac{\check{S}_n}{n} - \bar{X}\right| &= \left|\left(\frac{\check{S}_n}{n} - \mathbb{E}[\check{X}]\right) - (\bar{X} - \mathbb{E}[\check{X}])\right| \\ &\leq \left|\frac{\check{S}_n}{n} - \mathbb{E}[\check{X}]\right| + |\mathbb{E}[X - \check{X}]|. \end{aligned}$$

The final term on the right is at most $\epsilon/2$ for all sufficiently large b . Thus, for such b , every sample point for which $|\check{S}_n/n - \bar{X}| \geq \epsilon$ must also satisfy $|\check{S}_n/n - \mathbb{E}[\check{X}]| \geq \epsilon/2$. When we take the probability of this set of terms then,

$$\Pr\left\{\left|\frac{\check{S}_n}{n} - \bar{X}\right| \geq \epsilon\right\} \leq \Pr\left\{\left|\frac{\check{S}_n}{n} - \mathbb{E}[\check{X}]\right| \geq \epsilon/2\right\} \quad \text{for all sufficiently large } b.$$

Combining this with (A.18) gives the desired result.

f) Use the following equation to justify (1.96).

$$\begin{aligned} \Pr\left\{\left|\frac{S_n}{n} - \mathbb{E}[X]\right| > \epsilon\right\} &= \Pr\left\{\left|\frac{S_n}{n} - \mathbb{E}[X]\right| > \epsilon\right\} \cap \{S_n = \check{S}_n\} \\ &\quad + \Pr\left\{\left|\frac{S_n}{n} - \mathbb{E}[X]\right| > \epsilon\right\} \cap \{S_n \neq \check{S}_n\}. \end{aligned}$$

Solution: The first term on the right above is unchanged if the first S_n is replaced by $\check{S}_n$, and therefore it is upper-bounded by (A.19). The second term is upper-bounded by $\Pr\{S_n \neq \check{S}_n\}$, which is the probability that one or more of the X_j are outside of $[\bar{X} - b, \bar{X} + b]$. The union bound then results in (1.96).

Exercise 1.42: Let $\{X_i; i \geq 1\}$ be IID rv's with mean 0 and infinite variance. Assume that $\mathbb{E}[|X_i|^{1+h}] = \beta$ for some given h , $0 < h < 1$ and some finite β . Let $S_n = X_1 + \dots + X_n$.

a) Show that for any $y > 0$, $\Pr\{|X_i| \geq y\} \leq \beta y^{-1-h}$.

Solution: Since $\mathbb{E}[|X_i|^{1+h}]$ is finite, we can apply the Markov inequality to $|X_i|^{1+h}$, getting

$$\Pr\{|X_i| > y\} = \Pr\{|X_i|^{1+h} > y^{1+h}\} \leq \frac{\mathbb{E}[|X_i|^{1+h}]}{y^{1+h}} = \beta y^{-1-h}.$$

Note that this converges faster as $y \rightarrow \infty$ than the Markov inequality applied to $|X_i|$.

b) Let $\{\check{X}_i; i \geq 1\}$ be truncated variables $\check{X}_i = \begin{cases} b & : X_i \geq b \\ X_i & : -b \leq X_i \leq b \\ -b & : X_i \leq -b \end{cases}$

Show that $\mathbb{E}[\check{X}^2] \leq \frac{2\beta b^{1-h}}{1-h}$. Hint: For a nonnegative rv Z , $\mathbb{E}[Z^2] = \int_0^\infty 2z \Pr\{Z \geq z\} dz$ (To establish this, view Z^2 as a rv, so $\mathbb{E}[Z^2] = \int_0^\infty \Pr\{Z^2 > z^2\} dz^2$).

Solution: We know that $\mathbb{E}[\check{X}_i^2]$ is finite since it is truncated. Upper bounding it by first using the hint and next using (a) (for $0 < h < 1$), we get

$$\mathbb{E}[\check{X}_i^2] = \int_0^b 2x \Pr\{|X_i| \geq x\} dx \leq \int_0^b 2x \beta x^{-1-h} dx = \frac{2\beta b^{1-h}}{1-h}.$$

c) Let $\check{S}_n = \check{X}_1 + \dots + \check{X}_n$. Show that $\Pr\{S_n \neq \check{S}_n\} \leq n\beta b^{-1-h}$.

Solution:

$$\Pr\{S_n \neq \check{S}_n\} \leq \Pr\left\{\bigcup_{i=1}^n \{X_i \neq \check{X}_i\}\right\} \leq \sum_{i=1}^n \Pr\{X_i \neq \check{X}_i\} \leq n\beta b^{-1-h},$$

where we have used the union bound and then the fact that $X_i \neq \check{X}_i$ occurs if and only if $|X_i| > b$, which is bounded in (a).

d) Show that $\Pr\left\{\left|\frac{S_n}{n}\right| \geq \epsilon\right\} \leq \beta \left[\frac{2b^{1-h}}{(1-h)n\epsilon^2} + \frac{n}{b^{1+h}}\right]$.

Solution: $\Pr\{|S_n| \geq n\epsilon\} \leq \Pr\{S_n \neq \check{S}_n\} + \Pr\{|\check{S}_n| \geq n\epsilon\}$. Bounding the first term by (c) and the second term by the Markov inequality,

$$\begin{aligned} \Pr\{|S_n| \geq n\epsilon\} &\leq n\beta b^{-1-h} + \mathbb{E}[\check{S}_n^2] / n^2 \epsilon^2 = n\beta b^{-1-h} + \mathbb{E}[\check{X}_i^2] / n \epsilon^2 \\ &\leq n\beta b^{-1-h} + \frac{2\beta b^{1-h}}{(1-h)n\epsilon^2} = \beta \left[\frac{2b^{1-h}}{(1-h)n\epsilon^2} + \frac{n}{b^{1+h}} \right]. \end{aligned}$$

e) Optimize your bound with respect to b . How fast does this optimized bound approach 0 with increasing n ?

Solution: To minimize this bound over b , we take the derivative of the term in brackets, getting

$$\frac{\partial}{\partial b} \left[\frac{2b^{1-h}}{(1-h)n\epsilon^2} + \frac{n}{b^{1+h}} \right] = b^{-h} \left[\frac{2}{n\epsilon^2} - \frac{(1+h)n}{b^2} \right].$$

The derivative is 0 at $b = n\epsilon\sqrt{(1+h)/2}$. This must be the minimum since the coefficient b^{-h} is positive for all $b > 0$ and the term in brackets is increasing in b , being negative for

$b < n\epsilon\sqrt{(1+h)/2}$ and positive for larger b . Substituting this value of b into the result of (d) and simplifying with a fair amount of algebra,

$$\Pr\left\{\frac{S_n}{n} \geq \epsilon\right\} \leq \frac{4\beta}{n^h\epsilon^{1+h}(1-h^2)} \left[\frac{1+h}{2}\right]^{(1-h)/2}.$$

This goes to 0 with $1/n^h$, which is slower (as expected) than the $1/n$ decrease for the WLLN with a finite variance, but much faster than the WLLN assuming only $\mathbf{E}[|X|] < \infty$. In this latter case, there is no known rate of convergence.

Exercise 1.43: (MS convergence $\implies$ convergence in probability) Assume that $\{Z_n; n \geq 1\}$ is a sequence of rv's and α is a number with the property that $\lim_{n \rightarrow \infty} \mathbf{E}[(Z_n - \alpha)^2] = 0$.

a) Let $\epsilon > 0$ be arbitrary and show that for each $n \geq 0$,

$$\Pr\{|Z_n - \alpha| \geq \epsilon\} \leq \frac{\mathbf{E}[(Z_n - \alpha)^2]}{\epsilon^2}. \quad (\text{A.20})$$

Solution: It is possible for $\mathbf{E}[(Z_n - \alpha)^2]$ to be infinite for some n , although because of the limit, it is finite for all large enough n . We consider only such large enough n in what follows. The statement to be shown is almost the Chebyshev inequality, except that α need not be $\mathbf{E}[Z_n]$. However, the desired statement follows immediately from applying the Markov inequality to $(Z_n - \alpha)^2$. This gives us a further reminder that it is the Markov inequality that is the more fundamental and useful between Markov and Chebyshev.

b) For the ϵ above, let $\delta > 0$ be arbitrary. Show that there is an integer m such that $\mathbf{E}[(Z_n - \alpha)^2] \leq \epsilon^2\delta$ for all $n \geq m$.

Solution: By the definition of a limit, $\lim_{n \rightarrow \infty} \mathbf{E}[(Z_n - \alpha)^2] = 0$ means that for all $\epsilon_1 > 0$, there is an m large enough that $\mathbf{E}[(Z_n - \alpha)^2] < \epsilon_1$ for all $n \geq m$. Choosing $\epsilon_1 = \epsilon^2\delta$ establishes the desired result.

c) Show that this implies convergence in probability.

Solution: Substituting $\mathbf{E}[(Z_n - \alpha)^2] \leq \epsilon^2\delta$ into (A.20), we see that for all $\epsilon, \delta > 0$, there is an m such that $\Pr\{|S_n/n - \alpha|^2 \geq \epsilon\} \leq \delta$ for all $n \geq m$. This is convergence in probability.

Exercise 1.44: Let $X_1, X_2, \dots$ be a sequence of IID rv's each with mean 0 and variance σ^2 . Let $S_n = X_1 + \dots + X_n$ for all n and consider the random variable $S_n/\sigma\sqrt{n} - S_{2n}/\sigma\sqrt{2n}$. Find the limiting CDF for this sequence of rv's as $n \rightarrow \infty$. The point of this exercise is to see clearly that the CDF of $S_n/\sigma\sqrt{n} - S_{2n}/\sigma\sqrt{2n}$ is converging in n but that the sequence of rv's is not converging in any reasonable sense.

Solution: If we write out the above expression in terms of the X_i , we get

$$\frac{S_n}{\sigma\sqrt{n}} - \frac{S_{2n}}{\sigma\sqrt{2n}} = \sum_{i=1}^n X_i \left[\frac{1}{\sigma\sqrt{n}} - \frac{1}{\sigma\sqrt{2n}} \right] - \sum_{i=n+1}^{2n} \frac{X_i}{\sigma\sqrt{2n}}.$$

The first sum above approaches a Gaussian distribution of variance $(1 - 1/\sqrt{2})^2$ and the second sum approaches a Gaussian distribution of variance $1/2$. Since these two terms are

independent, the difference approaches a Gaussian distribution of variance $1 + (1 - 1/\sqrt{2})^2$. This means that the distribution of the difference does converge to this Gaussian rv as $n \rightarrow \infty$. As n increases, however, this difference slowly changes, and each time n is doubled, the new difference is only weakly correlated with the old difference.

Note that $S_n/n\sigma_X$ behaves in this same way. The CDF converges as $n \rightarrow \infty$, but the rv's themselves do not approach each other in any reasonable way. The point of the problem was to emphasize this property.

Exercise 1.45: Verify (1.57), *i.e.*, verify that $\lim_{y \rightarrow \infty} y \Pr\{Y \geq y\} = 0$ if $Y \geq 0$ and $\bar{Y} < \infty$. Hint: Show that $y \Pr\{Y \geq y\} \leq \int_{z \geq y} z dF_Y(z)$ and show that $\lim_{y \rightarrow \infty} \int_{z \geq y} z dF_Y(z) = 0$.

Solution: We can write $\Pr\{Y \geq y\}$ as the Stieltjes integral $\int_y^\infty dF_Y(z)$. Since $z \geq y$ over the range of integration,

$$y \Pr\{Y \geq y\} = \int_y^\infty y dF_Y(z) \leq \int_y^\infty z dF_Y(z).$$

We can express $\bar{Y}$ as the Stieltjes integral

$$\int_0^\infty z dF_Y(z) = \lim_{y \rightarrow \infty} \int_0^y z dF_Y(z).$$

Since $\bar{Y} < \infty$, the limit above exists, so that $\lim_{y \rightarrow \infty} \int_y^\infty z dF_Y(z) = 0$. Combining these results, $\lim_{y \rightarrow \infty} \int_{z \geq y} y \Pr\{Y \geq y\} = 0$.

The result can also be proven without using Stieltjes integration. Since $\bar{Y} = \lim_{y \rightarrow \infty} \int_0^y F_Y^c dy$ and $\bar{Y} < \infty$, the above limit exists, so

$$\lim_{y \rightarrow \infty} \int_y^\infty F_Y^c(z) dz = 0.$$

For any $\epsilon > 0$, we can thereby choose y_o such that

$$\int_{y_o}^\infty F_Y^c(z) dz \leq \epsilon/2. \quad (\text{A.21})$$

For $y > y_o$, we can express $y \Pr\{Y \geq y\}$ as

$$\begin{aligned} y \Pr\{Y \geq y\} &= (y - y_o) \Pr\{Y \geq y\} + y_o \Pr\{Y \geq y\} \\ &= \int_{y_o}^y \Pr\{Y \geq y\} dz + y_o \Pr\{Y \geq y\} \end{aligned} \quad (\text{A.22})$$

$$\leq \int_{y_o}^y F_Y^c(z) dz + y_o \Pr\{Y \geq y\}. \quad (\text{A.23})$$

The first term in (A.23) upper bounds the first term in (A.22) since $F_Y^c(z) \geq \Pr\{Y \geq y\}$ for $z < y$. From (A.21), the first term in (A.23) is upper bounded by $\epsilon/2$. Since $\lim_{y \rightarrow \infty} \Pr\{Y \geq y\} = 0$, $y_o \Pr\{Y \geq y\} \leq \epsilon/2$ for all sufficiently large y . Thus $y \Pr\{Y \geq y\} \leq \epsilon$ for all sufficiently large y . Since ϵ is arbitrary, $\lim_{y \rightarrow \infty} y \Pr\{Y \geq y\} = 0$.

Exercise 1.46: Show that $\prod_{m \geq n} (1 - 1/m) = 0$. Hint: Show that

$$\left(1 - \frac{1}{m}\right) = \exp\left(\ln\left(1 - \frac{1}{m}\right)\right) \leq \exp\left(-\frac{1}{m}\right).$$

Solution: A particularly useful inequality, for $x > -1$, is $\ln(1+x) \leq x$, with equality at $x = 0$. This can be trivially derived by maximizing $\ln(1+x) - x$. This verifies the above hint.

$$\prod_{m \geq n} (1 - 1/m) \leq \prod_{m \geq n} \exp(-1/m) = \exp\left(\sum_{m \geq n} -1/m\right) = \exp(-\infty) = 0,$$

where we have used the fact that the harmonic series diverges

Exercise 1.47: Consider a discrete rv X with the PMF

$$\begin{aligned} p_X(-1) &= (1 - 10^{-10})/2, \\ p_X(1) &= (1 - 10^{-10})/2, \\ p_X(10^{12}) &= 10^{-10}. \end{aligned}$$

a) Find the mean and variance of X . Assuming that $\{X_m; m \geq 1\}$ is an IID sequence with the distribution of X and that $S_n = X_1 + \cdots + X_n$ for each n , find the mean and variance of S_n . (no explanations needed.)

Solution: $\bar{X} = 100$ and $\sigma_X^2 = 10^{14} + (1 - 10^{-10}) - 10^4 \approx 10^{14}$. Thus $\bar{S}_n = 100n$ and $\sigma_{S_n}^2 \approx n \times 10^{14}$.

b) Let $n = 10^6$ and describe the event $\{S_n \leq 10^6\}$ in words. Find an exact expression for $\Pr\{S_n \leq 10^6\} = F_{S_n}(10^6)$.

Solution: This is the event that all 10^6 trials result in ± 1 . That is, there are no occurrences of 10^{12} . Thus $\Pr\{S_n \leq 10^6\} = (1 - 10^{-10})^{10^6}$

c) Find a way to use the union bound to get a simple upper bound and approximation of $1 - F_{S_n}(10^6)$.

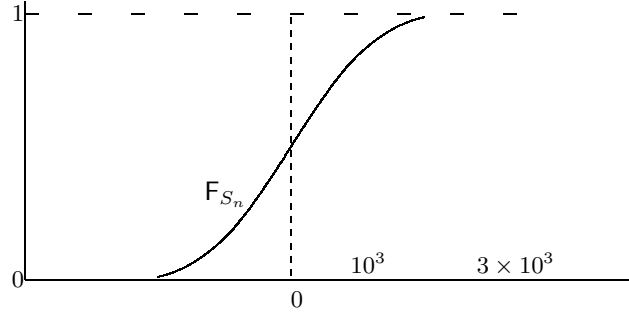
Solution: From the union bound, the probability of one or more occurrences of the sample value 10^{12} out of 10^6 trials is bounded by a sum of 10^6 terms, each equal to 10^{-10} , i.e., $1 - F_{S_n}(10^6) \leq 10^{-4}$. This is also a good approximation, since we can write

$$(1 - 10^{-10})^{10^6} = \exp\left(10^6 \ln(1 - 10^{-10})\right) \approx \exp(10^6 \cdot 10^{-10}) \approx 1 - 10^{-4}.$$

d) Sketch the CDF of S_n for $n = 10^6$. You can choose the horizontal axis for your sketch to go from -1 to $+1$ or from -3×10^3 to 3×10^3 or from -10^6 to 10^6 or from 0 to 10^{12} , whichever you think will best describe this CDF.

Solution: Conditional on no occurrences of 10^{12} , S_n simply has a binomial distribution. We know from the central limit theorem for the binomial case that S_n will be approximately Gaussian with mean 0 and standard deviation 10^3 . Since one or more occurrences of 10^{12} occur only with probability 10^{-4} , this can be neglected in the sketch, so the CDF is approximately Gaussian with 3 sigma points at $\pm 3 \times 10^3$. There is a little blip, in the 4th

decimal place out at 10^{12} which doesn't show up well in the sketch, but of course could be important for some purposes such as calculating σ^2 .



e) Now let $n = 10^{10}$. Give an exact expression for $\Pr\{S_n \leq 10^{10}\}$ and show that this can be approximated by e^{-1} . Sketch the CDF of S_n for $n = 10^{10}$, using a horizontal axis going from slightly below 0 to slightly more than 2×10^{12} . Hint: First view S_n as conditioned on an appropriate rv.

Solution: First consider the PMF $p_B(j)$ of the number $B = j$ of occurrences of the value 10^{12} . We have

$$p_B(j) = \binom{10^{10}}{j} p^j (1-p)^{10^{10}-j} \quad \text{where } p = 10^{-10}$$

$$\begin{aligned} p_B(0) &= (1-p)^{10^{10}} = \exp\{10^{10} \ln[1-p]\} \approx \exp(-10^{10}p) = e^{-1} \\ p_B(1) &= 10^{10} p (1-p)^{10^{10}-1} = (1-p)^{10^{10}-1} \approx e^{-1} \\ p_B(2) &= \binom{10^{10}}{2} p^2 (1-p)^{10^{10}-2} \approx \frac{1}{2} e^{-1}. \end{aligned}$$

Conditional on $B = j$, S_n will be approximately Gaussian with mean $10^{12}j$ and standard deviation 10^5 . Thus $F_{S_n}(s)$ rises from 0 to e^{-1} over a range from about -3×10^5 to $+3 \times 10^5$. It then stays virtually constant up to about $10^{12} - 3 \times 10^5$. It rises to $2/e$ by $10^{12} + 3 \times 10^5$. It stays virtually constant up to $2 \times 10^{12} - 3 \times 10^5$ and rises to $2.5/e$ by $2 \times 10^{12} + 3 \times 10^5$. When we sketch this, it looks like a staircase function, rising from 0 to $1/e$ at 0, from $1/e$ to $2/e$ at 10^{12} and from $2/e$ to $2.5/e$ at 2×10^{12} . There are smaller steps at larger values, but they would not show up on the sketch.

d) Can you make a qualitative statement about how the distribution function of a rv X affects the required size of n before the WLLN and the CLT provide much of an indication about S_n .

Solution: It can be seen that for this peculiar rv, S_n/n is not concentrated around its mean even for $n = 10^{10}$ and $S_n/\sqrt{n}$ does not look Gaussian even for $n = 10^{10}$. For this particular distribution, n has to be so large that B , the number of occurrences of 10^{12} , is large, and this requires $n \gg 10^{10}$. This illustrates a common weakness of limit theorems. They say what happens as a parameter (n in this case) becomes sufficiently large, but it takes extra work to see how large that is.

Exercise 1.48: Let $\{Y_n; n \geq 1\}$ be a sequence of rv's and assume that $\lim_{n \rightarrow \infty} \mathbb{E}[|Y_n|] = 0$. Show that $\{Y_n; n \geq 1\}$ converges to 0 in probability. Hint 1: Look for the easy way. Hint 2: The easy way uses the Markov inequality.

Solution: Applying the Markov inequality to $|Y_n|$ for arbitrary n and arbitrary $\epsilon > 0$, we have

$$\Pr\{|Y_n| \geq \epsilon\} \leq \frac{\mathbb{E}[|Y_n|]}{\epsilon}.$$

Thus going to the limit $n \rightarrow \infty$ for the given ϵ ,

$$\lim_{n \rightarrow \infty} \Pr\{|Y_n| \geq \epsilon\} = 0.$$

Since this is true for every $\epsilon > 0$, this satisfies the definition for convergence to 0 in probability.

A.2 Solutions for Chapter 2

Exercise 2.1: a) Find the Erlang density $f_{S_n}(t)$ by convolving $f_X(x) = \lambda \exp(-\lambda x)$, $x \geq 0$ with itself n times.

Solution: For $n = 2$, we convolve $f_X(x)$ with itself.

$$f_{S_2}(t) = \int_0^t f_{X_1}(x) f_{X_2}(t-x) dx = \int_0^t \lambda e^{-\lambda x} \lambda e^{-\lambda(t-x)} dx = \lambda^2 t e^{-\lambda t}.$$

For larger n , convolving $f_X(x)$ with itself n times is found by taking the convolution $n-1$ times, *i.e.*, $f_{S_{n-1}}(t)$, and convolving this with $f_X(x)$. Starting with $n = 3$,

$$\begin{aligned} f_{S_3}(t) &= \int_0^t f_{S_2}(x) f_{X_3}(t-x) dx = \int_0^t \lambda^2 x e^{-\lambda x} \lambda e^{-\lambda(t-x)} dx = \frac{\lambda^3 t^2}{2} e^{-\lambda t} \\ f_{S_4}(t) &= \int_0^t \frac{\lambda^3 x^2}{2} e^{-\lambda x} \cdot \lambda e^{-\lambda(t-x)} dx = \frac{\lambda^4 t^3}{3!} e^{-\lambda t}. \end{aligned}$$

We now see the pattern; each additional integration increases the power of λ and t by 1 and multiplies the denominator by $n-1$. Thus we hypothesize that $f_{S_n}(t) = \frac{\lambda^n t^{n-1}}{n!} e^{-\lambda t}$. If one merely wants to verify the well-known Erlang density, one can simply use induction from the beginning, but it is more satisfying, and not that much more difficult, to actually derive the Erlang density, as done above.

b) Find the moment generating function of X (or find the Laplace transform of $f_X(x)$), and use this to find the moment generating function (or Laplace transform) of $S_n = X_1 + X_2 + \cdots + X_n$.

Solution: The formula for the MGF is almost trivial here,

$$g_X(r) = \int_0^\infty \lambda e^{-\lambda x} e^{rx} dx = \frac{\lambda}{\lambda - r} \quad \text{for } r < \lambda.$$

Since S_n is the sum of n IID rv's,

$$g_{S_n}(r) = [g_X(r)]^n = \left(\frac{\lambda}{\lambda - r} \right)^n.$$

c) Find the Erlang density by starting with (2.15) and then calculating the marginal density for S_n .

Solution: To find the marginal density, $f_{S_n}(s_n)$, we start with the joint density in (2.15) and integrate over the region of space where $s_1 \leq s_2 \leq \cdots \leq s_n$. It is a peculiar integral, since the integrand is constant and we are just finding the volume of the $n-1$ dimensional space in $s_1, \dots, s_{n-1}$ with the inequality constraints above. For $n = 2$ and $n = 3$, we have

$$\begin{aligned} f_{S_2}(s_2) &= \lambda^2 e^{-\lambda s_2} \int_0^{s_2} ds_1 = (\lambda^2 e^{-\lambda s_2}) s_2 \\ f_{S_3}(s_3) &= \lambda^3 e^{-\lambda s_3} \int_0^{s_3} \left[\int_0^{s_2} ds_1 \right] ds_2 = \lambda^3 e^{-\lambda s_3} \int_0^{s_3} s_2 ds_2 = (\lambda^3 e^{-\lambda s_3}) \frac{s_3^2}{2}. \end{aligned}$$

The critical part of these calculations is the calculation of the volume, and we can do this inductively by guessing from the previous equation that the volume, given s_n , of the $n - 1$ dimensional space where $0 < s_1 < \dots < s_{n-1} < s_n$ is $s_n^{n-1}/(n-1)!$. We can check that by

$$\int_0^{s_n} \left[\int_0^{s_{n-1}} \dots \int_0^{s_2} ds_1 \dots ds_{n-2} \right] ds_{n-1} = \int_0^{s_n} \frac{s_{n-1}^{n-2}}{(n-2)!} ds_{n-1} = \frac{s_n^{n-1}}{(n-1)!}.$$

This volume integral, multiplied by $\lambda^n e^{-\lambda s_n}$, is then the desired marginal density.

A more elegant and instructive way to calculate this volume is by first observing that the volume of the $n - 1$ dimensional cube, s_n on a side, is s_n^{n-1} . Each point in this cube can be visualized as a vector $(s_1, s_2, \dots, s_{n-1})$. Each component lies in $(0, s_n)$, but the cube doesn't have the ordering constraint $s_1 < s_2 < \dots < s_{n-1}$. By symmetry, the volume of points in the cube satisfying this ordering constraint is the same as the volume in which the components $s_1, \dots, s_{n-1}$ are ordered in any other particular way. There are $(n - 1)!$ different ways to order these $n - 1$ components (*i.e.*, there are $(n - 1)!$ permutations of the components), and thus the volume with the ordering constraints, is $s_n^{n-1}/(n - 1)!$.

Exercise 2.2: a) Find the mean, variance, and moment generating function of $N(t)$, as given by (2.17).

Solution:

$$\mathbb{E}[N(t)] = \sum_{n=0}^{\infty} \frac{n(\lambda t)^n e^{-\lambda t}}{n!} = \lambda t \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1} e^{-\lambda t}}{(n-1)!} = \lambda t \sum_{m=0}^{\infty} \frac{(\lambda t)^m e^{-\lambda t}}{m!} = \lambda t,$$

where in the last step, we recognized the terms in the sum as the Poisson PMF of parameter λt ; since the PMF must sum to 1 the mean is λt as shown. Using the same approach, the second moment is $(\lambda t)^2 + \lambda t$, so the variance is λt . For the MGF,

$$\mathbb{E}[e^{rN(t)}] = \sum_{n=0}^{\infty} \frac{(\lambda t e^r)^n e^{-\lambda t}}{n!} = e^{-\lambda t} \sum_{n=0}^{\infty} \left[\frac{(\lambda t e^r)^n e^{-\lambda t e^r}}{n!} \right] e^{\lambda t e^r}.$$

Recognizing the term in brackets as the PMF of a Poisson rv of parameter λe^r in this expression, we get

$$\mathbb{E}[e^{rN(t)}] = e^{-\lambda t} e^{\lambda t e^r} = \exp[\lambda t(e^r - 1)].$$

b) Show by discrete convolution that the sum of two independent Poisson rv's is again Poisson.

Solution: Let X and Y be independent Poisson with parameters λ and μ respectively. Then

$$\begin{aligned} p_{X+Y}(m) &= \sum_{n=0}^m p_X(n) p_Y(m-n) = e^{-\lambda-\mu} \sum_{n=0}^m \frac{\lambda^n \mu^{m-n}}{n!(m-n)!} \\ &= \frac{e^{-\lambda-\mu}}{m!} \sum_{n=0}^m \binom{m}{n} \lambda^n \mu^{m-n} = \frac{e^{-\lambda-\mu} (\lambda + \mu)^m}{m!}, \end{aligned}$$

where we recognized the final sum as a binomial sum.

c) Show by using the properties of the Poisson process that the sum of two independent Poisson rv's must be Poisson.

Solution: For any $t, \tau > 0$ in a Poisson process we know that $N(t + \tau) = N(t) + \tilde{N}(t, t + \tau)$ is Poisson and $\tilde{N}(t, t + \tau)$ is Poisson in τ . Since $N(t)$ and $\tilde{N}(t, \tau)$ are independent and t and τ are arbitrary positive numbers, the sum of 2 independent Poisson rv's is Poisson.

Exercise 2.3: The purpose of this exercise is to give an alternate derivation of the Poisson distribution for $N(t)$, the number of arrivals in a Poisson process up to time t . Let λ be the rate of the process.

a) Find the conditional probability $\Pr\{N(t) = n \mid S_n = \tau\}$ for all $\tau \leq t$.

Solution: The condition $S_n = \tau$ means that the epoch of the n th arrival is τ . Conditional on this, the event $\{N(t) = n\}$ for some $t > \tau$ means there have been no subsequent arrivals from τ to t . In other words, it means that the $(n + 1)$ th interarrival time, X_{n+1} exceeds $t - \tau$. This interarrival time is independent of S_n and thus

$$\Pr\{N(t) = n \mid S_n = \tau\} = \Pr\{X_{n+1} > t - \tau\} = e^{-\lambda(t-\tau)} \quad \text{for } t > \tau. \quad (\text{A.24})$$

b) Using the Erlang density for S_n , use (a) to find $\Pr\{N(t) = n\}$.

Solution: We find $\Pr\{N(t) = n\}$ simply by averaging (A.24) over S_n .

$$\begin{aligned} \Pr\{N(t)=n\} &= \int_0^t \Pr\{N(t)=n \mid S_n=\tau\} f_{S_n}(\tau) d\tau \\ &= \int_0^t e^{-\lambda(t-\tau)} \frac{\lambda^n \tau^{n-1} e^{-\lambda\tau}}{(n-1)!} d\tau \\ &= \frac{\lambda^n e^{-\lambda t}}{(n-1)!} \int_0^t \tau^{n-1} d\tau = \frac{(\lambda t)^n e^{-\lambda t}}{n!}. \end{aligned}$$

Exercise 2.4: Assume that a counting process $\{N(t); t \geq 0\}$ has the independent and stationary increment properties and satisfies (2.17) (for all $t > 0$). Let X_1 be the epoch of the first arrival and X_n be the interarrival time between the $(n-1)$ st and the n th arrival. Use only these assumptions in doing the following parts of this exercise.

a) Show that $\Pr\{X_1 > x\} = e^{-\lambda x}$.

Solution: The event $\{X_1 > x\}$ is the same as the event $\{N(x) = 0\}$. Thus, from (2.17), $\Pr\{X_1 > x\} = \Pr\{N(x) = 0\} = e^{-\lambda x}$.

b) Let S_{n-1} be the epoch of the $(n-1)$ st arrival. Show that $\Pr\{X_n > x \mid S_{n-1} = \tau\} = e^{-\lambda x}$.

Solution: The conditioning event $\{S_{n-1} = \tau\}$ is somewhat messy to deal with in terms of the parameters of the counting process, so we start by solving an approximation of the desired result and then go to a limit as the approximation becomes increasingly close. Let $\delta > 0$ be a small positive number which we later allow to approach 0. We replace the event $\{S_{n-1} = \tau\}$ with $\{\tau - \delta < S_{n-1} \leq \tau\}$. Since the occurrence of two arrivals in a small interval of size δ is very unlikely (of order $o(\delta)$), we also include the condition that $S_{n-2} \leq \tau - \delta$ and $S_n \geq \tau$. With this, the approximate conditioning event becomes

$$\{S_{n-2} \leq \tau - \delta < S_{n-1} \leq \tau < S_n\} = \{N(\tau - \delta) = n - 2, \tilde{N}(\tau - \delta, \tau) = 1\}.$$

Since we are irretrievably deep in approximations, we also replace the event $\{X_n > x\}$ (conditional on this approximating condition) with $\{\tilde{N}(\tau, \tau+x) = 0\}$. Note that this approximation is exact for $\delta = 0$, since in that case $S_{n-1} = \tau$, so $X_n > x$ means that no arrivals occur in $(\tau, \tau+x)$.

We can now solve this approximate problem precisely,

$$\begin{aligned} \Pr\left\{\tilde{N}(\tau, \tau+x) = 0 \mid N(\tau-\delta) = n-2, \tilde{N}(\tau-\delta, \tau) = 1\right\} &= \Pr\left\{\tilde{N}(\tau, \tau+x) = 0\right\} \\ &= e^{-\lambda x}. \end{aligned}$$

In the first step, we used the independent increment property and in the second, the stationary increment property along with (a).

In the limit $\delta \rightarrow 0$, the conditioning event becomes $S_{n-1} = \tau$ and the conditioned event becomes $X_n > x$. The argument is very convincing, and becomes more convincing the more one thinks about it. At the same time, it is somewhat unsatisfactory since both the conditioned and conditioning event are being approximated. One can easily upper and lower bound the probability that $X_n > x$ for each δ but the ‘proof’ then requires many un insightful and tedious details.

c) For each $n > 1$, show that $\Pr\{X_n > x\} = e^{-\lambda x}$ and that X_n is independent of S_{n-1} .

Solution: We have seen that $\Pr\{X_n > x \mid S_{n-1} = \tau\} = e^{-\lambda x}$. Since the value of this probability conditioned on $\{S_{n-1} = \tau\}$ does not depend on τ , X_n must be independent of S_{n-1} .

d) Argue that X_n is independent of $X_1, X_2, \dots, X_{n-1}$.

Solution: Equivalently, we show that X_n is independent of $\{S_1=s_1, S_2=s_2, \dots, S_{n-1}=s_{n-1}\}$ for all choices of $0 < s_1 < s_2 < \dots < s_{n-1}$. Using the same artifice as in (b), this latter event is the same as the limit as $\delta \rightarrow 0$ of the event

$$\{N(s_1-\delta)=0, \tilde{N}(s_1-\delta, s_1)=1, \tilde{N}(s_1, s_2-\delta)=0, \tilde{N}(s_2-\delta, s_2)=1, \dots, \tilde{N}(s_{n-1}-\delta, s_{n-1})=1\}.$$

From the independent increment property, the above event is then independent of the rv $\tilde{N}(s_{n-1}, s_{n-1}+x)$ for each $x > 0$. As in (b), this shows that X_n is independent of $S_1, \dots, S_{n-1}$ and thus of $X_1, \dots, X_{n-1}$.

The most interesting part of this entire exercise is that the Poisson CDF was used only to derive the fact that X_1 has an exponential CDF. In other words, we have shown quite a bit more than Definition 2 of a Poisson process. We have shown that if X_1 is exponential and the stationary and independent increment properties hold, then the process is Poisson. On the other hand, we have shown that a careful derivation of the properties of the Poisson process from this definition requires a great deal of intricate, un insightful, and tedious analysis.

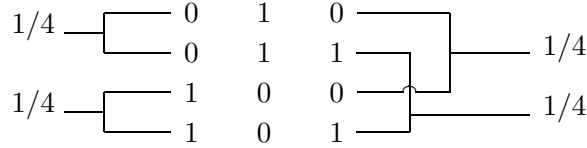
Exercise 2.5: The point of this exercise is to show that the sequence of PMF’s for the counting process of a Bernoulli process does not specify the process. In other words, knowing that $N(t)$ satisfies the binomial distribution for all t does not mean that the process is Bernoulli. This helps us understand why the second definition of a Poisson process requires stationary and independent increments along with the Poisson distribution for $N(t)$.

a) For a sequence of binary rv's $Y_1, Y_2, Y_3, \dots$, in which each rv is 0 or 1 with equal probability, find a joint distribution for Y_1, Y_2, Y_3 that satisfies the binomial distribution, $p_{N(t)}(k) = \binom{t}{k} 2^{-t}$ for $t = 1, 2, 3$ and $0 \leq k \leq t$, but for which Y_1, Y_2, Y_3 are not independent.

Your solution should contain four 3-tuples with probability $1/8$ each, two 3-tuples with probability $1/4$ each, and two 3-tuples with probability 0. Note that by making the subsequent arrivals IID and equiprobable, you have an example where $N(t)$ is binomial for all t but the process is not Bernoulli. Hint: Use the binomial for $t = 3$ to find two 3-tuples that must have probability $1/8$. Combine this with the binomial for $t = 2$ to find two other 3-tuples with probability $1/8$. Finally look at the constraints imposed by the binomial distribution on the remaining four 3-tuples.

Solution: The 3-tuples 000 and 111 each have probability $1/8$, and are the unique tuples for which $N(3) = 0$ and $N(3) = 3$ respectively. In the same way, $N(2) = 0$ only for $(Y_1, Y_2) = (0, 0)$, so $(0, 0)$ has probability $1/4$. Since $(0, 0, 0)$ has probability $1/8$, it follows that $(0, 0, 1)$ has probability $1/8$. In the same way, looking at $N(2) = 2$, we see that $(1, 1, 0)$ has probability $1/8$.

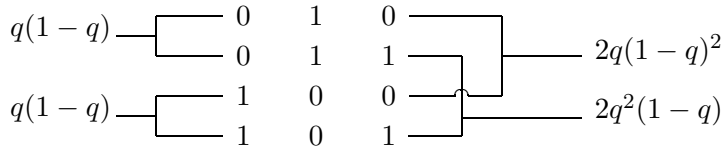
The four remaining 3-tuples are illustrated below, with the constraints imposed by $N(1)$ and $N(2)$ on the left and those imposed by $N(3)$ on the right.



It can be seen by inspection from the figure that if $(0, 1, 0)$ and $(1, 0, 1)$ each have probability $1/4$, then the constraints are satisfied. There is one other solution satisfying the constraints: choose $(0, 1, 1)$ and $(1, 0, 0)$ to each have probability $1/4$.

b) Generalize (a) to the case where Y_1, Y_2, Y_3 satisfy $\Pr\{Y_i = 1\} = q$ and $\Pr\{Y_i = 0\} = 1 - q$. Assume $q < 1/2$ and find a joint distribution on Y_1, Y_2, Y_3 that satisfies the binomial distribution, but for which the 3-tuple $(0, 1, 1)$ has zero probability.

Solution: Arguing as in (a), we see that $\Pr\{(0, 0, 0)\} = (1 - q)^3$, $\Pr\{(0, 0, 1)\} = (1 - q)^2 q$, $\Pr\{(1, 1, 1)\} = q^3$, and $\Pr\{(1, 1, 0)\} = q^2(1 - q)$. The remaining four 3-tuples are constrained as shown below.



If we set $\Pr\{(0, 1, 1)\} = 0$, then $\Pr\{(0, 1, 0)\} = q(1 - q)$, $\Pr\{(1, 0, 1)\} = 2q^2(1 - q)$, and $\Pr\{(1, 0, 0)\} = q(1 - q) - 2q^2(1 - q) = q(1 - q)(1 - 2q)$. This satisfies all the binomial constraints.

c) More generally yet, view a joint PMF on binary t -tuples as a nonnegative vector in a 2^t dimensional vector space. Each binomial probability $p_{N(t)}(k) = \binom{t}{k} q^k (1 - q)^{t-k}$ constitutes a linear constraint on this vector.

For each τ , show that one of these constraints may be replaced by the constraint that the components of the vector sum to 1.

Solution: There are 2^t binary n -tuples and each has a probability, so the joint PMF can be viewed as a vector of 2^t numbers. The binomial probability $p_{N(\tau)}(k) = \binom{\tau}{k} q^k (1-q)^{\tau-k}$ specifies the sum of the probabilities of the n -tuples in the event $\{N(\tau) = k\}$, and thus is a linear constraint on the joint PMF. Note: Mathematically, a linear constraint specifies that a given weighted sum of components is 0. The type of constraint here, where the weighted sum is a nonzero constant, is more properly called a first-order constraint. Engineers often refer to first order constraints as linear, and we follow that practice here.

Since $\sum_{k=0}^{\tau} \binom{\tau}{k} p^k q^{\tau-k} = 1$, one of these $\tau + 1$ constraints can be replaced by the constraint that the sum of all 2^t components of the PMF is 1.

d) Using (c), show that at most $(t+1)t/2 + 1$ of the binomial constraints are linearly independent. Note that this means that the linear space of vectors satisfying these binomial constraints has dimension at least $2^t - (t+1)t/2 - 1$. This linear space has dimension 1 for $t = 3$, explaining the results in parts a) and b). It has a rapidly increasing dimension for $t > 3$, suggesting that the binomial constraints are relatively ineffectual for constraining the joint PMF of a joint distribution. More work is required for the case of $t > 3$ because of all the inequality constraints, but it turns out that this large dimensionality remains.

Solution: We know that the sum of all the 2^t components of the PMF is 1, and we saw in (c) that for each integer τ , $1 \leq \tau \leq t$, there are τ additional linear constraints on the PMF established by the binomial terms $N(\tau = k)$ for $0 \leq k \leq \tau$. Since $\sum_{\tau=1}^t \tau = (t+1)t/2$, we see that there are $t(t+1)/2$ independent linear constraints on the joint PMF imposed by the binomial terms, in addition to the overall constraint that the components sum to 1. Thus the dimensionality of the 2^t vectors satisfying these linear constraints is at least $2^t - 1 - t(t+1)/2$.

Exercise 2.6: Let $h(x)$ be a positive function of a real variable that satisfies $h(x+t) = h(x) + h(t)$ and let $h(1) = c$.

a) Show that for integer $k > 0$, $h(k) = kc$.

Solution: We use induction. We know $h(1) = c$ and the inductive hypothesis is that $h(n) = nc$, which is satisfied for $n = 1$. We then have $h(n+1) = h(n) + h(1) = nc + c = (n+1)c$. Thus if the hypothesis is satisfied for n it is also satisfied for $n+1$, which verifies that it is satisfied for all positive integer n .

b) Show that for integer $j > 0$, $h(1/j) = c/j$.

Solution: Repeatedly adding $h(1/j)$ to itself, we get $h(2/j) = h(1/j) + h(1/j) = 2h(1/j)$, $h(3/j) = h(2/j) + h(1/j) = 3h(1/j)$ and so forth to $h(1) = h(j/j) = jh(1/j)$. Thus $h(1/j) = c/j$.

c) Show that for all positive integers k, j , $h(k/j) = ck/j$.

Solution: Since $h(1/j) = c/j$, for each positive integer j , we can use induction on positive integers k for any given $j > 0$ to get $h(k/j) = ck/j$.

d) The above parts show that $h(x)$ is linear in positive *rational* numbers. For very picky mathematicians, this does not guarantee that $h(x)$ is linear in positive *real* numbers. Show that if $h(x)$ is also monotonic in

x , then $h(x)$ is linear in $x > 0$.

Solution: Let $x > 0$ be a real number and let $x_1, x_2, \dots$ be a sequence of increasing rational numbers approaching x . Then $\lim_{i \rightarrow \infty} h(x_i) = c \lim_{i \rightarrow \infty} x_i = cx$. Thus $h(x) \geq cx$. If we look at a similarly decreasing sequence, we see that $h(x) \leq cx$, so $h(x) = cx$.

Exercise 2.7: Assume that a counting process $\{N(t); t \geq 0\}$ has the independent and stationary increment properties and, for all $t > 0$, satisfies

$$\begin{aligned} \Pr\{\tilde{N}(t, t + \delta) = 0\} &= 1 - \lambda\delta + o(\delta) \\ \Pr\{\tilde{N}(t, t + \delta) = 1\} &= \lambda\delta + o(\delta) \\ \Pr\{\tilde{N}(t, t + \delta) > 1\} &= o(\delta). \end{aligned} \tag{A.25}$$

a) Let $F_1^c(\tau) = \Pr\{N(\tau) = 0\}$ and show that $dF_1^c(\tau)/d\tau = -\lambda F_1^c(\tau)$.

Solution: Note that F_1^c is the complementary CDF of X_1 . Using the fundamental definition of a derivative,

$$\begin{aligned} \frac{dF_1^c(\tau)}{d\tau} &= \lim_{\delta \rightarrow 0} \frac{F_1^c(\tau + \delta) - F_1^c(\tau)}{\delta} = \lim_{\delta \rightarrow 0} \frac{\Pr\{N(\tau + \delta) = 0\} - \Pr\{N(\tau) = 0\}}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{\Pr\{N(\tau) = 0\} \left(\Pr\{\tilde{N}(\tau, \tau + \delta) = 0\} - 1 \right)}{\delta} \end{aligned} \tag{A.26}$$

$$\begin{aligned} &= \lim_{\delta \rightarrow 0} \frac{\Pr\{N(\tau) = 0\} (1 - \lambda\delta + o(\delta) - 1)}{\delta} \\ &= \Pr\{N(\tau) = 0\} (-\lambda) = -\lambda F_1^c(\tau), \end{aligned} \tag{A.27}$$

where (A.26) resulted from the independent increment property and (A.27) resulted from (A.25).

b) Show that X_1 , the time of the first arrival, is exponential with parameter λ .

Solution: The complementary CDF of X_1 is $F_1^c(\tau)$, which satisfies the first order linear differential equation in (a). The solution to that equation, with the boundary point $F_1^c(0) = 1$ is $e^{-\lambda\tau}$ for $\tau > 0$, showing that X_1 is exponential.

c) Let $F_n^c(\tau) = \Pr\{\tilde{N}(t, t + \tau) = 0 \mid S_{n-1} = t\}$ and show that $dF_n^c(\tau)/d\tau = -\lambda F_n^c(\tau)$.

Solution: By the independent increment property, $\{\tilde{N}(t, t + \tau) = 0\}$ is independent of $\{S_{n-1} = t\}$, and by the stationary increment property, it has the same probability as $N(\tau) = 0$. Thus $dF_n^c(\tau)/d\tau = -\lambda F_n^c(\tau)$ follows by the argument in (a).

d) Argue that X_n is exponential with parameter λ and independent of earlier arrival times.

Solution: Note that $F_n^c(\tau)$ is the complementary CDF of X_n conditional on $\{S_{n-1} = t\}$, and as shown in (c), it's distribution does not depend on S_{n-1} . In other words, $F_n^c(\tau)$ as found in (c) is the complementary CDF of X_n . It is exponential by the argument given in (b) for X_1 , and it is independent of earlier arrivals since $\{\tilde{N}(t, t + \tau) = 0\}$ is independent of arrivals before t .

This was also shown in the solution to Exercise 2.4 (c) and (d). In other words, definitions 2 and 3 of a Poisson process both follow from the assumptions that X_1 is exponential and that the stationary and independent increment properties hold.

Exercise 2.8: For a Poisson process, let $t > 0$ be arbitrary and let Z_1 be the duration of the interval from t until the next arrival after t . Let Z_m , for each $m > 1$, be the interarrival time from the epoch of the $(m-1)$ st arrival after t until the m th arrival after t .

a) Given that $N(t) = n$, explain why $Z_1 = X_{n+1} - t + S_n$ and, for each $m > 1$, $Z_m = X_{m+n}$.

Solution: Given $N(t) = n$, the m th arrival after t for $m \geq 1$ must be the $(n+m)$ th arrival overall. Its arrival epoch is then

$$S_{n+m} = S_{n+m-1} + X_{n+m}. \quad (\text{A.28})$$

By definition for $m > 1$, Z_m is the interval from S_{n+m-1} (the time of the $(m-1)$ st arrival after t) to S_{n+m} . Thus, from (A.28), $Z_m = X_{n+m}$ for $m > 1$. For $m = 1$, Z_1 is the interval from t until the next arrival, *i.e.*, $Z_1 = S_{n+1} - t$. Using $m = 1$ in (A.28), $Z_1 = S_n + X_{n+1} - t$.

b) Conditional on $N(t) = n$ and $S_n = \tau$, show that $Z_1, Z_2, \dots$ are IID.

Solution: The condition $N(t) = n$ and $S_n = \tau$ implies that $X_{n+1} > t - \tau$. Given this condition, $X_{n+1} - (t - \tau)$ is exponential, so $Z_1 = X_{n+1} - (t - \tau)$ is exponential. For $m > 1$, and for the given condition, $Z_m = X_{n+m}$. Since X_{n+m} is exponential and independent of $X_1, \dots, X_{n+m-1}$, we see that Z_m is also exponential and independent of $Z_1, \dots, Z_{m-1}$. Since these exponential distributions are the same, $Z_1, Z_2, \dots$, are IID conditional on $N(t) = n$ and $S_n = \tau$.

c) Show that $Z_1, Z_2, \dots$ are IID.

Solution: We have shown that $\{Z_m; m \geq 1\}$ are IID and exponential conditional on $N(t) = n$ and $S_n = \tau$. The joint distribution of $Z_1, \dots, Z_m$ is thus specified as a function of $N(t) = n$ and $S_n = \tau$. Since this function is constant in n and τ , the joint conditional distribution must be the same as the joint unconditional distribution, and therefore $Z_1, \dots, Z_m$ are IID for all $m > 0$.

Exercise 2.9: Consider a “shrinking Bernoulli” approximation $N_\delta(m\delta) = Y_1 + \dots + Y_m$ to a Poisson process as described in Subsection 2.2.5.

a) Show that

$$\Pr\{N_\delta(m\delta) = n\} = \binom{m}{n} (\lambda\delta)^n (1 - \lambda\delta)^{m-n}.$$

Solution: This is just the binomial PMF in (1.23)

b) Let $t = m\delta$, and let t be fixed for the remainder of the exercise. Explain why

$$\lim_{\delta \rightarrow 0} \Pr\{N_\delta(t) = n\} = \lim_{m \rightarrow \infty} \binom{m}{n} \left(\frac{\lambda t}{m}\right)^n \left(1 - \frac{\lambda t}{m}\right)^{m-n},$$

where the limit on the left is taken over values of δ that divide t .

Solution: This is the binomial PMF in (a) with $\delta = t/m$.

c) Derive the following two equalities:

$$\lim_{m \rightarrow \infty} \binom{m}{n} \frac{1}{m^n} = \frac{1}{n!}; \quad \text{and} \quad \lim_{m \rightarrow \infty} \left(1 - \frac{\lambda t}{m}\right)^{m-n} = e^{-\lambda t}.$$

Solution: Note that

$$\binom{m}{n} = \frac{m!}{n!(m-n)!} = \frac{1}{n!} \prod_{i=0}^{n-1} (m-i).$$

When this is divided by m^n , each term in the product above is divided by m , so

$$\binom{m}{n} \frac{1}{m^n} = \frac{1}{n!} \prod_{i=0}^{n-1} \frac{(m-i)}{m} = \frac{1}{n!} \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right). \quad (\text{A.29})$$

Taking the limit as $m \rightarrow \infty$, each of the n terms in the product approaches 1, so the limit is $1/n!$, verifying the first equality in (c). For the second,

$$\begin{aligned} \left(1 - \frac{\lambda t}{m}\right)^{m-n} &= \exp \left[(m-n) \ln \left(1 - \frac{\lambda t}{m}\right) \right] = \exp \left[(m-n) \left(\frac{-\lambda t}{m} + o(1/m) \right) \right] \\ &= \exp \left[-\lambda t + \frac{n\lambda t}{m} + (m-n)o(1/m) \right]. \end{aligned}$$

In the second equality, we expanded $\ln(1-x) = -x + x^2/2 \cdots$. In the limit $m \rightarrow \infty$, the final expression is $\exp(-\lambda t)$, as was to be shown.

If one wishes to see how the limit in (A.29) is approached, we have

$$\frac{1}{n!} \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right) = \frac{1}{n!} \exp \left(\sum_{i=0}^{n-1} \ln \left(1 - \frac{i}{m}\right) \right) = \frac{1}{n!} \exp \left(\frac{-n(n-1)}{2m} + o(1/m) \right).$$

d) Conclude from this that for every t and every n , $\lim_{\delta \rightarrow 0} \Pr\{N_\delta(t)=n\} = \Pr\{N(t)=n\}$ where $\{N(t); t > 0\}$ is a Poisson process of rate λ .

Solution: We simply substitute the results of (c) into the expression in (b), getting

$$\lim_{\delta \rightarrow 0} \Pr\{N_\delta(t) = n\} = \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

This shows that the Poisson PMF is the limit of shrinking Bernoulli PMF's, but recall from Exercise 2.5 that this is not quite enough to show that a Poisson process is the limit of shrinking Bernoulli processes. It is also necessary to show that the stationary and independent increment properties hold in the limit $\delta \rightarrow 0$. It can be seen that the Bernoulli process has these properties at each increment δ , and it is intuitively clear that these properties should hold in the limit, but it seems that carrying out all the analytical details to show this precisely is neither warranted or interesting.

Exercise 2.10: Let $\{N(t); t > 0\}$ be a Poisson process of rate λ .

a) Find the joint probability mass function (PMF) of $N(t)$, $N(t+s)$ for $s > 0$.

Solution: Note that $N(t+s)$ is the number of arrivals in $(0, t]$ plus the number in $(t, t+s)$. In order to find the joint distribution of $N(t)$ and $N(t+s)$, it makes sense to express $N(t+s)$

as $N(t) + \tilde{N}(t, t+s)$ and to use the independent increment property to see that $\tilde{N}(t, t+s)$ is independent of $N(t)$. Thus for $m > n$,

$$\begin{aligned} p_{N(t)N(t+s)}(n, m) &= \Pr\{N(t)=n\} \Pr\{\tilde{N}(t, t+s)=m-n\} \\ &= \frac{(\lambda t)^n e^{-\lambda t}}{n!} \times \frac{(\lambda s)^{m-n} e^{-\lambda s}}{(m-n)!}, \end{aligned}$$

where we have used the stationary increment property to see that $\tilde{N}(t, t+s)$ has the same distribution as $N(s)$. This solution can be rearranged in various ways, of which the most interesting is

$$p_{N(t)N(t+s)}(n, m) = \frac{(\lambda(t+s))^m e^{-\lambda(t+s)}}{m!} \times \binom{m}{n} \left(\frac{t}{t+s}\right)^n \left(\frac{s}{t+s}\right)^{m-n},$$

where the first term is $p_{N(t+s)}(m)$ (the probability of m arrivals in $(0, t+s]$) and the second, conditional on the first, is the binomial probability that n of those m arrivals occur in $(0, t)$.

b) Find $E[N(t) \cdot N(t+s)]$ for $s > 0$.

Solution: Again expressing $N(t+s) = N(t) + \tilde{N}(t, t+s)$,

$$\begin{aligned} E[N(t) \cdot N(t+s)] &= E[N^2(t)] + E[N(t)\tilde{N}(t, t+s)] \\ &= E[N^2(t)] + E[N(t)]E[N(s)] \\ &= \lambda t + \lambda^2 t^2 + \lambda t \lambda s. \end{aligned}$$

In the final step, we have used the fact (from Table 1.2 or a simple calculation) that the mean of a Poisson rv with PMF $(\lambda t)^n \exp(-\lambda t)/n!$ is λt and the variance is also λt (thus the second moment is $\lambda t + (\lambda t)^2$). This mean and variance was also derived in Exercise 2.2 and can also be calculated by looking at the limit of shrinking Bernoulli processes.

c) Find $E[\tilde{N}(t_1, t_3) \cdot \tilde{N}(t_2, t_4)]$ where $\tilde{N}(t, \tau)$ is the number of arrivals in $(t, \tau]$ and $t_1 < t_2 < t_3 < t_4$.

Solution: This is a straightforward generalization of what was done in (b). We break up $\tilde{N}(t_1, t_3)$ as $\tilde{N}(t_1, t_2) + \tilde{N}(t_2, t_3)$ and break up $\tilde{N}(t_2, t_4)$ as $\tilde{N}(t_2, t_3) + \tilde{N}(t_3, t_4)$. The interval $(t_2, t_3]$ is shared. Thus

$$\begin{aligned} E[\tilde{N}(t_1, t_3)\tilde{N}(t_2, t_4)] &= E[\tilde{N}(t_1, t_2)\tilde{N}(t_2, t_4)] + E[\tilde{N}^2(t_2, t_3)] + E[\tilde{N}(t_2, t_3)\tilde{N}(t_3, t_4)] \\ &= \lambda^2(t_2-t_1)(t_4-t_2) + \lambda^2(t_3-t_2)^2 + \lambda(t_3-t_2) + \lambda^2(t_3-t_2)(t_4-t_3) \\ &= \lambda^2(t_3-t_1)(t_4-t_2) + \lambda(t_3-t_2). \end{aligned}$$

Exercise 2.11: An elementary experiment is independently performed N times where N is a Poisson rv of mean λ . Let $\{a_1, a_2, \dots, a_K\}$ be the set of sample points of the elementary experiment and let p_k , $1 \leq k \leq K$, denote the probability of a_k .

a) Let N_k denote the number of elementary experiments performed for which the output is a_k . Find the PMF for N_k ($1 \leq k \leq K$). (Hint: no calculation is necessary.)

Solution: View the experiment as a combination of K Poisson processes where the k th has rate $p_k \lambda$ and the combined process has rate λ . At $t = 1$, the total number of experiments is

then Poisson with mean λ and the k th process is Poisson with mean $\mathbf{p}_k \lambda$. Thus $\mathbf{p}_{N_k}(n) = (\lambda \mathbf{p}_k)^n e^{-\lambda \mathbf{p}_k} / n!$.

b) Find the PMF for $N_1 + N_2$.

Solution: By the same argument,

$$\mathbf{p}_{N_1+N_2}(n) = \frac{[\lambda(\mathbf{p}_1 + \mathbf{p}_2)]^n e^{-\lambda(\mathbf{p}_1 + \mathbf{p}_2)}}{n!}.$$

c) Find the conditional PMF for N_1 given that $N = n$.

Solution: Each of the n combined arrivals over $(0, 1]$ is then a_1 with probability $\mathbf{p}_1$. Thus N_1 is binomial given that $N = n$,

$$\mathbf{p}_{N_1|N}(n_1|n) = \binom{n}{n_1} (\mathbf{p}_1)^{n_1} (1 - \mathbf{p}_1)^{n-n_1}.$$

d) Find the conditional PMF for $N_1 + N_2$ given that $N = n$.

Solution: Let the sample value of $N_1 + N_2$ be n_{12} . By the same argument in (c),

$$\mathbf{p}_{N_1+N_2|N}(n_{12}|n) = \binom{n}{n_{12}} (\mathbf{p}_1 + \mathbf{p}_2)^{n_{12}} (1 - \mathbf{p}_1 - \mathbf{p}_2)^{n-n_{12}}.$$

e) Find the conditional PMF for N given that $N_1 = n_1$.

Solution: Since N is then n_1 plus the number of arrivals from the other processes, and those additional arrivals are Poisson with mean $\lambda(1 - \mathbf{p}_1)$,

$$\mathbf{p}_{N|N_1}(n|n_1) = \frac{[\lambda(1 - \mathbf{p}_1)]^{n-n_1} e^{-\lambda(1-\mathbf{p}_1)}}{(n - n_1)!}.$$

Exercise 2.12: Starting from time 0, northbound buses arrive at 77 Mass. Avenue according to a Poisson process of rate λ . Customers arrive according to an independent Poisson process of rate μ . When a bus arrives, all waiting customers instantly enter the bus and subsequent customers wait for the next bus.

a) Find the PMF for the number of customers entering a bus (more specifically, for any given m , find the PMF for the number of customers entering the m th bus).

Solution: Since the customer arrival process and the bus arrival process are independent Poisson processes, the sum of the two counting processes is a Poisson counting process of rate $\lambda + \mu$. Each arrival for the combined process is a bus with probability $\lambda/(\lambda + \mu)$ and a customer with probability $\mu/(\lambda + \mu)$. The sequence of choices between bus or customer arrivals is an IID sequence. Thus, starting immediately after bus $m - 1$ (or at time 0 for $m = 1$), the probability of n customers in a row followed by a bus, for any $n \geq 0$, is $[\mu/(\lambda + \mu)]^n \lambda/(\lambda + \mu)$. This is the probability that n customers enter the m th bus, *i.e.*, defining N_m as the number of customers entering the m th bus, the PMF of N_m is

$$\mathbf{p}_{N_m}(n) = \left(\frac{\mu}{\lambda + \mu} \right)^n \frac{\lambda}{\lambda + \mu}. \quad (\text{A.30})$$

b) Find the PMF for the number of customers entering the m th bus given that the interarrival interval between bus $m - 1$ and bus m is x .

Solution: For any given interval of size x (*i.e.*, for the interval $(s, s+x]$ for any given s), the number of customer arrivals in that interval has a Poisson distribution of rate μ . Since the customer arrival process is independent of the bus arrivals, this is also the distribution of customer arrivals between the arrival of bus $m - 1$ and that of bus m given that the interval X_m between these bus arrivals is x . Thus letting X_m be the interval between the arrivals of bus $m - 1$ and m ,

$$p_{N_m|X_m}(n|x) = (\mu x)^n e^{-\mu x} / n!.$$

c) Given that a bus arrives at time 10:30 PM, find the PMF for the number of customers entering the next bus.

Solution: First assume that for some given m , bus $m - 1$ arrives at 10:30. The number of customers entering bus m is still determined by the argument in (a) and has the PMF in (A.30). In other words, N_m is independent of the arrival time of bus $m - 1$. From the formula in (A.30), the PMF of the number entering a bus is also independent of m . Thus the desired PMF is that on the right side of (A.30).

d) Given that a bus arrives at 10:30 PM and no bus arrives between 10:30 and 11, find the PMF for the number of customers on the next bus.

Solution: Using the same reasoning as in (b), the number of customer arrivals from 10:30 to 11 is a Poisson rv, say N' with PMF $p_{N'}(n) = (\mu/2)^n e^{-\mu/2} / n!$ (we are measuring time in hours so that μ is the customer arrival rate in arrivals per hour.) Since this is independent of bus arrivals, it is also the PMF of customer arrivals in (10:30 to 11] given no bus arrival in that interval.

The number of customers to enter the next bus is N' plus the number of customers N'' arriving between 11 and the next bus arrival. By the argument in (a), N'' has the PMF in (A.30). Since N' and N'' are independent, the PMF of $N' + N''$ (the number entering the next bus given this conditioning) is the convolution of the PMF's of N' and N'' , *i.e.*,

$$p_{N'+N''}(n) = \sum_{k=0}^n \left(\frac{\mu}{\lambda + \mu} \right)^k \frac{\lambda}{\lambda + \mu} \frac{(\mu/2)^{n-k} e^{-\mu/2}}{(n-k)!}.$$

This does not simplify in any nice way.

e) Find the PMF for the number of customers waiting at some given time, say 2:30 PM (assume that the processes started infinitely far in the past). Hint: think of what happens moving backward in time from 2:30 PM.

Solution: Let $\{Z_i; -\infty < i < \infty\}$ be the (doubly infinite) IID sequence of bus/customer choices where $Z_i = 0$ if the i th combined arrival is a bus and $Z_i = 1$ if it is a customer. Indexing this sequence so that -1 is the index of the most recent combined arrival before 2:30, we see that if $Z_{-1} = 0$, then no customers are waiting at 2:30. If $Z_{-1} = 1$ and $Z_{-2} = 0$, then one customer is waiting. In general, if $Z_{-n} = 0$ and $Z_{-m} = 1$ for $1 \leq m < n$, then n

customers are waiting. Since the Z_i are IID, the PMF of the number N_{past} waiting at 2:30 is

$$p_{N_{\text{past}}}(n) = \left(\frac{\mu}{\lambda + \mu} \right)^n \frac{\lambda}{\lambda + \mu}.$$

This is intuitive in one way, *i.e.*, the number of customers looking back toward the previous bus should be the same as the number of customers looking forward to the next bus since the bus/customer choices are IID. It is paradoxical in another way since if we visualize a sample path of the process, we see waiting customers gradually increasing until a bus arrival, then going to 0 and gradually increasing again, etc. It is then surprising that the number of customers at an arbitrary time is statistically the same as the number immediately before a bus arrival. This paradox is partly explained at the end of (f) and fully explained in Chapter 5.

Mathematically inclined readers may also be concerned about the notion of ‘starting infinitely far in the past.’ A more precise way of looking at this is to start the Poisson process at time 0 (in accordance with the definition of a Poisson process). We can then find the PMF of the number waiting at time t and take the limit of this PMF as $t \rightarrow \infty$. For very large t , the number M of combined arrivals before t is large with high probability. Given $M = m$, the geometric distribution above is truncated at m , which is a negligible correction for t large. This type of issue is handled more cleanly in Chapter 5.

f) Find the PMF for the number of customers getting on the next bus to arrive after 2:30. Hint: this is different from (a); look carefully at (e).

Solution: The number getting on the next bus after 2:30 is the sum of the number N_p waiting at 2:30 and the number of future customer arrivals N_f (found in (c)) until the next bus after 2:30. Note that N_p and N_f are IID. Convoluting these PMF’s, we get

$$\begin{aligned} p_{N_p + N_f}(n) &= \sum_{m=0}^n \left(\frac{\mu}{\lambda + \mu} \right)^m \frac{\lambda}{\lambda + \mu} \left(\frac{\mu}{\lambda + \mu} \right)^{n-m} \frac{\lambda}{\lambda + \mu} \\ &= (n+1) \left(\frac{\mu}{\lambda + \mu} \right)^n \left(\frac{\lambda}{\lambda + \mu} \right)^2. \end{aligned}$$

This is very surprising. It says that the number of people getting on the first bus after 2:30 is the sum of two IID rv’s, each with the same distribution as the number to get on the m th bus. This is an example of the ‘paradox of residual life,’ which we discuss very informally here and then discuss carefully in Chapter 5.

Consider a very large interval of time $(0, t_o]$ over which a large number of bus arrivals occur. Then choose a random time instant T , uniformly distributed in $(0, t_o]$. Note that T is more likely to occur within one of the larger bus interarrival intervals than within one of the smaller intervals, and thus, given the randomly chosen time instant T , the bus interarrival interval around that instant will tend to be larger than that from a given bus arrival, $m-1$ say, to the next bus arrival m . Since 2:30 is arbitrary, it is plausible that the interval around 2:30 behaves like that around T , making the result here also plausible.

g) Given that I arrive to wait for a bus at 2:30 PM, find the PMF for the number of customers getting on the next bus.

Solution: My arrival at 2:30 is in addition to the Poisson process of customers, and thus the number entering the next bus is $1 + N_p + N_f$. This has the sample value n if $N_p + N_f$ has the sample value $n - 1$, so from (f),

$$p_{1+N_p+N_f}(n) = n \left(\frac{\mu}{\lambda+\mu} \right)^{n-1} \left(\frac{\lambda}{\lambda+\mu} \right)^2.$$

Do not be discouraged if you made a number of errors in this exercise and if it still looks very strange. This is a first exposure to a difficult set of issues which will become clear in Chapter 5.

Exercise 2.13: a) Show that the arrival epochs of a Poisson process satisfy

$$f_{\mathbf{S}^{(n)}|S_{n+1}}(\mathbf{s}^{(n)}|s_{n+1}) = n!/s_{n+1}^n.$$

Hint: This is easy if you use only the results of Section 2.2.2.

Solution: Note that $\mathbf{S}^{(n)}$ is shorthand for $S_1, \dots, S_n$. Using Bayes' law,

$$f_{\mathbf{S}^{(n)}|S_{n+1}}(\mathbf{s}^{(n)}|s_{n+1}) = \frac{f_{\mathbf{S}^{(n)}}(\mathbf{s}^{(n)})f_{S_{n+1}|\mathbf{S}^{(n)}}(s_{n+1}|\mathbf{s}^{(n)})}{f_{S_{n+1}}(s_{n+1})}$$

From (2.15), $f_{\mathbf{S}^{(n)}}(\mathbf{s}^{(n)}) = \lambda^n e^{-\lambda s_n}$. Also, $f_{S_{n+1}|\mathbf{S}^{(n)}}(s_{n+1}|\mathbf{s}^{(n)}) = \lambda e^{-\lambda(s_{n+1}-s_n)}$. Finally $f_{S_{n+1}}(s_{n+1})$ is Erlang. Combining these terms,

$$f_{\mathbf{S}^{(n)}|S_{n+1}}(\mathbf{s}^{(n)}|s_{n+1}) = n!/s_{n+1}^n.$$

b) Contrast this with the result of Theorem 2.5.1

Solution: (a) says that $S_1, \dots, S_n$ are uniformly distributed (subject to the ordering constraint) between 0 and t for any sample value t for S_{n+1} . Theorem 2.5.1 says that they are uniformly distributed between 0 and t given that $N(t) = n$. The conditions $\{S_{n+1} = t\}$ and $\{N(t) = n\}$ each imply that there are n arrivals in $(0, t)$. The condition $\{S_{n+1} = t\}$ in addition specifies that arrival $n+1$ is at epoch t , whereas $\{N(t) = n\}$ specifies that arrival $n+1$ is in (t, ∞) . From the independent increment property, the arrival epochs in $(0, t)$ are independent of those in $[t, \infty)$ and thus the conditional joint distribution of $\mathbf{S}^{(n)}$ is the same for each conditioning event.

One might ask whether this equivalence of conditional distributions provides a rigorous way of answering (a). The answer is yes if the above argument is spelled out in more detail. It is simpler, however, to use the approach in (a). The equivalence approach is more insightful, on the other hand, so it is worthwhile to understand both approaches.

Exercise 2.14: Equation (2.42) gives $f_{S_i|N(t)}(s_i | n)$, which is the density of random variable S_i conditional on $N(t) = n$ for $n \geq i$. Multiply this expression by $\Pr\{N(t) = n\}$ and sum over n to find $f_{S_i}(s_i)$; verify that your answer is indeed the Erlang density.

Solution: It is almost magical, but of course it has to work out.

$$f_{S_i|N(t)}(s_i|n) = \frac{(s_i)^{i-1}}{(i-1)!} \frac{(t-s_i)^{n-i}}{(n-i)!} \frac{n!}{t^n}; \quad p_{N(t)}(n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

$$\begin{aligned}
\sum_{n=i}^{\infty} f(s_i|n) p(n) &= \frac{s_i^{i-1}}{(i-i)!} \sum_{n=i}^{\infty} \frac{(t-s_i)^{n-i}}{(n-i)!} \lambda^n e^{-\lambda t} \\
&= \frac{\lambda^i s_i^{i-1} e^{-\lambda s_i}}{(i-i)!} \sum_{n=i}^{\infty} \frac{\lambda^{n-i} (t-s_i)^{n-i} e^{-\lambda(t-s_i)}}{(n-i)!} \\
&= \frac{\lambda^i s_i^{i-1} e^{-\lambda s_i}}{(i-i)!}.
\end{aligned}$$

This is the Erlang distribution, and it follows because the preceding sum is the sum of terms in the PMF for the Poisson rv of rate $\lambda(t-s_i)$

Exercise 2.15: Consider generalizing the bulk arrival process in Figure 2.5. Assume that the epochs at which arrivals occur form a Poisson process $\{N(t); t > 0\}$ of rate λ . At each arrival epoch S_n , the number of arrivals Z_n , satisfies $\Pr\{Z_n=1\} = p$, $\Pr\{Z_n=2\} = 1-p$. The variables Z_n are IID.

a) Let $\{N_1(t); t > 0\}$ be the counting process of the epochs at which single arrivals occur. Find the PMF of $N_1(t)$ as a function of t . Similarly, let $\{N_2(t); t \geq 0\}$ be the counting process of the epochs at which double arrivals occur. Find the PMF of $N_2(t)$ as a function of t .

Solution: Since the process of arrival epochs is Poisson, and these epochs are split into single and double-arrival epochs by an IID splitting, the process of single-arrival epochs is Poisson with rate λp and the process of double-arrival epochs is Poisson with rate $\lambda(1-p)$. Thus, letting $q = 1-p$,

$$p_{N_1(t)}(n) = \frac{(\lambda p)^n e^{-\lambda p}}{n!}; \quad p_{N_2(t)}(m) = \frac{(\lambda q)^m e^{-\lambda q}}{m!}.$$

b) Let $\{N_B(t); t \geq 0\}$ be the counting process of the total number of arrivals. Give an expression for the PMF of $N_B(t)$ as a function of t .

Solution: Since there are two arrivals at each double-arrival epoch, we have $N_B(t) = N_1(t) + 2N_2(t)$, and as seen above $N_1(t)$ and $N_2(t)$ are independent. This can be done as a digital convolution of the PMF's for $N_1(t)$ and $2N_2(t)$, but this can be slightly confusing since $2N_2(t)$ is nonzero only for even integers. Thus we revert to the general approach, which also reminds you where convolution comes from and thus how it can be done in general (PDF's, PMF's, etc.)

$$\Pr\{N_B(t) = n, N_2(t) = m\} = p_{N_1(t)}(n-2m) p_{N_2(t)}(m).$$

The marginal PMF for $N_B(t)$ is then given by

$$\begin{aligned}
p_{N_B(t)}(n) &= \sum_{m=0}^{\lfloor n/2 \rfloor} p_{N_1(t)}(n-2m) p_{N_2(t)}(m) \\
&= \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(\lambda p t)^{n-2m} e^{-\lambda p t}}{(n-2m)!} \frac{(\lambda q t)^m e^{-\lambda q t}}{m!} \\
&= e^{-\lambda t} \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(\lambda p t)^{n-2m} (\lambda q t)^m}{(n-2m)! m!}.
\end{aligned}$$

Exercise 2.16: a) For a Poisson counting process of rate λ , find the joint probability density of $S_1, S_2, \dots, S_{n-1}$ conditional on $S_n = t$.

Solution: The joint density of $S_1, \dots, S_n$ is given in (2.15) as $f_{S_1, \dots, S_n}(s_1, \dots, s_n) = \lambda^n \exp(-\lambda s_n)$. The marginal density of S_n is the Erlang density. The conditional density is the ratio of these, *i.e.*,

$$f_{S_1, \dots, S_{n-1}|S_n}(s_1, \dots, s_{n-1}|s_n) = \frac{\lambda^n \exp(-\lambda s_n)}{\lambda^n s_n^{n-1} e^{-\lambda s_n} / (n-1)!} = \frac{(n-1)!}{s_n^{n-1}}.$$

b) Find $\Pr\{X_1 > \tau \mid S_n = t\}$.

Solution: We first use Bayes' law to find the density, $f_{X_1|S_n}(\tau|t)$.

$$\begin{aligned} f_{X_1|S_n}(\tau|t) &= \frac{f_{X_1}(\tau) f_{S_n|X_1}(t|\tau)}{f_{S_n}(t)} = \frac{\lambda e^{-\lambda \tau} \lambda^{n-1} (t-\tau)^{n-2} e^{-\lambda(t-\tau)} / (n-2)!}{\lambda^n t^{n-1} e^{-\lambda t} / (n-1)!} \\ &= \frac{(t-\tau)^{n-2} (n-1)}{t^{n-1}} \quad \text{for } \tau < t \end{aligned}$$

Integrating this with respect to τ , we get

$$\Pr\{X_1 > \tau \mid S_n = t\} = \left[\frac{t-\tau}{t} \right]^{n-1}.$$

c) Find $\Pr\{X_i > \tau \mid S_n = t\}$ for $1 \leq i \leq n$.

Solution: The condition here is $X_1 + \dots + X_n = t$. Since $X_1, \dots, X_n$ are IID without the condition, and the condition is symmetric in $X_1, \dots, X_n$, we see that $X_1, \dots, X_n$ are identically distributed conditional on $S_n = t$. Thus, from (b),

$$\Pr\{X_i > \tau \mid S_n = t\} = \left[\frac{t-\tau}{t} \right]^{n-1}; \quad \text{for } 1 \leq i \leq n.$$

d) Find the density $f_{S_i|S_n}(s_i|t)$ for $1 \leq i \leq n-1$.

Solution: We can use Bayes' law in the same way as (b), getting

$$f_{S_i|S_n}(s_i|t) = \frac{s_i^{i-1} (t-s_i)^{n-i-1} (n-1)!}{t^{n-1} (i-1)! (n-i-1)!}.$$

e) Give an explanation for the striking similarity between the condition $N(t) = n-1$ and the condition $S_n = t$.

Solution: The solutions to (c) and (d) are the same as (2.45) and (2.46) respectively for $N(t) = n-1$. The condition $N(t) = n-1$ and the condition $S_n = t$ both imply that the number of arrivals in $(0, t)$ is $n-1$. In addition, $N(t) = n-1$ implies that the first arrival after $(0, t)$ is strictly after t , whereas $S_n = t$ implies that the first arrival after $(0, t)$ is at t . Because of the independent increment property, this additional implication does not affect the distribution of $S_1, \dots, S_{n-1}$. See the solution to Exercise 2.13(b) for a further discussion of this point.

The important fact here is that the equivalence of the arrival distributions in $(0, t)$, given these slightly different conditions, is a valuable aid in problem solving, since either approach can be used.

Exercise 2.17: a) For a Poisson process of rate λ , find $\Pr\{N(t)=n \mid S_1=\tau\}$ for $t > \tau$ and $n \geq 1$.

Solution: Given that $S_1 = \tau$, the number, $N(t)$, of arrivals in $(0, t]$ is 1 plus the number in $(\tau, t]$. This latter number, $\tilde{N}(\tau, t)$ is Poisson with mean $\lambda(t - \tau)$. Thus,

$$\Pr\{N(t)=n \mid S_1=\tau\} = \Pr\{\tilde{N}(\tau, t) = n-1\} = \frac{[\lambda(t - \tau)]^{n-1} e^{-\lambda(t-\tau)}}{(n-1)!}.$$

b) Using this, find $f_{S_1}(\tau) \mid N(t)=n$.

Solution: Using Bayes' law,

$$f_{S_1|N(t)}(\tau|n) = \frac{n(t - \tau)^{n-1}}{t^n}.$$

c) Check your answer against (2.41).

Solution: Eq. (2.41) is $\Pr\{S_1 > \tau \mid N(t) = n\} = [(t - \tau)/t]^n$. The derivative of this with respect to τ is $-f_{S_1|N(t)}(\tau|t)$, which clearly checks with (b).

Exercise 2.18: Consider a counting process in which the rate is a rv Λ with probability density $f_\Lambda(\lambda) = \alpha e^{-\alpha\lambda}$ for $\lambda > 0$. Conditional on a given sample value λ for the rate, the counting process is a Poisson process of rate λ (i.e., nature first chooses a sample value λ and then generates a sample path of a Poisson process of that rate λ).

a) What is $\Pr\{N(t)=n \mid \Lambda=\lambda\}$, where $N(t)$ is the number of arrivals in the interval $(0, t]$ for some given $t > 0$?

Solution: Conditional on $\Lambda = \lambda$, $\{N(t); t > 0\}$ is a Poisson process, so

$$\Pr\{N(t)=n \mid \Lambda=\lambda\} = (\lambda t)^n e^{-\lambda t} / n!.$$

b) Show that $\Pr\{N(t)=n\}$, the unconditional PMF for $N(t)$, is given by

$$\Pr\{N(t)=n\} = \frac{\alpha t^n}{(t + \alpha)^{n+1}}.$$

Solution: The straightforward approach is to average the conditional distribution over λ ,

$$\begin{aligned} \Pr\{N(t) = n\} &= \int_0^\infty \frac{(\lambda t)^n e^{-\lambda t}}{n!} \alpha e^{-\lambda \alpha} d\lambda \\ &= \frac{\alpha t^n}{(t + \alpha)^n} \int_0^\infty \frac{[\lambda(t + \alpha)]^n e^{-\lambda(t + \alpha)}}{n!} d\lambda \\ &= \frac{\alpha t^n}{(t + \alpha)^{n+1}} \int_0^\infty \frac{x^n e^{-x}}{n!} dx = \frac{\alpha t^n}{(t + \alpha)^{n+1}}, \end{aligned}$$

where we changed the variable of integration from λ to $x = \lambda(t + \alpha)$ and then recognized the integral as the integral of an Erlang density of order $n+1$ with unit rate.

The solution can also be written as pq^n where $p = \alpha/(t+\alpha)$ and $q = t/(t+\alpha)$. This suggests a different interpretation for this result. $\Pr\{N(t) = n\}$ for a Poisson process (PP) of rate λ is a function only of λt and n . The rv $N(t)$ for a PP of rate λ thus has the same distribution as $N(\lambda t)$ for a PP of unit rate and thus $N(t)$ for a PP of variable rate Λ has the same distribution as $N(\Lambda t)$ for a PP of rate 1.

Since Λt is an exponential rv of parameter α/t , we see that $N(\Lambda t)$ is the number of arrivals of a PP of unit rate before the first arrival from an independent PP of rate α/t . This unit rate PP and rate α/t PP are independent and the combined process has rate $1 + \alpha/t$. The event $\{N(\Lambda t) = n\}$ then has the probability of n arrivals from the unit rate PP followed by one arrival from the α/t rate process, thus yielding the probability $q^n p$.

c) Find $f_{\Lambda}(\lambda \mid N(t)=n)$, the density of Λ conditional on $N(t)=n$.

Solution: Using Bayes' law with the answers in (a) and (b), we get

$$f_{\Lambda|N(t)}(\lambda \mid n) = \frac{\lambda^n e^{-\lambda(\alpha+t)} (\alpha+t)^{n+1}}{n!}.$$

This is an Erlang PDF of order $n+1$ and can be interpreted (after a little work) in the same way as (b).

d) Find $E[\Lambda \mid N(t)=n]$ and interpret your result for very small t with $n = 0$ and for very large t with n large.

Solution: Since Λ conditional on $N(t) = n$ is Erlang, it is the sum of $n+1$ IID rv's, each of mean $1/(t+\alpha)$. Thus

$$E[\Lambda \mid N(t)=n] = \frac{n+1}{\alpha+t}.$$

For $N(t) = 0$ and $t \ll \alpha$, this is close to $1/\alpha$, which is $E[\Lambda]$. This is not surprising since it has little effect on the distribution of Λ . For n large and $t \gg \alpha$, $E[\Lambda \mid N(t)=n] \approx n/t$.

e) Find $E[\Lambda \mid N(t)=n, S_1, S_2, \dots, S_n]$. (Hint: consider the distribution of $S_1, \dots, S_n$ conditional on $N(t)$ and Λ). Find $E[\Lambda \mid N(t)=n, N(\tau)=m]$ for some $\tau < t$.

Solution: From Theorem 2.5.1, $S_1, \dots, S_n$ are uniformly distributed, subject to $0 < S_1 < \dots < S_n < t$, given $N(t) = n$ and $\Lambda = \lambda$. Thus, conditional on $N(t) = n$, Λ is statistically independent of $S_1, \dots, S_n$.

$$E[\Lambda \mid N(t)=n, S_1, S_2, \dots, S_n] = E[\Lambda \mid N(t)=n] = \frac{n+1}{\alpha+t}.$$

Conditional on $N(t) = n$, $N(\tau)$ is determined by $S_1, \dots, S_n$ for $\tau < t$, and thus

$$E[\Lambda \mid N(t)=n, N(\tau)=m] = E[\Lambda \mid N(t)=n] = \frac{n+1}{\alpha+t}.$$

This corresponds to one's intuition; given the number of arrivals in $(0, t]$, it makes no difference where the individual arrivals occur.

Exercise 2.19: a) Use Equation (2.42) to find $E[S_i | N(t)=n]$. Hint: When you integrate $s_i f_{S_i}(s_i | N(t)=n)$, compare this integral with $f_{S_{i+1}}(s_i | N(t)=n+1)$ and use the fact that the latter expression is a probability density.

Solution: We can find $E[S_i | N(t)=n]$ from $f_{S_i|N(t)}(s_i|n)$ (as given in (2.42)) by integration,

$$E[S_i | N(t)=n] = \int_0^\infty x f_{S_i|N(t)}(x|n) dx = \int_0^\infty \frac{x^i (t-x)^{n-i} n!}{t^n (n-i)!(i-1)!} dx. \quad (\text{A.31})$$

Using the hint,

$$f_{S_{i+1}|N(t)}(x|n+1) = \frac{x^i (t-x)^{n-i} (n+1)!}{t^{n+1} (n-i)! i!}. \quad (\text{A.32})$$

The factors involving x are the same as in (A.31), and substituting (A.32) into (A.31)

$$E[S_i | N(t)=n] = \int_0^\infty \frac{it}{n+1} f_{S_{i+1}|N(t)}(x|n+1) dx = \frac{it}{n+1}.$$

As a check, we see that this agrees with the more elegant derivation in (2.43). The derivation in (2.43), however, does not generalize as easily to the second moment.

b) Find the second moment and the variance of S_i conditional on $N(t)=n$. Hint: Extend the previous hint.

Solution: Using the same approach as in (AP19)

$$E[S_i^2 | N(t)=n] = \int_0^\infty x^2 f_{S_i|N(t)}(x|n) dx = \int_0^\infty \frac{x^{i+1} (t-x)^{n-i} n!}{t^n (n-i)!(i-1)!} dx. \quad (\text{A.33})$$

This suggests comparing with the density of S_{i+2} conditional on $N(t) = n+2$,

$$f_{S_{i+2}|N(t)}(x|n+2) = \frac{x^{i+1} (t-x)^{n-i} (n+2)!}{t^{n+2} (n-i)!(i+1)!}. \quad (\text{A.34})$$

$$E[S_i^2 | N(t)=n] = \frac{(i+1) i t^2}{(n+2)(n+1)}.$$

Finally, calculating the variance as the second moment minus the mean squared,

$$\text{VAR}[S_i | N(t) = n] = \frac{(i+1) i t^2}{(n+2)(n+1)} - \left[\frac{it}{n+1} \right]^2 = \frac{it^2(n+1-i)}{(n+1)^2(n+2)}.$$

c) Assume that n is odd, and consider $i = (n+1)/2$. What is the relationship between S_i , conditional on $N(t)=n$, and the sample median of n IID uniform random variables.

Solution: We have seen that the first n arrival epochs of a Poisson process, conditional on $N(t) = n$ have the same joint probability distribution as the order statistics of n IID rv's that are uniform over $(0, t]$. Thus the sample value of the rv $S_{(n+1)/2}$ is the sample median of those rv's. From (a) and (b),

$$E[S_{(n+1)/2}] = t/2; \quad \text{VAR}[S_{(n+1)/2}] = \frac{t^2}{4(n+2)}.$$

d) Give a weak law of large numbers for the above median.

Solution: The median $S_{(n+1)/2}$ is a rv, just like the sample average of n rv's is a rv. By a WWLN for the median, we mean a result that says that $S_{(n+1)/2}$ converges in probability to a limit, in this case the mean of $S_{(n+1)/2}$ as $n \rightarrow \infty$. Using the Chebyshev inequality,

$$\Pr \left\{ \left| S_{(n+1)/2} - \frac{t}{2} \right| \geq \epsilon \right\} \leq \frac{\text{VAR} [S_{(n+1)/2}]}{\epsilon^2} = \frac{t^2}{4(n+2)\epsilon^2}.$$

Thus, for any fixed ϵ, t ,

$$\lim_{n \rightarrow \infty} \Pr \left\{ \left| S_{(n+1)/2} - \frac{t}{2} \right| \geq \epsilon \right\} = 0.$$

The limit here does not correspond to any limit in a given Poisson process, but it does correspond to a limit of the median of n uniformly distributed IID rv's. By combining this result with Exercise (1.28), it is possible to extend this result to the median of IID rv's with an arbitrary probability density.

Exercise 2.20: Suppose cars enter a one-way infinite length, infinite lane highway at a Poisson rate λ . The i th car to enter chooses a velocity V_i and travels at this velocity. Assume that the V_i 's are independent positive rv's having a common CDF F . Derive the distribution of the number of cars that are located in an interval $(0, a)$ at time t .

Solution: This is a thinly disguised variation of an M/G/ ∞ queue. The arrival process is the Poisson process of cars entering the highway. We then view the service time of a car as the time interval until the car reaches or passes point a . All cars then have IID service times, and service always starts at the time of arrival (*i.e.*, this can be viewed as infinitely many independent and identical servers). To avoid distractions, assume initially that V is a continuous rv. The CDF $G(\tau)$ of the service time X is then given by the equation

$$G(\tau) = \Pr\{X \leq \tau\} = \Pr\{a/V \leq \tau\} = \Pr\{V \geq a/\tau\} = F_V^c(a/\tau).$$

The PMF of the number $N_1(t)$ of cars in service at time t is then given by (2.36) and (2.37) as

$$p_{N_1(t)}(n) = \frac{m^n(t) \exp[-m(t)]}{n!},$$

where

$$m(t) = \lambda \int_0^t [1 - G(\tau)] d\tau = \lambda \int_0^t F_V(a/\tau) d\tau.$$

Since this depends only on the CDF, it can be seen that the answer is the same if V is discrete or mixed.

Exercise 2.21: Consider an M/G/ ∞ queue, *i.e.*, a queue with Poisson arrivals of rate λ in which each arrival i , independent of other arrivals, remains in the system for a time X_i , where $\{X_i; i \geq 1\}$ is a set of IID rv's with some given CDF $F(x)$.

You may assume that the number of arrivals in any interval $(t, t + \epsilon)$ that are still in the system at some later time $\tau \geq t + \epsilon$ is statistically independent of the number of arrivals in that same interval $(t, t + \epsilon)$ that have departed from the system by time τ .

a) Let $N(\tau)$ be the number of customers in the system at time τ . Find the mean, $m(\tau)$, of $N(\tau)$ and find $\Pr\{N(\tau) = n\}$.

Solution: The answer is worked out and explained in (2.36) and (2.37). We have

$$m(\tau) = \lambda \int_0^\tau [1 - F(t)] dt; \quad \Pr\{N(\tau) = n\} = \frac{m(\tau)^n \exp(-m(\tau))}{n!}.$$

Note that $m(\tau)$ is non-decreasing in τ and has the limit $\lim_{\tau \rightarrow \infty} m(\tau) = \lambda \bar{X}$.

b) Let $D(\tau)$ be the number of customers that have departed from the system by time τ . Find the mean, $E[D(\tau)]$, and find $\Pr\{D(\tau) = d\}$.

Solution: As illustrated in Figure 2.8, The number of arrivals that have departed and those that remain can be treated as a non-homogeneous splitting of the Poisson arrival process. Thus the departures can be handled in the same way as those still in the system. Let $\mu(\tau) = E[D(\tau)]$ be the mean of the number of arrivals that have departed by time t . Then $\mu(\tau)$ and $\Pr\{D(\tau) = d\}$ are given by

$$\mu(\tau) = \lambda \int_0^\tau F(t) dt; \quad \Pr\{D(\tau) = d\} = \frac{\mu(\tau)^d \exp(-\mu(\tau))}{d!}.$$

Note that $m(\tau) + \mu(\tau) = \lambda \int_0^\tau dt = \lambda\tau$ so that $\mu(\tau)$ tends to $\lambda\tau - \lambda\bar{X}$ as τ increases.

c) Find $\Pr\{N(\tau) = n, D(\tau) = d\}$.

Solution: By the same argument as used in Section 2.3 on the splitting of homogeneous Poisson processes, the processes $\{N(\tau); \tau > 0\}$ and the process $\{D(\tau); \tau > 0\}$ are statistically independent. The assumption in the exercise is one step in that argument. Combining (a) and (b), we then have

$$\Pr\{N(\tau) = n, D(\tau) = d\} = \frac{m(\tau)^n \mu(\tau)^d \exp(-m(\tau) - \mu(\tau))}{n! d!}. \quad (\text{A.35})$$

d) Let $A(\tau)$ be the total number of arrivals up to time τ . Find $\Pr\{N(\tau) = n \mid A(\tau) = a\}$.

Solution: Since $A(\tau) = N(\tau) + D(\tau)$, we can let $a = n + d$ for $a \geq n$ and rewrite (A.35) as

$$\Pr\{N(\tau) = n, A(\tau) = a\} = \frac{m(\tau)^n \mu(\tau)^{a-n} \exp(-m(\tau) - \mu(\tau))}{n! (a-n)!}. \quad (\text{A.36})$$

Since $A(\tau)$ is Poisson with rate $\lambda = [m(\tau) + \mu(\tau)]/\tau$, the conditional PMF is

$$\begin{aligned} \Pr\{N(\tau) = n \mid A(\tau) = a\} &= \frac{m(\tau)^n \mu(\tau)^{a-n} \exp(-\lambda\tau)}{n! (a-n)!} \times \frac{a!}{(\lambda\tau)^a \exp(-\lambda\tau)} \\ &= \binom{a}{n} \left[\frac{m(\tau)}{\lambda\tau} \right]^n \left[\frac{\mu(\tau)}{\lambda\tau} \right]^{a-n}. \end{aligned}$$

e) Find $\Pr\{D(\tau + \epsilon) - D(\tau) = d\}$.

Solution: Since $\{D(\tau); \tau > 0\}$ is a non-homogeneous Poisson process, $Y = D(\tau + \epsilon) - D(\tau)$ is a Poisson random variable of mean $\bar{Y} = \lambda \int_{\tau}^{\tau + \epsilon} F(t) dt$. Thus

$$\Pr\{Y = d\} = \frac{\bar{Y}^d \exp(-\bar{Y})}{d!}.$$

Note that Y is the number of departures in $(\tau, \tau + \epsilon]$. This is a Poisson rv which has a mean approaching $\lambda\epsilon$ for $\tau \rightarrow \infty$. The output process is a non-homogeneous Poisson process which becomes stationary (homogeneous) as $\tau \rightarrow \infty$.

Exercise 2.22: The voters in a given town arrive at the place of voting according to a Poisson process of rate $\lambda = 100$ voters per hour. The voters independently vote for candidate A and candidate B each with probability $1/2$. Assume that the voting starts at time 0 and continues indefinitely.

a) Conditional on 1000 voters arriving during the first 10 hours of voting, find the probability that candidate A receives n of those votes.

Solution: Intuitively, each of the 1000 voters votes independently for A with probability $1/2$. Thus, from the binomial formula,

$$\Pr\{n \text{ of } 1000 \text{ votes for } A\} = \binom{1000}{n} (1/2)^{1000}.$$

Being more careful, the number of votes for A in 10 hours is Poisson with mean 500 and the number for B is independent and Poisson with mean 500. The total number of votes in 10 hours is the sum of these and also Poisson. Using Bayes' law to find the votes for A conditional on the overall number, we get the same answer.

b) Again conditional on 1000 voters during the first 10 hours, find the probability that candidate A receives n votes in the first 4 hours of voting.

Solution: Intuitively, each of the 1000 voters comes in the first 4 hours independently with probability .4, and out of those, each independently votes for A with probability $1/2$. Thus each voter independently both comes in the first 4 hours and votes for A with probability 0.2. Thus

$$\Pr\{n \text{ votes for } A \text{ in first 4 hours} \mid 1000 \text{ in 10 hours}\} = \binom{1000}{n} (0.2)^n (0.8)^{1000-n}.$$

Being more careful (or first stating a general theorem about these conditional Poisson probabilities), the number in the first 4 hours is Poisson, the number of those who vote for A is Poisson and the overall number to vote in 10 hours is Poisson and the sum of those in the first 4 hours and those in the last 6 hours, further broken into A and B are independent Poisson. We can then find the conditional probability as in (a).

c) Let T be the epoch of the arrival of the first voter voting for candidate A . Find the density of T .

Solution: View the voters for A as a splitting of the overall arrival process. Thus the voters for A form a Poisson process of rate 50 and $f_T(t) = 50 \exp(-50t)$.

d) Find the PMF of the number of voters for candidate B who arrive before the first voter for A .

Solution: n B voters arrive before the first A if the first n voters are B and the $(n+1)$ st is A ; this is an event of probability $(1/2)^{n+1}$ for $n \geq 0$. Thus this number is a geometric rv.

e) Define the n th voter as a *reversal* if the n th voter votes for a different candidate than the $(n-1)$ st. For example, in the sequence of votes $AABAABB$, the third, fourth, and sixth voters are reversals; the third and sixth are A to B reversals and the fourth is a B to A reversal. Let $N(t)$ be the number of reversals up to time t (t in hours). Is $\{N(t); t > 0\}$ a Poisson process? Explain.

Solution: The first voter can not be a reversal. Every subsequent voter is a reversal with probability $1/2$. Thus, after the first arrival, each inter-reversal interval is exponential with mean $1/50$ hours. The process is not Poisson, however, because the interval until the first reversal is not exponential with mean $1/50$ hours.

When we study renewal processes, we denote renewal processes with a non-standard first renewal interval as ‘delayed renewal processes’ and find that most of the renewal results also apply to delayed renewal processes. Thus we might call this a delayed Poisson process.

f) Find the expected time (in hours) between reversals.

Solution: Starting from one reversal we can view subsequent reversals as an equi-probable splitting of the arrivals. Thus the expected time between reversals is $1/50$.

g) Find the probability density of the time between reversals.

Solution: As explained in (f), the time X between reversals is exponential with mean $1/50$, so $f_X(x) = 50 \exp(-50t)$.

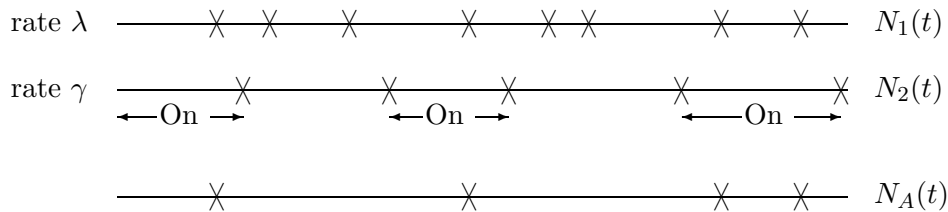
h) Find the density of the time from one A to B reversal to the next A to B reversal.

Solution: Since A to B reversals alternate with B to A reversals, the time Y from one A to B reversal to the next is the sum of 2 independent exponential rv’s. One can either calculate this density by convolution or recognize it as an Erlang rv of order 2.

$$f_Y(y) = \lambda^2 y \exp(-\lambda y) \quad \text{where } \lambda = 50 \text{ and } y \geq 0.$$

An alternate approach here is to see that after an A to B reversal, the time to the next A to B reversal is the sum of the time to the next A followed by the time to the next B .

Exercise 2.23: Let $\{N_1(t); t > 0\}$ be a Poisson counting process of rate λ . Assume that the arrivals from this process are switched on and off by arrivals from a second independent Poisson process $\{N_2(t); t > 0\}$ of rate γ .



Let $\{N_A(t); t > 0\}$ be the switched process; that is $N_A(t)$ includes the arrivals from $\{N_1(t); t > 0\}$ during periods when $N_2(t)$ is even and excludes the arrivals from $\{N_1(t); t > 0\}$ while $N_2(t)$ is odd.

a) Find the PMF for the number of arrivals of the first process, $\{N_1(t); t > 0\}$, during the n th period when the switch is on.

Solution: We have seen that the combined process $\{N_1(t) + N_2(t)\}$ is a Poisson process of rate $\lambda + \gamma$. For any even numbered arrival to process 2, subsequent arrivals to the combined process independently come from process 1 or 2, and come from process 1 with probability $\lambda/(\lambda + \gamma)$. The number N_s of such arrivals before the next arrival to process 2 is geometric with PMF $p_{N_s}(n) = [\lambda/(\lambda + \gamma)]^n [\gamma/(\lambda + \gamma)]$ for integer $n \geq 0$.

b) Given that the first arrival for the second process occurs at epoch τ , find the conditional PMF for the number of arrivals N_a of the first process up to τ .

Solution: Since processes 1 and 2 are independent, this is equal to the PMF for the number of arrivals of the first process up to τ . This number has a Poisson PMF, $(\lambda\tau)^n e^{-\lambda\tau}/n!$.

c) Given that the number of arrivals of the first process, up to the first arrival for the second process, is n , find the density for the epoch of the first arrival from the second process.

Solution: Let N_a be the number of process 1 arrivals before the first process 2 arrival and let X_2 be the time of the first process 2 arrival. In (a), we showed that $p_{N_a}(n) = [\lambda/(\lambda + \gamma)]^n [\gamma/(\lambda + \gamma)]$ and in (b) we showed that $p_{N_a|X_2}(n|\tau) = (\lambda\tau)^n e^{-\lambda\tau}/n!$. We can then use Bayes' law to find $f_{X_2|N_a}(\tau | n)$, which is the desired solution. We have

$$f_{X_2|N_a}(\tau | n) = f_{X_2}(\tau) \frac{p_{N_a|X_2}(n|\tau)}{p_{N_a}(n)} = \frac{(\lambda + \gamma)^{n+1} \tau^n e^{-(\lambda + \gamma)\tau}}{n!},$$

where we have used the fact that X_2 is exponential with PDF $\gamma \exp(-\gamma\tau)$ for $\tau \geq 0$. It can be seen that the solution is an Erlang rv of order $n + 1$. To interpret this (and to solve the exercise in a perhaps more elegant way), note that this is the same as the Erlang density for the epoch of the $(n+1)$ th arrival in the combined process. This arrival epoch is independent of the process 1/process 2 choices for these $n+1$ arrivals, and thus is the arrival epoch for the particular choice of n successive arrivals to process 1 followed by 1 arrival to process 2.

d) Find the density of the interarrival time for $\{N_A(t); t \geq 0\}$. Note: This part is quite messy and is done most easily via Laplace transforms.

Solution: The process $\{N_A(t); t > 0\}$ is not a Poisson process, but, perhaps surprisingly, it is a renewal process; that is, the interarrival times are independent and identically distributed. One might prefer to postpone trying to understand this until starting to study renewal processes, but we have the necessary machinery already.

Starting at a given arrival to $\{N_A(t); t > 0\}$, let X_A be the interval until the next arrival to $\{N_A(t); t > 0\}$ and let X be the interval until the next arrival to the combined process. Given that the next arrival in the combined process is from process 1, it will be an arrival to $\{N_A(t); t > 0\}$, so that under this condition, $X_A = X$. Alternatively, given that this next arrival is from process 2, X_A will be the sum of three independent rv's, first X , next, the interval X_2 to the following arrival for process 2, and next the interval from that point to the following arrival to $\{N_A(t); t > 0\}$. This final interarrival time will have the same

distribution as X_A . Thus the unconditional PDF for X_A is given by

$$\begin{aligned} f_{X_A}(x) &= \frac{\lambda}{\lambda+\gamma} f_X(x) + \frac{\gamma}{\lambda+\gamma} f_X(x) \otimes f_{X_2}(x) \otimes f_{X_A}(x) \\ &= \lambda \exp(-(\lambda+\gamma)x) + \gamma \exp(-(\lambda+\gamma)x) \otimes \gamma \exp(-\gamma x) \otimes f_{X_A}(x). \end{aligned}$$

where $\otimes$ is the convolution operator and all functions are 0 for $x < 0$.

Solving this by Laplace transforms is a mechanical operation of no real interest here. The solution is

$$f_{X_A}(x) = B \exp \left[-\frac{x}{2} \left(2\gamma + \lambda + \sqrt{4\gamma^2 + \lambda^2} \right) \right] + C \exp \left[-\frac{x}{2} \left(2\gamma + \lambda - \sqrt{4\gamma^2 + \lambda^2} \right) \right],$$

where

$$B = \frac{\lambda}{2} \left(1 + \frac{\lambda}{\sqrt{4\gamma^2 + \lambda^2}} \right); \quad C = \frac{\lambda}{2} \left(1 - \frac{\lambda}{\sqrt{4\gamma^2 + \lambda^2}} \right).$$

Exercise 2.24 : Let us model the chess tournament between Fisher and Spassky as a stochastic process. Let X_i , for $i \geq 1$, be the duration of the i th game and assume that $\{X_i; i \geq 1\}$ is a set of IID exponentially distributed rv's each with density $f_X(x) = \lambda e^{-\lambda x}$. Suppose that each game (independently of all other games, and independently of the length of the games) is won by Fisher with probability p , by Spassky with probability q , and is a draw with probability $1 - p - q$. The first player to win n games is defined to be the winner, but we consider the match up to the point of winning as being embedded in an unending sequence of games.

a) Find the distribution of time, from the beginning of the match, until the completion of the first game that is won (i.e., that is not a draw). Characterize the process of the number $\{N(t); t > 0\}$ of games won up to and including time t . Characterize the process of the number $\{N_F(t); t \geq 0\}$ of games won by Fisher and the number $\{N_S(t); t \geq 0\}$ won by Spassky.

Solution: The Poisson game process is split into 3 independent Poisson processes, namely the draw process of rate $(1 - p - q)\lambda$, the Fisher win process of rate $p\lambda$ and the Spassky win process of rate $q\lambda$. The process of wins is the sum of the Fisher and Spassky win processes, and is independent of the draw process. Thus the time to the first win is an exponential rv X_W of rate $(p + q)\lambda$. Thus the density and CDF are

$$f_{X_W}(x) = (p + q)\lambda \exp(-(p + q)\lambda x), \quad F_{X_W}(x) = 1 - \exp(-(p + q)\lambda x).$$

b) For the remainder of the problem, assume that the probability of a draw is zero; i.e., that $p + q = 1$. How many of the first $2n - 1$ games must be won by Fisher in order to win the match?

Solution: Note that (a) shows that the process of wins is Poisson within the process of games including draws, and thus the assumption that there are no draws (i.e., $p + q = 1$) only simplifies the notation slightly. Note also that it makes no difference to the probability of winning the match whether they continue playing beyond the game in which one of them first wins n games.

If Fisher wins n or more of the first $2n - 1$ games, then Spassky wins at most $n - 1$, so Fisher wins the match. Conversely if Fisher wins the match, he must win n or more of the first $2n - 1$ games. Thus Fisher wins if and only if he wins n or more of the first $2n - 1$ games.

c) What is the probability that Fisher wins the match? Your answer should not involve any integrals. Hint: consider the unending sequence of games and use (b).

Solution: The sequence of games is Bernoulli with probability p of a Fisher game win. Thus, using the binomial PMF for $2n-1$ plays, the probability that Fisher wins the match is

$$\Pr\{\text{Fisher wins match}\} = \sum_{k=n}^{2n-1} \binom{2n-1}{k} p^k q^{2n-1-k}.$$

Without the hint, the problem is more tricky, but no harder computationally. The probability that Fisher wins the match at the end of game k , for $n \leq k \leq 2n-1$, is the probability that he wins $n-1$ games out the first $k-1$ and then wins the k th. This is p times $\binom{k-1}{n-1} p^{n-1} q^{k-n}$. Thus

$$\Pr\{\text{Fisher wins match}\} = \sum_{k=n}^{2n-1} \binom{k-1}{n-1} p^n q^{k-n}.$$

It is surprising that these very different appearing expressions are the same.

d) Let T be the epoch at which the match is completed (i.e., either Fisher or Spassky wins). Find the CDF of T .

Solution: Let T_f be the time at which Fisher wins his n th game and T_s be the time at which Spassky wins his n th game (again assuming that playing continues beyond the winning of the match). The Poisson process of Fisher wins is independent of that of Spassky wins. Also, the time T at which the match ends is the minimum of T_f and T_s , so, for any $t > 0$, $\Pr\{T > t\} = \Pr\{T_f > t, T_s > t\}$. Thus

$$\Pr\{T > t\} = \Pr\{T_f > t\} \Pr\{T_s > t\}. \quad (\text{A.37})$$

Now T_f has an Erlang distribution so its complementary CDF is equal to the probability that fewer than n Fisher wins have occurred by time t . The number of Fisher wins is a Poisson rv, and Spassky wins are handled the same way. Thus,

$$\Pr\{T > t\} = \sum_{k=0}^{n-1} \frac{(\lambda p t)^k e^{-\lambda p t}}{k!} \sum_{j=0}^{n-1} \frac{(\lambda q t)^j e^{-\lambda q t}}{j!} = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda p t)^k}{k!} \sum_{j=0}^{n-1} \frac{(\lambda q t)^j}{j!}.$$

Finally $F_T(t) = 1 - \Pr\{T > t\}$.

e) Find the probability that Fisher wins and that T lies in the interval $(t, t + \delta)$ for arbitrarily small δ .

Solution: From (A.37), the PDF $f_T(t)$ is given by

$$f_T(t) = f_{T_f}(t) \Pr\{T_s > t\} + f_{T_s}(t) \Pr\{T_f > t\}.$$

The first term on the right is associated with a Fisher win, and the second with a Spassky win. Thus

$$\lim_{\delta \rightarrow 0} \frac{\Pr\{\text{Fisher wins}, T \in [t, t+\delta)\}}{\delta} = \frac{(\lambda p)^n t^{n-1} e^{-\lambda p t}}{(n-1)!} \sum_{j=0}^{n-1} \frac{(\lambda q t)^j e^{-\lambda q t}}{j!}. \quad (\text{A.38})$$

The main reason for calculating this is to point out that T and the event that Fisher wins the match are not independent events. The number of games that Fisher wins up to time t is independent of the number that Spassky wins up to t . Also, given that k games have been played by time t , the distribution of the number of those games won by Fisher does not vary with t . The problem here, given a Fisher win with $T = t$, the most likely number of Spassky wins is 0 when t is very small and $n - 1$ when t is very large. This can be seen by comparing the terms in the sum of (A.38). This is not something you could reasonably be expected to hypothesize intuitively; it is simply something that indicates that one must sometimes be very careful.

Exercise 2.25: a) For $1 \leq i < n$, find the conditional density of S_{i+1} , conditional on $N(t) = n$ and $S_i = s_i$.

Solution: Recall from (2.41) that

$$\Pr\{S_1 > \tau \mid N(t) = n\} = \Pr\{X_1 > \tau \mid N(t) = n\} = \left(\frac{t - \tau}{t}\right)^n.$$

Given that $S_i = s_i$, we can apply this same formula to $\tilde{N}(s_i, t)$ for the first arrival after s_i .

$$\Pr\{X_{i+1} > \tau \mid N(t) = n, S_i = s_i\} = \Pr\{X_{i+1} > \tau \mid \tilde{N}(s_i, t) = n - i, S_i = s_i\} = \left(\frac{t - s_i - \tau}{t - s_i}\right)^{n-i}.$$

Since $S_{i+1} = S_i + X_{i+1}$, we get

$$\begin{aligned} \Pr\{S_{i+1} > s_{i+1} \mid N(t) = n, S_i = s_i\} &= \left(\frac{t - s_{i+1}}{t - s_i}\right)^{n-i} \\ f_{S_{i+1}|N(t)S_i}(s_{i+1} \mid n, s_i) &= \frac{(n - i)(t - s_{i+1})^{n-i-1}}{(t - s_i)^{n-i}}. \end{aligned} \quad (\text{A.39})$$

b) Use (a) to find the joint density of $S_1, \dots, S_n$ conditional on $N(t) = n$. Verify that your answer agrees with (2.38).

Solution: For each i , the conditional probability in (a) is clearly independent of $S_{i-2}, \dots, S_1$. Thus we can use the chain rule to multiply (A35) by itself for each value of i . We must also include $f_{S_1|N(t)}(s_1 \mid n) = n(t - s_1)^{n-1}/t^n$. Thus

$$\begin{aligned} f_{S^{(n)}|N(t)}(s^{(n)} \mid n) &= \frac{n(t - s_1)^{n-1}}{t^n} \cdot \frac{(n-1)(t - s_2)^{n-2}}{(t - s_1)^{n-1}} \cdot \frac{(n-2)(t - s_3)^{n-3}}{(t - s_2)^{n-2}} \cdots \frac{(t - s_n)^0}{t - s_{n-1}} \\ &= \frac{n!}{t^n}. \end{aligned}$$

Note: There is no great insight to be claimed from this exercise. it is useful, however, in providing some additional techniques for working with such problems.

Exercise 2.26: A two-dimensional Poisson process is a process of randomly occurring special points in the plane such that (i) for any region of area A the number of special points in that region has a Poisson distribution with mean λA , and (ii) the number of special points in nonoverlapping regions is independent.

For such a process consider an arbitrary location in the plane and let X denote its distance from its nearest special point (where distance is measured in the usual Euclidean manner). Show that

a) $\Pr\{X > t\} = \exp(-\lambda\pi t^2)$.

Solution: Given an arbitrary location, $X > t$ if and only if there are no special points in the circle of radius t around the given point. The expected number in that circle is $\lambda\pi t^2$, and since the number in that circle is Poisson with expected value $\lambda\pi t^2$, the probability that number is 0 is $e^{-\lambda\pi t^2}$. Thus $\Pr\{X > t\} = e^{-\lambda\pi t^2}$.

b) $E[X] = 1/(2\sqrt{\lambda})$.

Solution: Since $X \geq 0$, we have

$$E[X] = \int_0^\infty \Pr\{X > t\} dt = \int_0^\infty \exp(-\lambda\pi t^2) dt.$$

We can look this up in a table of integrals, or recognize its resemblance to the Gaussian PDF. If we define $\sigma^2 = 1/(2\pi\lambda)$, the above integral is

$$E[X] = \sigma\sqrt{2\pi} \int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{t^2}{2\sigma^2}\right] dt = \frac{\sigma\sqrt{2\pi}}{2} = \frac{1}{2\sqrt{\lambda}}.$$

Exercise 2.27: This problem is intended to show that one can analyze the long term behavior of queueing problems by using just notions of means and variances, but that such analysis is awkward, justifying understanding the strong law of large numbers. Consider an M/G/1 queue. The arrival process is Poisson with $\lambda = 1$. The expected service time, $E[Y]$, is $1/2$ and the variance of the service time is given to be 1.

a) Consider S_n , the time of the n th arrival, for $n = 10^{12}$. With high probability, S_n will lie within 3 standard deviations of its mean. Find and compare this mean and the 3σ range.

Solution: Let X_i be the i th interarrival time. Then $S_n = \sum_{i=1}^n X_i$, so $E[S_n] = n$ and $\text{VAR}[S_n] = n$. Thus $E[S_{10^{12}}] = 10^{12}$ with 3×10^6 as the 3 sigma point.

b) Let V_n be the total amount of time during which the server is busy with these n arrivals (i.e., the sum of 10^{12} service times). Find the mean and 3σ range of V_n .

Solution: In the same way, $E[V_{10^{12}}] = 5 \times 10^{11}$ with 3×10^6 as the 3 sigma point.

c) Find the mean and 3σ range of I_n , the total amount of time the server is idle up until S_n (take I_n as $S_n - V_n$, thus ignoring any service time after S_n).

Solution: As before $E[I_{10^{12}}] = 5 \times 10^{11}$. Since interarrival times and service times are independent and $-V$ and V have the same variance, the 3 sigma point of I_n is 6×10^6 .

d) An idle period starts when the server completes a service and there are no waiting arrivals; it ends on the next arrival. Find the mean and variance of an idle period. Are successive idle periods IID?

Solution: The time from the beginning of an idle period to the next arrival is exponential with mean 1 and variance 1. Since the time at which the n th idle period starts is a function only of the first n interarrival times and service times, and since for each sample value of the beginning of the n th idle period, the interval until the next arrival is independent of those

arrivals (by the independent increment property) and independent of those service times, it follows that the interval until the next arrival is independent of those earlier interarrival intervals and service times. This interval (*i.e.*, this idle period) is thus independent of all earlier idle periods.

This is one of those peculiar situations where the independence of idle durations is virtually obvious but a careful demonstration is quite tricky. This type of argument will become clearer when we study renewal processes.

e) Combine (c) and (d) to estimate the total number of idle periods up to time S_n . Use this to estimate the total number of busy periods.

Solution: The aggregate idle time up to S_n can be expressed as $n/2 \pm 6\sqrt{n}$ and the aggregate duration of m idle periods can be expressed as $m \pm 3\sqrt{m}$. Letting $m(n)$ be the number of idle periods up to S_n , we then have $n/2 \pm 6\sqrt{n} \approx m \pm 3\sqrt{m}$. For n very large (10^{12}), the square roots are small relative to the linear terms, so we can argue that $m(n) \approx n/2 \pm [6\sqrt{n} + 3\sqrt{n/2}]$. One can do this more carefully by looking at the maximum with 3 sigma limits of one expression and the minimum of the other, but however this is expressed, the number of idle periods per arrival is increasingly close to $1/2$ with high probability as the number of arrivals is increased.

The period from 0 to S_n starts with an idle period and ends with a busy period, so that $m(n)$ is also the number of busy periods.

f) Combine (e) and (b) to estimate the expected length of a busy period.

Solution: Let B be the expected duration of a busy period. Then the expected aggregate duration of m busy periods is mB . Since the expected aggregate busy time for large m is equal to the expected aggregate idle time, we have $E[B] \approx 1$.

It is curious to note that this does not depend on the variance of the service time (beyond ensuring that the standard deviation of 10^{12} service times is small compared to the mean).

Exercise 2.28: The purpose of this problem is to illustrate that for an arrival process with independent but not identically distributed interarrival intervals, $X_1, X_2, \dots$, the number of arrivals $N(t)$ in the interval $(0, t]$ can be a defective rv. In other words, the ‘counting process’ is not a stochastic process according to our definitions. This illustrates that it is necessary to prove that the counting rv’s for a renewal process are actually rv’s.

a) Let the CDF of the i th interarrival interval for an arrival process be $F_{X_i}(x_i) = 1 - \exp(-\alpha^{-i}x_i)$ for some fixed $\alpha \in (0, 1)$. Let $S_n = X_1 + \dots + X_n$ and show that

$$E[S_n] = \frac{\alpha(1 - \alpha^n)}{1 - \alpha}.$$

Solution: Each X_i is an exponential rv, but the rate, α^{-i} , is rapidly increasing with i and the expected interarrival time, $E[X_i] = \alpha^i$, is rapidly decreasing with i . Thus

$$E[S_n] = \alpha + \alpha^2 + \dots + \alpha^n.$$

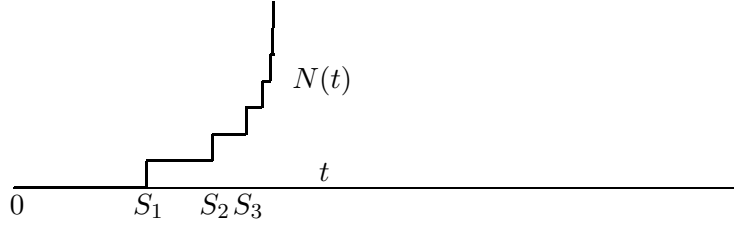
Recalling that $1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} = (1 - \alpha^n)/(1 - \alpha)$,

$$\begin{aligned} E[S_n] &= \alpha(1 + \alpha + \dots + \alpha^{n-1}) \\ &= \frac{\alpha(1 - \alpha^n)}{1 - \alpha} < \frac{\alpha}{1 - \alpha}. \end{aligned}$$

In other words, not only is $E[X_i]$ decaying to 0 geometrically with increasing i , but $E[S_n]$ is upper bounded, for all n , by $\alpha/(1-\alpha)$.

b) Sketch a ‘reasonable’ sample function for $N(t)$.

Solution: Since the expected interarrival times are decaying geometrically and the expected arrival epochs are bounded for all n , it is reasonable for a sample path to have the following shape:



Note that the question here is not precise (there are obviously many sample paths, and which are ‘reasonable’ is a matter of interpretation). The reason for drawing such sketches is to acquire understanding to guide the solution to the following parts of the problem.

c) Find $\sigma_{S_n}^2$.

Solution: Since X_i is exponential, $\sigma_{X_i}^2 = \alpha^{2i}$. Since the X_i are independent,

$$\begin{aligned} \sigma_{S_n}^2 &= \sigma_{X_1}^2 + \sigma_{X_2}^2 + \cdots + \sigma_{X_n}^2 \\ &= \alpha^2 + \alpha^4 + \cdots + \alpha^{2n} \\ &= \alpha^2(1 + \alpha^2 + \cdots + \alpha^{2(n-1)}) \\ &= \frac{\alpha^2(1 - \alpha^{2n})}{1 - \alpha^2} < \frac{\alpha^2}{1 - \alpha^2}. \end{aligned}$$

d) Use the Markov inequality on $\Pr\{S_n \geq t\}$ to find an upper bound on $\Pr\{N(t) \leq n\}$ that is smaller than 1 for all n and for large enough t . Use this to show that $N(t)$ is defective for large enough t .

Solution: The figure suggests (but does not prove) that for typical sample functions (and in particular for a set of sample functions of non-zero probability), $N(t)$ goes to infinity for finite values of t . If the probability that $N(t) \leq n$ (for a given t) is bounded, independent of n , by a number strictly less than 1, then that $N(t)$ is a defective rv rather than a true rv.

By the Markov inequality,

$$\begin{aligned} \Pr\{S_n \geq t\} &\leq \frac{\bar{S}_n}{t} \leq \frac{\alpha}{t(1-\alpha)} \\ \Pr\{N(t) < n\} &= \Pr\{S_n > t\} \leq \Pr\{S_n \geq t\} \leq \frac{\alpha}{t(1-\alpha)}. \end{aligned}$$

where we have used (2.3). Since this bound is independent of n , it also applies in the limit, *i.e.*,

$$\lim_{n \rightarrow \infty} \Pr\{N(t) \leq n\} \leq \frac{\alpha}{t(1-\alpha)}.$$

For any $t > \alpha/(1 - \alpha)$, we see that $\frac{\alpha}{t(1-\alpha)} < 1$. Thus $N(t)$ is defective for any such t , *i.e.*, for any t greater than $\lim_{n \rightarrow \infty} \mathbf{E}[S_n]$.

Actually, by working harder, it can be shown that $N(t)$ is defective for all $t > 0$. The outline of the argument is as follows: for any given t , we choose an m such that $\Pr\{S_m \leq t/2\} > 0$ and such that $\Pr\{S_\infty - S_m \leq t/2\} > 0$ where $S_\infty - S_m = \sum_{i=m+1}^{\infty} X_i$. The second inequality can be satisfied for m large enough by the Markov inequality. The first inequality is then satisfied since S_m has a density that is positive for $t > 0$.

A.3 Solutions for Chapter 3

Exercise 3.1: a) Let X, Y be IID rv's, each with density $f_X(x) = \alpha \exp(-x^2/2)$. In (b), we show that α must be $1/\sqrt{2\pi}$ in order for $f_X(x)$ to integrate to 1, but in this part, we leave α undetermined. Let $S = X^2 + Y^2$. Find the probability density of S in terms of α .

Solution: First we find the CDF of S .

$$\begin{aligned} F_S(s) &= \int \int_{x^2+y^2 \leq s} \alpha^2 e^{(-x^2-y^2)/2} dx dy \\ &= \int_0^{2\pi} \int_{r^2 \leq s} \alpha^2 r e^{-r^2/2} dr d\theta \\ &= \int_{r^2 \leq s} 2\pi \alpha^2 e^{-r^2/2} d(r^2/2) = 2\pi \alpha^2 (1 - e^{-s/2}), \end{aligned} \quad (\text{A.40})$$

where we first changed to polar coordinates and then integrated. The density is then

$$f_S(s) = \pi \alpha^2 e^{-s/2}; \quad \text{for } s \geq 0.$$

b) Prove from (a) that α must be $1/\sqrt{2\pi}$ in order for S , and thus X and Y , to be random variables. Show that $E[X] = 0$ and that $E[X^2] = 1$.

Solution: From (A.40) and the fact that $\lim_{s \rightarrow \infty} F_S(s) = 1$, we see that $2\pi \alpha^2 = 1$, so $\alpha = 1/\sqrt{2\pi}$. From the fact that $f_X(x) = f_X(-x)$ for all x , we see that $E[X] = 0$. Also, since S is exponential and is seen to have mean 2, and since X and Y must have the same second moment, we see that $E[X^2] = 1$. This also follows by using integration by parts.

c) Find the probability density of $R = \sqrt{S}$. R is called a *Rayleigh* rv.

Solution: Since $S \geq 0$ and $F_S(s) = 1 - e^{-s/2}$, we see that $R \geq 0$ and $F_R(r) = 1 - e^{-r^2/2}$. Thus the density is given by $f_R(r) = r e^{-r^2/2}$.

Exercise 3.2: a) By expanding in a power series in $(1/2)r^2\sigma^2$, show that

$$\exp\left(\frac{r^2\sigma^2}{2}\right) = 1 + \frac{r^2\sigma^2}{2} + \frac{r^4\sigma^4}{2(2^2)} + \cdots + \frac{r^{2k}\sigma^{2k}}{k!2^k} + \cdots. \quad (\text{A.41})$$

Solution: Recall that $e^x = 1 + x + x^2/2 + \cdots + x^n/n! + \cdots$. Using $r^2\sigma^2/2$ in place of x , we get (A.41).

b) By expanding e^{rZ} in a power series in rZ , show that

$$g_Z(r) = E[e^{rZ}] = 1 + rE[Z] + \frac{r^2E[Z^2]}{2} + \cdots + \frac{r^kE[Z^k]}{k!} + \cdots. \quad (\text{A.42})$$

Solution: The power series of e^{rZ} converges for each sample value of Z , so we have

$$E[e^{rZ}] = E\left[1 + rZ + \frac{r^2Z^2}{2} + \cdots + \frac{r^nZ^n}{n!} + \cdots\right].$$

If the moment generating function is finite, then the expectation can be interchanged with the sum, although we shall not prove that here. This exchange results in (A.42).

c) By matching powers of r between parts (a) and (b), show that for all integer $k \geq 1$,

$$\mathbb{E}[Z^{2k}] = \frac{(2k)! \sigma^{2k}}{k! 2^k} = (2k-1)(2k-3) \cdots (3)(1) \sigma^{2k} \quad ; \quad \mathbb{E}[Z^{2k+1}] = 0.$$

Solution: The unstated assumption in this exercise is that $Z \sim \mathcal{N}(0, \sigma^2)$. With this assumption, (3.6) shows that $\mathbf{g}_Z(r) = \exp(r^2 \sigma^2 / 2)$, which is given in (A.42). Since the power series is unique, the terms in (A.41) and (A.42) can be equated term by term so that for even powers of r ,

$$\frac{r^{2k} \sigma^{2k}}{k! 2^k} = \frac{r^{2k} \mathbb{E}[Z^{2k}]}{(2k)!}. \quad (\text{A.43})$$

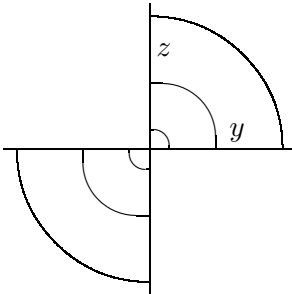
For odd powers of r , $\mathbb{E}[Z^r] = 0$, which we already know from symmetry. Solving (A.43), we get

$$\mathbb{E}[Z^{2k}] = \frac{(2k)! \sigma^{2k}}{k! 2^k}.$$

We can now separate $(2k)!$ into a product of even terms and a product of odd terms, $(2k)! = [(2k)(2k-2)(2k-4) \cdots (2)] [(2k-1)(2k-3) \cdots (1)]$. The product of the even terms is $k! 2^k$, so $\mathbb{E}[Z^{2k}] = (2k-1)(2k-3) \cdots (1) \sigma^{2k}$, which is the desired result.

Exercise 3.3: Let X and Z be IID normalized Gaussian random variables. Let $Y = |Z| \text{Sgn}(X)$, where $\text{Sgn}(X)$ is 1 if $X \geq 0$ and -1 otherwise. Show that X and Y are each Gaussian, but are not jointly Gaussian. Sketch the contours of equal joint probability density.

Solution: Note that Y has the magnitude of Z but the sign of X , so that X and Y are either both positive or both negative, *i.e.*, their joint density is nonzero only in the first and third quadrant of the X, Y plane. Conditional on a given X , the conditional density of Y is twice the conditional density of Z since both Z and $-Z$ are mapped into the same Y . Thus $f_{XY}(x, y) = (1/\pi) \exp(-x^2 - y^2)/2$ for all x, y in the first or third quadrant.



Exercise 3.4: a) Let $X_1 \sim \mathcal{N}(0, \sigma_1^2)$ and let $X_2 \sim \mathcal{N}(0, \sigma_2^2)$ be independent of X_1 . Convolve the density of X_1 with that of X_2 to show that $X_1 + X_2$ is Gaussian, $\mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$.

Solution: Let $Z = X_1 + X_2$. Since X_1 and X_2 are independent, the density of Z is the

convolution of the X_1 and X_2 densities. For initial simplicity, assume $\sigma_{X_1}^2 = \sigma_{X_2}^2 = 1$.

$$\begin{aligned}
 f_Z(z) &= f_{X_1}(z) * f_{X_2}(z) = \int_{-\infty}^{\infty} f_{X_1}(x) f_{X_2}(z-x) dx \\
 &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x^2 - xz + \frac{z^2}{2})} dx \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x^2 - xz + \frac{z^2}{4}) - \frac{z^2}{4}} dx \\
 &= \frac{1}{2\sqrt{\pi}} e^{-z^2/4} \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-(x - \frac{z}{2})^2} dx \\
 &= \frac{1}{2\sqrt{\pi}} e^{-z^2/4},
 \end{aligned}$$

since the last integral integrates a Gaussian pdf with mean $z/2$ and variance $1/2$, which evaluates to 1. As expected, Z is Gaussian with zero mean and variance 2.

The ‘trick’ used here in the fourth equation above is called *completing the square*. The idea is to take a quadratic expression such as $x^2 + \alpha x + \beta z^2$ and to add and subtract $\alpha^2 z^2/4$. Then $x^2 + \alpha x z + \alpha z^2/4$ is $(x + \alpha z/2)^2$, which leads to a Gaussian form that can be integrated.

Repeating the same steps for arbitrary $\sigma_{X_1}^2$ and $\sigma_{X_2}^2$, we get the Gaussian density with mean 0 and variance $\sigma_{X_1}^2 + \sigma_{X_2}^2$.

b) Let W_1, W_2 be IID normalized Gaussian rv's. Show that $a_1 W_1 + a_2 W_2$ is Gaussian, $\mathcal{N}(0, a_1^2 + a_2^2)$. Hint: You could repeat all the equations of (a), but the insightful approach is to let $X_i = a_i W_i$ for $i = 1, 2$ and then use (a) directly.

Solution: Following the hint $\sigma_{X_i}^2 = a_i^2$ for $i = 1, 2$, so $\alpha_1 W_1 + \alpha_2 W_2$ is $\mathcal{N}(0, \alpha_1^2 + \alpha_2^2)$.

c) Combine (b) with induction to show that all linear combinations of IID normalized Gaussian rv's are Gaussian.

Solution: The inductive hypothesis is that if $\{W_i; i \geq 1\}$ is a sequence of IID normal rv's, if $\{\alpha_i; i \geq 1\}$ is a sequence of numbers, and if $\sum_{i=1}^n \alpha_i W_i$ is $\mathcal{N}(0, \sum_{i=1}^n \alpha_i^2)$ for a given $n \geq 1$, then $\sum_{i=1}^{n+1} \alpha_i W_i$ is $\mathcal{N}(0, \sum_{i=1}^{n+1} \alpha_i^2)$. The basis for the induction was established in (b). For the inductive step, let $X = \sum_{i=1}^n \alpha_i W_i$. Now X is independent of W_{n+1} and by the inductive hypothesis $X \sim \mathcal{N}(0, \sum_{i=1}^n \alpha_i^2)$. From (a), $X + W_{n+1} \sim \mathcal{N}(0, \sum_{i=1}^{n+1} \alpha_i^2)$. This establishes the inductive step, so $\sum_{i=1}^n \alpha_i W_i \sim \mathcal{N}(0, \sum_{i=1}^n \alpha_i^2)$ for all n , i.e., all linear combinations of IID normalized Gaussian rv's are Gaussian.

Exercise 3.5: **a)** Let $\mathbf{U}$ be an n -rv with mean $\mathbf{m}$ and covariance $[K]$ whose MGF is given by (3.18). Let $X = \mathbf{r}^\top \mathbf{U}$ for an arbitrary real vector $\mathbf{r}$. Show that the MGF of X is given by $\mathbf{g}_X(r) = \exp[r\mathbf{E}[X] + r^2\sigma_X^2/2]$ and relate $\mathbf{E}[X]$ and σ_X^2 to $\mathbf{m}$ and $[K]$.

Solution: Note that X is defined in terms of a vector $\mathbf{r}$ and the argument of the MGF is an unrelated real number r . To reduce confusion, we use s as the argument of the MGF. Note also that it is not directly given whether either $\mathbf{U}$ or X are Gaussian. However, we

are given that (3.18) holds, so we have

$$\mathbf{g}_X(s) = \mathbb{E}[e^{sX}] = \mathbb{E}\left[e^{s\mathbf{r}^\top \mathbf{U}}\right] = \mathbf{g}_U(s\mathbf{r}) = \exp\left(s\mathbf{r}^\top \mathbf{m} + \frac{s^2 \mathbf{r}^\top [K] \mathbf{r}}{2}\right). \quad (\text{A.44})$$

Since $X = \mathbf{r}^\top \mathbf{U}$, we have $\mathbb{E}[X] = \mathbf{r}^\top \mathbf{m}$ and

$$\sigma_X^2 = \mathbb{E}[\mathbf{r}^\top (\mathbf{U} - \mathbf{m})(\mathbf{U} - \mathbf{m})^\top \mathbf{r}] = \mathbf{r}^\top [K] \mathbf{r}.$$

Substituting this into (A.44), we get $\mathbf{g}_X(s) = \exp[s\mathbb{E}[X] + s^2\sigma_X^2/2]$.

b) Show that $\mathbf{U}$ is a Gaussian rv.

Solution: A basic fact that we use (but do not prove) is that if a rv has the MGF of a Gaussian rv, then that rv is in fact Gaussian. The point of (b) is to show that this implies the same result for a Gaussian random vector. We showed in (a) that $X = \mathbf{r}^\top \mathbf{U}$ has the MGF of a Gaussian rv, and thus is Gaussian. This is true for all choices of $\mathbf{r}$, and thus shows that all linear combinations of $\mathbf{U}$ are Gaussian. Thus from Theorem 3.4.8 (trivially generalized to non-zero mean), $\mathbf{U}$ itself is Gaussian.

Exercise 3.6: a) Let $\mathbf{Z} \sim \mathcal{N}(0, [K])$ be n -dimensional. By expanding in a power series in $(1/2)\mathbf{r}^\top [K] \mathbf{r}$, show that

$$\mathbf{g}_Z(\mathbf{r}) = \exp\left[\frac{\mathbf{r}^\top [K] \mathbf{r}}{2}\right] = 1 + \frac{\sum_{j,k} r_j r_k K_{jk}}{2} + \cdots + \frac{\left(\sum_{j,k} r_j r_k K_{jk}\right)^m}{2^m m!} + \cdots. \quad (\text{A.45})$$

Solution: This looks complicated, but is really quite simple. The first equality above is the standard result in (3.14) for the MGF of jointly Gaussian rv's. For the second equality, note that ordinary vector/matrix multiplication shows that $\mathbf{r}^\top [K] \mathbf{r} = \sum_{j,k} r_j r_k K_{jk}$. For any given $\mathbf{r}$, this is simply a number, and we can use the series expansion of $\exp x = 1 + x + x^2/2 + \cdots$ (which always converges) on that number divided by 2, getting (A.45).

b) By expanding $e^{r_j Z_j}$ in a power series in $r_j Z_j$ for each j , show that

$$\mathbf{g}_Z(\mathbf{r}) = \mathbb{E}\left[\exp\left(\sum_j r_j Z_j\right)\right] = \sum_{\ell_1=0}^{\infty} \cdots \sum_{\ell_n=0}^{\infty} \frac{r_1^{\ell_1}}{(\ell_1)!} \cdots \frac{r_n^{\ell_n}}{(\ell_n)!} \mathbb{E}[Z_1^{\ell_1} \cdots Z_n^{\ell_n}]. \quad (\text{A.46})$$

Solution: Starting with the definition of $\mathbf{g}_Z(\mathbf{r})$,

$$\begin{aligned} \mathbf{g}_Z(\mathbf{r}) &= \mathbb{E}\left[\exp\sum_{j=1}^n r_j Z_j\right] = \mathbb{E}\left[\prod_{j=1}^n \exp(r_j Z_j)\right] \\ &= \mathbb{E}\left[\prod_{j=1}^n \left(\sum_{\ell_j=0}^{\infty} \frac{r_j^{\ell_j} Z_j^{\ell_j}}{\ell_j!}\right)\right], \end{aligned}$$

where we have used the power series expansion of e^x in the final equality. This is a product of n sums and (A.46) results from writing out each sum separately and then interchanging the expectation with the multiple sums. This interchange is valid but we won't justify it here.

c) Let $D = \{i_1, i_2, \dots, i_{2m}\}$ be a set of $2m$ distinct integers each between 1 and n and let $(j_1, \dots, j_{2m})$ be a permutation of D . Consider the term $r_{j_1} r_{j_2} \cdots r_{j_{2m}} \mathbb{E}[Z_{j_1} Z_{j_2} \cdots Z_{j_{2m}}]$ in (A.46). By comparing with the same set of terms in (A.45) containing the same product $r_{j_1} r_{j_2} \cdots r_{j_{2m}}$, show that

$$\mathbb{E}[Z_{i_1} Z_{i_2} \cdots Z_{i_{2m}}] = \frac{\sum_{j_1 j_2 \cdots j_{2m}} K_{j_1 j_2} K_{j_3 j_4} \cdots K_{j_{2m-1} j_{2m}}}{2^m m!}, \quad (\text{A.47})$$

where the sum is over all permutations $(j_1, j_2, \dots, j_{2m})$ of D .

Solution: This looks simple, but is really quite complicated. Let us start with $m = 1$ so that $D = (i_1, i_2)$ for some choice of i_1, i_2 , $i_1 \neq i_2$. The two permutations of D are (i_1, i_2) and (i_2, i_1) . The corresponding term in (A.46) is that with $\ell_{i_1} = 1, \ell_{i_2} = 1$, and $\ell_j = 0$ for $j \neq i_1, i_2$, i.e., it is the term $r_{i_1} r_{i_2} \mathbf{E}[Z_{i_1} Z_{i_2}] = r_{i_1} r_{i_2} K_{i_1, i_2}$. Both permutations correspond to the same term in (A.46). In (A.45) the two permutations each correspond to a separate but equal term, $r_{i_1} r_{i_2} K_{i_1 i_2}/2$ and $r_{i_2} r_{i_1} K_{i_2 i_1}/2$. Cancelling out the term $r_{i_1} r_{i_2}$, we get (A.47) for the case $m = 1$.

The analysis becomes more complex for $m > 1$. We observe three points from the above analysis for $m = 1$. First, each single term in (A.46) corresponds to multiple terms in (A.45). Second, we are only looking at terms in (A.46) for which each ℓ_i is either 1 or 0. In (e), we will find how to avoid analyzing the other terms. Finally, we are relying on the fact that $\mathbf{g}_Z(\mathbf{r})$ can be represented as a unique multinomial expansion in $\mathbf{r}$, so that we can compare different ways of calculating this multinomial.

For arbitrary $m \leq n/2$, let $i_1, \dots, i_{2m}$ be distinct integers between 1 and n . We order these integers $i_1 < i_2 < \dots < i_{2m}$ to be specific. As before, consider the term in (A.46) where $\ell_{i_1} = 1, \ell_{i_2} = 1, \dots, \ell_{i_{2m}} = 1$ and $\ell_i = 0$ otherwise. The specified term is then $r_{i_1} r_{i_2} \dots r_{i_{2m}} \mathbf{E}[Z_{i_1} Z_{i_2} \dots Z_{i_{2m}}]$.

Next consider the form $\left(\sum_{j,k} r_j r_k K_{j,k} \right)^m$ in (A.45) and in particular consider the terms in this sum for which $(j_1, k_1, j_2, k_2, \dots, j_m, k_m)$ is a permutation of $i_1, \dots, i_{2m}$. Each of these terms, as a function of $r_1, \dots, r_n$ is the product $r_{i_1}, \dots, r_{i_{2m}}$ times a product of correlation terms, and thus the sum of all these terms must be equal to $r_{i_1} r_{i_2} \dots r_{i_{2m}} \mathbf{E}[Z_{i_1} Z_{i_2} \dots Z_{i_{2m}}]$. Cancelling out $r_{i_1} r_{i_2} \dots r_{i_{2m}}$, we have (A.47).

d) Find the number of permutations of D containing the same set of unordered pairs $(\{j_1, j_2\}, \dots, \{j_{2m-1}, j_{2m}\})$. For example, $(\{1, 2\}, \{3, 4\})$ is the same set of unordered pairs as $(\{3, 4\}, \{2, 1\})$. Show that

$$\mathbf{E}[Z_{j_1} Z_{j_2} \dots Z_{j_{2m}}] = \sum_{j_1, j_2, \dots, j_{2m}} K_{j_1 j_2} K_{j_3 j_4} \dots K_{j_{2m-1} j_{2m}}, \quad (\text{A.48})$$

where the sum is over distinct sets of unordered pairs of the set D . Note: another way to say the same thing is that the sum is over the set of all permutations of D for which $j_{2k-1} < j_{2k}$ for $1 \leq k \leq m$ and $j_{2k-1} < j_{2k+1}$ for $1 \leq k \leq m-1$.

Solution: There are 2 ways to order the elements within each pair, so 2^m ways to order the elements within all the m pairs. There are then $m!$ ways to order the m pairs. Thus there are $2^m m!$ permutations in (A.47) that all correspond to the same set of unordered pairs. This cancels the denominator in (A.47), giving rise to (A.48). Note that this is valid only for $2m$ distinct rv's, although they can have arbitrary correlations. We see in (e) how to apply this to arbitrary powers of individual rv's.

e) To find $\mathbf{E}[Z_1^{j_1} \dots Z_n^{j_n}]$, where $j_1 + j_2 + \dots + j_n = 2m$, construct the random variables $U_1, \dots, U_{2m}$, where $U_1, \dots, U_{j_1}$ are all identically equal to Z_1 , where $U_{j_1+1}, \dots, U_{j_1+j_2}$ are identically equal to Z_2 , etc., and use (A.48) to find $\mathbf{E}[U_1 U_2 \dots U_{2m}]$. Use this formula to find $\mathbf{E}[Z_1^2 Z_2^2]$, $\mathbf{E}[Z_1^2 Z_2^2]$, and $\mathbf{E}[Z_1]^4$.

Solution: All of the expected values to be found involve $\mathbf{E}[U_1 U_2 \dots U_{2m}]$ for $m = 2$. From (A.48),

$$\mathbf{E}[U_1 U_2 U_3 U_4] = \mathbf{E}[U_1 U_2] \mathbf{E}[U_3 U_4] + \mathbf{E}[U_1 U_3] \mathbf{E}[U_2 U_4] + \mathbf{E}[U_1 U_4] \mathbf{E}[U_2 U_3].$$

Applying this to the 3 desired expectations,

$$\begin{aligned} \mathbb{E}[Z_1^2 Z_2 Z_3] &= \mathbb{E}[Z_1^2] \mathbb{E}[Z_2 Z_3] + 2\mathbb{E}[Z_1 Z_2] \mathbb{E}[Z_1 Z_3] \\ \mathbb{E}[Z_1^2 Z_2^2] &= \mathbb{E}[Z_1^2] \mathbb{E}[Z_2^2] + 2(\mathbb{E}[Z_1 Z_2])^2 \\ \mathbb{E}[Z_1^4] &= 3(\mathbb{E}[Z_1^2])^2. \end{aligned}$$

Exercise 3.7: Let $[Q]$ be an orthonormal matrix. Show that the squared distance between any two vectors $\mathbf{z}$ and $\mathbf{y}$ is equal to the squared distance between $[Q]\mathbf{z}$ and $[Q]\mathbf{y}$.

Solution: The squared distance, say d^2 between $\mathbf{z}$ and $\mathbf{y}$ is $d^2 = (\mathbf{z} - \mathbf{y})^\top (\mathbf{z} - \mathbf{y})$. Letting $\mathbf{x} = \mathbf{z} - \mathbf{y}$, we have $d^2 = \mathbf{x}^\top \mathbf{x}$. The squared distance, say d_1^2 , between $[Q]\mathbf{z}$ and $[Q]\mathbf{y}$ is

$$d_1^2 = ([Q]\mathbf{z} - [Q]\mathbf{y})^\top ([Q]\mathbf{z} - [Q]\mathbf{y}) = ([Q]\mathbf{x})^\top ([Q]\mathbf{x}) = \mathbf{x}^\top [Q]^\top [Q] \mathbf{x}.$$

From (3.26), $[Q]^\top = [Q]^{-1}$, so $[Q]^\top [Q] = [I]$ and $d_1^2 = d^2$.

Exercise 3.8: a) Let $[K] = \begin{bmatrix} .75 & .25 \\ .25 & .75 \end{bmatrix}$. Show that 1 and 1/2 are eigenvalues of $[K]$ and find the normalized eigenvectors. Express $[K]$ as $[Q]\Lambda Q^{-1}$ where $[\Lambda]$ is diagonal and $[Q]$ is orthonormal.

Solution: Note that $[K - \lambda I] = \begin{bmatrix} .75 - \lambda & .25 \\ .25 & .75 - \lambda \end{bmatrix}$. By inspection, this is singular for $\lambda = 1$ and $\lambda = 1/2$. We easily see that $(1, 1)^\top$ is an eigenvector for $\lambda = 1$, and this becomes $(\sqrt{1/2}, \sqrt{1/2})^\top$ when normalized. Similarly, $(1, -1)^\top$, normalized as $(\sqrt{1/2}, -\sqrt{1/2})^\top$, is an eigenvector for $\lambda = 1/2$. Thus $[\Lambda] = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$, $[Q] = \begin{bmatrix} \sqrt{1/2} & \sqrt{1/2} \\ \sqrt{1/2} & -\sqrt{1/2} \end{bmatrix}$ and $[Q]^{-1} = [Q]$.

b) Let $[K'] = \alpha[K]$ for real $\alpha \neq 0$. Find the eigenvalues and eigenvectors of $[K']$. Don't use brute force—think!

Solution: Note that $[K' - \lambda' I] = \alpha[K - \lambda' I/\alpha]$. Thus if λ'/α is an eigenvalue of $[K]$, then λ' is an eigenvalue of $[K']$. Since 1 and 1/2 are the eigenvalues of $[K]$, the eigenvalues of $[K']$ are α and $\alpha/2$. The eigenvectors remain unchanged.

c) Find the eigenvalues and eigenvectors of $[K^m]$, where $[K^m]$ is the m^{th} power of $[K]$.

Solution: Let $\mathbf{q}_j$, $j = 1, 2$ be an eigenvector of $[K]$. Then

$$[K^m]\mathbf{q}_j = \lambda_j [K^{m-1}]\mathbf{q}_j = \cdots = \lambda_j^m \mathbf{q}_j.$$

Thus the eigenvalues of $[K^m]$ are 1 and $(1/2)^m$. The eigenvectors are unchanged.

Exercise 3.9: Let X and Y be jointly Gaussian with means m_X , m_Y , variances σ_X^2 , σ_Y^2 , and normalized covariance ρ . Find the conditional density $f_{X|Y}(x|y)$.

Solution: This was calculated in a straightforward way in (3.37) for the case in which X and Y are zero mean. Thus, letting $\tilde{X}$ and $\tilde{Y}$ be the fluctuations of X and Y respectively, the conditional density of the fluctuations is given by

$$f_{\tilde{X}|\tilde{Y}}(\tilde{x}|\tilde{y}) = \frac{1}{\sigma_X \sqrt{2\pi(1-\rho^2)}} \exp \left[\frac{-[\tilde{x} - \rho(\sigma_X/\sigma_Y)\tilde{y}]^2}{2\sigma_X^2(1-\rho^2)} \right].$$

Since $X = \tilde{X} + \bar{X}$ and $Y = \tilde{Y} + \bar{Y}$, we have

$$f_{X|Y}(x|y) = \frac{1}{\sigma_X \sqrt{2\pi(1-\rho^2)}} \exp \left[\frac{-[x - \bar{X} - \rho(\sigma_X/\sigma_Y)(y - \bar{Y})]^2}{2\sigma_X^2(1-\rho^2)} \right].$$

Exercise 3.10: a) Let X and Y be zero-mean jointly Gaussian with variances σ_X^2 , σ_Y^2 , and normalized covariance ρ . Let $V = Y^3$. Find the conditional density $f_{X|V}(x|v)$. Hint: This requires no computation.

Solution: Note that $v = y^3$ for $y \in \mathbb{R}$ is a one-to-one mapping. It follows that if $V = v$, then $Y = v^{1/3}$. Thus $f_{X|V}(x|v)$ can be found directly from $f_{X|Y}(x|y)$ as given in (3.37) by substituting $v^{1/3}$ for y , i.e.,

$$f_{X|V}(x|v) = \frac{1}{\sigma_X \sqrt{2\pi(1-\rho^2)}} \exp \left[\frac{-[x - \rho(\sigma_X/\sigma_Y)v^{1/3}]^2}{2\sigma_X^2(1-\rho^2)} \right].$$

b) Let $U = Y^2$ and find the conditional density of $f_{X|U}(x|u)$. Hint: first understand why this is harder than (a).

Solution: Note that $u = y^2$ for $y \in \mathbb{R}$ is not one-to-one. If $U = u$, then Y is either $u^{1/2}$ or $-u^{1/2}$. We then have

$$f_{X|U}(x|u) = \frac{f_{XU}(x, u)}{f_U(u)} = \frac{f_{XY}(x, \sqrt{u}) + f_{XY}(x, -\sqrt{u})}{f_Y(\sqrt{u}) + f_Y(-\sqrt{u})}.$$

Since $f_Y(\sqrt{u}) = f_Y(-\sqrt{u})$, this can be rewritten as

$$f_{X|U}(x|u) = \frac{f_{X|Y}(x, \sqrt{y}) + f_{X|Y}(x, -\sqrt{y})}{2}.$$

Substituting these terms into (3.37) gives a rather ugly answer. The point here is not the answer but rather the approach to finding a conditional probability for a Gaussian problem when the conditioning rv is a non-one-to-one function of a Gaussian rv.

Exercise 3.11: a) Let $(\mathbf{X}^\top, \mathbf{Y}^\top)$ have a non-singular covariance matrix $[K]$. Show that $[K_X]$ and $[K_Y]$ are positive definite, and thus non-singular.

Solution: Let $\mathbf{X}$ have dimension n and $\mathbf{Y}$ have dimension m . If $[K_X]$ is not positive definite but only semi-definite, then there is a $\mathbf{b} \neq 0$ such that $\mathbf{b}^\top [K_X] \mathbf{b} = 0$. Defining $\hat{\mathbf{b}}$ as the $m+n$ dimensional vector with $\mathbf{b}$ as the first n components and zeros as the last m components, we see that $\hat{\mathbf{b}}^\top [K] \hat{\mathbf{b}} = 0$, so that $[K]$ is not positive definite. The contrapositive of this is that if $[K]$ is positive definite, then $[K_X]$ is also positive definite.

The same argument shows that if $[K]$ is positive definite, then $[K_Y]$ is also positive definite. In fact, this argument shows that all submatrices of $[K]$ that are centered around the main diagonal are positive definite.

b) Show that the matrices $[B]$ and $[D]$ in (3.39) are also positive definite and thus non-singular.

Solution: We can represent $[K]$ in the form $[Q\Lambda Q^{-1}]$ where $[Q]$ is a matrix whose columns are orthonormal eigenvectors of $[K]$ and $[\Lambda]$ is the diagonal matrix of eigenvalues. Assuming that $[K]$ is positive definite, the eigenvalues are all strictly positive. It is easily seen that the inverse of $[K]$ can be represented as $[K^{-1}] = [Q\Lambda^{-1}Q^{-1}]$ where $[\Lambda^{-1}]$ is the diagonal matrix of the reciprocals of the eigenvalues. Since the eigenvalues are positive, their reciprocals are also, so $[K^{-1}]$ is also positive definite. From (a), then, the matrices $[B]$ and $[D]$ are also positive definite.

Exercise 3.12: Let $\mathbf{X}$ and $\mathbf{Y}$ be jointly-Gaussian rv's with means $\mathbf{m}_X$ and $\mathbf{m}_Y$, covariance matrices $[K_X]$ and $[K_Y]$ and cross covariance matrix $[K_{X,Y}]$. Find the conditional probability density $f_{X|Y}(\mathbf{x} | \mathbf{y})$. Assume that the covariance of $(\mathbf{X}^\top, \mathbf{Y}^\top)$ is non-singular. Hint: think of the fluctuations of $\mathbf{X}$ and $\mathbf{Y}$.

Solution: Let $\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{m}_X$ and $\tilde{\mathbf{Y}} = \mathbf{Y} - \mathbf{m}_Y$ be the fluctuations of $\mathbf{X}$ and $\mathbf{Y}$. Then $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$ have the same covariances and cross covariance as $\mathbf{X}$ and $\mathbf{Y}$. They are also zero-mean and jointly Gaussian so their density is given by (3.40), *i.e.*,

$$f_{\tilde{\mathbf{X}}|\tilde{\mathbf{Y}}}(\tilde{\mathbf{x}}|\tilde{\mathbf{y}}) = \frac{\exp\left\{-\frac{1}{2}\left(\tilde{\mathbf{x}} + [B^{-1}C]\tilde{\mathbf{y}}\right)^\top [B] \left(\tilde{\mathbf{x}} + [B^{-1}C]\tilde{\mathbf{y}}\right)\right\}}{(2\pi)^{n/2}\sqrt{\det[B^{-1}]}}.$$

We now get $f_{X|Y}(\mathbf{x}|\mathbf{y})$ simply by translation,

$$f_{X|Y}(\mathbf{x}|\mathbf{y}) = \frac{\exp\left\{-\frac{1}{2}\left(\mathbf{x} - \mathbf{m}_X + [B^{-1}C](\mathbf{y} - \mathbf{m}_Y)\right)^\top [B] \left(\mathbf{x} - \mathbf{m}_X + [B^{-1}C](\mathbf{y} - \mathbf{m}_Y)\right)\right\}}{(2\pi)^{n/2}\sqrt{\det[B^{-1}]}}.$$

Exercise 3.13: a) Let $\mathbf{W}$ be a normalized IID Gaussian n -rv and let $\mathbf{Y}$ be a Gaussian m -rv. Suppose we would like to choose $\mathbf{Y}$ so that the joint covariance $\mathbf{E}[\mathbf{W}\mathbf{Y}^\top]$ is some arbitrary real-valued $n \times m$ matrix $[K]$. Find the matrix $[A]$ such that $\mathbf{Y} = [A]\mathbf{W}$ achieves the desired joint covariance. Note: this shows that any real-valued $n \times m$ matrix is the joint covariance matrix for some choice of random vectors.

Solution: $\mathbf{Y} = [A]\mathbf{W}$ means that $Y_i = \sum_{j=1}^n a_{ij}W_j$ for each i . Since the W_j are normalized and IID, $\mathbf{E}[Y_iW_j] = a_{ij}$. This in turn can be rewritten as $\mathbf{E}[\mathbf{Y}\mathbf{W}^\top] = [A]$. Thus we choose $[A] = [K^\top]$. Note that the desired matrix $[K]$ also determines the covariance $[K_Y]$, *i.e.*,

$$[K_Y] = \mathbf{E}[\mathbf{Y}\mathbf{Y}^\top] = \mathbf{E}[A\mathbf{W}\mathbf{W}^\top A^\top] = [AA^\top].$$

In other words, we can choose $[K]$ arbitrarily, but this also determines $[K_Y]$

b) Let $\mathbf{Z}$ be a zero-mean Gaussian n -rv with non-singular covariance $[K_Z]$, and let $\mathbf{Y}$ be a Gaussian m -rv. Suppose we would like the joint covariance $\mathbf{E}[\mathbf{Z}\mathbf{Y}^\top]$ to be some arbitrary $n \times m$ matrix $[K']$. Find the matrix $[B]$ such that $\mathbf{Y} = [B]\mathbf{Z}$ achieves the desired joint covariance. Note: this shows that any real valued $n \times m$ matrix is the joint covariance matrix for some choice of random vectors $\mathbf{Z}$ and $\mathbf{Y}$ where $[K_Z]$ is given (and non-singular).

Solution: For any given m by n matrix $[B]$, if we choose $\mathbf{Y} = [B]\mathbf{Z}$, then

$$\mathbf{E}[\mathbf{Z}\mathbf{Y}^\top] = \mathbf{E}[\mathbf{Z}\mathbf{Z}^\top[B^\top]] = [K_Z][B^\top].$$

Thus if we want to set this equal to some given matrix $[K']$, it is sufficient to choose $[B] = [K']^T [K_Z^{-1}]$.

c) Now assume that $\mathbf{Z}$ has a singular covariance matrix in (b). Explain the constraints this places on possible choices for the joint covariance $\mathbf{E}[\mathbf{Z}\mathbf{Y}^T]$. Hint: your solution should involve the eigenvectors of $[K_Z]$.

Solution: If $[K_Z]$ is singular, then there are one or more eigenvectors of $[K_Z]$ of eigenvalue 0, i.e., vectors $\mathbf{b}$ such that $\mathbf{E}[\mathbf{Z}\mathbf{Z}^T]\mathbf{b} = 0$. For such $\mathbf{b}$, we must have $\mathbf{E}[(\mathbf{b}^T\mathbf{Z})(\mathbf{Z}^T\mathbf{b})] = 0$, i.e., the first and second moments of $\mathbf{b}^T\mathbf{Z}$ must be 0. For each such $\mathbf{b}$, the transpose $\mathbf{b}^T$ must satisfy $\mathbf{b}^T\mathbf{E}[\mathbf{Z}\mathbf{Y}^T] = 0$. This means that each linearly independent eigenvector $\mathbf{b}$ of eigenvalue 0 provides a linear constraint $\mathbf{b}^T[K'] = 0$ on $[K']$.

Exercise 3.14: a) Let $\mathbf{W} = (W_1, W_2, \dots, W_{2n})^T$ be a $2n$ dimensional IID normalized Gaussian rv. Let $S_{2n} = W_1^2 + W_2^2 + \dots + W_{2n}^2$. Show that S_{2n} is an n th order Erlang rv with parameter $1/2$, i.e., that

$$f_{S_{2n}}(s) = \frac{2^{-n} s^{n-1} e^{-s/2}}{(n-1)!}. \quad (\text{A.49})$$

Hint: look at S_2 from Exercise 3.1.

Solution: Exercise 3.1 showed that S_2 is an exponentially distributed random variable of mean 2 (i.e., parameter $\lambda = 1/2$). Thus S_{2n} is the sum of n exponential rv's each of parameter $1/2$. The Erlang density (see Section 2.2) is defined as the density of the sum of n IID exponential rv's. Its density is given in Table 1.1 and is easily calculated by convolving n exponential densities. The result (for $\lambda = 1/2$) is given in (A.49).

b) Let $R_{2n} = \sqrt{S_{2n}}$. Find the probability density of R_{2n} .

Solution: Since $R_{2n}^2 = S_{2n}$, we have

$$\begin{aligned} f_{R_{2n}}(r) &= \frac{2^{-n} r^{2(n-1)} e^{-r^2/2}}{(n-1)!} 2r \\ &= \frac{2^{-(n-1)} r^{2n-1} e^{-r^2/2}}{(n-1)!}. \end{aligned}$$

c) Let $v_{2n}(r)$ be the volume of a $2n$ dimensional sphere of radius r and let $b_{2n}(r)$ be the surface area of that sphere, i.e., $b_{2n}(r) = dv_{2n}(r)/dr$. The point of this exercise is to show how to calculate these quantities. By considering an infinitesimally thin spherical shell of thickness δ at radius r , show that

$$f_{R_{2n}}(r) = b_{2n}(r) f_{\mathbf{W}}(\mathbf{w}) |_{\mathbf{W}: \mathbf{W}^T \mathbf{W} = r^2}. \quad (\text{A.50})$$

Solution: Note that $f_{R_{2n}}(r)\delta$ is the probability that R_{2n} lies between r and $r+\delta$ (more precisely, $f_{R_{2n}}(r)$ is $\lim_{\delta \rightarrow 0} (1/\delta) \Pr\{r \leq R_{2n} < r+\delta\}$). In the same infinitesimal sense $f_{\mathbf{W}}(\mathbf{w})$ is constant in this infinitesimal $2n$ dimensional spherical shell. Finally, in the same infinitesimal sense, $b_{2n}(r)\delta$ is the volume of this shell. We then see that (A.50) states that the probability of this shell is equal to the volume of the shell times the (constant) probability per unit volume of the shell. The equation is exact since δ has been cancelled out.

d) Calculate $b_{2n}(r)$ and $v_{2n}(r)$. Note that for any fixed $\delta \ll r$, the limiting ratio as $n \rightarrow \infty$ of the volume of a sphere of radius r to that of a spherical shell of width δ and outer radius r is 1.

Solution: Note that $f_{\mathbf{W}}(\mathbf{w}) = (\sqrt{2\pi})^{2n} \exp(-r^2/2)$ over the shell where $\mathbf{W}^T \mathbf{W} = r^2$.

Thus, from (A.50)

$$b_{2n}(r) = \frac{(2\pi)^n 2^{-(n-1)} r^{2n-1} e^{-r^2/2}}{(n-1)! e^{-r^2/2}} = \frac{2\pi^n r^{2n-1}}{(n-1)!}.$$

We find $v_{2n}(r)$ by integrating over $b_{2n}(r)$, getting

$$v_{2n}(r) = \frac{\pi^n r^{2n}}{n!}.$$

The final statement that the volume within δ of the surface to the entire volume requires us to look at the shell between $r - \delta$ and r , whose volume is $v_{2n}(r) - v_{2n}(r - \delta)$. The ratio of this shell to the volume is $v_{2n}(r)(1 - (1 - \delta/r)^{2n})$ which approaches 1 as $n \rightarrow \infty$. The shell between r and $r + \delta$ has a much larger volume than $v_{2n}(r)$ for large n . If you lived in large dimensional space, you could not store enough paint in your house to paint the outside.

Exercise 3.15: a) Solve directly for $[B]$, $[C]$, and $[D]$ in (3.39) for the one dimensional case where $n = m = 1$. Show that (3.40) agrees with (3.37)

Solution: For X and Y one-dimensional, the covariance matrix of $(X, Y)^\top$ is

$$[K] = \begin{bmatrix} [K_X] & [K_{X \cdot Y}] \\ [K_{X \cdot Y}^\top] & [K_Y] \end{bmatrix} = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}.$$

With the assumption in (3.39) that $[K]$ is positive definite, $[K^{-1}]$ must exist and we can solve directly for B, C in

$$[K^{-1}] = \begin{bmatrix} B & C \\ C & D \end{bmatrix}$$

by multiplying out the left two terms in $[K][K^{-1}] = [I]$. We get $\sigma_X^2 B + \rho\sigma_X\sigma_Y C = 1$ and $\rho\sigma_X\sigma_Y B + \sigma_Y^2 C = 0$. From the second equation, $C = -\rho\sigma_X B/\sigma_Y$. Substituting this into the first equation, $B = 1/[\sigma_X^2(1 - \rho^2)]$. Substituting this into the equation for C , $C = -\rho/[\sigma_X\sigma_Y(1 - \rho^2)]$. From symmetry or from the right two terms, we solve for D , getting

$$[K^{-1}] = \frac{1}{1 - \rho^2} \begin{bmatrix} \sigma_X^{-2} & -\rho\sigma_X^{-1}\sigma_Y^{-1} \\ -\rho\sigma_X^{-1}\sigma_Y^{-1} & \sigma_Y^{-2} \end{bmatrix}.$$

Finally, noting that $B^{-1}C = \rho\sigma_X/\sigma_Y$, it is simple but slightly tedious to substitute these terms into (3.40) to get (3.37).

Exercise 3.16: a) Express $[B]$, $[C]$, and $[D]$, as defined in (3.39), in terms of $[K_X]$, $[K_Y]$ and $[K_{X \cdot Y}]$ by multiplying the block expression for $[K]$ by that for $[K]^{-1}$. You can check your solutions against those in (3.46) to (3.48). Hint: You can solve for $[B]$ and $[C]$ by looking at only the left two of the four block equations in $[KK^{-1}]$. You can use the symmetry between $\mathbf{X}$ and $\mathbf{Y}$ to solve for $[D]$.

Solution: One reason for going through this exercise, for those not very familiar with matrix manipulations, is to realize that algebraic manipulations on matrices are very similar to those on equations of numbers and real variables. One major difference is that matrices are not in general commutative (*i.e.*, $AB \neq BA$ in many cases), and thus premultiplication and postmultiplication are different. Another is that invertibility involves much more than being non-zero. Recognizing this, we proceed with slightly guided plug and chug.

Multiplying out two of the block terms in $[KK^{-1}]$, we get $[K_{\mathbf{X}}B] + [K_{\mathbf{X} \cdot \mathbf{Y}}C^T] = [I]$ and $[K_{\mathbf{X} \cdot \mathbf{Y}}^T B] + [K_{\mathbf{Y}}C^T] = 0$. These involve only two of the unknown terms, and we now solve for those terms. Recognizing that $[K_{\mathbf{X}}]$ and $[K_{\mathbf{Y}}]$ are invertible, we can rearrange the second equation as

$$[C^T] = -[K_{\mathbf{Y}}^{-1}K_{\mathbf{X} \cdot \mathbf{Y}}^T B].$$

Substituting this into the first equation, we get

$$\left[[K_{\mathbf{X}}] - [K_{\mathbf{X} \cdot \mathbf{Y}}K_{\mathbf{Y}}^{-1}K_{\mathbf{X} \cdot \mathbf{Y}}^T] \right] [B] = [I].$$

Now $[B]$ must be invertible (See Exercise 3.11c) Thus the matrix preceeding $[B]$ above is also invertible, so

$$[B] = \left[[K_{\mathbf{X}}] - [K_{\mathbf{X} \cdot \mathbf{Y}}K_{\mathbf{Y}}^{-1}K_{\mathbf{X} \cdot \mathbf{Y}}^T] \right]^{-1}.$$

This agrees with the solution (derived very differently) in (3.46). Next, to solve for $[C]$, we take the transpose of $[C^T]$ above, leading to $[C] = -[BK_{\mathbf{X} \cdot \mathbf{Y}}K_{\mathbf{Y}}^{-1}]$. This agrees with (3.47).

We could solve for $[D]$ in the same way, but it is easier to use the symmetry and simply interchange the roles of $\mathbf{X}$ and $\mathbf{Y}$ to get (3.48).

b) Use your result in (a) for $[C]$ plus the symmetry between $\mathbf{X}$ and $\mathbf{Y}$ to show that

$$[BK_{\mathbf{X} \cdot \mathbf{Y}}K_{\mathbf{Y}}^{-1}] = [K_{\mathbf{X}}^{-1}K_{\mathbf{X} \cdot \mathbf{Y}}D].$$

Solution: The quantity on the left above is $-[C]$ as derived in (a). By using the symmetry between $\mathbf{X}$ and $\mathbf{Y}$, we see that $[DK_{\mathbf{X} \cdot \mathbf{Y}}^TK_{\mathbf{X}}^{-1}]$ is $-C^T$, and taking the transpose completes the argument.

c) Show that $[K_{\mathbf{V}}^{-1}G] = [H^TK_{\mathbf{Z}}^{-1}]$ for the formulations $\mathbf{X} = [G]\mathbf{Y} + \mathbf{V}$ and $\mathbf{Y} = [H]\mathbf{X} + \mathbf{Z}$ where $\mathbf{X}$ and $\mathbf{Y}$ are zero-mean, jointly Gaussian and have a non-singular combined covariance matrix. Hint: This is almost trivial from (b), (3.43), (3.44), and the symmetry.

Solution: From (3.43), $[K_{\mathbf{V}}] = [B^{-1}]$ and from (3.44), $[G] = [K_{\mathbf{X} \cdot \mathbf{Y}}K_{\mathbf{Y}}]$. Substituting this into the left side of (b) and the symmetric relations for $\mathbf{X}$ and $\mathbf{Y}$ interchanged into the right side completes the demonstration.

Exercise 3.17: Let $\mathbf{X}$ and $\mathbf{Z}$ be statistically independent Gaussian rv's of arbitrary dimension n and m respectively. Let $\mathbf{Y} = [H]\mathbf{X} + \mathbf{Z}$ where $[H]$ is an arbitrary real $n \times m$ matrix.

a) Explain why $X_1, \dots, X_n, Z_1, \dots, Z_m$ must be jointly Gaussian rv's and then why $X_1, \dots, X_n, Y_1, \dots, Y_m$ must be jointly Gaussian.

Solution: Let $\mathbf{a}$ be an arbitrary real n -vector and $\mathbf{b}$ be an arbitrary real m vector. Then $\mathbf{a}^\top \mathbf{X}$ is a Gaussian rv. Also $\mathbf{b}^\top \mathbf{Z}$ is a Gaussian rv and is independent of $\mathbf{a}^\top \mathbf{X}$. Thus $\mathbf{a}^\top \mathbf{X} + \mathbf{b}^\top \mathbf{Z}$ is a Gaussian rv. Since $\mathbf{a}$ and $\mathbf{b}$ are arbitrary, this shows that all linear combinations of $X_1, \dots, X_n, Z_1, \dots, Z_m$ are Gaussian, and thus that $X_1, \dots, X_n, Z_1, \dots, Z_m$ are jointly Gaussian. In the same way, any linear combination, $\mathbf{a}^\top \mathbf{X} + \mathbf{b}^\top \mathbf{Y} = \mathbf{a}^\top \mathbf{X} + \mathbf{b}^\top [H] \mathbf{X} + \mathbf{b}^\top \mathbf{Z}$ is the sum of two independent rv's, $(\mathbf{a}^\top + \mathbf{b}^\top [H]) \mathbf{X}$ and $\mathbf{b}^\top \mathbf{Z}$. Thus $X_1, \dots, X_n, Z_1, \dots, Z_m$ are jointly Gaussian.

b) Show that if $[K_X]$ and $[K_Z]$ are non-singular, then the combined matrix $[K]$ for $(X_1, \dots, X_n, Y_1, \dots, Y_m)^\top$ must be non-singular.

Solution: For any covariance matrix, $\mathbf{E}[\mathbf{U}\mathbf{U}^\top]$, the definition of positive definite is the same as the condition that any linear combination $\mathbf{c}^\top \mathbf{U}$ with $\mathbf{c} \neq 0$ is a rv with positive variance (*i.e.*, $\mathbf{c}^\top \mathbf{E}[\mathbf{U}\mathbf{U}^\top] \mathbf{c} > 0$). For the case here, if $\mathbf{b} \neq 0$, then $\text{VAR}[\mathbf{b}^\top \mathbf{z}] > 0$ and $\mathbf{b}^\top \mathbf{Z}$ is one of the independent rv's being added. If $\mathbf{b} = 0$, then the other rv is $\mathbf{a}^\top \mathbf{X}$, which has a positive variance since $\mathbf{a}$ must then be nonzero.

Exercise 3.18: a) Verify (3.56) for $k = 0$ by the use of induction on (3.54).

Solution: We start with $\mathbf{E}[X^2(1)] = \mathbf{E}[W^2(0)] \sigma^2$, which is the basis of the induction and which satisfies (3.56) for $n = 0$, $k = 0$. Next assume that $\mathbf{E}[X^2(n)] = \sigma^2(1 - \alpha^{2n})/(1 - \alpha^2)$ for given $n \geq 1$. Using (3.54) and the independence of X_n and W_n , we have

$$\begin{aligned} \mathbf{E}[X^2(n+1)] &= \alpha^2 \mathbf{E}[X^2(n)] + \mathbf{E}[W^2(n)] = \frac{\sigma^2 \alpha^2 (1 - \alpha^{2n})}{1 - \alpha^2} + \sigma^2 \\ &= \sigma^2 \left[\frac{\alpha^2 - \alpha^{2(n+1)}}{1 - \alpha^2} + \frac{1 - \alpha^2}{1 - \alpha^2} \right] = \frac{\sigma^2 (1 - \alpha^{2(n+1)})}{1 - \alpha^2}. \end{aligned}$$

b) Verify (3.56) for $k = 0$ directly from (3.55).

Solution: Since $W(0), W(1), \dots$, are independent, we can take the expected value of (3.55) squared to get

$$\mathbf{E}[X^2(n)] = \sigma^2 + \alpha^2 \sigma^2 + \dots + \alpha^{2(n-1)} \sigma^2 = \frac{\sigma^2 (1 - \alpha^{2n})}{1 - \alpha^2},$$

where the last inequality can be verified by dividing $1 - \alpha^{2n}$ by $1 - \alpha^2$.

c) Verify (3.56) for $k > 0$ by using induction on k .

Solution: We have demonstrated the base of the induction (with $k = 0$) for all n in (a) and (b). Now assume (3.56) for any given $k \geq 0$, *i.e.*,

$$\mathbf{E}[X(n)X(n+k)] = \frac{\sigma^2 \alpha^k (1 - \alpha^{2n})}{1 - \alpha^2}. \quad (\text{A.51})$$

For $k + 1$, we then have

$$\begin{aligned} \mathbf{E}[X(n)X(n+k+1)] &= \mathbf{E}\left[X(n)\left(\alpha X(n+k) + W(n+k)\right)\right] \\ &= \alpha \mathbf{E}[X(n)X(n+k)] = \frac{\sigma^2 \alpha^{k+1} (1 - \alpha^{2n})}{1 - \alpha^2}. \end{aligned}$$

d) Verify that

$$\lim_{n \rightarrow \infty} \mathbf{E}[X(n)X(n+k)] = \frac{\sigma^2 \alpha^k}{1 - \alpha^2}.$$

Solution: Since $|\alpha| < 1$ by assumption, $\lim_{n \rightarrow \infty} \alpha^{2n} = 0$. The result is then immediate from (A.51).

Exercise 3.19: Let $\{X(t); t \in \mathfrak{R}\}$ be defined by $X(t) = tA$ for all $t \in \mathfrak{R}$ where $A \sim \mathcal{N}(0, 1)$. Show that this is a Gaussian process. Find its mean for each t and find its covariance function. Note: The purpose of this exercise is to show that Gaussian processes can be very degenerate and trivial.

Solution: Since $X(t_1), \dots, X(t_n)$ are all linear combinations of A , which is $\mathcal{N}(0, 1)$, $\{X(t); t \in \mathfrak{R}\}$ is a Gaussian process. Since $\mathbf{E}[A] = 0$, we see that $\mathbf{E}[X(t)] = t\mathbf{E}[A] = 0$. Similarly, $\mathbf{E}[X(t)X(\tau)] = t\tau\mathbf{E}[A^2] = t\tau$. The sample functions of this process are straight lines through the origin with the slope of the line given by the sample value of A .

Exercise 3.20: Let $\{X(t); t \geq 0\}$ be a stochastic process with independent and stationary increments and let $X(0)$ be an arbitrary random variable. Show that $\mathbf{E}[X(t)] = \mathbf{E}[X(0)] + t\mathbf{E}[X(1) - X(0)]$ and that

$$K_X(t, \tau) = \text{VAR}[X(0)] + t[\text{VAR}[X(1)] - \text{VAR}[X(0)]].$$

Solution: For integer $n > 0$, we can express

$$X(n) = X(0) + \sum_{i=1}^n [X(i) - X(i-1)].$$

Using the stationary increment property, $\mathbf{E}[X(i) - X(i-1)] = \mathbf{E}[X(1) - X(0)]$, so $\mathbf{E}[X(n)] = \mathbf{E}[X(0)] + n\mathbf{E}[X(1) - X(0)]$. Extending this to increments of $1/m$ for arbitrary integers $n, m > 0$,

$$X(n/m) = X(0) + \sum_{i=1}^n X(\frac{i}{m}) - X(\frac{i-1}{m}).$$

Taking expected values and using the stationary increment property, $\mathbf{E}[X(n/m)] = \mathbf{E}[X(0)] + n\mathbf{E}[X(1/m) - X(0)]$. For $n = m$, this is $\mathbf{E}[X(1)] - \mathbf{E}[X(0)] = m\mathbf{E}[X(1/m) - X(0)]$, so we have

$$\mathbf{E}[X(n/m)] = \mathbf{E}[X(0)] + \frac{n}{m}\mathbf{E}[X(1) - X(0)].$$

This is the desired result for $\mathbf{E}[X(t)]$ for all positive rational t . Unfortunately this can not be extended to irrational t without making a further assumption about the process, and being concerned about it appears to be mathematical fetishism, especially since having a nonzero mean is somewhat silly for this kind of process.

For the covariance, we can express the fluctuations of the rv's as

$$\tilde{X}(n) = \tilde{X}(0) + \sum_{i=1}^n [\tilde{X}(i) - \tilde{X}(i-1)].$$

The increments are independent and stationary, and we also assume the increments are independent of $X(0)$. This should have been stated explicitly in the problem statement, but the exercise becomes more challenging when one has to figure that out. Assuming $k > n$, we then have

$$K_X(n, k) = \mathbb{E} [\tilde{X}(n)\tilde{X}(m)] = \text{VAR} [X(0)] + n\text{VAR} [X(1) - X(0)].$$

For $n = 1$, we see that $\text{VAR} [X(1) - X(0)] = \text{VAR} [X(1)] - \text{VAR} [X(0)]$, so the result follows for t and τ integer. The result for t and τ rational follows as before.

Exercise 3.21: a) Let $X(t) = R \cos(2\pi ft + \theta)$ where R is a Rayleigh rv and the rv θ is independent of R and uniformly distributed over the interval 0 to 2π . Show that $\mathbb{E} [X(t)] = 0$.

Solution: This can be done by standard (and quite tedious) manipulations, but if we first look at $t = 0$ and condition on a sample value of R , we are simply looking at $\cos(\theta)$, and since θ is uniform over $[0, 2\pi)$, it seems almost obvious that the mean should be 0. To capture this intuition, note that $\cos(\theta) = -\cos(\theta + \pi)$. Since θ is uniform between 0 and 2π , $\mathbb{E} [\cos(\theta)] = \mathbb{E} [\cos(\theta + \pi)]$, so that $\mathbb{E} [\cos(\theta)] = 0$. The same argument works for any t , so the result follows.

b) Show that $\mathbb{E} [X(t)X(t + \tau)] = \frac{1}{2}\mathbb{E} [R^2] \cos(2\pi f\tau)$.

Solution: Since θ and R are independent, we have

$$\begin{aligned} \mathbb{E} [X(t)X(t + \tau)] &= \mathbb{E} [R^2] \mathbb{E} [\cos(2\pi ft + \theta) \cos(2\pi f(t + \tau) + \theta)] \\ &= \mathbb{E} [R^2] \frac{1}{2} \mathbb{E} [\cos(4\pi ft + 2\pi f\tau + 2\theta) + \cos(2\pi f\tau)] \\ &= \frac{\mathbb{E} [R^2] \cos(2\pi f\tau)}{2}. \end{aligned}$$

where we used a standard trigonometric identity and then took the expectation over θ using the same argument as in (a).

c) Show that $\{X(t); t \in \mathbb{R}\}$ is a Gaussian process.

Solution: Let W_1, W_2 be IID normal Gaussian rv's. These can be expressed in polar coordinates as $W_1 = R \cos \theta$ and $W_2 = R \sin \theta$, where R is Rayleigh and θ is uniform. The rv $R \cos \theta$ is then $\mathcal{N}(0, 1)$. Similarly, $X(t)$ is a linear combination of W_1 and W_2 for each t , so each set $\{X(t_1), X(t_2), \dots, X(t_k)\}$ of rv's is jointly Gaussian. It follows that the process is Gaussian.

Exercise 3.22: Let $h(t)$ be a real square-integrable function whose Fourier transform is 0 for $|f| > B$ for some $B > 0$. Show that $\sum_n h^2(t - n/2B) = (1/2B) \int h^2(\tau) d\tau$ for all $t \in \mathbb{R}$. Hint: find the sampling theorem expansion for a time shifted sinc function.

Solution: We use a slightly simpler approach than that of the hint. The sampling theorem expansion of $h(t)$ is given by

$$h(t) = \sum_n h\left(\frac{n}{2B}\right) \text{sinc}(2Bt - n).$$

Since the functions $\{\sqrt{2B}\text{sinc}(2Bt - n); n \in \mathbb{Z}\}$ are orthonormal, the energy equation, (3.64), says that $\int_{-\infty}^{\infty} h^2(t)dt = \sum_n 2Bh^2(n/2B)$. For the special case of $t = 0$, this is the same as what is to be shown (summing over $-n$ instead of n). To generalize this, consider $h(t + \tau)$ as a function of t for fixed τ . Then using the same sampling expansion on this shifted function,

$$h(t + \tau) = \sum_n h\left(\frac{n}{2B} + \tau\right)\text{sinc}(2Bt - n).$$

The Fourier transform of $h(t + \tau)$ (as a function of t for fixed τ) is still 0 for $f > B$, and $\int h^2(t)dt = \int h^2(t + \tau)dt$. Thus,

$$\int h^2(t)dt = \sum_n 2Bh^2\left(\frac{n}{2B} + \tau\right).$$

Replacing n by $-n$ in the sum over $\mathbb{Z}$ and interchanging t and τ , we have the desired result.

Exercise 3.23: a) Let $\mathbf{Z} = (Z_1, \dots, Z_n)^T$ be a circularly-symmetric n -rv. Show that Z_k is circularly symmetric for each k , $1 \leq k \leq n$. Hint: use the definition directly (you cannot assume that $\mathbf{Z}$ is also Gaussian).

Solution: We carry out the argument for $n = 2$. The extension to arbitrary n will then be clear. Assume that $\mathbf{Z}$ has a probability density function and let $\mathbf{Y} = e^{i\theta}\mathbf{Z}$ for an arbitrary angle θ . Then $\mathbf{Z}$ is circularly symmetric if for all such $\mathbf{Y}$, $f_{Z_1, Z_2}(z_1, z_2) = f_{Y_1, Y_2}(z_1, z_2)$ for all $\mathbf{z}$. Integrating both sides over the real and imaginary parts of z_1 , we get the marginal density of Z_2 on one side and Y_2 on the other. That is $f_{Z_2}(z_2) = f_{Y_2}(z_2)$. Since $Y_2 = e^{i\theta}Z_2$, and this is true for all choices of θ , Z_2 is circularly symmetric.

Integrating over z_2 instead of z_1 shows that Z_1 is circularly symmetric. For $n > 2$, we integrate over all components other than k to show that Z_k is circularly symmetric. Although this shows that a circularly symmetric n -rv must have circularly symmetric components, the end of Example 3.7.7 shows that circularly-symmetric components do not necessarily combine into a circularly symmetric vector.

b) Show that $Z_1 + Z_2$ is circularly symmetric. For any complex n -vector $\mathbf{c}$, show that $\mathbf{c}^T \mathbf{Z}$ is a circularly symmetric rv.

Solution: Let $U = Z_1 + Z_2$. We can still find the PDF of U by convolving the PDF's of Z_1 and Z_2 , i.e., $f_U(u) = \int f_{Z_1}(z_1)f_{Z_2}(u - z_1)dz_1$. Here however, the integration is over the complex plane and the result for each complex value u is complex. Let $\mathbf{Y} = e^{i\theta}\mathbf{Z}$, so that $Y_1 = e^{i\theta}Z_1$ and $Y_2 = e^{i\theta}Z_2$. Let $V = Y_1 + Y_2 = e^{i\theta}U$. We then have $f_V(u) = \int f_{Y_1}(z_1)f_{Y_2}(u - z_1)dz_1$. From the circular symmetry of Z_1 and Z_2 , this integral is equal to $f_U(u)$, and this is true for all u and θ . Thus U is circularly symmetric. We can next show that $c_1Z_1 + c_2Z_2$ is circularly symmetric and then use induction to see that $\mathbf{c}^T \mathbf{Z}$ is circularly symmetric.

There are two lessons from this exercise. The first is that one can build up a set of results about circular symmetry and the second is that this is rather painful and time-consuming. The text (and the majority of applications) deal with complex random vectors that are both circularly symmetric and Gaussian, and here things are much more pleasant and elegant.

Exercise 3.24: Let A be a complex Gaussian rv, i.e., $A = A_1 + iA_2$ where A_1 and A_2 are zero-mean jointly-Gaussian real rv's with variances σ_1^2 and σ_2^2 respectively.

a) Show that $\mathbb{E}[AA^*] = \sigma_1^2 + \sigma_2^2$.

Solution:

$$AA^* = (A_1 + iA_2)(A_1 - iA_2) = A_1^2 + A_2^2. \quad (\text{A.52})$$

$$\mathbb{E}[AA^*] = \mathbb{E}[A_1^2 + A_2^2] = \sigma_1^2 + \sigma_2^2.$$

b) Show that

$$\mathbb{E}[(AA^*)^2] = 3\sigma_1^4 + 3\sigma_2^4 + 2\sigma_1^2\sigma_2^2 + 4(\mathbb{E}[A_1A_2])^2.$$

Solution: Using (A.52),

$$\mathbb{E}[(AA^*)^2] = \mathbb{E}[(A_1^2 + A_2^2)^2] = \mathbb{E}[A_1^4] + \mathbb{E}[A_2^4] + 2\mathbb{E}[A_1^2A_2^2]. \quad (\text{A.53})$$

From (3.7), $\mathbb{E}[A_\ell^4] = 3\sigma_\ell^4$ for $\ell = 1, 2$. The solution to Exercise 3.5(e) shows that $\mathbb{E}[A_1^2A_2^2] = \mathbb{E}[A_1^2]\mathbb{E}[A_2^2] + 2(\mathbb{E}[A_1A_2])^2$. Substituting these terms into (A.53) yields the result to be shown.

c) Show that $\mathbb{E}[(AA^*)^2] \geq 2(\mathbb{E}[AA^*])^2$ with equality if and only if A_1 and A_2 are IID. Hint: Lower bound $(\mathbb{E}[A_1A_2])^2$ by 0.

Solution: Using (a) and (b),

$$\begin{aligned} \mathbb{E}[(AA^*)^2] - 2(\mathbb{E}[AA^*])^2 &= 3\sigma_1^4 + 3\sigma_2^4 + 2\sigma_1^2\sigma_2^2 + 4(\mathbb{E}[A_1A_2])^2 - 2\sigma_1^4 - 2\sigma_2^4 - 4\sigma_1^2\sigma_2^2 \\ &= (\sigma_1^2 - \sigma_2^2)^2 + 4(\mathbb{E}[A_1A_2])^2 \geq 0. \end{aligned}$$

The condition for equality is that $\sigma_1^2 = \sigma_2^2$ and $(\mathbb{E}[A_1A_2]) = 0$, i.e., that A_1 and A_2 are IID, which means that A is circularly symmetric.

d Show that $\text{VAR}[AA^*] \geq (\mathbb{E}[AA^*])^2$.

Solution: This is immediate from (c), since $\text{VAR}[AA^*] = \mathbb{E}[(AA^*)^2] - (\mathbb{E}[AA^*])^2$. The condition for equality is that A is circularly symmetric.

Exercise 3.25: Let $K_X(t)$ be the covariance function of a WSS process $\{X(t); t \in \mathbb{R}\}$. Show that if $K_X(t)$ is continuous at $t = 0$, then it is continuous everywhere. Hint: Show that $\lim_{\delta \rightarrow 0} \mathbb{E}[X(0)(X(t-\delta) - X(t))] = 0$ for all t . Use the Schwarz inequality.

Solution: The Schwarz inequality applied to zero-mean rv's says that $|\mathbb{E}[XY]| \leq \sigma_X\sigma_Y$. Now consider $K_X(t) - K_X(t-\delta)$ for zero-mean rv's,

$$\begin{aligned} |K_X(t) - K_X(t-\delta)| &= |\mathbb{E}[X(0)(X(t) - X(t-\delta))]| \\ &\leq \sigma_X \left\{ \mathbb{E}[(X(t) - X(t-\delta))^2] \right\}^{1/2} \\ &= \sigma_X [2K_X(0) - 2K_X(\delta)]^{1/2}. \end{aligned}$$

Taking the limit on both sides as $\delta \rightarrow 0$, we see that if K_X is continuous at 0 it is continuous everywhere.

A.4 Solutions for Chapter 4

Exercise 4.1: Let $[P]$ be the transition matrix for a finite state Markov chain and let state i be recurrent. Prove that i is aperiodic if $P_{ii} > 0$.

Solution: Since $P_{11} > 0$, there is a walk of length 2 that starts in state 1, goes to state 1 at epoch 1 and goes from there to state 1 at epoch 2. In the same way, for each $n > 2$, a walk of length n exists, going through state 1 at each step. Thus $P_{11}^n > 0$, so the greatest common denominator of $\{n : P_{11}^n > 0\}$ is 1.

Exercise 4.2: Show that every Markov chain with $M < \infty$ states contains at least one recurrent set of states. Explaining each of the following statements is sufficient.

a) If state i_1 is transient, then there is some other state i_2 such that $i_1 \rightarrow i_2$ and $i_2 \not\rightarrow i_1$.

Solution: If there is no such state i_2 , then i_1 is recurrent by definition. That state is distinct from i_1 since otherwise $i_1 \rightarrow i_2$ would imply $i_2 \rightarrow i_1$.

b) If the i_2 of (a) is also transient, there is a third state i_3 such that $i_2 \rightarrow i_3$, $i_3 \not\rightarrow i_2$; that state must satisfy $i_3 \neq i_2$, $i_3 \neq i_1$.

Solution: The argument why i_3 exists with $i_2 \rightarrow i_3$, $i_3 \not\rightarrow i_2$ and with $i_3 \neq i_2$ is the same as (a). Since $i_1 \rightarrow i_2$ and $i_2 \rightarrow i_3$, we have $i_1 \rightarrow i_3$. We must also have $i_3 \not\rightarrow i_1$, since otherwise $i_3 \rightarrow i_1$ and $i_1 \rightarrow i_2$ would imply the contradiction $i_3 \rightarrow i_2$. Since $i_1 \rightarrow i_3$ and $i_3 \not\rightarrow i_1$, it follows as before that $i_3 \neq i_1$.

c) Continue iteratively to repeat (b) for successive states, $i_1, i_2, \dots$. That is, if $i_1, \dots, i_k$ are generated as above and are all transient, generate i_{k+1} such that $i_k \rightarrow i_{k+1}$ and $i_{k+1} \not\rightarrow i_k$. Then $i_{k+1} \neq i_j$ for $1 \leq j \leq k$.

Solution: The argument why i_{k+1} exists with $i_k \rightarrow i_{k+1}$, $i_{k+1} \not\rightarrow i_k$ and with $i_{k+1} \neq i_k$ is the same as before. To show that $i_{k+1} \neq i_j$ for each $j < k$, we use contradiction, noting that if $i_{k+1} = i_j$, then $i_{k+1} \rightarrow i_{j+1} \rightarrow i_k$.

d) Show that for some $k \leq M$, k is not transient, *i.e.*, it is recurrent, so a recurrent class exists.

Solution: For transient states $i_1, \dots, i_k$ generated in (c), state i_{k+1} found in (c) must be distinct from the distinct states $\{i_j; j \leq k\}$. Since there are only M states, there cannot be M transient states, since then, with $k = M$, a new distinct state i_{M+1} would be generated, which is impossible. Thus there must be some $k < M$ for which the extension to i_{k+1} leads to a recurrent state.

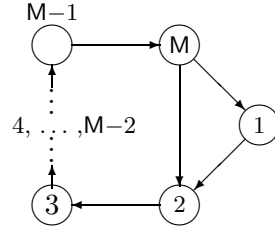
Exercise 4.3: Consider a finite-state Markov chain in which some given state, say state 1, is accessible from every other state. Show that the chain has exactly one recurrent class $\mathcal{R}$ of states and state $1 \in \mathcal{R}$.

Solution: Since $j \rightarrow 1$ for each j , there can be no state j for which $1 \rightarrow j$ and $j \not\rightarrow 1$. Thus state 1 is recurrent. Next, for any given j , if $1 \not\rightarrow j$, then j must be transient since $j \rightarrow 1$. On the other hand, if $1 \rightarrow j$, then 1 and j communicate and j must be in the same recurrent class as 1. Thus each state is either transient or in the same recurrent class as 1.

Exercise 4.4: Show how to generalize the graph in Figure 4.4 to an arbitrary number of states $M \geq 3$ with one cycle of M nodes and one of $M - 1$ nodes. For $M = 4$, let node 1 be the node not in the cycle of

$M - 1$ nodes. List the set of states accessible from node 1 in n steps for each $n \leq 12$ and show that the bound in Theorem 4.2.11 is met with equality. Explain why the same result holds for all larger M .

Solution: As illustrated below, the generalization has the set of nonzero transition probabilities $P_{i,i+1}$ for $1 \leq i < M$ and has $P_{M,1} > 0$ and $P_{M,2} > 0$.



For $M = 4$, let $\mathcal{T}(n)$ be the set of states accessible from state 1 in n steps.

$$\begin{array}{lll} \mathcal{T}(1) = \{2\} & \mathcal{T}(5) = \{2, 3\} & \mathcal{T}(9) = \{2, 3, 4\} \\ \mathcal{T}(2) = \{3\} & \mathcal{T}(6) = \{3, 4\} & \mathcal{T}(10) = \{1, 2, 3, 4\} \\ \mathcal{T}(3) = \{4\} & \mathcal{T}(7) = \{1, 2, 4\} & \mathcal{T}(11) = \{1, 2, 3, 4\} \\ \mathcal{T}(4) = \{1, 2\} & \mathcal{T}(8) = \{1, 2, 3\} & \mathcal{T}(12) = \{1, 2, 3, 4\}. \end{array}$$

What we observe above is that for the first three steps, $\mathcal{T}(n)$ is a singleton set, and for arbitrary M , $\mathcal{T}(n)$ for the first $M - 1$ steps are singleton states. There are then $M - 1$ doubleton states, and so forth up to $M - 1$ steps with $M - 1$ states included in $\mathcal{T}(n)$. Thus it is on step $(M - 1)^2 + 1$ that all states are first included.

Exercise 4.5: (Proof of Theorem 4.2.11) a) Show that an ergodic Markov chain with $M > 1$ states must contain a cycle with $\tau < M$ states. Hint: Use ergodicity to show that the smallest cycle cannot contain M states.

Solution: The states in any cycle (not counting the initial state) are distinct and thus the number of steps in a cycle is at most M . A recurrent chain must contain cycles, since for each pair of states $\ell \neq j$, there is a walk from ℓ to j and then back to ℓ ; if any state i other than ℓ is repeated in this walk, the first i and all subsequent states before the second i can be eliminated. This can be done repeatedly until a cycle remains.

Finally, suppose a cycle contains M states. If there is any transition $P_{ij} > 0$ for which (i, j) is not a transition on that cycle, then that transition can be added to the cycle and all the transitions between i and j on the existing cycle can be omitted, thus creating a cycle of fewer than M steps. If there are no nonzero transitions other than those in a cycle with M steps, then the Markov chain is periodic with period M and thus not ergodic.

b) Let ℓ be a fixed state on a fixed cycle of length $\tau < M$. Let $\mathcal{T}(m)$ be the set of states accessible from ℓ in m steps. Show that for each $m \geq 1$, $\mathcal{T}(m) \subseteq \mathcal{T}(m + \tau)$. Hint: For any given state $j \in \mathcal{T}(m)$, show how to construct a walk of $m + \tau$ steps from ℓ to j from the assumed walk of m steps.

Solution: Let j be any state in $\mathcal{T}(m)$. Then there is an m -step walk from ℓ to j . There is also a cycle of τ steps from state ℓ to ℓ . Concatenate this cycle (as a walk) with the above m step walk from ℓ to j , yielding a walk of $\tau + m$ steps from ℓ to j . Thus $j \in \mathcal{T}(m + \tau)$ and it follows that $\mathcal{T}(m) \subseteq \mathcal{T}(m + \tau)$.

c) Define $\mathcal{T}(0)$ to be the singleton set $\{\ell\}$ and show that

$$\mathcal{T}(0) \subseteq \mathcal{T}(\tau) \subseteq \mathcal{T}(2\tau) \subseteq \cdots \subseteq \mathcal{T}(n\tau) \subseteq \cdots$$

Solution: Since $\mathcal{T}(0) = \{\ell\}$ and $\ell \in \mathcal{T}(\tau)$, we see that $\mathcal{T}(0) \subseteq \mathcal{T}(\tau)$. Next, for each $n \geq 1$, use (b), with $m = n\tau$, to see that $\mathcal{T}(n\tau) \subseteq \mathcal{T}(n\tau + \tau)$. Thus each subset inequality above is satisfied.

d) Show that if one of the inclusions above is satisfied with equality, then all subsequent inclusions are satisfied with equality. Show from this that at most the first $M - 1$ inclusions can be satisfied with strict inequality and that $\mathcal{T}(n\tau) = \mathcal{T}((M - 1)\tau)$ for all $n \geq M - 1$.

Solution: We first show that if $\mathcal{T}((k+1)\tau) = \mathcal{T}(k\tau)$ for some k , then $\mathcal{T}(n\tau) = \mathcal{T}(k\tau)$ for all $n > k$. Note that $\mathcal{T}((k+1)\tau)$ is the set of states reached in τ steps from $\mathcal{T}(k\tau)$. Similarly $\mathcal{T}((k+2)\tau)$ is the set of states reached in τ steps from $\mathcal{T}((k+1)\tau)$. Thus if $\mathcal{T}((k+1)\tau) = \mathcal{T}(k\tau)$ then also $\mathcal{T}((k+2)\tau) = \mathcal{T}((k+1)\tau)$. Using induction,

$$\mathcal{T}(n\tau) = \mathcal{T}(k\tau) \quad \text{for all } n \geq k.$$

Now if k is the smallest integer for which $\mathcal{T}((k+1)\tau) = \mathcal{T}(k\tau)$, then the size of $\mathcal{T}(n\tau)$ must increase for each $n < k$. Since $|\mathcal{T}(0)| = 1$, we see that $|\mathcal{T}(n\tau)| \geq n + 1$ for $n \leq k$. Since M is the total number of states, we see that $k \leq M - 1$. Thus $\mathcal{T}(n\tau) = \mathcal{T}((M - 1)\tau)$ for all $n \geq M - 1$.

e) Show that all states are included in $\mathcal{T}((M - 1)\tau)$.

Solution: For any t such that $P_{\ell\ell}^t > 0$, we can repeat the argument in part (b), replacing τ by t to see that for any $m \geq 1$, $\mathcal{T}(m) \subseteq \mathcal{T}(m + t)$. Thus we have

$$\mathcal{T}((M-1)\tau) \subseteq \mathcal{T}((M-1)\tau + t) \subseteq \cdots \subseteq \mathcal{T}((M-1)\tau + t\tau) = \mathcal{T}((M-1)\tau),$$

where (d) was used in the final equality. This shows that all the inclusions above are satisfied with equality and thus that $\mathcal{T}((M-1)\tau) = \mathcal{T}((M-1)\tau + kt)$ for all $k \leq \tau$. Using t in place of τ in the argument in (d), this can be extended to

$$\mathcal{T}((M-1)\tau) = \mathcal{T}((M-1)\tau + kt) \quad \text{for all } k \geq 1.$$

Since the chain is ergodic, we can choose t so that both $P_{\ell\ell}^t > 0$ and $\gcd(t, \tau) = 1$. From elementary number theory, integers $k \geq 1$ and $j \geq 1$ can then be chosen so that $kt = j\tau + 1$. Thus

$$\mathcal{T}((M-1)\tau) = \mathcal{T}((M-1)\tau + kt) = \mathcal{T}((M-1) + j)\tau + 1) = \mathcal{T}((M-1)\tau + 1). \quad (4.5a)$$

As in (d), $\mathcal{T}((M-1)\tau + 2)$ is the set of states reachable in one step from $\mathcal{T}((M-1)\tau + 1)$. From (4.5a), this is the set of states reachable from $\mathcal{T}((M-1)\tau)$ in 1 step, *i.e.*,

$$\mathcal{T}((M-1)\tau + 2) = \mathcal{T}((M-1)\tau + 1) = \mathcal{T}((M-1)\tau).$$

Extending this,

$$\mathcal{T}((M-1)\tau) = \mathcal{T}((M-1)\tau + m) \quad \text{for all } m \geq 1.$$

This means that $\mathcal{T}((M-1)\tau)$ contains all states that can ever occur from time $((M-1)\tau)$ on, and thus must contain all states since the chain is recurrent.

f) Show that $P_{ij}^{(M-1)^2+1} > 0$ for all i, j .

Solution: We have shown that all states are accessible from state ℓ at all times $\tau(M-1)$ or later, and since $\tau \leq M-1$, all are accessible at all times $n \geq (M-1)^2$. The same applies to any state on a cycle of length at most $M-1$. It is possible (as in Figure 4.4), for some states to be only on a cycle of length M . Any such state can reach the cycle in the proof in at most $M-\tau$ steps. Using this path to reach a state on the cycle and following this by paths of length $\tau(M-1)$, all states can reach all other states at all times greater than or equal to

$$\tau(M-1) + M - \tau \leq (M-1)^2 + 1.$$

The above derivation assumed $M > 1$. The case $M = 1$ is obvious, so the theorem is proven.

Exercise 4.6: Consider a Markov chain with one ergodic class of r states, say $\{1, 2, \dots, r\}$ and $M-r$ other states that are all transient. Show that $P_{ij}^n > 0$ for all $j \leq r$ and $n \geq (r-1)^2 + 1 + M - r$.

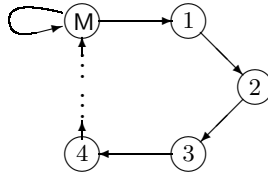
Solution: Applying Theorem 4.2.11 (proved in Exercise 4.5) to the ergodic class, We see that $P_{ij}^n > 0$ for all $i \leq r$ and $j \leq r$ and all $n \geq (r-1)^2 + 1$. Each transient state i has a path of some length $m \leq M-r$ to some state k in the ergodic class, and thus $P_{ij}^n \geq P_{ik}^m P_{kj}^{n-m} > 0$ for each $j \leq r$ if $n - m \geq (r-1)^2 + 1$, i.e., if $n \geq (r-1)^2 + 1 + M - r$.

Exercise 4.7: a) Let τ be the number of states in the smallest cycle of an arbitrary ergodic Markov chain of $M \geq 3$ states. Show that $P_{ij}^n > 0$ for all $n \geq (M-2)\tau + M$. Hint: Look at the proof of Theorem 4.2.11 in Exercise 4.5.

Solution: Exercise 4.5 (e) showed that if τ is the length of the smallest cycle, then for i in such a cycle, $P_{ij}^n > 0$ for each j and each $n \geq (M-1)\tau$. Since some state on a cycle of length τ can be reached from any state in at most $M-\tau$ steps, any state can be reached from any other state in n steps for all $n \geq (M-1)\tau + M - \tau = (M-2)\tau + M$.

b) For $\tau = 1$, draw the graph of an ergodic Markov chain (generalized for arbitrary $M \geq 3$) for which there is an i, j for which $P_{ij}^n = 0$ for $n = 2M - 3$. Hint: Look at Figure 4.4.

Solution: Figure 4.4 contains a cycle of $M-1$ states on the left side and an overall cycle of M states on the outside. If we shrink the left side cycle to a single state while maintaining the outer cycle, we get

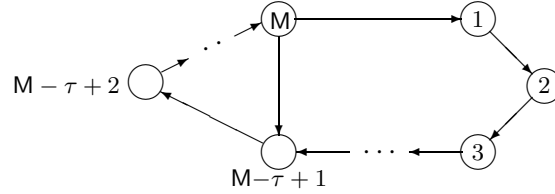


Starting with state 1, we see that state $M-1$ is reached for the first time at step $M-2$ and is reachable for the second time M steps later. One step before that, i.e., at time $2M-3$,

state $M - 1$ cannot be reached. At $2M - 2$, however, state $M - 1$ and all other states are reachable.

c) For arbitrary $\tau < M - 1$, draw the graph of an ergodic Markov chain (generalized for arbitrary M) for which there is an i, j for which $P_{ij}^n = 0$ for $n = (M - 2)\tau + M - 1$. Assume that M and τ are relatively prime.

Solution: We generalize what was done in (b), creating a cycle of length τ on the left while maintaining the cycle of length M on the outside.



This can be analyzed in the same way as Exercise 4.4. In particular, let $\mathcal{T}(n)$ be the set of states reachable from state 1 in n steps. Then $\mathcal{T}(n)$ is a singleton for $n < M$, a doubleton for $M \leq n < M + \tau$, and so forth up to containing $M - 1$ states for $M + \tau(M - 3) \leq n < M + \tau(M - 2)$. Thus there is a j for which $P_{1j}^{(M-2)\tau+M-1} = 0$. It can be seen that this j is state $M - \tau$.

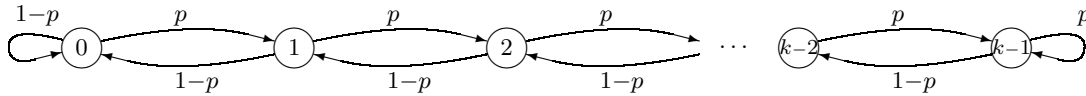
Exercise 4.8: A transition probability matrix $[P]$ is said to be doubly stochastic if

$$\sum_j P_{ij} = 1 \quad \text{for all } i; \quad \sum_i P_{ij} = 1 \quad \text{for all } j.$$

That is, each row sum and each column sum equals 1. If a doubly stochastic chain has M states and is ergodic (i.e., has a single class of states and is aperiodic), calculate its steady-state probabilities.

Solution: It is easy to see that if the row sums are all equal to 1, then $[P]e = e$. If the column sums are also equal to 1, then $e^T[P] = e^T$. Thus e^T is a left eigenvector of $[P]$ with eigenvalue 1, and it is unique within a scale factor since the chain is ergodic. Scaling e^T to be probabilities, $\pi = (1/M, 1/M, \dots, 1/M)$.

Exercise 4.9: a) Find the steady-state probabilities $\pi_0, \dots, \pi_{k-1}$ for the Markov chain below. Express your answer in terms of the ratio $\rho = p/q$ where $q = 1 - p$. Pay particular attention to the special case $\rho = 1$.



Solution: The steady-state equations, using the abbreviation $q = 1 - p$ are

$$\begin{aligned} \pi_0 &= q\pi_0 + q\pi_1 \\ \pi_j &= p\pi_{j-1} + q\pi_{j+1}; \quad \text{for } 1 \leq j \leq k-2 \\ \pi_{k-1} &= p\pi_{k-2} + p\pi_{k-1}. \end{aligned}$$

Simplifying the first equation, we get $p\pi_0 = q\pi_1$.

Substituting $q\pi_1$ for $p\pi_0$ in the second equation, we get $\pi_1 = q\pi_1 + q\pi_2$. Simplifying, we get $p\pi_1 = q\pi_2$.

We can then use induction. Using the inductive hypothesis $p\pi_{j-1} = q\pi_j$ (which has been verified for $j = 1, 2$) on $\pi_j = p\pi_{j-1} + q\pi_{j+1}$, we get

$$p\pi_j = q\pi_{j+1} \quad \text{for } 1 \leq j \leq k-2.$$

Combining these equations, $\pi_j = \rho\pi_{j-1}$ for $1 \leq j \leq k-1$, so $\pi_j = \rho^j\pi_0$ for $1 \leq j \leq k-1$. Normalizing by the fact that the steady-state probabilities sum to 1, and taking $\rho \neq 1$,

$$\pi_0 \left(\sum_{j=0}^{k-1} \rho^j \right) = 1 \quad \text{so } \pi_0 = \frac{1-\rho}{1-\rho^k}; \quad \pi_j = \rho^j \frac{1-\rho}{1-\rho^k}. \quad (\text{A.54})$$

For $\rho = 1$, $\rho^j = 1$ and $\pi_j = 1/k$ for $0 \leq j \leq k-1$. Note that we did not use the final steady-state equation. The reason is that $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ specifies $\boldsymbol{\pi}$ only within a scale factor, so the k equations can not be linearly independent and the k th equation can not provide anything new.

Note also that the general result here is most simply stated as $p\pi_{j-1} = q\pi_j$. This says that the steady-state probability of a transition from $j-1$ to j is the same as that from j to $j-1$. This is intuitively obvious since in any sample path, the total number of transitions from $j-1$ to j is within 1 of the transitions in the opposite direction. This important intuitive notion will become precise after studying the strong law of numbers.

b) Sketch $\pi_0, \dots, \pi_{k-1}$. Give one sketch for $\rho = 1/2$, one for $\rho = 1$, and one for $\rho = 2$.

Solution: We see from (A.54) that for $\rho \neq 1$, π_j is geometrically distributed for $1 \leq j \leq k-1$. For $\rho > 1$, it is geometrically increasing in j and for $\rho < 1$, it is geometrically decreasing. For $\rho = 1$, it is constant. If you can't visualize this without a sketch, you should draw the sketch yourself.

c) Find the limit of π_0 as k approaches ∞ ; give separate answers for $\rho < 1$, $\rho = 1$, and $\rho > 1$. Find limiting values of π_{k-1} for the same cases.

Solution:

$$\lim_{k \rightarrow \infty} \pi_0 = \begin{cases} \lim_{k \rightarrow \infty} \frac{1-\rho}{1-\rho^k} = 1-\rho & \text{for } \rho < 1, \\ \lim_{k \rightarrow \infty} \frac{1}{k} = 0 & \text{for } \rho = 1, \\ \lim_{k \rightarrow \infty} \frac{\rho-1}{\rho^k-1} = 0 & \text{for } \rho > 1. \end{cases}$$

Note that the approach to 0 for $\rho = 1$ is harmonic in k and that for $\rho > 1$ is geometric in k . For state $k-1$, the analogous result is

$$\lim_{k \rightarrow \infty} \pi_{k-1} = \begin{cases} \lim_{k \rightarrow \infty} \rho^{k-1} \frac{1-\rho}{1-\rho^k} = 0 & \text{for } \rho < 1, \\ \lim_{k \rightarrow \infty} \frac{1}{k} = 0 & \text{for } \rho = 1, \\ \lim_{k \rightarrow \infty} \rho^{k-1} \frac{\rho-1}{\rho^k-1} = 1 - \frac{1}{\rho} & \text{for } \rho > 1, \end{cases}$$

This shows us that for $\rho > 1$, the probability is still geometric in the states, but the large probabilities are clustered at the highest states. We will see in the next chapter that with a countable number of states, all states have probability zero when $\rho \geq 1$, and this shows us how that limit is approached with increasing k .

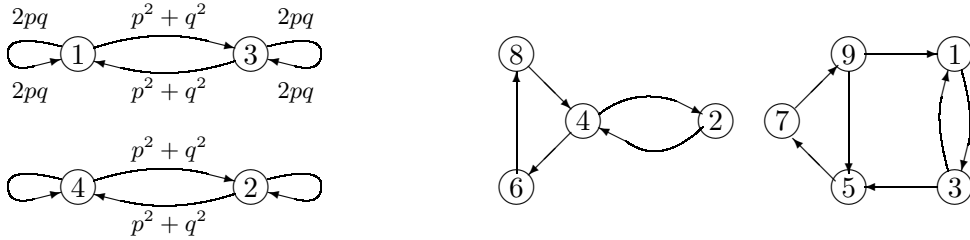
Exercise 4.10: a) Find the steady-state probabilities for each of the Markov chains in Figure 4.2. Assume that all clockwise probabilities in the first graph are the same, say p , and assume that $P_{4,5} = P_{4,1}$ in the second graph.

Solution: These probabilities can be found in a straightforward but tedious fashion by solving (4.8). Note that $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ is a set of M linear equations of which only $M-1$ are linearly independent and $\sum \pi_i = 1$ provides the needed extra equation. The solutions are $\pi_i = 1/4$ for each state in the first graph and $\pi_i = 1/10$ for all but state 4 in the second graph; $\pi_4 = 1/5$.

One learns more by trying to find $\boldsymbol{\pi}$ by inspection. For the first graph, the π_i are clearly equal by symmetry. For the second graph, states 1 and 5 are immediately accessible only from state 4 and are thus equally likely and each has half the probability of 4. The probabilities on the states of each loop should be the same, leading to the answer above. It would be prudent to check this answer by (4.8), but that is certainly easier than solving (4.8).

b) Find the matrices $[P^2]$ for the same chains. Draw the graphs for the Markov chains represented by $[P^2]$, i.e., the graph of two step transitions for the original chains. Find the steady-state probabilities for these two-step chains. Explain why your steady-state probabilities are not unique.

Solution: Let $q = 1 - p$ in the first graph. In the second graph, all transitions out of states 3, 4, and 9 have probability $1/2$. All other transitions have probability 1.



One steady-state probability for the first chain is $\pi_1 = \pi_3 = 1/2$ and the other is $\pi_2 = \pi_4 = 1/2$. These are the steady-state probabilities for the two recurrent classes of $[P^2]$. The second chain also has two recurrent classes. The steady-state probabilities for the first are $\pi_2 = \pi_6 = \pi_8 = 0.2$ and $\pi_4 = .4$. Those for the second are $\pi_1 = \pi_3 = \pi_5 = \pi_7 = \pi_9 = 0.2$.

c) Find $\lim_{n \rightarrow \infty} [P^{2n}]$ for each of the chains.

Solution: The limit for each chain is block diagonal with one block being the even numbers and the other the odd numbers. Within a block, the rows are the same. For the first chain, the blocks are (1, 3) and (2, 4). We have $\lim_{n \rightarrow \infty} P_{ij}^{2n} = 1/2$ for i, j both odd or both even; it is 0 otherwise. For the second chain, within the even block, $\lim_{n \rightarrow \infty} P_{ij}^n = 0.2$ for $j \neq 4$ and 0.4 for $j = 4$. For the odd block, $\lim_{n \rightarrow \infty} P_{ij}^n = 0.2$ for all odd i, j .

Exercise 4.11: a) Assume that $\boldsymbol{\nu}^{(i)}$ is a right eigenvector and $\boldsymbol{\pi}^{(j)}$ is a left eigenvector of an $M \times M$ stochastic matrix $[P]$ where $\lambda_i \neq \lambda_j$. Show that $\boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)} = 0$. Hint: Consider two ways of finding $\boldsymbol{\pi}^{(j)}[P]\boldsymbol{\nu}^{(i)}$.

Solution: If we first premultiply $[P]$ by $\boldsymbol{\pi}^{(j)}$, we get

$$\boldsymbol{\pi}^{(j)}[P]\boldsymbol{\nu}^{(i)} = \lambda_j \boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)}.$$

If we first postmultiply $[P]$ by $\boldsymbol{\nu}^{(i)}$, we get

$$\boldsymbol{\pi}^{(j)}[P]\boldsymbol{\nu}^{(i)} = \lambda_i \boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)}.$$

Thus $(\lambda_j - \lambda_i)\boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)} = 0$. Since $(\lambda_j - \lambda_i) \neq 0$, we must have $\boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)} = 0$.

b) Assume that $[P]$ has M distinct eigenvalues. The right eigenvectors of $[P]$ then span M space (see section 5.2 of Strang, [28]), so the matrix $[U]$ with those eigenvectors as columns is non-singular. Show that U^{-1} is a matrix whose rows are the M left eigenvectors of $[P]$. Hint: use (a).

Solution: Order the right eigenvectors so that the i th column of $[U]$ is $\boldsymbol{\nu}^{(i)}$ with eigenvalue λ_i . Let $[V]$ be a matrix whose rows are linearly independent left eigenvectors of $[P]$, ordered so that the i th row is $\boldsymbol{\pi}^{(i)}$ with eigenvalue λ_i .

Then $[VU]_{ji}$ is $\boldsymbol{\pi}^{(j)}\boldsymbol{\nu}^{(i)}$. From part (a), this is 0 for $i \neq j$, from which it follows that $[VU]$ is diagonal. We next show that the diagonal elements of $[VU]$ must be nonzero. Since $\boldsymbol{\pi}^{(i)}$ is an eigenvector, it is nonzero by definition. Since $[U]$ is nonsingular, $\boldsymbol{\pi}^{(i)}[U]$ is nonzero. The elements of $\boldsymbol{\pi}^{(i)}[U]$ are $\{\boldsymbol{\pi}^{(i)}\boldsymbol{\nu}^{(j)}; 1 \leq j \leq M\}$. These elements are 0 for all $j \neq i$, so $\boldsymbol{\pi}^{(i)}\boldsymbol{\nu}^{(i)}$ must be nonzero. Thus the i th element on the diagonal on $[VU]$ is nonzero for each i .

The eigenvector $\boldsymbol{\pi}^{(i)}$ of λ_i (assuming M distinct eigenvalues) is unique within a scale factor, but that scale factor can now be chosen so that $\boldsymbol{\pi}^{(i)}\boldsymbol{\nu}^{(i)} = 1$. With this scaling, $[VU] = [I]$, which means that $[V] = [U^{-1}]$. Thus, the rows of $[U^{-1}]$ are the left eigenvectors of $[P]$, scaled so that $\boldsymbol{\pi}^{(i)}\boldsymbol{\nu}^{(i)} = 1$ for each i .

c) For any given integer i , $1 \leq i \leq M$, let $[A]$ be a diagonal matrix with a single non-zero element, $A_{ii} = a \neq 0$. Using the assumptions and results of (b), show that

$$[UAU^{-1}] = a\boldsymbol{\nu}^{(i)}\boldsymbol{\pi}^{(i)}.$$

Hint: visualize straightforward vector/matrix multiplication.

Solution: Since each column of $[A]$ is zero other than the i th, each column of $[UA]$ is zero other than the i th. Since the i th column of $[U]$ is $\boldsymbol{\nu}^{(i)}$, the i th column of $[UA]$ is $a\boldsymbol{\nu}^{(i)}$. Thus we have

$$[UAU^{-1}]_{kj} = \sum_{\ell} [UA]_{k\ell} [U^{-1}]_{\ell j} = [UA]_{ki} [U^{-1}]_{ij} = a\boldsymbol{\nu}_k^{(i)} \pi_j^{(i)}.$$

Putting these terms together, $[UAU^{-1}] = a\boldsymbol{\nu}^{(i)}\boldsymbol{\pi}^{(i)}$.

d) Verify (4.30).

Solution: Note that $[\Lambda^n]$ is a diagonal matrix with $[\Lambda^n]_{ii} = \lambda_i^n$. Let $[\Lambda_i^n]$ be a diagonal matrix with a single non-zero element, λ_i^n , in position i, i . Then $[\Lambda] = \sum_{i=1}^M [\Lambda_i^n]$. It follows that

$$[U\Lambda^n U^{-1}] = [U] \left[\sum_{i=1}^M [\Lambda_i^n] \right] [U^{-1}] = \sum_{i=1}^M [U\Lambda_i^n U^{-1}] = \sum_{i=1}^M \lambda_i^n \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}.$$

where we have used the result of (c) in each term.

Exercise 4.12: a) Let λ_k be an eigenvalue of a stochastic matrix $[P]$ and let $\boldsymbol{\pi}^{(k)}$ be a left eigenvector for λ_k . Show that for each component $\pi_j^{(k)}$ of $\boldsymbol{\pi}^{(k)}$ and each n that

$$\lambda_k^n \pi_j^{(k)} = \sum_i \pi_i^{(k)} P_{ij}^n.$$

Solution: By the definition of an eigenvector, $\boldsymbol{\pi}^{(k)}[P] = \lambda_k \boldsymbol{\pi}^{(k)}$. By iterating this, $\boldsymbol{\pi}^{(k)}[P^2] = \lambda_k \boldsymbol{\pi}^{(k)}[P] = \lambda_k^2 \boldsymbol{\pi}^{(k)}$, and in general $\boldsymbol{\pi}^{(k)}[P^n] = \lambda_k^n \boldsymbol{\pi}^{(k)}$. The desired expression is the j th component of this vector equation.

b) By taking magnitudes of each side and looking at the appropriate j , show that

$$|\lambda_k|^n \leq M.$$

Solution: Choose j to maximize $|\pi_j^{(k)}|$ for the given k . Then from (a) and the fact that $P_{ij}^n \leq 1$,

$$|\lambda_k^n| \max_j |\pi_j^{(k)}| \leq \sum_i |\pi_i^{(k)}| P_{ij}^n \leq \sum_i [\max_j |\pi_j^{(k)}|] = M [\max_j |\pi_j^{(k)}|].$$

Cancelling the common term, $|\lambda_k|^n = |\lambda_k^n| \leq M$.

c) Show that $|\lambda_k| \leq 1$.

Solution: Note that $|\lambda_k|^n$ is exponential in n . Since $|\lambda_k|$ is nonnegative and real, this is exponential and increasing in n without bound if $|\lambda_k| > 1$. Since $|\lambda_k|^n$ is bounded by M , $|\lambda_k| \leq 1$.

Exercise 4.13: Consider a finite state Markov chain with matrix $[P]$ which has κ aperiodic recurrent classes, $\mathcal{R}_1, \dots, \mathcal{R}_\kappa$ and a set $\mathcal{T}$ of transient states. For any given recurrent class $\mathcal{R}_\ell$, consider a vector $\boldsymbol{\nu}$ such that $\nu_i = 1$ for each $i \in \mathcal{R}_\ell$, $\nu_i = \lim_{n \rightarrow \infty} \Pr\{X_n \in \mathcal{R}_\ell | X_0 = i\}$ for each $i \in \mathcal{T}$, and $\nu_i = 0$ otherwise. Show that $\boldsymbol{\nu}$ is a right eigenvector of $[P]$ with eigenvalue 1. Hint: Redraw Figure 4.5 for multiple recurrent classes and first show that $\boldsymbol{\nu}$ is an eigenvector of $[P^n]$ in the limit.

Solution: A simple example of this result is treated in Exercise 4.29 and a complete derivation (extended almost trivially to periodic as well as aperiodic recurrent classes) is given in the solution to Exercise 4.18. Thus we give a more intuitive and slightly less complete derivation here.

Number the states with the transient states first, followed by each recurrent class in order. Then $[P]$ has the following block structure

$$[P] = \begin{bmatrix} [P_{\mathcal{T}}] & [P_{\mathcal{T}\mathcal{R}_1}] & \ddots & \ddots & [P_{\mathcal{T}\mathcal{R}_\kappa}] \\ 0 & [P_{\mathcal{R}_1}] & 0 & \ddots & 0 \\ \ddots & \ddots & \ddots & \ddots & \ddots \\ 0 & 0 & \ddots & 0 & [P_{\mathcal{R}_\kappa}] \end{bmatrix}.$$

The ℓ th recurrent class has an $|\mathcal{R}_\ell|$ by $|\mathcal{R}_\ell|$ transition matrix which, viewed alone, has an eigenvalue $\lambda = 1$ of multiplicity 1, a corresponding unique (within a scale factor) left

eigenvector, say $\boldsymbol{\pi}(\mathcal{R}_\ell)$, and a corresponding unique (within a scale factor) right eigenvector, $\boldsymbol{\nu}(\mathcal{R}_\ell) = (1, \dots, 1)^\top$ (see Theorem 4.4.2).

Let $\boldsymbol{\pi}^{(\ell)}$ be an M dimensional row vector whose components are equal to those of $\boldsymbol{\pi}(\mathcal{R}_\ell)$ over the states of $\mathcal{R}_\ell$ and equal to 0 otherwise. Then it can be seen by visualizing elementary row/matrix multiplication on the block structure of $[P]$ that $\boldsymbol{\pi}^{(\ell)}[P] = \boldsymbol{\pi}^{(\ell)}$. This gives us κ left eigenvectors of eigenvalue 1, one for each recurrent class $\mathcal{R}_\ell$.

These κ left eigenvectors are clearly linearly independent and span the κ dimensional space of left eigenvectors of eigenvalue 1 (see Exercise 4.18).

If there are no transient states, then a set of κ right eigenvectors can be chosen in the same way as the left eigenvectors. That is, for each ℓ , $1 \leq \ell \leq \kappa$, the components of $\boldsymbol{\nu}^{(\ell)}$ can be chosen to be 1 for each state in $\mathcal{R}_\ell$ and 0 for all other states. This doesn't satisfy the eigenvector equation if there are transient states, however. We now show, instead, that for each ℓ , $1 \leq \ell \leq \kappa$, there is a right eigenvector $\boldsymbol{\nu}^{(\ell)}$ of eigenvalue 1 that can be nonzero both on $\mathcal{R}_\ell$ and $\mathcal{T}$. such that $\nu_i^{(\ell)} = 0$ for all $i \in \mathcal{R}_m$, for each $m \neq \ell$. Finally we will show that these κ vectors are linearly independent and have the properties specified in the problem statement.

The right eigenvector equation that must be satisfied by $\boldsymbol{\nu}^{(\ell)}$ assuming that $\nu_i^{(\ell)} \neq 0$ only for $i \in \mathcal{R}_\ell \cup \mathcal{T}$ can be written out component by component, getting

$$\begin{aligned} \nu_i^{(\ell)} &= \sum_{j \in \mathcal{R}_\ell} P_{ij} \nu_j^{(\ell)} && \text{for } i \in \mathcal{R}_\ell \\ \nu_i^{(\ell)} &= \sum_{j \in \mathcal{T}} P_{ij} \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij} \nu_j^{(\ell)} && \text{for } i \in \mathcal{T}. \end{aligned}$$

The first set of equations above are simply the usual right eigenvector equations for eigenvalue 1 over the recurrent submatrix $[P_{\mathcal{R}_\ell}]$. Thus $\nu_j^{(\ell)} = 1$ for $j \in \mathcal{R}_\ell$ and this solution (over $\mathcal{R}_\ell$) is unique within a scale factor. Substituting this solution into the second set of equations, we get

$$\nu_i^{(\ell)} = \sum_{j \in \mathcal{T}} P_{ij} \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij} \quad \text{for } i \in \mathcal{T}.$$

This has a unique solution for each ℓ (see Exercise 4.18). These eigenvectors, $\{\boldsymbol{\nu}^{(\ell)}, 1 \leq \ell \leq \kappa\}$ must be linearly independent since $\nu_i^{(\ell)} = 1$ for $i \in \mathcal{R}_\ell$ and $\nu_i^{(\ell)} = 0$ for $i \in \mathcal{R}_m$, $m \neq \ell$. They then form a basis for the κ dimensional space of eigenvectors of $[P]$ of eigenvalue 1.

These eigenvectors are also eigenvectors of eigenvalue 1 for $[P^n]$ for each $n > 1$. Thus

$$\nu_i^{(\ell)} = \sum_{j \in \mathcal{T}} P_{ij}^n \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij}^n \quad \text{for } i \in \mathcal{T}.$$

Now recall that for $i, j \in \mathcal{T}$, we have $\lim_{n \rightarrow \infty} P_{ij}^n = 0$. Also $\sum_{j \in \mathcal{R}_\ell} P_{ij}^n$ is the probability that $X_n \in \mathcal{R}_\ell$ given $X_0 \in \mathcal{T}$. Since there is no exit from $\mathcal{R}_\ell$, this quantity is non-decreasing in n and bounded by 1, so it has a limit. This limit is the probability of ever going from i to $\mathcal{R}_\ell$, completing the derivation.

Exercise 4.14: Answer the following questions for the following stochastic matrix $[P]$

$$[P] = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix}.$$

a) Find $[P^n]$ in closed form for arbitrary $n > 1$.

Solution: There are several approaches here. We first give the brute-force solution of simply multiplying $[P]$ by itself multiple times (which is reasonable for a first look), and then give the elegant solution.

$$[P^2] = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/4 & 2/4 & 1/4 \\ 0 & 1/4 & 3/4 \\ 0 & 0 & 1 \end{bmatrix}.$$

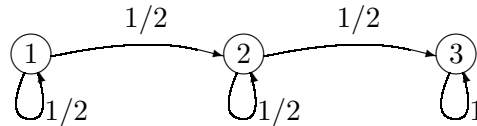
$$[P^3] = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 2/4 & 1/4 \\ 0 & 1/4 & 3/4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/8 & 3/8 & 4/8 \\ 0 & 1/8 & 7/8 \\ 0 & 0 & 1 \end{bmatrix}.$$

We could proceed to $[P^4]$, but it is natural to stop and think whether this is telling us something. The bottom row of $[P^n]$ is clearly $(0, 0, 1)$ for all n , and we can easily either reason or guess that the first two main diagonal elements are 2^{-n} . The final column is whatever is required to make the rows sum to 1. The only questionable element is P_{12}^n . We guess that it is $n2^{-n}$ and verify it by induction,

$$\begin{aligned} [P^{n+1}] &= [P][P^n] = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2^{-n} & n2^{-n} & 1 - (n+1)2^{-n} \\ 0 & 2^{-n} & 1 - 2^{-n} \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2^{-n-1} & (n+1)2^{-n-1} & 1 - (n+2)2^{-n-1} \\ 0 & 2^{-n} & 1 - 2^{-n} \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

This solution is not very satisfying, first because it is tedious, second because it required a somewhat unmotivated guess, and third because no clear rhyme or reason emerged.

The elegant solution, which can be solved with no equations, requires looking at the graph of the Markov chain,



It is now clear that $P_{11}^n = 2^{-n}$ is the probability of taking the lower loop for n successive steps starting in state 1. Similarly $P_{22}^n = 2^{-n}$ is the probability of taking the lower loop at state 2 for n successive steps.

Finally, P_{12}^n is the probability of taking the transition from state 1 to 2 exactly once out the n transitions starting in state 1 and of staying in the same state (first 1 and then 2)

for the other $n - 1$ transitions. There are n such paths, corresponding to the n possible steps at which the $1 \rightarrow 2$ transition can occur, and each path has probability 2^{-n} . Thus $P_{12}^n = n2^{-n}$, and we ‘see’ why this factor of n appears. The transitions P_{i3}^n are then chosen to make the rows sum to 1, yielding the same solution as above.

b) Find all distinct eigenvalues and the multiplicity of each distinct eigenvalue for $[P]$.

Solution: Note that $[P]$ is an upper triangular matrix, and thus $[P - \lambda I]$ is also upper triangular. Thus its determinant is the product of the terms on the diagonal, $\det[P - \lambda I] = (\frac{1}{2} - \lambda)^2(1 - \lambda)$. It follows that $\lambda = 1$ is an eigenvalue of multiplicity 1 and $\lambda = 1/2$ is an eigenvalue of multiplicity 2.

c) Find a right eigenvector for each distinct eigenvalue, and show that the eigenvalue of multiplicity 2 does not have 2 linearly independent eigenvectors.

Solution: For any Markov chain, $\mathbf{e} = (1, \dots, 1)^\top$ is a right eigenvector. For the given chain, this is unique within a scale factor, since $\lambda = 1$ has multiplicity 1. For $\boldsymbol{\nu}$ to be a right eigenvector of eigenvalue $1/2$, it must satisfy

$$\begin{aligned} \frac{1}{2}\nu_1 + \frac{1}{2}\nu_2 + 0\nu_3 &= \frac{1}{2}\nu_1 \\ 0\nu_1 + \frac{1}{2}\nu_2 + \frac{1}{2}\nu_3 &= \frac{1}{2}\nu_2 \\ \nu_3 &= \frac{1}{2}\nu_3. \end{aligned}$$

From the first equation, $\nu_2 = 0$ and from the third $\nu_3 = 0$, so $\boldsymbol{\nu} = (1, 0, 0)$ is the right eigenvector of $\lambda = 1/2$, unique within a scale factor. Thus $\lambda = 1/2$ does not have 2 linearly independent eigenvectors.

d) Use (c) to show that there is no diagonal matrix $[\Lambda]$ and no invertible matrix $[U]$ for which $[P][U] = [U][\Lambda]$.

Solution: Letting $\boldsymbol{\nu}_1, \boldsymbol{\nu}_2, \boldsymbol{\nu}_3$ be the columns of an hypothesized matrix $[U]$, we see that $[P][U] = [U][\Lambda]$ can be written out as $[P]\boldsymbol{\nu}_i = \lambda_i\boldsymbol{\nu}_i$ for $i = 1, 2, 3$. For $[U]$ to be invertible, $\boldsymbol{\nu}_1, \boldsymbol{\nu}_2, \boldsymbol{\nu}_3$ must be linearly independent eigenvectors of $[P]$. However part (c) showed that 3 such eigenvectors do not exist.

e) Rederive the result of (d) using the result of a) rather than c).

Solution: If the $[U]$ and $[\Lambda]$ of (d) exist, then $[P^n] = [U][\Lambda^n][U^{-1}]$. Then, as in (4.30), $[P^n] = \sum_{i=1}^3 \lambda_i^n \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}$ where $\boldsymbol{\nu}^{(i)}$ is the i th column of $[U]$ and $\boldsymbol{\pi}^{(i)}$ is the i th row of $[U^{-1}]$. Since $P_{12}^n = n(1/2)^n$, the factor of n means that it cannot have the form $a\lambda_1^n + b\lambda_2^n + c\lambda_3^n$ for any choice of $\lambda_1, \lambda_2, \lambda_3, a, b, c$.

Note that the argument here is quite general. If $[P^n]$ has any terms containing a polynomial in n times λ_i^n , then the eigenvectors can't span the space and a Jordan form decomposition is required.

Exercise 4.15: a) Let $[J_i]$ be a 3 by 3 block of a Jordan form, *i.e.*,

$$[J_i] = \begin{bmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ 0 & 0 & \lambda_i \end{bmatrix}.$$

Show that the n th power of $[J_i]$ is given by

$$[J_i^n] = \begin{bmatrix} \lambda_i^n & n\lambda_i^{n-1} & \binom{n}{2}\lambda_i^{n-2} \\ 0 & \lambda_i^n & n\lambda_i^{n-1} \\ 0 & 0 & \lambda_i^n \end{bmatrix}.$$

Hint: Perhaps the easiest way is to calculate $[J_i^2]$ and $[J_i^3]$ and then use iteration.

Solution: It is probably worthwhile to multiply $[J_i]$ by itself one or two times to gain familiarity with the problem, but note that the proposed formula for $[J_i^n]$ is equal to $[J_i]$ for $n = 1$, so we can use induction to show that it is correct for all larger n . That is, for each $n \geq 1$, we assume the given formula for $[J_i^n]$ is correct and demonstrate that $[J_i][J_i^n]$ is the given formula evaluated at $n + 1$. We suppress the subscript i in the evaluation.

We start with the elements on the first row, *i.e.*,

$$\begin{aligned} \sum_j J_{1j} J_{j1}^n &= \lambda \cdot \lambda^n = \lambda^{n+1} \\ \sum_j J_{1j} J_{j2}^n &= \lambda \cdot n\lambda^{n-1} + 1 \cdot \lambda^n = (n+1)\lambda^n \\ \sum_j J_{1j} J_{j3}^n &= \binom{n}{2}\lambda \cdot \lambda^{n-2} + 1 \cdot n\lambda^{n-1} = \binom{n+1}{2}\lambda^{n-1}. \end{aligned}$$

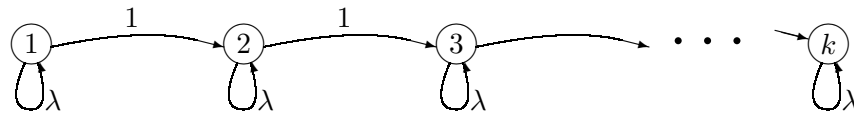
In the last equation, we used

$$\binom{n}{2} + n = \frac{n(n-1)}{2} + n = \frac{n(n+1)}{2} = \binom{n+1}{2}.$$

The elements in the second and third row can be handled the same way, although with slightly less work. The solution in (b) provides a more elegant and more general solution.

b) Generalize (a) to a $k \times k$ block of a Jordan form J . Note that the n th power of an entire Jordan form is composed of these blocks along the diagonal of the matrix.

Solution: This is rather difficult unless looked at in just the right way, but the right way is quite instructive and often useful. Thus we hope you didn't spin your wheels too much, and hope you will learn from the solution. The idea comes from the elegant way to solve (a) of Exercise 4.14, namely taking a graphical approach rather than a matrix multiplication approach. Forgetting for the moment that J is not a stochastic matrix, we can draw its graph, *i.e.*, the representation of the non-zero elements of the matrix, as



Each edge, say from $i \rightarrow j$ in the graph, represents a non-zero element of J and is labelled by the value of J_{ij} . If we look at $[J^2]$, then $J_{ij}^2 = \sum_k J_{ik} J_{kj}$ is the sum over the walks of length 2 in the graph of the 'measure' of that walk, where the measure of a walk is the product of the labels on the edges of that walk. Similarly, we can see that J_{ij}^n is the sum over walks of length n of the product of the labels for that walk. In short, J_{ij}^n can be found

from the graph in the same way as for stochastic matrices; we simply ignore the fact that the outgoing edges from each node do not sum to 1.

We can now see immediately how to evaluate these elements. For J_{ij}^n , we must look at all the walks of length n that go from i to j . For $i > j$, the number is 0, so we assume $j \geq i$. Each such walk must include $j - i$ rightward transitions and thus must include $n - j + i$ self loops. Thus the measure of each such walk is λ^{n-j+i} . The number of such walks is the number of ways in which the $j - i$ rightward transition times can be selected out of the n transitions altogether, *i.e.*, $\binom{n}{j-i}$. Thus, the general formula is

$$J_{ij}^n = \binom{n}{j-i} \lambda^{j-i} \quad \text{for } j \geq i.$$

c) Let $[P]$ be a stochastic matrix represented by a Jordan form $[J]$ as $[P] = U[J]U^{-1}$ and consider $[U^{-1}][P][U] = [J]$. Show that any repeated eigenvalue λ of $[P]$ that is represented by a Jordan block of 2 by 2 or more must satisfy $|\lambda| < 1$. Hint: Upper bound the element magnitudes of $[U^{-1}][P^n][U]$ by taking the magnitude of the elements of $[U]$ and $[U^{-1}]$ and by upper bounding each element of $[P^n]$ by 1.

Solution: Each element of the matrix $[U^{-1}][P^n][U]$ is a sum of products of terms and the magnitude of that sum is upper bounded by the sum of products of the magnitudes of the terms. Representing the matrix whose elements are the magnitudes of a matrix $[U]$ as $[|U|]$, and recognizing that $[P^n] = [|P^n|]$ since its elements are nonnegative, we have

$$[|J^n|] \leq [|U^{-1}|][|P^n|][|U|] \leq [|U^{-1}|][|M|][|U|],$$

where $[M]$ is a matrix whose elements are all equal to 1.

The matrix on the right is independent of n , so each of its elements is simply a finite number. If $[P]$ has an eigenvalue λ of magnitude 1 whose multiplicity exceeds its number of linearly independent eigenvectors, then $[J^n]$ contains an element $n\lambda^{n-1}$ of magnitude n , and for large enough n , this is larger than the finite number above, yielding a contradiction. Thus, for any stochastic matrix, any repeated eigenvalue with a defective number of linearly independent eigenvectors has a magnitude strictly less than 1.

d) Let λ_s be the eigenvalue of largest magnitude less than 1. Assume that the Jordan blocks for λ_s are at most of size k . Show that each ergodic class of $[P]$ converges at least as fast as $n^k \lambda_s^n$.

Solution: What we want to show is that for each element i, j of the matrix, $|[P^n] - e\pi|_{ij} \leq bn^k |\lambda_s^n|$ for some sufficiently large constant b depending on $[P]$ but not n .

The states of an ergodic class have no transitions out, and for questions of convergence within the class we can ignore transitions in. Thus we will get rid of some excess notation by simply assuming an ergodic Markov chain. Thus there is one eigenvalue equal to 1 and it has unit multiplicity. All other eigenvalues are strictly less than 1. The largest such in magnitude is denoted as λ_s . Assume that the eigenvalues are arranged in $[J]$ in decreasing magnitude, so that $J_{11} = 1$.

Since $[J]$ is in general a Jordan form, its upper triangular block structure is illustrated

below. The corresponding structure for $[J^n]$, using part (b), is also shown.

$$[J] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_s & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_s & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_s & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_4 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_4 \end{bmatrix} \quad [J^n] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_s^n & n\lambda_s^{n-1} & \binom{n}{2}\lambda_s^{n-2} & 0 & 0 & 0 \\ 0 & 0 & \lambda_s^n & n\lambda_s^{n-1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_s^n & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_3^n & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_4^n & \lambda_4^{n-1} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_4^n \end{bmatrix}.$$

For all n , we have $[P^n] = [U][J^n][U^{-1}]$. Note that $\lim_{n \rightarrow \infty} [J^n]$ is a matrix with $J_{ij}^n \rightarrow 0$ except for $i = j = 1$ where $J_{11}^n = 1$ for all n . Let $\mathbf{u}_1$ be the first column of $[U]$ and let $\mathbf{v}_1$ be the first row of $[U^{-1}]$. Then since only the top left element of $[J^n]$ is nonzero in the limit, we can see that $\lim_{n \rightarrow \infty} [UJ^nU^{-1}] = \mathbf{u}_1\mathbf{v}_1$. We have seen that $\lim_{n \rightarrow \infty} [P^n] = \mathbf{e}\boldsymbol{\pi}$. Thus it follows that $\mathbf{u}_1 = \mathbf{e}$ and $\mathbf{v}_1 = \boldsymbol{\pi}$.

It follows from this that $[P^n] - \mathbf{e}\boldsymbol{\pi} = [U][\tilde{J}^n][U^{-1}]$ where $[\tilde{J}^n]$ is the same as $[J^n]$ except that the eigenvalue 1 has been replaced by 0. This means, however, that the magnitude of each element of $[\tilde{J}^n]$ is upper bounded by $n^k|\lambda_s^{n-k}|$. It follows that the magnitude of each element in $[U][\tilde{J}^n][U^{-1}]$ is upper bounded by $bn^k|\lambda_s|^n$ for large enough b (the value of b takes account of $[U]$, $[U^{-1}]$, and λ_s^{-k}).

Exercise 4.16: a) Let λ be an eigenvalue of a matrix $[A]$, and let $\boldsymbol{\nu}$ and $\boldsymbol{\pi}$ be right and left eigenvectors respectively of λ , normalized so that $\boldsymbol{\pi}\boldsymbol{\nu} = 1$. Show that

$$[[A] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}]^2 = [A^2] - \lambda^2\boldsymbol{\nu}\boldsymbol{\pi}.$$

Solution: We simply multiply out the original square,

$$\begin{aligned} [[A] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}]^2 &= [A^2] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}[A] - \lambda[A]\boldsymbol{\nu}\boldsymbol{\pi} + \lambda^2\boldsymbol{\nu}\boldsymbol{\pi}\boldsymbol{\nu}\boldsymbol{\pi} \\ &= [A^2] - \lambda^2\boldsymbol{\nu}\boldsymbol{\pi} - \lambda^2\boldsymbol{\nu}\boldsymbol{\pi} + \lambda^2\boldsymbol{\nu}\boldsymbol{\pi} = [A^2] - \lambda^2\boldsymbol{\nu}\boldsymbol{\pi}. \end{aligned}$$

b) Show that $[[A^n] - \lambda^n\boldsymbol{\nu}\boldsymbol{\pi}][[A] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}] = [A^{n+1}] - \lambda^{n+1}\boldsymbol{\nu}\boldsymbol{\pi}$.

Solution: This is essentially the same as (a)

$$\begin{aligned} [[A^n] - \lambda^n\boldsymbol{\nu}\boldsymbol{\pi}][[A] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}] &= [A^{n+1}] - \lambda^n\boldsymbol{\nu}\boldsymbol{\pi}[A] - \lambda[A^n]\boldsymbol{\nu}\boldsymbol{\pi} + \lambda^{n+1}\boldsymbol{\nu}\boldsymbol{\pi}\boldsymbol{\nu}\boldsymbol{\pi} \\ &= [A^{n+1}] - \lambda^{n+1}\boldsymbol{\nu}\boldsymbol{\pi}. \end{aligned}$$

c) Use induction to show that $[[A] - \lambda\boldsymbol{\nu}\boldsymbol{\pi}]^n = [A^n] - \lambda^n\boldsymbol{\nu}\boldsymbol{\pi}$.

Solution: (a) gives the base of the induction and (b) gives the inductive step.

Exercise 4.17: Let $[P]$ be the transition matrix for an aperiodic Markov unichain with the states numbered as in Figure 4.5.

a) Show that $[P^n]$ can be partitioned as

$$[P^n] = \begin{bmatrix} [P_T^n] & [P_x^n] \\ 0 & [P_R^n] \end{bmatrix}.$$

That is, the blocks on the diagonal are simply products of the corresponding blocks of $[P]$, and the upper right block is whatever it turns out to be.

Solution: If we imagine multiplying $[P][P]$ by straightforward block multiplication, we get

$$[P^2] = \begin{bmatrix} [P_{\mathcal{T}}P_{\mathcal{T}}] & [P_{\mathcal{T}}P_{\mathcal{T}\mathcal{R}} + P_{\mathcal{T}\mathcal{R}}P_{\mathcal{R}}] \\ [0] & [P_{\mathcal{R}}P_{\mathcal{R}}] \end{bmatrix}.$$

In the same way, using induction, and not tracking the upper right term

$$[P^n] = \begin{bmatrix} [P_{\mathcal{T}}^n] & [P_{\mathcal{T}}^n] \\ [0] & [P_{\mathcal{R}}^n] \end{bmatrix}.$$

Another approach is to look at P_{ij}^n . If $i, j \in \mathcal{T}$, then any n step walk from i to j stays in $\mathcal{T}$. Similarly, if $i, j \in \mathcal{R}$, any walk from i to j stays in $\mathcal{R}$. Finally, there are no walks from $\mathcal{R}$ to $\mathcal{T}$, giving rise to the matrix $[0]$ in the lower left.

b) Let q_i be the probability that the chain will be in a recurrent state after t transitions, starting from state i , i.e., $q_i = \sum_{j \in \mathcal{R}} P_{ij}^t$. Show that $q_i > 0$ for all transient i .

Solution: As specified in Figure 4.5, t is the number of transient states. As explained in Section 4.3.4, each transient state has a path of at most t steps to a recurrent state. All extensions of that path remain in recurrent states and thus if the original path has fewer than t steps, any extension of that path to a walk with t steps still ends at a recurrent state and has positive probability. Thus q_i is positive.

c) Let q be the minimum q_i over all transient i and show that $P_{ij}^{nt} \leq (1 - q)^n$ for all transient i, j (i.e., show that $[P_{\mathcal{T}}^n]$ approaches the all zero matrix $[0]$ with increasing n).

Solution: Since $\sum_{k \in \mathcal{R}} P_{ik}^t \geq q$, it follows that $\sum_{j \in \mathcal{T}} P_{ij}^t \leq 1 - q$ for all $i \in \mathcal{T}$. Using the Chapman-Kolmogorov equations for $i \in \mathcal{T}$,

$$\sum_{j \in \mathcal{T}} P_{ij}^{2t} = \sum_{k, j \in \mathcal{T}} P_{ik}^t P_{kj}^t \leq \sum_{k \in \mathcal{T}} P_{ik}^t (1 - q) \leq (1 - q)^2.$$

Iterating this equation, we get $\sum_{j \in \mathcal{T}} P_{ij}^{nt} \leq (1 - q)^n$ and thus $P_{ij}^{nt} \leq (1 - q)^n$ for all $j \in \mathcal{T}$.

d) Let $\boldsymbol{\pi} = (\boldsymbol{\pi}_{\mathcal{T}}, \boldsymbol{\pi}_{\mathcal{R}})$ be a left eigenvector of $[P]$ of eigenvalue 1. Show that $\boldsymbol{\pi}_{\mathcal{T}} = \mathbf{0}$ and show that $\boldsymbol{\pi}_{\mathcal{R}}$ must be positive and be a left eigenvector of $[P_{\mathcal{R}}]$. Thus show that $\boldsymbol{\pi}$ exists and is unique (within a scale factor).

Solution: Note that $\boldsymbol{\pi}$ is also an eigenvector of $[P^n]$ for all n , so that $\boldsymbol{\pi}_{\mathcal{T}}[P_{\mathcal{T}}]^n = \boldsymbol{\pi}_{\mathcal{T}}$. Since $\lim_{n \rightarrow \infty} [P_{\mathcal{T}}^n] = 0$, this shows that $\boldsymbol{\pi}_{\mathcal{T}} = 0$. We then have $\boldsymbol{\pi}_{\mathcal{R}} = \boldsymbol{\pi}_{\mathcal{R}}[P_{\mathcal{R}}]$, which has a unique solution.

e) Show that $\mathbf{e}$ is the unique right eigenvector of $[P]$ of eigenvalue 1 (within a scale factor).

Solution: $\mathbf{e}$ must be a right eigenvector of $[P]$ since $[P]$ is stochastic (i.e., its rows all sum to 1, which is what $[P]\mathbf{e} = \mathbf{e}$ means). Since $[P]$ is a unichain, it has a single eigenvalue equal to 1 and therefore a unique right eigenvector within a scale factor.

Exercise 4.18: Generalize Exercise 4.17 to the case of a Markov chain $[P]$ with κ recurrent classes and one or more transient classes. In particular,

a) Show that $[P]$ has exactly κ linearly independent left eigenvectors, $\boldsymbol{\pi}^{(1)}, \boldsymbol{\pi}^{(2)}, \dots, \boldsymbol{\pi}^{(\kappa)}$ of eigenvalue 1, and that the m th can be taken as a probability vector that is positive on the m th recurrent class and zero elsewhere.

Solution: See Exercise 4.28 for a very simple example of these results, and see the solution to Exercise 4.13 for a more intuitive but less precise explanation.

Number the states so that the transient states come first, then the states in recurrent class 1, and so forth up to the recurrent states in class κ . Then $[P]$ has the following block structure

$$[P] = \begin{bmatrix} [P_{\mathcal{T}}] & [P_{\mathcal{T}\mathcal{R}_1}] & \ddots & \ddots & [P_{\mathcal{T}\mathcal{R}_\kappa}] \\ 0 & [P_{\mathcal{R}_1}] & 0 & \ddots & 0 \\ \ddots & \ddots & \ddots & \ddots & \ddots \\ 0 & 0 & \ddots & 0 & [P_{\mathcal{R}_\kappa}] \end{bmatrix}.$$

The ℓ th recurrent class has an $|\mathcal{R}_\ell|$ by $|\mathcal{R}_\ell|$ transition matrix which, viewed alone, has an eigenvalue $\lambda = 1$ of multiplicity 1, a corresponding unique (within a scale factor) left eigenvector, say $\boldsymbol{\pi}(\mathcal{R}_\ell)$, and a corresponding unique (within a scale factor) right eigenvector, $\boldsymbol{\nu}(\mathcal{R}_\ell) = (1, \dots, 1)^\top$ (see Theorem 4.4.2).

Let $\boldsymbol{\pi}^{(\ell)}$ be an M dimensional row vector whose components are equal to those of $\boldsymbol{\pi}(\mathcal{R}_\ell)$ over the states of $\mathcal{R}_\ell$ and equal to 0 elsewhere. Then it can be seen by visualizing elementary row/matrix multiplication on the block structure of $[P]$ that $\boldsymbol{\pi}^{(\ell)}[P] = \boldsymbol{\pi}^{(\ell)}$. This gives us κ left eigenvectors of eigenvalue 1, one for each recurrent class $\mathcal{R}_\ell$.

If we look at the determinant equation, $\det[P - \lambda I]$, the upper triangular block structure of $[P]$ implies that $\det[P - \lambda I]$ breaks into the product of $\kappa + 1$ determinants, one for the transient states and one for each recurrent class. Thus we can identify the eigenvalues with the classes, $|\mathcal{T}|$ eigenvalues for $\mathcal{T}$ and, for each ℓ , $|\mathcal{R}_\ell|$ eigenvalues for $\mathcal{R}_\ell$. Each recurrent class has one eigenvalue equal to 1 and the transient set has no eigenvalues equal to 1. To verify this latter statement, assume the contrary, *i.e.*, assume there is a row vector $\boldsymbol{\pi}^{(T)}$ that is nonzero only on the components of $\mathcal{T}$ and which satisfies $\boldsymbol{\pi}^{(T)} = \boldsymbol{\pi}^{(T)}[P]$. Then we must also have $\boldsymbol{\pi}^{(T)} = \boldsymbol{\pi}^{(T)}[P^n]$ for all $n > 1$. However, we have seen that $P_{ij}^n \rightarrow 0$ as $n \rightarrow \infty$ for all $i, j \in \mathcal{T}$. This yields the desired contradiction, since there can then be no nonzero solution to $\boldsymbol{\pi}^{(T)} = \boldsymbol{\pi}^{(T)}[P^n]$ for large n .

We have thus shown that $\lambda = 1$ has multiplicity κ for matrix $[P]$ and we have identified all the left eigenvectors of eigenvalue 1 (or more strictly, we have identified a basis for the vector space of left eigenvectors of eigenvalue 1).

b) Show that $[P]$ has exactly κ linearly independent right eigenvectors, $\boldsymbol{\nu}^{(1)}, \boldsymbol{\nu}^{(2)}, \dots, \boldsymbol{\nu}^{(\kappa)}$ of eigenvalue 1, and that the m th can be taken as a vector with $\nu_i^{(m)}$ equal to the probability that recurrent class m will ever be entered starting from state i .

If there are no transient states, then a set of κ right eigenvectors can be chosen in the same way as the left eigenvectors. That is, for each ℓ , $1 \leq \ell \leq \kappa$, the components of $\boldsymbol{\nu}^{(\ell)}$ can be chosen to be 1 for each state in $\mathcal{R}_\ell$ and 0 for all other states. This doesn't satisfy the eigenvector equation if there are transient states, however. We now show, instead, that for each ℓ , $1 \leq \ell \leq \kappa$, there is a right eigenvector $\boldsymbol{\nu}^{(\ell)}$ of eigenvalue 1 such that $\nu_i^{(\ell)} = 0$ for all $i \in \mathcal{R}_m$, for each $m \neq \ell$. Finally we will show that these κ vectors are linearly independent and have the properties specified in the problem statement.

The right eigenvector equation that must be satisfied by $\boldsymbol{\nu}^{(\ell)}$ with the conditions above can be written out component by component, getting

$$\begin{aligned}\nu_i^{(\ell)} &= \sum_{j \in \mathcal{R}_\ell} P_{ij} \nu_j^{(\ell)} && \text{for } i \in \mathcal{R}_\ell \\ \nu_i^{(\ell)} &= \sum_{j \in \mathcal{T}} P_{ij} \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij} \nu_j^{(\ell)} && \text{for } i \in \mathcal{T}.\end{aligned}$$

The first set of equations above are simply the usual right eigenvector equations for eigenvalue 1 over the recurrent submatrix $[P_{\mathcal{R}_\ell}]$. Thus $\nu_j^{(\ell)} = 1$ for $j \in \mathcal{R}_\ell$ and this solution (over $\mathcal{R}_\ell$) is unique within a scale factor. Substituting this solution into the second set of equations, we get

$$\nu_i^{(\ell)} = \sum_{j \in \mathcal{T}} P_{ij} \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij} \quad \text{for } i \in \mathcal{T}. \quad (\text{A.55})$$

We can view this as a vector/matrix equation by letting $\boldsymbol{\nu}_{\mathcal{T}}^{(\ell)}$ be the vector $\boldsymbol{\nu}^{(\ell)}$ restricted to the transient states. Thus, $\boldsymbol{\nu}_{\mathcal{T}}^{(\ell)} = [P_{\mathcal{T}}] \boldsymbol{\nu}_{\mathcal{T}}^{(\ell)} + \mathbf{x}$ where $x_i = \sum_{j \in \mathcal{R}_\ell} P_{ij}$. Since $\mathcal{T}$ is a set of transient states, $[P_{\mathcal{T}}]$ does not have 1 as an eigenvalue, so $[P_{\mathcal{T}} - I]$ is non-singular. Thus this set of equations has a unique solution (we have already specified the scale factor on $\boldsymbol{\nu}^{(\ell)}$ by setting $\nu_i^{(\ell)} = 1$ for $i \in \mathcal{R}_\ell$).

We have now established that for each ℓ , $1 \leq \ell \leq \kappa$, $[P]$ has a right eigenvector of eigenvalue 1 that is nonzero only on $\mathcal{R}_\ell$ and $\mathcal{T}$. Since the components of $\boldsymbol{\nu}^{(\ell)}$ are 1 on $\mathcal{R}_\ell$ and 0 on all other recurrent classes, these κ vectors are clearly linearly independent. Since the eigenvalue 1 has multiplicity κ , these eigenvectors form a basis for the eigenvectors of eigenvalue 1.

It remains only to show that $\nu_i^{(\ell)} = \lim_{n \rightarrow \infty} \Pr\{X_n \in \mathcal{R}_\ell \mid X_0 = i\}$. We have seen many times that if $\boldsymbol{\nu}^{(\ell)}$ is a right eigenvector of $[P]$ of eigenvalue 1, then it is also a right eigenvector of $[P^n]$ of eigenvalue 1. Since $[P^n]$ has the same block structure as $[P]$, we can repeat the argument leading to (A.55). For each $n > 1$, then, we have

$$\nu_i^{(\ell)} = \sum_{j \in \mathcal{T}} P_{ij}^n \nu_j^{(\ell)} + \sum_{j \in \mathcal{R}_\ell} P_{ij}^n \quad \text{for } i \in \mathcal{T}.$$

In the limit $n \rightarrow \infty$, each P_{ij}^n for $i, j \in \mathcal{T}$ goes to 0. Also $\sum_{j \in \mathcal{R}_\ell} P_{ij}^n$ is the probability that $X_n \in \mathcal{R}_\ell$ given $X_0 = i$. Since there is no exit from $\mathcal{R}_\ell$, this quantity is non-decreasing in n and bounded by 1, so it has a limit. This limit is the probability of ever going from i to $\mathcal{R}_\ell$, completing the derivation.

c) Show that

$$\lim_{n \rightarrow \infty} [P^n] = \sum_m \boldsymbol{\nu}^{(m)} \boldsymbol{\pi}^{(m)}.$$

Solution: Unfortunately, this result is not true in general. The additional restriction that each recurrent class is ergodic must be added. In this case, P_{ij}^n for $i \in \mathcal{T}$ and $j \in$

$\mathcal{R}_m$ converges to the probability that class $\mathcal{R}_m$ is entered from i times the steady-state probability of j within class $\mathcal{R}_m$.

Exercise 4.19: Suppose a recurrent Markov chain has period d and let $\mathcal{S}_m$, $0 \leq m \leq d-1$, be the m th subset in the sense of Theorem 4.2.9. Assume the states are numbered so that the first s_0 states are the states of $\mathcal{S}_0$, the next s_1 are those of $\mathcal{S}_1$, and so forth. Then the matrix $[P]$ for the chain has the block form given by

$$[P] = \begin{bmatrix} 0 & [P_0] & \ddots & \ddots & 0 \\ 0 & 0 & [P_1] & \ddots & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots \\ 0 & 0 & \ddots & \ddots & [P_{d-2}] \\ [P_{d-1}] & 0 & \ddots & \ddots & 0 \end{bmatrix},$$

where $[P_m]$ has dimension $s_m \times s_{m+1}$ for $0 \leq m \leq d-1$, where the index $(d-1)+1$ is interpreted as 0 throughout (*i.e.*, mod d arithmetic is used on indices). In what follows it is often more convenient to express $[P_m]$ as an $M \times M$ matrix $[P'_m]$ whose entries are 0 except for the rows of $\mathcal{S}_m$ and the columns of $\mathcal{S}_{m+1}$, where the entries are equal to those of $[P_m]$. In this view, $[P] = \sum_{m=0}^{d-1} [P'_m]$.

a) Show that $[P^d]$ has the form

$$[P^d] = \begin{bmatrix} [Q_0] & 0 & \ddots & 0 \\ 0 & [Q_1] & \ddots & \ddots \\ 0 & 0 & \ddots & [Q_{d-1}] \end{bmatrix},$$

where $[Q_m] = [P_m][P_{m+1}] \cdots [P_{d-1}][P_0] \cdots [P_{m-1}]$. Expressing $[Q_m]$ as an $M \times M$ matrix $[Q'_m]$ whose entries are 0 except for the rows and columns of $\mathcal{S}_m$ where the entries are equal to those of $[Q_m]$, this becomes $[P^d] = \sum_{m=0}^{d-1} [Q'_m]$.

Solution: As explained in Section 4.3.5, P_{ij}^d must be 0 for (i, j) not in the same subset. Since the subsets are arranged as contiguous sets of state numbers, $[Q_j]$ is the d th order transition matrix of the states in class $\mathcal{S}_j$. To see this more clearly and to understand why $[Q_m] = [P_m][P_{m+1}] \cdots [P_{d-1}][P_0] \cdots [P_{m-1}]$, note that $[P^2]$ is given by

$$[P^2] = \begin{bmatrix} 0 & 0 & [P_0 P_1] & \ddots & \ddots \\ 0 & 0 & 0 & [P_1 P_2] & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots \\ [P_{d-2} P_{d-1}] & 0 & \ddots & \ddots & \ddots \\ 0 & [P_{d-1} P_0] & \ddots & \ddots & \ddots \end{bmatrix}.$$

Similarly, $[P^m]$ for $m < d$ has a block structure where states in subset i can go only to states in subset $(i+m) \bmod d$ and go there with probabilities determined by going first to $i+1$, then $i+2$ up to $i+m$, all mod d .

b) Show that $[Q_m]$ is the matrix of an ergodic Markov chain and let its left eigenvector of eigenvalue 1 (scaled to be a probability vector) be $\hat{\pi}_m$ and let $\hat{\nu}_m$ be the corresponding right eigenvector scaled so that $\hat{\pi}_m \hat{\nu}_m = 1$. Let $\hat{\pi}'_m$ and $\hat{\nu}'_m$ be the corresponding M -vectors. Show that $\lim_{n \rightarrow \infty} [P^{nd}] = \sum_{m=0}^{d-1} \hat{\nu}'_m \hat{\pi}'_m$.

Solution: Note that $[Q_m]$ for $0 \leq m \leq d-1$ must be recurrent. This follows since $[P]$ is recurrent (so that each pair of states in $\mathcal{S}_m$ communicates) and is periodic (so that each pair in $\mathcal{S}_m$ communicates only over periods of d). Also $[Q_m]$ is aperiodic since otherwise P_{ii}^k for $i \in \mathcal{S}_m$ would be nonzero only for $k = \ell d$ for some integer $\ell > 1$, which is contrary to the given fact that the chain, and thus i , is periodic with period d . Thus each $[Q_m]$ is the matrix of an ergodic Markov chain. It follows that each $[Q_m]$ has a unique left eigenvector $\hat{\pi}_m$ whose components sum to 1, i.e., $\hat{\pi}_m \mathbf{e} = 1$ where $\mathbf{e}$ is the unit vector of s_m dimensions (i.e., $\hat{\nu}_m = \mathbf{e}$). Thus $\lim_{n \rightarrow \infty} [Q_m^n] = \hat{\nu}_m \hat{\pi}_m$. If we now consider $[Q'_m]$ as the M by M matrix that is equal to $[Q_m]$ on $i, j \in \mathcal{S}_m$ and zero elsewhere, we see that $\lim_{n \rightarrow \infty} [Q'_m]^n = \nu'_m \pi'_m$. Thus $\lim_{n \rightarrow \infty} [P^{nd}] = \sum_{m=0}^{d-1} \hat{\nu}'_m \hat{\pi}'_m$.

c) Show that $\hat{\pi}'_m[P'_m] = \hat{\pi}'_{m+1}$. Note that $\hat{\pi}'_m$ is an M -tuple that is non-zero only on the components of $\mathcal{S}_m$.

Solution: $[Q_m] = [P_m P_{m+1} \cdots P_{d-1} P_0 \cdots P_{m-1}]$. We saw in (b) that each Q_m has a unique (within scale factor) left eigenvector $\hat{\pi}_m$ of eigenvalue 1. Thus $\hat{\pi}_m[Q_m] = \hat{\pi}_m$ and

$$\hat{\pi}_m[P_m P_{m+1} \cdots P_{d-1} P_0 \cdots P_{m-1}] = \hat{\pi}_m.$$

For $m < d-1$, we can postmultiply both sides of this equation by $[P_m]$ to get

$$\hat{\pi}_m[P_m][Q_{m+1}] = \hat{\pi}_m[P_m].$$

This means that $\hat{\pi}_m[P_m]$ is a left eigenvector (with eigenvalue 1) of $[Q_{m+1}]$. Normalizing each π_m to be a probability vector, the components of $\hat{\pi}_m[P_m]$ sum to 1, so $\pi_m[P_m] = \pi_{m+1}$. For $m = d-1$, the same argument shows that $\hat{\pi}_{d-1}[P_{d-1}]$ is a left eigenvector of $[Q_0]$. In what follows, we use mod d arithmetic on indices without special mention.

As in parts (a) and (b), we can express these results in terms of the M by M matrices. Thus, $[Q'_m] = [P'_m P'_{m+1} \cdots P'_{d-1} P'_0 \cdots P'_{m-1}]$ and it follows that $\hat{\pi}'_m[P'_m] = \hat{\pi}'_{m+1}$.

d) Let $\phi = \frac{2\pi\sqrt{-1}}{d}$ and let $\pi^{(k)} = \sum_{m=0}^{d-1} \hat{\pi}'_m e^{mk\phi}$. Show that $\pi^{(k)}$ is a left eigenvector of $[P]$ of eigenvalue $e^{-\phi k}$.

Solution: Note that $\{k\phi; k = 0, 1, \dots, d-1\}$ is the set of d equi-spaced points around the complex unit circle. We want to show that these points are eigenvalues of $[P]$ and that $\pi^{(k)}$ is a corresponding left eigenvector of eigenvalue $k\phi$. We have

$$\begin{aligned} \pi^{(k)}[P] &= \sum_{m=0}^{d-1} e^{mk\phi} \pi'_m[P] \\ &= \sum_{m=0}^{d-1} e^{mk\phi} \pi'_m[P'_m] \\ &= \sum_{m=0}^{d-1} e^{mk\phi} \pi'_{m+1} \\ &= e^{-k\phi} \sum_{m=0}^{d-1} e^{(m+1)k\phi} \pi'_{m+1} \\ &= e^{-k\phi} \sum_{m=1}^d e^{mk\phi} \pi'_m = e^{-k\phi} \pi^{(k)}. \end{aligned}$$

In the second step, we used the block structure of $[P]$ and in the third step, we used (c). In the fourth step, we multiplied and divided by $e^{-k\phi}$, and in the fifth, we substituted the index m for $m+1$ and used the mod d summation to recognize that summing from 1 to d is equivalent to summing from 0 to $d-1$. Finally we used the definition of $\pi^{(k)}$.

Exercise 4.20: (continuation of Exercise 4.19) **a)** Show that, with the eigenvectors defined in Exercise 4.19,

$$\lim_{n \rightarrow \infty} [P^{nd}][P] = \sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i+1)}, \quad (\text{A.56})$$

where, as before, $(d-1)+1$ is taken to be 0.

Solution: The quantities $\boldsymbol{\nu}'_i$ and $\boldsymbol{\pi}'_i$ defined in Exercise 4.19 are being referred to here as $\boldsymbol{\nu}^{(i)}$ and $\boldsymbol{\pi}^{(i)}$. We showed there that $\lim_{n \rightarrow \infty} [P^{nd}] = \sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}$ and $\boldsymbol{\pi}^{(i)}[P] = \boldsymbol{\pi}^{(i+1)}$. Thus (again using mod d indices)

$$\lim_{n \rightarrow \infty} [P^{nd}][P] = \sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}[P] = \sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i+1)}.$$

b) Show that, for $1 \leq j < d$,

$$\lim_{n \rightarrow \infty} [P^{nd}][P^j] = \sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i+j)}. \quad (\text{A.57})$$

Solution: This follows by successively postmultiplying (A.56) by $[P]$ $j-1$ times.

c) Show that

$$\lim_{n \rightarrow \infty} [P^{nd}] \left\{ I + [P] + \cdots + [P^{d-1}] \right\} = \left(\sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} \right) \left(\sum_{j=0}^{d-1} \boldsymbol{\pi}^{(i+j)} \right). \quad (\text{A.58})$$

Solution: This results immediately by summing (A.57) over j .

d) Show that

$$\lim_{n \rightarrow \infty} \frac{1}{d} \left([P^n] + [P^{n+1}] + \cdots + [P^{n+d-1}] \right) = \mathbf{e} \boldsymbol{\pi}, \quad (\text{A.59})$$

where $\boldsymbol{\pi}$ is the steady-state probability vector for $[P]$. Hint: Show that $\mathbf{e} = \sum_m \boldsymbol{\nu}^{(m)}$ and $\boldsymbol{\pi} = (1/d) \sum_m \boldsymbol{\pi}^{(m)}$.

Solution: Because of the mod d indices, the last term in (A.58) can be replaced by $\sum_{j=0}^{d-1} \boldsymbol{\pi}^{(j)}$. Now since $\boldsymbol{\pi}^{(j)}[P] = \boldsymbol{\pi}^{(j+1)}$, we see that $\sum_{j=0}^{d-1} \boldsymbol{\pi}^{(j)}[P] = \sum_{j=0}^{d-1} \boldsymbol{\pi}^{(j)}$, again because of the mod d indices. Thus $\sum_{j=0}^{d-1} \boldsymbol{\pi}^{(j)}$ is a left eigenvector of $[P]$. However, each $\boldsymbol{\pi}^{(j)}$ was chosen as a steady-state vector for a periodic subset of states and thus each sums to 1. Thus $\boldsymbol{\pi} = (1/d) \sum_{j=0}^{d-1} \boldsymbol{\pi}^{(j)}$ is a left eigenvector of $[P]$ for which the components sum to 1. This result also follows from Exercise 4.19(d) by setting $k = 0$.

We also note that $\boldsymbol{\nu}^{(i)}$ is a vector whose components are equal to 1 over the i th periodic subset of states and 0 elsewhere. Thus $\sum_{i=0}^{d-1} \boldsymbol{\nu}^{(i)} = \mathbf{e}$. We have thus shown that

$$\frac{1}{d} \lim_{n \rightarrow \infty} [P^{nd}] \left\{ [I] + [P] + \cdots + [P^{d-1}] \right\} = \mathbf{e} \boldsymbol{\pi}.$$

This equation remains valid if we postmultiply it by $[P]$ or $[P^m]$ for any m , and thus we get (A.59).

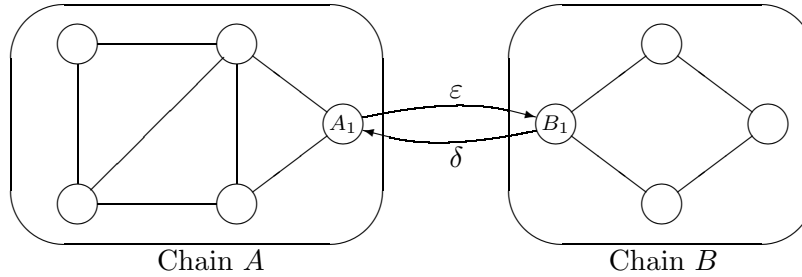
e) Show that the above result is also valid for periodic unichains.

Solution: Each transient state can join each periodic subset with some arbitrary probability, but this makes no difference as far as (A.59) is concerned since these probabilities are summed over the periodic subsets. We omit the details.

Exercise 4.21: Suppose A and B are each ergodic Markov chains with transition probabilities $\{P_{A_i, A_j}\}$ and $\{P_{B_i, B_j}\}$ respectively. Denote the steady-state probabilities of A and B by $\{\pi_{A_i}\}$ and $\{\pi_{B_i}\}$ respectively. The chains are now connected and modified as shown below. In particular, states A_1 and B_1 are now connected and the new transition probabilities P' for the combined chain are given by

$$\begin{aligned} P'_{A_1, B_1} &= \varepsilon, & P'_{A_1, A_j} &= (1 - \varepsilon)P_{A_1, A_j} && \text{for all } A_j \\ P'_{B_1, A_1} &= \delta, & P'_{B_1, B_j} &= (1 - \delta)P_{B_1, B_j} && \text{for all } B_j. \end{aligned}$$

All other transition probabilities remain the same. Think intuitively of ε and δ as being small, but do not make any approximations in what follows. Give your answers to the following questions as functions of ε , δ , $\{\pi_{A_i}\}$ and $\{\pi_{B_i}\}$.



a) Assume that $\varepsilon > 0$, $\delta = 0$ (i.e., that A is a set of transient states in the combined chain). Starting in state A_1 , find the conditional expected time to return to A_1 given that the first transition is to some state in chain A .

Solution: Conditional on the first transition from state A_1 being to a state $A_i \neq A_1$, these conditional transition probabilities are the same as the original transition probabilities for A . If we look at a long sequence of transitions in chain A alone, the relative frequency of state A_1 tends to π_{A_1} so we might hypothesize that, within A , the expected time to return to A_1 starting from A_1 is $1/\pi_{A_1}$. This hypothesis is correct and we will verify it by a simple argument when we study renewal theory. Here, however, we verify it by looking at first-passage times within the chain A . For now, label the states in A as $(1, 2, \dots, M)$ where 1 stands for A_1 . For $2 \leq i \leq M$, let v_i be the expected time to first reach state 1 starting in state i . As in (4.31),

$$v_i = 1 + \sum_{j \neq 1} P_{ij} v_j; \quad 2 \leq i \leq M. \quad (\text{A.60})$$

We can use these equations to write an equation for the expected time T to return to state 1 given that we start in state 1. The first transition goes to each state j with probability P_{1j} and the remaining time to reach state 1 from state j is v_j . We define $v_1 = 0$ since if the first transition from 1 goes to 1, there is no remaining time required to return to state A_1 . We then have

$$T = 1 + \sum_{j=1}^M P_{1j} v_j. \quad (\text{A.61})$$

Note that this is very different from (4.32) where $[P]$ is a Markov chain in which 1 is a trapping state. We can now combine (A.61) (for component 1) and (A.60) (for components 2 to M) into the following vector equation:

$$T\mathbf{e}_1 + \mathbf{v} = \mathbf{e} + [P]\mathbf{v},$$

where $\mathbf{e}_1 = (1, 0, 0, \dots, 0)^\top$ and $\mathbf{e} = (1, 1, \dots, 1)^\top$. Motivated by the hypothesis that $T = 1/\pi_1$, we premultiply this vector equation by the steady-state row vector $\boldsymbol{\pi}$, getting

$$T\pi_1 + \boldsymbol{\pi}\mathbf{v} = 1 + \boldsymbol{\pi}[P]\mathbf{v} = 1 + \boldsymbol{\pi}\mathbf{v}.$$

Cancelling $\boldsymbol{\pi}\mathbf{v}$ from each side, we get $T = 1/\pi_1$ as hypothesized.

b) Assume that $\epsilon > 0$, $\delta = 0$. Find $T_{A,B}$, the expected time to first reach state B_1 starting from state A_1 . Your answer should be a function of ϵ and the original steady-state probabilities $\{\pi_{A_i}\}$ in chain A .

Solution: Starting in state A_1 , we reach B_1 in a single step with probability ϵ . With probability $1 - \epsilon$, we wait for a return to A_1 and then have expected time $T_{A,B}$ remaining. Thus $T_{A,B} = \epsilon + (1 - \epsilon)\left(\frac{1}{\pi_{A_1}} + T_{A,B}\right)$. Solving this equation,

$$T_{A,B} = 1 + \frac{1 - \epsilon}{\epsilon\pi_{A_1}}.$$

c) Assume $\epsilon > 0$, $\delta > 0$. Find $T_{B,A}$, the expected time to first reach state A_1 , starting in state B_1 . Your answer should depend only on δ and $\{\pi_{B_i}\}$.

Solution: The fact that $\epsilon > 0$ here is irrelevant since that transition can never be used in the first passage from B_1 to A_1 . Thus the answer is the reversed version of the answer to (b), where now π_{B_1} is the steady-state probability of B_1 for chain B alone.

$$T_{B,A} = 1 + \frac{1 - \delta}{\delta\pi_{B_1}}.$$

d) Assume $\epsilon > 0$ and $\delta > 0$. Find $P'(A)$, the steady-state probability that the combined chain is in one of the states $\{A_j\}$ of the original chain A .

Solution: With $0 < \epsilon < 1$ and $0 < \delta < 1$, the combined chain is ergodic. To see this, note that all states communicate with each other so the combined chain is recurrent. Also, all walks in A are still walks in the combined chain, so the gcd of their lengths is 1. Thus A , and consequently B , are still aperiodic.

We can thus use the steady state equations to find the unique steady-state vector $\boldsymbol{\pi}$. In parts (d), (e), and (f), we are interested in those probabilities in chain A . We denote those states, as before, as $(1, \dots, M)$ where 1 is state A_1 . Steady-state probabilities for A in the combined chain for given ϵ, δ are denoted π'_j , whereas they are denoted as π_j in the original chain. We first find π'_1 and then π'_j for $2 \leq j \leq M$.

As we saw in (a), the expected first return time from a state to itself is the reciprocal of the steady-state probability, so we first find T_{AA} , the expected time of first return from A_1 to A_1 . Given that the first transition from state 1 goes to a state in A , the expected

first-return time is $1/\pi_1$ from (a). If the transition goes to B_1 , the expected first-return time is $1 + T_{BA}$, where T_{BA} is found in (c). Combining these with the a priori probabilities of going to A or B ,

$$T_{AA} = (1 - \epsilon)/\pi_1 + \epsilon[1 + T_{BA}] = \frac{1 - \epsilon}{\pi_1} + 2\epsilon + \frac{\epsilon(1 - \delta)}{\delta\pi_{B_1}}.$$

Thus

$$\pi'_1 = \left[\frac{1 - \epsilon}{\pi_1} + 2\epsilon + \frac{\epsilon(1 - \delta)}{\delta\pi_{B_1}} \right]^{-1}.$$

Next we find π'_j for the other states in A in terms of the π_j for the uncombined chains and π'_1 . The original and the combined steady-state equations for $2 \leq j \leq M$ are

$$\pi_j = \sum_{i \neq 1} \pi_i P_{ij} + \pi_1 P_{1j}; \quad \pi'_j = \sum_{i \neq 1} \pi'_i P_{ij} + \pi'_1(1 - \epsilon)P_{1j}.$$

These equations, as $M - 1$ equations in the unknowns π_j ; $j \geq 2$ for given π_1 , uniquely specify $\pi_2, \dots, \pi_M$ and they differ in π_1 being replaced by $(1 - \epsilon)\pi'_1$. From this, we see that the second set of equations is satisfied if we choose

$$\pi'_j = \pi_j \frac{(1 - \epsilon)\pi'_1}{\pi_1}.$$

We can now sum the steady-state probabilities in A to get

$$\Pr\{A\} = \sum_{j=1}^M \pi'_j = \pi'_1 \left[\epsilon + \frac{1 - \epsilon}{\pi_1} \right].$$

e) Assume $\epsilon > 0$, $\delta = 0$. For each state $A_j \neq A_1$ in A , find v_{A_j} , the expected number of visits to state A_j , starting in state A_1 , before reaching state B_1 . Your answer should depend only on ϵ and $\{\pi_{A_i}\}$.

Solution: We use a variation on the first passage time problem in (a) to find the expected number of visits to state j , $E[N_j]$, in the original chain starting in state 1 before the first return to 1. Here we let $v_i(j)$ be the expected number of visits to j , starting in state i , before the first return to 1. The equations are

$$E[N_j] = \sum_{k \neq 1} P_{1k} v_k(j) + P_{1j}; \quad v_i(j) = \sum_{k \neq 1} P_{ik} v_k(j) + P_{ij} \quad \text{for } i \neq 1.$$

The first equation represents $E[N(j)]$ in terms of the expected number of visits to j conditional on each first transition k from the initial state 1. in the special case where that first transition is to j , the expected number of visits to j includes both 1 for the initial visit plus $v_j(j)$ for the subsequent visits. The second set of equations are similar, but give the expected number of visits to j (before a return to 1) starting from each state other than 1.

Writing this as a vector equation, with $\mathbf{v}(j) = (0, v_2(j), v_3(j), \dots, v_M(j))^T$, we get

$$E[N_j] \mathbf{e}_1 + \mathbf{v}(j) = [P] \mathbf{v}(j) + [P] \mathbf{e}_j.$$

Note that $[P]e_j$ is the column vector $(p_{1j}, \dots, p_{Mj})^T$. Premultiplying by π , we see that $E[N_j] = \pi_j/\pi_1$. Finally, to find v_{A_j} , the expected number of visits to state j before the first to B_1 , we have

$$v_{A_j} = (1 - \epsilon)[E[N_j] + v_{A_j}] = \frac{(1 - \epsilon)E[N_j]}{\epsilon} = \frac{(1 - \epsilon)\pi_j}{\epsilon\pi_i}.$$

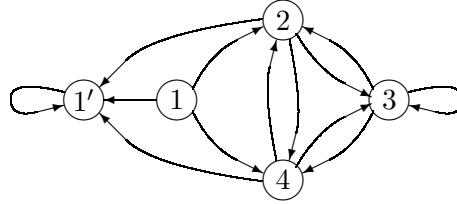
f) Assume $\epsilon > 0$, $\delta > 0$. For each state A_j in A , find π'_{A_j} , the steady-state probability of being in state A_j in the combined chain. Hint: Be careful in your treatment of state A_1 .

Solution: This was solved in (d). Readers might want to come back to this exercise later and re-solve it more simply using renewal theory.

Exercise 4.22: Section 4.5.1 showed how to find the expected first passage times to a fixed state, say 1, from all other states. It is often desirable to include the expected first recurrence time from state 1 to return to state 1. This can be done by splitting state 1 into 2 states, first an initial state with no transitions coming into it but the original transitions going out, and second, a final trapping state with the original transitions coming in.

a) For the chain on the left side of Figure 4.6, draw the graph for the modified chain with 5 states where state 1 has been split into 2 states.

Solution: We split state 1 into states $1'$ and 1 , where $1'$ is the trapping state and 1 can be an initial state.



b) Suppose one has found the expected first-passage-times v_j for states $j = 2$ to 4 (or in general from 2 to M). Find an expression for v_1 , the expected first recurrence time for state 1 in terms of $v_2, v_3, \dots, v_M$ and $P_{12}, \dots, P_{1M}$.

Solution: Note that v_2, v_3 , and v_4 are unchanged by the addition of state 1 , since no transitions can go to 1 from states $2, 3$, or 4 . We then have

$$v_1 = 1 + \sum_{j=2}^4 P_{1j}v_j.$$

Exercise 4.23: a) Assume throughout that $[P]$ is the transition matrix of a unichain (and thus the eigenvalue 1 has multiplicity 1). Show that a solution to the equation $[P]w - w = r - ge$ exists if and only if $r - ge$ lies in the column space of $[P - I]$ where $[I]$ is the identity matrix.

Solution: Let $\mathcal{C}[P - I]$ be the column space of $[P - I]$. A vector x is in $\mathcal{C}[P - I]$ if x is a linear combination of columns of $[P - I]$, i.e., if x is w_1 times the first column of $[P - I]$ plus w_2 times the second column, etc. More succinctly, $x \in \mathcal{C}[P - I]$ if and only if $x = [P - I]w$ for some vector w . Thus $r - ge \in \mathcal{C}[P - I]$ if and only if $[P - I]w = r - ge$ for some w , which after rearrangement is what is to be shown.

b) Show that this column space is the set of vectors $\mathbf{x}$ for which $\boldsymbol{\pi}\mathbf{x} = 0$. Then show that $\mathbf{r} - g\mathbf{e}$ lies in this column space.

Solution: We know that $[P]$ has a single eigenvalue equal to 1. Thus $[P - I]$ is singular and the steady-state vector $\boldsymbol{\pi}$ satisfies $\boldsymbol{\pi}[P - I] = 0$. Thus for every $\mathbf{w}$, $\boldsymbol{\pi}[P - I]\mathbf{w} = 0$ so $\boldsymbol{\pi}\mathbf{x} = 0$ for every $\mathbf{x} \in \mathcal{C}[P - I]$. Furthermore, $\boldsymbol{\pi}$ (and its scalar multiples) are the only vectors to satisfy $\boldsymbol{\pi}[P - I] = 0$ and thus $\mathcal{C}[P - I]$ is an $M - 1$ dimensional vector space. Since the vector space of vectors $\mathbf{x}$ that satisfy $\boldsymbol{\pi}\mathbf{x} = 0$ is $M - 1$ dimensional, this must be the same as $\mathcal{C}[P - I]$. Finally, since g is defined as $\boldsymbol{\pi}\mathbf{r}$, we have $\boldsymbol{\pi}(\mathbf{r} - g\mathbf{e}) = 0$, so $\mathbf{r} - g\mathbf{e} \in \mathcal{C}[P - I]$.

c) Show that, with the extra constraint that $\boldsymbol{\pi}\mathbf{w} = 0$, the equation $[P]\mathbf{w} - \mathbf{w} = \mathbf{r} - g\mathbf{e}$ has a unique solution.

Solution: For any $\mathbf{w}'$ that satisfies $[P]\mathbf{w}' - \mathbf{w}' = \mathbf{r} - g\mathbf{e}$, it is easy to see that $\mathbf{w}' + \alpha\mathbf{e}$ also satisfies this equation for any real α . Furthermore, since the column space of $[P - I]$ is $M - 1$ dimensional, this set of solutions, namely $\{\mathbf{w}' + \alpha\mathbf{e}; \alpha \in \mathbb{R}\}$ is the entire set of solutions. The additional constraint that $\boldsymbol{\pi}(\mathbf{w}' + \alpha\mathbf{e}) = \boldsymbol{\pi}\mathbf{w}' + \alpha = 0$ specifies a unique element in this set.

Exercise 4.24: For the Markov chain with rewards in Figure 4.8,

a) Find the solution to (4.37) and find the gain g .

Solution: The symmetry in the transition probabilities shows that $\boldsymbol{\pi} = (1/2, 1/2)^\top$, and thus $g = \boldsymbol{\pi}\mathbf{r} = 1/2$. The first component of (4.37), *i.e.*, of $\mathbf{w} + g\mathbf{e} = [P]\mathbf{w} + \mathbf{r}$ is then $w_1 + 1/2 = P_{11}w_1 + P_{12}w_2 + r_1$. This simplifies to $w_1 - w_2 = -50$. The second component is redundant, and $\boldsymbol{\pi}\mathbf{w} = 0$ simplifies to $w_1 + w_2 = 0$. Thus $w_1 = -25$ and $w_2 = 25$.

b) Modify Figure 4.8 by letting P_{12} be an arbitrary probability. Find g and $\mathbf{w}$ again and give an intuitive explanation of why P_{12} effects w_2 .

Solution: With arbitrary P_{12} , the steady-state probabilities become

$$\pi_1 = \frac{0.01}{P_{12} + 0.01}; \quad \pi_2 = \frac{P_{12}}{P_{12} + 0.01}.$$

The steady-state gain, $g = \boldsymbol{\pi}\mathbf{r}$, then becomes $g = P_{12}/(P_{12} + 0.01)$. Solving for $\mathbf{w}$ as before, we get

$$w_1 = \frac{-P_{12}}{(P_{12} + 0.01)^2} \quad w_2 = \frac{0.01}{(P_{12} + 0.01)^2}.$$

As P_{12} increases, the mean duration of each visit to state 1 decreases so that the fraction of time spent in state 1 also decreases, thus increasing the expected gain. At the same time, the relative advantage of starting in state 2 decreases since the interruptions to having a reward on each transition become shorter. For example, if $P_{12} = 1/2$, the mean time to leave state 1, starting in state 1, is 2.

Exercise 4.25: (Proof of Corollary 4.5.6) a) Show that the gain per stage g is 0. Hint: Show that $\mathbf{r}$ is zero where the steady-state vector $\boldsymbol{\pi}$ is non-zero.

Solution: The steady-state vector $\boldsymbol{\pi}$ is non-zero only for the recurrent states, and the reward is 0 on each of those states, so the steady-state gain $g = \boldsymbol{\pi}\mathbf{r}$ is 0.

b) Let $[P_{\mathcal{R}}]$ be the transition matrix for the recurrent states and let $\mathbf{r}_{\mathcal{R}} = 0$ be the reward vector and $\mathbf{w}_{\mathcal{R}}$ the relative-gain vector for $[P_{\mathcal{R}}]$. Show that $\mathbf{w}_{\mathcal{R}} = 0$. Hint: Use Theorem 4.5.4.

Solution: Treating the recurrent states in isolation, (4.37) becomes $\mathbf{w}_{\mathcal{R}} = [P_{\mathcal{R}}]\mathbf{w}_{\mathcal{R}}$ and $\pi_{\mathcal{R}}\mathbf{w}_{\mathcal{R}} = 0$. The first part of this is simply the equation for the right eigenvector of $[P_{\mathcal{R}}]$ of eigenvalue 1. This is unique within a scale factor, and the second part specifies that the scale factor is 0. As a matter of terminology, $\mathbf{w}_{\mathcal{R}}$ is then not an eigenvector, since eigenvectors cannot be 0, but $\mathbf{w}_{\mathcal{R}} = 0$ does satisfy the eigenvector equation aside from the nonzero condition.

c) Show that $w_i = 0$ for all $i \in \mathcal{R}$. Hint: Compare the relative-gain equations for $[P]$ to those for $[P_{\mathcal{R}}]$.

Solution: Use the block upper triangular form of $[P]$ in Figure 4.5 and represent the transient and recurrent parts of $\mathbf{w}$ as $(\mathbf{w}_{\mathcal{T}}, \mathbf{w}_{\mathcal{R}})$. Then $[P]\mathbf{w}$ over the recurrent states is $[P_{\mathcal{R}}]\mathbf{w}_{\mathcal{R}}$, and (4.37) restricted to these components becomes the same as in (b).

d) Show that for each $n \geq 0$, $[P^n]\mathbf{w} = [P^{n+1}]\mathbf{w} + [P^n]\mathbf{r}$. Hint: Start with the relative-gain equation for $[P]$.

Solution: The relative gain equation, (4.37), with $g = 0$, is $\mathbf{w} = [P]\mathbf{w} + \mathbf{r}$. Premultiplying this by $[P]$ yields $[P]\mathbf{w} = [P^2]\mathbf{w} + [P]\mathbf{r}$, and, for $n > 1$, premultiplying it by $[P^n]$ yields $[P^n]\mathbf{w} = [P^{n+1}]\mathbf{w} + [P^n]\mathbf{r}$. For the special case of $n = 0$, we interpret $[P^0]$ as the identity matrix, so $[P^n]\mathbf{w} = [P^{n+1}]\mathbf{w} + [P^n]\mathbf{r}$ becomes $\mathbf{w} = [P]\mathbf{w} + \mathbf{r}$, which is the relative gain equation.

e) Show that $\mathbf{w} = [P^{n+1}]\mathbf{w} + \sum_{m=0}^n [P^m]\mathbf{r}$. Hint: Sum the result in (d).

Solution: Summing the result of (d) for the first $n + 1$ terms,

$$\sum_{m=0}^n [P^m]\mathbf{w} = \sum_{m=1}^{n+1} [P^m]\mathbf{w} + \sum_{m=0}^n [P^m]\mathbf{r}.$$

Canceling the repeated terms from each side, we have the desired result.

f) Show that $\lim_{n \rightarrow \infty} [P^{n+1}]\mathbf{w} = 0$ and that $\lim_{n \rightarrow \infty} \sum_{m=0}^n [P^m]\mathbf{r}$ is finite, nonnegative, and has positive components for $r_i > 0$. Hint: Use lemma 4.3.6.

Solution: Since $\mathbf{w}_{\mathcal{R}} = 0$, we have $[P^{n+1}]\mathbf{w} = [P_{\mathcal{T}}^{n+1}]\mathbf{w}_{\mathcal{T}}$. From Lemma 4.3.6, the components of $[P_{\mathcal{T}}^{n+1}]$ all approach 0 geometrically in n . Thus $\lim_{n \rightarrow \infty} \sum_{m=0}^n [P^m]\mathbf{r}$ is finite. It is nonnegative since $\mathbf{r}$ is nonnegative. Finally, if $r_i > 0$, then $w_i = \sum_{n=0}^{\infty} P_{ij}^n r_j > 0$ because $P_{ii}^0 = 1$.

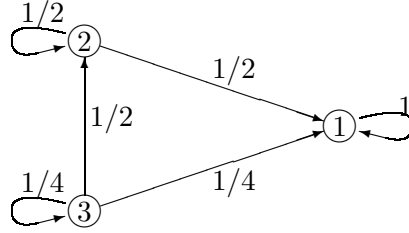
g) Demonstrate the final result of the corollary by using the previous results on $\mathbf{r} = \mathbf{r}' - \mathbf{r}''$.

Solution: This final result is obvious by linearity.

Exercise 4.26: Consider the Markov chain below:

a) Suppose the chain is started in state i and goes through n transitions; let $v_i(n)$ be the expected number of transitions (out of the total of n) until the chain enters the trapping state, state 1. Find an expression for $\mathbf{v}(n) = (v_1(n), v_2(n), v_3(n))^T$ in terms of $\mathbf{v}(n-1)$ (take $v_1(n) = 0$ for all n). (Hint: view the system as a Markov reward system; what is the value of $\mathbf{r}$?)

Solution: We use essentially the same approach as in Section 4.5.1, but we are explicit here about the number of transitions. Starting in any state $i \neq 1$ with n transitions to go,



the expected number of transitions until the first that either enters the trapping state or completes the n th transition is given by

$$v_i(n) = 1 + \sum_{j=1}^3 P_{ij} v_j(n-1). \quad (\text{A.62})$$

Since $i = 1$ is the trapping state, we can express $v_1(n) = 0$ for all n , since no transitions are required to enter the trapping state. Thus (A.62) is modified to $v_1(n) = \sum_{j=1}^3 P_{1j} v_j(n-1)$ for $i = 1$. Viewing $\mathbf{r}$ as a reward vector, with one unit of reward for being in any state other than the trapping state, we have $\mathbf{r} = (0, 1, 1)^\top$. Thus (A.62) can be expressed in vector form as

$$\mathbf{v}(n) = \mathbf{r} + [P]\mathbf{v}(n-1).$$

b) Solve numerically for $\lim_{n \rightarrow \infty} \mathbf{v}(n)$. Interpret the meaning of the elements v_i in the solution of (4.32).

Solution: We have already seen that $v_1(n) = 0$ for all n , and thus, since $P_{23} = 0$, we have $v_2(n) = 1 + (1/2)v_2(n-1)$. Since $v_2(0) = 0$, this iterates to

$$v_2(n) = 1 + \frac{1}{2}(1 + v_2(n-2)) = 1 + \frac{1}{2} + \frac{1}{4}(1 + v_2(n-3)) = \dots = 2 - 2^{-(n-1)}.$$

For $v_3(n)$, we use the same approach, but directly use the above solution for $v_2(n)$ for each n .

$$\begin{aligned}
 v_3(n) &= 1 + P_{32}v_2(n-1) + P_{33}v_3(n-1) = 1 + \frac{1}{2}(1 - 2^{-(n-2)}) + \frac{1}{4}v_3(n-1) \\
 &= (2 - 2^{n-1}) + \frac{1}{4}\left[2 - 2^{n-2} + \frac{1}{4}v_3(n-2)\right] = \dots \\
 &= (2 - 2^{n-1}) + \frac{1}{4}\left[2 - 2^{n-2}\right] + \frac{1}{16}\left[2 - 2^{n-3}\right] + \dots + \frac{1}{4^{n-1}}[2 - 2^0] \\
 &= 2\left[1 + \frac{1}{4} + \frac{1}{4^2} + \dots + \frac{1}{4^{n-1}}\right] - \left[2^{-(n-1)} + 2^{-n} + \dots + 2^{-2n+2}\right] \\
 &= 2\left[\frac{1 - 4^{-n}}{1 - 4^{-1}}\right] - 2^{-(n-1)}\left[\frac{1 - 2^{-n}}{1 - 2^{-1}}\right] = \frac{8}{3} - \frac{2^{-2n+3}}{3} - 2^{-n+2} + 2^{-2n+2}.
 \end{aligned}$$

Taking the limit,

$$\lim_{n \rightarrow \infty} v_2(n) = 2; \quad \lim_{n \rightarrow \infty} v_3(n) = \frac{8}{3}.$$

c) Give a direct argument why (4.32) provides the solution directly to the expected time from each state to enter the trapping state.

Solution: The limit as $n \rightarrow \infty$ gives the expected time to enter the trapping state with no limit on the required number of transitions. This limit exists in general since the transient states persist with probabilities decreasing exponentially with n .

Exercise 4.27: a) Show that (4.48) can be rewritten in the more compact form

$$\mathbf{v}^*(n, \mathbf{u}) = \mathbf{v}^*(1, \mathbf{v}^*(n-1, \mathbf{u})).$$

Solution: From (4.45), $\mathbf{v}^*(1, \mathbf{u}) = \max_{\mathbf{k}} (\mathbf{r}^{\mathbf{k}} + [P^{(\mathbf{k})}]\mathbf{u})$. This is valid for an arbitrary choice of $\mathbf{u}$, so choosing $\mathbf{v}^*(n-1, \mathbf{u})$ in place of $\mathbf{u}$, we get

$$\mathbf{v}^*(1, \mathbf{v}^*(n-1, \mathbf{u})) = \max_{\mathbf{k}} (\mathbf{r}^{\mathbf{k}} + [P^{(\mathbf{k})}]\mathbf{v}^*(n-1, \mathbf{u})).$$

This is equal to $\mathbf{v}^*(n, \mathbf{u})$ from (4.48).

b) Explain why it is also true that

$$\mathbf{v}^*(2n, \mathbf{u}) = \mathbf{v}^*(n, \mathbf{v}^*(n, \mathbf{u})). \quad (\text{A.63})$$

Solution: We can see that $\mathbf{v}^*(n, \mathbf{v}^*(n, \mathbf{u}))$ is the maximum gain over n stages using a final gain of $\mathbf{v}^*(n, \mathbf{u})$ where the final gain $\mathbf{v}^*(n, \mathbf{u})$ can be interpreted as the maximum gain over the next n stages (with a final gain at time $2n$ of $\mathbf{u}$). This expression makes sense since the gain over the next n stages depends only on the state at the beginning of that next n stages. It depends on that initial state i at time n only as the i th component of the vector gain.

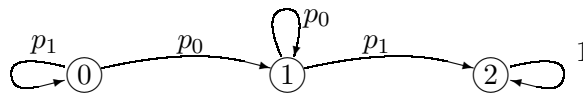
c) One might guess that (A.63) could be used iteratively, finding $\mathbf{v}^*(2^{n+1}, \mathbf{u})$ from $\mathbf{v}^*(2^n, \mathbf{u})$. Explain why this is not possible in any straightforward way. Hint: Think through explicitly how one might calculate $\mathbf{v}^*(n, \mathbf{v}^*(n, \mathbf{u}))$ from $\mathbf{v}^*(n, \mathbf{u})$.

Solution: If one could find $\mathbf{v}^*(n, \mathbf{u})$ as a convenient formula in $\mathbf{u}$, then by substituting $\mathbf{v}^*(n, \mathbf{u})$ for $\mathbf{u}$, one could immediately find $\mathbf{v}^*(2n, \mathbf{u})$. The trouble is that the iteration for finding $\mathbf{v}^*(n, \mathbf{u})$ is an iteration requiring n steps for any given $\mathbf{u}$. When $\mathbf{u}$ is changed to $\mathbf{v}^*(n, \mathbf{u})$, then one needs another n steps to find $\mathbf{v}^*(n, \mathbf{v}^*(n, \mathbf{u}))$. This is exactly the same computation that is required to find $\mathbf{v}^*(2n, \mathbf{u})$ in the first place.

Exercise 4.28: Consider finding the expected time until a given string appears in an IID binary sequence with $\Pr\{X_i=1\} = p_1$, $\Pr\{X_i=0\} = p_0 = 1 - p_1$.

a) Following the procedure in Example 4.5.1, draw the 3 state Markov chain for the string (0, 1). Find the expected number of trials until the first occurrence of that string.

Solution:



Let v_i be the expected first-passage time from node i to node 2. Then

$$\begin{aligned} v_0 &= 1 + p_1 v_0 + p_0 v_1; & v_1 &= 1 + p_0 v_1 + p_1 v_2 \\ v_0 &= 1/p_0 + v_1; & v_1 &= 1/p_1 + v_2. \end{aligned}$$

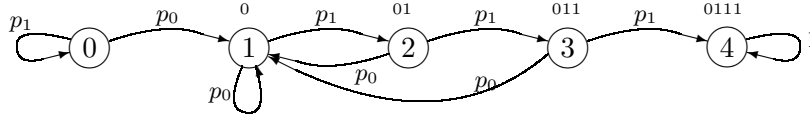
Combining these equations to eliminate v_1 ,

$$v_0 = 1/p_0 + 1/p_1 + v_2 = 1/p_0 p_1 + v_2.$$

Finally, the trapping state, 2, is reached when the string 01 first occurs, so $v_2 = 0$ and $v_0 = 1/p_0 p_1$.

b) For parts b) to d), let $(a_1, a_2, a_3, \dots, a_k) = (0, 1, 1, \dots, 1)$, i.e., zero followed by $k - 1$ ones. Draw the corresponding Markov chain for $k = 4$.

Solution:



c) Let v_i , $1 \leq i \leq k$ be the expected first-passage time from state i to state k . Note that $v_k = 0$. For each i , $1 \leq i < k$, show that $v_i = \alpha_i + v_{i+1}$ and $v_0 = \beta_i + v_{i+1}$ where α_i and β_i are each expressed as a product of powers of p_0 and p_1 . Hint: use induction on i using $i = 1$ as the base. For the inductive step, first find β_{i+1} as a function of β_i starting with $i = 1$ and using the equation $v_0 = 1/p_0 + v_1$.

Solution: (a) solved the problem for $i = 1$. The fact that the string length was 2 there was of significance only at the end where we set $v_2 = 0$. We found that $\alpha_1 = 1/p_1$ and $\beta_1 = 1/p_0 p_1$.

For the inductive step, assume $v_i = \alpha_i + v_{i+1}$ and $v_0 = \beta_i + v_{i+1}$ for a given $i \geq 1$. Using the basic first-passage-time equation,

$$\begin{aligned} v_{i+1} &= 1 + p_0 v_1 + p_1 v_{i+2} \\ &= p_0 v_0 + p_1 v_{i+2} \\ &= p_0 \beta_i + p_0 v_{i+1} + p_1 v_{i+2}. \end{aligned}$$

The second equality uses the basic equation for v_0 , i.e., $v_0 = 1 + p_1 v_0 + p_0 v_1$ which reduces to $p_0 v_1 + 1 = p_0 v_0$ and the third equality uses the inductive hypothesis $v_0 = \beta_i + v_{i+1}$. Combining the terms in v_{i+1} ,

$$v_{i+1} = \frac{p_0 \beta_i}{p_1} + v_{i+2}.$$

This completes half the inductive step, showing that $\alpha_{i+1} = p_0 \beta_i / p_1$. Now we use this result plus the inductive hypothesis $v_0 = \beta_i + v_{i+1}$ to get

$$v_0 = \beta_i + v_{i+1} = \beta_i + \frac{p_0 \beta_i}{p_1} + v_{i+2} = \frac{\beta_i}{p_1} + v_{i+2}.$$

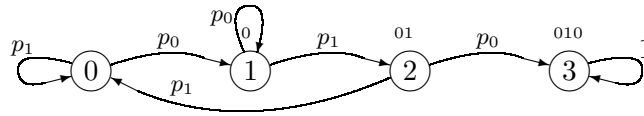
This completes the second half of the induction, showing that $\beta_{i+1} = \beta_i/p_1$. Iterating on these equations for β_i and α_i , we find the explicit expression

$$\alpha_i = \frac{1}{p_1^i}; \quad \beta_i = \frac{1}{p_0 p_1^i}.$$

Note that the algebra here was quite simple, but if one did not follow the hints precisely, one could get into a terrible mess. In addition, the whole thing was quite unmotivated. We view the occurrence of these strings as renewals in Exercise 5.35 and find a more intuitive way to derive the same answer.

d) Let $\mathbf{a} = (0, 1, 0)$. Draw the corresponding Markov chain for this string. Evaluate v_0 , the expected time for $(0, 1, 0)$ to occur.

Solution:



The solution for v_0 and v_1 in terms of v_2 is the same as (a). The basic equation for v_2 in terms of its outward transitions is

$$\begin{aligned} v_2 &= 1 + p_0 v_0 + p_1 v_3 \\ &= 1 + p_0 \left[\frac{1}{p_0 p_1} + v_2 \right]. \end{aligned}$$

Combining the terms in v_2 , we get

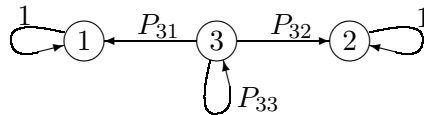
$$p_1 v_2 = 1 + 1/p_1.$$

Using $v_0 = 1/p_0 p_1 + v_2$,

$$v_0 = \frac{1}{p_1^2 p_0} + \frac{1}{p_1} + \frac{1}{p_1^2} = \frac{1}{p_1} + \frac{1}{p_0 p_1^2}.$$

This solution will become more transparent after doing Exercise 5.35.

Exercise 4.29: a) Find $\lim_{n \rightarrow \infty} [P^n]$ for the Markov chain below. Hint: Think in terms of the long term transition probabilities. Recall that the edges in the graph for a Markov chain correspond to the positive transition probabilities.



Solution: The chain has 2 recurrent states, each in its own class, and one transient state. Thus $\lim_{n \rightarrow \infty} P_{11}^n = 1$ and $\lim_{n \rightarrow \infty} P_{22}^n = 1$. Let $q_1 = \lim_{n \rightarrow \infty} P_{31}^n$ and $q_2 = \lim_{n \rightarrow \infty} P_{32}^n$. Since $q_1 + q_2 = 1$ and since in each transition starting in state 3, P_{31} and P_{32} give the

probabilities of moving to 1 or 2, $q_1 = P_{31}/(P_{31} + P_{32})$ and $q_2 = P_{32}/(P_{31} + P_{32})$. The other long term transition probabilities are zero, so

$$\lim_{n \rightarrow \infty} [P^n] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ q_1 & q_2 & 0 \end{bmatrix}.$$

The general case here, with an arbitrary set of transient states and an arbitrary number of recurrent classes is solved in Exercise 4.18.

b) Let $\pi^{(1)}$ and $\pi^{(2)}$ denote the first two rows of $\lim_{n \rightarrow \infty} [P^n]$ and let $\nu^{(1)}$ and $\nu^{(2)}$ denote the first two columns of $\lim_{n \rightarrow \infty} [P^n]$. Show that $\pi^{(1)}$ and $\pi^{(2)}$ are independent left eigenvectors of $[P]$, and that $\nu^{(1)}$ and $\nu^{(2)}$ are independent right eigenvectors of $[P]$. Find the eigenvalue for each eigenvector.

Solution: $\pi^{(1)} = (1, 0, 0)$ and $\pi^{(2)} = (0, 1, 0)$. Multiplying $\pi^{(i)}$ by $\lim_{n \rightarrow \infty} [P^n]$ for $i = 1, 2$ we see that these are the left eigenvectors of eigenvalue 1 of the two recurrent classes. We also know this from Section 4.4. Similarly, $\nu^{(1)} = (1, 0, q_1)^T$ and $\nu^{(2)} = (0, 1, q_2)^T$ are right eigenvectors of eigenvalue 1, both of $\lim_{n \rightarrow \infty} [P^n]$ and also of $[P]$.

c) Let $\mathbf{r}$ be an arbitrary reward vector and consider the equation

$$\mathbf{w} + g^{(1)}\nu^{(1)} + g^{(2)}\nu^{(2)} = \mathbf{r} + [P]\mathbf{w}. \quad (\text{A.64})$$

Determine what values $g^{(1)}$ and $g^{(2)}$ must have in order for (A.64) to have a solution. Argue that with the additional constraints $w_1 = w_2 = 0$, (A.64) has a unique solution for $\mathbf{w}$ and find that $\mathbf{w}$.

Solution: In order for this equation to have a solution, it is necessary, first, for a solution to exist when both sides are premultiplied by $\pi^{(1)}$. This results in $g^{(1)} = r_1$. Similarly, premultiplying by $\pi^{(2)}$ results in $g^{(2)} = r_2$. In other words, $g^{(1)}$ is the gain per transition when in state 1 or when the chain moves from state 3 to state 1. Similarly $g^{(2)}$ is the gain per transition in state 2 or after a transition from 3 to 2. Since there are two recurrent sets of states, there is no common meaning to a gain per transition.

Setting $w_1 = w_2 = 0$ makes a certain amount of sense, since starting in state 1 or starting in state 2, the reward increases by r_1 or r_2 per transition with no initial transient. With this choice, the first two components of the vector equation in (A.64) are $0=0$ and the third is

$$w_3 + r_3q_1 + r_2q_2 = r_3 + P_{33}w_3.$$

Solving for w_3 ,

$$w_3 = \frac{1}{P_{31} + P_{32}} \left[r_3 - r_1 \frac{P_{31}}{P_{31} + P_{32}} - r_2 \frac{P_{32}}{P_{31} + P_{32}} \right].$$

This can be interpreted as the relative gain of starting in state 3 relative to the ‘average’ of starting in 1 or 2. The interpretation is quite limited, since the gain per transition depends on which recurrent class is entered, and thus relative gain has nothing definitive for comparison.

Exercise 4.30: Let $\mathbf{u}$ and $\mathbf{u}'$ be arbitrary final reward vectors with $\mathbf{u} \leq \mathbf{u}'$.

a) Let $\mathbf{k}$ be an arbitrary stationary policy and prove that $\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}) \leq \mathbf{v}^{\mathbf{k}}(n, \mathbf{u}')$ for each $n \geq 1$.

Solution: For $n = 1$,

$$\mathbf{v}^{\mathbf{k}}(1, \mathbf{u}) = \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{u} \leq \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{u}' = \mathbf{v}^{\mathbf{k}}(1, \mathbf{u}'),$$

where the inequality has used the fact that $[P^{\mathbf{k}}]$ is nonnegative. We use this as the basis in the following inductive step: assume that $\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}) \leq \mathbf{v}^{\mathbf{k}}(n, \mathbf{u}')$. Then

$$\mathbf{v}^{\mathbf{k}}(n+1, \mathbf{u}) = \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}) \leq \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}') = \mathbf{v}^{\mathbf{k}}(n+1, \mathbf{u}').$$

Thus $\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}) \leq \mathbf{v}^{\mathbf{k}}(n, \mathbf{u}')$. Note that this does not depend on whether $[P^{\mathbf{k}}]$ is the matrix of a unichain; it is valid for any Markov chain and any policy.

b) For the optimal dynamic policy, prove that $\mathbf{v}^*(n, \mathbf{u}) \leq \mathbf{v}^*(n, \mathbf{u}')$ for each $n \geq 1$. This is known as the monotonicity theorem.

Solution: Let $\mathbf{k}'$ maximize $\mathbf{v}(1, \mathbf{k})$ over $\mathbf{k}$. Then

$$\mathbf{v}^*(1, \mathbf{u}) = \mathbf{v}^{\mathbf{k}'}(1, \mathbf{u}) \leq \mathbf{v}^{\mathbf{k}'}(1, \mathbf{u}') \leq \mathbf{v}^*(1, \mathbf{u}').$$

We use this as the basis of the following inductive step over n . Assume that $\mathbf{v}^*(n, \mathbf{u}) \leq \mathbf{v}^*(n, \mathbf{u}')$. Let $\mathbf{k}'$ maximize $\mathbf{v}^{\mathbf{k}}(n+1, \mathbf{u})$ over $\mathbf{k}$.

$$\begin{aligned} \mathbf{v}^*(n+1, \mathbf{u}) &= \mathbf{v}^{\mathbf{k}'}(n+1, \mathbf{u}) = \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{v}^*(n, \mathbf{u}) \\ &\leq \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{v}^*(n, \mathbf{u}') \\ &\leq \mathbf{v}^*(n+1, \mathbf{u}'), \end{aligned}$$

where in the next to last step, we used the inductive hypothesis plus the nonnegativity of $[P^{\mathbf{k}'}]$ and in the last step, we re-optimized over the policy for the new final reward $\mathbf{u}'$.

c) Now let $\mathbf{u}$ and $\mathbf{u}'$ be arbitrary. Let $\alpha = \max_i (u_i - u'_i)$. Show that

$$\mathbf{v}^*(n, \mathbf{u}) \leq \mathbf{v}^*(n, \mathbf{u}') + \alpha \mathbf{e}. \quad (\text{A.65})$$

Solution: Note that $u_i - u'_i \leq \alpha$ for all i , so $\mathbf{u} \leq \mathbf{u}' + \alpha \mathbf{e}$. Thus, using (b),

$$\mathbf{v}^*(n, \mathbf{u}) \leq \mathbf{v}^*(n, \mathbf{u}' + \alpha \mathbf{e}).$$

Letting $\mathbf{k}'$ maximize $\mathbf{v}^{\mathbf{k}}(n, \mathbf{u}' + \alpha \mathbf{e})$, we have

$$\begin{aligned} \mathbf{v}^*(n, \mathbf{u}' + \alpha \mathbf{e}) &= \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}](\mathbf{u}' + \alpha \mathbf{e}) \\ &= \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{u}' + \alpha \mathbf{e} \leq \mathbf{v}^*(n, \mathbf{u}') + \alpha \mathbf{e}, \end{aligned}$$

which gives us (A.65). The final inequality here is actually an equality, since adding a fixed quantity α to the final reward in all states can not change the optimization over states.

Exercise 4.31: Consider a Markov decision problem with M states in which some state, say state 1, is inherently reachable from each other state.

a) Show that there must be some other state, say state 2, and some decision, k_2 , such that $P_{21}^{(k_2)} > 0$.

Solution: Assume that $M > 1$ and let j be any state other than 1. The definition of inherent reachability says that there is a policy, say $\mathbf{k}(1)$, for which there is a path from j to 1. Denote the last state on this path before state 1 as state 2 and let k_2 be the decision in state 2 under policy $\mathbf{k}(1)$. Then $P_{21}^{(k_2)} > 0$.

b) Show that there must be some other state, say state 3, and some decision, k_3 , such that either $P_{31}^{(k_3)} > 0$ or $P_{32}^{(k_3)} > 0$.

Solution: Assume that $M > 2$ and let j be any state other than 1 or 2. There is a policy, say $\mathbf{k}(2)$ under which there is a path from j to 1. Denote state 3 as the last state on this path that differs from 1 and 2. Let k_3 be the decision in state 3 under policy $\mathbf{k}(2)$. Then if this path goes directly from 3 to 1, we have $P_{31}^{(k_3)} > 0$. Alternatively, if this path goes from 3 to 2 to 1, we have $P_{32}^{(k_3)} > 0$ (and, of course, $P_{21}^{(k_2)} > 0$).

c) Assume, for some i , and some set of decisions $k_2, \dots, k_i$ that, for each j , $2 \leq j \leq i$, $P_{j\ell}^{(k_j)} > 0$ for some $\ell < j$ (i.e., that each state from 2 to j has a non-zero transition to a lower numbered state). Show that there is some state (other than 1 to i), say $i+1$ and some decision k_{i+1} such that $P_{i+1,\ell}^{(k_{i+1})} > 0$ for some $\ell \leq i$.

Solution: For those confused by all the indices, we are constructing a tree, rooted at node (state) 1, with ingoing branches corresponding to transitions with positive probability under the selected decisions. The numbering of the states is not unique, but is used simply to show that such a tree can be constructed. Each higher numbered node is connected by an inward going transition to the part of the tree already constructed.

As before, assume $M > i$ and choose some state j other than 1 to i . There is a policy $\mathbf{k}(i)$ for which there is a path from j to state 1. Denote $i+1$ as the last state on this path that differs from 1 to i . Let ℓ be the next state on this path (which must satisfy $1 \leq \ell \leq i$) and let k_{i+1} be the decision in policy $\mathbf{k}(i)$ used in state $i+1$. Then $P_{i+1,\ell}^{(k_{i+1})} > 0$.

d) Use parts a), b), and c) to observe that there is a stationary policy $\mathbf{k} = k_1, \dots, k_M$ for which state 1 is accessible from each other state.

Solution: When each state has been chosen (i.e., when $i = M$, then every state j has a path to state 1 where k_j is the decision used in state j . This set of decisions $\{k_j; 1 \leq j \leq M\}$ constitutes a policy under which state 1 is accessible from each other node.

Exercise 4.32: George drives his car to the theater, which is at the end of a one-way street. There are parking places along the side of the street and a parking garage that costs \$5 at the theater. Each parking place is independently occupied or unoccupied with probability 1/2. If George parks n parking places away from the theater, it costs him n cents (in time and shoe leather) to walk the rest of the way. George is myopic and can only see the parking place he is currently passing. If George has not already parked by the time he reaches the n th place, he first decides whether or not he will park if the place is unoccupied, and then observes the place and acts according to his decision. George can never go back and must park in the parking garage if he has not parked before.

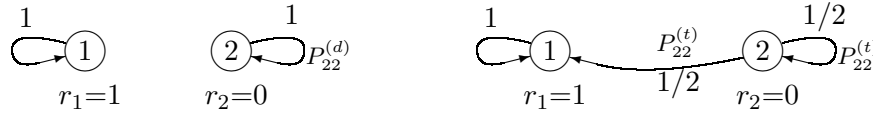
a) Model the above problem as a 2 state dynamic programming problem. In the “driving” state, state 2, there are two possible decisions: park if the current place is unoccupied or drive on whether or not the current place is unoccupied.

Solution: View the two states as walking (state 1) and driving (state 2). The state transitions correspond to passing successive possibly idle parking spaces, which are passed either by car or foot. Thus $r_1 = 1$ (the cost in shoe leather and time) and $r_2 = 0$. There

is no choice of policy in state 1, but in state 2, policy t is to try to park and policy d is continue to drive.

First consider stage 1 where George is between the second and first parking space from the end. If George is in the walk state, there is one unit of cost going from 2 to 1 and 1 unit of final cost going from 1 to 0. If George is in the drive state (2) and uses the try policy, $r_2 = 0$ and with probability $1/2$, he parks and has one unit final cost getting to the theatre. With probability $1/2$, he doesn't park and the final cost is 500. For policy d , $r_2 = 0$ and with probability 1, the final cost is 500.

In summary, $\mathbf{u} = (1, 500)^\top$, $\mathbf{r} = (0, 1)^\top$ and the Markov chain with the t and d policies in state 2 is



b) Find $v_i^*(n, \mathbf{u})$, the *minimum* expected aggregate cost for n stages (i.e., immediately before observation of the n th parking place) starting in state $i = 1$ or 2; it is sufficient to express $v_i^*(n, \mathbf{u})$ in terms of $v_i^*(n-1)$. The final costs, in cents, at stage 0 should be $v_2(0) = 500$, $v_1(0) = 0$.

Solution: We start with stage 1.

$$\mathbf{v}^*(1, \mathbf{u}) = \mathbf{r} + \min_{\mathbf{k}} [P^{\mathbf{k}}] \mathbf{u} = (2, 250.5)^\top,$$

where policy t is clearly optimal. At an arbitrary stage n , the cost is found iteratively from stage $n-1$ by (4.48), i.e.,

$$\mathbf{v}^*(n, \mathbf{u}) = \min_{\mathbf{k}} \left(\mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}] \mathbf{v}^*(n-1, \mathbf{u}) \right). \quad (\text{A.66})$$

c) For what values of n is the optimal decision the decision to drive on?

Solution: The straightforward approach (particularly with computational aids is to simply calculate (A.66). The cost drops sharply with increasing n and for $n \geq 8$, the optimal decision is to drive on, whereas for $n \leq 7$, the optimal decision is to park if possible.

As with many problems, the formalism of dynamic programming makes hand calculation and understanding more awkward than it need be. The simple approach, for any distance n away from the theatre, is to simply calculate the cost of parking if a place is available, namely n , and to calculate the expected cost if one drives on. This expected cost, E_n is easily seen to be

$$E_n = \frac{1}{2}(m-1) + \frac{1}{4}(m-2) + \cdots + \frac{1}{2^{m-1}}(1) + \frac{500}{2^{m-1}}.$$

With some patience, this is simplified to $E_n = m - 2 + (501)/2^{m-1}$, from which it is clear that one should drive for $n \geq 8$ and park for $n \leq 7$.

d) What is the probability that George will park in the garage, assuming that he follows the optimal policy?

Solution: George will park in the garage if the last 7 potential parking spaces are full, an event of probability 2^{-7} .

Exercise 4.33: (Proof of Corollary 4.6.9) **a)** Show that if two stationary policies $\mathbf{k}'$ and $\mathbf{k}$ have the same recurrent class $\mathcal{R}'$ and if $k'_i = k_i$ for all $i \in \mathcal{R}'$, then $w'_i = w_i$ for all $i \in \mathcal{R}'$. Hint: See the first part of the proof of Lemma 4.6.7.

Solution: For all $i \in \mathcal{R}'$, $P_{ij}^{(k_i)} = P_{ij}^{(k'_i)}$. The steady-state vector $\boldsymbol{\pi}$ is determined solely by the transition probabilities in the recurrent class, so $\pi_i = \pi'_i$ for all $i \in \mathcal{R}'$. Since $r_i = r'_i$ for all $i \in \mathcal{R}'$, it also follows that $g = g'$. The equations for the components of the relative gain vector $\mathbf{w}$ for $i \in \mathcal{R}'$ are

$$w_i + g = r_i^{(k_i)} + \sum_j P_{ij}^{(k_i)} w_j; \quad i \in \mathcal{R}.$$

These equations, along with $\boldsymbol{\pi}\mathbf{w} = 0$ have a unique solution if we look at them only over the recurrent class $\mathcal{R}'$. Since all components are the same for $\mathbf{k}$ and $\mathbf{k}'$, there is a unique solution over $\mathcal{R}'$, i.e., $w_i = w'_i$ for all $i \in \mathcal{R}'$.

b) Assume that $\mathbf{k}'$ satisfies 4.50 (i.e., that it satisfies the termination condition of the policy improvement algorithm) and that $\mathbf{k}$ satisfies the conditions of (a). Show that (4.64) is satisfied for all states ℓ .

Solution: The implication from $\mathcal{R}'$ being the recurrent class of $\mathbf{k}$ and $\mathbf{k}'$ is that each of them are unichains. Thus $\mathbf{w} + g\mathbf{e} = \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{w}$ with $\boldsymbol{\pi}\mathbf{w} = 0$ has a unique solution, and there is a unique solution for the primed case. Rewriting these equations for the primed and unprimed case, and recognizing from (a) that $g\mathbf{e} = g'\mathbf{e}$ and $\boldsymbol{\pi}\mathbf{w} = \boldsymbol{\pi}\mathbf{w}'$,

$$\mathbf{w} - \mathbf{w}' = [P^{\mathbf{k}}](\mathbf{w} - \mathbf{w}') + \left\{ \mathbf{r}^{\mathbf{k}} - \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}}]\mathbf{w}' - [P^{\mathbf{k}'}]\mathbf{w}' \right\}. \quad (\text{A.67})$$

This is the vector form of (4.64).

c) Show that $\mathbf{w} \leq \mathbf{w}'$. Hint: Follow the reasoning at the end of the proof of Lemma 4.6.7.

Solution: The quantity in braces in (A.67) is non-negative because of the termination condition of the policy improvement algorithm. Also $\mathbf{w} = \mathbf{w}'$ over the recurrent components of $\mathbf{k}$. Viewing the term in braces as the non-negative difference between two reward vectors, Corollary 4.5.6 shows that $\mathbf{w} \leq \mathbf{w}'$.

Exercise 4.34: Consider the dynamic programming problem below with two states and two possible policies, denoted $\mathbf{k}$ and $\mathbf{k}'$. The policies differ only in state 2.



a) Find the steady-state gain per stage, g and g' , for stationary policies $\mathbf{k}$ and $\mathbf{k}'$. Show that $g = g'$.

Solution: Denote the steady-state vectors as $\boldsymbol{\pi}$ and $\boldsymbol{\pi}'$. Then $\boldsymbol{\pi} = (1/5, 4/5)$ so $g = 4$. Similarly, $\boldsymbol{\pi}' = (1/3, 2/3)$ so $g' = 4$.

b) Find the relative-gain vectors, $\mathbf{w}$ and $\mathbf{w}'$, for stationary policies $\mathbf{k}$ and $\mathbf{k}'$.

Solution: With an M state unichain, $\mathbf{w} + g\mathbf{e} = [P]\mathbf{w} + \mathbf{r}$ provides $M - 1$ independent equations for $\mathbf{w}$; $\boldsymbol{\pi}\mathbf{w} = 0$ provides the final equation. Thus we solve $w_1 + g = P_{11}w_1 + P_{12}w_2 + r_1$ and $\pi_1w_1 + \pi_2w_2$. The solution are $\mathbf{w} = (-6.4, 1.6)^\top$ and $\mathbf{w}' = (-16/3, 8/3)^\top$.

c) Suppose the final reward, at stage 0, is $u_1 = 0$, $u_2 = u$. For what range of u does the dynamic programming algorithm use decision $\mathbf{k}$ in state 2 at stage 1?

Solution: It will simplify the following sections if we do this for arbitrary u_1, u_2 . Then

$$v_1^*(1, \mathbf{u}) = \frac{u_1 + u_2}{2}; \quad v_2^*(1, \mathbf{u}) = \max \left(5 + \frac{u_1 + 7u_2}{8}, 6 + \frac{u_1 + 3u_2}{4} \right).$$

Expressing the difference $v_2^*(1, \mathbf{u}) - v_1^*(1, \mathbf{u})$ in terms of $u_2 - u_1$, this becomes

$$v_2^*(1, \mathbf{u}) - v_1^*(1, \mathbf{u}) = 8 + \frac{u_2 - u_1 - 8}{8} + \max \left(\frac{u_2 - u_1 - 8}{8}, 0 \right). \quad (\text{A.68})$$

Thus $\mathbf{k}$ is used for $u_2 - u_1 > 8$ and $\mathbf{k}'$ for $u_2 - u_1 < 8$. Either can be used for $u = 8$.

d) For what range of u does the dynamic programming algorithm use decision $\mathbf{k}$ in state 2 at stage 2? at stage n ? You should find that (for this example) the dynamic programming algorithm uses the same decision at each stage n as it uses in stage 1.

Solution: We find $v_1^*(2, \mathbf{u})$ and $v_2^*(2, \mathbf{u})$ in the same way, simply replacing u_1, u_2 by $v_1^*(1, \mathbf{u})$ and $v_2^*(1, \mathbf{u})$. We have found that decision $\mathbf{k}$ is used if $u_2 - u_1 > 8$, and it is seen from (A.68) that in this case $v_2^*(1, \mathbf{u}) - v_1^*(1, \mathbf{u}) > 8$, so that $\mathbf{k}$ is used in stage 2. The same argument works inductively for all n , so if $u_2 - u_1 > 8$ that $\mathbf{k}$ is used at all stages. Clearly, $\mathbf{k}'$ is used in all stages if $u_2 - u_1 < 8$.

e) Find the optimal gain $v_2^*(n, \mathbf{u})$ and $v_1^*(n, \mathbf{u})$ as a function of stage n assuming $u = 10$.

Solution: In this case, the stationary policy $\mathbf{k}$ is used at all stages and from (4.40),

$$\mathbf{v}^*(n, \mathbf{u}) = n\mathbf{g}\mathbf{e} + \mathbf{w} + [P^{\mathbf{k}}]^n(\mathbf{u} - \mathbf{w}).$$

f) Find $\lim_{n \rightarrow \infty} [v^*(n, \mathbf{u}) - n\mathbf{g}\mathbf{e}]$ and show how it depends on $\mathbf{u}$.

Solution: From (e), using the limiting form for $[P^{\mathbf{k}}]^n$,

$$\mathbf{v}^*(n, \mathbf{u}) - n\mathbf{g}\mathbf{e} = \mathbf{w} + \mathbf{e}\boldsymbol{\pi}(\mathbf{u} - \mathbf{w}) = \mathbf{w} + \mathbf{e}\boldsymbol{\pi}\mathbf{u},$$

where $\mathbf{w}$ and $\boldsymbol{\pi}$ should be primed if $u_2 - u_1 < 8$. As a sanity check, $\mathbf{w} + \mathbf{e}\boldsymbol{\pi}\mathbf{u} > \mathbf{w}' + \mathbf{e}\boldsymbol{\pi}'\mathbf{u}'$ if $u_2 - u_1 > 8$ with the equality reversed if $u_2 - u_1 < 8$.

Exercise 4.35: Consider a Markov decision problem in which the stationary policies $\mathbf{k}$ and $\mathbf{k}'$ each satisfy (4.50) and each correspond to ergodic Markov chains.

a) Show that if $\mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{w}' \geq \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{w}'$ is not satisfied with equality, then $g' > g$.

Solution: The solution is very similar to the proof of Lemma 4.6.5. Since $[P^{\mathbf{k}}]$ is ergodic, $\boldsymbol{\pi}^{\mathbf{k}}$ is strictly positive. Now $\mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{w}' \geq \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{w}'$ must be satisfied because of (4.50), and if one or more components are not satisfied with equality, then

$$\boldsymbol{\pi}^{\mathbf{k}}(\mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{w}') > \boldsymbol{\pi}^{\mathbf{k}}(\mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}]\mathbf{w}') = g + \boldsymbol{\pi}^{\mathbf{k}}\mathbf{w}', \quad (\text{A.69})$$

where in the equality, we used the definition of g and the fact that $\boldsymbol{\pi}^{\mathbf{k}}$ is an eigenvector of $[P^{\mathbf{k}}]$. Since $\mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}]\mathbf{w}' = \mathbf{w}' + g'\mathbf{e}$ from (4.37), (A.69) simplifies to

$$\boldsymbol{\pi}^{\mathbf{k}}\mathbf{w}' + g'\boldsymbol{\pi}^{\mathbf{k}}\mathbf{e} > g + \boldsymbol{\pi}^{\mathbf{k}}\mathbf{w}'.$$

After canceling $\pi^k \mathbf{w}'$ from each side, it is clear that $g' > g$.

b) Show that $\mathbf{r}^{k'} + [P^{k'}]\mathbf{w}' = \mathbf{r}^k + [P^k]\mathbf{w}'$ (Hint: use (a)).

Solution: The ergodicity of $\mathbf{k}$ and $\mathbf{k}'$ assures the inherently reachable assumption of Theorem 4.6.8, and thus we know from the fact that $\mathbf{k}$ satisfies the termination condition of the policy improvement algorithm that $g \geq g'$. Thus $g' > g$ is impossible and none of the components of $\mathbf{r}^{k'} + [P^{k'}]\mathbf{w}' \geq \mathbf{r}^k + [P^k]\mathbf{w}'$ can be satisfied with inequality, *i.e.*,

$$\mathbf{r}^{k'} + [P^{k'}]\mathbf{w}' \geq \mathbf{r}^k + [P^k]\mathbf{w}'.$$

c) Find the relationship between the relative gain vector $\mathbf{w}$ for policy $\mathbf{k}$ and the relative-gain vector $\mathbf{w}'$ for policy $\mathbf{k}'$. (Hint: Show that $\mathbf{r}^k + [P^k]\mathbf{w}' = g\mathbf{e} + \mathbf{w}'$; what does this say about $\mathbf{w}$ and $\mathbf{w}'$?)

Solution: Using (4.37) and (b), we have

$$\mathbf{w}' + g'\mathbf{e} = \mathbf{r}^{k'} + [P^{k'}]\mathbf{w}' = \mathbf{r}^k + [P^k]\mathbf{w}'.$$

We also have

$$\mathbf{w} + g\mathbf{e} = \mathbf{r}^k + [P^k]\mathbf{w}.$$

Subtracting the second equation from the first, and remembering that $g' = g$, we get

$$\mathbf{w}' - \mathbf{w} = [P^k](\mathbf{w}' - \mathbf{w}).$$

Thus $\mathbf{w}' - \mathbf{w}$ is a right eigenvector (of eigenvalue 1) of $[P^k]$. This implies that $\mathbf{w}' - \mathbf{w} = \alpha\mathbf{e}$ for some $\alpha \in \mathbb{R}$. Note that this was seen in the special case of Exercise 4.34(d).

d) Suppose that policy $\mathbf{k}$ uses decision 1 in state 1 and policy $\mathbf{k}'$ uses decision 2 in state 1 (*i.e.*, $k_1 = 1$ for policy $\mathbf{k}$ and $k_1 = 2$ for policy $\mathbf{k}'$). What is the relationship between $r_1^{(k)}, P_{11}^{(k)}, P_{12}^{(k)}, \dots, P_{1J}^{(k)}$ for k equal to 1 and 2?

Solution: The first component of the equation $\mathbf{r}^{k'} + [P^{k'}]\mathbf{w}' = \mathbf{w}' + g\mathbf{e}$ is $r_1^{(2)} + \sum_{j=1}^M P_{1j}^{(2)} w'_j = w_1^{(1)} g$. From (c), $\mathbf{w}' = \mathbf{w} + \alpha\mathbf{e}$, so we have

$$r_1^{(2)} + \sum_{j=1}^M P_{1j}^{(2)} (w_j + \alpha) = g(w_1 + \alpha).$$

Since α cancels out,

$$r_1^{(2)} + \sum_{j=1}^M P_{1j}^{(2)} w_j = r_1^{(1)} + \sum_{j=1}^M P_{1j}^{(1)} w_j.$$

e) Now suppose that policy $\mathbf{k}$ uses decision 1 in each state and policy $\mathbf{k}'$ uses decision 2 in each state. Is it possible that $r_i^{(1)} > r_i^{(2)}$ for all i ? Explain carefully.

Solution: It seems a little surprising, since $\pi^k \mathbf{r}^k = \pi^{k'} \mathbf{r}^{k'}$, but Exercise 4.34 essentially provides an example. In that example, r_1 is fixed over both policies, but by providing a

choice in state 1 like that provided in state 2, one can create a policy with $g = g'$ where $r_i^{(1)} > r_i^{(2)}$ for all i .

f) Now assume that $r_i^{(1)}$ is the same for all i . Does this change your answer to (e)? Explain.

Solution: If $r_i^{(1)}$ is the same, say r for all i , then $g = r$. The only way to achieve $g' = r$ for all i is for $r_i^{(2)} \geq r$ for at least one i . Thus we cannot have $r_i^{(1)} > r_i^{(2)}$ for all i .

Exercise 4.36: Consider a Markov decision problem with three states. Assume that each stationary policy corresponds to an ergodic Markov chain. It is known that a particular policy $\mathbf{k}' = (k_1, k_2, k_3) = (2, 4, 1)$ is the unique optimal stationary policy (i.e., the gain per stage in steady state is maximized by always using decision 2 in state 1, decision 4 in state 2, and decision 1 in state 3). As usual, $r_i^{(k)}$ denotes the reward in state i under decision k , and $P_{ij}^{(k)}$ denotes the probability of a transition to state j given state i and given the use of decision k in state i . For each of the following ways of modifying the rewards in the Markov decision problem, find first whether the gain per stage of policy $\mathbf{k}'$ is increased, decreased, or unchanged, and second, whether it is possible that another policy becomes optimal after the change. The changes in each part are to be considered in the absence of the changes in the other parts:

a) $r_1^{(1)}$ is replaced by $r_1^{(1)} - 1$.

Solution: The gain g' for policy $\mathbf{k}'$ is unchanged since $\mathbf{k}'$ uses decision 2 rather than 1 in state 1. Each other policy, say $\mathbf{k}$ initially has a gain $g < g'$, and if $\mathbf{k}$ uses decision 1 in state 1, its gain decreases and otherwise its gain remains the same. Thus $\mathbf{k}'$ must still be optimal.

b) $r_1^{(2)}$ is replaced by $r_1^{(2)} + 1$.

Solution: Since the chain for policy $\mathbf{k}'$ is ergodic, state 1 is used with positive probability so g' must increase, say to $\tilde{g}'$. Surprisingly, there might be a policy $\mathbf{k}$ whose modified gain $\tilde{g}$ is greater than $\tilde{g}'$. To see this, suppose the rewards using policy $\mathbf{k}'$ are all zero, and the rewards for all other decisions in all states are slightly negative. Then $\mathbf{k}'$ is obviously optimal with $g' = 0$. When $r_1^{(2)}$ is increased to 1, there is the possibility of added gain per stage by using decisions in the other states that increase the steady-state probability of state 1, since the slight negative gain in the other states is negligible. Note that if there is such a new optimal policy $\mathbf{k}$, it still uses decision 2 in state 1; it is the decisions in the other states that are changed.

c) $r_1^{(k)}$ is replaced by $r_1^{(k)} + 1$ for all state 1 decisions k .

Solution: As in (b), g' increases and some other policy might become optimal. Here it is possible that only the decision in state 1 changes, since under the same type of example as before, another decision in state 1 might increase the probability of being in state 1 with only a negligible loss of gain in each visit to state 1.

d) For all i , $r_i^{(k_i)}$ is replaced by $r_i^{(k_i)} + 1$ for the decision k_i of policy $\mathbf{k}'$.

Solution: Here the gain using policy $\mathbf{k}'$ is increased by 1 in each stage, so g' increases by exactly 1. As we now show, $\mathbf{k}'$ is still the optimal policy after the change. For any policy $\mathbf{k}$, the increase in gain is 1 in each state in which $\mathbf{k}$ uses the same decision as $\mathbf{k}'$ and the increase is 0 in each other state. Thus the increase in gain per stage, g with policy $\mathbf{k}$ is at most 1. Since $g < g'$ before the change, $g < g'$ after the change.

Exercise 4.37: (The Odoni Bound) Let $\mathbf{k}'$ be the optimal stationary policy for a Markov decision problem and let g' and $\boldsymbol{\pi}'$ be the corresponding gain and steady-state probability respectively. Let $v_i^*(n, \mathbf{u})$ be the optimal dynamic expected reward for starting in state i at stage n with final reward vector $\mathbf{u}$.

a) Show that $\min_i [v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})] \leq g' \leq \max_i [v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})]$; $n \geq 1$. Hint: Consider premultiplying $v^*(n, \mathbf{u}) - v^*(n-1, \mathbf{u})$ by $\boldsymbol{\pi}'$ or $\boldsymbol{\pi}$ where $\mathbf{k}$ is the optimal dynamic policy at stage n .

Solution: For the upper bound,

$$\begin{aligned} \mathbf{v}^*(n, \mathbf{u}) &= \max_{\mathbf{k}} \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}] \mathbf{v}^*(n-1, \mathbf{u}) \\ &\geq \mathbf{r}^{\mathbf{k}'} + [P^{\mathbf{k}'}] \mathbf{v}^*(n-1, \mathbf{u}), \end{aligned} \quad (\text{A.70})$$

where $\mathbf{k}'$ is the optimal stationary policy. Premultiplying both sides by $\boldsymbol{\pi}^{\mathbf{k}'}$,

$$\boldsymbol{\pi}^{\mathbf{k}'} \mathbf{v}^*(n, \mathbf{u}) \geq g' + \boldsymbol{\pi}^{\mathbf{k}'} \mathbf{v}^*(n-1, \mathbf{u}).$$

$$\boldsymbol{\pi}^{\mathbf{k}'} \left(\mathbf{v}^*(n, \mathbf{u}) - \mathbf{v}^*(n-1, \mathbf{u}) \right) \geq g'.$$

Thus it is impossible for all components of $\mathbf{v}^*(n, \mathbf{u}) - \mathbf{v}^*(n-1, \mathbf{u})$ to be less than g' , so $g' \leq \max_i v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})$.

For the lower bound, premultiply both sides of (A.70) by $\boldsymbol{\pi}^{\mathbf{k}}$ where $\mathbf{k}$ maximizes (A.70). Then

$$\boldsymbol{\pi}^{\mathbf{k}} \left(\mathbf{v}^*(n, \mathbf{u}) - \mathbf{v}^*(n-1, \mathbf{u}) \right) = g,$$

where $g = \boldsymbol{\pi}^{\mathbf{k}} \mathbf{r}^{\mathbf{k}}$. Assuming inherent reachability, as we do throughout, Theorem 4.6.8 shows that $g < g'$. Thus $\min_i [v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})] \leq g'$.

b) Show that the lower bound is non-decreasing in n and the upper bound is non-increasing in n and both converge to g' with increasing n .

Solution: Let $\mathbf{k}$ and $\mathbf{k}_1$ be the optimal dynamic policies at stages n and $n-1$ respectively for final gain $\mathbf{u}$. Then

$$\mathbf{v}^*(n, \mathbf{u}) = \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}] \mathbf{v}^*(n-1, \mathbf{u}) \geq \mathbf{r}^{\mathbf{k}_1} + [P^{\mathbf{k}_1}] \mathbf{v}^*(n-1, \mathbf{u}) \quad (\text{A.71})$$

$$\mathbf{v}^*(n-1, \mathbf{u}) = \mathbf{r}^{\mathbf{k}_1} + [P^{\mathbf{k}_1}] \mathbf{v}^*(n-2, \mathbf{u}) \geq \mathbf{r}^{\mathbf{k}} + [P^{\mathbf{k}}] \mathbf{v}^*(n-2, \mathbf{u}). \quad (\text{A.72})$$

Taking the difference of these equations, using the equality in (A.71) and the inequality in (A.72), we get the upper bound ,

$$\begin{aligned} \mathbf{v}^*(n, \mathbf{u}) - \mathbf{v}^*(n-1, \mathbf{u}) &\leq [P^{\mathbf{k}}] \left(\mathbf{v}^*(n-1, \mathbf{u}) - \mathbf{v}^*(n-2, \mathbf{u}) \right) \\ &\leq \max_i (v_i^*(n-1, \mathbf{u}) - v_i^*(n-2, \mathbf{u})) [P^{\mathbf{k}}] \mathbf{e} \\ &= \max_i (v_i^*(n-1, \mathbf{u}) - v_i^*(n-2, \mathbf{u})) \mathbf{e}. \end{aligned}$$

Each component on the left is upper bounded by the coefficient of $\mathbf{e}$ on the right, so

$$\max_i (v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})) \leq \max_i (v_i^*(n-1, \mathbf{u}) - v_i^*(n-2, \mathbf{u})).$$

The lower bound is virtually the same. We start with the difference of the inequality in (A.71) and the equality in (A.72) to get

$$\mathbf{v}^*(n, \mathbf{u}) - \mathbf{v}^*(n-1, \mathbf{u}) \geq [P^{k_1}] \left(\mathbf{v}^*(n-1, \mathbf{u}) - \mathbf{v}^*(n-2, \mathbf{u}) \right).$$

In the same way as before, then,

$$\min_i (v_i^*(n, \mathbf{u}) - v_i^*(n-1, \mathbf{u})) \geq \min_i (v_i^*(n-1, \mathbf{u}) - v_i^*(n-2, \mathbf{u})).$$

The upper and lower bounds do not necessarily approach each other. A counter example, without even any alternative policies, is given by a two state periodic chain with 0 reward in each state and a final reward of $(1, 0)^\top$. They do approach each other with the extra condition that the optimal policy is unique and ergodic, but we don't prove that here.

Exercise 4.38: Consider an integer-time queueing system with a finite buffer of size 2. At the beginning of the n^{th} time interval, the queue contains at most two customers. There is a cost of one unit for each customer in queue (i.e., the cost of delaying that customer). If there is one customer in queue, that customer is served. If there are two customers, an extra server is hired at a cost of 3 units and both customers are served. Thus the total immediate cost for two customers in queue is 5, the cost for one customer is 1, and the cost for 0 customers is 0. At the end of the n th time interval, either 0, 1, or 2 new customers arrive (each with probability 1/3).

a) Assume that the system starts with $0 \leq i \leq 2$ customers in queue at time -1 (i.e., in stage 1) and terminates at time 0 (stage 0) with a final cost $\mathbf{u}$ of 5 units for each customer in queue (at the beginning of interval 0). Find the expected aggregate cost $v_i(1, \mathbf{u})$ for $0 \leq i \leq 2$.

Solution: The aggregate cost at stage 1 is the immediate cost plus the expected final cost at time 0, i.e.,

$$v_i(1, \mathbf{u}) = r_i + \sum_j P_{ij} u_j,$$

where $r_0 = 0, r_1 = 1$, and $r_2 = 5$, where the cost of hiring an extra server is included as part of the immediate cost. Since $P_{ij} = 1/3$ for all i, j , the expected final cost is 5 for all i . Thus,

$$v_0(1, \mathbf{u}) = 5, \quad v_1(1, \mathbf{u}) = 6, \quad v_2(1, \mathbf{u}) = 10.$$

b) Assume now that the system starts with i customers in queue at time -2 with the same final cost at time 0. Find the expected aggregate cost $v_i(2, \mathbf{u})$ for $0 \leq i \leq 2$.

Solution: As in dynamic programming (simplified by having no policy choice),

$$v_i(2, \mathbf{u}) = r_i + \sum_j P_{ij} v_j(1, \mathbf{u}) = r_i + 7.$$

$$v_0(2, \mathbf{u}) = 7, \quad v_1(2, \mathbf{u}) = 8, \quad v_2(2, \mathbf{u}) = 12.$$

c) For an arbitrary starting time $-n$, find the expected aggregate cost $v_i(n, \mathbf{u})$ for $0 \leq i \leq 2$.

Solution: Using the same approach as in (b),

$$v_i(n, \mathbf{u}) = r_i + \sum_j P_{ij} v_j(n-1, \mathbf{u}). \quad (\text{A.73})$$

We note that in going from $n = 1$ to $n = 2$, $v_i(2, \mathbf{u}) = 2 + v_i(1, \mathbf{u})$, and thus

$$\sum_j P_{ij} v_j(2, \mathbf{u}) = 2 + \sum_{ij} P_{ij} v_j(1, \mathbf{u}).$$

We can take this as the basis for an inductive proof that

$$\sum_j P_{ij} v_j(n, \mathbf{u}) = 2 + \sum_{ij} P_{ij} v_j(n-1, \mathbf{u}).$$

The inductive step follows from (A.73) by multiplying by P_{ij} and summing over i . We thus see that

$$v_i(n, \mathbf{u}) = r_i + 3 + 2n.$$

To be a little more honest and a little less elegant, one would calculate $v_i(n, \mathbf{u})$ for $n = 3$ (and perhaps $n = 4$), see the pattern, and then look for the inductive proof.

d) Find the cost per stage and find the relative cost (gain) vector.

Solution: This was essentially done in (c). The gain (cost) per stage is 2 and the relative gain (cost) vector is $(3, 4, 8)$.

e) Now assume that there is a decision maker who can choose whether or not to hire the extra server when there are two customers in queue. If the extra server is not hired, the 3 unit fee is saved, but only one of the customers is served. If there are two arrivals in this case, assume that one is turned away at a cost of 5 units. Find the minimum dynamic aggregate expected cost $v_i^*(1)$, $0 \leq i \leq$, for stage 1 with the same final cost as before.

Solution: We have analyzed the stationary policy where the extra server is hired in state 2. The possible new policy involves a decision in state 2 which we denote by primes. For the decision not to hire an additional server, the cost of the server is eliminated but the cost of throwing away an arrival given 2 new arrivals is 5. It is more convenient to view the cost of throwing away an arrival as an immediate cost rather than a final cost (the student is welcome to pursue the other alternative, but it becomes unwieldy). Thus $r'_2 = 2 + 5/3 = 11/3$. Also $P'_{20} = 0$, $P'_{21} = 1/3$ and $P'_{22} = 2/3$.

We see that the minimum cost decision at stage 1 in states 0 and 1 does not involve the decision in state 2, so $v_0^*(1, \mathbf{u}) = 1$ and $v_1^*(1, \mathbf{u}) = 2$ as before. For state 2,

$$v_2^*(1, \mathbf{u}) = \min \left(10, \quad r'_2 + \sum_j P'_{2j} u_j \right) = \min \left(10, \quad \frac{11}{3} + \frac{5}{3} + \frac{20}{3} \right).$$

This is the minimum of 10 and 12, which is 10, showing that the original decision to hire an extra server is optimal at stage 1.

f) Find the minimum dynamic aggregate expected cost $v_i^*(n, \mathbf{u})$ for stage n , $0 \leq i \leq 2$.

Solution: Repeating (b) - (d) with this additional decision, we see that the decision to hire an extra server is optimal at each stage.

g) Now assume a final cost $\mathbf{u}$ of one unit per customer rather than 5, and find the new minimum dynamic aggregate expected cost $v_i^*(n, \mathbf{u})$, $0 \leq i \leq 2$.

Solution: The decision to hire the extra server in state 2 at stage 1 has immediate cost 5 (as before) but average final cost 1. Thus $v_2(1, \mathbf{u}') = 6$ where $\mathbf{u}' = (0, 1, 2)^\top$. The cost under the decision to not hire is

$$v'_2(1, \mathbf{u}') = r'_2 + \sum_j P'_{ij} u'_j = \frac{11}{3} + \frac{1}{3} + \frac{4}{3} = \frac{16}{3}.$$

Thus the optimal policy at stage 1 is to not hire the extra server in state and

$$v_0^*(1, \mathbf{u}') = 1; \quad v_1^*(1, \mathbf{u}') = 2; \quad v_2^*(1, \mathbf{u}') = \frac{16}{3}.$$

Using these results for the stage 2 computation, we find that $v_2(2, \mathbf{u}') = 70/9$ and $v'_2(2, \mathbf{u}') = 71/9$, so that hiring the extra server is the optimal decision and $v_2^*(2) = 70/9$.

$$v_0^*(2, \mathbf{u}') = \frac{25}{9}; \quad v_1^*(2, \mathbf{u}') = \frac{34}{9}; \quad v_2^*(2, \mathbf{u}') = \frac{70}{9}.$$

This can be rewritten as $v^*(2, \mathbf{u}') = 4 + \mathbf{w} - \frac{38e}{9}$. We can use this to hypothesize that $v^*(n, \mathbf{u}') = 2n + \mathbf{w} - \frac{38e}{9}$. This can be checked for $n \geq 2$ in general by induction, comparing $v_2(n, \mathbf{u}')$ with $v'_2(n, \mathbf{u}')$ to see that hiring an extra server is the minimum cost decision for each $n \geq 2$ when in state 2.

A.5 Solutions for Chapter 5

Exercise 5.1: The purpose of this exercise is to show that for an arbitrary renewal process, $N(t)$, the number of renewals in $(0, t]$ is a (non-defective) random variable.

a) Let $X_1, X_2, \dots$ be a sequence of IID inter-renewal rv's. Let $S_n = X_1 + \dots + X_n$ be the corresponding renewal epochs for each $n \geq 1$. Assume that each X_i has a finite expectation $\bar{X} > 0$ and, for any given $t > 0$, use the weak law of large numbers to show that $\lim_{n \rightarrow \infty} \Pr\{S_n \leq t\} = 0$.

Solution: From the WLLN, (1.75),

$$\lim_{n \rightarrow \infty} \Pr\left\{\left|\frac{S_n}{n} - \bar{X}\right| > \epsilon\right\} = 0 \quad \text{for every } \epsilon > 0.$$

Choosing $\epsilon = \bar{X}/2$, say, and looking only at the lower limit above, $\lim_{n \rightarrow \infty} \Pr\{S_n < n\bar{X}/2\} = 0$. For any given t and all large enough n , $t < n\bar{X}/2$, so $\lim_{n \rightarrow \infty} \Pr\{S_n \leq t\} = 0$.

b) Use (a) to show that $\lim_{n \rightarrow \infty} \Pr\{N(t) \geq n\} = 0$ for each $t > 0$ and explain why this means that $N(t)$ is a rv, *i.e.*, is not defective.

Solution: Since $\{S_n \leq t\} = \{N(t) \geq n\}$, we see that $\lim_{n \rightarrow \infty} \Pr\{N(t) \geq n\} = 0$ for each $t > 0$. Since $N(t)$ is nonnegative, this shows that it is a rv.

c) Now suppose that the X_i do not have a finite mean. Consider truncating each X_i to $\check{X}_i$, where for any given $b > 0$, $\check{X}_i = \min(X_i, b)$. Let $\check{N}(t)$ be the renewal counting process for the inter-renewal intervals $\check{X}_i$. Show that $\check{N}(t)$ is non-defective for each $t > 0$. Show that $N(t) \leq \check{N}(t)$ and thus that $N(t)$ is non-defective. Note: Large inter-renewal intervals create small values of $N(t)$, and thus $E[X] = \infty$ has nothing to do with potentially large values of $N(t)$, so the argument here was purely technical.

Solution: Since $\Pr\{\check{X} > b\} = 0$, we know that $\check{X}$ has a finite mean, and consequently, from (b), $\check{N}(t)$ is a rv for each $t > 0$. Since $\check{X}_n \leq X_n$ for each n , we also have $\check{S}_n \leq S_n$ for all $n \geq 1$. Thus if $\check{S}_n > t$, we also have $S_n > t$. Consequently, if $\check{N}(t) < n$, we also have $N(t) < n$. It follows that $\Pr\{N(t) \geq n\} \leq \Pr\{\check{N}(t) \geq n\}$, so $N(t)$ is also a rv.

Exercise 5.2: This exercise shows that, for an arbitrary renewal process, $N(t)$, the number of renewals in $(0, t]$, has finite expectation.

a) Let the inter-renewal intervals have the CDF $F_X(x)$, with, as usual, $F_X(0) = 0$. Using whatever combination of mathematics and common sense is comfortable for you, show that for any ϵ , $0 < \epsilon < 1$, there is a $\delta > 0$ such that $F_X(\delta) \leq 1 - \epsilon$. In other words, you are to show that a positive rv must lie in some range of positive values bounded away from 0 with positive probability.

Solution: Consider the sequence of events $\{X > 1/k; k \geq 1\}$. The union of these events is the event $\{X > 0\}$. Since $\Pr\{X \leq 0\} = 0$, $\Pr\{X > 0\} = 1$. The events $\{X > 1/k\}$ are nested in k , so that, from (1.9),

$$1 = \Pr\left\{\bigcup_k \{X > 1/k\}\right\} = \lim_{k \rightarrow \infty} \Pr\{X > 1/k\}.$$

Thus, for any $0 < \epsilon < 1$, and any k large enough, $\Pr\{X > 1/k\} > \epsilon$. Taking δ to be $1/k$ for that value of k shows that $\Pr\{X \leq \delta\} \leq 1 - \epsilon$. Another equally good approach is to use the continuity from the right of F_X .

b) Show that $\Pr\{S_n \leq \delta\} \leq (1 - \epsilon)^n$.

Solution: S_n is the sum of n interarrival times, and, bounding very loosely, $S_n \leq \delta$ implies that $X_i \leq \delta$ for each i , $1 \leq i \leq n$. Using the ϵ and δ of (a), $\Pr\{X_i \leq \delta\} \leq (1 - \epsilon)$ for $1 \leq i \leq n$. Since the X_i are independent, we then have $\Pr\{S_n \leq \delta\} \leq (1 - \epsilon)^n$.

c) Show that $\mathbb{E}[N(\delta)] \leq 1/\epsilon$.

Solution: Since $N(t)$ is nonnegative and integer,

$$\begin{aligned} \mathbb{E}[N(\delta)] &= \sum_{n=1}^{\infty} \Pr\{N(\delta) \geq n\} \\ &= \sum_{n=1}^{\infty} \Pr\{S_n \leq \delta\} \\ &\leq \sum_{n=1}^{\infty} (1 - \epsilon)^n \\ &= \frac{1 - \epsilon}{1 - (1 - \epsilon)} = \frac{1 - \epsilon}{\epsilon} \leq \frac{1}{\epsilon}. \end{aligned}$$

d) For the ϵ, δ of (a), show that for every integer k , $\mathbb{E}[N(k\delta)] \leq k/\epsilon$ and thus that $\mathbb{E}[N(t)] \leq \frac{t+\delta}{\epsilon\delta}$ for any $t > 0$.

Solution: The solution of (c) suggests breaking the interval $(0, k\delta]$ into k intervals each of size δ . Letting $\tilde{N}_i = N(i\delta) - N((i-1)\delta)$ be the number of arrivals in the i th of these intervals, we have $\mathbb{E}[N(k\delta)] = \sum_{i=1}^k \mathbb{E}[\tilde{N}_i]$.

For the first of these intervals, we have shown that $\mathbb{E}[\tilde{N}_1] \leq 1/\epsilon$, but that argument does not quite work for the subsequent intervals, since the first arrival in an interval $i > 1$ might correspond to an interarrival interval that starts early in the interval $i-1$ and ends late in interval i . Subsequent arrivals in interval i must correspond to interarrival intervals that both start and end in interval i and thus have duration at most δ . Thus if we let $S_n^{(i)}$ be the number of arrivals in the i th interval, we have

$$\Pr\{S_n^{(i)} \leq \delta\} \leq (1 - \epsilon)^{n-1}.$$

Repeating the argument in (c) for $i > 1$, then,

$$\begin{aligned} \mathbb{E}[\tilde{N}_i] &= \sum_{n=1}^{\infty} \Pr\{\tilde{N}_i \geq n\} = \sum_{n=1}^{\infty} \Pr\{S_n^{(i)} \leq \delta\} \\ &\leq \sum_{n=1}^{\infty} (1 - \epsilon)^{n-1} = \frac{1}{1 - (1 - \epsilon)} = \frac{1}{\epsilon}. \end{aligned}$$

Since $E[N(\delta k)] = \sum_{i=1}^k E[\tilde{N}_i]$, we then have

$$E[N(k\delta)] \leq k/\epsilon.$$

Since $N(t)$ is non-decreasing in t , $N(t)$ can be upper bounded by replacing t by the smallest integer multiple of $1/\delta$ that is t or greater, *i.e.*,

$$E[N(t)] \leq E[N(\delta \lceil t/\delta \rceil)] \leq \frac{\lceil t/\delta \rceil}{\epsilon} \leq \frac{(t/\delta) + 1}{\epsilon}.$$

e) Use the result here to show that $N(t)$ is non-defective.

Solution: Since $N(t)$, for each t , is nonnegative and has finite expectation, it is obviously non-defective. One way to see this is that $E[N(t)]$ is the integral of the complementary CDF, $F_{N(t)}^c(n)$ of $N(t)$. Since this integral is finite, $F_{N(t)}^c(n)$ must approach 0 with increasing n .

Exercise 5.3: Let $\{X_i; i \geq 1\}$ be the inter-renewal intervals of a renewal process generalized to allow for inter-renewal intervals of size 0 and let $\Pr\{X_i=0\} = \alpha$, $0 < \alpha < 1$. Let $\{Y_i; i \geq 1\}$ be the sequence of non-zero interarrival intervals. For example, if $X_1 = x_1 > 0$, $X_2 = 0$, $X_3 = x_3 > 0, \dots$, then $Y_1 = x_1$, $Y_2 = x_3, \dots$

a) Find the CDF of each Y_i in terms of that of the X_i .

Solution: The intuitive approach here is to note that each positive inter-renewal interval has the distribution of a generalized renewal, conditional on the generalized renewal being positive. Thus the positive inter-renewal intervals are IID with the distribution $\Pr\{Y > y\} = \Pr\{X > y\} / \Pr\{X > 0\}$. Rearranging this in terms of the CDF,

$$F_Y(y) = 1 - \frac{1 - F_X(y)}{1 - F_X(0)} = \frac{F_X(y) - F_X(0)}{1 - F_X(0)}. \quad (\text{A.74})$$

If one wishes to see this worked out with all the steps, we can start with Y_1 . Conditional on $X_1 > 0$, Y_1 has the CDF in (A.74). Next, conditional on $X_1 = 0$ and $X_2 > 0$, Y_1 again has the CDF in (A.74). Continuing with $X_1 = 0$, $X_2 = 0$ and $X_3 > 0$ etc. we get (A.74) in general. We can then continue with $Y_2, Y_3, \dots$ in the same way.

b) Find the PMF of the number of arrivals of the generalized renewal process at each epoch at which arrivals occur.

Solution: There is no arrival at $t = 0$ if $X_1 > 0$. There is one arrival if $X_1 = 0$, $X_2 > 0$, etc. Letting the rv K be the number of arrivals at 0, $p_K(k) = (\Pr\{X = 0\})^k \Pr\{X > 0\}$. If there are arrivals at $t = 0$ (*i.e.*, given one or more arrivals at $t = 0$), then $\Pr\{K = k | K \geq 1\} = (\Pr\{X = 0\})^{k-1} \Pr\{X > 0\}$. This also gives the probability of k arrivals at any epoch at which 1 or more arrivals occur.

c) Explain how to view the generalized renewal process as an ordinary renewal process with inter-renewal intervals $\{Y_i; i \geq 1\}$ and bulk arrivals at each renewal epoch.

Solution: The generalized renewal process here is essentially a special case of bulk arrivals at the epochs of an ordinary renewal process; the number of bulk arrivals at each ordinary arrival epoch is geometrically distributed (with one or more arrivals). There is a peculiarity here about arrivals at $t = 0$. For an ordinary renewal process with geometric bulk arrivals, there are no arrivals at time zero, whereas here there are geometrically many (with 0 or more arrivals) at $t = 0$. It is difficult to imagine an application in which arrivals at time 0 are desirable, let alone being geometric (with zero or more arrivals) rather than with 1 or more arrivals.

d) When a generalized renewal process is viewed as an ordinary renewal process with bulk arrivals, what is the distribution of the bulk arrivals? (The point of this part is to illustrate that bulk arrivals on an ordinary renewal process are considerably more general than generalized renewal processes.)

Solution: As seen in (b), the number of bulk arrivals at each ordinary arrival epoch is geometrically distributed with one or more arrivals.

Exercise 5.4: Is it true for a renewal process that:

- a) $N(t) < n$ if and only if $S_n > t$?
- b) $N(t) \leq n$ if and only if $S_n \geq t$?
- c) $N(t) > n$ if and only if $S_n < t$?

Solution: Part (a) is true, as pointed out in (5.1) and more fully explained in (2.2) and (2.3).

Parts b) and c) are false, as seen by any situation where $S_n < t$ and $S_{n+1} > t$. In these cases, $N(t) = n$. In other words, one must be careful about strict versus non-strict inequalities when using $\{S_n \leq t\} = \{N(t) \geq n\}$

Exercise 5.5: (This shows that convergence WP1 implies convergence in probability.) Let $\{Y_n; n \geq 1\}$ be a sequence of rv's that converges to 0 WP1. For any positive integers m and k , let

$$A(m, k) = \{\omega : |Y_n(\omega)| \leq 1/k \text{ for all } n \geq m\}.$$

a) Show that if $\lim_{n \rightarrow \infty} Y_n(\omega) = 0$ for some given ω , then (for any given k) $\omega \in A(m, k)$ for some positive integer m .

Solution: Note that for a given ω , $\{Y_n(\omega); n \geq 1\}$ is simply a sequence of real numbers. The definition of convergence of such a sequence to 0 says that for any ϵ (or any $1/k$ where $k > 0$ is an integer), there must be an m large enough that $Y_n(\omega) \leq 1/k$ for all $n \geq m$. In other words, the given ω is contained in $A(m, k)$ for that m .

b) Show that for all $k \geq 1$

$$\Pr\left\{\bigcup_{m=1}^{\infty} A(m, k)\right\} = 1.$$

Solution: The set of ω for which $\lim_{n \rightarrow \infty} Y_n(\omega) = 0$ has probability 1. For any such ω and any given k , part (a) showed that $\omega \in A(m, k)$ for some integer $m > 0$. Thus each such ω lies in the above union, implying that the union has probability 1.

c) Show that, for all $m \geq 1$, $A(m, k) \subseteq A(m+1, k)$. Use this (plus (1.9)) to show that

$$\lim_{m \rightarrow \infty} \Pr\{A(m, k)\} = 1.$$

Solution: Note that if $|Y_n(\omega)| \leq 1/k$ for all $n \geq m$, then also $|Y_n(\omega)| \leq 1/k$ for all $n \geq m+1$. This means that $A(m, k) \subseteq A(m+1, k)$. From (1.9) then

$$1 = \Pr\left\{\bigcup_m A(m, k)\right\} = \lim_{m \rightarrow \infty} \Pr\{A(m, k)\}.$$

d) Show that if $\omega \in A(m, k)$, then $|Y_m(\omega)| \leq 1/k$. Use this (plus (c)) to show that

$$\lim_{m \rightarrow \infty} \Pr\{|Y_m| > 1/k\} = 0.$$

Since $k \geq 1$ is arbitrary, this shows that $\{Y_n; n \geq 1\}$ converges in probability.

Solution: Note that if $|Y_n(\omega)| \leq 1/k$ for all $n \geq m$, then certainly $|Y_n(\omega)| \leq 1/k$ for $n = m$. It then follows from (c) that $\lim_{m \rightarrow \infty} \Pr\{|Y_m| \leq 1/k\} = 1$, which is equivalent to the desired statement. This shows that $\{Y_n; n \geq 1\}$ converges in probability.

Exercise 5.6: A town starts a mosquito control program and the rv Z_n is the number of mosquitoes at the end of the n th year ($n = 0, 1, 2, \dots$). Let X_n be the growth rate of mosquitoes in year n ; i.e., $Z_n = X_n Z_{n-1}$; $n \geq 1$. Assume that $\{X_n; n \geq 1\}$ is a sequence of IID rv's with the PMF $\Pr\{X=2\} = 1/2$; $\Pr\{X=1/2\} = 1/4$; $\Pr\{X=1/4\} = 1/4$. Suppose that Z_0 , the initial number of mosquitos, is some known constant and assume for simplicity and consistency that Z_n can take on non-integer values.

a) Find $E[Z_n]$ as a function of n and find $\lim_{n \rightarrow \infty} E[Z_n]$.

Solution: First we find the expected value of X_n (which is the same for each n).

$$E[X_n] = \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} = \frac{19}{16}.$$

$$E[Z_n] = E[X_n] E[Z_{n-1}] = E[X_n] E[X_{n-1}] \cdots E[X_1] Z_0 = \left(\frac{19}{16}\right)^n Z_0.$$

Assuming that $Z_0 > 0$, we see that $\lim_{n \rightarrow \infty} E[Z_n] = \infty$.

b) Let $W_n = \log_2 X_n$. Find $E[W_n]$ and $E[\log_2(Z_n/Z_0)]$ as a function of n .

Solution:

$$E[W_n] = \frac{1}{2} \log_2(2) + \frac{1}{4} \log_2\left(\frac{1}{2}\right) + \frac{1}{4} \log_2\left(\frac{1}{4}\right) = -\frac{1}{4}.$$

$$\log_2(Z_n/Z_0) = \log_2(X_1 X_2 \cdots X_n) = \sum_{j=1}^n W_j.$$

Thus $E[\log(Z_n/Z_0)] = -n/4$.

c) There is a constant α such that $\lim_{n \rightarrow \infty} (1/n)[\log_2(Z_n/Z_0)] = \alpha$ with probability 1. Find α and explain how this follows from the strong law of large numbers.

Solution: From (b), we see that $(1/n) \log_2 (Z_n/Z_0)$ is the sample average of n IID rv's. The SLLN applies since $\mathbf{E}[|W_n|] < \infty$. Thus

$$\lim_{n \rightarrow \infty} (1/n) [\log_2 (Z_n/Z_0)] = -\frac{1}{4} \quad \text{WP1.} \quad (\text{A.75})$$

d) Using (c), show that $\lim_{n \rightarrow \infty} Z_n = \beta$ with probability 1 for some β and evaluate β .

Solution: For any sample value ω satisfying (A.75), and any $\epsilon > 0$, there must be an m for which

$$\frac{1}{n} \log (Z_n(\omega)/Z_0) < -\frac{1}{4} + \epsilon; \quad \text{for all } n \geq m.$$

For each such ω , $Z_n(\omega) < Z_0 2^{-n(0.25-\epsilon)}$ for all $n \geq m$. Thus (choosing $\epsilon < 0.25$), $\lim_{n \rightarrow \infty} Z_n(\omega) = 0$. It follows that $\lim_{n \rightarrow \infty} Z_n = 0$ WP1.

e) Explain carefully how the result in (a) and the result in (d) are compatible. What you should learn from this problem is that the expected value of the log of a product of IID rv's might be more significant than the expected value of the product itself.

Solution: After a very large number of years, the mosquito population will increase in about half the years and decrease in about half the years, but half the decreases lead to quartering of the population rather than halving. Thus we expect the mosquito population to gradually die out with high probability, and (d) states this precisely.

When we look at the expected number of mosquitoes, however, very rare events have an extreme effect on the average. For example, doubling each year for n years in a row has probability 2^{-n} , but leads to 2^n mosquitoes, hiding the large probability of a gradually decreasing population.

We will see a number of situations (including gambling and investing) that involve products of IID rv's. This same sort of effect is typical in these cases.

Exercise 5.7: In this exercise, you will find an explicit expression for $\{\omega : \lim_n Y_n(\omega) = 0\}$. You need not be mathematically precise.

a) Let $\{Y_n; n \geq 1\}$ be a sequence of rv's. Using the definition of convergence for a sequence of numbers, justify the following set equivalences:

$$\{\omega : \lim_n Y_n(\omega) = 0\} = \bigcap_{k=1}^{\infty} \{\omega : \text{there exists an } m \text{ such that } |Y_n(\omega)| \leq 1/k \text{ for all } n \geq m\} \quad (\text{A.76})$$

$$= \bigcap_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \{\omega : Y_n(\omega) \leq 1/k \text{ for all } n \geq m\} \quad (\text{A.77})$$

$$= \bigcap_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{\omega : Y_n(\omega) \leq 1/k\}. \quad (\text{A.78})$$

Solution: For any given sample point ω , $\{Y_n(\omega); n \geq 1\}$ is a sample sequence of the random sequence $\{Y_n; n \geq 1\}$ and is simply a sequence of real numbers. That sequence converges to 0 if for every integer $k > 0$, there is an $m(k)$ such that $Y_n(\omega) \leq 1/k$ for $n \geq k$. The set of ω that satisfies this test is given by (A.76). A set theoretic way to express the existence of an $m(k)$ is to take the union over $m \geq 1$, giving us (A.77). Finally, the set theoretic way to express 'for all $n \geq m$ ' is to take the intersection over $n \geq m$, giving us (A.78).

b) Explain how this shows that $\{\omega : \lim_n Y_n(\omega) = 0\}$ must be an event.

Solution: Since Y_n is a rv, $\{\omega : Y_n(\omega) \leq 1/k\}$ is an event for all n, k . The countable intersection over $n \geq m$ is then an event (by the axioms of probability), the countable union of these events is an event for each k , and the final intersection is an event.

c) Use De Morgan's laws to show that the complement of the above equivalence is

$$\{\omega : \lim_n Y_n(\omega) = 0\}^c = \bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\omega : Y_n(\omega) > 1/k\}. \quad (\text{A.79})$$

Solution: For a sequence of sets $A_1, A_2, \dots$, de Morgan's laws say that $\bigcup_n A_n^c = \{\bigcap_n A_n\}^c$ and also that $\bigcap_n A_n^c = \{\bigcup_n A_n\}^c$. Applying the second form of de Morgan's laws to the right side of the complement of each side of (A.78), we get

$$\begin{aligned} \{\omega : \lim_n Y_n(\omega) = 0\}^c &= \bigcup_{k=1}^{\infty} \left\{ \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{\omega : Y_n(\omega) \leq 1/k\} \right\}^c \\ &= \bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \left\{ \bigcap_{n=m}^{\infty} \{\omega : Y_n(\omega) \leq 1/k\} \right\}^c \\ &= \bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\omega : Y_n(\omega) \leq 1/k\}^c \\ &= \bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\omega : Y_n(\omega) > 1/k\} \end{aligned}$$

In the second line, the first form of de Morgan's laws were applied to the complemented term in braces on the first line. The third line applied the second form to the complemented term in the second line.

d) Show that for $\{Y_n; n \geq 1\}$ to converge to zero WP1, it is necessary and sufficient to satisfy

$$\Pr\left\{\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{Y_n > 1/k\}\right\} = 0 \quad \text{for all } k \geq 1. \quad (\text{A.80})$$

Solution: Applying the union bound to (A.79),

$$\Pr\left\{\{\omega : \lim_n Y_n(\omega) = 0\}^c\right\} \leq \sum_{k=1}^{\infty} \Pr\left\{\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{Y_n > 1/k\}\right\}.$$

If (A.80) is satisfied for all $k \geq 1$, then the above sum is 0 and $\Pr\{\{\omega : \lim_n Y_n(\omega) = 0\}^c\} = 0$. This means that $\Pr\{\lim_{n \rightarrow \infty} Y_n = 0\} = 1$, i.e., that $\lim_{n \rightarrow \infty} Y_n = 0$ WP1.

If (A.80) is not satisfied for some given k , on the other hand, then the probability of the event $\{\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{Y_n > 1/k\}\}$ must be positive, say $\epsilon > 0$, so, using (A.79) and lower bounding the probability of the union over k by the single k above,

$$\begin{aligned} \Pr\left\{\{\omega : \lim_n Y_n(\omega) = 0\}^c\right\} &= \Pr\left\{\bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\omega : Y_n(\omega) > 1/k\}\right\} \\ &\geq \Pr\left\{\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\omega : Y_n(\omega) > 1/k\}\right\} \geq \epsilon. \end{aligned}$$

Thus there is a set of ω , whose probability is at least ϵ , for which $\lim_n Y_n(\omega)$ either does not exist or does not equal 0.

e) Show that for $\{Y_n; n \geq 1\}$ to converge WP1, it is necessary and sufficient to satisfy

$$\lim_{m \rightarrow \infty} \Pr\left\{\bigcup_{n=m}^{\infty} \{Y_n > 1/k\}\right\} = 0 \quad \text{for all } k \geq 1.$$

Hint: Use (a) of Exercise 5.8. Note: (e) provides an equivalent condition that is often useful in establishing convergence WP1. It also brings out quite clearly the difference between convergence WP1 and convergence in probability.

Solution: As in Exercise 5.8, let $B_m = \bigcup_{n \geq m} \{Y_n > 1/k\}$. Then the condition in (e) is $\lim_{m \rightarrow \infty} \Pr\{B_m\} = 0$ and the necessary and sufficient condition established in (d) is $\Pr\left\{\bigcap_{m \geq 1} B_m\right\} = 0$. The equivalence of these conditions is implied by (1.10).

The solution to this exercise given here is mathematically precise and has the added benefit of showing that the various sets used in establishing the SLLN are in fact events since they are expressed as countable intersections and unions of events.

Exercise 5.8: (The Borel-Cantelli lemma) Consider the event $\bigcap_{m \geq 1} \bigcup_{n \geq m} A_n$ where $A_1, A_2, \dots$ are arbitrary events.

a) Show that

$$\lim_{m \rightarrow \infty} \Pr\left\{\bigcup_{n \geq m} A_n\right\} = 0 \quad \Longleftrightarrow \quad \Pr\left\{\bigcap_{m \geq 1} \bigcup_{n \geq m} A_n\right\} = 0.$$

Hint: Apply (1.10).

Solution: Let $B_m = \bigcup_{n \geq m} A_n$. Then we are to show that

$$\lim_{m \rightarrow \infty} \Pr\{B_m\} = 0 \quad \Longleftrightarrow \quad \Pr\left\{\bigcap_{m \geq 1} B_m\right\} = 0.$$

Since $B_m \supseteq B_{m+1}$ for each $m \geq 1$, this is implied by (1.10).

b) Show that $\sum_{m=1}^{\infty} \Pr\{A_m\} < \infty$ implies that $\lim_{m \rightarrow \infty} \Pr\left\{\bigcup_{n \geq m} A_n\right\} = 0$. Hint: First explain why $\sum_{m=1}^{\infty} \Pr\{A_m\} < \infty$ implies that $\lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} \Pr\{A_n\} = 0$.

Solution: Let $S_m = \sum_{n=1}^m \Pr\{A_n\}$. By definition of an infinite limit,

$$\sum_{m=1}^{\infty} \Pr\{A_m\} = \lim_{m \rightarrow \infty} \sum_{n=1}^m \Pr\{A_n\} = \lim_{m \rightarrow \infty} \Pr\{S_m\}.$$

Since $\Pr\{A_n\} \geq 0$, $\{\Pr\{S_m\}; m \geq 1\}$ is non-decreasing. Assume in what follows that $\sum_{m=1}^{\infty} \Pr\{A_m\} < \infty$ with some given value α . Then $\lim_{m \rightarrow \infty} \Pr\{S_m\}$ exists with the value α . We then have $\lim_{m \rightarrow \infty} (\alpha - \Pr\{S_m\}) = 0$, which means that $\lim_{m \rightarrow \infty} \sum_{n=m+1}^{\infty} \Pr\{A_n\} = 0$. Shifting the index m , $\lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} \Pr\{A_n\} = 0$. Now, using the union bound,

$$\Pr\left\{\bigcup_{n \geq m} A_n\right\} \leq \sum_{n \geq m} \Pr\{A_n\}.$$

Taking the limit as $m \rightarrow \infty$, the right side is 0 so $\lim_{m \rightarrow \infty} \Pr\left\{\bigcup_{n \geq m} A_n\right\} \leq 0$.

c) Show that if $\sum_{m=1}^{\infty} \Pr\{A_m\} < \infty$, then $\Pr\left\{\bigcap_{m \geq 1} \bigcup_{n \geq m} A_n\right\} = 0$. This well-known result is called the *Borel-Cantelli lemma*.

Solution: If $\sum_{m=1}^{\infty} \Pr\{A_m\} < \infty$, then from (b), $\lim_{m \rightarrow \infty} \Pr\left\{\bigcup_{n \geq m} A_n\right\} = 0$. From (a), this implies that $\Pr\left\{\bigcap_{m \geq 1} \bigcup_{n \geq m} A_n\right\} = 0$.

d) The set $\left\{ \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n \right\}$ is often referred to as the set of sample points ω that are contained in infinitely many of the A_n . Consider an ω that is contained in a finite number k of the sets A_n and argue that there must be an integer m such that $\omega \notin A_n$ for all $n > m$. Conversely, if ω is contained in infinitely many of the A_n , show that there cannot be such an m .

Solution: If a given ω is in any of the sets A_n , let b_1 be the smallest n for which $\omega \in A_n$. If it is in other sets, let b_2 be the next smallest, and so forth. Thus $\omega \in A_{b_i}$ for $1 \leq i \leq k$. If k is finite, then $\omega \notin A_n$ for all $n > b_k$. If k is infinite, then for all m , $\omega \in A_n$ for some $n > m$. This type of set looks very complicated at first, but it arises in many situations, and it is useful intuitively to interpret it as the event of infinitely recurring sample points.

Exercise 5.9: (Strong law for renewals where $\bar{X} = \infty$) Let $\{X_i; i \geq 1\}$ be the inter-renewal intervals of a renewal process and assume that $E[X_i] = \infty$. Let $b > 0$ be an arbitrary number and $\check{X}_i$ be a truncated random variable defined by $\check{X}_i = X_i$ if $X_i \leq b$ and $\check{X}_i = b$ otherwise.

a) Show that for any constant $M > 0$, there is a b sufficiently large so that $E[\check{X}_i] \geq M$.

Solution: Since $E[X] = \int_0^\infty F_X^c(x) dx = \infty$, we know from the definition of an integral over an infinite limit that

$$E[X] = \lim_{b \rightarrow \infty} \int_0^b F_X^c(x) dx = \infty.$$

For $\check{X} = \min(X, b)$, we see that $F_{\check{X}}(x) = F_X(x)$ for $x < b$ and $F_{\check{X}}(x) = 1$ for $x \geq b$. Thus $E[\check{X}] = \int_0^b F_X^c(x) dx$. We have just seen that the limit of this as $b \rightarrow \infty$ is ∞ , so that for any $M > 0$, there is a b sufficiently large that $E[\check{X}] \geq M$.

b) Let $\{\check{N}(t); t \geq 0\}$ be the renewal counting process with inter-renewal intervals $\{\check{X}_i; i \geq 1\}$ and show that for all $t > 0$, $\check{N}(t) \geq N(t)$.

Solution: Note that $X - \check{X}$ is a non-negative rv, i.e., it is 0 for $X \leq b$ and greater than b otherwise. Thus $\check{X} \leq X$. It follows then that for all $n \geq 1$,

$$\check{S}_n = \check{X}_1 + \check{X}_2 + \cdots + \check{X}_n \leq X_1 + X_2 + \cdots + X_n = S_n.$$

Since $\check{S}_n \leq S_n$, it follows for all $t > 0$ that if $S_n \leq t$ then also $\check{S}_n \leq t$. This then means that if $N(t) \geq n$, then also $\check{N}(t) \geq n$. Since this is true for all n , $\check{N}(t) \geq N(t)$, i.e., the number of renewals after truncation is greater than or equal to the number before truncation.

c) Show that for all sample functions $N(t, \omega)$, except a set of probability 0, $N(t, \omega)/t \leq 2/M$ for all sufficiently large t . Note: Since M is arbitrary, this means that $\lim_{t \rightarrow \infty} N(t)/t = 0$ with probability 1.

Solution: Let M and $b < \infty$ such that $E[\check{X}] \geq M$ be fixed in what follows. Since $\check{X} \leq b$, we see that $E[\check{X}] < \infty$, so we can apply Theorem 5.3.1, which asserts that

$$\Pr \left\{ \omega : \lim_{t \rightarrow \infty} \frac{\check{N}(t, \omega)}{t} = \frac{1}{E[\check{X}]} \right\} = 1.$$

Let $A(M)$ denote the set of sample points for which the above limit exists, *i.e.*, for which $\lim_{t \rightarrow \infty} \check{N}(t, \omega)/t = 1/\mathbb{E}[\check{X}]$; Thus $A(M)$ has probability 1 from Theorem 5.3.1. We will show that, for each $\omega \in A(M)$, $\lim_t N(t, \omega)/t \leq 1/(2M)$. We know that for every $\omega \in A(M)$, $\lim_t \check{N}(t, \omega)/t = 1/\mathbb{E}[\check{X}]$. The definition of the limit of a real-valued function states that for any $\epsilon > 0$, there is a $\tau(\epsilon)$ such that

$$\left| \frac{\check{N}(t, \omega)}{t} - \frac{1}{\mathbb{E}[\check{X}]} \right| \leq \epsilon \quad \text{for all } t \geq \tau(\epsilon).$$

Note that $\tau(\epsilon)$ depends on b and ω as well as ϵ , so we denote it as $\tau(\epsilon, b, \omega)$. Using only one side of this inequality, $N(t, \omega)/t \leq \epsilon + 1/\mathbb{E}[\check{X}]$ for all $t \geq \tau(\epsilon, b, \omega)$. Since we have seen that $N(t, \omega) \leq \check{N}(t, \omega)$ and $\mathbb{E}[\check{X}] \geq M$, we have

$$\frac{N(t, \omega)}{t} \leq \epsilon + \frac{1}{M} \quad \text{for all } t \geq \tau(\epsilon, b, \omega).$$

Since ϵ is arbitrary, we can choose it as $1/M$, so $N(t, \omega)/t \leq 1/(2M)$ for all sufficiently large t for each $\omega \in A(M)$. Now consider the intersection $\bigcap_M A(M)$ over integer $M \geq 1$. Since each $A(M)$ has probability 1, the intersection has probability 1 also. For all ω in this intersection, $\lim_{n \rightarrow \infty} \Pr\{N(t)/t\} \leq 1/(2M)$ for all integer $M \geq 1$, so $\lim_{n \rightarrow \infty} \Pr\{N(t)/t\} = 0$ WP1.

Exercise 5.10: The first part of the proof of the SLLN showed that if $\mathbb{E}[X^4] < \infty$, then $\mathbb{E}[X^2] < \infty$. Here we generalize this to show that for $1 \leq m < n$, if $\mathbb{E}[|X^n|] < \infty$, then $\mathbb{E}[|X^m|] < \infty$.

a) For any $x > 0$, show that $x^m \leq 1 + x^n$.

Solution: This is virtually the same as the argument used in the proof of the SLLN. If $x \leq 1$, then $x^m \leq 1$ for any $m \geq 1$ and if $x > 1$, then $x^m < x^n$ for any $m \leq n$. Thus for all $x > 0$, $x^m \leq 1 + x^n$.

b) For any rv X , show that if $\mathbb{E}[|X^n|] < \infty$, then $\mathbb{E}[|X^m|] \leq 1 + \mathbb{E}[|X^n|]$.

Solution: For any sample value $X(\omega)$ of X , (a) shows that $|X^m(\omega)| \leq 1 + |X^n(\omega)|$. Taking the expectation yields $\mathbb{E}[|X^m|] \leq 1 + \mathbb{E}[|X^n|]$.

Exercise 5.11: Let $Y(t) = S_{N(t)+1} - t$ be the residual life at time t of a renewal process. First consider a renewal process in which the interarrival time has density $f_X(x) = e^{-x}$; $x \geq 0$, and next consider a renewal process with density

$$f_X(x) = \frac{3}{(x+1)^4}; \quad x \geq 0. \quad (\text{A.81})$$

For each of the above densities, use renewal-reward theory to find:

i) the time average of $Y(t)$

Solution: From (5.17),

$$\lim_{t \rightarrow \infty} \frac{\int_{\tau=0}^t Y(\tau) d\tau}{t} = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]} \quad \text{WP1.}$$

For $f_X(x) = e^{-x}$ for $x > 0$, $E[X] = 1$ and $E[X^2] = 2$. Thus the time-average residual life is 1. This is something we already know, since if the interarrival time is exponential, the renewal process is Poisson.

For the density in (A.81), we have $E[X] = \int_0^\infty 3x(x+1)^{-4} dx = 1/2$ and $E[X^2] = \int_0^\infty 3x^2(x+1)^{-4} dx = 1$, so the time-average residual life is again 1.

ii) the second moment in time of $Y(t)$ (i.e., $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T Y^2(t) dt$).

Solution: We regard $Y^2(t)$ as simply another reward function. Given an inter-renewal interval x_n , the reward at each residual life r from 0 to x_n is r^2 , so the accumulated reward over that interval is $r_n = \int_0^{x_n} r^2 dr = x_n^3/3$. Thus the expected reward over an interval is $E[R_n] = E[X^3]/3$. Using Theorem 5.4.5, then, the time-average second moment is

$$\lim_{t \rightarrow \infty} \frac{\int_{\tau=0}^t Y^2(\tau) d\tau}{t} = \frac{E[X^3]}{3E[X]} \quad \text{WP1 if } E[X^3] < \infty.$$

For $f_X(x) = e^{-x}$, $E[X^3] = 6$ so the time-average second moment is 2.

For the density in (A.81),

$$E[X^3] = \int_0^\infty 3x^3(x+1)^{-4} dx = \infty.$$

so the time-average second moment is infinite. This is somewhat shocking, but the distribution of X has tails that go to 0 rather slowly, and the enormous acculated reward over large sample values of X_n cause this infinite reward.

For the exponential density, verify your answers by finding $E[Y(t)]$ and $E[Y^2(t)]$ directly.

Solution: The exponential density yields a Poisson process and the residual life at any time is exponential, independent of the previous arrival. Thus $E[Y(t)] = 1$ for all t and $E[Y^2(t)] = 2$ for all t .

Exercise 5.12: Consider a variation of an M/G/1 queueing system in which there is no facility to save waiting customers. Assume customers arrive according to a Poisson process of rate λ . If the server is busy, the customer departs and is lost forever; if the server is not busy, the customer enters service with a service time CDF denoted by $F_Y(y)$.

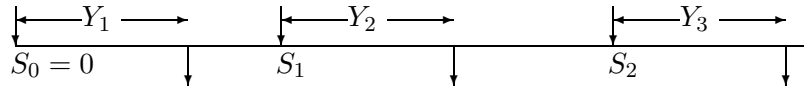
Successive service times (for those customers that are served) are IID and independent of arrival times. Assume that customer number 0 arrives and enters service at time $t = 0$.

a) Show that the sequence of times $S_1, S_2, \dots$ at which successive customers enter service are the renewal times of a renewal process. Show that each inter-renewal interval $X_i = S_i - S_{i-1}$ (where $S_0 = 0$) is the sum of two independent random variables, $Y_i + U_i$ where Y_i is the i th service time; find the probability density of U_i .

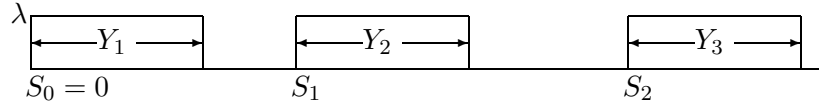
Solution: Let Y_1 be the first service time, i.e., the time spent serving customer 0. Customers who arrive during $(0, Y_1]$ are lost, and, given that $Y_1 = y$, the residual time until the next customer arrives is memoryless and exponential with rate λ . Thus the time $X_1 = S_1$ at which the next customer enters service is $Y_1 + U_1$ where U_1 is exponential with rate λ , i.e., $f_{U_1}(u) = \lambda \exp(-\lambda u)$.

At time X_1 , the arriving customer enters service, customers are dropped until $X_1 + Y_2$, and after an exponential interval U_2 of rate λ a new customer enters service at time $X_1 + X_2$ where $X_2 = Y_2 + U_2$. Both Y_2 and U_2 are independent of X_1 , so X_2 and X_1 are independent. Since the Y_i are IID and the U_i are IID, X_1 and X_2 are IID. In the same way, the sequence $X_1, X_2, \dots$ are IID intervals between successive services. Thus $\{X_i; i \geq 1\}$ is a sequence of inter-renewal intervals for a renewal process and $S_1, S_2, \dots$ are the renewal epochs.

b) Assume that a reward (actually a cost in this case) of one unit is incurred for each customer turned away. Sketch the expected reward function as a function of time for the sample function of inter-renewal intervals and service intervals shown below; the expectation is to be taken over those (unshown) arrivals of customers that must be turned away.



Solution: Customers are turned away at rate λ during the service times, so if we let $R(t)$ be the rate at which customers are turned away for a given sample path of services and arrivals, we have $R(t) = \lambda$ for t in a service interval and $R(t) = 0$ otherwise.



Note that the number of arrivals within a service interval is dependent on the length of the service interval but independent of arrivals outside of that interval and independent of other service intervals.

To be more systematic, we would define $\{Z_m; m \geq 1\}$ as the sequence of customer inter-arrival intervals. This sequence along with the service sequence $\{Y_n; n \geq 1\}$ specifies the sequences $\{S_n, Y_n; n \geq 1\}$. The discarded customer arrivals within each service interval is then Poisson of rate λ . The function $\{R(t); t > 0\}$ above would then better be described as $E[R(t) | \{S_n, Y_n; n \geq 1\}]$ where $R(t)$ would then become a unit impulse at each discarded arrival.

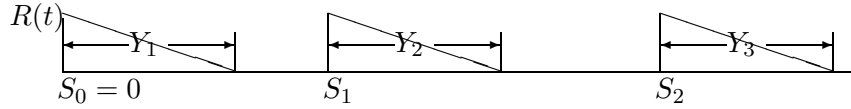
c) Let $\int_0^t R(\tau) d\tau$ denote the accumulated reward (i.e., cost) from 0 to t and find the limit as $t \rightarrow \infty$ of $(1/t) \int_0^t R(\tau) d\tau$. Explain (without any attempt to be rigorous or formal) why this limit exists with probability 1.

Solution: The reward within the n th inter-renewal interval (averaged over the discarded arrivals, but conditional on Y_n) is $R_n = \lambda Y_n$. Assuming that $E[Y_n] < \infty$, Theorem 5.4.5 asserts that average reward, WP1, is $\frac{\lambda E[Y]}{E[Y] + 1/\lambda}$. Thus the theorem asserts that the limit exists with probability 1. If $E[Y_n] = \infty$, one could use a truncation argument to show that the average reward, WP1, is λ .

d) In the limit of large t , find the expected reward from time t until the next renewal. Hint: Sketch this expected reward as a function of t for a given sample of inter-renewal intervals and service intervals; then find the time average.

Solution: For the sample function above, the reward to the next inter-renewal (again

averaged over dropped arrivals) is given by

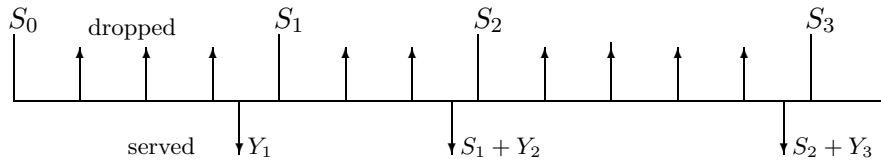


The reward over the n th inter-renewal interval is then $\lambda Y_n^2/2$ so the sample path average of the expected reward per unit time is

$$\frac{\mathbb{E}[R(t)]}{\bar{X}} = \frac{\lambda \mathbb{E}[Y^2]}{2(\bar{Y} + 1/\lambda)}.$$

e) Now assume that the arrivals are deterministic, with the first arrival at time 0 and the n th arrival at time $n - 1$. Does the sequence of times $S_1, S_2, \dots$ at which subsequent customers start service still constitute the renewal times of a renewal process? Draw a sketch of arrivals, departures, and service time intervals. Again find $\lim_{t \rightarrow \infty} (\int_0^t R(\tau) d\tau) / t$.

Solution: Since the arrivals are deterministic at unit intervals, the customer to be served at the completion of Y_1 is the customer arriving at $\lceil Y_1 \rceil$ (the problem statement was not sufficiently precise to specify what happens if a service completion and a customer arrival are simultaneous, so we assume here that such a customer is served). The customers arriving from time 1 to $\lceil Y_1 \rceil - 1$ are then dropped as illustrated below.



We see that the interval between the first and second service is $\lceil Y_2 \rceil$ and in general between service $n - 1$ and n is $\lceil Y_n \rceil$. These intervals are IID and thus form a renewal process.

Finding the sample path average as before,

$$\frac{\mathbb{E}[R(t)]}{\bar{X}} = \frac{\mathbb{E}[\lceil Y \rceil - 1]}{\mathbb{E}[\lceil Y \rceil]}.$$

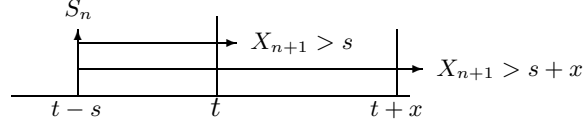
Exercise 5.13: Let $Z(t) = t - S_{N(t)}$ be the age of a renewal process and $Y(t) = S_{N(t)+1} - t$ be the residual life. Let $F_X(x)$ be the CDF of the inter-renewal interval and find the following as a function of $F_X(x)$:

a) $\Pr\{Y(t) > x \mid Z(t) = s\}$

Solution: First assume that X is discrete, and thus S_n is also discrete for each n . We first find $\Pr\{Z(t) = s\}$ for any given s , $0 < s \leq t$. Since $\{Z(t) = s\} = \{S_{N(t)} = t - s\}$, we have

$$\begin{aligned} \Pr\{Z(t) = s\} &= \Pr\{S_{N(t)} = t - s\} = \sum_{n=0}^{\infty} \Pr\{N(t) = n, S_n = t - s\} \\ &= \sum_{n=0}^{\infty} \Pr\{S_n = t - s, X_{n+1} > s\} = F_X^c(s) \sum_{n=0}^{\infty} \Pr\{S_n = t - s\} \quad (\text{A.82}) \end{aligned}$$

where in the next to last step we noted that if the n th arrival comes at $t - s$ and $N(t) = n$, then no arrivals occur in $(t - s, t]$, so $X_{n+1} > t - s$. The last step used the fact that X_{n+1} is IID over n and statistically independent of S_n . The figure below illustrates these relationships and also illustrates how to find $\Pr\{Y(t) > x, Z(t) = s\}$.



$$\begin{aligned}
 \Pr\{Y(t) > x, Z(t) = s\} &= \sum_{n=0}^{\infty} \Pr\{N(t) = n, S_n = t - s, S_{n+1} > t + x\} \\
 &= \sum_{n=0}^{\infty} \Pr\{N(t) = n, S_n = t - s, X_{n+1} > s + x\} \\
 &= F_X^c(s + x) \sum_{n=0}^{\infty} \Pr\{N(t) = n, S_n = t - s\} \quad (\text{A.83})
 \end{aligned}$$

We can now find $\Pr\{Y(t) > x \mid Z(t) = s\}$ by dividing (A.83) by (A.82) and cancelling out the common summation. Thus

$$\Pr\{Y(t) > x \mid Z(t) = s\} = \frac{F_X^c(s + x)}{F_X^c(s)}.$$

We can see the intuitive reason for this answer from the figure above: $Z(t) = s$ specifies that there is an n such that $S_n = t - s$ and there are no arrivals in $(t - s, t]$; $Y(t) > x$ extends this interval of no arrivals to $(t - s, t + x]$, and these probabilities do not depend on n .

The argument above assumed that X is discrete, but this is a technicality that could be handled by a limiting argument. In this case, however, it is cleaner to modify the above argument by working directly with the conditional probabilities.

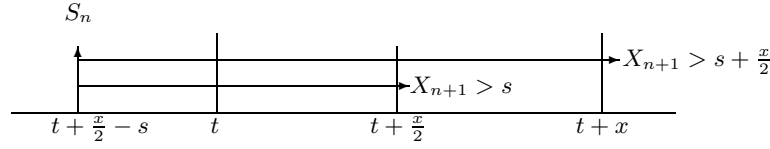
$$\begin{aligned}
 \Pr\{Y(t) > x \mid Z(t) = s\} &= \sum_{n=0}^{\infty} \Pr\{N(t) = n\} \Pr\{Y(t) > x \mid S_{N(t)} = t - s, N(t) = n\} \\
 &= \sum_{n=0}^{\infty} \Pr\{N(t) = n\} \Pr\{X_{n+1} > t + s \mid S_n = t - s, X_{n+1} > s\} \\
 &= \frac{F_X^c(s + x)}{F_X^c(s)} \sum_{n=0}^{\infty} \Pr\{N(t) = n\} = \frac{F_X^c(s + x)}{F_X^c(s)},
 \end{aligned}$$

where the second line again uses the fact that $\{S_n = t - s, N(t) = n\} = \{S_n = t - s, X_{n+1} > s\}$.

b) $\Pr\{Y(t) > x \mid Z(t + x/2) = s\}$.

Solution: The conditioning event $\{Z(t + x/2) = s\}$ means that there is an arrival at $t + x/2 - s$ and no further arrival until after $t + x/2$. We must look at two cases, first where the arrival at $t + x/2 - s$ comes at or before t (i.e., $s \geq x/2$) and second where it comes after t . The solution where $s \geq x/2$ is quite similar to (a), and we repeat the argument

there for general X modifying the terms in (a) as needed. The diagram below will clarify the various terms, assuming $s \geq x/2$.



As in (a),

$$\Pr\left\{Y(t) > x \mid Z(t + \frac{x}{2}) = s\right\} = \sum_{n=0}^{\infty} \Pr\left\{N(t + \frac{x}{2}) = n\right\} \Pr\left\{Y(t) > x \mid Z(t + \frac{x}{2}) = s, N(t + \frac{x}{2}) = n\right\}.$$

The conditioning event above is

$$\left\{Z(t + \frac{x}{2}) = s, N(t + \frac{x}{2}) = n\right\} = \left\{S_n = t + \frac{x}{2} - s, X_{n+1} > s\right\}. \quad (\text{A.84})$$

Given this conditioning event, the diagram shows that $Y(t) > x$ implies that $X_{n+1} > s + \frac{x}{2}$. Thus

$$\begin{aligned} \Pr\left\{Y(t) > x \mid Z(t + \frac{x}{2}) = s\right\} &= \sum_{n=0}^{\infty} \Pr\left\{N(t + \frac{x}{2}) = n\right\} \Pr\left\{X_{n+1} > s + \frac{x}{2} \mid S_n = t + \frac{x}{2} - s, X_{n+1} > s\right\} \\ &= \frac{F_X^c(s + \frac{x}{2})}{F_X^c(s)} \quad \text{for } s \geq x/2. \end{aligned} \quad (\text{A.85})$$

For the other case, where $s < x/2$, the condition asserts that $S_{N(t + \frac{x}{2})} > t$. Thus there must be an arrival in the interval from t to $t + \frac{x}{2}$, so $\Pr\{Y(t) > x \mid Z(t + \frac{x}{2}) = s\} = 0$.

c) $\Pr\{Y(t) > x \mid Z(t+x) > s\}$ for a Poisson process.

Solution: The event $\{Z(t+x) > s\}$ is the event that there are no arrivals in the interval $(t+x-s, t+x]$. We consider two cases separately, first $s \geq x$ and next $s < x$. If $s \geq x$, then there are no arrivals from t to $t+x$, so that $Y(t)$ must exceed x . Thus

$$\Pr\{Y(t) > x \mid Z(t+x) > s\} = 1; \quad \text{for } s \geq x.$$

Alternatively, if $s < x$, any arrival between t and $t+x$ must arrive between t and $t+x-s$, so the probability of no arrival between t and $t+x$ given no arrival between $t+x-s$ and $t+x$ is the probability of no arrival between t and $t+x-s$.

$$\Pr\{Y(t) > x \mid Z(t+x) > s\} = \exp(-\lambda(x-s)); \quad \text{for } s < x,$$

where λ is the rate of the Poisson process.

Exercise 5.14: Let $F_Z(z)$ be the fraction of time (over the limiting interval $(0, \infty)$) that the age of a renewal process is at most z . Show that $F_Z(z)$ satisfies

$$F_Z(z) = \frac{1}{\bar{X}} \int_{x=0}^z \Pr\{X > x\} dx \quad \text{WP1.}$$

Hint: Follow the argument in Example 5.4.7.

Solution: We want to find the time-average CDF of the age $Z(t)$ of an arbitrary renewal process, and do this by following Example 5.4.7. For any given $z > 0$, define the reward function $R(t)$ to be 1 for $Z(t) \leq z$ and to be 0 otherwise, *i.e.*,

$$R(t) = \mathcal{R}(Z(t), \tilde{X}(t)) = \begin{cases} 1; & Z(t) \leq z \\ 0; & \text{otherwise} \end{cases}.$$

Note that $R(t)$ is positive only over the first z units of a renewal interval. Thus $R_n = \min(z, X_n)$. It follows that

$$\mathbb{E}[R_n] = \mathbb{E}[\min(z, X_n)] = \int_0^\infty \Pr\{\min(X, z) > x\} dx \quad (\text{A.86})$$

$$= \int_0^z \Pr\{X > x\} dx. \quad (\text{A.87})$$

Let $F_Z(z) = \lim_{t \rightarrow \infty} \int_0^t (1/t) R(\tau) d\tau$ denote the fraction of time that the age is less than or equal to z . From Theorem 5.4.5 and (A.87),

$$F_Z(z) = \frac{\mathbb{E}[R_n]}{\overline{X}} = \frac{1}{\overline{X}} \int_0^z \Pr\{X > x\} dx \quad \text{WP1.}$$

Exercise 5.15: a) Let J be a stopping rule and $\mathbb{I}_{\{J \geq n\}}$ be the indicator random variable of the event $\{J \geq n\}$. Show that $J = \sum_{n \geq 1} \mathbb{I}_{\{J \geq n\}}$.

Solution: Given that $J = m$ for any given $m \geq 1$, we see that $\mathbb{I}_{\{J \geq n\}}$ is 1 for each $n \leq m$ and 0 for $n > m$. Thus $\sum_{n \geq 1} \mathbb{I}_{\{J \geq n\}} = m$ given $J = m$. Since this is true for all sample values m of J , $J = \sum_{n \geq 1} \mathbb{I}_{\{J \geq n\}}$.

b) Show that $\mathbb{I}_{\{J \geq 1\}} \geq \mathbb{I}_{\{J \geq 2\}} \geq \dots$, *i.e.*, show that for each $n \geq 1$, $\mathbb{I}_{\{J \geq n\}}(\omega) \geq \mathbb{I}_{\{J \geq n+1\}}(\omega)$ for each $\omega \in \Omega$ (except perhaps for a set of probability 0).

Solution: Given $J = m$, $\mathbb{I}_{\{J \geq n\}}$ is 1 for $n \leq m$ and 0 for $n > m$. Thus $\mathbb{I}_{\{J \geq n\}} \geq \mathbb{I}_{\{J \geq n+1\}}$ with strict inequality when $n = m$. The statement $\mathbb{I}_{\{J \geq n\}} \geq \mathbb{I}_{\{J \geq n+1\}}$ is true for each event $\{J = m\}$, however and thus true in general. The strange exception of a set of probability 0 is because rv's can be undefined on a set of 0 probability, which is something we don't usually concern ourselves with and need not concern ourselves with here.

Exercise 5.16: a) Use Wald's equality to compute the expected number of trials of a Bernoulli process up to and including the k th success.

Solution: In a Bernoulli process $\{X_n; n \geq 1$, we call trial n a success if $X_n = 1$. Define a stopping trial J as the first trial n at which $\sum_{m=1}^n X_m = k$. This constitutes a stopping rule since $\mathbb{I}_{J=n}$ is a function of $X_1, \dots, X_n$. Given $J = n$, $S_J = X_1 + \dots + X_n = k$, and since this is true for all n , $S_J = k$ unconditionally. Thus $\mathbb{E}[S_J] = k$. From Wald's equality, $\mathbb{E}[S_J] = \overline{X}\mathbb{E}[J]$ so that $\mathbb{E}[J] = k/\overline{X}$.

We should have shown that $\mathbb{E}[J] < \infty$ to justify the use of Wald's equality, but we will show that in (b).

b) Use elementary means to find the expected number of trials up to and including the first success. Use this to find the expected number of trials up to and including the k th success. Compare with (a).

Solution: Let $\Pr\{X_n = 1\} = p$. Then the first success comes at trial 1 with probability p , and at trial n with probability $(1-p)^{n-1}p$. The expected time to the first success is then $1/p = 1/\bar{X}$. The expected time to the k th success is then $k/\bar{X}$, which agrees with the result in (a).

The reader might question the value of Wald's equality in this exercise, since the demonstration that $E[J] < \infty$ was most easily accomplished by solving the entire problem by elementary means. In typical applications, however, the condition that $E[J] < \infty$ is essentially trivial.

Exercise 5.17: A gambler with an initial finite capital of $d > 0$ dollars starts to play a dollar slot machine. At each play, either his dollar is lost or is returned with some additional number of dollars. Let X_i be his change of capital on the i th play. Assume that $\{X_i; i=1, 2, \dots\}$ is a set of IID random variables taking on integer values $\{-1, 0, 1, \dots\}$. Assume that $E[X_i] < 0$. The gambler plays until losing all his money (i.e., the initial d dollars plus subsequent winnings).

a) Let J be the number of plays until the gambler loses all his money. Is the weak law of large numbers sufficient to argue that $\lim_{n \rightarrow \infty} \Pr\{J > n\} = 0$ (i.e., that J is a random variable) or is the strong law necessary?

Solution: We show below that the weak law is sufficient. The event $\{J > n\}$ is the event that $S_i > -d$ for $1 \leq i \leq n$. Thus $\Pr\{J > n\} \leq \Pr\{S_n > -d\}$. Since $E[S_n] = n\bar{X}$ and $\bar{X} < 0$, we see that the event $\{S_n > -d\}$ for large n is an event in which S_n is very far above its mean. Putting this event in terms of distance from the sample average to the mean,

$$\Pr\{S_n > -d\} = \Pr\left\{\frac{S_n}{n} - \bar{X} > \frac{-d}{n} - \bar{X}\right\}.$$

The WLLN says that $\lim_{n \rightarrow \infty} \Pr\left\{\left|\frac{S_n}{n} - \bar{X}\right| > \epsilon\right\} = 0$ for all $\epsilon > 0$, and this implies the same statement with the absolute value signs removed, i.e., $\Pr\left\{\frac{S_n}{n} - \bar{X} > \epsilon\right\} = 0$ for all $\epsilon > 0$. If we choose $\epsilon = -\bar{X}/2$ in the equation above, it becomes

$$\lim_{n \rightarrow \infty} \Pr\left\{S_n > \frac{n\bar{X}}{2}\right\} = 0.$$

Since $\bar{X} < 0$, we see that $-d > n\bar{X}/2$ for $n > 2|d/\bar{X}|$, and thus $\lim_{n \rightarrow \infty} \Pr\{S_n > -d\} = 0$.

b) Find $E[J]$. Hint: The fact that there is only one possible negative outcome is important here.

Solution: One stops playing on trial $J = n$ if one's capital reaches 0 for the first time on the n th trial, i.e., if $S_n = -d$ for the first time at trial n . This is clearly a function of $X_1, X_2, \dots, X_n$, so J is a stopping rule. Note that stopping occurs exactly on reaching $-d$ since S_n can decrease with n only in increments of -1 and S_n is always integer. Thus $S_J = -d$

Using Wald's equality, we then have

$$E[J] = \frac{-d}{\bar{X}},$$

which is positive since $\bar{X}$ is negative. You should note from the exercises we have done with Wald's equality that it is often used to solve for $E[J]$ after determining $E[S_J]$.

For mathematical completeness, we should also verify that $E[J] < \infty$ to use Wald's equality. If X is bounded, this can be done by using the Chernoff bound on $\Pr\{S_n > -d\}$. This is exponentially decreasing in n , thus verifying that $E[J] < \infty$, and consequently that $E[J] = -d/\bar{X}$. If X is not bounded, a truncation argument can be used. Letting $\check{X}_b$ be X truncated to b , we see that $E[\check{X}_b]$ is increasing with b toward $\bar{X}$, and is less $E[X]$ for all b . Thus the expected stopping time, say $E[J_b]$ is upper bounded by $-d/\bar{X}$ for all b . It follows that $\lim_{b \rightarrow \infty} E[J_b]$ is finite (and equal to $-d/\bar{X}$). Most students are ill-advised to worry too much about such details at first.

Exercise 5.18: Let $\{X_i; i \geq 1\}$ be IID binary random variables with $P_X(0) = P_X(1) = 1/2$. Let J be a positive integer-valued random variable defined on the above sample space of binary sequences and let $S_J = \sum_{i=1}^J X_i$. Find the simplest example you can in which J is not a stopping trial for $\{X_i; i \geq 1\}$ and where $E[X]E[J] \neq E[S_J]$. Hint: Try letting J take on only the values 1 and 2.

Solution: We start with a brief discussion of how to find examples, since this is a major part of creative research, but is practically ignored in engineering, science, and mathematics curricula. One should start by trying to combine whatever insight one has about the problem with the simplest possible cases, where simplicity might involve very little choice (such as here with $J = 1$ or 2), or great symmetry, or very extreme cases. One then tries some simple examples which might not provide the desired result, and tries to understand what has gone wrong. One also, of course, uses any available hints.

In this case, using the hint, we need only think about the four pairs of binary numbers and how to map each into J . Since we are looking for an example that is not a stopping rule, choosing $J = 1$ must sometimes depend on X_2 . We also want our example to satisfy $E[S_J] \neq E[J]E[X]$. We have no control over $E[X]$, so we want to either make $E[S_J]$ small or large for given $E[J]$. If $E[S_J]$ represents winnings and $E[J]$ represents effort to get those winnings, we want to get large winnings with little effort.

A good strategy is now almost self evident. When $X_2 = 1$, we choose $J = 2$ and when $X_2 = 0$, we choose $J = 1$. Thus, we get a win when $X_2 = 1$ and don't waste the effort when $X_2 = 0$. This requires clairvoyance, or peeking, etc., but this just means we are not using a stopping rule. With this rule $E[S_J] = E[X_1] + E[X_2] = 1$. Also, $E[J] = 3/2$ and $E[J]E[X] = 3/4$, so Wald's equality does not hold (which is not surprising since J is not a stopping rule).

Exercise 5.19: Let $J = \min\{n | S_n \leq b \text{ or } S_n \geq a\}$, where a is a positive integer, b is a negative integer, and $S_n = X_1 + X_2 + \cdots + X_n$. Assume that $\{X_i; i \geq 1\}$ is a set of zero-mean IID rv's that can take on only the set of values $\{-1, 0, +1\}$, each with positive probability.

a) Is J a stopping rule? Why or why not? Hint: The more difficult part of this is to argue that J is a random variable (*i.e.*, non-defective); you do not need to construct a proof of this, but try to argue why it must be true.

Solution: For J to be a stopping trial, it must be a random variable and also, for each n , $\mathbb{I}_{J=n}$ must be a function of $X_1, \dots, X_n$. For the case here, $S_n = X_1 + \cdots + X_n$ is

clearly a function of $X_1, \dots, X_n$, so the event that $S_n \geq a$ or $S_n \leq b$ is a function of $X_1, \dots, X_n$. The first n at which this occurs is a function of $S_1, \dots, S_n$, which is a function of $X_1, \dots, X_n$. Thus $\mathbb{I}_{J=n}$ must be a function of $X_1, \dots, X_n$. For J to be a rv, we must show that $\lim_{n \rightarrow \infty} \Pr\{J \leq n\} = 1$. The central limit theorem states that $(S_n - n\bar{X})/\sqrt{n}\sigma$ approaches a normalized Gaussian rv in distribution as $n \rightarrow \infty$. Since X is given to be zero-mean, $S_n/\sqrt{n}\sigma$ must approach normal. Now both $a/\sqrt{n}\sigma$ and $b/\sqrt{n}\sigma$ approach 0 as $n \rightarrow \infty$, so the probability that $\{S_n; n > 0\}$ (*i.e.*, the process in the absence of a stopping rule) remains between these limits goes to 0 as $n \rightarrow \infty$. Thus the probability that the process has not stopped by time n goes to 0 as $n \rightarrow \infty$.

An alternate approach here is to model $\{S_n; n \geq 1\}$ for the stopped process as a Markov chain where a and b are recurrent states and the other states are transient. Then we know that one of the recurrent states are reached eventually with probability 1.

b) What are the possible values of S_J ?

Solution: Since $\{S_n; n > 0\}$ can change only in integer steps, S_n cannot exceed a without some S_m , $m < n$ first equaling a and it cannot be less than b without some S_k , $k < n$ first equaling b . Thus S_J is only a or b .

c) Find an expression for $\mathbb{E}[S_J]$ in terms of p , a , and b , where $p = \Pr\{S_J \geq a\}$.

Solution: $\mathbb{E}[S_J] = a\Pr\{S_J = a\} + b\Pr\{S_J = b\} = pa + (1 - p)b$

d) Find an expression for $\mathbb{E}[S_J]$ from Wald's equality. Use this to solve for p .

Solution: We have seen that J is a stopping trial for the IID rv's $\{X_i; i > 0\}$. Assuming for the moment that $\mathbb{E}[J] < \infty$, Wald's equality holds and $\mathbb{E}[S_J] = \bar{X}\mathbb{E}[J]$. Since $\bar{X} = 0$, we conclude that $\mathbb{E}[S_J] = 0$. Combining this with (c), we have $0 = pa + (1 - p)b$, so $p = -b/(a - b)$. This is easier to interpret as

$$p = |b|/(a + |b|). \quad (\text{A.88})$$

The assumption that $\mathbb{E}[J] < \infty$ is more than idle mathematical nitpicking, since we saw in Example 5.5.4 that $\mathbb{E}[J] = \infty$ for $b = -\infty$. The CLT does not resolve this issue, since the probability that S_n is outside the limits $[-b, a]$ approaches 0 with increasing n only as n^{-1} . Viewing the process as a finite-state Markov chain with a and b viewed as a common trapping state does resolve the issue, since, as seen in Theorem 4.5.4 applied to expected first passage times, the expected time to reach the trapping state is finite. As will be seen when we study Markov chains with countably many states, this result is no longer valid, as illustrated by Example 5.5.4. This issue will become more transparent when we study random walks in Chapter 9.

The solution in (A.88) applies only for $\bar{X} = 0$, but Chapter 9 shows how to solve the problem for an arbitrary distribution on X . We also note that the solution is independent of $\mathbf{p}_X(0)$, although $\mathbf{p}_X(0)$ is obviously involved in finding $\mathbb{E}[J]$. Finally we note that this helps explain the peculiar behavior of the 'stop when you're ahead' example. The negative threshold b represents the capital of the gambler in that example and shows that as $b \rightarrow -\infty$, the probability of reaching the threshold a increases, but at the expense of a larger catastrophe if the gamblers capital is wiped out.

Exercise 5.20: Show that the interchange of expectation and sum in (5.34) is valid if $E[J] < \infty$. Hint: First express the sum as $\sum_{n=1}^{k-1} X_n \mathbb{I}_{J \geq n} + \sum_{n=k}^{\infty} (X_n^+ + X_n^-) \mathbb{I}_{J \geq n}$ and then consider the limit as $k \rightarrow \infty$.

Solution: We first prove Theorem 5.5.3 in a different and more elegant way, which eliminates the need to justify the exchange of expectation and sum in (5.34).

Let J be a stopping trial for the sequence $\{X_n; n \geq 1\}$. Consider a renewal process in which the stopping trial is viewed as a renewal. That is, for a sample path $\{x_n; n \geq 1\}$ for $\{X_n; n \geq 1\}$, let J_1 be the n at which the stopping rule is satisfied. Consider using the same stopping rule again, now starting at $x_{n+1}, x_{n+2}, \dots$. Let J_2 be the value of m for which $x_{n+1}, x_{n+2}, \dots, x_{n+m}$ satisfies the stopping rule. In this same way, consider the successive trials at which the stopping rule is satisfied as an integer arrival process. Since $\mathbb{I}_{J=n}$ is a function of $X_1, \dots, X_n$, and since the $\{X_n; n \geq 1\}$ are IID, this arrival process is actually a renewal process. Now let $R(t)$ be a renewal reward process where at each positive integer time t , $R(t) = \sum_{n=1}^t X_n$. For non-integer t , let $R(t) = R(\lceil t \rceil)$. By the SLLN, $\lim_{t \rightarrow \infty} (1/t) \int_0^t R(\tau) d\tau = \bar{X}$ with probability 1.

Now J is the length of an interrenewal interval, so the expected interrenewal interval is $E[J]$, which is given to be finite. Finally the expected reward in an interrenewal interval is $E[S_J] = E\left[\sum_{n=1}^J X_n\right]$. The renewal reward theorem, Theorem 5.4.5 (generalized to cover the reward function here as discussed at the end of Section 5.4.1), says that $\bar{X} = E[S_J] / E[J]$, so that $E[S_J] = \bar{X} E[J]$, which is Theorem 5.5.3.

It is quite surprising to realize that Wald's equality is really just a special case of the renewal reward theorem. Wald's equality is used to prove the elementary renewal theorem, so this illustrates the advantage of understanding both time averages and ensemble averages, since they are quite intimately related.

We now return to solving the problem as intended in the hint. We must pull an important theorem, called the Lebesgue monotone convergence theorem, out of the hat. This theorem is a simple but fundamental result in measure theory, so we accept it without proof. It provides an almost generic justification for interchanging limits and expectations. Stated in terms of rv's and expectations, it says the following:

Lebesgue monotone convergence theorem: Let $Y_1, Y_2, \dots$, be a sequence of rv's with the properties (a) that $Y_1(\omega) \leq Y_2(\omega) \leq \dots$ for each sample point ω and (b) that $\lim_{k \rightarrow \infty} Y_k(\omega)$ converges (is finite) for each ω (WP1). Let $Y(\omega) = \lim_{k \rightarrow \infty} Y_k(\omega)$ for each ω . Then Y is a rv and $\lim_{k \rightarrow \infty} E[Y_k] = E[Y]$.

In the context of (5.34), we initially assume that X (i.e., each X_n) is nonnegative. Then let $Y_k = \sum_{n=1}^k X_n \mathbb{I}_{J \geq n}$. Since $X_n \mathbb{I}_{J \geq n}$ is nonnegative, we see that $Y_k(\omega) \leq Y_{k+1}(\omega)$ for each $k \geq 1$ and each ω . Finally, since $J(\omega)$ is finite WP1, we see that $\lim_{k \rightarrow \infty} Y_k(\omega) = Y_{J(\omega)}(\omega)$, so the second condition of the theorem is satisfied. The theorem then says that

$$E\left[\lim_{k \rightarrow \infty} Y_k\right] = E[Y] = \lim_{k \rightarrow \infty} E[Y_k],$$

which is (5.34).

Finally, if X_n has both a positive and negative part, say $X_n^+ = \max[X_n, 0]$ and $X_n^- = \min[X_n, 0]$, we apply the result above to $\{X_n^+; n \geq 1\}$ and $\{X_n^-; n \geq 1\}$ separately.

Exercise 5.21: Consider a G/G/1 queue with inter-arrival sequence $\{X_i; i \geq 1\}$ and service time sequence $\{V_i; i \geq 0\}$. Assume that $E[X] < \infty$ and that $E[V] < E[X]$.

a) Let $Z_n = \sum_{i=1}^n (X_i - V_{i-1})$ and let I_n be the sum of the intervals between 0 and $S_n = \sum_{i=1}^n X_i$ during which the server is empty. Show that $I_n \geq Z_n$ and explain the difference between them. Show that if arrival n enters an empty system, then $I_n = Z_n$.

Solution: $\sum_{i=0}^{n-1} V_i$ is the aggregate time required by the server to serve customers 0 to $n-1$. The amount of time the server has been busy is equal to this sum less the unfinished work still in the queue and server. The aggregate time the server has been idle over $(0, S_n]$ is thus Z_n plus the unfinished work. If customer n enters an empty system, then there is no unfinished work and $I_n = Z_n$.

b) Assume that the inter-arrival distribution is bounded in the sense that for some $b < \infty$, $\Pr\{X > b\} = 0$. Show that the duration of each idle period for the G/G/1 queue has a duration less than or equal to b .

Solution: Note that $S_n - S_{n-1} \leq b$. The server was busy just after the arrival at S_{n-1} and thus the idle period was at most b .

c) Show that the number of idle intervals for the server up to time S_n is at least Z_n/b . Show that the number of renewals $N^r(S_n)$ up to the time S_n at which an arrival sees an empty system is at least (Z_n/b) .

Solution: The sequence of arrivals to an empty system constitute the renewal process of interest, so $N^r(S_n)$ is the number M_n of idle intervals over $(0, S_n]$. Note that S_n is either the arrival time of an arrival that enters the queue or an arrival that ends an idle period, so the number of idle intervals and the number of renewals in $(0, S_n]$ is the same. Since the duration of each idle interval is at most b , $M_n b \geq I_n \geq Z_n$, so $M_n \geq Z_n/b$.

d) Show that the renewal counting process satisfies $\lim_{t \rightarrow \infty} \frac{N^r(t)}{t} = a$ WP1 for some $a > 0$. Show that the mean number of arrivals per arrival to an empty system is $J = 1/a < \infty$.

Solution: It was shown in Section 5.5.3 that $\{N^r(t); t > 0\}$ is a renewal counting process, so $\lim_{t \rightarrow \infty} \frac{N^r(t)}{t}$ exists WP1 and its value a is the reciprocal of the interval between renewals. We must show that $a > 0$. We have

$$\begin{aligned} a &= \lim_{t \rightarrow \infty} \frac{N^r(t)}{t} = \lim_{n \rightarrow \infty} \frac{N^r(S_n)}{S_n} = \lim_{n \rightarrow \infty} \frac{M_n}{S_n} \\ &= \lim_{n \rightarrow \infty} \frac{M_n}{n} \lim_{n \rightarrow \infty} \frac{n}{S_n} \\ &\geq \lim_{n \rightarrow \infty} \frac{Z_n}{bn} \lim_{n \rightarrow \infty} \frac{n}{S_n} = \frac{E[X] - E[V]}{b} \cdot \frac{1}{E[X]} > 0, \end{aligned}$$

where we have used the SLLN and the strong law for renewal processes.

e) Assume everything above except that $\Pr\{X > b\} = 0$ for some b . Use the truncation method on X to establish (d) for this new case.

Solution: Let $E[X] - E[V] = \epsilon$ and choose b large enough that when we choose $\tilde{X} = \min\{X, b\}$, we have $E[\tilde{X}] - E[V] \geq \epsilon/2$. Now compare an arbitrary sample function of $\{X_i; i \geq 1\}$, $\{V_i; i \geq 0\}$ with the truncated version $\{\tilde{X}_i; i \geq 1\}$, $\{V_i; i \geq 0\}$. For each n for which the sample values $\tilde{X}_n = \tilde{x}_n < X_n - x_n$, we see that the unfinished work in the

truncated system at $\tilde{S}_n = \tilde{s}_n$ increases relative to $S_n = s_n$. When the original system has a period where the server is idle, the unfinished work in the truncated system decreases toward 0 while that in the original system remains equal to 0. Thus, if the truncated system has an idle period in the interval before $\tilde{S}_n$, the untruncated system has an idle period there also. In other words, the number of idle intervals in the truncated system up to $\tilde{S}_n$ must be less than or equal to the number of idle periods in the original system up to S_n . Thus $\tilde{N}^r(\tilde{S}_n) \leq N^r(S_n)$ for all n . Since the rate of arrivals seeing an empty system in the truncated version exceeds 0, that in the original system does also.

Exercise 5.22: A company has three promising approaches to the design of a new wireless application. Unknown to the company, the first approach requires 2 weeks, the second, 4 weeks, and the third 8 weeks, but only the first will be successful. The company hires one programmer after another for the design, stopping when a design is successful. Let X_i be the time taken by the i th programmer, assuming that each programmer independently chooses an approach with equal probability (*i.e.*, $\mathbf{p}_{X_i}(x) = 1/3$ for $x = 2, 4$, and 8). Let J be the number of the first programmer who is successful and let $T = \sum_{i=1}^J X_i$ be the time of the first success.

a) Use Wald's equality to find $\mathbf{E}[T]$.

Solution: Let $\{X_i; i \geq 1\}$ be the IID sequence of programmer times, visualizing this sequence as continuing beyond the time of the first successful program. The stopping trial J is the smallest i for which $X_i = 2$. Note that J is a function only of $X_1, \dots, X_J$, so J is a (possibly defective) stopping rule. We see that $J = 1$ with probability $1/3$, $J = 2$ with probability $(2/3)(1/3)$, and in general $J = n$ with probability $(2/3)^{n-1}(1/3)$. Thus J is a geometric rv (and thus non-defective) and $\mathbf{E}[J] = 3$ (which is finite). This shows that the Wald equality holds, so

$$\mathbf{E}[T] = \mathbf{E}[X] \mathbf{E}[J] = \frac{1}{3}(2+4+8) \mathbf{E}[J] = 14.$$

b) Compute $\mathbf{E}\left[\sum_{i=1}^J X_i \mid J=n\right]$ and show that it is not equal to $\mathbf{E}\left[\sum_{i=1}^n X_i\right]$.

Solution: The conditioning event $\{J = n\}$ is the same as the event $\{X_1 \neq 2, \dots, X_{n-1} \neq 2, X_n = 2\}$. Given this event, $X_1, \dots, X_{n-1}$ are IID and take on the values 4 and 8 with equal probability. Thus

$$\mathbf{E}\left[\sum_{i=1}^n X_i \mid J = n\right] = (n-1)\frac{1}{2}(4+8) + 2 = 6n - 4. \quad (\text{A.89})$$

On the other hand, viewing the sequence $\{X_i; i \geq 1\}$ as continuing beyond the stopping time,

$$\mathbf{E}\left[\sum_{i=1}^n X_i\right] = n\mathbf{E}[X] = \frac{14n}{3}. \quad (\text{A.90})$$

The fact that (A.89) increases with n faster than (A.90) should not be surprising, since when J is large, the X_i for $i < J$ must all be large (4 or 8), thus enlarging the expected value.

c) Use (b) for a second derivation of $\mathbf{E}[T]$.

Solution: Note that (A.89) is $E[T \mid J=n]$, so

$$E[T] = \sum_{n=1}^{\infty} E[T \mid J=n] p_J(n) = \sum_{n=1}^{\infty} (6n - 4) p_J(n) = 6E[J] - 4.$$

Since $E[J] = 3$, this results in $E[T] = 14$ as before.

d) Find $E[T]$ if the company tells successive programmers which approaches have been tried unsuccessfully earlier.

Solution: We assume that the first programmer chooses the approaches with equal probability, that the second chooses the two remaining choices with equal probability if the first is not successful, and the third chooses the final choice if the first two are unsuccessful. This can be calculated by brute force, but the easy way is to note the two week approach must always be taken before stopping. The 4 week approach will be tried before the two week approach with probability $1/2$, and the 8 week approach will be tried before the two week approach with probability $1/2$. Thus $E[T] = 2 + 4/2 + 8/2 = 8$. It is not surprising that this requires much less time than continuing to use failed approaches.

Exercise 5.23: **a)** Consider a renewal process for which the inter-renewal intervals have the PMF $p_X(1) = p_X(2) = 1/2$. Let $m(t) = E[N(t)]$ and use elementary combinatorics to show that $m(1) = 1/2$, $m(2) = 5/4$, and $m(3) = 15/8$.

Solution: Note that $m(n) = \sum_{i=1}^n E[W_i]$ where $W_i = 1$ if an arrival occurs at time i and $W_i = 0$ otherwise. Now $W_1 = 1$ if and only if (iff) $X_1 = 1$, so $E[W_1] = 1/2$ and $m(1) = 1/2$. Next $W_2 = 1$ iff either $X_1 = 2$ or both $X_1 = 1$ and $X_2 = 1$. Thus $E[W_2] = 3/4$ and $m(2) = 5/4$. To calculate $m(3)$, we get a little fancier and note that if $W_2 = 0$, then $W(3)$ must be 1 and if $W(2) = 1$, then $W(3)$ is 1 with probability $1/2$. Thus

$$\Pr\{W_3 = 1\} = \Pr\{W_2 = 0\} + \frac{1}{2}\Pr\{W_2 = 1\} = \frac{5}{8}.$$

It follows that $m(3) = E[W_1] + E[W_2] + E[W_3] = 15/8$.

b) Use elementary means to show that $E[S_{N(1)}] = 1/2$ and $E[S_{N(1)+1}] = 9/4$. Verify (5.37) in this case (*i.e.*, for $t = 1$) and show why $N(1)$ is not a stopping trial. Note also that the expected duration, $E[S_{N(1)+1} - S_{N(1)}]$ is not equal to $\bar{X}$.

Solution: Note that $N(1)$ is the number of arrivals up to and including time 1, so $N(1) = 1$ with probability $1/2$ and $N(1) = 0$ with probability $1/2$. Now $S_{N(1)}$ is the epoch of arrival $N(1)$. If $N(1) = 1$, then that arrival must have occurred at epoch 1. If $N(1) = 0$, no arrival has occurred and by convention $S_0 = 0$. Thus $S_{N(1)}$ is 1 with probability $1/2$ and is 0 with probability $1/2$; so $E[S_{N(1)}] = 1/2$.

The first arrival epoch after time 1, *i.e.*, $S_{N(1)+1}$, is at time 2 if $X_1 = 2$ and is at time 2 or 3 with equal probability if $X_1 = 1$. Thus it is 2 with probability $3/4$ and 3 with probability $1/4$, so $E[S_{N(1)+1}] = 9/4$.

Eq. (5.37) says that $E[S_{N(1)+1}] = E[X](E[N(1)] + 1)$. Since $\bar{X} = 3/2$ and $E[N(1)] + 1 = 3/2$, (5.37) is satisfied.

The event $\{N(1) = 0\}$ has non-zero probability but is not determined before the observation of X_1 . Thus $N(1)$ cannot be a stopping trial for $\{X_n; n \geq 1\}$. Also, Wald's equality (if it applied) would equate $\mathbf{E}[S_{N(1)}] = 1/2$ with $\overline{X}\mathbf{E}[N(1)] = 3/4$, so it is not satisfied.

c) Generalize (a) so that $\Pr\{X = 1\} = 1 - p$ and $\Pr\{X = 2\} = p$. Let $W_n = N(n) - N(n-1)$, i.e., W_n is 1 if an arrival occurs at time n and is 0 if no arrival occurs at n . Show that $\overline{W}_n$ satisfies the difference equation $\overline{W}_n = 1 - p\overline{W}_{n-1}$ for $n \geq 1$ where by convention $\overline{W}_0 = 1$. Use this to show that

$$\overline{W}_n = \frac{1 - (-p)^{n+1}}{1 + p}. \quad (\text{A.91})$$

From this, solve for $m(n)$ for $n \geq 1$ and verify the result in (a).

Solution: Note that if $W_{n-1} = 0$, then, since each interarrival interval is either 1 or 2, we must have $W_n = 1$. Alternatively, if $W_{n-1} = 1$, then W_n is 1 with probability $1 - p$. Thus

$$\Pr\{W_n = 1\} = \Pr\{W_{n-1} = 0\} + (1 - p)\Pr\{W_{n-1} = 1\}.$$

Since $\overline{W}_n = \Pr\{W_n = 1\}$, this can be rewritten as

$$\overline{W}_n = 1 - \overline{W}_{n-1} + (1 - p)\overline{W}_{n-1} = 1 - p\overline{W}_{n-1}.$$

Expanding this expression,

$$\overline{W}_n = 1 - p(1 - p\overline{W}_{n-2}) = \cdots = 1 - p + p^2 - p^3 + \cdots + (-p)^n.$$

Now for any number y , note that

$$(1 - y)(1 + y + y^2 + \cdots + y^n) = 1 - y^{n+1}. \quad (\text{A.92})$$

Taking $-p$ for y , we get (A.91). We can find $m(n) = \sum_{i=1}^n \overline{W}_i$ by using (A.92) to sum the powers of $-p$ in (A.91). The result is

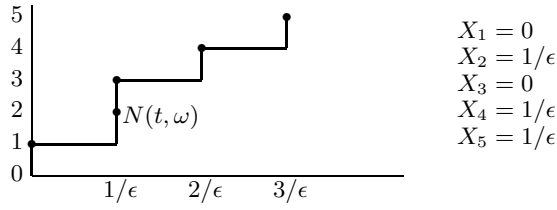
$$m(n) = \frac{n}{1 + p} - \frac{p^2(1 - (-p)^n)}{(1 + p)^2}.$$

With some patience, this agrees with the solution in (a). This same approach, combined with more tools about difference equations, can be used to solve for $m(n)$ for other integer valued renewal processes. Finally note that we can solve for $\mathbf{E}[S_{N(n)+1}]$ for all n by using Wald's equality.

Exercise 5.24: Let $\{N(t); t > 0\}$ be a renewal counting process generalized to allow for inter-renewal intervals $\{X_i\}$ of duration 0. Let each X_i have the PMF $\Pr\{X_i = 0\} = 1 - \epsilon$; $\Pr\{X_i = 1/\epsilon\} = \epsilon$.

a) Sketch a typical sample function of $\{N(t); t > 0\}$. Note that $N(0)$ can be non-zero (i.e., $N(0)$ is the number of zero interarrival times that occur before the first non-zero interarrival time).

Solution:



b) Evaluate $\mathbf{E}[N(t)]$ as a function of t .

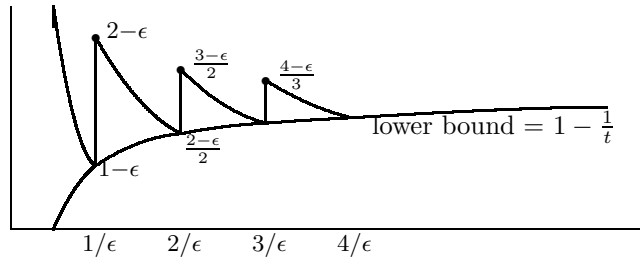
Solution: Arrivals occur only at nonnegative-integer multiples of $1/\epsilon$. Epoch 0 is different than the positive multiples of $1/\epsilon$ in that the nominal initial arrival that starts the process is not counted as part of $N(t)$. Thus $\mathbf{p}_{N(0)}(j) = (1 - \epsilon)^j \epsilon$ for integer $j \geq 0$. Thus $\mathbf{E}[N(0)] = (1 - \epsilon)/\epsilon$.

At each positive multiple ℓ of $1/\epsilon$, the initial arrival is counted, so the probability of $j \geq 1$ arrivals at time $\ell\epsilon$ is $(1 - \epsilon)^{j-1} \epsilon$ for all $j \geq 1$. Thus the expected number of arrivals at epoch ℓ/ϵ for each $\ell \geq 1$ is $1/\epsilon$. We can now evaluate $\mathbf{E}[N(k/\epsilon)]$ as the expected number of arrivals at 0 plus the expected number at each ℓ/ϵ , $1 \leq \ell \leq k$. Thus $\mathbf{E}[N(k\epsilon)] = k/\epsilon + (1 - \epsilon)/\epsilon$. Finally, since arrivals occur only at multiples of $1/\epsilon$, $\mathbf{E}[N(t)]$ for arbitrary $t > 0$ is

$$\mathbf{E}[N(t)] = \frac{1}{\epsilon} \lfloor t\epsilon \rfloor + \frac{1 - \epsilon}{\epsilon}. \quad (\text{A.93})$$

c) Sketch $\mathbf{E}[N(t)]/t$ as a function of t .

Solution:



d) Evaluate $\mathbf{E}[S_{N(t)+1}]$ as a function of t (do this directly, and then use Wald's equality as a check on your work).

Solution: At each epoch k/ϵ with integer $k \geq 1$ there are one or more arrivals, and there are no arrivals at other times. Thus for any $t \geq 0$, the epoch of the first arrival after t is $(1/\epsilon)\lfloor \epsilon t \rfloor + 1/\epsilon$. Thus

$$S_{N(t)+1} = \frac{1}{\epsilon} \lfloor \epsilon t \rfloor + \frac{1}{\epsilon}.$$

Since $N(t)$ is the number of arrivals up to and including t , $N(t) + 1$ is the first arrival after t , so the formula above works for arbitrary t and particularly for $t = k/\epsilon$. Note that for each $t > 0$, $S_{N(t)+1}$ is deterministic, so

$$\mathbf{E}[S_{N(t)+1}] = \frac{1}{\epsilon} \lfloor \epsilon t \rfloor + \frac{1}{\epsilon}. \quad (\text{A.94})$$

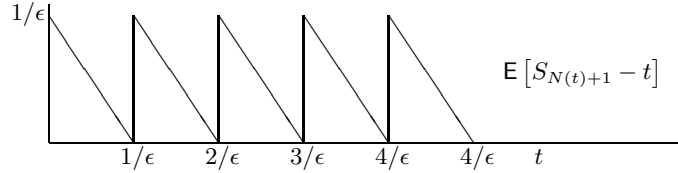
The Wald equality says $\mathbf{E}[S_{N(t)+1}] = \bar{X}\mathbf{E}[N(t) + 1]$. Since $\bar{X} = 1$, this checks with (A.93) and (A.94).

e) Sketch the lower bound $\mathbf{E}[N(t)]/t \geq 1/\mathbf{E}[X] - 1/t$ on the same graph with (c).

Solution: See (c).

f) Sketch $\mathbf{E}[S_{N(t)+1} - t]$ as a function of t and find the time average of this quantity.

Solution:



By inspection, $\lim_{\tau \rightarrow \infty} (1/\tau) \int_0^\tau \mathbf{E}[S_{N(t)+1} - t] dt = 1/2\epsilon$.

g) Evaluate $\mathbf{E}[S_{N(t)}]$ as a function of t ; verify that $\mathbf{E}[S_{N(t)}] \neq \mathbf{E}[X]\mathbf{E}[N(t)]$.

Solution: From the same argument as in (d), $\mathbf{E}[S_{N(t)}] = \frac{1}{\epsilon} \lfloor \epsilon t \rfloor$. From (A.93), this is not equal to $\bar{X}\mathbf{E}[N(t)]$.

Exercise 5.25: Let $\{N(t); t > 0\}$ be a renewal counting process and let $m(t) = \mathbf{E}[N(t)]$ be the expected number of arrivals up to and including time t . Let $\{X_i; i \geq 1\}$ be the sequence of inter-renewal intervals and assume that $F_X(0) = 0$.

a) For all $x > 0$ and $t > x$ show that $\mathbf{E}[N(t)|X_1=x] = \mathbf{E}[N(t-x)] + 1$.

Solution: Given that the first arrival occurs at x , the number of arrivals in $(x, t]$ forms a renewal process starting at x , and thus the expected number of arrivals in $(x, t]$ is $\mathbf{E}[N(t-x)]$. The total number of arrivals in $(0, t]$ is then $\mathbf{E}[N(t-x)] + 1$ as was to be shown.

b) Use (a) to show that $m(t) = F_X(t) + \int_0^t m(t-x) dF_X(x)$ for $t > 0$. This equation is the renewal equation, derived differently in (5.55).

Solution: Assume that X has a density, $f_X(x)$, for initial simplicity. Then

$$\begin{aligned} m(t) &= \int_0^t \mathbf{E}[N(t)|X_1=x] f_X(x) dx \\ &= \int_0^t (m(t-x) + 1) f_X(x) dx \\ &= F_X(t) + \int_0^t m(t-x) f_X(x) dx. \end{aligned} \tag{A.95}$$

This same argument works for arbitrary X by using Stieltjes integration in place of the integrals above.

c) Suppose that X is an exponential random variable of parameter λ . Evaluate $L_m(s)$ from (5.57); verify that the inverse Laplace transform is λt ; $t \geq 0$.

Solution: Evaluating the Laplace transform of both sides of (A.95) and recognizing that $F_X(x)$ is the integral of $f_X(x)$ and that $\int_0^t m(t-x) f_X(x) dx$ is the convolution of $m(t)$ and

$f_X(x)$, we get

$$L_m(s) = \frac{L_X(s)}{s} + L_m(s)L_X(s).$$

Solving this equation for $L_m(s)$,

$$L_m(s) = \frac{L_X(s)}{s(1 - L_X(s))}. \quad (\text{A.96})$$

The Laplace transform of an exponential with density $\lambda e^{-\lambda x}$ for $x \geq 0$ is $L_X(s) = \lambda/(\lambda + s)$. Substituting this into (A.96) yields $L_m(s) = \lambda/s^2$. The inverse Laplace transform of this is λt for $t \geq 0$.

Exercise 5.26: a) Let the inter-renewal interval of a renewal process have a second order Erlang density, $f_X(x) = \lambda^2 x \exp(-\lambda x)$. Evaluate the Laplace transform of $m(t) = \mathbf{E}[N(t)]$.

Solution: The Laplace transform of $f_X(s)$ is

$$L_X(s) = \int_0^\infty \lambda^2 x \exp(-(\lambda+s)x) dx = \frac{\lambda^2}{(\lambda+s)^2}.$$

We then evaluate $L_m(s)$ from (5.57) to get

$$L_m(s) = \frac{L_X(s)}{s(1 - L_X(s))} = \frac{\lambda^2}{s^3 + 2\lambda s^2} = \frac{\lambda^2}{s^2(s + 2\lambda)}.$$

b) Use this to evaluate $m(t)$ for $t \geq 0$. Verify that your answer agrees with (5.59).

Solution: Note that $L_m(s)$ has a second order pole at $s = 0$ and the residue (the value of $s^2 L_m(s)$ at $s = 0$) is $\lambda/2$. The remainder leads to another single order pole at $s = -2\lambda$, leading to the Heaviside expansion

$$L_m(s) = \frac{\lambda}{2s^2} - \frac{1}{4s} + \frac{1}{4(s + 2\lambda)}.$$

The inverse Laplace transform is then

$$m(t) = \frac{\lambda t}{2} - \frac{1}{4} + \frac{1}{4}e^{-2\lambda t} \quad \text{for } t \geq 0.$$

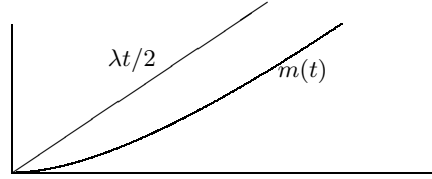
For the second order Erlang, $\overline{X} = 2/\lambda$ and $\sigma^2 = 2/\lambda^2$. Substituting this in (5.59) yields the non-vanishing terms $\lambda t/2 - 1/4$, in agreement with the result here.

c) Evaluate the slope of $m(t)$ at $t = 0$ and explain why that slope is not surprising.

Solution: The slope of $m(t)$ at 0, i.e., $\lim_{t \rightarrow 0+} \frac{dm(t)}{dt}$, is 0, which means that the expected number of renewals in $(0, \delta]$ is $o(\delta)$. This is not surprising since $f_X(\delta)$ is also $o(\delta)$.

d) View the renewals here as being the even numbered arrivals in a Poisson process of rate λ . Sketch $m(t)$ for the process here and show one half the expected number of arrivals for the Poisson process on the same sketch. Explain the difference between the two.

Solution: The first arrival of a second order Erlang distribution has the distribution of the second arrival in a Poisson process, and successive arrivals from the Erlang distribution then have the distribution of the even numbered Poisson arrivals. To sketch $m(t)$ for the process here, note that $m(0) = 0$ with an initial slope of 0. Asymptotically, $m(t) = \lambda/2t - 1/4$ and the slope is increasing from 0 to $\lambda/2$. This gives rise to the sketch below.



The number of even arrivals in $(0, t]$ is always either one less than the number of odd arrivals or the same, asymptotically each half the time. Since the even plus the odd arrivals sum to the arrivals of a Poisson process of rate λt , the expected number of even arrivals lags slightly behind half the overall expected number, as shown in the above curve.

Exercise 5.27: a) Let $N(t)$ be the number of arrivals in the interval $(0, t]$ for a Poisson process of rate λ . Show that the probability that $N(t)$ is even is $[1 + \exp(-2\lambda t)]/2$. Hint: Look at the power series expansion of $\exp(-\lambda t)$ and that of $\exp(\lambda t)$, and look at the sum of the two. Compare this with $\sum_{n \text{ even}} \Pr\{N(t) = n\}$.

Solution: The power series expansions are

$$\begin{aligned} e^{-\lambda t} &= 1 - \lambda t + \frac{(\lambda t)^2}{2} - \frac{(\lambda t)^3}{3!} + \cdots + \frac{(-\lambda t)^n}{n!} + \cdots \\ e^{\lambda t} &= 1 + \lambda t + \frac{(\lambda t)^2}{2} + \frac{(\lambda t)^3}{3!} + \cdots + \frac{(\lambda t)^n}{n!} + \cdots \end{aligned}$$

Summing these two equations and dividing by 2,

$$\frac{1}{2} (e^{-\lambda t} + e^{\lambda t}) = \sum_{n \text{ even}} \frac{(\lambda t)^n}{n!}.$$

Multiplying both sides by $e^{-\lambda t}$,

$$\frac{1}{2} [1 + e^{-2\lambda t}] = \sum_{n \text{ even}} \frac{(\lambda t)^n e^{-\lambda t}}{n!} = \sum_{n \text{ even}} \Pr\{N(t) = n\}. \quad (\text{A.97})$$

b) Let $\tilde{N}(t)$ be the number of even numbered arrivals in $(0, t]$. Show that $\tilde{N}(t) = N(t)/2 - \mathbb{I}_{\text{odd}}(t)/2$ where $\mathbb{I}_{\text{odd}}(t)$ is a random variable that is 1 if $N(t)$ is odd and 0 otherwise.

Solution: When $\mathbb{I}_{\text{odd}}(t) = 0$, $N(t)$ is even and the number of odd numbered arrivals is equal to the number of even-numbered arrivals; thus $2\tilde{N}(t) = N(t)$. When $\mathbb{I}_{\text{odd}}(t) = 1$, the number of odd-numbered arrivals is one more than the number of even-numbered arrivals; thus $2\tilde{N}(t) + 1 = N(t)$. In both cases, then, $2\tilde{N}(t) + \mathbb{I}_{\text{odd}}(t) = N(t)$. Dividing by 2 gives the desired result.

c) Use (a) and (b) to find $\mathbb{E}[\tilde{N}(t)]$. Note that this is $m(t)$ for a renewal process with second-order Erlang inter-renewal intervals.

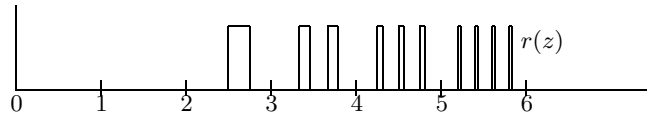
Solution: From (b), $\mathbb{E}[\tilde{N}(t)] = \frac{1}{2}\mathbb{E}[N(t)] - \frac{1}{2}\mathbb{E}[\mathbb{I}_{\text{odd}}(t)]$. But $\mathbb{E}[I_{\text{odd}}(t)]$ is 1 minus the probability that $N(t)$ is even, so from (A.97), $\mathbb{E}[I_{\text{odd}}(t)] = \frac{1}{2}[1 - e^{-2\lambda t}]$. Thus

$$\mathbb{E}[\tilde{N}(t)] = \frac{\lambda t}{2} - \frac{1}{4} + \frac{1}{4}e^{-2\lambda t}.$$

Note that this is the expected number of arrivals for the Erlang second-order arrival process derived by Lagrange multiplier methods in Exercise 5.26

Exercise 5.28: a) Consider a function $r(z) \geq 0$ defined as follows for $0 \leq z < \infty$: For each integer $n \geq 1$ and each integer k , $1 \leq k < n$, $r(z) = 1$ for $n + k/n \leq z \leq n + k/n + 2^{-n}$. For all other z , $r(z) = 0$. Sketch this function and show that $r(z)$ is not directly Riemann integrable.

Solution:



For any δ , and all $n > 2/\delta$, the increments of size δ all contain points equal to 0 and points equal to 1. Thus the Riemann upper sum converges to ∞ and the lower sum is zero. This shows that the function is not directly Riemann integrable. The problem here is not the discontinuities, but rather the increasingly fast variations as $t \rightarrow \infty$.

b) Evaluate the Riemann integral $\int_0^\infty r(z) dz$.

Solution: By definition, the Riemann integral $\int_0^\infty r(z) dz = \lim_{t \rightarrow \infty} \int_0^t r(z) dz$. For integer t ,

$$\int_0^{t+1} r(z) dz = \sum_{k=1}^t (k-1)2^{-k}.$$

This is clearly finite for all integer $t > 0$ and the integral is non-decreasing for t both integer and non-integer. The limit as $t \rightarrow \infty$ is 1. Note that for direct Riemann integration, we take the limit over $t \rightarrow \infty$ first and then the limit over δ . For ordinary Riemann integration, the limit over δ is taken first and the limit over t second.

c) Suppose $r(z)$ is decreasing, i.e., that $r(z) \geq r(y)$ for all $y > z > 0$. Show that if $r(z)$ is Riemann integrable, it is also directly Riemann integrable.

Solution: Assume $r(z)$ is non-increasing and Riemann integrable over $[0, \infty)$. We are assuming $r(z) \geq 0$ throughout the problem (and if not, r cannot be Riemann integrable in this part). Thus $\int_0^\infty r(z) dz = \lim_{t \rightarrow \infty} \int_0^t r(z) dz$ exists. If we define $\int_t^\infty r(z) dz$ to be $\int_0^\infty r(z) dz - \int_0^t r(z) dz$, we see that $\lim_{t \rightarrow \infty} \int_t^\infty r(z) dz = 0$. Note that $\int_t^\infty r(z) dz$ is also given by $\lim_{t_1 \rightarrow \infty} \int_t^{t_1} r(z) dz$.

Consider any partition $P_\delta^t = z_0, z_1, \dots$, of $[t, \infty)$ where $z_0 = t$ and $0 < z_{i+1} - z_i \leq \delta$ for $i \geq 0$. Then $r(z_i) \geq r(z) \geq r(z_{i+1})$ for $z_i \leq z \leq z_{i+1}$. It follows that the upper Riemann

sum, say $U(P_\delta^t)$ and the lower Riemann sum $L(P_\delta^t)$ are given by

$$U(P_\delta^t) = \sum_{i=0}^{\infty} r(z_i)(z_{i+1} - z_i); \quad L(P_\delta^t) = \sum_{i=0}^{\infty} r(z_{i+1})(z_{i+1} - z_i). \quad (\text{A.98})$$

The difference $U(P_\delta^t) - L(P_\delta^t)$ is then

$$\begin{aligned} U(P_\delta^t) - L(P_\delta^t) &= \sum_{i=0}^{\infty} (r(z_i) - r(z_{i+1}))(z_{i+1} - z_i) \\ &\leq \sum_{i=0}^{\infty} (r(z_i) - r(z_{i+1}))\delta = r(t)\delta \end{aligned}$$

Since r is Riemann integrable and non-increasing, $\lim_{t \rightarrow \infty} r(t) = 0$. Thus $\lim_{t \rightarrow \infty} [U(P_\delta^t) - L(P_\delta^t)] = 0$. Taking the limit $t \rightarrow \infty$, we see that the direct Riemann integral from t to ∞ exists. This implies that the direct Riemann integral over $[0, t)$ exists also.

d) Suppose $f(z) \geq 0$, defined for $z \geq 0$, is decreasing and Riemann integrable and that $f(z) \geq r(z)$ for all $z \geq 0$. Show that $r(z)$ is Directly Riemann integrable.

Solution: For any t and any partition P_δ^t for t to ∞ , we can apply (A.98) to f , showing that $U_f(P_\delta^t) - L_f(P_\delta^t) \leq \delta f(t)$. The Riemann and direct Riemann integral of f over t to ∞ exist, are the same, and lie between the upper and lower Riemann sums, so

$$U_f(P_\delta^t) \leq \delta f(t) + \int_t^\infty f(z) dz.$$

Since $r(z) \leq f(z)$, it is easy to see that with any partition P_δ^t , $U_r(P_\delta^t) \leq U_f(P_\delta^t)$. Similarly, $L_r(P_\delta^t) \geq 0$, so

$$U_r(P_\delta^t) - L_r(P_\delta^t) \leq \delta f(t) + \int_t^\infty f(z) dz.$$

Again, taking the limit $t \rightarrow \infty$, we see that $\lim_{t \rightarrow \infty} [U_r(P_\delta^t) - L_r(P_\delta^t)] = 0$, so the direct Riemann integral exists.

e) Let X be a nonnegative rv for which $\mathbb{E}[X^2] < \infty$. Show that $yF_X^c(x)$ is directly Riemann integrable. Hint: Consider $yF_X^c(y) + \int_y^\infty F_X^c(x) dx$ and use Figure 1.7 (or integration by parts) to show that this expression is decreasing in y .

Solution: The fact that $yF_X^c(y) + \int_y^\infty F_X^c(x) dx$ is non-increasing in y is obvious geometrically from Figure 1.7. It can be shown using integration by parts that

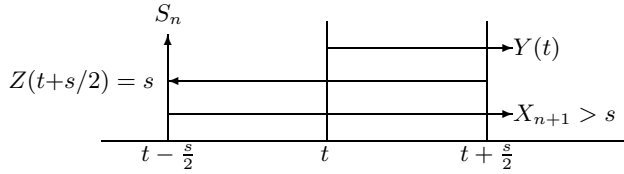
$$\mathbb{E}[X^2] = \int_0^\infty \left[yF_X^c(y) + \int_y^\infty F_X^c(x) dx \right] dy.$$

Thus the above Riemann integral exists. The argument above is an upper bound to $yF_X^c(y)$ for each y , and thus by (d), $yF_X^c(y)$ is directly Riemann integrable. It is curious that the integral of $yF_X^c(y)$ from 0 to ∞ is $\mathbb{E}[X^2]/2$.

Exercise 5.29: Let $Z(t), Y(t), \tilde{X}(t)$ denote the age, residual life, and duration at time t for a renewal counting process $\{N(t); t > 0\}$ in which the interarrival time has a density given by $f(x)$. Find the following probability densities; assume steady state.

a) $f_{Y(t)}(y | Z(t+s/2)=s)$ for given $s > 0$.

Solution: The condition $\{Z(t+s/2)=s\}$ is the event that an arrival, say S_n , occurred at $(t+s/2)-s = t-s/2$ and that no further arrival occurred in the interval $(t-s/2, t+s/2]$ (this assumes that $t \geq s/2$, which is implicit in the notion of steady state). Thus the conditional residual life $Y(t)$ at t must be greater than $s/2$ and, for $t > s/2$, $Y(t) = X_{n+1}(t+s/2)$. (see the diagram below).



Thus, adding the condition that $X_{n+1} > s$,

$$f_{Y(t)}(y | Z(t+s/2)=s) = \frac{f_X(y + s/2)}{F_X^c(s)}; \quad \text{for } y > s/2.$$

This is valid without the steady-state assumption so long as $t \geq s/2$.

b) $f_{Y(t), Z(t)}(y, z)$.

Solution: First review the joint distribution of age and duration given in (5.77) for the non-arithmetic case. In steady state, the arrival rate is $1/\bar{X}$, so for given $z > 0$, the probability of an arrival in the incremental interval $[t-z, t-z+dz)$ is $dz/\bar{X} + o(dz)$. Given such an arrival at $t-z$, the probability that the next arrival is in $[t-z+x, t-z+x+dx)$ is $f_X(x)dx$. If $x > z$, this corresponds to an age z and duration x . Thus, putting (5.77) in PDF form, the joint density is 0 except for $\tilde{X}(t) > Z(t) \geq 0$, where it is

$$f_{Z(t)\tilde{X}(t)}(z, x) = \frac{f_X(x)}{\bar{X}}; \quad \text{for } x > z \geq 0. \quad (\text{A.99})$$

Since $Y(t) = \tilde{X}(t) - Z(t)$, the joint density of $Y(t), Z(t)$ is 0 except for $Y(t) > 0, Z(t) \geq 0$, where it is

$$f_{Z(t)Y(t)}(z, y) = \frac{f_X(y+z)}{\bar{X}}; \quad \text{for } y > 0, z \geq 0.$$

An alternate approach is to observe that for $z \geq 0, y > 0, t > z$, the joint density of $f_{Z(t)Y(t)}(z, y)$ is the joint density of ‘an arrival’ at $t-z$ and the next arrival at $t+y$; more specifically, this is the sum over $n \geq 1$ of the density of the n th arrival at $t-z$ and the $(n+1)$ th arrival at $t+y$, i.e.,

$$f_{Z(t)Y(t)}(z, y) = \sum_{n \geq 1} f_{S_n}(t-z) f_X(y+z).$$

In the limit $t \rightarrow \infty$, Blackwell essentially says that $\sum_{n \geq 1} f_{S_n}(t-z) = 1/\bar{X}$. This same approach can be used throughout the exercise and clarifies the meaning of ‘arrival density.’

c) $f_{Y(t)|\tilde{X}(t)}(y|x)$.

Solution: Replacing $Z(t)$ in (A.99) by $Y(t) = \tilde{X}(t) - Z(t)$ (where the dependence of $Z(t)$ appears only in the condition $\tilde{X}(t) > Z(t) \geq 0$), the joint density of $Y(t), \tilde{X}(t)$ is 0 except for $\tilde{X}(t) \geq Y(t) > 0$, where it is

$$f_{Y(t)\tilde{X}(t)}(z, x) = \frac{f_X(x)}{\bar{X}}; \quad \text{for } 0 < y \leq x.$$

Conditional on $\tilde{X}(t) = x$, we see that the joint density is constant for $0 \leq Y(t) < x$, so

$$f_{Y(t)|\tilde{X}(t)}(y|x) = \frac{1}{x}; \quad \text{for } 0 < y \leq x.$$

d) $f_{Z(t)|Y(t-s/2)}(z|s)$ for given $s > 0$.

Solution: This is the essentially the same as (a), with the roles of $Z(t)$ and $Y(t)$ interchanged (but the steady state assumption is needed here). Using the same argument,

$$f_{Z(t)|Y(t-s/2)}(z|s) = \frac{f_X(z+s/2)}{F_X^c(s)}; \quad \text{for } z \geq s/2.$$

e) $f_{Y(t)}(y|Z(t+s/2) \geq s)$ for given $s > 0$.

Solution: By a slight modification of (a) and (b), we see that the joint density of $Y(t)$ and $Z(t+s/2)$ is

$$f_{Y(t), Z(t+s/2)}(y, z) = \frac{1}{\bar{X}} f_X(y+z-s/2); \quad \text{for } z \geq s/2, y > s/2.$$

By integrating this over $z \geq s/2$, we get the joint probability that $Z(t+s/2) \geq s/2$ and probability density that $Y(t) = y$.

$$\int_s^\infty f_{Y(t), Z(t+s/2)}(y, z) dz = \frac{1}{\bar{X}} \int_s^\infty f_X(y+z-s/2) dz = \frac{F_X^c(y+s/2)}{\bar{X}}.$$

The conditional density of $Y(t)$ conditional on $Z(t+s/2) \geq s$ is this result, divided by $\Pr\{Z(t+s/2) \geq s\}$. From the steady state assumption, $\Pr\{Z(t+s/2) \geq s\} = \Pr\{Z(t) \geq s\}$. Thus, using (5.88),

$$\Pr\{Z(t+s/2) \geq s\} = \Pr\{Z(t) \geq s\} = \frac{1}{\bar{X}} \int_s^\infty F_X^c(\tau) d\tau.$$

Thus

$$f_{Y(t)}(y|Z(t+s/2) \geq s) = \frac{F_X^c(y+s/2)}{\int_s^\infty F_X^c(\tau) d\tau}; \quad \text{for } y > s/2.$$

Exercise 5.30: a) Find $\lim_{t \rightarrow \infty} \{E[N(t)] - t/\bar{X}\}$ for a renewal counting process $\{N(t); t > 0\}$ with inter-renewal times $\{X_i; i \geq 1\}$. Hint: use Wald's equation.

Solution: Use Wald's equation, stopping on the first arrival after t . Thus $\mathbb{E}[S_{N(t)+1}] = \bar{X}\mathbb{E}[N(t) + 1]$ and

$$\mathbb{E}[N(t) - t/\bar{X}] = \frac{1}{\bar{X}}(\mathbb{E}[S_{N(t)+1}] - t) - 1 = \frac{1}{\bar{X}}\mathbb{E}[Y(t)] - 1,$$

where $Y(t) = S_{N(t)+1} - t$ is the residual life at t . As seen in detail in Section 5.7.4 for age and in less detail for residual life in Section 5.7.5, this has a limit if X is non-arithmetic (and typically does not have a limit otherwise). Assuming the nonarithmetic case, $\lim_{t \rightarrow \infty} \mathbb{E}[Y(t)] = \mathbb{E}[X^2]/2\bar{X}$. Thus

$$\lim_{t \rightarrow \infty} \mathbb{E}[N(t) - t/\bar{X}] = \frac{\mathbb{E}[X^2]}{2\bar{X}^2} - 1. \quad (\text{A.100})$$

b) Evaluate your result for the case in which X is an exponential random variable (you already know what the result should be in this case).

Solution: For an exponential rv X with $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, we have $\bar{X} = 1/\lambda$ and $\mathbb{E}[X^2] = 2/\lambda^2$. Thus the expression in (A.100) is zero, and we already know from the Poisson process that $\mathbb{E}[N(t)] = \lambda t$ for all $t > 0$. Thus this case serves as a sanity check on the general result.

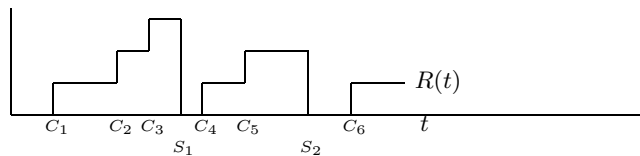
c) Evaluate your result for a case in which $\mathbb{E}[X] < \infty$ and $\mathbb{E}[X^2] = \infty$. Explain (very briefly) why this does not contradict the elementary renewal theorem.

Solution: If $\mathbb{E}[X] < \infty$ and $\mathbb{E}[X^2] = \infty$, then the limit in (A.100) is infinite, but the result does not show how $\mathbb{E}[N(t) - t/\bar{X}]$ approaches ∞ as $t \rightarrow \infty$. The elementary renewal theorem says that $\lim_{t \rightarrow \infty} (1/t)(\mathbb{E}[N(t) - t/\bar{X}]) = 0$, which shows that $\mathbb{E}[N(t) - t/\bar{X}]$ must approach ∞ as $t \rightarrow \infty$ more slowly than linearly in t .

Exercise 5.31: Customers arrive at a bus stop according to a Poisson process of rate λ . Independently, buses arrive according to a renewal process with the inter-renewal interval CDF $F_X(x)$. At the epoch of a bus arrival, all waiting passengers enter the bus and the bus leaves immediately. Let $R(t)$ be the number of customers waiting at time t .

a) Draw a sketch of a sample function of $R(t)$.

Solution: Let $S_n = X_1 + X_2 + \dots$ be the epoch of the n th bus arrival and C_m the epoch of the m th customer arrival. Then $\{C_m; m \geq 1\}$ is the sequence of arrival times in a Poisson process of rate λ and $\{S_n; n \geq 1\}$ is the renewal time sequence of a renewal process.



b) Given that the first bus arrives at time $X_1 = x$, find the expected number of customers picked up; then find $\mathbb{E}[\int_0^x R(t)dt]$, again given the first bus arrival at $X_1 = x$.

Solution: The expected number of customers picked up, given $S_1 = x$ is the expected number of arrivals in the Poisson process by x , i.e., λx . For $t < x$, $R(t)$ is the number of

customers that have arrived by time t . $R(t)$, for $t < S_1$, is independent of S_1 , so

$$\mathbb{E} \left[\int_0^x R(t) dt \right] = \int_0^x \lambda x dx = \frac{\lambda x^2}{2}.$$

c) Find $\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t R(\tau) d\tau$ (with probability 1). Assuming that F_X is a non-arithmetic distribution, find $\lim_{t \rightarrow \infty} \mathbb{E}[R(t)]$. Interpret what these quantities mean.

Solution: In (b), we found the expected reward in an inter-renewal interval of size x , *i.e.*, we took the expected value over the Poisson process given a specific value of X_1 . Taking the expected value of this over the bus renewal interval, we get $(1/2)\lambda \mathbb{E}[X^2]$. This is the expected accumulated reward over the first bus inter-renewal period, denoted $\mathbb{E}[R_1]$ in (5.22). It is the expected number of customers at each time, integrated over the time until the first bus arrival. This integrated value is not the expected number waiting for a bus, but rather will be used as a step in finding the time-average number waiting over time. Since the bus arrivals form a renewal process, this is equal to $\mathbb{E}[R_n]$ for each inter-renewal period n . By (5.24),

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t R(\tau) d\tau = \frac{\mathbb{E}[R_n]}{\bar{X}} = \frac{\lambda \mathbb{E}[X^2]}{2\bar{X}} \quad \text{WP1.} \quad (\text{A.101})$$

Since the renewals are non-arithmetic, this is the same as $\lim_{t \rightarrow \infty} \mathbb{E}[R(t)]$.

Note that this is not the same as the expected number of customers per bus. It is the expected number of waiting customers, averaged over time. If a bus is highly delayed, a large number of customers accumulate, but also, because of the large interval waiting for the bus, the contribution to the time average grows as the square of the wait.

d) Use (c) to find the time-average expected wait per customer.

Solution: Little's theorem can be applied here since (A.101) gives the limiting time-average number of customers in the system, L . The time-average wait per customer W is $W = L/\lambda$, which from (A.101) is $\mathbb{E}[X^2]/2\bar{X}$. The system here does not satisfy the conditions of Little's theorem, but it is easy to check that the proof of the theorem applies in this case.

e) Find the fraction of time that there are no customers at the bus stop. (Hint: this part is independent of (a), (b), and (c); check your answer for $\mathbb{E}[X] \ll 1/\lambda$).

Solution: There are no customers at the bus stop at the beginning of each renewal period. Let U_n be the interval from the beginning of the n th renewal period until the first customer arrival. It is possible that no customer will arrive before the next bus arrives, so the interval within the n th inter-renewal period when there is no customer waiting is $\min(U_n, X_n)$. Consider a reward function $R(t)$ equal to 1 when no customer is in the system and 0 otherwise. The accumulated reward R_n within the n th inter-renewal period is then $R_n = \min(U_n, X_n)$. Thus, using the independence of U_n and X_n ,

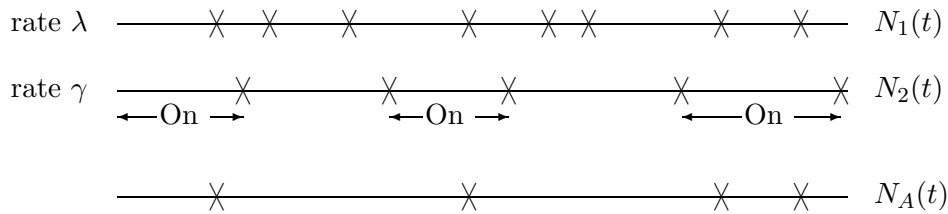
$$\begin{aligned} \mathbb{E}[R_n] &= \int_0^\infty \Pr\{\min(U_n, X_n) > t\} dt = \int_0^\infty \Pr\{U_n > t\} \Pr\{X_n > t\} dt \\ &= \int_0^\infty \Pr\{X_n > t\} e^{-\lambda t} dt. \end{aligned}$$

Using (5.24), the limiting time-average fraction of time when no customers are waiting is $(1/\bar{X}) \int_0^t \Pr\{X_n > t\} e^{-\lambda t} dt$. Checking when $\bar{X} \ll 1/\lambda$, we see that the above integral is close to $\bar{X}$. In particular, if X_n is exponential with rate μ , we see that the fraction above is $\mu/(\mu + \lambda)$.

Exercise 5.32: Consider the same setup as in Exercise 5.31 except that now customers arrive according to a non-arithmetic renewal process independent of the bus arrival process. Let $1/\lambda$ be the expected inter-renewal interval for the customer renewal process. Assume that both renewal processes are in steady state (i.e., either we look only at $t \gg 0$, or we assume that they are equilibrium processes). Given that the n th customer arrives at time t , find the expected wait for customer n . Find the expected wait for customer n without conditioning on the arrival time.

Solution: The expected time customer n waits for a bus, given that $C_n = t$, where C_n is n 's arrival time, is the residual life of the bus process at time t . Since the bus process is in equilibrium, this is the same as the time-average residual life of the bus process, which is $E[X^2]/2\bar{X}$. This does not depend on t , and is thus the unconditional expected wait.

Exercise 5.33: Let $\{N_1(t); t > 0\}$ be a Poisson counting process of rate λ . Assume that the arrivals from this process are switched on and off by arrivals from a non-arithmetic renewal counting process $\{N_2(t); t > 0\}$ (see figure below). The two processes are independent.



Let $\{N_A(t); t \geq 0\}$ be the switched process; that is $N_A(t)$ includes arrivals from $\{N_1(t); t > 0\}$ while $N_2(t)$ is even and excludes arrivals from $\{N_1(t); t > 0\}$ while $N_2(t)$ is odd.

a) Is $N_A(t)$ a renewal counting process? Explain your answer and if you are not sure, look at several examples for $N_2(t)$.

Solution: $N_A(t)$ is not a renewal process. An easy way to see this is to start by letting $N_2(t)$ be deterministic with $\gamma = 1$ and let λ be very large. Then $N_A(t)$ contains large numbers of arrivals from each even integer to the subsequent odd integer and none from odd to even. The inter-renewal interval for $N_A(t)$ then depends on the placement of each arrival relative to the next even integer.

This argument doesn't quite work since $N_2(t)$ is arithmetic here. This is cured by letting the inter-renewal interval for process 2 be uniform between 0.999 and 1.001. Then the interarrivals for $N_A(t)$ are still bunched into groups separated by approximately 1 and it is clear that the interarrivals close to the end of a group have a different distribution than those close to the beginning, i.e., the interarrival times of $N_A(t)$ are not IID.

Note, as a comment on doing research, that if one doesn't start with a simple arithmetic process here, then it is difficult to get the insight to construct a correct non-arithmetic process.

b) Find $\lim_{t \rightarrow \infty} \frac{1}{t} N_A(t)$ and explain why the limit exists with probability 1. Hint: Use symmetry—that

is, look at $N_1(t) - N_A(t)$. To show why the limit exists, use the renewal-reward theorem. What is the appropriate renewal process to use here?

Solution: Let $\{N_{2e}(t); t > 0\}$ be the arrival process of the even numbered arrivals of $\{N_2(t); t > 0\}$. This is seen to be another non-arithmetic renewal process since each inter-renewal interval is the sum $X_1 + X_2$ of 2 inter-renewal intervals from $N_2(t)$. Then consider a reward $R(t)$ of 1 for the initial part of an inter-renewal interval of the N_{2e} process until the subsequent odd renewal of $\{N_2(t); t > 0\}$. Let $R(t) = 0$ from this odd renewal until the subsequent renewal of $N_{2e}(t)$. The time-average renewal reward theorem then shows that $\lim_{t \rightarrow \infty} (1/t) \int_0^t R(\tau) d\tau$ exists WP1. The symmetry between $R(t)$ and $1 - R(t)$ shows that this limit is $1/2$. From the memoryless property of the Poisson process, then

$$\lim_{t \rightarrow \infty} \frac{1}{t} N_A(t) = \lambda/2 \quad \text{WP1.}$$

c) Now suppose that $\{N_1(t); t > 0\}$ is a non-arithmetic renewal counting process but not a Poisson process and let the expected inter-renewal interval be $1/\lambda$. For any given δ , find $\lim_{t \rightarrow \infty} \mathbf{E}[N_A(t + \delta) - N_A(t)]$ and explain your reasoning. Why does your argument in (b) fail to demonstrate a time average for this case?

Solution: The argument in (b) fails here because the arrivals in $N_1(t)$ are not memoryless here and thus the number of process 1 arrivals in a given even to odd period of $N_2(t)$ might depend on the duration of other renewal intervals from $N_2(t)$.

Applying Blackwell's theorem to $N_2(t)$, $\lim_{t \rightarrow \infty} \mathbf{E}[N_2(t + \delta) - N_2(t)] = \gamma\delta$. For all large enough t for the given δ , $\mathbf{E}[N_2(t + \delta) - N_2(t)] = \gamma\delta \pm o(\delta)$. From the Markov inequality, then, $\Pr\{N_2(t + \delta) - N_2(t) \geq 1\} \leq \gamma\delta \pm o(\delta)$. This means that $N_2(\tau)$ is even throughout $t \leq \tau < t + \delta$ with probability $(1/2)(1 - \delta\gamma \pm o(\delta))$.

If $N_2(\tau)$ is even for $[t, t + \delta)$, then $N_A(t + \delta) - N_A(t) = N_1(t + \delta) - N_1(t)$. If $N_2(\tau)$ is odd for $[t, t + \delta)$, then $N_A(t + \delta) - N_A(t) = 0$. The remaining case, where $N_2(\tau)$ changes is a low probability event, but $N_A(t + \delta) - N_A(t) = 0$ is between the above two values in this event. Thus

$$\begin{aligned} \mathbf{E}[N_A(t + \delta) - N_A(t)] &= \frac{(1 - \delta\gamma \pm o(\delta))}{2} \mathbf{E}[N_1(t + \delta) - N_1(t)] \\ &= \frac{(1 - \delta\gamma \pm o(\delta))(\lambda\delta \pm o(\delta))}{2} = \frac{\lambda\delta \pm o(\delta)}{2}, \end{aligned}$$

where in the last step we have used Blackwell's theorem on $N_1(t)$. By adding together an arbitrarily large number of these incremental intervals and going to the limit, we get

$$\lim_{t \rightarrow \infty} \mathbf{E}[N_A(t + \delta) - N_A(t)] = \lambda\delta/2.$$

Exercise 5.34: An M/G/1 queue has arrivals at rate λ and a service time distribution given by $F_Y(y)$. Assume that $\lambda < 1/\mathbf{E}[Y]$. Epochs at which the system becomes empty define a renewal process. Let $F_Z(z)$ be the CDF of the inter-renewal intervals and let $\mathbf{E}[Z]$ be the mean inter-renewal interval.

a) Find the fraction of time that the system is empty as a function of λ and $\mathbf{E}[Z]$. State carefully what you mean by such a fraction.

Solution: Consider a renewal reward process where the reward is 1 when the system is empty and 0 otherwise. At the beginning of each renewal period, the system remains empty until the first subsequent arrival; since the arrival process is Poisson, that interval is exponential with a mean time $1/\lambda$. Thus the expected aggregate reward per renewal interval is $1/\lambda$ and the time-average reward, WP1, is $1/\lambda \mathbb{E}[Z]$. Thus f_{idle} , the fraction of time (WP1) that the server is idle is

$$f_{\text{idle}} = \frac{1}{\lambda \mathbb{E}[Z]}. \quad (\text{A.102})$$

This means that for all ω , except perhaps a set of probability 0, $\lim_{t \rightarrow \infty} (1/t)$ [aggregate time that sample path ω is idle from 0 to t] = $1/\lambda \bar{Z}$.

b) Apply Little's theorem, not to the system as a whole, but to the number of customers in the server (i.e., 0 or 1). Use this to find the fraction of time that the server is busy.

Solution: Little's theorem says that $\bar{L} = \lambda \bar{W}$. Here $\bar{L}$ is the time-average number of customers in the server and $\bar{W}$ is the time-average service time WP1. Little also says that each of these converge WP1. Since the server is busy when it contains one customer and idle when it contains none, $\bar{L}$ is f_{busy} , the fraction of time (WP1) that the server is busy.. Similarly, since the expected service time is $\bar{Y}$ and successive services are IID, $\bar{W} = \bar{Y}$. Thus $f_{\text{busy}} = \lambda \bar{Y}$.

c) Combine your results in (a) and (b) to find $\mathbb{E}[Z]$ in terms of λ and $\mathbb{E}[Y]$; give the fraction of time that the system is idle in terms of λ and $\mathbb{E}[Y]$.

Solution: Since $f_{\text{busy}} = 1 - f_{\text{idle}}$ and $f_{\text{busy}} = \lambda \bar{Y}$, we have $f_{\text{idle}} = 1 - \lambda \bar{Y}$, Combining this with (A.102), we get

$$\mathbb{E}[Z] = \frac{1}{\lambda(1 - \lambda \bar{Y})}.$$

d) Find the expected duration of a busy period.

Solution: Each renewal interval contains one idle period and one busy period. Thus the expected duration of a busy period is $\mathbb{E}[Z] - 1/\lambda = \frac{\bar{Y}}{1 - \lambda \bar{Y}}$.

Exercise 5.35: Consider a sequence $X_1, X_2, \dots$ of IID binary rv's with $\Pr\{X_n = 1\} = p_1$ and $\Pr\{X_n = 0\} = p_0 = 1 - p_1$. A *renewal* is said to occur at time $n \geq 2$ if $X_{n-1} = 0$ and $X_n = 1$.

a) Show that $\{N(n); n > 0\}$ is a renewal counting process where $N(n)$ is the number of renewals up to and including time n .

Solution: Let $Y_1, Y_2, \dots$ be the intervals between successive occurrences of the pair (0,1). Note that $Y_1 \geq 2$ since the first binary rv occurs at time 1. Also $Y_n \geq 2$ for each $n > 1$, since if there is an occurrence at time n (i.e., if $X_{n-1} = 0, X_n = 1$, then the pair $X_n = 0, X_{n+1} = 1$ is impossible. The length of each inter-arrival interval is determined solely by the bits within that interval, which are IID both among themselves and between intervals. Thus the inter-arrival intervals are IID and are thus renewals, so the counting process is also a renewal process.

b) What is the probability that a renewal occurs at time $n, n \geq 2$?

Solution: The probability of a renewal at time $n \geq 2$ is $\Pr\{X_{n-1} = 0, X_n = 1\} = p_0 p_1$.

c) Find the expected inter-renewal interval; use Blackwell's theorem.

Solution: The inter-renewal distribution is arithmetic with span 1 and the expected number of renewals in the interval $(n-1, n]$ is simply the probability of a renewal at time n , i.e., $\Pr\{X_{n-1} = 0, X_n = 1\} = p_0 p_1$. Thus, from Blackwell's theorem,

$$\mathbf{E}[N(n)] - \mathbf{E}[N(n-1)] = p_0 p_1 = \frac{1}{\mathbf{E}[Y]}.$$

The expected inter-renewal interval is then $1/p_0 p_1$; this is also the expected time until the first occurrence of (0,1). It is not hard to find $\mathbf{E}[Y]$ by elementary means without using Blackwell, but the advantage of using Blackwell's theorem is clear in (g) where the string of interest has an arbitrary length.

d) Now change the definition of renewal; a renewal now occurs at time n if $X_{n-1} = 1$ and $X_n = 1$. Show that $\{N^*(n); n \geq 0\}$ is a delayed renewal counting process where N_n^* is the number of renewals up to and including n for this new definition of renewal.

Solution: Let $Y_1, Y_2, \dots$ now be the inter-arrival intervals between successive occurrences of the pair (1,1). Note that $Y_1 \geq 2$ as before since the first binary rv occurs at time 1, so the string (1,1) cannot appear until time 2. In this case, however, it is possible to have $Y_i = 1$ for $i \geq 2$ since if $(X_{m-1}, X_m, X_{m+1}) = (1, 1, 1)$, then a renewal occurs at time m and also at $m+1$.

Since successive occurrences of (1,1) can overlap, we must be a little careful to show that $Y_1, Y_2, \dots$ are independent. Given $Y_1, \dots, Y_i$, we see that $Y_{i+1} = 1$ with probability p_1 , and this is independent of $Y_1, \dots, Y_i$, so the event $\{Y_{i+1} = 1\}$ is independent of $Y_1, \dots, Y_i$. In the same way $\{Y_{i+1} = k\}$ is independent of $Y_1, \dots, Y_i$, so Y_{i+1} is independent of $Y_1, \dots, Y_i$ for all $i \geq 2$, establishing the independence. The fact that $Y_2, Y_3, \dots$ are equally distributed is obvious as before. Thus N^* is a delayed renewal process.

e) Find $\mathbf{E}[Y_i]$ for $i \geq 2$ for the case in (d).

Solution: Using the same argument as in (c), $\mathbf{E}[Y_i] = \frac{1}{p_1^2}$ for $i \geq 2$.

f) Find $\mathbf{E}[Y_1]$ for the case in (d), Hint: Show that $\mathbf{E}[Y_1|X_1 = 1] = 1 + \mathbf{E}[Y_2]$ and $\mathbf{E}[Y_1|X_1 = 0] = 1 + \mathbf{E}[Y_1]$.

Solution: If $X_1 = 1$, then the wait for a renewal, starting at time 1, has the same probabilities as the wait for a renewal starting at the previous renewal. If $X_1 = 0$, then the wait, starting at time 1 is the same as it was before. Thus

$$\mathbf{E}[Y_1] = 1 + p_1 \mathbf{E}[Y_2] + p_0 \mathbf{E}[Y_1].$$

Combining the terms in $\mathbf{E}[Y_1]$, we have $p_1 \mathbf{E}[Y_1] = 1 + p_1 \mathbf{E}[Y_2]$, so

$$\mathbf{E}[Y_1] = \frac{1}{p_1} + \mathbf{E}[Y_2] = \frac{1}{p_1} + \frac{1}{p_1^2}.$$

g) Looking at your results above for the strings (0,1) and (1,1), show that for an arbitrary string $\mathbf{a} = (a_1, \dots, a_k)$, the arrival process of successive occurrences of the string is a renewal process if no proper suffix of $\mathbf{a}$ is a prefix of $\mathbf{a}$. Otherwise it is a delayed renewal process.

Solution: For the 2 strings looked at, (0,1) has the property that no proper suffix is a prefix, and (1,1) has the property that the suffix (1) is equal to the prefix (1). In general, if there is a proper suffix of length i equal to a prefix of length i , then, when the string occurs, the final i bits of the string provide a prefix for the next string, which can then have a length as small as $k - i$. Since the first occurrence of the string can not appear until time k , The arrival process is a delayed renewal process, where the independence and the identical distribution after the first occurrence follow the arguments for the strings of length 1 above. If no proper suffix of $\mathbf{a}$ is equal to a prefix of $\mathbf{a}$, then the end of one string provides no beginning for a new string and the first renewal is statistically identical to the future renewals; *i.e.*, the process is a renewal process.

h) Suppose a string $\mathbf{a} = (a_1, \dots, a_k)$ of length k has no proper suffixes equal to a prefix. Show that the time to the first renewal satisfies

$$\mathbb{E}[Y_1] = \frac{1}{\prod_{\ell=1}^k p_{a_\ell}}. \quad (\text{A.103})$$

Solution: As shown in (g), the successive occurrences of $\mathbf{a}$ constitute a renewal process (rather than a delayed renewal process). The denominator in (A.103) is then the probability of a renewal at any given time $n \geq k$, and the result follows from Blackwell's theorem as in the earlier parts.

i) Suppose the string $\mathbf{a} = (a_1, \dots, a_k)$ has at least one proper suffix equal to a prefix, and suppose i is the length of the longest such suffix. Show that the expected time until the first occurrence of $\mathbf{a}$ is given by

$$\mathbb{E}[Y_1] = \frac{1}{\prod_{\ell=1}^k p_{a_\ell}} + \mathbb{E}[U_i], \quad (\text{A.104})$$

where $\mathbb{E}[U_i]$ is the expected time until the first occurrence of the string $(a_1, \dots, a_i)$.

Solution: The first term above is the expected inter-renewal interval, which follows from Blackwell. If a renewal occurs at time n , then, since $(a_1, \dots, a_i)$ is both a prefix and suffix of $\mathbf{a}$, we must have $(X_{n-i+1}, \dots, X_n) = (a_1, \dots, a_i)$. This means that an inter-renewal interval is the time from an occurrence of $(a_1, \dots, a_i)$ until the first occurrence of $(a_1, \dots, a_k)$. Starting from time 0, it is necessary for $(a_1, \dots, a_i)$ to occur before $(a_1, \dots, a_k)$, so we can characterize the time until the first occurrence of $\mathbf{a}$ as the sum of the time to the first occurrence of $(a_1, \dots, a_i)$ plus the time from that first occurrence to the first occurrence of $(a_1, \dots, a_k)$; as explained in greater detail in Example 4.5.1, this is statistically the same as an inter-renewal time.

j) Show that the expected time until the first occurrence of $\mathbf{a} = (a_1, \dots, a_k)$ is given by

$$\mathbb{E}[Y_1] = \sum_{i=1}^k \frac{\mathbb{I}_i}{\prod_{\ell=1}^i p_{a_\ell}}, \quad (\text{A.105})$$

where, for $1 \leq i \leq k$, $\mathbb{I}_i$ is 1 if the prefix of $\mathbf{a}$ of length i is equal to the suffix of length i . Hint: Use (A.104) recursively. Also show that if $\mathbf{a}$ has a suffix of length i equal to the prefix of length i and also a suffix of length j equal to a prefix of length j where $j < i$, then the suffix of $(a_1, \dots, a_i)$ of length j is also equal to the prefix of both $\mathbf{a}$ and $(a_1, \dots, a_i)$ of length j .

Solution: Note that the suffix of $\mathbf{a}$ of length k is not a proper suffix and is simply equal to $\mathbf{a}$ itself and thus also equal to the prefix of $\mathbf{a}$ of length k . This is the final term of the expression above and also the first term in (A.104).

The next smaller value of i for which $\mathbb{I}_i = 1$ is the largest i for which a proper suffix of $\mathbf{a}$ of length i is equal to a prefix. If we apply (A.104) to finding the expected first occurrence of $a_1, \dots, a_i$ for that i , we see that it is the next to final non-zero term in the above sum plus the expected time for the first occurrence of the largest j for which the proper suffix of $(a_1, \dots, a_i)$ is equal to the prefix of $(a_1, \dots, a_i)$ of length j . Now the first j bits of $(a_1, \dots, a_i)$ are also the first j bit of $\mathbf{a}$. Also, since $(a_1, \dots, a_i) = (a_{k-i+1}, \dots, a_k)$, we see that the last j bits of $(a_1, \dots, a_i)$ are also the last j bits of $\mathbf{a}$. Thus j is the third from last value for which $\mathbb{I}_i = 1$.

This argument continues (recursively) through all the terms in the above sum. We can also interpret the partial sums above as the expected times of occurrence for each prefix that is equal to a suffix.

k) Use (i) to find, first, the expected time until the first occurrence of $(1,1,1,1,1,0)$ and, second, that of $(1,1,1,1,1,1)$. Use (4.31) to check the relationship between these answers.

Solution: The first string has no proper suffixes equal to a prefix, so the expected time is $p_1^{-6}p_0^{-1}$. For the second string, all suffixes are also prefixes, so the expected time is

$$\begin{aligned} \sum_{j=1}^6 p_1^{-j} &= p_1^{-6}(1 + p_1 + \dots + p_1^5) = p_1^{-6}(1 - p_1^6)/(1 - p_1) \\ &= p_1^{-6}p_0^{-1} - p_0^{-1}. \end{aligned}$$

Adapting (4.31) to the notation here,

$$\begin{aligned} \mathbb{E}[\text{time to } 111111] &= 1 + p_1 \mathbb{E}[\text{time to } 111111] + p_0 \mathbb{E}[\text{time to } 1111110] \\ p_0 \mathbb{E}[\text{time to } 111111] &= 1 + p_0 \mathbb{E}[\text{time to } 1111110], \end{aligned}$$

which checks with the answer above.

Exercise 5.36: A large system is controlled by n identical computers. Each computer independently alternates between an operational state and a repair state. The duration of the operational state, from completion of one repair until the next need for repair, is a random variable X with finite expected duration $\mathbb{E}[X]$. The time required to repair a computer is an exponentially distributed random variable with density $\lambda e^{-\lambda t}$. All operating durations and repair durations are independent. Assume that all computers are in the repair state at time 0.

a) For a single computer, say the i th, do the epochs at which the computer enters the repair state form a renewal process? If so, find the expected inter-renewal interval.

Solution: For the i th computer, let $\{X_{i,m}; m > 0\}$ be the sequence of successive operating durations between repairs. Let $V_{i,1}$ be the completion time of the first repair. Since the repair times are each exponential at rate λ , this first repair completion time is also exponential, independent of how long computer i has been in the repair state before time 0. Let $\{V_{i,m}; m > 1\}$ be the subsequent repair intervals of the i th computer; these are also exponential at rate λ and are independent of each other and of $V_{i,1}$. Thus the sequence $\{V_{i,m}; m > 0\}$ is IID. Let $\{U_{i,m} = V_{i,m} + X_{i,m}; m > 0\}$ be the sequence of successive repair plus operation intervals. From the specified independence of repair and operational intervals, $\{U_{i,m}; m > 0\}$ is a sequence of IID rv's. Thus this sequence of rv's is the sequence

of interarrival intervals of a renewal process, and the successive epochs at which computer i enters the repair state after time 0 are the epochs of a renewal process. The expected inter-renewal interval is $\bar{U} = \bar{X} + 1/\lambda$.

Note that this is essentially obvious, except for the need to be clear about the first repair time of computer i starting at time 0.

b) Do the epochs at which it enters the operational state form a renewal process?

Solution: The first epoch at which it enters the operational state is $V_{i,1}$ and each successive inter-epoch interval is $X_{i,n} + V_{i,n+1}$ for $n > 0$. Thus the inter-epoch intervals are independent, but the first has a different distribution than the rest. This means that these epochs do not form a renewal process, but instead form a delayed renewal process.

c) Find the fraction of time over which the i th computer is operational and explain what you mean by fraction of time.

Solution: Let $R_i(t)$ be a renewal reward function for the renewal process in (a). Let $R_i(t) = 1$ when computer i is in the operational state and let $R_i(t) = 0$ otherwise. The time average of $R_i(t)$, *i.e.*, $\lim_{t \rightarrow \infty} (1/t) \int_0^t R_i(\tau) d\tau$ is the fraction of time computer i is operational. From Theorem 5.4.5, this limit converges WP1 to the value $\bar{X}/(\bar{X} + 1/\lambda)$.

Those of you following the derivations of theorems very carefully might notice that Theorem 5.4.5 does not quite apply here, since $R_i(t)$ is a function of $\{V_i(t); i > 0\}$ as well as the inter-renewal intervals. As discussed at the end of Section 5.4 and in 5.5.3-5, however, the theorem does apply to these more general cases.

d) Let $Q_i(t)$ be the probability that the i th computer is operational at time t and find $\lim_{t \rightarrow \infty} Q_i(t)$.

Solution: Using the notation of (c),

$$Q_i(t) = \Pr\{R_i(t) = 1\} = E[R_i(t)].$$

Thus $\lim_{t \rightarrow \infty} Q_i(t) = \lim_{t \rightarrow \infty} E[R_i(t)]$, assuming the limit exists. Since $U_i(n) = V_{i,n} + X_{i,n}$ is non-arithmetic, Section 5.7 of the text essentially shows that this limiting ensemble average exists and equals the time average, *i.e.*, that $\lim_{t \rightarrow \infty} Q_i(t) = \bar{X}/(\bar{X} + 1/\lambda)$.

As in (c), the argument in Section 5.7 has to be extended slightly, but going through the details does not seem to be warranted here.

e) The system is in failure mode at a given time if all computers are in the repair state at that time. Do the epochs at which system failure modes begin form a renewal process?

Solution: At time 0, all computers are in the repair state, and thus each requires an independent exponentially distributed repair time (of rate λ) which is independent of all subsequent repair intervals and operational intervals. At each subsequent entry to a failure mode, each computer requires an independent exponentially distributed repair time (of rate λ), independent of all subsequent repair intervals and operational intervals. Thus the epochs of entering a failure mode constitute a renewal process.

f) Let $\Pr\{t\}$ be the probability that the the system is in failure mode at time t . Find $\lim_{t \rightarrow \infty} \Pr\{t\}$. Hint: look at (d).

Solution: The repair and operation intervals of each computer are independent of those of all other computers, and thus, for each t the event that computer i is operational at time t is independent of the events that the other computers are operational at time t . Thus

$$\Pr\{t\} = [1 - Q_i(t)]^n.$$

Using (d) to take the limit as $t \rightarrow \infty$,

$$\lim_{t \rightarrow \infty} \Pr\{t\} = \left[1 - \frac{\bar{X}}{\bar{X} + 1/\lambda}\right]^n = \left[\frac{1}{\lambda\bar{X} + 1}\right]^n.$$

g) For δ small, find the probability that the system enters failure mode in the interval $(t, t + \delta]$ in the limit as $t \rightarrow \infty$.

Solution: For δ sufficiently small, the probability of two or more computers entering or leaving the repair mode in $(t, t + \delta]$ is negligible compared with that of one computer. Thus the system will enter failure mode in $(t, t + \delta]$ if $n - 1$ computers are in repair mode at time t and the remaining computer enters repair mode in $(t, t + \delta]$. From (f), the limiting probability that a given set of $n - 1$ computers are in repair mode at t is $[1/(1 + \lambda\bar{X})]^{n-1}$. Applying Blackwell's theorem to the renewal process of the remaining computer, the limiting probability that the remaining computer enters repair mode (*i.e.*, that computer renewal process has a renewal) in $(t, t + \delta]$ is $\delta/(\bar{X} + 1/\lambda)$. Since there are n choices for the remaining computer, the probability that the system enters failure mode in $(t, t + \delta]$ is $n\lambda\delta/(\lambda\bar{X} + 1)^n$.

h) Find the expected time between successive entries into failure mode.

Solution: Let $\{W_i; i > 0\}$ be the sequence of inter-renewal intervals for the renewal process of system failures. Thus we are asked to find $E[W]$. From Blackwell's theorem, applied to this failure renewal process, the expected number of failures in $(t, t + \delta]$, in the limit $t \rightarrow \infty$, is $\delta/E[W]$. For δ small, this expected number is the same as the probability of a failure as given in (g). Thus,

$$E[W] = (\lambda\bar{X} + 1)^n / \lambda n.$$

An alternate approach is to first observe that the expected duration of a failure is the time until the first of the n computers is repaired, *i.e.*, $1/\lambda n$. Then, letting $R(t)$ be a reward of 1 during failure mode and 0 elsewhere, we see that $1/\lambda n$ is the expected reward during an inter-renewal interval. Then, using Corollary 5.4.6, the time-average reward is $\lambda n/E[W]$ WP1. But now using the equality of time and limiting ensemble average, the time-average reward is $\lim_{t \rightarrow \infty} \Pr\{t\}$ as found in (f).

i) Next assume that the repair time of each computer has an arbitrary density rather than exponential, but has a mean repair time of $1/\lambda$. Do the epochs at which system failure modes begin form a renewal process?

Solution: For each computer, the set of epochs at which that computer enters the repair state form a renewal process or a delayed renewal process, depending on the assumption about whether a repair interval starts at time 0 or is in process at time 0. When a system failure occurs, however, each computer (except the last to fail) has been in the repair state for some random period, and the set of those periods affects the interval until the next

failure. Thus the epochs at which system failures start do not form a renewal process and also do not form a delayed renewal process.

j) Repeat (f) for the assumption in (i).

Solution: Despite the fact that the sequence of failure epochs do not form a renewal process, (f) still gives the limit, as $t \rightarrow \infty$, of the probability of a failure at time t , $\Pr\{t\}$. The reason is that the independence between computers still holds, so that $\Pr\{t\} = (1 - Q_i(t))^n$ as in (f), where now $Q_i(t)$ is the probability that computer i is in the operational mode at time t . Since the times at which computer i enters the repair state is still a renewal process (or a delayed renewal process), the limit $\lim_{t \rightarrow \infty} Q_i(t) = \bar{X}/(\bar{X} + 1/\lambda)$ as shown in (d). Thus the answer in (f) is still valid.

Exercise 5.37 : Let $\{N_1(t); t > 0\}$ and $\{N_2(t); t > 0\}$ be independent renewal counting processes. Assume that each has the same CDF $F(x)$ for interarrival intervals and assume that a density $f(x)$ exists for the interarrival intervals.

a) Is the counting process $\{N_1(t) + N_2(t); t > 0\}$ a renewal counting process? Explain.

Solution: For the special case in which $\{N_1(t); t > 0\}$ and $\{N_2(t); t > 0\}$ are Poisson, the sum $N_1(t) + N_2(t)$ is also Poisson and thus renewal. Otherwise, when a renewal from one process occurs, the residual life of the other process (*i.e.*, the time until the next arrival) is statistically dependent on the age of that process, which is not the same for each arrival of the other process. Thus $N_1(t) + N_2(t)$ is not a renewal process. For example, if the inter-renewal interval for $\{N_1(t); t > 0\}$ is the sum of two IID exponentials (*i.e.*, if $N_1(t)$ is the number of even arrivals in a Poisson process), then, on a renewal at time t for process 1, the time until the next renewal depends on whether process 2 has experienced an even or odd number of Poisson arrivals at time t .

b) Let $Y(t)$ be the interval from t until the first arrival (from either process) after t . Find an expression for the CDF of $Y(t)$ in the limit $t \rightarrow \infty$ (you may assume that time averages and ensemble averages are the same).

Solution: Let $Y_1(t)$ be the residual life for process 1 and let $Y_2(t)$ be the residual life for process 2. Then $Y(t) = \min[Y_1(t), Y_2(t)]$. Since the two processes are independent, $Y_1(t)$ and $Y_2(t)$ are independent rv's, and, of course, are non-negative. It follows that for any $y > 0$, $\Pr\{Y(t) > y\} = \Pr\{Y_1(t) > y\} \Pr\{Y_2(t) > y\}$. Thus, for the CDF's,

$$\begin{aligned} F_{Y(t)}(y) &= 1 - (1 - F_{Y_1(t)}(y))(1 - F_{Y_2(t)}(y)) \\ &= 1 - (1 - F_{Y_1(t)}(y))^2, \end{aligned}$$

where we have used the fact that $Y_1(t)$ and $Y_2(t)$ are IID. Since $\{Y_1(t); t > 0\}$ is a renewal process, the CDF of the residual life is given (as a time average WP1) by (5.30) as

$$F_{Y_1}(y) = \frac{1}{\bar{X}} \int_0^y \Pr\{X > x\} dx \quad \text{WP1.}$$

Given that time and ensemble averages are the same, *i.e.*, given that each process is non-arithmetic, this is also the limiting ensemble average as $t \rightarrow \infty$ for $Y_1(t)$.

c) Assume that a reward R of rate 1 unit per second starts to be earned whenever an arrival from process 1 occurs and ceases to be earned whenever an arrival from process 2 occurs.

Assume that $\lim_{t \rightarrow \infty} (1/t) \int_0^t R(\tau) d\tau$ exists with probability 1 and find its numerical value.

Solution: Assuming that the above limit exists, it must have the value $1/2$ from symmetry. To see this, let R^* be a reward of rate 1 unit per second that starts whenever an arrival from 2 occurs and ceases whenever an arrival from process 1 occurs. Then at any time τ , either $R(\tau) = 1$ (if the last arrival was from 1) or $R^*(\tau) = 1$ (if the last arrival was from 2). Thus $R(\tau) + R^*(\tau) = 1$ for all τ . Since the interarrival intervals for the two processes are IID, the two processes are identically distributed. Thus $R(\tau)$ and its integral over τ are identically distributed to $R^*(t)$ and its integral over τ . Since the sum integrates to 1, each individually sum to $1/2$. Although the above explanation was somewhat lengthy, it is essentially obvious.

d) Let $Z(t)$ be the interval from t until the first time after t that $R(t)$ (as in (c)) changes value. Find an expression for $E[Z(t)]$ in the limit $t \rightarrow \infty$. Hint: Make sure you understand why $Z(t)$ is not the same as $Y(t)$ in (b). You might find it easiest to first find the expectation of $Z(t)$ conditional on both the duration of the $\{N_1(t); t > 0\}$ interarrival interval containing t and the duration of the $\{N_2(t); t \geq 0\}$ interarrival interval containing t ; draw pictures!

Solution: Rename $Z(t)$ as $U(t)$ to avoid confusion with the age variables $Z_1(t)$ and $Z_2(t)$ and leave out t since we are dealing with a fixed arbitrarily large t throughout. Note that if $Z_1 > Z_2$, then the next change in R after t is at $t + Y_1$. if $Z_1 < Z_2$, R changes at $t + Y_2$. We assume that X is continuous and ignore whether or not inequalities are strict. We only outline a solution since the arithmetic is messy and of no real interest.

Consider $Z_{\max} = \max(Z_1, Z_2)$. Since Z_1 and Z_2 are IID, we have

$$F_{Z_{\max}}(z) = F_{Z_1}^2(z).$$

From (5.93), the CDF of Z_i , $i = 1, 2$ is $F_{Z_i}(z) = (1/\bar{X}) \int_0^z F_X^c(x) dx$, so

$$\begin{aligned} F_{Z_{\max}}(z) &= \frac{1}{\bar{X}^2} \left(\int_0^z F_X^c(x) dx \right)^2 \\ f_{Z_{\max}}(z) &= \frac{2f_X(x)}{\bar{X}^2} \left(\int_0^z F_X^c(x) dx \right). \end{aligned} \quad (\text{A.106})$$

Now consider the joint duration and age, $\tilde{X}_i$, Z_i for either process, $i = 1, 2$. Assuming that X has a density, and converting the incremental expression in (5.77) to a density form,

$$f_{Z_i, \tilde{X}_i}(z, x) = \frac{f_X(x)}{\bar{X}}; \quad \text{for } 0 < z < x.$$

The marginal density of Z_i is then found by integrating over $x > z$ above, giving $f_{Z_i}(z) = F_X^c(z)/\bar{X}$. The conditional density of $\tilde{X}_i$ conditional on Z_i can then be found as

$$f_{\tilde{X}_i|Z_i}(x|z) = \frac{f_{Z_i, \tilde{X}_i}(z, x)}{f_{Z_i}(z)} = \frac{f_X(x)}{F_X^c(z)} \quad \text{for } x > z > 0.$$

Now let $\tilde{X}_{\max}(x)$ denote $\tilde{X}_i$ for the i that maximizes (Z_1, Z_2) . Then

$$f_{\tilde{X}_{\max}|Z_{\max}}(x|z) = \frac{f_X(x)}{F_X^c(z)} \quad \text{for } x > z > 0. \quad (\text{A.107})$$

The joint PDF of $(Z_{\max}, \tilde{X}_{\max})$ is now given by the product of (A.106) and (A.107). We can then calculate the marginal PDF of $\tilde{X}_{\max}$ from this and then calculate the expectation of $\tilde{X}_{\max}$. Alternatively, we can find the conditional mean of $\tilde{X}_{\max}$ and then find the unconditional mean. Neither lead to much insight. At any rate, after finding $E[\tilde{X}_{\max}]$, we can find $E[U] = E[\tilde{X}_{\max}] - E[Z_{\max}]$.

Exercise 5.38: This problem provides another way of treating ensemble averages for renewal-reward problems. Assume for notational simplicity that X is a continuous random variable.

a) Show that $\Pr\{\text{one or more arrivals in } (\tau, \tau + \delta)\} = m(\tau + \delta) - m(\tau) - o(\delta)$ where $o(\delta) \geq 0$ and $\lim_{\delta \rightarrow 0} o(\delta)/\delta = 0$.

Solution: Recall that $m(t) = E[N(t)]$ and consequently, $m(t + \delta) - m(t)$ is the expected number of arrivals in $(t, t + \delta]$. Thus

$$m(t + \delta) - m(t) = \sum_{i=1}^{\infty} i \Pr\{i \text{ arrivals in } (t, t + \delta)\} \geq \Pr\{\text{one or more arrivals in } (t, t + \delta)\}.$$

Unfortunately, the difference between the two sides of this inequality can be greater than $o(\delta)$. An example occurs when $F_X(x) = \sqrt{x}$ for $0 \leq x \leq 1$. Then the probability of two arrivals in $(0, \delta)$ is greater than the probability of one arrival in $(0, \delta/2)$ and a second arrival in the interval of duration $\delta/2$ following the first. The probability of this is $(\sqrt{\delta/2})^2 = \delta/2$, which is not $o(\delta)$.

When you can't prove something in your research, a good policy is to alternate between looking for a proof and looking for a counterexample. In the remainder of the exercise, we restrict attention to usual case where (a) is satisfied.

b) Show that $\Pr\{Z(t) \in [z, z + \delta), \tilde{X}(t) \in (x, x + \delta)\}$ is equal to $[m(t - z) - m(t - z - \delta) - o(\delta)][F_X(x + \delta) - F_X(x)]$ for $x \geq z + \delta$.

Solution: The clean way to do this is given in the proof of Theorem 5.7.2, and it is also shown there that the $o(\delta)$ term is unnecessary. Here we take another approach that is less general and rigorous, but perhaps more intuitive. Assuming δ to be negligibly small, $\Pr\{Z(t) \in [z, z + \delta)\}$ is essentially the probability of, first, an arrival in $(t - z - \delta, t - z]$ and, second, *no* arrival in $(t - z, t]$. The probability of no arrival in $(t - z, t]$, given an arrival essentially at $t - z$ is $F_X^c(z)$. Thus

$$\Pr\{Z(t) \in [z, z + \delta)\} \approx [m(t - z) - m(t - z - \delta)]F_X^c(z). \quad (\text{A.108})$$

Now, given $Z(t) \in [z, z + \delta)$, the probability that the next arrival after $t - z$ is in $(x, x + \delta]$ is $(F_X(x + \delta) - F_X(x))/F_X^c(z)$. This together with (A.108) gives the desired result.

c) Assuming that $m'(\tau) = dm(\tau)/d\tau$ exists for all τ , show that the joint density of $Z(t)$, $\tilde{X}(t)$ is $f_{Z(t), \tilde{X}(t)}(z, x) = m'(t - z)f_X(x)$ for $x > z$.

Solution: Ignoring the $o(\delta)$ term, the probability in (b) is $m'(t-z)f_X(x)\delta^2$. The δ^2 term disappears in going to the joint density.

d) Show that $E[R(t)] = \int_{z=0}^t \int_{x=z}^{\infty} \mathcal{R}(z, x) f_X(x) dx m'(t-z) dz$.

Solution: Letting $\mathcal{R}(z, x)$ be the reward for $Z = z$, $\tilde{X} = x$, and using the joint density in (c), we get the desired result.

Exercise 5.39: In this problem, we show how to calculate the distribution of the residual life $Y(t)$ as a transient in t . Let $\mu(t) = dm(t)/dt$ where $m(t) = E[N(t)]$, and let the interarrival distribution have the density $f_X(x)$. Let $Y(t)$ have the density $f_{Y(t)}(y)$.

a) Show that these densities are related by the integral equation

$$\mu(t+y) = f_{Y(t)}(y) + \int_{u=0}^y \mu(t+u) f_X(y-u) du. \quad (\text{A.109})$$

Solution: Assuming that all these densities exist, and ignoring the bizarre example in Exercise 5.38 (a), $\delta\mu(t+y) + o(\delta)$ is the probability of an arrival in $(t+y, t+y+\delta]$. Such an arrival event can be expressed as a disjoint union of subevents depending on the epoch of the most recent arrival on or before $t+y$. One of these subevents is that of no arrivals in $(t, t+y]$ and an arrival in $(t+y, t+y+\delta]$. This subevent is the event that $Y(t) \in (y, y+\delta)$ and has probability $\delta f_Y(y) + o(\delta)$. The other subevents where the previous arrival occurs in one of the discrete δ sized intervals between t and $t+y$. Thus

$$\delta\mu(t+y) + o(\delta) = \delta f_{Y(t)}(y) + \sum_{i=1}^{\lfloor y/\delta \rfloor} \delta^2 \mu(t+\delta i) f_X(y-\delta i) + o(\delta).$$

Cancelling δ from both sides and going to the limit $\delta \rightarrow 0$, we get (A.109).

b) Let $L_{\mu,t}(r) = \int_{y \geq 0} \mu(t+y) e^{-ry} dy$ and let $L_{Y(t)}(r)$ and $L_X(r)$ be the Laplace transforms of $f_{Y(t)}(y)$ and $f_X(x)$ respectively. Find $L_{Y(t)}(r)$ as a function of $L_{\mu,t}$ and L_X .

Solution: For fixed t and y , the integral in (A.109) is the convolution of μ over $\tau > t$ and f_X for $0 \leq \tau \leq y$. Thus,

$$L_{\mu,t}(r) = L_{Y(t)}(r) + L_{\mu,t}(r) L_X(r); \quad L_{Y(t)}(r) = [1 - L_X(r)] L_{\mu,t}(r). \quad (\text{A.110})$$

c) Consider the inter-renewal density $f_X(x) = (1/2)e^{-x} + e^{-2x}$ for $x \geq 0$ (as in Example 5.6.1). Find $L_{\mu,t}(r)$ and $L_{Y(t)}(r)$ for this example.

Solution: From Example 5.6.1, we have

$$\begin{aligned} L_X(r) &= \frac{(3/2)r + 2}{(r+1)(r+2)} \\ L_m(s) &= \frac{4}{3r^2} + \frac{1}{9r} - \frac{1}{9(r+3/2)} \\ m(t) &= \frac{4t}{3} + \frac{1 - \exp[-(3/2)t]}{9}. \end{aligned}$$

We see that $\mu(t) = \frac{d}{dt}m(t) = 4/3 + (1/6)\exp(-3t/2)$. From this, we calculate

$$L_{\mu,t}(r) = \int_0^{\infty} \mu(t+y) e^{-ry} dy = \frac{4}{3r} + \frac{1}{6(r+3/2)} e^{-3t/2}.$$

Substituting this and the expression for $L_X(r)$ into (A.110), we get

$$L_{Y(t)}(r) = \frac{8(r + 3/2) + re^{-3t/2}}{6(r + 1)(r + 2)}.$$

d) Find $f_{Y(t)}(y)$. Show that your answer reduces to that of (5.30) in the limit as $t \rightarrow \infty$.

Solution: Using the Heaviside expansion formula on $L_{Y(t)}(r)$,

$$\begin{aligned} f_{Y(t)}(y) &= \frac{2}{3}(e^{-y} + e^{-2y}) - \frac{1}{6}(e^{-y} - 2e^{-2y}) \exp\left(\frac{-3t}{2}\right) \\ \lim_{t \rightarrow \infty} f_{Y(t)}(y) &= \frac{2}{3}(e^{-y} + e^{-2y}). \end{aligned} \quad (\text{A.111})$$

The time-average result in (5.30), when applied to this example, yields $\bar{X} = 3/4$ and $F_X^c(x) = (1/2)(e^{-x} + e^{-2x})$. Thus $F_Y^c(y) = (2/3)e^{-y} + (1/3)e^{-2y}$. Differentiating, we get the density in (A.111).

e) Explain how to go about finding $f_{Y(t)}(y)$ in general, assuming that f_X has a rational Laplace transform.

Solution: One simply goes through the same procedure as in Example 5.6.1 and here. Recognizing that the Laplace transforms remain rational in r throughout the procedure, one finally gets $f_{Y(t)}(y)$. In fact the procedure could easily be automated. The reason we do not stress this more is that one learns more about Laplace transforms than renewal theory from these standard manipulations.

Exercise 5.40: This problem is designed to give you an alternate way of looking at ensemble averages for renewal-reward problems. First we find an exact expression for $\Pr\{S_{N(t)} > s\}$. We find this for arbitrary s and t , $0 < s < t$.

a) By breaking the event $\{S_{N(t)} > s\}$ into subevents $\{S_{N(t)} > s, N(t) = n\}$, explain each of the following steps:

$$\Pr\{S_{N(t)} > s\} = \sum_{n=1}^{\infty} \Pr\{t \geq S_n > s, S_{n+1} > t\} \quad (\text{A.112})$$

$$= \sum_{n=1}^{\infty} \int_{y=s}^t \Pr\{S_{n+1} > t \mid S_n = y\} dF_{S_n}(y) \quad (\text{A.113})$$

$$= \int_{y=s}^t F_X^c(t-y) d \sum_{n=1}^{\infty} F_{S_n}(y) \quad (\text{A.114})$$

$$= \int_{y=s}^t F_X^c(t-y) dm(y) \quad \text{where } m(y) = \mathbf{E}[N(y)]. \quad (\text{A.115})$$

Solution: $N(t)$ is the (random) number of arrivals up to and including t , and $S_{N(t)}$ is the epoch of the last of these arrivals. Thus $S_{N(t)+1}$ must be greater than t . The joint event $N(t) = n$ and $S_{N(t)} > s$ then has probability $\Pr\{t \geq S_n > s, S_{n+1} > t\}$. These joint events are disjoint over n and do not include $n = 0$ since $S_0 = 0$ by convention and, by assumption, $s > 0$. Thus summing over the probabilities of these subevents for $n \geq 1$ gives $\Pr\{S_n > s\}$ establishing (A.112). We then see that (A.113) is simply an alternate way of expressing the probability of the subevents above.

In (A.114), we recognized that $X_{n+1} = S_{n+1} - S_n$ and that X_{n+1} is independent of S_n . We also interchanged the order of summation and integration, which is permissible since all the terms are positive. Finally in (A.115), we used (5.54).

b) Show that for $0 < s < t < u$,

$$\Pr\{S_{N(t)} > s, S_{N(t)+1} > u\} = \int_{y=s}^t F_X^c(u-y) dm(y). \quad (\text{A.116})$$

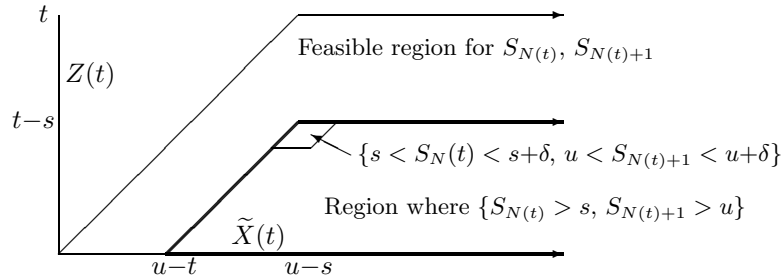
Solution: This is virtually the same as (a), except (A.112) is replaced by

$$\Pr\{S_{N(t)} > s, S_{N(t)+1} > u\} = \sum_{n=1}^{\infty} \Pr\{t \geq S_n > s, S_{n+1} > u\}.$$

In (A.113-A.115), t is simply replaced by u .

c) Draw a two dimensional sketch, with age and duration as the axes, and show the region of (age, duration) values corresponding to the event $\{S_{N(t)} > s, S_{N(t)+1} > u\}$.

Solution:



d) Assume that for large t , $dm(y)$ can be approximated (according to Blackwell) as $(1/\bar{X})dy$, where $\bar{X} = E[X]$. Assuming that X also has a density, use the result in parts b) and c) to find the joint density of age and duration.

Solution: We can use the expression in (A.116) to evaluate the probability of the trapezoidal region in (c). Assuming that δ is vanishingly small,

$$\Pr\{s < S_{N(t)} < s+\delta, u < S_{N(t)+1} < u+\delta\} = f_X(u-s)(\bar{X})^{-1}\delta^2.$$

Going to the limit $\delta \rightarrow 0$, the joint probability density is

$$f_{S_{N(t)}S_{N(t)+1}}(s, u) = \frac{f_X(u-s)}{\bar{X}}.$$

Exercise 5.41: Show that for a G/G/1 queue, the time-average wait in the system is the same as $\lim_{n \rightarrow \infty} E[W_n]$. Hint: Consider an integer renewal counting process $\{M(n); n \geq 0\}$ where $M(n)$ is the number of renewals in the G/G/1 process of Section 5.5.3 that have occurred by the n th arrival. Show that this renewal process has a span of 1. Then consider $\{W_n; n \geq 1\}$ as a reward within this renewal process.

Solution: The time-average wait in the system is what we have called the sample-path-average waiting time in Little's theorem. It is $\lim_{t \rightarrow \infty} \left[\left(\sum_{i=0}^{N(t)} W_i \right) / N(t) \right]$, and according to Little's theorem, the limit exists and is equal to a given number $\bar{W}$ WP1. The question is whether this sample-path average is the same as the limiting ensemble average $\lim_{n \rightarrow \infty} E[W_n]$. The hint shows that $\lim_{n \rightarrow \infty} E[W_n]$ exists, and Little's theorem implies that this ensemble average is equal to the sample-path average.

Exercise 5.42: If one extends the definition of renewal processes to include inter-renewal intervals of duration 0, with $\Pr\{X=0\} = \alpha$, show that the expected number of simultaneous renewals at a renewal epoch is $1/(1 - \alpha)$, and that, for a non-arithmetic process, the probability of 1 or more renewals in the interval $(t, t + \delta]$ tends to $(1 - \alpha)\delta/E[X] + o(\delta)$ as $t \rightarrow \infty$.

Solution: A renewal epoch is the time at which one or more renewals occur. Such an epoch consists of one renewal with probability $1 - \alpha$, two with probability $\alpha(1 - \alpha)$, etc. Summing the series, the expected number of renewals at a renewal epoch is $1/(1 - \alpha)$.

As in the solution to Exercise 5.3, we can consider the related ordinary renewal process in which only the renewals after a nonzero interval are counted. In other words, the inter-renewal intervals form the sequence $\{Y_n; n \geq 1\}$ generated by crossing out all inter-renewal intervals equal to 0. Thus $F_Y^c(y) = F_X^c(y)/F_X^c(0) = F_X^c(y)/(1 - \alpha)$. Integrating the complementary CDF, we get the expectation, $E[Y] = E[X]/(1 - \alpha)$. Applying Blackwell's theorem to $\{Y_n; n \geq 1\}$ the expected number of renewal epochs in $(t, t + \delta]$ tends to $\delta/\bar{Y}$ as $t \rightarrow \infty$, and thus the probability of a renewal epoch tends to $\delta/\bar{Y} + o(\delta) = \delta(1 - \alpha)/\bar{X}$. Note that we could also apply Blackwell to $\{X_n; n \geq 1\}$, but we couldn't get from there to the probability of an arrival in a δ interval.

Exercise 5.43:

The purpose of this exercise is to show why the interchange of expectation and sum in the proof of Wald's equality is justified when $E[J] < \infty$ but not otherwise. Let $X_1, X_2, \dots$, be a sequence of IID rv's, each with the CDF F_X . Assume that $E[|X|] < \infty$.

a) Show that $S_n = X_1 + \dots + X_n$ is a rv for each integer $n > 0$. Note: S_n is obviously a mapping from the sample space to the real numbers, but you must show that it is finite with probability 1. Hint: Recall the additivity axiom for the real numbers.

Solution: The additivity axiom for real numbers says that the sum of 2 real numbers is a real number (*i.e.*, finite real number). By induction the sum of n real numbers is a real number. For n rv's, the sample values of each are finite WP1, and hence the sum is finite WP1. Exercise 1.13 treats this in more depth

b) Let J be a stopping trial for $X_1, X_2, \dots$. Show that $S_J = X_1 + \dots + X_J$ is a rv. Hint: Represent $\Pr\{S_J \leq A\}$ as $\sum_{n=1}^{\infty} \Pr\{J = n, S_n \leq A\}$.

Solution: Since $\{S_J \leq A\}$ is clearly an event for each real A , it suffices to show that $\lim_{A \rightarrow \infty} \Pr\{S_J \leq A\} = 1$. Taking the complementary event,

$$\Pr\{S_J > A\} = \sum_{n=1}^{\infty} \Pr\{J = n, S_n > A\} \leq \sum_{n=1}^k \Pr\{S_n > A\} + \Pr\{J > k\}.$$

This is valid for all $k \geq 1$, so for any given k , we can take the limit $A \rightarrow \infty$ above to see that $\lim_{A \rightarrow \infty} \Pr\{S_J > A\} < \Pr\{J > k\}$ for that k . Since J is a rv, we can now take the limit $k \rightarrow \infty$ to see that $\lim_{A \rightarrow \infty} \Pr\{S_J > A\} = 0$. Thus S_J is a rv.

c) For the stopping trial J above, let $J^{(k)} = \min(J, k)$ be the stopping trial J truncated to integer k . Explain why the interchange of sum and expectation in the proof of Wald's equality is justified in this case, so $E[S_{J^{(k)}}] = \bar{X}E[J^{(k)}]$.

Solution: A finite sum and expectation can always be interchanged if the expectations exist (see Exercise 1.14).

d) In parts (d), (e), and (f), assume, in addition to the assumptions above, that $F_X(0) = 0$, i.e., that the X_i are positive rv's. Show that $\lim_{k \rightarrow \infty} E[S_{J^{(k)}}] < \infty$ if $E[J] < \infty$ and $\lim_{k \rightarrow \infty} E[S_{J^{(k)}}] = \infty$ if $E[J] = \infty$.

Solution: Since the X_i are positive rv's with $E[|X|] < \infty$, it follows that $\bar{X} > 0$. The rv $J^{(k)}$ is non-decreasing in k and upper bounded by J . Thus $E[J^{(k)}]$ is non-decreasing in k and upper bounded by $E[J]$. If $E[J] < \infty$, then $\lim_{k \rightarrow \infty} E[J^{(k)}]$ is also finite, so, from (c), $\lim_{k \rightarrow \infty} E[S_{J^{(k)}}] < \infty$. If $E[J] = \infty$, then for any B , there is a k large enough that $E[J^{(k)}] > B$, which means that $\lim_{k \rightarrow \infty} E[J^{(k)}] = \infty$. Thus, since $\bar{X} > 0$,

$$\lim_{k \rightarrow \infty} E[S_{J^{(k)}}] = \lim_{k \rightarrow \infty} \bar{X} E[J^{(k)}] = \infty \quad \text{for } E[J] = \infty.$$

e) Show that

$$\Pr\{S_{J^{(k)}} > x\} \leq \Pr\{S_J > x\} \quad \text{for all } k, x. \quad (\text{A.117})$$

Solution: Since $S_n = S_{n-1} + X_n$, we see that $S_n > S_{n-1}$ for all $n \geq 1$ and thus $S_n > S_m$ for all $n > m > 0$. Since $J^{(k)} \leq J$, it follows that $S_J \geq S_{J^{(k)}}$. The desired result follows.

f) Show that $E[S_J] = \bar{X}E[J]$ if $E[J] < \infty$ and $E[S_J] = \infty$ if $E[J] = \infty$.

Solution: Integrating (A.117) over $x > 0$, we see that $E[S_{J^{(k)}}] \leq E[S_J]$ and also that $E[S_{J^{(k)}}]$ is increasing in k , which provides an independent derivation of some of the results in (d).

g) Now assume that X has both negative and positive values with non-zero probability and let $X^+ = \max(0, X)$ and $X^- = \min(X, 0)$. Express S_J as $S_J^+ + S_J^-$ where $S_J^+ = \sum_{i=1}^J X_i^+$ and $S_J^- = \sum_{i=1}^J X_i^-$. Show that $E[S_J] = \bar{X}E[J]$ if $E[J] < \infty$ and that $E[S_J]$ is undefined otherwise.

Solution: The result in (f) can be applied independently to $\{X_i^+; i \geq 1\}$ and $\{X_i^-; i \geq 1\}$, giving the desired result.

Exercise 5.44: This is a very simple exercise designed to clarify confusion about the roles of past, present, and future in stopping rules. Let $\{X_n; n \geq 1\}$ be a sequence of IID binary rv's, each with the pmf $p_X(1) = 1/2$, $p_X(0) = 1/2$. Let J be a positive integer-valued rv that takes on the sample value n of the first trial for which $X_n = 1$. That is, for each $n \geq 1$,

$$\{J = n\} = \{X_1=0, X_2=0, \dots, X_{n-1}=0, X_n=1\}.$$

a) Use the definition of stopping trial, Definition 5.5.1, to show that J is a stopping trial for $\{X_n; n \geq 1\}$.

Solution: Note that $\mathbb{I}_{J=n} = 1$ if $X_n = 1$ and $X_i = 0$ for all $i < n$. $\mathbb{I}_{J=n} = 0$ otherwise. Thus $\mathbb{I}_{J=n}$ is a function of $X_1, \dots, X_n$. We also see that J is a positive integer-valued rv. That is, each sample sequence of $\{X_i; i \geq 1\}$ (except the zero probability sequence of all zeros) maps into a positive integer.

b) Show that for any given n , the rv's X_n and $\mathbb{I}_{J=n}$ are *statistically dependent*.

Solution: Note that $\Pr\{\mathbb{I}_{J=n} = 1\} = 2^{-n}$ and $\Pr\{X_n = 1\} = 1/2$. The product of these is $2^{-(n+1)}$, whereas $\Pr\{X_n = 1, \mathbb{I}_{J=n} = 1\} = 2^{-n}$, demonstrating statistical dependence. More intuitively $\Pr\{X_n = 1 \mid \mathbb{I}_{J=n} = 1\} = 1$ and $\Pr\{X_n = 1\} = 1/2$.

c) Show that for every $m > n$, X_n and $\mathbb{I}_{J=m}$ are *statistically dependent*.

Solution: For $m > n$, $\mathbb{I}_{J=m}=1$ implies that $X_i = 0$ for $i < m$, and thus that $X_n = 0$. Thus $\Pr\{X_n = 1 \mid \mathbb{I}_{J=m}=1\} = 0$. Since $\Pr\{X_n = 1\} = 1/2$, $\mathbb{I}_{J=m}$ and X_n are dependent.

d) Show that for every $m < n$, X_n and $\mathbb{I}_{J=m}$ are *statistically independent*.

Solution: Since $\mathbb{I}_{J=m}$ is a function of $X_1, \dots, X_m$ and X_n is independent of $X_1, \dots, X_m$, it is clear that X_n is independent of $\mathbb{I}_{J=m}$.

e) Show that X_n and $\mathbb{I}_{J \geq n}$ are *statistically independent*. Give the simplest characterization you can of the event $\{J \geq n\}$.

Solution: For the same reason as $\mathbb{I}_{J=m}$ is independent of X_n for $m < n$, we see that $\mathbb{I}_{J < n}$ is independent of X_n . Now $\{J \geq n\} = \{J < n\}^c$, so $\mathbb{I}_{J \geq n} = 1 - \mathbb{I}_{J < n}$. Thus $\{J \geq n\}$ is also independent of X_n . We give an intuitive explanation after (f).

f) Show that X_n and $\mathbb{I}_{J > n}$ are *statistically dependent*.

Solution: The event $\{J > n\}$ implies that $X_1, \dots, X_n$ are all 0, so $\Pr\{X_n = 1 \mid J > n\} = 0$. Since $\Pr\{X_n=1\} = 1/2$, it follows that X_n and $\mathbb{I}_{J > n}$ are dependent.

It is important (and essentially the central issue of the exercise) to understand why (e) and (f) are different. Note that $\{J \geq n\} = \{J = n\} \cup \{J > n\}$. Now $\{J = n\}$ implies that $X_n = 1$ whereas $\{J > n\}$ implies that $X_n = 0$. The union, however, implies nothing about X_n , and is independent of X_n . The union, $\mathbb{I}_{J \geq n}$ is the event that $X_1, \dots, X_{n-1}$ are all 0 and X_n then determines whether $J = n$ or $J > n$.

Note: The results here are characteristic of most sequences of IID rv's. For most people, this requires some realignment of intuition, since $\{J \geq n\}$ is the union of $\{J=m\}$ for all $m \geq n$, and each of these events are highly dependent on X_n . The right way to think of this is that $\{J \geq n\}$ is the complement of $\{J < n\}$, which is determined by $X_1, \dots, X_{n-1}$. Thus $\{J \geq n\}$ is also determined by $X_1, \dots, X_{n-1}$ and is thus independent of X_n . The moral of the story is that thinking of stopping rules as rv's independent of the future is very tricky, even in totally obvious cases such as this.

Exercise 5.45: Assume a friend has developed an excellent program for finding the steady-state probabilities for finite-state Markov chains. More precisely, given the transition matrix $[P]$, the program returns $\lim_{n \rightarrow \infty} P_{ii}^n$ for each i . Assume all chains are aperiodic.

a) You want to find the expected time to first reach a given state k starting from a different state m for a Markov chain with transition matrix $[P]$. You modify the matrix to $[P']$ where $P'_{km} = 1$, $P'_{kj} = 0$ for $j \neq m$, and $P'_{ij} = P_{ij}$ otherwise. How do you find the desired first-passage time from the program output given $[P']$ as an input? (Hint: The times at which a Markov chain enters any given state can be considered as renewals in a (perhaps delayed) renewal process).

Solution: First assume k, m are in the same recurrent class in $[P']$. Successive entries to state k form a delayed renewal process (delayed because the process starts in state m). By (5.106), *i.e.*, Blackwell's theorem for delayed arithmetic renewal processes, the probability of an arrival at a limiting time n (*i.e.*, $\lim_{n \rightarrow \infty} [P']_{kk}^n$) is $1/E[X_{kk}]$ where X_{kk} is the inter-renewal time between successive arrivals to state k . The first passage time is $E[X_{kk}] - 1$ since the first transition after an arrival at k is the transition back to m . The friend's program gives us $\lim_{n \rightarrow \infty} [P']_{kk}^n$, so the first passage time is $\left\{ \lim_{n \rightarrow \infty} [P']_{kk}^n \right\}^{-1} - 1$. Note

that successive arrivals to state m also form a renewal process, but not every return to state m corresponds to a first passage time to k (as detailed in (b)).

If k and m are not in the same recurrence class in $[P']$, then the expected first passage time is infinite, *i.e.*, there is a positive probability of taking some path from m for which there is no return to m or path to k . Such a path must exist in the original chain also.

b) Using the same $[P']$ as the program input, how can you find the expected number of returns to state m before the first passage to state k ?

Solution: Consider a (delayed) renewal reward process with unit reward on each arrival to state m and a renewal on each arrival to state k . Then the limiting probability of an arrival to m at time n is $\lim_{n \rightarrow \infty} [P']_{mm}^n$, which can be found from the program. From the renewal reward theorem, the expected reward per renewal is

$$E[\text{reward per renewal}] = \frac{\lim_{n \rightarrow \infty} [P']_{mm}^n}{\lim_{n \rightarrow \infty} [P']_{kk}^n}.$$

This is the expected number of visits to m per visit to k , and we are asked to find the number of *returns* to m per visit to k , which is one less than the value above. There is always at least one visit to m per visit to k , so this expected number of returns is always non-negative.

c) Suppose, for the same Markov chain $[P]$ and the same starting state m , you want to find the probability of reaching some given state n before the first-passage to k . Modify $[P]$ to some $[P'']$ so that the above program with P'' as an input allows you to easily find the desired probability.

Solution: One solution is to modify the chain so that n always goes to k and k always goes to m , *i.e.*, $P''_{km} = 1$, $P''_{kj} = 0$ for $j \neq m$ and $P''_{nk} = 1$ and $P''_{nj} = 0$ for $j \neq k$. We see that this doesn't alter any first passage path from m to k that doesn't pass through n and doesn't alter any first passage path from m to n that doesn't first pass through k . Assigning a unit reward for each visit to n and noting that at most one visit to n can occur per inter-renewal interval, we see that the probability of visiting n before k is the same as the expected reward per inter-renewal. Thus

$$\Pr\{n \text{ before } k\} = \frac{\lim_{\ell \rightarrow \infty} [P'']_{nn}^\ell}{\lim_{\ell \rightarrow \infty} [P'']_{kk}^\ell}.$$

d) Let $\Pr\{X(0) = i\} = Q_i$, $1 \leq i \leq M$ be an arbitrary set of initial probabilities for the same Markov chain $[P]$ as above. Show how to modify $[P]$ to some $[P''']$ for which the steady-state probabilities allow you to easily find the expected time of the first-passage to state k .

Solution: Essentially, we want to modify the matrix $[P]$ so that after each visit to k , the chain reverts to its initial state. This suggests $P'''_{ki} = Q_i$ for each i , with the other transitions unchanged. This doesn't quite work, since k might be one of the initial states, and in finding the first passage time from k we must use P_{ki} rather than P'''_{ki} . The solution to this is the same as was used in Exercise 4.22. The state k is split into two states, one with the original transition probabilities and the other with $P'''_{ki} = Q_i$.

Exercise 5.46: Consider a ferry that carries cars across a river. The ferry holds an integer number k of cars and departs the dock when full. At that time, a new ferry immediately appears and begins loading

newly arriving cars ad infinitum. The ferry business has been good, but customers complain about the long wait for the ferry to fill up.

a) Assume that cars arrive according to a renewal process. The IID interarrival times have mean $\bar{X}$, variance σ^2 and moment generating function $g_X(r)$. Does the sequence of departure times of the ferries form a renewal process? Explain carefully.

Solution: Yes, the ferry departure times form a renewal process. The reason is that the ℓ th ferry departure is immediately after the $k\ell$ th customer arrival. The time from the ℓ th to $\ell + 1$ ferry departure is the time from the $(k\ell)$ th to $((k + 1)\ell)$ th customer arrival, which is clearly independent of all previous ferry departure times.

b) Find the expected time that a customer waits, starting from its arrival at the ferry terminal and ending at the departure of its ferry. Note 1: Part of the problem here is to give a reasonable definition of the expected customer waiting time. Note 2: It might be useful to consider $k = 1$ and $k = 2$ first.

Solution: For $k = 1$, the ferry leaves immediately when a customer arrives, so the expected waiting time for each customer is 0. For $k = 2$, odd numbered customers wait for the following even numbered customer, and even number customers don't wait at all, so the average waiting time over customers (which we take as the only sensible definition of expected waiting time) is $\bar{X}/2$.

We next find the expected waiting time, averaged over customers, for the ℓ th ferry. To simplify notation, we look at the first ferry. Later ferries could be shown to be the same by renewal reward theory, but more simply they are the same simply by adding extra notation. The average expected wait over the k customers is the sum of their expected waits divided by k (recall that this is true even if (as here) the wait of different customers are statistically dependent. The expected wait of customer 1 is $(k - 1)\bar{X}$, that of customer 2 $(k - 2)\bar{X}$, etc. Recall (or derive) that $1 + 2 + \dots + (k - 1) = (k - 1)k/2$. Thus the expected wait per customer is $(k - 1)\bar{X}/2$, which checks with the result for $k = 1$ and $k = 2$.

c) Is there a 'slow truck' phenomenon (a dependence on $E[X^2]$) here? Give an intuitive explanation. Hint: Look at $k = 1$ and $k = 2$ again.

Solution: Clearly, there is no 'slow truck' phenomenon for the ferry wait here since the expected wait per customer is $(k-1)\bar{X}$. The reason is most evident for $k = 1$, where the wait is 0. The arrival of a car at the ferry terminal could have been delayed by an arbitrary amount by a slow truck in *getting to* the ferry terminal, but is not delayed at all in *getting on* the ferry since the slow truck took an earlier ferry. For larger k , a vehicle could be delayed by a later arriving truck, but at most $k - 1$ vehicles could be delayed that way, while the $E[X^2]$ slow truck effect arises from the potentially unbounded number of customers delayed by a slow truck.

d) In an effort to decrease waiting, the ferry managers institute a policy where no customer ever has to wait more than one hour. Thus, the first customer to arrive after a ferry departure waits for either one hour or the time at which the ferry is full, whichever comes first, and then the ferry leaves and a new ferry starts to accumulate new customers. Does the sequence of ferry departures form a renewal process under this new system? Does the sequence of times at which each successive empty ferry is entered by its first customer form a renewal process? You can assume here that $t = 0$ is the time of the first arrival to the first ferry. Explain carefully.

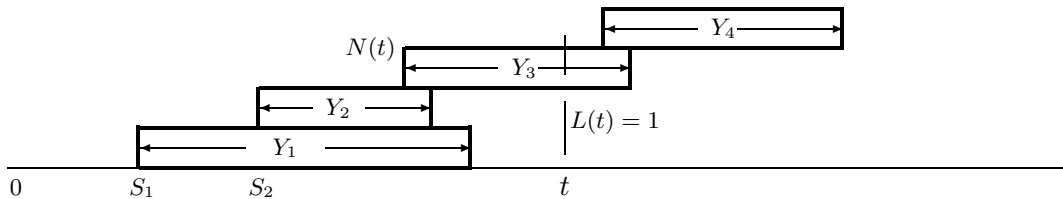
Solution: The sequence of ferry departures does not form a renewal process (nor a delayed renewal process). As an example, suppose that $k = 2$ and X is 1 minute with probability $9/10$ and 2.1 hours with probability $1/10$. Then a ferry interdeparture interval of 2 minutes followed by one of an hour and one minute implies that the second departure was generated by a one-minute inter-arrival followed by a 2.1-hour inter-arrival. The second ferry departure then left 1 hour after the 1 minute-interarrival, with 1.1 hour remaining until the 2.1-hour interarrival. Thus the third ferry departure is at least 1.2 hour after the second departure. The times at which each successive ferry is entered by its first customer is a renewal process since customer arrivals form a renewal process.

e) Give an expression for the expected waiting time of the first new customer to enter an empty ferry under this new strategy.

Solution: The first new customer waits for a time T in hours which is the minimum of 1 hour and $X_2 + \dots, X_k$. This latter term has the CDF of the sum S_{k-1} of IID rv's with the distribution of X . Since the expected value of a rv is the integral of its complementary CDF,

$$\mathbb{E}[T] = \int_0^1 [1 - F_{S_{k-1}}(s)] ds.$$

Exercise 5.47: Consider a $(G/G/\infty)$ 'queueing' system. That is, the arriving customers form a renewal process, *i.e.*, the interarrival intervals $\{X_n; n \geq 1\}$ are IID. You may assume throughout that $\mathbb{E}[X] < \infty$. Each arrival immediately enters service; there are infinitely many servers, so one is immediately available for each arrival. The service time Y_i of the i th customer is a rv of expected value $\bar{Y} < \infty$ and is IID with all other service times and independent of all inter-arrival times. There is no queueing in such a system, and one can easily intuit that the number in service never becomes infinite since there are always available servers.



a) Give a simple example of distributions for X and Y in which this system never becomes empty. Hint: Deterministic rv's are fair game.

Solution: Suppose, for example that the arrivals are deterministic, one arrival each second. Suppose that each service takes 2 seconds. Then when the second arrival occurs, the first service is only half over. When the third arrival arrives, the first service is just finishing, but the second arrival is only half finished, etc. This also illustrates the unusual behavior of systems with an unlimited number of servers. If the arrival rate is increased, there are simply more servers at work simultaneously; there is no possibility of the system becoming overloaded. The example here can be made less trivial by making the interarrival distribu-

tion arbitrary but upper bounded by 1 second. The service distribution could similarly be arbitrary but lower bounded by 2 seconds.

b) We want to prove Little's theorem for this type of system, but there are no renewal instants for the entire system. As illustrated above, let $N(t)$ be the renewal counting process for the arriving customers and let $L(t)$ be the number of customers in the system (*i.e.*, receiving service) at time t . In distinction to our usual view of queueing systems, assume that there is no arrival at time 0 and the first arrival occurs at time $S_1 = X_1$. The n th arrival occurs at time $S_n = X_1 + \cdots + X_n$.

Carefully explain why, for each sample point ω and each time $t > 0$,

$$\int_0^t L(t, \omega) dt \leq \sum_{i=1}^{N(t, \omega)} Y_i(\omega).$$

Solution: For the case in the above figure, $\int_0^t L(t, \omega) dt$ is Y_1 plus Y_2 plus the part of Y_3 on which service has been completed by t . Thus $\int_0^t L(t, \omega) dt \leq Y_1 + Y_2 + Y_3$. Since $N(t) = 3$ for this example, $\int_0^t L(t, \omega) dt \leq \sum_{i=1}^{N(t)} Y_i$.

In general, $\sum_{i=1}^{N(t)} Y_i$ is the total amount of service required by all customers arriving before or at t . $\int_0^t L(t, \omega) dt$ is the amount of service already provided by t . Since not all the required service has necessarily been provided by t , the inequality is as given.

c) Find the limit as $t \rightarrow \infty$ of $\frac{1}{t} \sum_{i=1}^{N(t, \omega)} Y_i(\omega)$ and show that this limit exists WP1.

Solution: This is virtually the same as many limits we have taken. The trick is to multiply and divide by $N(t, \omega)$ and then use the SLLN and the strong law for renewals, *i.e.*,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=1}^{N(t, \omega)} Y_i(\omega) = \lim_{t \rightarrow \infty} \frac{N(t, \omega)}{t} \frac{\sum_{i=1}^{N(t, \omega)} Y_i(\omega)}{N(t, \omega)}.$$

For a given ω , $N(t, \omega)$ is a real valued function of t which approaches $1/\bar{X}$ for all ω except perhaps a set of probability 0. For each of those sample points, $\lim_{t \rightarrow \infty} N(t, \omega) = \infty$ and thus the second term above has the following limit WP1.

$$\lim_{t \rightarrow \infty} \frac{\sum_{i=1}^{N(t, \omega)} Y_i(\omega)}{N(t, \omega)} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Y_i(\omega).$$

Since the Y_i are IID, the SLLN says that this limit is $\bar{Y}$ WP1. The limit $\lim_{t \rightarrow \infty} N(t, \omega)/t = 1/\bar{X}$ WP1. The set of ω for which both limits exist then has probability 1, so

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=1}^{N(t, \omega)} Y_i(\omega) = \frac{\bar{Y}}{\bar{X}} \quad \text{for a set of } \omega \text{ of probability 1.}$$

d) Assume that the service time distribution is bounded between 0 and some $b > 0$, *i.e.*, that $F_Y(b) = 1$. Carefully explain why

$$\int_0^{t+b} L(\tau, \omega) d\tau \geq \sum_{i=1}^{N(t, \omega)} Y_i(\omega).$$

Solution: This is almost the same as (b). All customers that have entered service by t have completed service by $t + b$, and thus the difference between the two terms above is the service provided to customers that have arrived between t and $t + b$.

e) Find the limit as $t \rightarrow \infty$ of $\frac{1}{t} \int_0^t L(\tau, \omega) d\tau$ and indicate why this limit exists WP1.

Solution: Combining the results of b) and d),

$$\sum_{i=1}^{N(t, \omega)} Y_i(\omega) \leq \int_0^{t+b} L(\tau, \omega) d\tau \leq \sum_{i=1}^{N(t+b, \omega)} Y_i(\omega).$$

Dividing by t and going to the limit as $t \rightarrow \infty$, the limit on the left is $\bar{Y}/\bar{X}$ WP1 and the limit on the right, after multiplying and dividing by $t + b$, is the same. Thus,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^{t+b} L(\tau, \omega) d\tau = \lim_{t \rightarrow \infty} \frac{t+b}{t} \frac{1}{t+b} \int_0^{t+b} L(\tau, \omega) d\tau = \bar{Y}/\bar{X}.$$

This is Little's theorem for the case of a rv X that is truncated to b and a truncation argument establishes Little's theorem for arbitrary X .

A.6 Solutions for Chapter 6

Exercise 6.1: Let $\{P_{ij}; i, j \geq 0\}$ be the set of transition probabilities for a countable-state Markov chain. For each i, j , let $F_{ij}(n)$ be the probability that state j occurs sometime between time 1 and n inclusive, given $X_0 = i$. For some given j , assume that $\{x_i; i \geq 0\}$ is a set of nonnegative numbers satisfying $x_i = P_{ij} + \sum_{k \neq j} P_{ik}x_k$ for all $i \geq 0$. Show that $x_i \geq F_{ij}(n)$ for all n and i , and hence that $x_i \geq F_{ij}(\infty)$ for all i . Hint: use induction.

Solution: We use induction on n . As the basis for the induction, we know that $F_{ij}(1) = P_{ij}$. Since the x_i are by assumption nonnegative, it follows for all i that

$$F_{ij}(1) = P_{ij} \leq P_{ij} + \sum_{k \neq j} P_{ik}x_k = x_i.$$

For the inductive step, assume that $F_{ij}(n) \leq x_i$ for a given n and all i . Using (6.8),

$$\begin{aligned} F_{ij}(n+1) &= P_{ij} + \sum_{k \neq j} P_{ik}F_{kj}(n) \\ &\leq P_{ij} + \sum_{k \neq j} P_{ik}x_k = x_i \quad \text{for all } i. \end{aligned}$$

By induction, it then follows that $F_{ij}(n) \leq x_i$ for all i, n . From (6.7), $F_{ij}(n)$ is non-decreasing in n and upper bounded by 1. It thus has a limit, $F_{ij}(\infty)$, which satisfies $F_{ij}(\infty) \leq x_i$ for all i .

One solution to the set of equations $x_i = P_{ij} + \sum_{k \neq j} P_{ik}x_k$ is $x_i = 1$ for all i . Another (from 6.9)) is $x_i = F_{ij}(\infty)$. These solutions are different when $F_{ij}(\infty) < 1$, and this exercise then shows that the solution $\{F_{ij}(\infty); i \geq 0\}$ is the smallest of all possible solutions.

Exercise 6.2: a) For the Markov chain in Figure 6.2, show that, for $p \geq 1/2$, $F_{00}(\infty) = 2(1-p)$ and show that $F_{i0}(\infty) = [(1-p)/p]^i$ for $i \geq 1$. Hint: first show that this solution satisfies (6.9) and then show that (6.9) has no smaller solution (Exercise 6.1 shows that $F_{ij}(\infty)$ is the smallest solution to (6.9)). Note that you have shown that the chain is transient for $p > 1/2$ and that it is recurrent for $p = 1/2$.

Solution: Note that the diagram in Figure 6.2 implicitly assumes $p < 1$, and we assume that here, since the $p = 1$ case is trivial.

The hint provides a straightforward algebraic solution, but provides little insight. It may be helpful, before carrying that solution out, to solve the problem by the more intuitive method used in the ‘stop when you’re ahead’ example, Example 5.4.4. This method also derives the values of $F_{i0}(\infty)$ rather than just verifying that they form a solution.

Let $F_{10}(\infty)$ be denoted by α . This is the probability of ever reaching state 0 starting from state 1. This probability is unchanged by converting state 0 to a trapping state, since when starting in state 1, transitions from state 0 can occur only after reaching state 0.

In the same way, $F_{i,i-1}(\infty)$ for any $i > 1$ is unchanged by converting state $i-1$ into a trapping state, and the modified Markov chain, for states $k \geq i-1$ is then the same as the modified chain that uses state 0 as a trapping state. Thus $F_{i,i-1}(\infty) = \alpha$.

We can proceed from this to calculate $F_{i,i-2}$ for any $i > 2$. In order to access state $i - 2$ from state i , it is necessary to first access state $i - 1$. Given a first passage from i to $i - 1$, (an event of probability α), it is necessary to have a subsequent first passage from $i - 1$ to $i - 2$, so $F_{i,i-2}(\infty) = \alpha^2$. Iterating on this, $F_{i0}(\infty) = \alpha^i$.

We can now solve for α by using (6.9) for $i = 1, j = 0$: $F_{10}(\infty) = q + pF_{20}(\infty)$. Thus $\alpha = q + p\alpha^2$. This is a quadratic equation in α with the solutions $\alpha = 1$ and $\alpha = q/p$. We then know that $F_{10}(\infty)$ has either the value 1 or q/p . For whichever value of α is correct, we also have $P_{i0}(\infty) = \alpha^i$ for $i \geq 1$. Finally, from (6.9) for $j = 0, i = 0$, $F_{00}(\infty) = q + p\alpha$.

From the reasoning above, we know that these two possible solutions are the only possibilities. If both of them satisfy (6.9), then, from Exercise 6.1, the one with $\alpha = q/p$ is the correct one since it is the smaller solution to (6.9). We now show that the solution with $\alpha = q/p$ satisfies (6.9). This is the first part of what the question asks, but it is now unnecessary to also show that this is the smallest solution.

If we substitute the hypothesized solution,

$$F_{00}(\infty) = 2q; \quad F_{i0}(\infty) = (q/p)^i \quad \text{for } i > 0, \quad (\text{A.118})$$

into (6.9) for $j = 0$, we get the hypothesized set of equalities,

$$\begin{aligned} 2q &= q + p(q/p) && \text{for } i = 0 \\ q/p &= q + p(q/p)^2 && \text{for } i = 1 \\ (q/p)^i &= q(q/p)^{i-1} + p(q/p)^{i+1} && \text{for all } i \geq 2. \end{aligned}$$

The first of these is clearly an identity, and the third is seen to be an identity by rewriting the right side as $p(q/p)^i + q(q/p)^i$. The second is an identity by almost the same argument. Thus (A.118) satisfies (6.9) and thus gives the correct solution.

For those who dutifully took the hint directly, it is still necessary to show that (A.118) is the smallest solution to (6.9). Let x_i abbreviate $F_{i0}(\infty)$ for $i \geq 0$. Then (6.9) for $j = 0$ can be rewritten as

$$\begin{aligned} x_0 &= q + px_1 \\ x_1 &= q + px_2 \\ x_i &= qx_{i-1} + px_{i+1} \quad \text{for } i \geq 2. \end{aligned}$$

The equation for each $i \geq 2$ can be rearranged to the alternate form,

$$x_{i+1} - (q/p)x_i = x_i - (q/p)x_{i-1}. \quad (\text{A.119})$$

For $i = 1$, the similar rearrangement is $x_2 - (q/p)x_1 = x_1 - q/p$. Now consider the possibility of a solution to these equations with $x_1 < q/p$, say $x_1 = (q/p) - \delta$ with $\delta > 0$. Recursing on the equations (A.119), we have

$$x_{i+1} - (q/p)x_i = -\delta \quad \text{for } i \geq 1. \quad (\text{A.120})$$

For the case $q = p = 1/2$, this becomes $x_{i+1} - x_i = -\delta$. Thus for sufficiently large i , x_i becomes negative, so there can be no non-negative solution of (6.9) with $F_{10} < q/p$.

For the case $1/2 < p < 1$, (A.120) leads to

$$x_{i+1} = (q/p)x_i - \delta \leq (q/p)^2 x_{i-1} - \delta \leq \cdots \leq (q/p)^i - \delta.$$

For large enough i , this shows that x_{i+1} is negative, showing that no non-negative solution exists with $x_1 < q/p$.

b) Under the same conditions as (a), show that $F_{ij}(\infty)$ equals $2(1-p)$ for $j = i$, equals $[(1-p)/p]^{i-j}$ for $i > j$, and equals 1 for $i < j$.

Solution: In the first part of the solution to (a), we used a trapping state argument to show that $F_{i,i-1}(\infty) = F_{10}(\infty)$ for each $i > 1$. That same argument shows that $F_{ij} = F_{i-j,0}(\infty)$ for all $i > j$. Thus $F_{ij}(\infty) = (q/p)^{i-j}$ for $i > j$.

Next, for $i < j$, consider converting state j into a trapping state. This does not alter $F_{ij}(\infty)$ for $i < j$, but converts the states $0, 1, \dots, j$ into a finite-state Markov chain with a single recurrent state, j . Thus $F_{ij}(\infty) = 1$ for $i < j$.

Finally, for $i = j > 1$, (6.9) says that

$$F_{ii}(\infty) = qF_{i-1,i}(\infty) + pF_{i+1,i}(\infty) = q + pF_{10}(\infty) = 2q,$$

where we have used the earlier results for $i < j$ and $i > j$. The case for $F_{11}(\infty)$ is virtually the same.

Exercise 6.3 a): Show that the n th order transition probabilities, starting in state 0, for the Markov chain in Figure 6.2 satisfy

$$P_{0j}^n = pP_{0,j-1}^{n-1} + qP_{0,j+1}^{n-1}, \quad \text{for } j \neq 0; \quad P_{00}^n = qP_{00}^{n-1} + qP_{01}^{n-1}.$$

Hint: Use the Chapman-Kolmogorov equality, (4.7).

Solution: This is the Chapman-Kolmogorov equality in the form $P_{ij}^n = \sum_k P_{ik}^{n-1} P_{kj}$ where $P_{00} = q$, $P_{k,k+1} = p$ for all $k \geq 0$ and $P_{k,k-1} = q$ for all $k \geq 1$; $P_{kj} = 0$ otherwise.

b) For $p = 1/2$, use this equation to calculate P_{0j}^n iteratively for $n = 1, 2, 3, 4$. Verify (6.3) for $n = 4$ and then use induction to verify (6.3) in general. Note: this becomes an absolute mess for $p \neq 1/2$, so don't attempt this in general.

Solution: This is less tedious if organized as an array of terms. Each term (except P_{00}^n) for each n is then half the term diagonally above and to the left, plus half the term diagonally above and to the right. P_{00}^n is half the term above plus half the term diagonally above and to the right.

j	0	1	3	3	4
$P_{0,j}^1$	1/2	1/2			
$P_{0,j}^2$	1/2	1/4	1/4		
$P_{0,j}^3$	3/8	3/8	1/8	1/8	
$P_{0,j}^4$	3/8	1/4	1/4	1/16	1/16

terms are similar. We also see (with less work) that (6.3) is valid for $1 \leq n \leq 3$ for all $j, 0 \leq j \leq n$. We can avoid fussing with the constraint $j \leq n$ by following the convention that $\binom{n}{k} = 0$ for all $n \geq 1$ and $k > n$. We have then shown that (6.3) is valid for $1 \leq n \leq 4$ and all $j \geq 0$.

We next use induction to validate (6.3) in general. From the previous calculation, any $n, 1 \leq n \leq 4$ can be used as the basis of the induction and then, given the assumption that (6.3) is valid for any given $n > 0$ and all $j \geq 0$, we will prove that (6.3) is also valid for $n+1$ and all $j \geq 0$. Initially, for given n , we assume $j > 0$; the case $j = 0$ is a little different since it has self transitions, so we consider it later.

For the subcase where $n + j$ is even, we have

$$\begin{aligned} P_{0j}^{n+1} &= \frac{1}{2} [P_{0,j-1}^n + P_{0,j+1}^n] \\ &= \frac{1}{2} \left[\binom{n}{(j+n)/2} 2^{-n} + \binom{n}{((j+n)/2) + 1} 2^{-n} \right] \\ &= \binom{n+1}{((j+n)/2) + 1} 2^{-(n+1)}. \end{aligned}$$

The first equality comes from part (a) with $p = q = 1/2$. The second equality uses (6.3) for the given n and uses the fact that $n+j-1$ and $n+j+1$ are odd. The final equality follows immediately from the combinatorial identity

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}.$$

This identity follows by viewing $\binom{n+1}{k+1}$ as the number of ways to arrange $k+1$ ones in a binary $n+1$ tuple. These arrangements can be separated into those that start with a one followed by k ones out of n and those that start with a zero followed by $k+1$ ones out of n . The final result for P_{0j}^{n+1} above is for $n+1+j$ odd and agrees with the odd case in (6.3) after substituting $n+1$ for n in (6.3). This validates (6.3) for this case. The subcase where $n+j$ is odd is handled in the same way.

For the case $j = 0$, we use the second portion of part (a), namely $P_{00}^{n+1} = \frac{1}{2}P_{00}^n + \frac{1}{2}P_{01}^n$. For the subcase where n is even, we then have

$$\begin{aligned} P_{00}^{n+1} &= \frac{1}{2} \left[\binom{n}{n/2} 2^{-n} + \binom{n}{(n/2) + 1} 2^{-n} \right] \\ &= \binom{n+1}{(n/2)+1} 2^{-(n+1)}. \end{aligned}$$

In the first equality, we have used (6.3) with $n+j$ even for $j=0$ and $n+j$ odd for $j=1$. The second equality uses the combinatorial identity above. This result agrees with the odd case in (6.3) after substituting $n+1$ for n . Finally, the subcase where n is odd is handled in almost the same way, except that one must recognize that $\binom{n}{(n+1)/2} = \binom{n}{(n-1)/2}$.

(c) As a more interesting approach, which brings out the relationship of Figures 6.2 and 6.1, note that (6.3), with $j + n$ even, is the probability that $S_n = j$ for the chain in 6.1. Similarly, (6.3) with $j + n$ odd is the

probability that $S_n = -j - 1$ for the chain in 6.1. By viewing each transition over the self loop at state 0 as a sign reversal for the chain in 6.1, explain why this surprising result is true. (Again, this doesn't work for $p \neq 1/2$, since the sign reversals also reverse the $+1, -1$ transitions.)

Solution: The meaning of the hint will be less obscure if we redraw Figure 6.1 in the following way:

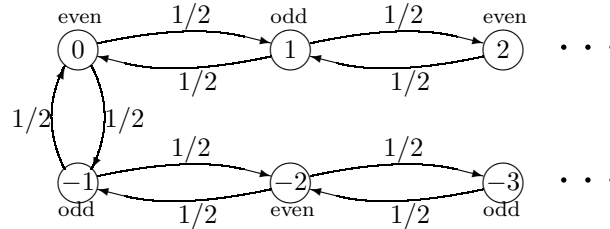


Figure 6.1 redrawn

Compare this with the discrete M/M/1 chain of Figure 6.2,

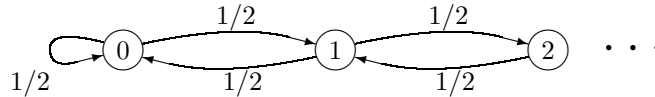


Figure 6.2 redrawn

Note that Figure 6.2 can be viewed as the result of combining each nonnegative state i in the redrawn Figure 6.1 with the state $-i-1$ lying directly beneath it. To be more specific, the transition probability from state 0 in Fig. 6.1 to the aggregated states 1 and -2 is $1/2$, and the transition probability to the aggregated states 0 and -1 is $1/2$. The same transition probabilities hold for state -1. Similarly, starting from any state i in Fig 6.1, there is a transition probability $1/2$ to the aggregated states $i+1, -i-2$ and $1/2$ to the aggregated states $i-1$ and i . The same aggregated transition probabilities hold starting from state $-i-1$.

What this means is that the set of aggregated pairs forms a Markov chain in its own right, and this Markov chain is the M/M/1 chain of Fig 6.2. The n th order transition probabilities P_{0i}^n for the M/M/1 chain are thus the same as the n th order aggregate transition probabilities say $Q_{0i}^n + Q_{0,-i-1}^n$ for the Bernoulli chain. Since the Bernoulli chain is periodic with period 2, however, and each pair of states consists of one even and one odd term, only one of these aggregated terms is nonzero. This helps explain the strange looking difference in part (b) between $n+i$ even and odd.

Exercise 6.4: Let j be a transient state in a Markov chain and let j be accessible from i . Show that i is transient also. Interpret this as a form of Murphy's law (if something bad can happen, it will, where the bad thing is the lack of an eventual return). Note: give a direct demonstration rather than using Theorem 6.2.5.

Solution: It is instructive to start off using Theorem 6.2.5, and after that, we give two solutions without it. First, using the given path from state i to j , we use contradiction, showing that the assumption that state i is recurrent implies the contradiction that j is recurrent. This contradiction will imply that i is not recurrent, *i.e.*, transient. From Lemma 6.2.3 (with the roles of i and j reversed, i recurrent implies that $F_{ji}(\infty) = 1$. This in turn implies that there is a path from j to i , so that i and j are in the same class. From Theorem 6.2.5, this implies that j is recurrent, which is a contradiction. Thus state i must

be transient.

The second proof will use contradiction in the same way. If i is recurrent, then the Markov chain generates a renewal process with renewals on occurrences of state i . From Theorem 6.2.6, part 3, $\lim_{t \rightarrow \infty} E[N_{ii}(t)] = \infty$. For each renewal (occurrence of i), there is an occurrence of j before the next renewal with at least the positive probability, say $\alpha > 0$, that the path from i to j is taken immediately after the renewal (and thus before the next renewal). Thus the expected number of occurrences of j per occurrence of i is at least α and $E[N_{jj}(t)] \geq \alpha E[N_{ii}(t)] - 1$. Thus $\lim_{t \rightarrow \infty} E[N_{jj}(t)] = \infty$, so, from theorem 6.2.6, j must be recurrent. This is a contradiction, so i is transient.

The third proof is direct (does not use contradiction) and uses a technique that can be useful in many contexts. Since state j is transient, it follows by definition that $F_{jj}(\infty) < 1$. This implies that the number of occurrences of state j over the infinite time interval is a geometric rv and the interval between occurrences is a degenerate rv (see Theorem 6.2.6). Consider the sequence of occurrences of states i and j , ignoring the time intervals between them and considering only the order in which they occur, such as, for example i, i, j, i, j, j where after the last occurrence of j , no more occurrences of either i or j occur. After each i , the next occurrence in this sequence will be either i , j , or nothing. The probability of each of these, conditional on i , is independent of position in the sequence and independent of the terms before that i . The same situation (with different probabilities) occurs after each j , and ‘nothing’ can be viewed as a trapping state. Thus this sequence (including the ‘nothings’) forms a 3-state Markov chain. Since state j is transient in the original chain, it is also transient in this 3-state chain, since j appears the same finite number of times in each.

It is given that there is a path from i to j in the original chain, so there is a nonzero transition probability from i to j in the 3-state chain. If there is also a path from j to i in the original chain, then there is a nonzero transition from j to i in the 3-state chain, so j and i are in the same class (both for the original and the 3-state chain). If there is no such transition from j to i , then they are not in the same class, but both are still transient (as seen from the analysis of classes for finite-state chains). Thus i is transient in the 3-state chain, so the number of occurrences of state i in the original chain is finite with probability 1. Thus i is transient also.

It is not necessary, in this argument, to be able to calculate the transition probabilities in the 3-state chain, but it might seem a little less abstract to see how to calculate them. For example, to find the transition probability from i to i in the 3-state chain, consider changing the original Markov chain by making j a trapping state. This does not change $F_{ij}(\infty)$, but $F_{ii}(\infty)$ then becomes the probability that an i follows an i in the 3-state chain. The other transition probabilities are found in a similar way.

Murphy’s law is not a law, but rather a popular saying used to express pessimism. In terms of Murphy’s law, if there is a nonzero probability that i ever gets to a state from which there is the possibility of no return, then that possibility will eventually happen.

Exercise 6.5: This exercise treats a mathematical detail from the proof of Lemma 6.2.4. Assume that j is recurrent, there is a path of probability $\alpha > 0$ from j to i , and $n \geq 1$ is an integer. Let T_{ji} be the possibly defective rv giving the first-passage time from j to i .

Verify the following equations for any integer $t > 0$:

$$\Pr\{T_{ji} > t\} = \Pr\{T_{ji} > t, T_{jj}^{(n)} > t\} + \Pr\{T_{ji} > t, T_{jj}^{(n)} \leq t\} \quad (\text{A.121})$$

$$\leq \Pr\{T_{jj}^{(n)} > t\} + \Pr\{T_{ji} > T_{jj}^{(n)}\} \quad (\text{A.122})$$

$$\leq \Pr\{T_{jj}^{(n)} > t\} + (1 - \alpha)^n \quad (\text{A.123})$$

$$\Pr\{T_{ji} = \infty\} \leq (1 - \alpha)^n. \quad (\text{A.124})$$

Solution: (A.121) simply splits the event $\{T_{ji} > t\}$ into two disjoint terms; (A.122) upper bounds the probability of the joint event in the first term by the probability of the first of those events. The second term can be rewritten as $\Pr\{T_{ji} > t \geq T_{jj}^{(n)}\}$ and upper bounded by eliminating t . Note that $\{T_{ji} > T_{jj}^{(n)}\}$ is the event that the first visit to i comes after the first n visits to j . This is upper bounded in (A.123) by the probability of not taking the given path of probability α from j to i in each of the first n visits to j . Finally, (A.124) takes the limit as $t \rightarrow \infty$, first using the fact that $T_{jj}^{(n)}(t)$ is a rv, so that $\lim_{t \rightarrow \infty} \Pr\{T_{jj}^{(n)} > t\} = 0$. Since n is arbitrary, this shows that $\lim_{t \rightarrow \infty} \Pr\{T_{ji} > t\} = 0$, which is the result claimed in Lemma 6.2.4.

Exercise 6.6: (Renewal theory proof that a class is all recurrent or all transient.) Assume that state j in a countable-state Markov chain is recurrent and i communicates with j . More specifically, assume that there is a path of length m from i to j and one of length k from j to i .

a) Show that for any $n > 0$,

$$P_{ii}^{m+n+k} \geq P_{ij}^m P_{jj}^n P_{ji}^k. \quad (\text{A.125})$$

Solution: First, if $P_{jj}^n = 0$, then the right side of (A.125) is zero, so (A.125) is satisfied in that case. Next, for any n such that $P_{jj}^n > 0$, there is one or more walks of length m from i to j , one or more walks of length n from j to j and one or more walks of length k from j to i . Concatenating these, we have one or more walks of length $m + n + k$ from i to i . The set of such walks, which go from i to j in m steps, j to j in n steps, and j to i in k steps has positive probability $P_{ij}^m P_{jj}^n P_{ji}^k$ and is a subset of all walks that go from i to i in $m + n + k$ steps, establishing (A.125).

b) By summing over n , show that state i is also recurrent. Hint: Use Theorem 6.2.6.

Solution: Using the fourth part of Theorem 6.2.6 and the fact that j is recurrent, we have $\sum_{n=1}^{\infty} P_{jj}^{(n)} = \infty$. Using (A.125),

$$\sum_{n=1}^{\ell} P_{ii}^{(m+n+k)} = P_{ij}^m P_{ji}^k \sum_{n=1}^{\ell} P_{jj}^n.$$

Since $P_{ij}^m P_{ji}^k > 0$, this has the limit ∞ as $\ell \rightarrow \infty$. Applying the if and only if aspect of the fourth part of the theorem shows that i is recurrent.

c) Explain why this shows that all states in a class are recurrent or all are transient.

Solution: We have shown that if a state j is recurrent, then each state i that communicates with j (i.e., is in the same class as j) is also recurrent. Thus if any one state in the class is

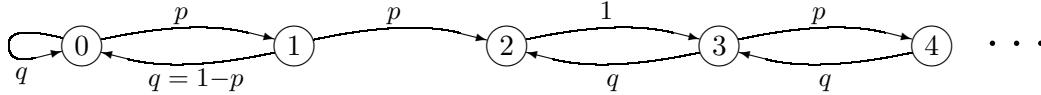
recurrent, then all states in that class are, which also means that if any state is recurrent, no state can be transient. Thus if any state is transient, all states must be transient.

Exercise 6.7: Consider an irreducible Markov chain that is positive recurrent. Recall the technique used to find the expected first passage time from j to i , i.e., $\bar{T}_{ji}$, in Section 4.5. The state i was turned into a trapping state by turning all the transitions out of i into a single transition $P_{ii} = 1$. Here, in order to preserve the positive recurrence, we instead move all transitions out of state i into the single transition $P_{ij} = 1$.

a) Use Figure 6.2 to illustrate that the above strategy can turn an irreducible chain into a reducible chain. Also explain why states i and j are still positive recurrent and still in the same class.

Solution: In the figure below, we have modified state 2 so that it has only one outgoing transition and that transition goes to state 3. As seen, states 0 and 1 have become transient, but 2 and 3 (and all higher numbered states) are in the same recurrent class.

More generally, if any state i in this chain (other than state 0) is modified to have only one outgoing transition, then if that one remaining transition is upward, all paths downward from that state are severed. If $j > i+1$, then the transition from i to $i+1$ is also severed, but all the states between i and j are reachable via j . Similarly, if the one severed transition is downward, all paths from i upward are severed. Either way, the chain is no longer irreducible.



b) Let $\{\pi'_k; k \geq 0\}$ be the steady-state probabilities for the positive-recurrent class in the modified Markov chain. Show that the expected first passage time $\bar{T}'_{ji}$ from j to i in the modified Markov chain is $(1/\pi'_i) - 1$.

Solution: We showed in (a) that states i and j are positive recurrent in the modified chain for Figure 6.2.2, and we show in part (c) that they are positive recurrent in the modified chain whenever the original chain is positive recurrent. Since the only outgoing transition from state i in the modified chain goes to state j , we see that every first passage from i back to i goes first to j and from there back to i via a first passage from j to i . Thus for the modified chain, the expected first passage time from j to i is 1 less than the expected first passage time from i to i . That expected first passage time from i to i is $1/\pi'_i$ by renewal theory (assuming that j is positive recurrent and thus $\pi'_i > 0$.)

c) Show that the expected first passage time from j to i is the same in the modified and unmodified chain.

Solution: Each sample walk for a first passage from j to i contains state i only as the final state in the path, and thus cannot include an outward transition from i . Thus each such walk in the original chain is a walk in the modified chain and vice versa. The probability of each such walk is the same in the modified chain as in the original chain. Thus $\bar{T}'_{ji} = \bar{T}_{ji}$.

It follows that $\bar{T}'_{ii} = \bar{T}'_{ji} + 1 = \bar{T}_{ji} + 1$. Since $\pi'_i = 1/\bar{T}'_{ii}$, we can show that $\pi'_i > 0$ by showing that $\bar{T}_{ji} < \infty$. Since i and j communicate in the modified chain, this will show that i and j are positive recurrent in the modified chain.

For each state j in the original positive-recurrent chain, we have $\pi_j = 1/\bar{T}_{jj} > 0$ and thus

$\bar{T}_{jj} < \infty$. We can express

$$\bar{T}_{jj} = P_{jj} + \sum_{k \neq j} P_{jk} [\bar{T}_{kj} + 1].$$

Thus $\bar{T}_{kj} < \infty$ for each k such that there is a path of length 1 from j to k . In the same way, for every path ℓ, k, j of length 2 to state j , we have

$$\bar{T}_{kj} = P_{kj} + \sum_{\ell \neq k} P_{k\ell} [\bar{T}_{\ell j} + 1].$$

Since $\bar{T}_{kj} < \infty$, this shows that $\bar{T}_{\ell j} < \infty$ for all ℓ such that there is a path of length 2 from j to ℓ . By iteration to longer paths, we see that for an arbitrary path length from j to i , $\bar{T}_{ij} < \infty$. Since the chain is recurrent, there is a path from i to j for each i and j , so $\bar{T}_{ij} < \infty$ for all i, j . Thus the class of states including i and j in the modified chain is positive recurrent. Note that the modified chain is a function of both i and j , so π'_i is a function of j as well as i .

d) Show by example that after the modification above, two states i and j that were not positive recurrent before the modification can become positive recurrent and the above technique can again be used to find the expected first passage time.

Solution: Let the modified chain in (a) be the original chain in this part. Consider finding the expected first passage time from state 1 to 2. Then the modified chain has a transition of probability 1 from state 2 to 1 and states 0, 1, and 2 become a positive recurrent class. The expected time from 1 to 2 is then $1/\pi'_2 - 1$.

Exercise 6.8: Let $\{X_n; n \geq 0\}$ be a branching process with $X_0 = 1$. Let $\bar{Y}, \sigma^2$ be the mean and variance of the number of offspring of an individual.

a) Argue that $\lim_{n \rightarrow \infty} X_n$ exists with probability 1 and either has the value 0 (with probability $F_{10}(\infty)$) or the value ∞ (with probability $1 - F_{10}(\infty)$).

Solution: We consider 2 special, rather trivial, cases before considering the important case (the case covered in the text). Let $\{p_i; i \geq 0\}$ be the PMF of the number of offspring of each individual. Then if $p_1 = 1$, we see that $X_n = 1$ for all n , so the statement to be argued is simply false. You should be proud of yourself if you noticed the need for ruling this case out before constructing a proof.

The next special case is where $p_0 = 0$ and $p_1 < 1$. Then $X_{n+1} \geq X_n$ (i.e., the population never shrinks but can grow). Since $X_n(\omega)$ is non-decreasing for each sample path, either $\lim_{n \rightarrow \infty} X_n(\omega) = \infty$ or $\lim_{n \rightarrow \infty} X_n(\omega) = j$ for some $j < \infty$. The latter case is impossible, since $P_{jj} = p_1^j$ and thus $P_{jj}^m = p_1^{mj} \rightarrow 0$ as $m \rightarrow \infty$.

Ruling out these two trivial cases, we have $p_0 > 0$ and $p_1 < 1 - p_0$. In this case, state 0 is recurrent (i.e., it is a trapping state) and states 1, 2, ... are in a transient class. To see this, note that $P_{10} = p_0 > 0$, so $F_{11}(\infty) \leq 1 - p_0 < 1$, which means by definition that state 1 is transient. All states $i > 1$ communicate with state 1, so by Theorem 6.2.5, all states $j \geq 1$ are transient. Thus one can argue that the process has 'no place to go' other than 0 or ∞ .

The following tedious analysis makes this precise. Each $j > 0$ is transient, so from Theorem 6.2.6 part 3,

$$\lim_{t \rightarrow \infty} \mathbf{E}[N_{jj}(t)] < \infty.$$

Note that $N_{1j}(t)$ is the number of visits to j in the interval $[1, t]$ starting from state 1 at time 0. This is one more than the number of returns to j after the first visit to j . The expected number of such returns is upper bounded by the number in t steps starting in j , so $\mathbf{E}[N_{1j}(t)] \leq 1 + \mathbf{E}[N_{jj}(t)]$. It follows that

$$\lim_{t \rightarrow \infty} \mathbf{E}[N_{1j}(t)] < \infty \quad \text{for each } j > 0.$$

Now note that the expected number of visits to j in $[1, t]$ can be rewritten as $\sum_{n=1}^t P_{1j}^n$. Since this sum in the limit $t \rightarrow \infty$ is finite, the remainder in the sum from t to ∞ must approach 0 as $t \rightarrow \infty$, so

$$\lim_{t \rightarrow \infty} \sum_{n>t} P_{1j}^n = 0.$$

From this, we see that for every finite integer ℓ ,

$$\lim_{t \rightarrow \infty} \sum_{n>t} \sum_{j=1}^{\ell} P_{1j}^n = 0.$$

This says that for every $\epsilon > 0$, there is a t sufficiently large that the probability of ever entering states 1 to ℓ on or after step t is less than ϵ . Since $\epsilon > 0$ is arbitrary, all sample paths (other than a set of probability 0) never enter states 1 to ℓ after some finite time. Since ℓ is arbitrary, $\lim_{n \rightarrow \infty} X_n$ exists WP1 and is either 0 or ∞ . By definition, it is 0 with probability $F_{10}(\infty)$.

b) Show that $\text{VAR}[X_n] = \sigma^2 \bar{Y}^{n-1}(\bar{Y}^n - 1)/(\bar{Y} - 1)$ for $\bar{Y} \neq 1$ and $\text{VAR}[X_n] = n\sigma^2$ for $\bar{Y} = 1$.

Solution: We will demonstrate the case for $\bar{Y} \neq 1$ and $\bar{Y} = 1$ together by showing that

$$\text{VAR}[X_n] = \sigma^2 \bar{Y}^{n-1} [1 + \bar{Y} + \bar{Y}^2 + \cdots + \bar{Y}^{n-1}]. \quad (\text{A.126})$$

First express $\mathbf{E}[X_n^2]$ in terms of $\mathbf{E}[X_{n-1}^2]$. Note that, conditional on $X_{n-1} = \ell$, X_n is the sum of ℓ IID rv's each with mean $\bar{Y}$ and variance σ^2 , so

$$\begin{aligned} \mathbf{E}[X_n^2] &= \sum_{\ell} \Pr\{X_{n-1} = \ell\} \mathbf{E}[X_n^2 | X_{n-1} = \ell] \\ &= \sum_{\ell} \Pr\{X_{n-1} = \ell\} [\ell\sigma^2 + \ell^2\bar{Y}^2] \\ &= \sigma^2 \mathbf{E}[X_{n-1}] + \bar{Y}^2 \mathbf{E}[X_{n-1}^2] \\ &= \sigma^2 \bar{Y}^{n-1} + \bar{Y}^2 \mathbf{E}[X_{n-1}^2]. \end{aligned}$$

We also know from (6.50) (or by simple calculation) that $E[X_n] = \bar{Y}E[X_{n-1}]$. Thus,

$$\text{VAR}[X_n] = E[X_n^2] - [E[X_n]]^2 = \sigma^2 \bar{Y}^{n-1} + \bar{Y}^2 \text{VAR}[X_{n-1}]. \quad (\text{A.127})$$

We now use induction to derive (A.126) from (A.127). For the base of the induction, we see that X_1 is the number of progeny from the single element X_0 , so $\text{VAR}[X_1] = \sigma^2$. For the inductive step, we assume that

$$\text{VAR}[X_{n-1}] = \sigma^2 \bar{Y}^{n-2} [1 + \bar{Y} + \bar{Y}^2 + \cdots + \bar{Y}^{n-2}].$$

Substituting this into (A.127),

$$\begin{aligned} \text{VAR}[X_n] &= \sigma^2 \bar{Y}^{n-1} + \sigma^2 \bar{Y}^n [1 + \bar{Y} + \cdots + \bar{Y}^{n-2}] \\ &= \sigma^2 \bar{Y}^{n-1} [1 + \bar{Y} + \cdots + \bar{Y}^{n-1}], \end{aligned}$$

completing the induction.

Exercise 6.9: There are n states and for each pair of states i and j , a positive number $d_{ij} = d_{ji}$ is given. A particle moves from state to state in the following manner: Given that the particle is in any state i , it will next move to any $j \neq i$ with probability P_{ij} given by

$$P_{ij} = \frac{d_{ij}}{\sum_{k \neq i} d_{ik}}. \quad (\text{A.128})$$

Assume that $P_{ii} = 0$ for all i . Show that the sequence of positions is a reversible Markov chain and find the limiting probabilities.

Solution: From Theorem 6.5.3, $\{P_{ij}\}$ is the set of transition probabilities and $\boldsymbol{\pi}$ is the steady-state probability vector of a reversible chain if $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j . Thus, given $\{d_{ij}\}$, we attempt to find $\boldsymbol{\pi}$ to satisfy the equations

$$\pi_i P_{ij} = \frac{\pi_i d_{ij}}{\sum_k d_{ik}} = \frac{\pi_j d_{ji}}{\sum_k d_{jk}} = \pi_j P_{ji}; \quad \text{for all } i, j.$$

We have taken $d_{ii} = 0$ for all i here. Since $d_{ij} = d_{ji}$ for all i, j , we can cancel d_{ij} and d_{ji} from the inner equations, getting

$$\frac{\pi_i}{\sum_k d_{ik}} = \frac{\pi_j}{\sum_k d_{jk}}; \quad \text{for all } i, j.$$

Thus π_i must be proportional to $\sum_k d_{ik}$, so normalizing to make $\sum_i \pi_i = 1$, we get

$$\pi_i = \frac{\sum_k d_{ik}}{\sum_\ell \sum_k d_{\ell k}}; \quad \text{for all } i.$$

If the chain has a countably infinite number of states, this still works if the sums exist.

This exercise is not quite as specialized as it sounds, since given any reversible Markov chain, we can define $d_{ij} = \pi_i P_{ij}$ and get this same set of equations (normalized in a special way).

Exercise 6.10: Consider a reversible Markov chain with transition probabilities P_{ij} and limiting probabilities π_i . Also consider the same chain truncated to the states $0, 1, \dots, M$. That is, the transition probabilities $\{P'_{ij}\}$ of the truncated chain are

$$P'_{ij} = \begin{cases} \frac{P_{ij}}{\sum_{k=0}^M P_{ik}} & ; \quad 0 \leq i, j \leq M \\ 0 & ; \quad \text{elsewhere.} \end{cases}.$$

Show that the truncated chain is also reversible and has limiting probabilities given by

$$\bar{\pi}_i = \frac{\pi_i \sum_{j=0}^M P_{ij}}{\sum_{k=0}^M \left(\pi_k \sum_{m=0}^M P_{km} \right)}. \quad (\text{A.129})$$

Solution: The steady-state probabilities $\{\pi_i; i \geq 0\}$ of the original chain must be positive. Thus $\pi_i > 0$ for each i . By summing $\bar{\pi}_i$ over i in (A.129), it is seen that the numerator is the same as the denominator, so $\sum_{i=0}^M \bar{\pi}_i = 1$. Finally,

$$\bar{\pi}_i P'_{ij} = \frac{\pi_i P_{ij}}{\sum_{k=0}^M \left(\pi_k \sum_{m=0}^M P_{km} \right)}.$$

Since $\pi_i P_{ij} = \pi_j P_{ji}$ for each i, j , it is clear that $\bar{\pi}_i P'_{ij} = \bar{\pi}_j P'_{ji}$ for each $i, j \leq M$. Thus, by Theorem 6.5.3, the truncated chain is reversible.

Exercise 6.11: A Markov chain (with states $\{0, 1, 2, \dots, J-1\}$ where J is either finite or infinite) has transition probabilities $\{P_{ij}; i, j \geq 0\}$. Assume that $P_{0j} > 0$ for all $j > 0$ and $P_{j0} > 0$ for all $j > 0$. Also assume that for all i, j, k , we have $P_{ij}P_{jk}P_{ki} = P_{ik}P_{kj}P_{ji}$.

a) Assuming also that all states are positive recurrent, show that the chain is reversible and find the steady-state probabilities $\{\pi_i; i \geq 0\}$ in simplest form.

Solution: Section 6.5.3 shows that reversibility implies the three-cycle condition $P_{ij}P_{jk}P_{ki} = P_{ik}P_{kj}P_{ji}$, and the exercise asks us to show that, with the addition of another condition about positive transition probabilities, the three cycle condition also implies reversibility.

We first show why there is a need for some condition on positive probabilities, then demonstrate the result for finitely many states, and finally, after quite a bit of work, give a demonstration for the countable case; this final demonstration also yields quite a few other results.

To see the need for certain positive probabilities, consider a four-state chain where $P_{01} = P_{12} = P_{23} = P_{30} = 2/3$ and $P_{32} = P_{21} = P_{10} = P_{03} = 1/3$. Assume that $P_{ij} = 0$ for all other i, j . This is not reversible, since (from symmetry) $\pi_i = 1/4$ for all i , but, for example, $\pi_0 P_{01} \neq \pi_1 P_{10}$. The trouble here is that the 3-cycle condition is satisfied, but only satisfied vacuously since there are no 3-cycles with positive probability in this chain. This example can be extended to show the irreversibility of a chain with any number of states if three states, say 1, 2, 3, have the same transition probabilities as above (and thus communicate with states $i > 3$ only through state 0) and $P_{01} = 2\alpha/3$ and $P_{03} = \alpha/3$, where $1 - \alpha > 0$ is the aggregate probability of going from state 0 to states $i > 3$.

Next assume the positivity conditions in the problem statement again, and suppose the number of states J is finite. From the guessing theorem, Theorem 6.5.3, the chain is

reversible if $\{\pi_i; i \geq 0\}$ exists such that $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j . To select a set of π_i that satisfy this condition, let π_0 be arbitrary for the moment and let $\pi_i = \pi_0 P_{0i}/P_{i0}$ for all $i > 0$ (each π_i , $i \geq 1$ exists since $P_{i0} > 0$). This ensures that $\pi_i P_{i0} = \pi_0 P_{0i}$ is satisfied for all $i > 1$. Since $\sum_i \pi_i$ must be 1 for the guessing theorem conditions to be satisfied, we have

$$1 = \sum_i \pi_i = \pi_0 \left[1 + \sum_{i>0} \frac{P_{0i}}{P_{i0}} \right]. \quad (11.a)$$

The sum has finitely many terms and by assumption each P_{i0} is positive. Thus the sum is finite and for the guessing theorem conditions to be satisfied, we must have

$$\pi_0 = \left[1 + \sum_{i>0} \frac{P_{0i}}{P_{i0}} \right]^{-1} \quad \text{and thus } \pi_i = \frac{P_{0i}}{P_{i0}} \left[1 + \sum_{i>0} \frac{P_{0i}}{P_{i0}} \right]^{-1}.$$

This satisfies all the guessing theorem conditions except $\pi_i P_{ij} = \pi_j P_{ji}$ for each $i, j > 0$. For any such $i, j > 0$, consider the 3-cycle condition $P_{0i} P_{ij} P_{j0} = P_{0j} P_{ji} P_{i0}$. Since $P_{0j} > 0$ and $P_{0i} > 0$, we can solve for P_{ji} ,

$$P_{ji} = P_{ij} \frac{P_{0i}}{P_{i0}} \frac{P_{j0}}{P_{0j}} = P_{ij} \frac{\pi_i}{\pi_0} \frac{\pi_0}{\pi_j} = P_{ij} \frac{\pi_i}{\pi_j}.$$

This demonstrates the reversibility condition for each $i, j > 0$, and thus, by the guessing theorem, the chain is reversible with the indicated steady-state probabilities.

This does not quite work when the number of states is countably infinite, since then the sum $\sum_{i>0} P_{0i}/P_{i0}$ might be infinite. Since the chain is positive recurrent, the true π_0 is positive, but the π_0 in the above argument is only an hypothesized probability and thus requires a different approach to show that it must be positive.

We now use the following 3 lemmas to demonstrate reversibility when the number of states is countably infinite.

Lemma 1: For the conditions in the exercise, and for arbitrary states i, j, k , we have $P_{0i} P_{ij} P_{jk} P_{k0} = P_{0k} P_{kj} P_{ji} P_{i0}$,

Proof: If this is restricted to distinct nonzero i, j, k , it says that all 4-cycles that include state 0 are reversible. The lemma does not restrict i, j, k to be distinct and nonzero. Similarly the condition $P_{0i} P_{ij} P_{j0} = P_{0j} P_{ji} P_{i0}$ in the problem statement does not require i and j to be distinct and nonzero. It makes no difference whether the problem statement applies when i, j are equal to each other or to 0, since in those cases, for example $P_{00} P_{0j} P_{j0} = P_{0j} P_{j0} P_{00}$, the equality holds automatically without any need for an assumption. We prove the lemma, however, without the assumption that i, j, k are nonzero and unique.

For arbitrary states i, j, k , we now have

$$\begin{aligned} P_{0i} P_{ij} P_{jk} P_{k0} &= \frac{P_{0i} P_{ij} P_{j0} P_{0j} P_{jk} P_{k0}}{P_{j0} P_{0j}} \\ &= \frac{P_{0j} P_{ji} P_{i0} P_{0k} P_{kj} P_{j0}}{P_{j0} P_{0j}} \\ &= P_{ji} P_{i0} P_{0k} P_{kj} = P_{0k} P_{kj} P_{ji} P_{i0}. \end{aligned}$$

where the first line multiplies and divides by the positive term $P_{j0}P_{0j}$; the second uses the given condition on $P_{0i}P_{ij}P_{j0}$ (and $P_{0j}P_{jk}P_{k0}$); and the third rearranges terms. $\square$

The next lemma generalizes Lemma 1 from three states, i, j, k to an arbitrary number.

Lemma 2: For arbitrary $m \geq 2$, let $i_1, i_2, \dots, i_m$ be arbitrary states (not necessarily unique or nonzero).. Then

$$P_{0i_1}P_{i_1i_2} \cdots P_{i_{m-1}i_m}P_{i_m0} = P_{0i_m}P_{i_mi_{m-1}} \cdots P_{i_2i_1}P_{i_10}. \quad (11.b)$$

Proof: We use induction. The given condition, $P_{0i}P_{ij}P_{j0} = P_{0j}P_{ji}P_{i0}$ is the desired result for $m = 2$ and Lemma 1 gives the desired result for $m = 3$. The inductive proof below assumes the result to be true for any $m-1 \geq 1$ and uses this to establish the result for m . Thus Lemma 1 is not really necessary except as a less abstract version of the proof below.

$$\begin{aligned} P_{0i_1}P_{i_1i_2} \cdots P_{i_{m-1}i_m}P_{i_m0} &= \frac{(P_{0i_1}P_{i_1i_2} \cdots P_{i_{m-2}i_{m-1}}P_{i_{m-1}0})(P_{0i_{m-1}}P_{i_{m-1}i_m}P_{i_m0})}{P_{i_{m-1}0}P_{0,i_{m-1}}} \\ &= \frac{(P_{0,i_{m-1}}P_{i_{m-1}i_{m-2}} \cdots P_{i_2i_1}P_{i_10})(P_{0,i_m}P_{i_mi_{m-1}}P_{i_{m-1},0})}{P_{i_{m-1},0}P_{0,i_{m-1}}} \\ &= P_{0i_m}P_{i_mi_{m-1}} \cdots P_{i_2i_1}P_{i_10}, \end{aligned}$$

where the steps above are the same as those in Lemma 1. $\square$

Lemma 3: For each $m \geq 2$ and each state $i > 0$,

$$P_{0i}P_{i0}^m = P_{0i}^mP_{i0}.$$

Proof: Note that $P_{i_1i_2}, P_{i_2i_3}, \dots, P_{i_m0}$ is the probability of a given path of length m going from state i_1 to 0. If this is summed over all choices for $i_2, i_3, \dots, i_m$, we get the probability of going from state i_1 to 0 in m steps, i.e.,

$$\sum_{i_2, i_3, \dots, i_m} P_{i_1i_2}, P_{i_2i_3}, \dots, P_{i_m0} = P_{i_10}^m.$$

Thus the left side of (11.b), summed over $i_2, \dots, i_m$ is $P_{0i_1}P_{i_10}^m$. In the same way, the right side of (11.b), summed over $i_2, i_3, \dots, i_m$ is $P_{0,i_1}^mP_{i_10}$. Since each term being summed on the left is equal to the corresponding term on the right, we have

$$P_{0i_1}P_{i_10}^{m-1} = P_{0i_1}^mP_{i_10}.$$

Since $i_1 > 0$ is arbitrary, the lemma is proved. $\square$

We can now prove that the chain is reversible. There are two cases; first, the chain is aperiodic and second, the chain is periodic. If the chain is aperiodic, then by Blackwell (see Section 6.3.3), $\lim_{m \rightarrow \infty} P_{0i}^m = \pi_i$ and $\lim_{m \rightarrow \infty} P_{i0}^m = \pi_0$. Thus $\pi_0P_{0i} = \pi_iP_{i0}$ for all i . It then follows by the same argument as in the finite Markov chain case that $\pi_iP_{ij} = \pi_jP_{ji}$ for all i, j , so the chain is reversible.

Finally, the periodic case can arise only if $P_{ij} = 0$ for all $i > 0$ and $j > 0$, and if in addition $P_{00} = 0$. In this case, the chain is obviously reversible. This reversibility also follows from Blackwell's theorem.

Lemma 2 above can also be modified slightly to show that the 3-cycle condition (where i, j are distinct and nonzero) implies the m -cycle condition (again with distinct and nonzero states).

b) Find a condition on $\{P_{0j}; j \geq 0\}$ and $\{P_{j0}; j \geq 0\}$ that is sufficient to ensure that all states are positive recurrent.

Solution: Positive recurrence was one of the given conditions in part (a) for demonstrating reversibility. However, (11.b) was derived without using the positive recurrence assumption. Thus, if $\sum_{i>0} \frac{P_{0i}}{P_{i0}} < \infty$ is satisfied, then (11.a) specifies all the π_i . The rest of the argument used above for the finite-state case then shows that the chain is both positive recurrent and reversible. Thus $\sum_{i>0} \frac{P_{0i}}{P_{i0}} < \infty$, in addition to the other conditions of part (a) except positive recurrence, is sufficient to demonstrate positive recurrence (as well as reversibility).

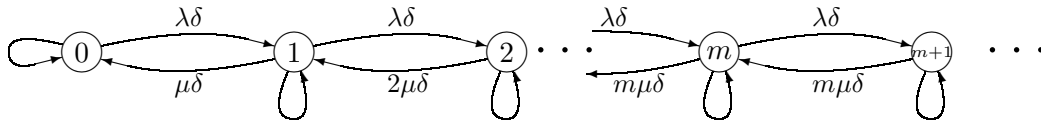
Exercise 6.12: a) Use the birth and death model described in figure 6.4 to find the steady-state probability mass function for the number of customers in the system (queue plus service facility) for the following queues:

- i) M/M/1 with arrival probability $\lambda\delta$, service completion probability $\mu\delta$.
- ii) M/M/m with arrival probability $\lambda\delta$, service completion probability $i\mu\delta$ for i servers busy, $1 \leq i \leq m$.
- iii) M/M/ ∞ with arrival probability $\lambda\delta$, service probability $i\mu\delta$ for i servers. Assume δ so small that $i\mu\delta < 1$ for all i of interest.

Assume the system is positive recurrent.

Solution: M/M/1: This is worked out in detail in section 6.6. From (6.45), $\pi_i = (1 - \rho)\rho^i$ where $\rho = \lambda/\mu < 1$.

M/M/m: The Markov chain for this is



This is a birth-death chain and the steady-state probabilities are given by (6.33) where $\rho_i = \lambda/((i+1)\mu)$ for $i < m$ and $\rho_i = \lambda/m\mu$ for $i \geq m$. Evaluating this,

$$\pi_i = \frac{\pi_0(\lambda/\mu)^i}{i!} \quad \text{for } i < m; \quad \pi_i = \frac{\pi_0(\lambda/\mu)^i}{m!(m^{i-m})} \quad \text{for } i \geq m,$$

where π_0 is given by

$$\pi_0^{-1} = 1 + \sum_{i=1}^{m-1} \frac{(\lambda/\mu)^i}{i!} + \frac{(m\rho_m)^m}{m!(1 - \rho_m)}. \quad (\text{A.130})$$

M/M/ ∞ : The assumption that δ is so small that $i\mu\delta < 1$ for all ‘ i of interest’ is rather strange, since we don’t know what is of interest. When we look at the solution to the M/M/1 and M/M/m sample-time queues above, however, we see that they do not depend on δ . It is necessary for $m\mu\delta \leq 1$ for the Markov chain to be defined, and $m\mu\delta \ll 1$ for it to be a reasonable approximation to the behavior of the continuous time queue, but the

results turn out to be the same as the continuous time queue in any case, as will be seen in Chapter 7. What was intended here was to look at the limit of the M/M/m queue in the limit $m \rightarrow \infty$. When we do this,

$$\pi_i = \frac{\pi_0(\lambda/\mu)^i}{i!} \quad \text{for all } i,$$

where

$$\pi_0^{-1} = 1 + \sum_{i=1}^{\infty} \frac{(\lambda/\mu)^i}{i!}.$$

Recognizing this as the power series of an exponential, $\pi_0 = e^{-\lambda/\mu}$.

b) For each of the queues above give necessary conditions (if any) for the states in the chain to be i) transient, ii) null recurrent, iii) positive recurrent.

Solution: The M/M/1 queue is transient if $\lambda/\mu > 1$, null recurrent if $\lambda/\mu = 1$, and positive recurrent if $\lambda/\mu < 1$ (see Section 6.6). In the same way, the M/M/m queue is transient if $\lambda/m\mu > 1$, null recurrent if $\lambda/m\mu = 1$ and positive recurrent if $\lambda/m\mu < 1$. The M/M/ ∞ queue is positive recurrent in all cases. As the arrival rate speeds up, the number of servers in use increases accordingly.

c) For each of the queues find:

L = (steady-state) mean number of customers in the system.

L^q = (steady-state) mean number of customers in the queue.

W = (steady-state) mean waiting time in the system.

W^q = (steady-state) mean waiting time in the queue.

Solution: The above parameters are related in a common way for the three types of queues. First, applying Little's law first to the system and then to the queue, we get

$$W = L/\lambda; \quad W^q = L^q/\lambda. \quad (\text{A.131})$$

Next, define L^v as the steady-state mean number of customers in service and W^v as the steady state waiting time per customer in service. We see that for each system, $W^v = 1/\mu$, since when a customer enters service the mean time to completion is $1/\mu$. From Little's law applied to service, $L^v = \lambda/\mu$. Now $L = L^q + L^v$ and $W = W^q + W^v$. Thus

$$L = L^q + \lambda/\mu; \quad W = W^q + 1/\mu. \quad (\text{A.132})$$

Thus for each of the above queue types, we need only compute one of these four quantities and the others are trivially determined. We assume positive recurrence in all cases.

For M/M/1, we compute

$$L = \sum_i i\pi_i = (1-\rho) \sum_{i=0}^{\infty} i\rho^i = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu-\lambda}.$$

For M/M/m, we compute L^q since queueing occurs only in states $i > m$.

$$\begin{aligned} L^q &= \sum_{i>m} (i-m)\pi_i = \sum_{i>m} \frac{(i-m)\pi_0(\lambda/\mu)^i}{m!(m^{i-m})} \\ &= \sum_{j>0} \frac{j\pi_0\rho_m^j(\lambda/\mu)^m}{m!} = \frac{\pi_0\rho_m(\lambda/\mu)^m}{(1-\rho_m)^2m!}, \end{aligned}$$

where π_0 is given in (A.130).

Finally, for M/M/ ∞ , there is no queueing, so $L^q = 0$.

Exercise 6.13: a) Given that an arrival occurs in the interval $(n\delta, (n+1)\delta)$ for the sampled-time M/M/1 model in figure 6.5, find the conditional PMF of the state of the system at time $n\delta$ (assume n arbitrarily large and assume positive recurrence).

Solution: The probability of an arrival in $(n\delta, (n+1)\delta)$ is the same for all states at time $n\delta$ and thus is independent of the state at $n\delta$. Thus

$$\Pr\{X_n=i \mid (X_{n+1}-X_n)=1\} = \Pr\{X_n=i\} = (1-\rho)\rho^i,$$

where the final equality assumes steady state.

b) For the same model, again in steady state but not conditioned on an arrival in $(n\delta, (n+1)\delta)$, find the probability $Q(i, j)$ for $(i \geq j > 0)$ that the system is in state i at $n\delta$ and that $i-j$ departures occur before the next arrival.

Solution: We do this directly for the sample-time Markov model and then briefly describe the same argument for the underlying continuous time process. The marginal probability $\Pr\{X_n=i\}$ in steady state is $(1-\rho)\rho^i$. The probability that the next state change is a departure (given that $X_n > 0$) is $\mu/(\mu+\lambda)$. Similarly, the probability that the next $i-j$ changes are departures (given that $X_n > j$) is $[\mu/(\mu+\lambda)]^{i-j}$. This is because each of these changes is independent of the others. Finally the probability that the next change after these $i-j$ departures is an arrival is $\lambda/(\lambda+\mu)$. Thus $Q(i, j)$, the probability that the state is i and exactly $i-j$ departures occur before the next arrival, is

$$Q(i, j) = (1-\rho)\rho^i \left[\frac{\mu}{\mu+\lambda} \right]^{i-j} \frac{\lambda}{\mu+\lambda}; \quad \text{for } i \geq j > 0.$$

c) Find the expected number of customers seen in the system by the first arrival after time $n\delta$. (Note: the purpose of this exercise is to make you cautious about the meaning of “the state seen by a random arrival”).

Solution: Note that $Q(i, j)$ is the probability that $X_n=i$ and that $X_A=j$ where A is the time at which the next arrival occurs in the transition from A to $A+\delta$. In other words, X_A is the state seen by the next arrival after time n . We have

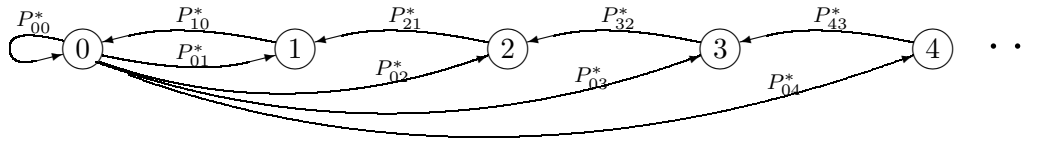
$$\begin{aligned} \Pr\{X_A=j\} &= \sum_{i=j}^{\infty} Q(i, j) = \sum_{i-j \geq 0} \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^j \left(\frac{\lambda}{\mu+\lambda}\right)^{i-j} \left(\frac{\lambda}{\mu+\lambda}\right) \\ &= \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^{j+1}. \end{aligned}$$

This says that the next arrival after time n does not see a steady-state distribution. What is happening is that if the state at time n is 0, then the next change must be an arrival. If the state is positive, then the next change might be a departure, decreasing the state so that when the next arrival occurs, the state tends to have a smaller value than it has in steady state.

The same argument applies for the continuous queue, where now we can view arrivals and departures as competing Poisson processes. There is a famous principle called PASTA, meaning Poisson arrivals see time averages. It is a useful principle, but the meaning of ‘see’ is very subtle. We have just shown that the first arrival after an arbitrary time $n\delta$ in steady state does not see a steady-state distribution.

Exercise 6.14: Find the backward transition probabilities for the Markov chain model of age in Figure 6.3. Draw the graph for the backward Markov chain, and interpret it as a model for residual life.

Solution: The backward transition probabilities are by definition given by $P_{ij}^* = \pi_j P_{ji} / \pi_i$. Since the steady-state probabilities π_i are all positive, $P_{ij}^* > 0$ if and only if $P_{ji} > 0$. Thus the graph for the backward chain is the same as the forward chain except that all the arrows are reversed, and, of course, the labels are changed accordingly. The graph shows that there is only one transition coming out of each positive numbered state in the backward chain. Thus $P_{i,i-1}^* = 1$ for all $i > 0$. This can be easily verified algebraically from 6.27. It is also seen that $P_{i0}^* = \Pr\{W = i + 1\}$.



We can interpret this Markov chain as representing residual life for an integer renewal process. The number of the state represents the residual life immediately before a transition, *i.e.*, state 0 means that a renewal will happen immediately, state 1 means that it will happen in one time unit, etc. When a renewal occurs, the next state will indicate the residual life at the end of that unit interval. Thus when a transition from 0 to i occurs, the corresponding renewal interval is $i + 1$.

For those confused by both comparing the chain above as the backward chain of age in Figure 6.3 and as the residual life chain of the same renewal process, consider a sample path for the renewal process in which an inter-renewal interval of 5 is followed by one of 2. The sample path for age in Figure 6.3 is then (0, 1, 2, 3, 4, 0, 1, 2, 0). The sample path for residual life for the same sample path of inter-renewals, for the interpretation above, is (0, 4, 3, 2, 1, 0, 2, 1, 0). On the other hand, the backward sample path for age is (0, 2, 1, 0, 4, 3, 2, 1, 0). In other words, the residual life sample path runs backward from age within each renewal interval, but forward between renewals.

Exercise 6.15: Consider the sample time approximation to the M/M/1 queue in Figure 6.5.

a) Give the steady-state probabilities for this chain (no explanations or calculations required—just the answer).

Solution: $\pi_i = (1 - \rho)\rho^i$ for $i \geq 0$ where $\rho = \lambda/\mu$.

In parts b) to g) do not use reversibility and do not use Burke's theorem. Let X_n be the state of the system at time $n\delta$ and let D_n be a random variable taking on the value 1 if a departure occurs between $n\delta$ and $(n+1)\delta$, and the value 0 if no departure occurs. Assume that the system is in steady state at time $n\delta$.

b) Find $\Pr\{X_n = i, D_n = j\}$ for $i \geq 0, j = 0, 1$.

Solution: The purpose of the following parts is to prove Burke's theorem by elementary means with no reference to reversibility. Given $X_n = 0$, a departure cannot occur in $(n\delta, (n+1)\delta)$ since there is no customer to depart. Thus $D_n = 0$. Given $X_n = i$ for any $i > 0$, $D_n = 1$ with probability $\mu\delta$. Thus

$$\begin{aligned} \Pr\{X_n=0, D_n=1\} &= 0; & \Pr\{X_n=0, D_n=0\} &= 1-\rho \\ \Pr\{X_n=i, D_n=1\} &= \mu\delta(1-\rho)\rho^i; & \Pr\{X_n=i, D_n=0\} &= (1-\mu\delta)(1-\rho)\rho^i; \quad \text{for } i > 0. \end{aligned}$$

c) Find $\Pr\{D_n = 1\}$.

Solution: Summing over the values of X_n above,

$$\Pr\{D_n = 1\} = \sum_{i \geq 1} \mu\delta(1-\rho)\rho^i = \mu\delta\rho = \lambda\delta.$$

This is initially counter-intuitive. We are used to thinking of departures conditional on the state of the system, where, so long as the state is positive, departures are independent and occur with probability $\mu\delta$. This says that in the marginal distribution for departures a departure occurs in an increment with probability $\lambda\delta$. Part (g) shows that the marginal distribution (averaged over state) for a sequence of departures is IID at rate $\lambda\delta$.

d) Find $\Pr\{X_n = i \mid D_n = 1\}$ for $i \geq 0$.

Solution: From (b) and (c), we see that $\Pr\{X_n = 0 \mid D_n = 1\} = 0$. For $i \geq 1$,

$$\Pr\{X_n = i \mid D_n = 1\} = \frac{\mu\delta(1-\rho)\rho^i}{\lambda\delta} = (1-\rho)\rho^{i-1}.$$

e) Find $\Pr\{X_{n+1} = i \mid D_n = 1\}$ and show that X_{n+1} is statistically independent of D_n . Hint: Use (d); also show that $\Pr\{X_{n+1} = i\} = \Pr\{X_{n+1} = i \mid D_n = 1\}$ for all $i \geq 0$ is sufficient to show independence.

Solution: Given $D_n = 1$, *i.e.*, a departure in $(n\delta, (n+1)\delta)$, we have $X_{n+1} = X_n - 1$.

$$\Pr\{X_{n+1} = i \mid D_n = 1\} = \Pr\{X_n = i+1 \mid D_n = 1\} = (1-\rho)\rho^i,$$

where we have used (d). From (a), this is $\Pr\{X_{n+1} = i\}$, so the event $\{X_{n+1} = i\}$ is independent of the event $\{D_n = 1\}$ and thus also independent of the complementary event $\{D_n = 0\}$. Thus the rv X_{n+1} is independent of the rv D_n .

f) Find $\Pr\{X_{n+1} = i, D_{n+1} = j \mid D_n\}$ and show that the pair of variables (X_{n+1}, D_{n+1}) is statistically independent of D_n .

Solution:

$$\begin{aligned} \Pr\{X_{n+1}=i, D_{n+1}=j \mid D_n\} &= \Pr\{D_{n+1}=j \mid X_{n+1}=i, D_n\} \Pr\{X_{n+1}=i \mid D_n\} \\ &= \Pr\{D_{n+1}=j \mid X_{n+1}=i\} \Pr\{X_{n+1}=i \mid D_n\} \quad (\text{A.133}) \\ &= \Pr\{D_{n+1}=j \mid X_{n+1}=i\} \Pr\{X_{n+1}=i\} \quad (\text{A.134}) \\ &= \Pr\{D_{n+1}=j, X_{n+1}=i\}. \end{aligned}$$

In (A.133), we used the fact $\{X_n; n \geq 0\}$ is a Markov chain, so that, given $X_{n=1}$, the rv's X_n and X_{n+2} are independent. Since D_{n+1} is a function of X_{n+1} and X_{n+2} and D_n is a function of X_{n+1} and X_n , it follows that D_{n+1} and D_n are independent conditional of X_{n+1} . In (A.134) we used the independence between X_{n+1} and D_n established in (e).

g) For each $k > 1$, find $\Pr\{X_{n+k} = i, D_{n+k} = j \mid D_{n+k-1}, D_{n+k-2}, \dots, D_n\}$ and show that the pair (X_{n+k}, D_{n+k}) is statistically independent of $(D_{n+k-1}, D_{n+k-2}, \dots, D_n)$. Hint: use induction on k ; as a substep, find $\Pr\{X_{n+k} = i \mid D_{n+k-1} = 1, D_{n+k-2}, \dots, D_n\}$ and show that X_{n+k} is independent of $D_{n+k-1}, D_{n+k-2}, \dots, D_n$.

Solution: We use induction on k , noting that (f) establishes the base of the induction at $k = 1$. We compress the notation by letting D_n^{n+k-1} represent $(D_n, D_{n+1}, \dots, D_{n+k-1})$. Then, using the same approach as in (f),

$$\begin{aligned} \Pr\{X_{n+k}, D_{n+k} \mid D_n^{n+k-1}\} &= \Pr\{D_{n+k} \mid X_{n+k}, D_n^{n+k-1}\} \Pr\{X_{n+k} \mid D_n^{n+k-1}\} \\ &= \Pr\{D_{n+k} \mid X_{n+k}\} \Pr\{X_{n+k} \mid D_n^{n+k-1}\} \quad (\text{A.135}) \end{aligned}$$

$$\begin{aligned} &= \Pr\{D_{n+k} \mid X_{n+k}\} \Pr\{X_{n+k}\} \quad (\text{A.136}) \\ &= \Pr\{D_{n+k}, X_{n+k}\}. \end{aligned}$$

The argument in (A.135) is almost the same as that in (A.133). The only difference is that here, D_n^{n+k-1} is a function of $X_n, X_{n+1}, \dots, X_{n+k}$, which, given X_{n+k} , is independent of X_{n+k+1} and thus D_{n+k} . In (A.136), we have claimed that $\Pr\{X_{n+k} \mid D_n^{n+k-1}\} = \Pr\{X_{n+k}\}$, and that is the step where we must use the hint. We look at this first for the case $D_{n+k-1} = 1$ and use the argument in (e).

$$\Pr\{X_{n+k}=i \mid D_{n+k-1}=1, D_n^{n+k-2}\} = \Pr\{X_{n+k-1}=i+1 \mid D_{n+k-1}=1, D_n^{n+k-2}\}.$$

From the inductive hypothesis, this is independent of D_n^{n+k-2} , and from (e), it is equal to $\Pr\{X_{n+k} = i\}$.

h) What do your results mean relative to Burke's theorem.

Solution: We have in fact proven Burke's theorem for sampled time M/M/1 queues. Since X_{n+k}, D_{n+k} is independent of the prior sequence of departures, D_{n+k} is also independent of prior departures. Thus the departure process running backward in time is a Bernoulli process of rate $\lambda\delta$ and the state is independent of previous departures. The argument was not particularly difficult, except for being unmotivated and somewhat tricky. The use of reversibility provides a much more elegant and insightful way of proving this result.

Exercise 6.16: Let $\{X_n, n \geq 1\}$ denote an irreducible recurrent Markov chain having a countable state space. Now consider a new stochastic process $\{Y_n, n \geq 0\}$ that only accepts values of the Markov chain that are between 0 and some integer m . For instance, if $m = 3$ and $X_1 = 1, X_2 = 3, X_3 = 5, X_4 = 6, X_5 = 2$, then $Y_1 = 1, Y_2 = 3, Y_3 = 2$.

a) Is $\{Y_n, n \geq 0\}$ a Markov chain? Explain briefly.

Solution: $\{Y_n; n \geq 1\}$ is a Markov chain. The transition probability, say Q_{ij} from i to j in this modified chain is P_{ij} (the probability of a direct transition in the original chain), plus the aggregate probability of all distinct walks in the original chain starting in i , ending

in j , and remaining in states greater than m otherwise. The probability of each such walk, conditional on the initial state i , is independent of states previous to that initial state, providing the Markov property.

b) Let p_j denote the proportion of time that $\{X_n, n \geq 1\}$ is in state j . If $p_j > 0$ for all j , what proportion of time is $\{Y_n, n \geq 0\}$ in each of the states $0, 1, \dots, m$?

Solution: The assumption that $p_j > 0$ for all j implies that $\{X_n; n \geq 1\}$ is positive recurrent. Thus, if the X chain starts in any given state i , the fraction of time spent by the X process in state j is $\lim_{t \rightarrow \infty} (1/t)N_{ij}(t)$ where the limit exists and is equal to p_j WP1.

Time in the Y chain is different from time in the X chain. The clock ‘ticks’ in the Y chain only when the state of the X chain is between 0 and m . Thus the ratio of time on the Y clock to time on the X clock as X time goes to infinity is $\sum_{k=1}^m p_k$ WP1.

Thus if we let $M_{ij}(\tau)$ be the number of visits to state j (starting in state i) up to time τ on the Y clock, we see that time τ on the Y clock, as a function of sample path and time t on the X clock is given by $\tau(t) = \sum_{k=0}^m N_{ik}(t)$. Thus

$$(1/\tau)M_{ij}(\tau) = \frac{N_{ij}(t)}{\sum_{k=0}^m N_{ik}(t)}.$$

Taking the limit as τ (and thus t) approaches ∞ , we get

$$\lim_{\tau \rightarrow \infty} (1/\tau)M_{ij}(\tau) = \lim_{t \rightarrow \infty} \frac{N_{ij}(t)}{t} \frac{t}{\sum_{k=0}^m N_{ik}(t)} = \frac{p_j}{\sum_{k=0}^m p_k}.$$

c) Suppose $\{X_n\}$ is null recurrent and let $p_i(m), i = 0, 1, \dots, m$ denote the long-run proportions for $\{Y_n, n \geq 0\}$. Show that $p_j(m) = p_i(m)E[\text{time the } X \text{ process spends in } j \text{ between returns to } i]$, for $j \neq i$.

Solution: We interpret $E[\text{time the } X \text{ process spends in } j \text{ between returns to } i]$ as expected visits to j between visits to i . Successive returns to i form a renewal process (or delayed renewal process, depending on the starting state) in the X process (albeit with an infinite expected renewal interval). For each sample path, the number of visits to j between visits to i is the same in the Y process as the X process. Also, the Y process is a finite-state Markov chain, and thus $p_i(m) > 0$ for $0 \leq i \leq m$. The result then follows by viewing visits to i as unit rewards in the renewal process of visits to j in the Y process.

Exercise 6.17: Verify that (6.58) is satisfied by the hypothesized solution to p in (6.62). Also show that the equations involving the idle state ϕ are satisfied.

Solution: Let us first review where these equations come from. We started by making some plausible guesses about the backward transition probabilities. If these guesses are correct, then (6.56)-(6.59) must be satisfied, by the definition of backward probabilities. Then using the guess in (6.57) about departures in the forward direction, we derived a general formula for arbitrary states in (6.62), repeated here as

$$\pi_s = \left(\frac{\lambda}{1-\lambda} \right)^m \left(\prod_{j=1}^m \bar{F}(z_j) \right) \pi_\phi. \quad (\text{A.137})$$

where the state s is $s = (m, z_1, \dots, z_m)$. We have also normalized time so that $\delta = 1$ is the unit service time. This formula satisfies (6.57) since it was derived from (6.57). As shown in the text, it also satisfies (6.56), *i.e.*, the relationship between rotation transitions assuming that the guesses were correct. If the formula also satisfies (6.58) (for forward arrival transitions) and if it satisfies the transitions from the empty state, then, by the guessing theorem, Theorem 6.5.4, the guess must be correct. We then use (A.137) to calculate $\pi_s P_{s,a(s)}$ and $\pi_{a(s)} P_{a(s),s}^*$ and show they are the same.

$$\pi_s P_{s,a(s)} = \lambda(1 - f(1))\pi_s = \lambda(1 - f(1)) \left(\frac{\lambda}{1 - \lambda} \right)^m \left(\prod_{j=1}^m \bar{F}(z_j) \right) \pi_\phi.$$

Since $a(s) = (m+1, z_1, \dots, z_m, 1)$ and $P_{a(s),s}^* = 1 - \lambda$,

$$\begin{aligned} \pi_{a(s)} P_{a(s),s}^* &= (1 - \lambda) \left(\frac{\lambda}{1 - \lambda} \right)^{m+1} \left(\prod_{j=1}^m \bar{F}(z_j) \right) \bar{F}(1) \pi_\phi \\ &= \lambda \left(\frac{\lambda}{1 - \lambda} \right)^m \left(\prod_{j=1}^m \bar{F}(z_j) \right) \bar{F}(1) \pi_\phi. \end{aligned}$$

Noting that $1 - f(1) = \bar{F}(1)$, we see that these two expressions are the same and (6.58) is satisfied.

There are two kinds of transitions from the empty state, first self transitions from ϕ to ϕ and second transitions from ϕ to $(1, 1)$. We clearly have $P_{\phi,\phi} = P_{\phi,\phi}^*$ so $\pi_\phi P_{\phi,\phi} = \pi_\phi P_{\phi,\phi}^*$. Finally,

$$\pi_\phi P_{\phi,(1,1)} = \pi_\phi \lambda \bar{F}(1); \quad \pi_{1,1} P_{(1,1),\phi}^* = \frac{\lambda}{1 - \lambda} \bar{F}(1) \pi_\phi (1 - \lambda),$$

so $\pi_\phi P_{\phi,(1,1)} = \pi_{1,1} P_{(1,1),\phi}^*$.

Exercise 6.18: Replace the state $\mathbf{m} = (m, z_1, \dots, z_m)$ in Section 6.8 with an expanded state $\mathbf{m} = (m, z_1, w_1, z_2, w_2, \dots, z_m, w_m)$ where m and $\{z_i; 1 \leq i \leq m\}$ are as before and $w_1, w_2, \dots, w_m$ are the original service requirements of the m customers.

a) Hypothesizing the same backward round-robin system as hypothesized in Section 6.8, find the backward transition probabilities and give the corresponding equations to (6.56-6.59) for the expanded state description.

Solution: It might be helpful to do Exercise 6.17 before this to increase familiarity with this type of argument. We normalize the time reference to make $\delta = 1$ and use s to refer to the expanded state here. If an arrival with $m > 1$ required units of service occurs in state $s = (m, w_1, z_1, \dots, w_m, z_m)$, then the next state is $a(s, m) = (m+1, w_1, z_1, \dots, w_m, z_m, w, w-1)$. The probability of this transition is $P_{s,a(s(w))} = \lambda f(w)$.

The hypothesis we are adopting for the backward system is the same hypothesis as in Section 6.8, namely the backward system is also round robin with Bernoulli arrivals at rate

λ per time unit. The state is denoted using the same notation as the forward system, but the customer at the right side of the queue (customer m) is served and then rotates to the left side and z_m represents the remaining service rather than the service already applied. Thus an arrival in the forward system corresponds to a departure in the backward system. A departure occurs only if the service still required by the customer at the right end of the queue is one less than that of the original requirement and if there is no arrival in the backward system.

The tricky part is to look at the state $a(s, w)$, $w > 1 = (m+1, w_1, z_1, \dots, w_m, a_m, w, w-1)$ and realize that when the backward system ‘sees’ this state, it sees only the state and does not ‘know’ that the forward system just had an arrival. The state $a(s, w)$; $w > 1$ satisfies the backward requirement for departure, *i.e.*, the rightmost element satisfies $z = w-1$. Also, with probability $1 - \lambda$ (from the backward systems view), no backward arrival occurs at that increment. Thus $P_{a(s, w), s}^* = 1 - \lambda$. The analogy to (6.56), assuming $w > 1$, with this new way of representing states is then

$$P_{s, a(s, w)} = \lambda f(w); \quad P_{a(s, w), s}^* = 1 - \lambda \quad \lambda f(w) \pi_s = (1 - \lambda) \pi_{a(s, w)}.$$

Using the same type of analysis for a departure in the forward system, a departure can only occur if $s = (m, w_1, w_1-1, w_2, z_2, \dots, w_m, z_m)$. If in addition no arrival occurs, a departure occurs and the new state is $d(s) = (m-1, w_2, z_2, \dots, w_m, z_m)$. In the backward system, this corresponds to an arrival with service requirement w_m ; this causes a backward transition from $d(s)$ to s . The probability of these transitions and the resulting hypothesized balance equation is given by the analogy to 6.57,

$$P_{s, d(s)} = (1 - \lambda); \quad P_{d(s), s}^* = \lambda f(w_1); \quad (1 - \lambda) \pi_s = \lambda f(w_1) \pi_{d(s)},$$

where we remember that this applies only for $s = (m, w_1, w_1-1, w_2, z_2, \dots, w_m, z_m)$, *i.e.*, for $m \geq 1$ where $z_1 = w_1-1$.

Finally, rotation occurs only for s such that $m \geq 1$ and $z_1 < w_1 - 1$, and it then occurs with probability $1 - \lambda$. The analogy to (6.58) is

$$P_{s, r(s)} = 1 - \lambda; \quad P_{r(s), s}^* = 1 - \lambda; \quad \pi_s = \pi_{r(s)}.$$

Finally a transition from s to s occurs if there is an arrival of requirement 1. The corresponding backward transition is also an arrival of requirement 1. The analogy to 6.59 is

$$P_{s, s} = \lambda f(1); \quad P_{s, s}^* = \lambda f(1); \quad \pi_s = \pi_s.$$

b) Solve the resulting equations to show that

$$\pi_{\mathbf{m}} = \pi_{\phi} \left(\frac{\lambda \delta}{1 - \lambda \delta} \right)^m \prod_{j=1}^m f(w_j). \quad (\text{A.138})$$

Solution: For any state s , consider the string of transitions with no arrivals until the system becomes empty. The product of these transition probabilities consist of 1’s for each rotation and $\lambda f(w_i)/(1 - \lambda)$ for each departure, leading to (A.138). There is a slight mathematical question here in verifying that each state can be reduced to ϕ in this way. Note that after

each arrival, the new (w, z) pair satisfies $z = 1, w > 1$ and each rotation increases one of the z 's, but only up to $w-1$ for that pair. Thus each feasible state contains only pairs for which $z_i < w_i$. The z_i 's are increasing with time and the customer departs when $z_1 = w_1 - 1$. It is easy to check that (A.138) is consistent with each of the balance equations above, and therefore, by Theorem 6.5.4, both the backward round robin hypothesis and (A.138) are verified.

c) Show that the probability that there are m customers in the system, and that those customers have original service requirements given by $w_1, \dots, w_m$, is

$$\Pr\{m, w_1, \dots, w_m\} = \pi_\phi \left(\frac{\lambda\delta}{1 - \lambda\delta} \right)^m \prod_{j=1}^m (w_j - 1)f(w_j). \quad (\text{A.139})$$

Solution: We saw in (b) that all feasible states satisfy $z_i < w_i$ for each $i, 1 \leq i \leq m$. Thus if all states satisfying $1 \leq z_i \leq w_i - 1$ are feasible, then there are $\prod_{i=1}^{m-1} (w_i - 1)$ different states with the same values of $m, w_1, \dots, w_m$. To show that these states are all feasible, start with an arbitrary hypothesized state $s = m, w_1, z_1, \dots, w_m, z_m$ with $1 \leq z_i < w_i, 1 \leq i \leq m$. The argument in (c) finds the probability of that state (if it is feasible), and the corresponding backward sequence then shows that that state is feasible. Adding the probability of each of these $\prod_{i=1}^{m-1} (w_i - 1)$ and using (A.138) gives us (A.139).

d) Given that a customer has original service requirement w , find the expected time that the customer spends in the system.

Solution: The probability that m customers are in the system in steady state is given in (6.66) as $P_m = (1 - \rho)\rho^m$ where $\rho = [\lambda/(1 - \lambda)](\overline{W} - 1)$ (recall that the notation for states in this exercise does not change the underlying system). Thus the probability that the m customers in the system have original requirements $w_1, \dots, w_m$ given m customers in the system is $\Pr\{w_1, \dots, w_m \mid m\} = \prod_{i=1}^m (w_i - 1)f(w_i)[\overline{W} - 1]^m$. Thus the requirements of the m customers are independent given m . Let $N(w)$ be the number of customers in the system in steady state that have requirement w . Conditional on m in the system, $E[N(w) \mid m] = m(w - 1)f(w)/[\overline{W} - 1]$. Averaging over m , then

$$E[N(w)] = \frac{\rho}{1 - \rho} \frac{1}{[\overline{W} - 1]}.$$

Finally using Little's law on the customers requiring w units of service, the expected delay of those customers is $(w - 1)/[1 - \lambda(\overline{W} - 1)]$.

Note that as the increment of service becomes small, $\overline{W}$ and typical values of w become large since we have normalized time to the increment of service. The fact that w and $\overline{W}$ are reduced by 1 here is primarily an anomaly of the state labeling process. For example, if all requirements were unit requirements, then the system would always be in state 0. In the same way, if there is only one customer in the system with requirement w , the state includes that customer only for $w - 1$ time units. There is nothing wrong in this; the state accurately represents the system at each integer time at the end of the service during that increment. However, quantities like delay must be interpreted with some care.

A.7 Solutions for Chapter 7

Exercise 7.1: Consider an M/M/1 queue as represented in Figure 7.4. Assume throughout that $X_0 = i$ where $i > 0$. The purpose of this exercise is to understand the relationship between the holding interval until the next state transition and the interval until the next arrival to the M/M/1 queue. Your explanations in the following parts can and should be very brief.

a) Explain why the expected holding interval $E[U_1|X_0 = i]$ until the next state transition is $1/(\lambda + \mu)$.

Solution: By definition of a countable state Markov process, U_n , conditional on X_{n-1} is an exponential. For the M/M/1 queue, the rate of the exponential out of state $i > 0$ is $\lambda + \mu$, and thus the expected interval U_1 is $1/(\lambda + \mu)$.

b) Explain why the expected holding interval U_1 , conditional on $X_0 = i$ and $X_1 = i + 1$, is

$$E[U_1|X_0 = i, X_1 = i + 1] = 1/(\lambda + \mu).$$

Show that $E[U_1|X_0 = i, X_1 = i - 1]$ is the same.

Solution: The holding interval U_1 , again by definition of a countable state Markov process, conditional on $X_0 = i > 0$, is independent of the next state X_1 (see Figure 7.1). Thus

$$E[U_1|X_0 = i, X_1 = i + 1] = E[U_1|X_0 = i] = \frac{1}{\lambda + \mu}.$$

This can be visualized by viewing the arrival process and the departure process (while the queue is busy) as independent Poisson processes of rate λ and μ respectively. From Section 2.3, on splitting and combining of Poisson processes, U_1 (the time until the first occurrence in the combined Poisson process) is independent of which split process (arrival or departure) that first occurrence comes from.

Since this result is quite unintuitive for most people, we explain it in yet another way. Quantizing time into very small increments of size δ , the probability of a customer arrival in each increment is $\lambda\delta$ and the probability of a customer departure (assuming the server is busy) is $\mu\delta$. This is the same for every increment and is independent of previous increments (so long as the server is busy). Thus X_1 (which is $X_0 + 1$ for an arrival and $X_0 - 1$ for a departure) is independent of the time of that first occurrence. Thus given $X_0 > 0$, the time of the next occurrence (U_1) is independent of X_1 .

c) Let V be the time of the first arrival after time 0 (this may occur either before or after the time W of the first departure.) Show that

$$E[V|X_0 = i, X_1 = i + 1] = \frac{1}{\lambda + \mu}. \quad (\text{A.140})$$

$$E[V|X_0 = i, X_1 = i - 1] = \frac{1}{\lambda + \mu} + \frac{1}{\lambda}. \quad (\text{A.141})$$

Hint: In the second equation, use the memorylessness of the exponential rv and the fact that V under this condition is the time to the first departure plus the remaining time to an arrival.

Solution: Given $X_0 = i > 0$ and $X_1 = i + 1$, the first transition is an arrival, and from (b), the interval until that arrival is $1/(\lambda + \mu)$, verifying (A.140). Given $X_0 = i > 0$ and $X_1 = i - 1$, the first transition is a departure, and from (b) the conditional expected time until this first transition is $1/(\lambda + \mu)$. The expected time after this first transition to an

arrival is simply the expected interval until an arrival, starting at that first transition, verifying (A.141).

d) Use your solution to (c) plus the probability of moving up or down in the Markov chain to show that $E[V] = 1/\lambda$. (Note: you already know that $E[V] = 1/\lambda$. The purpose here is to show that your solution to (c) is consistent with that fact.)

Solution: This is a sanity check on (A.140) and (A.141). For $i > 0$,

$$\Pr\{X_1=i+1 \mid X_0=i\} = \frac{\lambda}{\lambda+\mu}; \quad \Pr\{X_1=i-1 \mid X_0=i\} = \frac{\mu}{\lambda+\mu}.$$

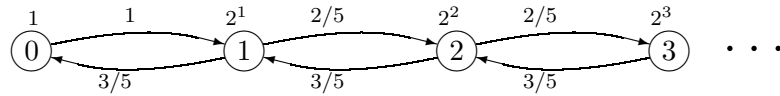
Using this with (A.140) and (A.141),

$$\begin{aligned} E[V \mid X_0=i] &= \frac{\lambda}{\lambda+\mu} \cdot \frac{1}{\lambda+\mu} + \frac{\mu}{\lambda+\mu} \left[\frac{1}{\lambda+\mu} + \frac{1}{\lambda} \right] \\ &= \frac{1}{\lambda+\mu} + \frac{\mu}{\lambda(\lambda+\mu)} = \frac{1}{\lambda}. \end{aligned}$$

Exercise 7.2: Consider a Markov process for which the embedded Markov chain is a birth-death chain with transition probabilities $P_{i,i+1} = 2/5$ for all $i \geq 1$, $P_{i,i-1} = 3/5$ for all $i \geq 1$, $P_{01} = 1$, and $P_{ij} = 0$ otherwise.

a) Find the steady-state probabilities $\{\pi_i; i \geq 0\}$ for the embedded chain.

Solution: The embedded chain is given below. Note that it is the same embedded chain as in Example 7.2.8 and Exercise 7.3, but the process behavior will be very different when the holding times are brought into consideration.



The embedded chain is a birth/death chain, and thus the steady-state probabilities are related by $(2/5)\pi_i = (3/5)\pi_{i+1}$ for $i \geq 1$. Thus, as we have done many times, $\pi_i = \pi_1(2/3)^{i-1}$. The transitions between state 0 and 1 are different, and $\pi_1 = (5/3)\pi_0$. Thus for $i \geq 1$, we have

$$\pi_i = \frac{5}{3} \cdot \left(\frac{2}{3}\right)^{i-1} \pi_0 = \frac{5}{2} \cdot \left(\frac{2}{3}\right)^i \pi_0.$$

Setting the sum of π_0 plus $\sum_{i \geq 1} \pi_i$ to 1, we get $\pi_0 = 1/6$. Thus

$$\pi_i = (5/12)(2/3)^i \quad \text{for } i \geq 1; \quad \pi_0 = \frac{1}{6}. \quad (\text{A.142})$$

b) Assume that the transition rate ν_i out of state i , for $i \geq 0$, is given by $\nu_i = 2^i$. Find the transition rates $\{q_{ij}\}$ between states and find the steady-state probabilities $\{p_i\}$ for the Markov process. Explain heuristically why $\pi_i \neq p_i$.

Solution: The transition rate q_{ij} is given by $\nu_i P_{ij}$. Thus for $i > 0$,

$$q_{i,i+1} = \frac{2}{5} \cdot 2^i; \quad q_{i,i-1} = \frac{3}{5} \cdot 2^i; \quad q_{01} = 1.$$

The simplest way to evaluate p_i is by (7.7), i.e., $p_i = \pi_i / [\nu_i \sum_{j \geq 0} \pi_j / \nu_j]$.

$$\sum_{j \geq 0} \frac{\pi_j}{2^j} = \frac{1}{6} + \frac{5}{12} \sum_{i > 0} \left(\frac{1}{3}\right)^i = \frac{3}{8}.$$

Thus $p_0 = 4/9$ and, for $i > 0$, $p_i = (10/9)3^{-i}$. Note that π_i is the steady-state fraction of *transitions* going into state i and p_i is the steady-state fraction of *time* in state i . Since $\nu_i^{-1} = 2^{-i}$ is the expected time-interval in state i per transition into state i , we would think (correctly) that p_i would approach 0 much more rapidly as $i \rightarrow \infty$ than π_i does.

c) Explain why there is no sampled-time approximation for this process. Then truncate the embedded chain to states 0 to m and find the steady-state probabilities for the sampled-time approximation to the truncated process.

Solution: In order for a sampled-time approximation to exist, one must be able to choose the time-increment δ small enough so that the conditional probability of a transition out of the state in that increment is small. Since the transition rates are unbounded, this cannot be done here. If we truncate the chain to states 0 to m , then the ν_i are bounded by 2^m , so choosing a time-increment less than 2^{-m} will allow a sampled-time approximation to exist.

There are two sensible ways to do this truncation. In both, $q_{m,m+1}$ must be changed to 0. but then $q_{m,m-1}$ can either be kept the same (thus reducing ν_m) or ν_m can be kept the same (thus increasing $q_{m,m-1}$). We keep $q_{m,m-1}$ the same since it simplifies the answer slightly. Let $\{p_i^{(m)}; 0 \leq i \leq m\}$ be the steady-state process PMF in the truncated chain. Since these truncated chains (as well as the untruncated chain) are birth-death chains, we can use (7.38), $p_i^{(m)} = p_0^{(m)} \prod_{j < i} \rho_j$, where $\rho_j = q_{j,j+1}/q_{j+1,j}$. Thus, $\rho_0 = 5/6$ and $\rho_i = 1/3$ for $1 < i < m$. Thus

$$p_i^{(m)} = p_0^{(m)} \frac{5}{6} \left(\frac{1}{3}\right)^{i-1} \quad \text{for } i \leq m.$$

Since $p_0^{(m)} + p_1^{(m)} + \dots + p_m^{(m)} = 1$, we can solve for $p_0^{(m)}$ from

$$1 = p_0^{(m)} \left[1 + \frac{5}{6} (1 + 3^{-1} + 3^{-2} + \dots + 3^{-m+1}) \right] = p_0^{(m)} \left[1 + \frac{5}{4} (1 - 3^{-m}) \right].$$

Combining these equations.

$$p_0^{(m)} = \frac{4}{9 - 5(3^{-m})}; \quad p_i^{(m)} = \frac{10(3^{-i})}{9 - 5(3^{-m})} \quad \text{for } 1 \leq i \leq m. \quad (\text{A.143})$$

For $\delta < 2^{-m}$, these are also the sampled-time ‘approximation’ to the truncated chain.

d) Show that as $m \rightarrow \infty$, the steady-state probabilities for the sequence of sampled-time approximations approach the probabilities p_i in (b).

Solution: For each i , we see from (A.143) that $\lim_{m \rightarrow \infty} p_i^{(m)} = p_i$. It is important to recognize that the convergence is not uniform in i since (A.143) is defined only for $i \leq m$. Similarly, for each i , the approximation is close only when both $m \geq i$ and δ is small relative to 3^{-m} .

Exercise 7.3: Consider a Markov process for which the embedded Markov chain is a birth-death chain with transition probabilities $P_{i,i+1} = 2/5$ for all $i \geq 1$, $P_{i,i-1} = 3/5$ for all $i \geq 1$, $P_{01} = 1$, and $P_{ij} = 0$ otherwise.

a) Find the steady-state probabilities $\{\pi_i; i \geq 0\}$ for the embedded chain.

Solution: This is the same embedded chain as in Exercise 7.2. The steady-state embedded probabilities were calculated there in (A.142) to be

$$\pi_i = (5/12)(2/3)^i \quad \text{for } i \geq 1; \quad \pi_0 = \frac{1}{6}.$$

b) Assume that the transition rate out of state i , for $i \geq 0$, is given by $\nu_i = 2^{-i}$. Find the transition rates $\{q_{ij}\}$ between states and show that there is no probability vector solution $\{p_i; i \geq 0\}$ to (7.23).

Solution: This particular Markov process was discussed in Example 7.2.8. The transition rates q_{ij} are given by $q_{01} = 1$ and, for all $i > 0$,

$$q_{i,i+1} = P_{i,i+1}\nu_i = \frac{2}{5} \cdot 2^{-i}; \quad P_{i,i-1}\nu_i = \frac{3}{5} \cdot 2^{-i}.$$

The other transition rates are all 0. We are to show that there is no probability vector $\{p_i : i \geq 0\}$ solution to (7.23), *i.e.*, no solution to

$$p_j\nu_j = \sum_i p_i q_{ij} \quad \text{for all } j \geq 0 \quad \text{with } \sum_i p_i = 1. \quad (7.23')$$

Let $\alpha_j = p_j\nu_j$, so that any hypothesized solution to (7.23') becomes $\alpha_j = \sum_i \alpha_i P_{ij}$ for all j and $\sum_i \alpha_i/\nu_i = 1$.

Since the embedded chain is positive recurrent, there is a unique solution to $\{\pi_i; i \geq 0\}$ such that $\pi_j = \sum_i \pi_i P_{ij}$ for all j and $\sum_i \pi_i = 1$. Thus there must be some $0 < \beta < \infty$ such that any hypothesized solution to (7.23') satisfies $\alpha_j = \beta\pi_j$ for all $j \geq 0$. Thus $\sum_i \pi_j/\nu_j < \infty$ for this hypothesized solution.

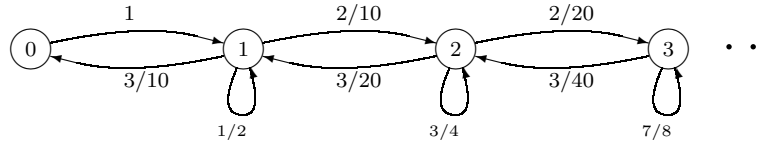
Now note that $\pi_j/\nu_j = (5/12)(4/3)^j$. Since this increases exponentially with j , we must have $\sum_j \pi_j/\nu_j = \infty$. Thus there cannot be a probability vector solution to (7.23). We discuss this further after (c) and (d).

c) Argue that the expected time between visits to any given state i is infinite. Find the expected number of transitions between visits to any given state i . Argue that, starting from any state i , an eventual return to state i occurs with probability 1.

Solution: From Theorem 7.2.6 (which applies since the embedded chain is positive recurrent), the expected time between returns to state i is $\overline{W}(i) = (1/\pi_i) \sum_k \pi_k/\nu_k$. Since $\sum_k \pi_k/\nu_k$ is infinite and π_i is positive for all i , $\overline{W}(i) = \infty$. As explained in Section 7.2, $W(i)$ is a rv (*i.e.*, non-defective), and is thus finite with probability 1 (guaranteeing an eventual return with probability 1). Under the circumstances here, however, it is a rv with infinite expectation for each i . The expected number of transitions between returns to state i is finite (since the embedded chain is positive recurrent), but the increasingly long intervals spent in high numbered states causes the expected renewal time from i to i to be infinite.

d) Consider the sampled-time approximation of this process with $\delta = 1$. Draw the graph of the resulting Markov chain and argue why it must be null recurrent.

Solution:



Note that the approximation is not very good at the low-numbered states, but it gets better and better for higher numbered states since ν_i becomes increasingly negligible compared to $\delta = 1$ and thus there is negligible probability of more than one arrival or departure in a unit increment. Thus it is intuitively convincing that the mean interval between returns to any given state i must be infinite, but that a return must happen eventually with probability 1. This is a convincing argument why the chain is null recurrent.

At the same time, the probability of an up-transition from i to $i + 1$ is $4/3$ of the probability of a down-transition from $i + 1$ to i , making the chain look like the classical example of a transient chain in Figure 6.2. Thus there is a need for a more convincing argument.

One could truncate both the process and the sample-time chain to states 0 to m and then go to the limit as $m \rightarrow \infty$, but this would be very tedious.

We next outline a procedure that is mathematically rigorous to show that the chain above is null recurrent; the procedure can also be useful in other circumstances. Each sample sequence of states for the chain above consists of runs of the same state separated on each side by a state either one larger or one smaller. Consider creating a new Markov chain by replacing each such run of repeated states with a single copy of that state. It is easy to see that this new chain is Markov. Also since there can be no repetition of a state, the new transition probabilities, say $Q_{i,i+1}$ and $Q_{i,i-1}$ for $i > 0$ satisfy $Q_{i,i+1} = P_{i,i+1}/(P_{i,i+1} + P_{i,i-1}) = 2/5$ and $Q_{i,i-1} = 3/5$. Thus this new chain is the same as the embedded chain of the original process, which is already known to be positive recurrent.

At this point, we can repeat the argument in Section 7.2.2, merely replacing the exponentially distributed reward interval in Figure 7.7 with a geometrically distributed interval. The expected first passage time from i to i (in the approximation chain) is then infinite as before, and the return occurs eventually with probability 1. Thus the approximation chain is null recurrent.

The nice thing about the above procedure is that it can be applied to any birth death chain with self transitions.

Exercise 7.4: Consider the Markov process for the M/M/1 queue, as given in Figure 7.4.

a) Find the steady-state process probabilities (as a function of $\rho = \lambda/\mu$) from (7.15) and also as the solution to (7.23). Verify that the two solutions are the same.

Solution: For the M/M/1 queue, the embedded steady-state probabilities are calculated in (7.4) to be $\pi_0 = (1 - \rho)/2$ and $\pi_i = (1/2)(1 - \rho^2)\rho^{i-1}$ for $i > 0$ where $\rho = \lambda/\mu$ and $\rho < 1$ is assumed. Since the holding parameters $\nu_0 = \lambda$ and $\nu_i = \lambda + \mu$ for $i > 0$, it is easy but somewhat tedious to calculate p_j from (7.7), i.e., from $p_j = (\pi_j/\nu_j)/(\sum_k \pi_k/\nu_k)$. We find that $\sum_k \pi_k/\nu_k = 1/2\lambda$, and finally $p_j = (1 - \rho)\rho^j$. This is clearly not the simple approach for the M/M/1 queue. Substituting this into (7.23), we easily see that this solution also

satisfies (7.23). It also satisfies (7.37), which of course is the easy way to find the p_j .

b) For the remaining parts of the exercise, assume that $\rho = 0.01$, thus ensuring (for aiding intuition) that states 0 and 1 are much more probable than the other states. Assume that the process has been running for a very long time and is in steady state. Explain in your own words the difference between π_1 (the steady-state probability of state 1 in the embedded chain) and p_1 (the steady-state probability that the process is in state 1). More explicitly, what experiments could you perform (repeatedly) on the process to measure π_1 and p_1 .

Solution: π_1 is the long term fraction of transitions going into state 1, whereas p_1 is the long term fraction of time in state 1. With $\rho = 0.01$, the process is in state zero 99% of the time and state one .99% $\approx 1\%$ of the time. Since for every transition into state 0, the following transition must go to state 1, it is not surprising that $\pi_1 = (1/2)(1 - \rho)^2 \approx 1/2$.

One could measure the π_i 's by tabulating the fraction of transitions going into each state over a very long period. One could measure the p_i 's by tabulating the fraction of time in each state over a very long period. One could also measure the p_i 's by averaging the number of samples in that state over a long interval, using either evenly spaced or randomly spaced samples. One could not measure either the π_i 's or p_i 's, however, by sampling the next transition after an evenly spaced or randomly spaced sample time. With $\rho = 0.01$, the process is in state zero 99% of the time, and the next transition would then be state 1 with probability slightly more than 0.99.

The p_i 's are called the steady-state process probabilities, and this terminology makes sense because, from (7.11), $\lim_{t \rightarrow \infty} \Pr\{X(t) = i\} = p_i$. Thus after the process has been running a long time t , p_i gives the probability that the process is in state i at time t . The π_i can only be considered as steady-state probabilities within the sequence of state transitions in the embedded Markov chain.

c) Now suppose you want to start the process in steady state. Show that it is impossible to choose initial probabilities so that both the process and the embedded chain start in steady state. Which version of steady state is closest to your intuitive view? (There is no correct answer here, but it is important to realize that the notion of steady state is not quite as simple as you might imagine).

Solution: The embedded chain is in steady state when the state distribution is given by $\{\pi_i; i \geq 0\}$ and the process is in steady state when the state distribution is $\{p_i; i \geq 0\}$. We have seen that these distributions are different. Most people would view the process to be started in steady state if $\Pr\{X(0) = i\} = p_i$, since then $\Pr\{X(t) = i\} = p_i$ for all t . This is why the p_i are denoted as steady-state process probabilities.

d) Let $M(t)$ be the number of transitions (counting both arrivals and departures) that take place by time t in this Markov process and assume that the embedded Markov chain starts in steady state at time 0. Let $U_1, U_2, \dots$ be the sequence of holding intervals between transitions (with U_1 being the time to the first transition). Show that these rv's are identically distributed. Show by example that they are not independent (*i.e.*, $M(t)$ is not a renewal process).

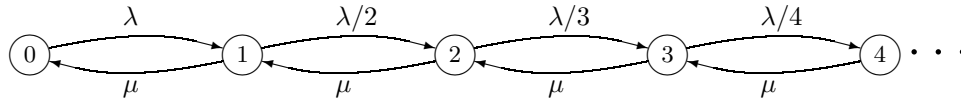
Solution: If the embedded Markov chain is in steady state, then the state at the next transition is also in steady state, so U_2 has the same distribution as U_1 . If we visualize this process as an M/M/1 queue, we see occasional arrivals usually followed very quickly by a departure, so arrivals and departures almost alternate, with rather rare periods when a customer actually waits in a queue. Analytically, since $\pi_0 = 0.99/2$ and $1 - \pi_0 = 1.01/2$,

we see that $F_{U_1}^c(t) \approx (1/2)e^{-\lambda t} + (1/2)e^{-(\lambda+\mu)t}$. If we take $\lambda = 0.01$, $\mu = 1$, and $t = 10$, then $F_{U_1}^c(t) \approx 1/2$ and we get

$$\Pr\{U_1 > t, U_2 > t \mid X_0 = 0, X_1 = 1\} = \exp[-\lambda t - (\lambda + \mu)t] = \exp[-10.2] \approx 0.$$

We get the same answer conditioning on $X_0 = 1, X_1 = 0$, and the condition $X_0 = X_1$ is an impossible event. The event $\{U_1 > 10, U_2 > 10\}$ is more unlikely than above for all other combinations of X_0, X_1 , so $\Pr\{U_1 > 10, U_2 > 10\} \approx 0$, demonstrating a lack of statistical independence. The fact that $M(t)$ is not a renewal process (for this process or others) makes the analysis in Section 7.2 more complex than it might appear to be.

Exercise 7.5: Consider the Markov process illustrated below. The transitions are labelled by the rate q_{ij} at which those transitions occur. The process can be viewed as a single server queue where arrivals become increasingly discouraged as the queue lengthens. The word *time-average* below refers to the limiting time-average over each sample-path of the process, except for a set of sample paths of probability 0.



a) Find the time-average fraction of time p_i spent in each state $i > 0$ in terms of p_0 and then solve for p_0 . Hint: First find an equation relating p_i to p_{i+1} for each i . It also may help to recall the power series expansion of e^x .

Solution: The p_i , $i \geq 0$ for a birth-death Markov process are related by $p_{i+1}q_{i+1,i} = p_i q_{i,i+1}$, which in this case is $p_{i+1}\mu = p_i\lambda/(i+1)$. Iterating this equation,

$$p_i = p_{i-1} \frac{\lambda}{\mu i} = p_{i-2} \frac{\lambda^2}{\mu^2 i(i-1)} = \cdots = p_0 \frac{\lambda^i}{\mu^i i!}.$$

Denoting λ/μ by ρ ,

$$1 = \sum_{i=0}^{\infty} p_i = p_0 \left[\sum_{i=0}^{\infty} \frac{\rho^i}{i!} \right] = p_0 e^{\rho}.$$

Thus,

$$p_0 = e^{-\rho}; \quad p_i = \frac{\rho^i e^{-\rho}}{i!}.$$

b) Find a closed form solution to $\sum_i p_i \nu_i$ where ν_i is the rate at which transitions out of state i occur. Show that the embedded chain is positive recurrent for all choices of $\lambda > 0$ and $\mu > 0$ and explain intuitively why this must be so.

Solution: The embedded chain steady-state probabilities π_i can be found from the steady-state process probabilities by (7.11), *i.e.*,

$$\pi_j = \frac{p_j \nu_j}{\sum_i p_i \nu_i} \quad (7.11').$$

We start by finding $\sum_i p_i \nu_i$. The departure rate from state i is

$$\nu_0 = \lambda; \quad \nu_i = \mu + \frac{\lambda}{i+1} \quad \text{for all } i > 0.$$

We now calculate $\sum_i p_i \nu_i$ by separating out the $i = 0$ term and then, for $i \geq 1$, sum separately over the two terms, μ and $\lambda/(i+1)$, of ν_i .

$$\sum_{i=0}^{\infty} p_i \nu_i = e^{-\rho} \lambda + \sum_{i=1}^{\infty} e^{-\rho} \frac{\rho^i \mu}{i!} + \sum_{i=1}^{\infty} e^{-\rho} \frac{\rho^i \lambda}{i!(i+1)}.$$

Substituting $\mu\rho$ for λ and combining the first and third term,

$$\begin{aligned} \sum_{i=0}^{\infty} p_i \nu_i &= \sum_{i=1}^{\infty} e^{-\rho} \frac{\rho^i \mu}{i!} + \sum_{i=0}^{\infty} e^{-\rho} \frac{\rho^{i+1} \mu}{(i+1)!} \\ &= 2 \sum_{i=1}^{\infty} e^{-\rho} \frac{\rho^i \mu}{i!} = 2\mu(1 - e^{-\rho}). \end{aligned}$$

Since $\sum_i p_i \nu_i < \infty$, we see from (7.11') that each π_i is strictly positive and that $\sum_i \pi_i = 1$. Thus the embedded chain is positive recurrent. Intuitively, $\{\pi_i; i \geq 0\}$ is found from $\{p_i; i \geq 0\}$ by finding $\pi'_i = p_i \nu_i$ and then normalizing $\{\pi'_i; i \geq 0\}$ to sum to 1. Since $\lambda \leq \nu_i \leq \lambda + \mu$, i.e., the ν_i all lie within positive bounds, this normalization must work (i.e., $\sum_i \pi_i \nu_i < \infty$).

c) For the embedded Markov chain corresponding to this process, find the steady-state probabilities π_i for each $i \geq 0$ and the transition probabilities P_{ij} for each i, j .

Solution: Since $\pi_j = p_j \nu_j / [\sum_i p_i \nu_i]$, we simply plug in the values for p_i, ν_i , and $\sum_i p_i \nu_i$ found in (a) and (b) to get

$$\pi_0 = \frac{\rho}{2(e^\rho - 1)}; \quad \pi_i = \frac{\rho^i}{2i!(e^\rho - 1)} \left(\frac{\rho}{i+1} + 1 \right); \quad \text{for } i > 1.$$

There are many forms for this answer. One sanity check is to observe that the embedded chain probabilities do not change if λ and μ are both multiplied by the same constant, and thus the π_i must be a function of ρ alone. Another sanity check is to observe that in the limit $\rho \rightarrow 0$, the embedded chain is dominated by an alternation between states 0 and 1, so that in this limit $\pi_0 = \pi_1 = 1/2$.

d) For each i , find both the time-average interval and the time-average number of overall state transitions between successive visits to i .

Solution: The time-average interval between visits to state i is $\bar{W}_i = 1/(p_i \nu_i)$. This is explained in detail in Section 7.2.6, but the essence of the result is that for renewals at successive entries to state i , p_i must be the ratio of the expected time $1/\nu_i$ spent in state i to the overall expected renewal interval $\bar{W}_i$. Thus $\bar{W}_i = 1/(\nu_i p_i)$.

$$\bar{W}_0 = \frac{e^\rho}{\lambda}; \quad \bar{W}_i = \frac{(i+1)! e^\rho}{\rho^i [\lambda + (i+1)\mu]}; \quad \text{for } i \geq 1.$$

The time-average number of state transitions per visit to state i is $\bar{T}_{ii} = 1/\pi_i$. This is proven in Theorem 6.3.8.

Exercise 7.6: (Detail from proof of Theorem 7.2.6 a) Let U_n be the n th holding interval for a Markov process starting in steady state, with $\Pr\{X_0 = i\} = \pi_i$. Show that $E[U_n] = \sum_k \pi_k/\nu_k$ for each integer n .

Solution: There are two reasonable definitions of starting in steady state for a Markov process; the first is for the embedded chain to start in steady state and the second is for the process to start with $X(0)$ having a PMF equal to the process steady-state probabilities p_i . (see Exercise 7.4). Here it is explicitly stated that the embedded chain starts in steady state, and thus at each transition, X_n is in steady state with the PMF $\Pr\{X_n = k\} = \pi_k$. Since the expected holding interval given $X_n = k$ is $1/\nu_k$, the marginal expected holding interval at each transition n is $E[U_n] = \sum_k \pi_k/\nu_k$.

b) Let S_n be the epoch of the n th transition. Show that $\Pr\{S_n \geq n\beta\} \leq \frac{\sum_k \pi_k/\nu_k}{\beta}$ for all $\beta > 0$.

Solution: $E[S_n] = E[U_1] + E[U_2] + \cdots + E[U_n] = n \sum_k \pi_k/\nu_k$. Applying the Markov inequality, we get the desired result.

c) Let $M(t)$ be the number of transitions of the Markov process up to time t , given that X_0 is in steady state. Show that $\Pr\{M(n\beta) \geq n\} \geq 1 - \frac{\sum_k \pi_k/\nu_k}{\beta}$.

Solution: Using (b),

$$\Pr\{S_n \leq n\beta\} = 1 - \Pr\{S_n > n\beta\} \geq 1 - \Pr\{S_n \geq n\beta\} \geq 1 - \frac{\sum_k \pi_k/\nu_k}{\beta}.$$

The event $\{S_n \leq n\beta\}$ is the event that the n th transition occurs before time $n\beta$. This is the same as the event $\{M(n\beta) \geq n\}$, i.e., the event that the number of transitions by time $n\beta$ is n or more. Thus for any $\beta > 0$ and every n ,

$$\Pr\{M(n\beta) \geq n\} \geq 1 - \frac{\sum_k \pi_k/\nu_k}{\beta}.$$

d) Show that if $\sum_k \pi_k/\nu_k$ is finite, then $\lim_{t \rightarrow \infty} M(t)/t = 0$ WP1 is impossible. (Note that this is equivalent to showing that $\lim_{t \rightarrow \infty} M(t)/t = 0$ WP1 implies $\sum_k \pi_k/\nu_k = \infty$).

Solution: If $\sum_k \pi_k/\nu_k$ is finite and β is, say, $2 \sum_k \pi_k/\nu_k$, then $\Pr\{M(n\beta) \geq n\} \geq 1/2$ for all n . Thus $(1/n)M(n\beta)$ does not converge to 0 in probability and thus not with probability 1.

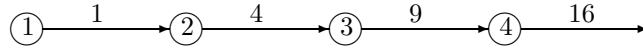
e) Let $M_i(t)$ be the number of transitions by time t starting in state i . Show that if $\sum_k \pi_k/\nu_k$ is finite, then $\lim_{t \rightarrow \infty} M_i(t)/t = 0$ WP1 is impossible.

Solution: Eq. (7.18) shows that $\lim_{t \rightarrow \infty} M_i(t)/t$ is the same for all i and thus the same as $\lim_{t \rightarrow \infty} M(t)/t$. Thus none of these can be equal to 0 WP1.

Exercise 7.7: a) Consider the process in the figure below. The process starts at $X(0) = 1$, and for all $i \geq 1$, $P_{i,i+1} = 1$ and $\nu_i = i^2$ for all i . Let S_n be the epoch when the n th transition occurs. Show that

$$E[S_n] = \sum_{i=1}^n i^{-2} < 2 \text{ for all } n.$$

Hint: Upper bound the sum from $i = 2$ by integrating x^{-2} from $x = 1$.



Solution:

$$\mathbb{E}[S_n] \leq \sum_{i=1}^{\infty} i^{-2} \leq 1 + \int_1^{\infty} x^{-2} dx = 2.$$

b) Use the Markov inequality to show that $\Pr\{S_n > 4\} \leq 1/2$ for all n . Show that the probability of an infinite number of transitions by time 4 is at least $1/2$.

Solution: By the Markov inequality, $\Pr\{S_n \geq 4\} \leq 2/4 = 1/2$ for all n . Thus $\Pr\{S_n < 4\} > 1/2$ for all n . Since $\{S_n < 4\} \supseteq \{S_{n+1} < 4\}$ for all n , $\Pr\{\lim_{n \rightarrow \infty} S_n < 4\} \geq 1/2$. This means that infinitely many transitions have occurred by time 4 with probability at least $1/2$. Thus $\{X(t); t > 0\}$ is not really a stochastic process since $X(t)$ is not defined as a rv for $t > 4$, (and in fact not for smaller t either) since $X(t) = \infty$ with positive probability.

Exercise 7.8: a) Consider a Markov process with the set of states $\{0, 1, \dots\}$ in which the transition rates $\{q_{ij}\}$ between states are given by $q_{i,i+1} = (3/5)2^i$ for $i \geq 0$, $q_{i,i-1} = (2/5)2^i$ for $i \geq 1$, and $q_{ij} = 0$ otherwise. Find the transition rate ν_i out of state i for each $i \geq 0$ and find the transition probabilities $\{P_{ij}\}$ for the embedded Markov chain.

Solution: For $i > 0$,

$$\nu_i = q_{i,i+1} + q_{i,i-1} = 2^i.$$

For $i = 0$, $\nu_0 = q_{01} = 3/5$. It follows that $P_{ij} = q_{ij}/\nu_i$ is given by

$$P_{i,i+1} = 3/5; \quad P_{i,i-1} = 2/5 \quad \text{for } i > 0; \quad P_{01} = 1.$$

b) Find a solution $\{p_i; i \geq 0\}$ with $\sum_i p_i = 1$ to (7.23).

Solution: This is a birth-death chain so the $\{p_i; i \geq 0\}$ satisfying 7.23 exist if there is a solution to $p_i q_{i,i+1} = p_{i+1} q_{i+1,i}$ for $i \geq 0$ and $\sum_i p_i = 1$. Thus

$$p_{i+1} = \frac{(3/5)2^i p_i}{(2/5)2^{i+1}} = (3/4)p_i.$$

Thus, defining $\rho = 3/4$, the desired probabilities are $p_i = (1 - \rho)\rho^i$. These are called steady-state process probabilities because, if the embedded chain is positive recurrent, then each p_i gives the long term average fraction of time the sample functions of the process are in state i WP1. As we see in part (c), this embedded chain is transient; the point of the problem is to discover what this solution for $\{p_i; i \geq 0\}$ means, if anything.

c) Show that all states of the embedded Markov chain are transient.

Solution: The embedded Markov chain is the embedded Markov chain of an M/M/1 queue. Since the steady-state balance equations are $\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i}$, we see that if a steady

state exists, it satisfies $\pi_{i+1} = (3/2)\pi_i$ for each $i \geq 1$. We have seen before (see Exercise 6.2) that this means that the the embedded Markov chain is transient.

d) Explain in your own words why your solution to (b) is not in any sense a set of steady-state probabilities.

Solution: The transient embedded Markov chain shows that for each state i , there is a positive probability that i will never be re-entered after the first visit, and more precisely that the aggregate time spent in i after the first entry is a geometrically distributed sum of IID exponential rv's. This in turn (see Example 2.3.3) implies that the aggregate time in state i is an exponentially distributed rv. This can certainly not be viewed in any reasonable sense as a steady-state fraction of time in i .

We next show that an infinite number of states are visited with arbitrarily large probability in a finite time. This indicates that this chain is similar to the one in Exercise 7.7, where it is obvious that time averages can't have any real meaning. To carry out this argument, define $\bar{Y}_m$ as the expected first passage time from state m to state $m+1$. Then $\bar{S}_m = \bar{Y}_0 + \bar{Y}_1 + \dots + \bar{Y}_{m-1}$ is the expected first passage time from state 0 to state m . We denote p as $3/5$ and q as $2/5$, thus allowing the argument to be used with different transition probabilities such that $p > q$.

$$\bar{Y}_m = p 2^{-m} + q [2^{-m} + \bar{Y}_{m-1} + \bar{Y}_m]; \quad \text{for all } m \geq 1.$$

To understand this, the expected time to the first transition, starting from state m , is 2^{-m} . With probability p , that transition goes directly to state $m+1$, whereas with probability q , that transition must be followed first by a first passage from $m-1$ to m and next by a first passage from m to $m+1$. Simplifying this equation, we get

$$\bar{Y}_m = \frac{1}{p} 2^{-m} + \frac{q}{p} \bar{Y}_{m-1}; \quad \text{for all } m \geq 1. \quad (i)$$

Recurring on this equation back to $\bar{Y}_0$, we get

$$\begin{aligned} \bar{Y}_m &= \frac{2^{-m}}{p} \left[1 + \frac{2q}{p} + \left(\frac{2q}{p} \right)^2 + \dots + \left(\frac{2q}{p} \right)^{m-1} \right] + \left(\frac{q}{p} \right)^m \bar{Y}_0 \\ &= \frac{2^{-m}}{p} \left[1 + \frac{2q}{p} + \left(\frac{2q}{p} \right)^2 + \dots + \left(\frac{2q}{p} \right)^{m-1} \right] + \frac{2^{-m}}{p} \left(\frac{2q}{p} \right)^m \quad (ii) \\ &= \frac{2^{-m}}{p} \left[\frac{(2q/p)^{m+1} - 1}{(2q/p) - 1} \right] \\ &= \frac{2(q/p)^{m+1} - (1/2)^m}{2q - p} \quad (iii) \end{aligned}$$

We have used the fact that $\bar{Y}_0$ is the expected duration of the first passage time from state 0 to 1, which is simply $1/p$. The result in (iii) thus holds for all $m \geq 1$, and, as can be easily checked from the formula, it also holds for $m=0$, where $\bar{Y}_0 = 1/p$.

Each sample function of this process (starting at state 0) can be viewed as a sequence of segments, the m th of which starts at the first entry to state m and ends at the first entry to $m+1$. The duration of the m th segment is a rv Y_m . It is independent of all the other

segment durations, and must be non-degenerate since it has finite expectation, $\bar{Y}_m$, as given in (iii).

Since $q < p$, the second term in (iii) becomes negligible with respect to the first term with increasing m , so that $\bar{Y}_m$ essentially approaches 0 geometrically as $(q/p)^m$. Thus the expected aggregate duration of this countable sequence of intervals is finite. The partial sums of this sequence of segments, and thus the partial sums of transition intervals, become so closely spaced that the infinite aggregate is bounded. The process does not continue forever, but ends at a limit point with infinitely many segments, and thus also transitions, piled up at the limit point.

To make this more precise, we now find $\bar{S}_n$, the expected first passage time from 0 to n , by summing $\bar{Y}_m$ from 0 to $n-1$. Thus

$$\begin{aligned}\bar{S}_n &= \frac{2q}{p(2q-p)} \left[\frac{1 - (q/p)^n}{1 - q/p} \right] - \frac{2}{2q-p} [1 - (1/2)^n] \\ &= \frac{2}{p-q} - \left(\frac{2}{2q-p} \right) \left[\frac{q(q/p)^n}{p-q} - (1/2)^n \right] < \frac{2}{p-q}\end{aligned}$$

The important thing here is that $\bar{S}_n$ is $2/(p-q)$, reduced by a quantity that essentially decreases geometrically to zero as $n \rightarrow \infty$. Thus $\lim_{n \rightarrow \infty} \bar{S}_n = 1/(p-q)$. Now S_n , the first passage time from state 0 to m is increasing with n and thus $\lim_{n \rightarrow \infty} S_n$ must be a rv since its expected value is finite. This is the limit referred to above where successive process transitions pile up. The limit depends on the sample path, but is a rv with expected value $1/(p-q)$. This pile up of transitions at a finite limit is much like the simpler process in Exercise 7.7.

For $p = 3/5$ and $q = 2/5$, $\lim_{n \rightarrow \infty} \bar{S}_n = 10$. By the Markov inequality, $\Pr\{S_n \geq 2\alpha/(p-q)\} \leq 1/\alpha$ for all $\alpha > 1$. Since this is true for all $n \geq 1$ and $S_n \geq S_{n-1}$ for all $n \geq 1$, we see that

$$\Pr\left\{\lim_{n \rightarrow \infty} S_n \geq \alpha/(p-q)\right\} \leq 1/\alpha \quad \text{for all } \alpha > 1.$$

In other words, the limiting point at which transitions pile up into a limit might vary with the sample process, but that limiting point is finite WP1. For the case here, for example, $\Pr\{\lim_{n \rightarrow \infty} S_n \geq 20\} \leq 1/2$.

Since we have come this far, it makes sense to also interpret the meaning in part (b) of the solution, $p_i = (1-\rho)\rho^i$, of (7.23) where $\rho = p/2q$. Note in (i) that $\bar{Y}_m$ is the expected first passage time to state $m+1$ from state m . The first term, $2^{-m}/p$, is the expected time (perhaps including more than one visit) in state m and the other, $(q/p)\bar{Y}_{m-1}$, is the expected time in states 0 to $m-1$. The recursion leading to (ii) carries this further, so that the term $(q/p)^k(2^{-m}/p)$ is the expected time in state $m-k$ during the duration of the first passage from m to $m+1$. Thus the expected time in state i , $0 \leq i \leq m$ within this first passage time from m to $m+1$ is $\rho^i(2^{-m}\rho^{-m}/p)$. This is simply the solution to (7.23) normalized to be a fraction of $\bar{Y}_m$. In other words, during each segment m of the process, the expected time in each state $0 \leq i \leq m$ is $\bar{Y}_m(p_i/\sum_{j \leq m} p_j)$.

One can interpret this as a type of average if one chooses, but it is not very close to the usual time average of a positive recurrent Markov process. First, there is no SLLN to rely

on and second the process terminates after a finite time. It is still quite surprising that the p_i have so much meaning given the extreme peculiarity of this process.

Exercise 7.9: Let $q_{i,i+1} = 2^{i-1}$ for all $i \geq 0$ and let $q_{i,i-1} = 2^{i-1}$ for all $i \geq 1$. All other transition rates are 0.

a) Solve the steady-state equations and show that $p_i = 2^{-i-1}$ for all $i \geq 0$.

Solution: The process is a birth/death process, so we can find the steady-state probabilities (if they exist) from the equations $p_i q_{i,i+1} = p_{i+1} q_{i+1,i}$ for $i \geq 0$. Thus $p_{i+1} = p_i/2$. Normalizing to $\sum_i p_i = 1$, we get $p_i = 2^{-i-1}$.

b) Find the transition probabilities for the embedded Markov chain and show that the chain is null recurrent.

Solution: First assume, for the purpose of finding a contradiction, that the embedded Markov chain is positive recurrent. Then by (7.21), $\pi_j = p_j \nu_j / \sum_i p_i \nu_i$. Note that $\nu_i = q_{i,i+1} + q_{i,i-1} = 2^i$ for all $i \geq 1$. Thus $p_i \nu_i = 1/2$ for $i \geq 1$ and $\sum_i p_i \nu_i = \infty$. Thus $\pi_j = 0$ for $j \geq 0$ and the chain must be either null recurrent or transient. We show that the chain is null recurrent in (c)

c) For any state i , consider the renewal process for which the Markov process starts in state i and renewals occur on each transition to state i . Show that, for each $i \geq 1$, the expected inter-renewal interval is equal to 2. Hint: Use renewal-reward theory.

Solution: We use the same argument as in Section 7.2.2 with unit reward when the process is in state i . The limiting fraction of time in state i given that $X_0 = i$ is then $p_i(i) = 1/(\nu_i \bar{W}(i))$ where $\bar{W}(i)$ is the mean renewal time between entries to state i . This is the same for all starting states, and by Blackwell's theorem it is also $\lim_{t \rightarrow \infty} \Pr\{X(t) = i\}$. This $p_i(i)$ must also satisfy the steady-state process equations and thus be equal to $p_i = 2^{-i-1}$. Since $\nu_i = 2^i$, we have $\bar{W}(i) = 2$ for all $i \geq 1$. Finally, this means that a return to i must happen in a finite number of transitions. Since the embedded chain cannot be positive recurrent, it thus must be null recurrent.

The argument here has been a little tricky since the development in the text usually assumes that the embedded chain is positive recurrent, but the use of renewal theory above gets around that.

d) Show that the expected number of transitions between each entry into state i is infinite. Explain why this does *not* mean that an infinite number of transitions can occur in a finite time.

Solution: We have seen in (b) and (c) that the embedded chain is null-recurrent. This means that, given $X_0 = i$, for any given i , a return to i must happen in a finite number of transitions (i.e., $\lim_{n \rightarrow \infty} F_{ii}(n) = 1$), but the expected number of such transitions is infinite. We have seen many rv's that have an infinite expectation, but, being rv's, have a finite sample value WP1.

Exercise 7.10: a) Consider the two state Markov process of Example 7.3.1 with $q_{01} = \lambda$ and $q_{10} = \mu$. Find the eigenvalues and eigenvectors of the transition rate matrix $[Q]$.

Solution: The matrix $[Q]$ is $\begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}$. For any transition rate matrix, the rows all sum to 0, and thus $[Q]e = 0$, establishing that 0 is an eigenvalue with the right eigenvector

$\mathbf{e} = (1, 1)^\top$. The left eigenvector, normalized to be a probability vector, is the steady-state vector $\mathbf{p} = (\mu/(\lambda + \mu), \lambda/(\lambda + \mu))$. The other eigenvalue is then easily calculated as $-(\lambda + \mu)$ with left eigenvector $(-1, 1)$ and right eigenvector $(-\lambda/(\lambda + \mu), \mu/(\lambda + \mu))$. This answer is not unique, since the eigenvectors can be scaled differently while still maintaining $\mathbf{p}_i \mathbf{v}_j^\top = \delta_{ij}$.

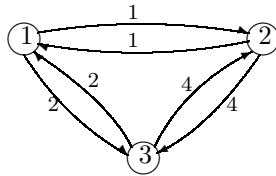
b) If $[Q]$ has M distinct eigenvalues, the differential equation $d[P(t)]/dt = [Q][P(t)]$ can be solved by the equation

$$[P(t)] = \sum_{i=1}^M \mathbf{v}_i e^{t\lambda_i} \mathbf{p}_i^\top,$$

where $\mathbf{p}_i$ and $\mathbf{v}_i$ are the left and right eigenvectors of eigenvalue λ_i . Show that this equation gives the same solution as that given for Example 7.3.1.

Solution: This follows from substituting the values above. Note that the eigenvectors above are the same as those for the sample time approximations and the eigenvalues are related as explained in Example 7.3.1.

Exercise 7.11: Consider the three state Markov process below; the number given on edge (i, j) is q_{ij} , the transition rate from i to j . Assume that the process is in steady state.



a) Is this process reversible?

Solution: Since $q_{ij} = q_{ji}$ for all states i, j , we can satisfy the condition for reversibility in (7.52) by choosing $p_i = p_j$ for all i, j . Thus the chain is reversible and $p_1 = p_2 = p_3 = 1/3$

b) Find p_i , the time-average fraction of time spent in state i for each i .

Solution: This is solved in (a), but is somewhat surprising because of the different transition rates. One can rationalize this by starting with all q_{ij} equal, so that $p_i = 1/3$ for all i is obvious from symmetry. Then if q_{23} and q_{32} are each increased but kept equal, we can see that this increases the oscillation rate between 2 and 3, but wouldn't be expected to change p_2 or p_3 . The same argument applies if either of the other pairs are then changed.

c) Given that the process is in state i at time t , find the mean delay from t until the process leaves state i .

Solution: This mean delay is $(\nu_i)^{-1} = (q_{ij} + q_{ik})^{-1}$ where k, j, i are distinct. Thus $\nu_1^{-1} = 1/3$, $\nu_2^{-1} = 1/5$, $\nu_3^{-1} = 1/6$.

d) Find π_i , the time-average fraction of all transitions that go into state i for each i .

Solution: From (7.21), $\pi_i = p_i \nu_i / (\sum_j p_j \nu_j)$. Using (b) and (c), $\pi_1 = 3/14$, $\pi_2 = 5/14$, $\pi_3 = 6/14$.

e) Suppose the process is in steady state at time t . Find the steady-state probability that the next state to be entered is state 1.

Solution: In order for 1 to be the next state to be entered, the state at time t must be 2 or 3. Given $X(t) = 2$ the probability of going next to 1 is $1/5$. Given $X(t) = 3$, the probability of going next to 1 is $1/3$. Thus $\Pr\{\text{next state} = 1\} = (1/3)(1/5) + (1/3)(1/3) = 8/45$.

f) Given that the process is in state 1 at time t , find the mean delay until the process first returns to state 1.

Solution: From Theorem 7.2.6, $\overline{W}_i = (\nu_i p_i)^{-1} = 1$. Another instructive approach is to use the renewal-reward process (in time) for which renewals occur on entries to state 1 and unit reward occurs while in state 1. The expected reward during a visit to state 1 is the mean time ν_1^{-1} until an exit from 1, *i.e.*, $1/3$. The overall fraction of time spent in state 1 is also $1/3$ WP1. By the renewal reward theorem, Theorem 5.4.5, the mean time between visits to state 1 is $\frac{1/3}{1/3} = 1$. Since the time spent in state 1 is exponential, this is also the time to a renewal starting in state 1 at an arbitrary time t .

g) Consider an arbitrary irreducible finite-state Markov process in which $q_{ij} = q_{ji}$ for all i, j . Either show that such a process is reversible or find a counter example.

Solution: The argument in (a) is general for finite state processes, so any such process is reversible and the steady-state probabilities are again equal. For a countable state space, each steady-state process probability p_i must be equal, but then they must all be 0, and they cannot sum to 1. Thus steady-state probabilities can not exist, and the definition of reversibility in Section 7.6 does not apply since the backward transition probabilities q_{ij}^* are undefined.

Exercise 7.12: a) Consider an M/M/1 queueing system with arrival rate λ and service rate $\mu > \lambda$. Assume that the queue is in steady state. Given that an arrival occurs at time t , find the probability that the system is in state i immediately *after* time t .

Solution: One must be careful interpreting these statements. We assume that the system is in steady state for times $\tau < t$. The arrival process is memoryless and for times $\tau \geq t$, the probability of an arrival in any increment is independent of the state at each $\tau < t$. Thus given an arrival at t , the state increases by 1 from $X(t^-)$, the steady-state state immediately before t . Thus, defining $\rho = \lambda/\mu$, $\Pr\{X(t) = i\} = (1 - \rho)\rho^{i-1}$ for all $i > 0$. Of course $\Pr\{X(t) = 0\} = 0$.

b) Assuming FCFS service, and conditional on i customers in the system immediately after the above arrival, characterize the time until the above customer departs as a sum of random variables.

Solution: Let Y_1 be the service time of the entering customer, let $Y_2, \dots, Y_{i-1}$ be the service times of the $i - 2$ earlier customers in the queue, and let Y_i be the remaining required service of the customer being served. The time until the arriving customer departs is then $Y_1 + Y_2 + \dots + Y_i$. These are IID exponential rv's of rate μ each, so the sum is Erlang.

c) Find the unconditional probability density of the time until the above customer departs. **Hint:** You know (from splitting a Poisson process) that the sum of a geometrically distributed number of IID exponentially distributed random variables is exponentially distributed. Use the same idea here.

Solution: From (a), the number X of customers in the system at time t has the PMF $p_X(i) = (1 - \rho)\rho^{i-1}$ for $i \geq 1$. This is geometric starting at 1. The time W until the given customer departs is $W = Y_1 + \dots + Y_X$ where the Y_i are IID and each is exponential with parameter μ . As explained in Example 2.3.3, the distribution of such a geometric sum of

IID exponentially distributed rv's is itself exponential with rate $(1 - \rho)\mu$, i.e.,

$$f_W(w) = \mu(1 - \rho) \exp[-\mu(1 - \rho)w].$$

In order to better understand this equation for the problem at hand, note that the sequence of departing customers (at least up to time $t + W$) can be viewed as a Poisson process of rate μ . View this Poisson process as the combination of two independent sub-Poisson processes, the first of rate $(1 - \rho)\mu$ and the second of rate $\rho\mu$. These sub-processes are mathematical constructs with no immediate interpretation for the given problem. The probability that the first departure from the combined process is from sub-process 1 is then $1 - \rho$. For each $i > 1$, the probability that the first $i - 1$ departures are from process two and the i th is from process 1 is $(1 - \rho)\rho^{i-1}$. Thus the probability that the number in the original system at time t is $X = i$ is the same as the probability that the first $i - 1$ departures are from sub-process 2 and departure i is from sub-process 1. Thus the marginal probability density for W is the same in each case, and is thus the density of the first departure in the first sub-process, which is the PDF given above. Since $\rho = \lambda/\mu$, this result is usually stated as $f_W(w) = (\mu - \lambda)e^{-(\mu - \lambda)w}$.

Exercise 7.13: a) Consider an M/M/1 queue in steady state. Assume $\rho = \lambda/\mu < 1$. Find the probability $Q(i, j)$ for $i \geq j > 0$ that the system is in state i at time t and that $i - j$ departures occur before the next arrival.

Solution: The probability that the queue is in state i at time t is $p_i = (1 - \rho)\rho^i$ (see (7.40)). Given state $i > 0$, successive departures (so long as the state remains positive) are Poisson at rate μ and arrivals are Poisson at rate λ . Thus (conditional on $X(t) = i$) the probability of exactly $i - j$ departures before the next arrival is $(\mu/(\lambda + \mu))^{i-j} \lambda/(\lambda + \mu)$. Thus

$$Q(i, j) = (1 - \rho)\rho^i \left(\frac{\mu}{\lambda + \mu} \right)^{i-j} \frac{\lambda}{\lambda + \mu}.$$

b) Find the PMF of the state immediately before the first arrival after time t .

Solution: $Q(i, j)$ is the probability that $X(t) = i$ and that $X(\tau^-) = j$ where τ is the time of the next arrival after t and $j > 0$. Thus for $j > 0$,

$$\begin{aligned} \Pr\{X(\tau^-) = j\} &= \sum_{i \geq j} Q(i, j) = \sum_{i \geq j} (1 - \rho) \left(\frac{\lambda}{\lambda + \mu} \right)^i \left(\frac{\mu}{\lambda + \mu} \right)^{-j} \frac{\lambda}{\lambda + \mu} \\ &= (1 - \rho) \left(\frac{\lambda}{\mu} \right)^j \left(\frac{1}{1 - \lambda/(\lambda + \mu)} \right) \left(\frac{\lambda}{\lambda + \mu} \right) = (1 - \rho)\rho^{j+1}. \end{aligned}$$

For $j = 0$, a similar but simpler calculation leads to $\Pr\{X(\tau^-) = 0\} = (1 - \rho)(1 + \rho)$. In other words, the system is not in steady state immediately before the next arrival. This is not surprising, since customers can depart but not arrive in the interval (t, τ) .

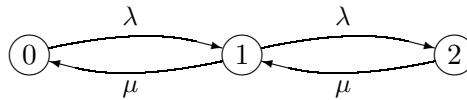
c) There is a well-known queueing principle called PASTA, standing for “Poisson arrivals see time averages”. Given your results above, give a more precise statement of what that principle means in the case of the M/M/1 queue.

Solution: The PASTA principle requires being so careful about a precise statement of the conditions under which it holds that it is better treated as an hypothesis to be considered rather than a principle. One plausible meaning for what arrivals ‘see’ is given in (b), and that is not steady state. Another plausible meaning is to look at the fraction of arrivals that arise from each state; that fraction is the steady-state probability. Perhaps the simplest visualization of PASTA is to look at the discrete-time model of the M/M/1 queue. There the steady-state fraction of arrivals that come from state i is equal to the steady-state probability p_i .

Exercise 7.14: A small bookie shop has room for at most two customers. Potential customers arrive at a Poisson rate of 10 customers per hour; they enter if there is room and are turned away, never to return, otherwise. The bookie serves the admitted customers in order, requiring an exponentially distributed time of mean 4 minutes per customer.

a) Find the steady-state distribution of number of customers in the shop.

Solution: The system can be modeled as a Markov process with 3 states, representing 0, 1, or 2 customers in the system. Arrivals in state 2 are turned away, so they do not change the state and are not shown. In transitions per hour, $\lambda = 10$ and $\mu = 15$. Since the process is a birth-death process, the steady-state equations are $\lambda p_i = \mu p_{i+1}$ for $i = 0, 1$. Thus $p_1 = (2/3)p_0$ and $p_2 = 4/9p_0$. Normalizing, $p_0 = 9/19$, $p_1 = 6/19$, $p_2 = 4/19$.



b) Find the rate at which potential customers are turned away.

Solution: All arrivals in state 2 are turned away, so the average rate at which customers are turned away is $40/19$ per hour.

c) Suppose the bookie hires an assistant; the bookie and assistant, working together, now serve each customer in an exponentially distributed time of mean 2 minutes, but there is only room for one customer (*i.e.*, the customer being served) in the shop. Find the new rate at which customers are turned away.

Solution: Now $\lambda = 10$ and $\mu = 30$. There are two states, with $p_0 = 3/4$ and $p_1 = 1/4$. Customers are turned away at rate $10/4$, which is somewhat higher than the rate without the assistant.

Exercise 7.15: This exercise explores a continuous time version of a simple branching process.

Consider a population of primitive organisms which do nothing but procreate and die. In particular, the population starts at time 0 with one organism. This organism has an exponentially distributed lifespan T_0 with rate μ (*i.e.*, $\Pr\{T_0 \geq \tau\} = e^{-\mu\tau}$). While this organism is alive, it gives birth to new organisms according to a Poisson process of rate λ . Each of these new organisms, while alive, gives birth to yet other organisms. The lifespan and birthrate for each of these new organisms are independently and identically distributed to those of the first organism. All these and subsequent organisms give birth and die in the same way, again independently of all other organisms.

a) Let $X(t)$ be the number of (live) organisms in the population at time t . Show that $\{X(t); t \geq 0\}$ is a Markov process and specify the transition rates between the states.

Solution: If $X(t) = j > 0$, then the state will change at the next instant when either a birth or death occurs among the j living organisms. The first birth or death among the j

organisms can be viewed as the smallest of j independent exponential rv's of rate λ and j of rate μ . That first birth or death is a birth with probability $\lambda/(\lambda + \mu)$, so the transition rate from state j to state $j + 1$ is $j\lambda$. Similarly, the transition rate from j to $j - 1$ is $j\mu$. If $X(t) = 0$, then no deaths or births can occur at any $\tau > t$. Thus in general, state transitions are exponential (with no transitions from state 0), and the rate of each transition at time t depends only on the state at time t . Thus the process is a Markov process.

b) Find the embedded Markov chain $\{X_n; n \geq 0\}$ corresponding to the Markov process in (a). Find the transition probabilities for this Markov chain.

Solution: Let X_n be the state at the end of the n th transition. Given that $X_n = j$ for any given $j > 0$, X_{n+1} can be only $j + 1$ or $j - 1$. Also

$$\Pr\{X_{n+1} = j+1 \mid X_n = j\} = \frac{j\lambda}{j\lambda + j\mu} = \frac{\lambda}{\lambda + \mu}.$$

There are no transitions from state 0 in the Markov process, and thus there are no transitions from state 0 in the embedded chain.

c) Explain why the Markov process and Markov chain above are not irreducible. Note: The majority of results you have seen for Markov processes assume the process is irreducible, so be careful not to use those results in this exercise.

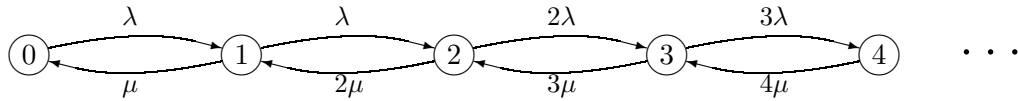
Solution: State 0 is a trapping state in the Markov chain, *i.e.*, it communicates with no other state. Thus the Markov chain is not irreducible. By definition, the Markov process is then also not irreducible.

d) For purposes of analysis, add an additional transition of rate λ from state 0 to state 1. Show that the Markov process and the embedded chain are irreducible. Find the values of λ and μ for which the modified chain is positive recurrent, null recurrent, and transient.

Solution: With the additional transition from state 0 to 1, there is a path from each state to each other state (both in the process and the embedded chain), and thus both are irreducible. Curiously enough, the embedded chain is now the same as the embedded chain of an M/M/1 queue. It is positive recurrent for $\rho < 1$, null recurrent for $\rho = 1$, and transient for $\rho > 1$.

e) Assume that $\lambda < \mu$. Find the steady-state process probabilities for the modified Markov process.

Solution: The transitions in the Markov process are illustrated below.



Let q_j be the steady-state process probability of state $j > 0$. Then

$$q_1 = \frac{\lambda q_0}{\mu}; \quad q_j = \frac{(j-1)\lambda q_{j-1}}{j\mu} \quad \text{for } j > 1. \quad (\text{A.144})$$

Let $\rho = \lambda/\mu$ (so $0 < \rho < 1$). Iterating (A.144), q_j can be expressed in terms of q_0 as

$$q_j = \frac{(j-1)\rho}{j} \frac{(j-2)\rho}{j-1} \cdots \frac{\rho}{2} \frac{\rho q_0}{1} = \frac{\rho^j q_0}{j}. \quad (\text{A.145})$$

The simplicity of (A.145) is quite surprising, but there is another way to derive the same result that helps explain it. This process has the same embedded chain as the M/M/1 process for which the steady-state process probabilities are $p_j = \rho^j p_0$. For the M/M/1 case, the rate of transitions out of each state j is $\nu_j = \lambda + \mu$, whereas here the corresponding rate is $\nu'_j = j(\lambda + \mu)$. From the relationship between the steady-state probabilities for the embedded chain and the process, however, $p_j = \alpha \pi_j / \nu_j$ and $q_j = \beta \pi_j / \nu'_j$ where α and β are normalizing constants. From this, we see that $q_j = (\alpha/\beta) p_j / j = (\alpha/\beta) p_0 \rho^j / j$. The normalizing constants can be ignored, since they merely insure that $\sum_j q_j = 1$, and thus this provides another derivation of (A.145).

We still need to solve for q_0 to find the steady-state process probabilities. We have

$$1 = \sum_j q_j = q_0 \left[1 + \sum_{j=1}^{\infty} \frac{\rho^j}{j} \right] = q_0 [1 - \ln(1 - \rho)].$$

To see the equality of $\sum_{j \geq 1} \rho^j / j$ and $-\ln(1 - \rho)$, note that they agree at $\rho = 0$ and, for each $\rho > 0$, the derivative of each with respect to ρ is $1/(1 - \rho)$. We then have

$$q_0 = \frac{1}{1 - \ln(1 - \rho)}; \quad q_j = \frac{\rho^j}{j[1 - \ln(1 - \rho)]} \quad \text{for } j \geq 1.$$

f) Find the mean recurrence time between visits to state 0 for the modified Markov process.

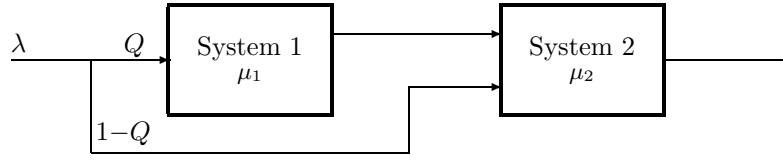
Solution: The mean recurrence time between visits to state 0 in the modified Markov process is

$$\overline{W}_0 = \frac{1}{\nu_0 q_0} = \frac{1 - \ln[1 - \rho]}{\lambda}.$$

g) Find the mean time $\overline{T}$ for the population in the original Markov process to die out. Note: We have seen before that removing transitions from a Markov chain or process to create a trapping state can make it easier to find mean recurrence times. This is an example of the opposite, where adding an exit from a trapping state makes it easy to find the recurrence time.

Solution: The mean recurrence time in the modified chain for state 0 consists of two parts — the mean time $(1/\lambda)$ to first reach state 1 from state 0 and then the mean time to first reach state 0 again. This latter time is the same whether or not the extra transition from 0 to 1 is present and is thus given by $-(1/\lambda) \ln(1 - \rho)$.

Exercise 7.16: Consider the job sharing computer system illustrated below. Incoming jobs arrive from the left in a Poisson stream. Each job, independently of other jobs, requires pre-processing in system 1 with probability Q . Jobs in system 1 are served FCFS and the service times for successive jobs entering system 1 are IID with an exponential distribution of mean $1/\mu_1$. The jobs entering system 2 are also served FCFS and successive service times are IID with an exponential distribution of mean $1/\mu_2$. The service times in the two systems are independent of each other and of the arrival times. Assume that $\mu_1 > \lambda Q$ and that $\mu_2 > \lambda$. Assume that the combined system is in steady state.



a) Is the input to system 1 Poisson? Explain.

Solution: The input to system 1 is the splitting of a Poisson process and is thus Poisson of rate λQ .

b) Are each of the two input processes coming into system 2 Poisson? Explain.

Solution: The output of system 1 is Poisson of rate λQ by Burke's theorem. The other process entering system 2 is also Poisson with rate $(1 - Q)\lambda$, since it is a splitting of the original Poisson process of rate λ . This input to system 2 is independent of the departures from system 1 (since it is independent both of the arrivals and departures from system 1.) Thus the combined process into system 2 is Poisson with rate λ

c) Give the joint steady-state PMF of the number of jobs in the two systems. Explain briefly.

Solution: The state $X_2(t)$ of system 2 at time t is dependent on the inputs to system 2 up to time t and the services in system 2 up to time t . By Burke's theorem, the outputs from system 1 at times up to time t are independent of the state $X_1(t)$ of system 1 at time t , and thus the inputs to system 2 from system 1 at times up to t are independent of $X_1(t)$. The other input to system 2 is also independent entirely of system 1. Thus $X_1(t)$ and $X_2(t)$ are independent. Thus $X_1(t)$ and $X_2(t)$ are independent and are the states of M/M/1 queues in steady state at time t .

$$\Pr\{X_1(t), X_2(t) = i, j\} = (1 - \rho_1)\rho_1^i(1 - \rho_2)\rho_2^j,$$

where $\rho_1 = \lambda Q/\mu_1$ and $\rho_2 = \lambda/\mu_2$. Note that this independence applies to the states of system 1 and 2 at the same instant t . The processes are not independent, and, for example, $X_1(t)$ and $X_2(t + \tau)$ for $\tau > 0$ are not independent.

d) What is the probability that the first job to leave system 1 after time t is the same as the first job that entered the entire system after time t ?

Solution: The first job out of system 1 after t is the same as the first job into system 1 after t if and only if system 1 is empty at time t . This is also the same as the first job into the overall system if the first arrival to the entire system goes into system 1. The probability of both of these events (which are independent) is thus $P_1 = Q(1 - \rho_1)$.

e) What is the probability that the first job to leave system 2 after time t both passed through system 1 and arrived at system 1 after time t .

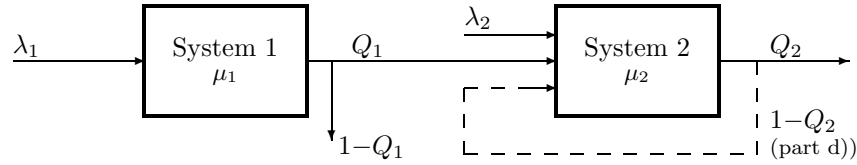
Solution: Let P_2 be the probability of this event. This requires, first, that the event in (d) is satisfied, second that system 2 is empty at time t , and, third, that the first job to bypass system 1 after the first arrival to system 1 occurs after the service of that first arrival.

This third event is the event that an exponential rv of rate μ_1 has a smaller sample value

than one of rate $\lambda(1 - Q)$. The probability of this is $\mu_1/(\mu_1 + \lambda(1 - Q))$. Thus

$$P_2 = P_1(1 - \rho_2) \frac{\mu_1}{\mu_1 + \lambda(1 - Q)}.$$

Exercise 7.17: Consider the following combined queueing system. The first queue system is M/M/1 with service rate μ_1 . The second queue system has IID exponentially distributed service times with rate μ_2 . Each departure from system 1 independently leaves the entire system with probability $1 - Q_1$. System 2 has an additional Poisson input of rate λ_2 , independent of inputs and outputs from the first system. Each departure from the second system independently leaves the combined system with probability Q_2 and re-enters system 2 with probability $1 - Q_2$. For (a), (b), (c) assume that $Q_2 = 1$ (i.e., there is no feedback).



a) Characterize the process of departures from system 1 that enter system 2 and characterize the overall process of arrivals to system 2.

Solution: The departures from system 1 are Poisson of rate λ_1 from Burke's theorem. After splitting into two streams, the stream entering system 2 is Poisson of rate $Q_1\lambda_1$. The additional Poisson input to system 2 is independent of the first, and thus the combined arrivals to system 2 are Poisson of rate $Q_1\lambda_1 + \lambda_2$.

b) Assuming FCFS service in each system, find the steady state distribution of time that a customer spends in each system.

Solution: As determined in Example 2.3.3 and rederived in Exercise 7.12, the PDF of system time in system 1 is $f_1(t) = (\mu_1 - \lambda_1) \exp(-(\mu_1 - \lambda_1)t)$. For system 2, (since it is M/M/1), the PDF is $f_2(t) = (\mu_2 - \lambda_{2T}) \exp[-(\mu_2 - \lambda_{2T})t]$, where $\lambda_{2T} = (Q_1\lambda_1 + \lambda_2)$.

c) For a customer that goes through both systems, show why the time in each system is independent of that in the other; find the distribution of the combined system time for such a customer.

Solution: This is explained in the discussion of Burke's theorem, but we repeat it briefly here. By (c) of Burke's theorem, the arrival time (and thus the service time) corresponding to a departure from system 1 at time t is independent of the departures before t . Those departures from system 1 that enter system 2 (plus the independent arrivals at rate λ_2 and the service times of system 2) determine the state of system 2 at t , which is thus independent of the system 1 service time. Thus the overall service time density, $f_T(t)$ is the convolution of the individual service time densities.

$$f_T(t) = \frac{\alpha\beta}{\alpha - \beta} [e^{-\beta t} - e^{-\alpha t}],$$

where $\alpha = \mu_1 - \lambda_1$ and $\beta = \mu_2 - \lambda_{2T}$. If $\alpha = \beta$, this must be replaced with the Erlang density of order 2.

d) Now assume that $Q_2 < 1$. Is the departure process from the combined system Poisson? Which of the three input processes to system 2 are Poisson? Which of the input processes are independent? Explain your reasoning, but do not attempt formal proofs.

Solution: If we look only at system 2 with the feedback loop around it, we see exogenous Poisson inputs at rate $\lambda_{2T} = \lambda_1 Q_1 + \lambda_2$. This is the same system as in Figure 7.14. From the analysis there, the overall output (that at rate $Q_2 \lambda_{2T}$) is Poisson. The overall input to system 2, including the feedback link is not Poisson (see Figure 7.15).

Exercise 7.18: Suppose a Markov chain with transition probabilities $\{P_{ij}\}$ is reversible. Suppose we change the transition probabilities out of state 0 from $\{P_{0j}; j \geq 0\}$ to $\{P'_{0j}; j \geq 0\}$. Assuming that all P_{ij} for $i \neq 0$ are unchanged, what is the most general way in which we can choose $\{P'_{0j}; j \geq 0\}$ so as to maintain reversibility? Your answer should be explicit about how the steady-state probabilities $\{\pi_i; i \geq 0\}$ are changed. Your answer should also indicate what this problem has to do with uniformization of reversible Markov processes, if anything. Hint: Given P_{ij} for all i, j , a single additional parameter will suffice to specify P'_{0j} for all j .

Solution: Let π'_i denote the new steady-state probabilities. To maintain reversibility, $\pi'_0 P'_{0j} = \pi'_j P_{j0}$ for all $j \neq 0$ and $\pi'_i P_{ij} = \pi'_j P_{ji}$ for all $i, j \neq 0$. We consider the latter set of equations first and divide each of these equations by the original reversibility conditions, $\pi_i P_{ij} = \pi_j P_{ji}$. This gives us $\pi'_i / \pi_i = \pi'_j / \pi_j$ for all $i, j \neq 0$. This means that all the steady-state probabilities for $i \neq 0$ scale by the same amount, *i.e.*, that for some α , $\pi'_i = \alpha \pi_i$ for all $i \neq 0$. By summing over i , we see that $1 - \pi'_0 = \alpha(1 - \pi_0)$.

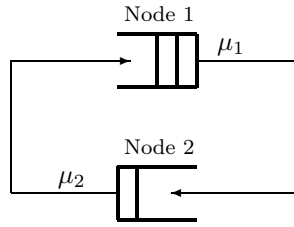
We can substitute this equation into $\pi'_0 P'_{0j} = \pi'_j P_{j0}$ to get

$$P'_{0j} = \frac{\pi'_j P_{j0}}{\pi'_0} = \frac{\alpha \pi_j P_{j0}}{1 - \alpha + \alpha \pi_0} = \frac{\alpha \pi_0 P_{0j}}{1 - \alpha + \alpha \pi_0}.$$

Thus the single parameter α , along with the original probabilities, specifies all of the changed transition probabilities P'_{0j} , and those changes simply scale each P_{0j} , $j \neq 0$, up or down in proportion. Note also that instead of specifying the parameter α , we could simply specify the new desired π'_0 or the desired scaling of P_{0j} , $j \neq 0$. Finally, since $\sum_j P_{0j} = 1$ where the sum includes $j = 0$, specifying the change in P_{00} specifies all the other changes.

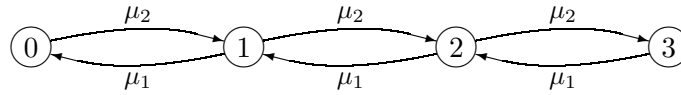
What this has to do with normalization is that when one inserts self transition rates into a given state, say 0, of a reversible Markov process, the embedded Markov chain changes in exactly this way. The process transitions q_{ij} except for $i=j=0$ remain the same, but the $\{P_{0j}; j \neq 0\}$ in the embedded chain each change in proportion to accomodate the new P_{00} . Each such change preserves the reversibility of the embedded chain, so that if a Markov process is reversible, the uniformized chain is also reversible.

Exercise 7.19: Consider the closed queueing network in the figure below. There are three customers who are doomed forever to cycle between node 1 and node 2. Both nodes use FCFS service and have exponentially distributed IID service times. The service times at one node are also independent of those at the other node and are independent of the customer being served. The server at node i has mean service time $1/\mu_i$, $i = 1, 2$. Assume to be specific that $\mu_2 < \mu_1$.



a) The system can be represented by a four state Markov process. Draw its graphical representation and label it with the individual states and the transition rates between them.

Solution: The three customers are statistically identical, so we can take the number of customers in node 1 (which can be 0, 1, 2, 3) to be the state. In states 1, 2, 3, departures from node 1 take place at rate μ_1 . In states 0, 1, 2, departures occur from node 2 at rate μ_2 , and these serve as arrivals to node 1. Thus the process has the following graphical representation.



b) Find the steady-state probability of each state.

Solution: Perhaps surprisingly, this is the same as an M/M/1 queue in which arrivals are turned away when $X(t) = 3$. In the queue representation, we lose the identity of the three customers, but we need not keep track of them since they are statistically identical. As we see in the rest of the exercise, the customer identities can be tracked supplementally, since the arrivals to node 1 rotate from customer 1 to 2 to 3. Thus each third arrival in the queue representation corresponds to the same customer.

Let $\rho = \mu_2/\mu_1$. Then the steady-state process probabilities are $p_1 = p_0\rho$; $p_2 = \rho^2 p_0$, $p_3 = \rho^3 p_0$, where $p_0 = 1/(1 + \rho + \rho^2 + \rho^3)$.

c) Find the time-average rate at which customers leave node 1.

Solution: Each customer departure from node 1 corresponds to a downward transition from the queue representation, and thus occurs at rate μ_1 from each state except 0 of the queue. Thus in steady state, customers leave node 1 at rate $r = (1 - p_0)\mu_1$. This is also the time-average rate at which customers leave node 1.

d) Find the time-average rate at which a given customer cycles through the system.

Solution: Each third departure from node 1 is a departure of the same customer, and this corresponds to each third downward transition in the queue. Thus the departure rate of a given customer from queue 1, which is the same as the rotation rate of that customer, is $r/3$.

e) Is the Markov process reversible? Suppose that the backward Markov process is interpreted as a closed queueing network. What does a departure from node 1 in the forward process correspond to in the backward process? Can the transitions of a single customer in the forward process be associated with transitions of a single customer in the backward process?

Solution: The queueing process is reversible since it is a birth-death process. The backward process from the two node system, however, views each forward departure from node 1 as an arrival to node 1, *i.e.*, a departure from node 2. Customers 1, 2, 3 then rotate in backward order, 1, 3, 2, in the backward system. Thus if one wants to maintain the customer order, the system is not reversible. The system, viewed as a closed queueing system in the sense of Section 7.7.1, is in essence the same as the queueing network here and also abstracts away the individual customer identities, so it is again reversible.

Exercise 7.20: Consider an M/G/1 queueing system with last come first serve (LCFS) pre-emptive resume service. That is, customers arrive according to a Poisson process of rate λ . A newly arriving customer interrupts the customer in service and enters service itself. When a customer is finished, it leaves the system and the customer that had been interrupted by the departing customer resumes service from where it had left off. For example, if customer 1 arrives at time 0 and requires 2 units of service, and customer 2 arrives at time 1 and requires 1 unit of service, then customer 1 is served from time 0 to 1; customer 2 is served from time 1 to 2 and leaves the system, and then customer 1 completes service from time 2 to 3. Let X_i be the service time required by the i^{th} customer; the X_i are IID random variables with expected value $E[X]$; they are independent of customer arrival times. Assume $\lambda E[X] < 1$.

a) Find the mean time between busy periods (*i.e.*, the time until a new arrival occurs after the system becomes empty).

Solution: The arrival process is Poisson, and thus the residual life until the next arrival, starting at any given time, is exponential at rate λ . Thus the mean period separating busy periods is $1/\lambda$.

b) Find the time-average fraction of time that the system is busy.

Solution: Applying Little's law to the service facility, the time-average number of customers in service (which is the fraction of time the server is busy) is equal to λ times the sample-path average service time per customer, $E[X]$. Since the server is busy if and only if the system is busy, $\lambda E[X]$ is the time-average fraction of time the system is busy.

c) Find the mean duration, $E[B]$, of a busy period. Hint: use (a) and b).

Solution: Consider the beginning of each busy period as a renewal and consider a reward rate of one unit while the system is busy. Then the mean reward per renewal is $E[B]$ and the mean duration of a renewal period is $E[B] + 1/\lambda$. The ratio of these, by the renewal reward theorem, is $\lambda E[X]$. Thus

$$\lambda E[X] = \frac{E[B]}{E[B] + 1/\lambda}; \quad E[B] = \frac{E[X]}{1 - \lambda E[X]}. \quad (\text{A.146})$$

d) Explain briefly why the customer that starts a busy period remains in the system for the entire busy period; use this to find the expected system time of a customer given that that customer arrives when the system is empty.

Solution: The customers in the system form a stack (where the top of the stack is regarded as the customer in service. If a second customer arrives while the first is in service, it is placed above the first in the stack, and the first can never complete service until after the second. Successive arrivals within the busy period are also above the first (and also above the second if it is still there), so the busy period finishes when the first customer finishes.

e) Is there any statistical dependence between the system time of a given customer (*i.e.*, the time from the customer's arrival until departure) and the number of customers in the system when the given customer arrives?

Solution: There is no such dependence. The newly arriving customer is on the stack above all previously arrived but not completed customers, so the new customer's system time depends only on future arrivals and their service times, all of which are independent of past arrivals and service times.

f) Show that a customer's expected system time is equal to $E[B]$. Hint: Look carefully at your answers to d) and e).

Solution: Since a customer's system time is independent of the number in the system when it arrives, the customer's system time has the same expectation as a customer who finds the system empty. Thus $E[B]$ is the expected system time of each arrival.

g) Let C be the expected system time of a customer conditional on the service time X of that customer being 1. Find (in terms of C) the expected system time of a customer conditional on $X = 2$; (Hint: compare a customer with $X = 2$ to two customers with $X = 1$ each); repeat for arbitrary $X = x$.

Solution: For a customer with service requirement 2, consider the time required to complete one unit of service. This time depends only on the epochs and service requirements of future arrivals, and is thus identically distributed to that of a customer with one unit of service requirement. At the end of that unit of service, the second unit of service is on the top of the stack, and its expected system time is the same as a new customer entering at that instant. The system time of the original customer with service requirement 2 is the sum, $2C$, of these two system times, each of which is C .

Similarly, for integer m service requirement, the expected system time is mC . Doing the same argument backwards, if a system has service requirement $1/k$, an arrival with one unit of service requirement has k times as much expected system time. Thus for a service requirement equal to any rational number m/k , the expected system time is mC/k . Finally, since the expected system time is monotonic in the service requirement, a customer with service requirement x has an expected system time equal to xC .

h) Find the constant C . Hint: use f) and g); don't do any tedious calculations.

Solution: Since xC is the expected service time conditional on $X = x$, the unconditional expected service time is $CE[X] = E[B]$. Using (A.146), we then have $C = 1/(1 - \lambda\bar{X})$. Note that the LCFS service discipline avoids the slow truck effect of FCFS for MG1 queues. The reason is that only customers already in the system are held up by a customer with large service requirement. This number is independent of the service requirements for LCFS, whereas for FCFS, the expected number held up by a long service-time customer is proportional to the service requirement of that long service-time customer.

Exercise 7.21: Consider a queueing system with two classes of customers, type A and type B . Type A customers arrive according to a Poisson process of rate λ_A and type B customers arrive according to an independent Poisson process of rate λ_B . The queue has a FCFS server with exponentially distributed IID service times of rate $\mu > \lambda_A + \lambda_B$. Characterize the departure process of class A customers; explain carefully. Hint: Consider the combined arrival process and be judicious about how to select between A and

B types of customers.

Solution: The combined arrival process is Poisson, since it is the addition of two independent Poisson processes. The service discipline serves customers in order of arrival and therefore the departure process is the same as that for a single Poisson input process. Thus the output process is Poisson also. Since the arriving customers are independently A type with probability $\lambda_A/(\lambda_A + \lambda_B)$, the departing customers, being in the same order as the arriving customers, are independently A type with the same probability. Thus the departing A customers form a Poisson process of rate λ_A , and this process is independent of the type B departures.

Exercise 7.22: Consider a pre-emptive resume last come first serve (LCFS) queueing system with two classes of customers. Type A customer arrivals are Poisson with rate λ_A and Type B customer arrivals are Poisson with rate λ_B . The service time for type A customers is exponential with rate μ_A and that for type B is exponential with rate μ_B . Each service time is independent of all other service times and of all arrival epochs.

Define the “state” of the system at time t by the string of customer types in the system at t , in order of arrival. Thus state AB means that the system contains two customers, one of type A and the other of type B ; the type B customer arrived later, so is in service. The set of possible states arising from transitions out of AB is as follows:

ABA if another type A arrives.

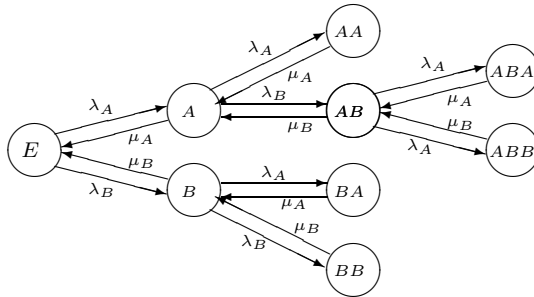
ABB if another type B arrives.

A if the customer in service (B) departs.

Note that whenever a customer completes service, the next most recently arrived resumes service, so the state changes by eliminating the final element in the string.

a) Draw a graph for the states of the process, showing all states with 2 or fewer customers and a couple of states with 3 customers (label the empty state as E). Draw an arrow, labelled by the rate, for each state transition. Explain why these are states in a Markov process.

Solution:



For each state, the interval to the next transition is exponential and the next state, conditional on the present state is independent of all earlier states. Thus by definition it is a Markov process.

b) Is this process reversible? Explain. Assume positive recurrence. Hint: If there is a transition from one state S to another state S' , how is the number of transitions from S to S' related to the number from S' to S ?

Solution: It is reversible for the same reason as a birth/death chain. The graph (viewing each pair of transitions joining two nodes as a single link) is a tree, and thus each transition in one direction on a link, must be followed by a transition in the opposite direction before the next transition in the first direction. Thus if states s and s' are joined by a link, the steady-state process probabilities are related by $p_s q_{ss'} = p_{s'} q_{s's}$.

Note that this is an important generalization of the result that a birth-death process is reversible. Any process where the graph of nonzero transitions has the form of a tree is also reversible under the same positive-recurrence conditions as a birth-death chain.

c) Characterize the process of type A departures from the system (*i.e.*, are they Poisson?; do they form a renewal process?; at what rate do they depart?; etc.)

Solution: Since the process is reversible, the backward process is statistically identical to the forward process. Type A departures in the forward process correspond to type A arrivals in the backward process, and thus the process of A departures is Poisson with rate λ_A . Similarly, type B departures are Poisson with rate λ_B and are independent of type A departures.

d) Express the steady-state probability $\Pr\{A\}$ of state A in terms of the probability of the empty state $\Pr\{E\}$. Find the probability $\Pr\{AB\}$ and the probability $\Pr\{ABBA\}$ in terms of $\Pr\{E\}$. Use the notation $\rho_A = \lambda_A/\mu_A$ and $\rho_B = \lambda_B/\mu_B$.

Solution: From (c), we have $\Pr\{E\} \lambda_A = \Pr\{A\} \mu_A$, so $\Pr\{A\} = \rho_A \Pr\{E\}$. Similarly, $\Pr\{AB\} = \rho_B \Pr\{A\}$, so $\Pr\{AB\} = \rho_B \rho_A \Pr\{E\}$. In the same way, $\Pr\{ABBA\} = \rho_A^2 \rho_B^2 \Pr\{E\}$.

e) Let Q_n be the probability of n customers in the system, as a function of $Q_0 = \Pr\{E\}$. Show that $Q_n = (1 - \rho) \rho^n$ where $\rho = \rho_A + \rho_B$.

Solution: Note that

$$Q_1 = \Pr\{A\} + \Pr\{B\} = \rho_A Q_0 + \rho_B Q_0 = \rho Q_0.$$

Similarly, consider any state s with n customers in the system. Let $\Pr\{s\}$ be the probability of s . The probability of state sA (where sA is the state consisting of s followed by A) is $\rho_A \Pr\{s\}$. It follows that the probability of s followed by the single letter A or B is $\rho \Pr\{s\}$. Summing this over all states for which n customers are in the system, $Q_{n+1} = \rho Q_n$. By induction then, $Q_n = Q_0 \rho^n$. Finally since $\sum_n Q_n = 1$, we have $Q_n = (1 - \rho) \rho^n$.

Exercise 7.23: a) Generalize Exercise 7.22 to the case in which there are m types of customers, each with independent Poisson arrivals and each with independent exponential service times. Let λ_i and μ_i be the arrival rate and service rate respectively of the i^{th} user. Let $\rho_i = \lambda_i/\mu_i$ and assume that $\rho = \rho_1 + \rho_2 + \dots + \rho_m < 1$. In particular, show, as before that the probability of n customers in the system is $Q_n = (1 - \rho) \rho^n$ for $0 \leq n < \infty$.

Solution: The argument is exactly the same as that in Exercise 7.22.

b) View the customers in (a) as a single type of customer with Poisson arrivals of rate $\lambda = \sum_i \lambda_i$ and with a service density $\sum_i (\lambda_i/\lambda) \mu_i \exp(-\mu_i x)$. Show that the expected service time is ρ/λ . Note that you have shown that, if a service distribution can be represented as a weighted sum of exponentials, then the

distribution of customers in the system for LCFS service is the same as for the M/M/1 queue with equal mean service time.

Solution: Note that λ_i/λ is the probability that an arrival will be a type i arrival. Thus the expected service time is $(1/\mu) = \sum_i (\lambda_i/\lambda)(1/\mu_i) = \rho/\lambda$. Thus $\rho = \lambda/\mu$. The result in (a) then says that the distribution of the number of customers in the system is the same as that of customers of an M/M/1 queue with the given λ and μ .

Exercise 7.24: Consider a k node Jackson-type network with the modification that each node i has s servers rather than one server. Each server at i has an exponentially distributed service time with rate μ_i . The exogenous input rate to node i is $\rho_i = \lambda_0 Q_{i0}$ and each output from i is switched to j with probability Q_{ij} and switched out of the system with probability Q_{i0} (as in the text). Let $\lambda_i, 1 \leq i \leq k$, be the solution, for given λ_0 , to

$$\lambda_j = \sum_{i=0}^k \lambda_i Q_{ij}; \quad 1 \leq j \leq k,$$

and assume that $\lambda_i < s\mu_i; 1 \leq i \leq k$. Show that the steady-state probability of state $\mathbf{m}$ is

$$\Pr\{\mathbf{m}\} = \prod_{i=1}^k p_i(m_i), \quad (\text{A.147})$$

where $p_i(m_i)$ is the probability of state m_i in an (M, M, s) queue. Hint: simply extend the argument in the text to the multiple server case.

Solution: The Jackson network with $M/M/s$ servers is a special case of the generalization of Jackson networks in which the service rate at each node is still exponential, but exponential with a rate depending on the number in service. The steady-state distribution in (A.147) is given in (7.76). In the special case for $M/M/s$ queues, ρ_{ij} in (7.76) and (7.77) is given by $\rho_{ij} = \lambda_i/\mu_{ij}$ where $\mu_{ij} = j\mu_i$ for $j \leq s$ and $\mu_{ij} = s\mu_i$ for $j \geq s$.

It is important to observe here that the solution for the overall arrival rate at each node λ_i is independent of the number of servers (assuming that the network is stable, *i.e.*, that the arrival rate is less than $s\mu_i$ at each node). To be more explicit, $\{Q_{ij}\}$ forms a stochastic matrix which has a unique steady-state probability distribution $\boldsymbol{\pi}$. This can be scaled to the arrival rate λ_i at each node by the fact that λ_0 is known.

Exercise 7.25: Suppose a Markov process with the set of states A is reversible and has steady-state probabilities $p_i; i \in A$. Let B be a subset of A and assume that the process is changed by setting $q_{ij} = 0$ for all $i \in B, j \notin B$. Assuming that the new process (starting in B and remaining in B) is irreducible, show that the new process is reversible and find its steady-state probabilities.

Solution: For the given transition rates $\{Q_{ij}\}$ on A , the modified transition rates on B are $q'_{ij} = q_{ij}$ for $i, j \in B$ and $q'_{ij} = 0$ otherwise. For $i \in B$, define $p'_i = p_i / \sum_{j \in B} p_j$. Then, for all $i, j \in B$, we have $p'_i q'_{ij} = p'_j q'_{ji}$. Since $\sum_i p_i \nu_i$ is clearly finite, Theorem 7.6.2 asserts that the new Markov process on B is reversible.

This, along with a number of other exercises here, emphasizes the importance of the guessing theorem, Theorem 7.6.3, with its special case for reversible processes in Theorem 7.6.2.

Exercise 7.26: Consider a queueing system with two classes of customers. Type A customer arrivals are Poisson with rate λ_A and Type B customer arrivals are Poisson with rate λ_B . The service time for type

A customers is exponential with rate μ_A and that for type B is exponential with rate μ_B . Each service time is independent of all other service times and of all arrival epochs.

a) First assume there are infinitely many identical servers, and each new arrival immediately enters an idle server and begins service. Let the state of the system be (i, j) where i and j are the numbers of type A and B customers respectively in service. Draw a graph of the state transitions for $i \leq 2, j \leq 2$. Find the steady-state PMF, $\{p(i, j); i, j \geq 0\}$, for the Markov process. Hint: Note that the type A and type B customers do not interact.

Solution: It is clear that type A and B customers do not interact, since the arrivals are independent, there is no queueing, and each new arrival has a service time μ_A or μ_B , depending on the type of arrival. Theorem 7.6.2 (the guessing theorem) verifies this. Thus $p(i, j) = p_A(i)p_B(j)$ where $p_A(i)$ is the PDF of the number of customers in the system (in steady state) for an $M/M/\infty$ queue for system A and $p_B(j)$ is the PDF for system B . From the $M/M/\infty$ case of Exercise 6.12, $p_A(i) = \rho_A^i \exp(-\rho_A)/i!$ and $p_B(i) = \rho_B^i \exp(-\rho_B)/i!$ where $\rho_A = \lambda_A/\mu_A$ and $\rho_B = \lambda_B/\mu_B$. Thus, with $\rho = \rho_A + \rho_B$,

$$p(i, j) = \frac{\rho_A^i \rho_B^j \exp -\rho}{i!j!}.$$

b) Assume for the rest of the exercise that there is some finite number m of servers. Customers who arrive when all servers are occupied are turned away. Find the steady-state PMF, $\{p(i, j); i, j \geq 0, i + j \leq m\}$, in terms of $p(0, 0)$ for this Markov process. Hint: Combine (a) with the result of Exercise 7.25.

Solution: The Markov process in (a) is reversible since $p(i, j)\lambda_A = p(i+1, j)\mu_A$ with a similar relation for B . If the number of servers is reduced to m , the corresponding Markov process is the same as that in (a) except that all transitions from states (m, j) to $(m+1, j)$ are removed, all transitions from states (i, m) to $(i, m+1)$ are removed, and we consider only the class of states with $i, j \leq m$. Using the result from Exercise 7.25, the resulting process is reversible and the steady-state probabilities are scaled versions of those in (a). Thus, letting $p_m(i, j)$ be the resulting steady-state probabilities,

$$p_m(i, j) = \frac{\rho_A^i \rho_B^j p_m(0, 0)}{i!j!}.$$

c) Let Q_n be the probability that there are n customers in service at some given time in steady state. Show that $Q_n = p(0, 0)\rho^n/n!$ for $0 \leq n \leq m$ where $\rho = \rho_A + \rho_B$, $\rho_A = \lambda_A/\mu_A$, and $\rho_B = \lambda_B/\mu_B$. Solve for $p(0, 0)$.

Solution: We use convolution, getting

$$Q_n = \sum_{i=0}^n p_m(i, n-i) = \sum_{i=0}^n \frac{\rho_A^i \rho_B^{n-i} p_m(0, 0)}{i!(n-i)!}.$$

Using the binomial expansion for $(\rho_A + \rho_B)^n$, this is equal to $p_m(0, 0)\rho^n/n!$.

Using the fact that $\sum_{n=0}^m Q_n = 1$, we see that

$$p_m(0, 0) = 1 / \left(\sum_{n=0}^m \rho^n / n! \right).$$

Exercise 7.27: a) Generalize Exercise 7.26 to the case in which there are K types of customers, each with independent Poisson arrivals and each with independent exponential service times. Let λ_k and μ_k be the arrival rate and service rate respectively for the k th user type, $1 \leq k \leq K$. Let $\rho_k = \lambda_k/\mu_k$ and $\rho = \rho_1 + \rho_2 + \dots + \rho_K$. In particular, show, as before, that the probability of n customers in the system is $Q_n = p(0, \dots, 0)\rho^n/n!$ for $0 \leq n \leq m$.

Solution: This is an exercise in manipulating a large number of indices, and really does little new beyond Exercise 7.26. Let a state be represented by $(i_1, \dots, i_k, \dots, i_K)$, where for $1 \leq k \leq K$, i_k is the number of type k customers in the system. The steady-state process probability, $p_m(i_1, \dots, i_K)$ is then

$$p_m(i_1, \dots, i_K) = p_m(0, \dots, 0) \prod_{k=1}^K \frac{\rho_k^{i_k}}{i_k!}.$$

This is derived by starting with $m = \infty$, where each class of customer is independent of the other classes by the same argument as in Exercise 6.26. Then Exercise 7.25 is used to see that the finite m case is a truncation of $m = \infty$. It is thus reversible and simply a scaling of the $m = \infty$ case over the truncated set of states. We next sum over all states for which the sum of customers of each type is n . The multinomial expansion works in the same way as the binomial expansion in Exercise 7.26 to yield

$$Q_n = p_m(0, \dots, 0)\rho^n/n! = (1 - \rho)\rho^n/n! \quad \text{where } p_m(0, 0) = 1/\left(\sum_{n \leq m} \rho^n/n!\right).$$

b) View the customers in (a) as a single type of customers with Poisson arrivals of rate $\lambda = \sum_k \lambda_k$ and with a service density $\sum_k (\lambda_k/\lambda)\mu_k \exp(-\mu_k x)$. Show that the expected service time is ρ/λ . Note that what you have shown is that, if a service distribution can be represented as a weighted sum of exponentials, then the distribution of customers in the system is the same as for the M/M/m/m queue with equal mean service time.

Solution: In (a), the type k customers have exponential service at rate μ_k and thus density $\mu_k \exp(-\mu_k t)$ and mean service time $1/\mu_k$. Viewing the customers as a Poisson splitting of a Poisson process of rate λ , The marginal density of the service requirement of a customer (averaged over types) is $\sum_k (\lambda_k/\lambda)\mu_k \exp(-\mu_k x)$. The mean service requirement is then the average of the mean service requirement over types, *i.e.*, $\sum_k (\lambda_k/\lambda)(1/\mu_k) = \rho/\lambda$.

Exercise 7.28: Consider a sampled-time M/D/m/m queueing system. The arrival process is Bernoulli with probability $\lambda \ll 1$ of arrival in each time unit. There are m servers; each arrival enters a server if a server is not busy and otherwise the arrival is discarded. If an arrival enters a server, it keeps the server busy for d units of time and then departs; d is some integer constant and is the same for each server.

Let n , $0 \leq n \leq m$ be the number of customers in service at a given time and let x_i be the number of time units that the i^{th} of those n customers (in order of arrival) has been in service. Thus the state of the system can be taken as $(n, \mathbf{x}) = (n, x_1, x_2, \dots, x_n)$ where $0 \leq n \leq m$ and $1 \leq x_1 < x_2 < \dots < x_n \leq d$.

Let $A(n, \mathbf{x})$ denote the next state if the present state is $(n, \mathbf{x})$ and a new arrival enters service. That is,

$$A(n, \mathbf{x}) = (n+1, 1, x_1+1, x_2+1, \dots, x_n+1) \quad \text{for } n < m \text{ and } x_n < d \quad (\text{A.148})$$

$$A(n, \mathbf{x}) = (n, 1, x_1+1, x_2+1, \dots, x_{n-1}+1) \quad \text{for } n \leq m \text{ and } x_n = d. \quad (\text{A.149})$$

That is, the new customer receives one unit of service by the next state time, and all the old customers receive one additional unit of service. If the oldest customer has received d units of service, then it leaves

the system by the next state time. Note that it is possible for a customer with d units of service at the present time to leave the system and be replaced by an arrival at the present time (*i.e.*, (A.149) with $n = m$, $x_n = d$). Let $B(n, \mathbf{x})$ denote the next state if either no arrival occurs or if a new arrival is discarded.

$$B(n, \mathbf{x}) = (n, x_1 + 1, x_2 + 1, \dots, x_n + 1) \quad \text{for } x_n < d \quad (\text{A.150})$$

$$B(n, \mathbf{x}) = (n - 1, x_1 + 1, x_2 + 1, \dots, x_{n-1} + 1) \quad \text{for } x_n = d. \quad (\text{A.151})$$

a) Hypothesize that the *backward* chain for this system is also a sampled-time M/D/m/m queueing system, but that the state $(n, x_1, \dots, x_n)$, $(0 \leq n \leq m, 1 \leq x_1 < x_2 < \dots < x_n \leq d)$ has a different interpretation: n is the number of customers as before, but x_i is now the remaining service required by customer i . Explain how this hypothesis leads to the following steady-state equations:

$$\lambda \pi_{n, \mathbf{x}} = (1 - \lambda) \pi_{A(n, \mathbf{x})} \quad ; n < m, x_n < d \quad (\text{A.152})$$

$$\lambda \pi_{n, \mathbf{x}} = \lambda \pi_{A(n, \mathbf{x})} \quad ; n \leq m, x_n = d \quad (\text{A.153})$$

$$(1 - \lambda) \pi_{n, \mathbf{x}} = \lambda \pi_{B(n, \mathbf{x})} \quad ; n \leq m, x_n = d \quad (\text{A.154})$$

$$(1 - \lambda) \pi_{n, \mathbf{x}} = (1 - \lambda) \pi_{B(n, \mathbf{x})} \quad ; n \leq m, x_n < d. \quad (\text{A.155})$$

Solution: Given a state $(n, \mathbf{x})$ at time t from which an arrival can be admitted, λ is the probability of an arrival, and thus a transition into $A(n, \mathbf{x})$ at time $t+1$ occurs if there is an arrival from n to $n+1$. Since $A(n, \mathbf{x})$ is the result of an arrival, the new customer has received only one unit of service. Thus for the backward system in state $A(n, \mathbf{x})$ at time $t+1$, in the backward system, that customer is going to depart in the transition from $t+1$ to t .

Now consider the case in which $x_n < d$. Then, the number of customers in the system in state $A(n, \mathbf{x})$ is $n + 1$. In order for the backward system to transit from $A(n, \mathbf{x})$ at $t+1$ to $(n, \mathbf{x})$ at time t , it is necessary and sufficient that no backward arrival occur. Thus the probability of a backward transition from $A(n, \mathbf{x})$ to $(n, \mathbf{x})$ is $1 - \lambda$, justifying (A.152) under the given hypothesis.

Next consider the case $x_n = d$. Then one customer arrives and another departs in making a transition from $(n, \mathbf{x})$ to $A(n, \mathbf{x})$. In the backward system, then there is the necessity for a backward arrival, as well as the backward departure we have already justified. Thus the probability of this backward transition is λ , justifying (A.153). (A.154) and (A.155) are verified in the same way.

b) Using this hypothesis, find $\pi_{n, \mathbf{x}}$ in terms of π_0 , where π_0 is the probability of an empty system. Hint: Use (A.154) and (A.155); your answer should depend on n , but not $\mathbf{x}$.

Solution: Let $(n, \mathbf{x})$ be any fixed state and consider a (forward) sample path in which no arrivals occur for d time units. At the end of these d time units, the system must be empty. Use (A.154) for each transition in which a departure occurs and use (A.155) where a departure does not occur (at most one departure can occur at each time unit). We then have

$$\pi_{n, \mathbf{x}} = \left(\frac{\lambda}{1 - \lambda} \right)^n \pi_0. \quad (\text{A.156})$$

c) Verify that the above hypothesis is correct.

Solution: Substituting (A.156) into both sides of (A.152) to (A.155), we see that each is satisfied for all states and both a forward arrival (if possible) or non-arrival. These equations

also include all backward transitions of backward arrivals (if possible) and backward non-arrivals. Thus by the guessing theorem, Theorem 7.6.3, these equations provide the unique solution for the steady-state probabilities.

d) Find an expression for π_0 .

Solution: Each state $(n, \mathbf{x})$ for a given n can be alternatively viewed as a d -vector of 1's and 0's where the i th component is 1 if one of the customers, say the j th in order in $\mathbf{x}$, satisfies $x_j = i$. Note that at most one arrival can occur in each time unit, so each customer has received a different amount of service. Thus there are $\binom{d}{n}$ different states $(n, \mathbf{x})$ for each given n . Summing (A.156) over $\mathbf{x}$, the probability p_n that there are n customers in the system is

$$p_n = \pi_0 \binom{d}{n} \left(\frac{\lambda}{1-\lambda} \right)^n. \quad (\text{A.157})$$

Summing this over n to find π_0 ,

$$\sum_{n=1}^m \pi_0 \binom{d}{n} \left(\frac{\lambda}{1-\lambda} \right)^n = 1; \quad \pi_0 = \left[\sum_{n=1}^m \binom{d}{n} \left(\frac{\lambda}{1-\lambda} \right)^n \right]^{-1}.$$

e) Find an expression for the steady-state probability that an arriving customer is discarded.

Solution: This can be interpreted either as the probability that an arrival occurs at a given time and is discarded or as the probability it is discarded given that it arrives at a given time. The latter is the probability of the set of states $(n, \mathbf{x})$ for which $n = m$ and $x_n < d$. There are $\binom{d-1}{m}$ such states, so the probability of turning away a customer is

$$\pi_0 \binom{d-1}{m} \left(\frac{\lambda}{1-\lambda} \right)^m.$$

Multiplying by λ , we get the probability that a customer arrives at a given time and is discarded.

It is interesting to look at this in the limit as the time increment goes to 0. Let Λ be the arrival rate in arrivals per second, let δ be the time increment, and $\lambda = \Lambda\delta$ be the arrival rate per time increment. Let D be the service duration in seconds and $d = D/\delta$ be the service duration in time increments. (A.157) then becomes

$$p_n = \pi_0 \binom{D/\delta}{n} \left(\frac{\delta\Lambda}{1-\delta\Lambda} \right)^n \longrightarrow \frac{\pi_0 D^n \Lambda^n}{n!}.$$

Recognizing that D is the expected service time, we see that this is the same result as the $M/M/m/m$ queue. Exercise 7.27 shows that the same result holds if the service distribution is a sum of exponentials. The same result holds in general for $M/G/m/m$. This can be derived as a limit of discrete distributions using the same arguments as here, with much more complex notation and tricky limiting operations.

Exercise 7.29: A taxi alternates between three locations. When it reaches location 1 it is equally likely to go next to either 2 or 3. When it reaches 2 it will next go to 1 with probability 1/3 and to 3 with

probability $2/3$. From 3 it always goes to 1. The mean time between locations i and j are $t_{12} = 20$, $t_{13} = 30$, $t_{23} = 30$. Assume exponential service and assume $t_{ij} = t_{ji}$.

Find the (limiting) probability that the taxi's most recent stop was at location i , $i = 1, 2, 3$.

What is the (limiting) probability that the taxi is heading for location 2?

What fraction of time is the taxi traveling from 2 to 3. Note: Upon arrival at a location the taxi immediately departs.

Solution: The taxi location is best modeled as a semi-Markov process $\{X(t); t > 0\}$ where $X(t)$ is the most recent site visited up to time t . (it cannot be modeled as a Markov process since the holding time in each state i depends both on i and the next state j). For the embedded Markov chain, we are given that $P_{12} = 1/2$, $P_{21} = 1/3$, $P_{13} = 1/2$, $P_{31} = 1$, $P_{23} = 2/3$. From this, the steady-state probabilities in the embedded Markov chain are

$$\pi_1 = 3/7, \quad \pi_2 = 3/14, \quad \pi_3 = 5/14.$$

The holding times, *i.e.*, the intervals between sites, are $\bar{U}(1, 2) = \bar{U}(2, 1) = 20$ and $\bar{U}(1, 3) = \bar{U}(3, 1) = \bar{U}(2, 3) = 30$. We can then use (7.88) to find the expected holding time in each state, $\bar{U}(j) = \sum_k P_{jk} \bar{U}(j, k)$ so

$$\bar{U}(1) = 25, \quad \bar{U}(2) = 80/3, \quad \bar{U}(3) = 30.$$

The time-average fraction of time in state i , which is also the limiting probability of being in state i is then given by (7.90) as $p_j = \pi_j \bar{U}(j) / (\sum_k \pi_k \bar{U}(k))$. Thus we have

$$p_1 = 15/38, \quad p_2 = 8/38; \quad p_3 = 15/38.$$

In terms of the taxi problem, p_i is the probability that $X(t) = i$, *i.e.*, the probability that the most recent site visited up to time t is location i .

Next, the probability that the next site to be visited is j is the sum over i of the probabilities that the state is going from i to j at time t . Letting $Q(i, j)$ be the probability that $X(t) = i$ and that the next state is j , we see from (7.94) that $Q(i, j) = p_i P_{ij} \bar{U}(i, j) / \bar{U}(i)$. Thus the probability that the next state is 2 is

$$\Pr\{\text{next state} = 2\} = Q(1, 2) = p_1 P_{12} \bar{U}(1, 2) / \bar{U}(1) = 3/19.$$

Finally, the limiting fraction of time spent traveling from 2 to 3 is $Q(2, 3)$, which from (7.94) is $3/19$.

Exercise 7.30: a) Assume that the Markov process in Exercise 7.5 is changed in the following way: whenever the process enters state 0, the time spent before leaving state 0 is now a *uniformly distributed* rv, taking values from 0 to $2/\lambda$. All other transitions remain the same. For this new process, determine whether the successive epochs of entry to state 0 form renewal epochs, whether the successive epochs of exit from state 0 form renewal epochs, and whether the successive entries to any other given state i form renewal epochs.

Solution: On each successive entry to state 0, the embedded chain is in state 0 and the duration in that state is a uniform rv independent of all earlier states and their durations.

This duration in state 0 is finite and IID between successive visits to 0. All subsequent states and durations are independent of those after each subsequent visit to state 0 and the time to return to state 0 is finite with probability 1 from the same argument used for the Markov process. Thus successive returns to state 0 form a renewal process. The same argument works for exits from state 0.

For successive entries to some arbitrary given state j , the argument is slightly more complex, but it also forces us to understand the issue better. The argument that the epochs of successive entries to a given state in a Markov process form a renewal process does not depend at all on the exponential holding times. It depends only on the independence of the holding times and on the Markov property of the embedded chain. Thus changing one holding time from an exponential to a uniform rv of the same expected value does not change the independence of holding times and transitions between successive entries into any given state. Thus successive visits to any given state form renewal epochs.

b) For each i , find both the time-average interval and the time-average number of overall state transitions between successive visits to i .

Solution: The expected holding time in each state is unchanged by the modification and thus the time average interval between successive visits to any given state i is the same as before the modification. Also, the modification did not change the embedded chain, so the overall time-average number of state transitions between visits to i is also unchanged. These averages are calculated in part (d) of Exercise 7.5.

c) Is this modified process a Markov process in the sense that $\Pr\{X(t) = i \mid X(\tau) = j, X(s) = k\} = \Pr\{X(t) = i \mid X(\tau) = j\}$ for all $0 < s < \tau < t$ and all i, j, k ? Explain.

Solution: The modified process is not a Markov process. Essentially, the fact that the holding time in state 0 is no longer exponential, and thus not memoryless, means that if $X(t) = 0$, the time at which state 0 was entered provides information about when state 0 will be left.

As an example, suppose that $\mu \ll \lambda$ so we can ignore multiple visits to state 0 within an interval of length $2/\lambda$. We then see that $\Pr\{X(t+\lambda^{-1}) = 0 \mid X(t) = 0, X(t-\lambda^{-1}) = 0\} = 0$ since there is 0 probability of a visit to state 0 lasting for more than time $2/\lambda$. On the other hand,

$$\Pr\{X(t+\lambda^{-1}) = 0 \mid X(t) = 0, X(t-\lambda^{-1}) = 1\} > 0.$$

In fact, this latter probability is $1/8$ in the limit of large t and small μ .

Exercise 7.31: Consider an M/G/1 queueing system with Poisson arrivals of rate λ and expected service time $E[X]$. Let $\rho = \lambda E[X]$ and assume $\rho < 1$. Consider a semi-Markov process model of the M/G/1 queueing system in which transitions occur on departures from the queueing system and the state is the number of customers immediately following a departure.

a) Suppose a colleague has calculated the steady-state probabilities $\{p_i\}$ of being in state i for each $i \geq 0$. For each $i \geq 0$, find the steady-state probability π_i of state i in the embedded Markov chain. Give your solution as a function of ρ , p_i , and p_0 .

Solution: Note that the embedded chain is not a birth/death chain, since multiple arrivals can occur without a state change. Eq. (7.90) gives $\{p_i; i \geq 0\}$ in terms of $\{\pi_i; i \geq 0\}$ and

we can use that to solve for $\{\pi_i; i \geq 0$ in terms of $\{p_i; i \geq 0\}$,

$$\pi_i = \frac{p_i/\bar{U}(i)}{\sum_j p_j/\bar{U}(j)}.$$

For all states $i > 0$, $\bar{U}_i$ is simply the expected service time, $E[X]$. For $i = 0$, however, the state does not change until both an arrival occurs and then that arrival is served. Thus $\bar{U}(0) = E[X] + 1/\lambda$. Thus $\sum_j p_j/\bar{U}(j) = (1 - p_0)/E[X] + p_0/(E[X] + 1/\lambda)$ and for $i > 0$,

$$\pi_i = \frac{p_i}{1 - p_0 + p_0/(1 + 1/\rho)} = \frac{p_i(\rho + 1)}{1 + \rho - p_0}. \quad (\text{A.158})$$

Similarly, $\pi_0 = \rho p_0/(1 + \rho - p_0)$.

b) Calculate p_0 as a function of ρ .

Solution: Consider the renewal process where renewals occur on each entry to state 0 and let $\bar{W}$ be the expected time between renewals. We first show that $\bar{W} = \lambda/(1 - \rho)$ and then use this to find p_0 . Consider a reward function equal to 1 when the system is idle. Then the expected reward per renewal is $1/\lambda$. Since the time-average reward is the fraction of time the system is idle, it is $1 - \rho$. Thus by the renewal reward theorem, $1 - \rho = (1/\lambda)(1/\bar{W})$, so $\bar{W} = 1/(\lambda(1 - \rho))$.

Now consider a different reward function which is 1 both while the system is empty and while the first new arrival is being processed. The expected reward per renewal is $1/\lambda + E[X] = (1 + \rho)/\lambda$. Now p_0 is the time-average reward, so using the renewal-reward theorem again,

$$p_0 = \frac{1 + \rho}{\lambda} \cdot \frac{1}{\bar{W}} = (1 + \rho)(1 - \rho). \quad (\text{A.159})$$

c) Find π_i as a function of ρ and p_i .

Solution: Substituting (A.159) into (A.158), we get $\pi_i = p_i/\rho$ for $i > 0$. Similarly for $i = 0$, $\pi_0 = 1 - \rho$.

d) Is p_i the same as the steady-state probability that the queueing system contains i customers at a given time? Explain carefully.

Solution: There is no reason to believe that p_i , which is the probability of state i in the embedded Markov chain, should be the same as the steady-state probability of i in the $M/G/1$ queue. The state in the $M/G/1$ queue changes on each arrival and departure, whereas that in the chain changes only on departures. For example, $p_0 = (1 + \rho)(1 - \rho)$, whereas the time-average probability of an empty system is $(1 - \rho)$.

Exercise 7.32: Consider an $M/G/1$ queue in which the arrival rate is λ and the service time distribution is uniform $(0, 2W)$ with $\lambda W < 1$. Define a semi-Markov chain following the framework for the $M/G/1$ queue in Section 7.8.

a) Find P_{0j} for $j \geq 0$.

Solution: There is little to do here, since (7.99) gives P_{0j} in terms of the density $g(u)$ of the service time. Here $g(u) = 1/2W$ for $0 \leq u \leq 2W$, so P_{0j} is

$$P_{0j} = \int_0^{2W} \frac{(\lambda u)^j \exp(-\lambda u)}{2W j!} du; \quad j \geq 0.$$

Note that the integrand here is the probability of j additional arrivals in the interval of size u between the first arrival (*i.e.*, the beginning of the first service) and the end of that service. This expression can be manipulated in various ways, but does not materially simplify.

b) Find P_{ij} for $i > 0$; $j \geq i - 1$.

Solution:

$$P_{ij} = \int_0^{2W} \frac{(\lambda u)^{j-i+1} \exp(-\lambda u)}{2W(j-i+1)} du; \quad j \geq i-1, i > 0.$$

Here, with $i > 0$, the integrand is the probability of $j - i + 1$ arrivals in an interval of size u . Note that starting a service in state i with $j - i + 1$ arrivals and one departure leaves the system in state j .

Exercise 7.33: Consider a semi-Markov process for which the embedded Markov chain is irreducible and positive recurrent. Assume that the distribution of inter-renewal intervals for one state j is arithmetic with span d . Show that the distribution of inter-renewal intervals for all states is arithmetic with the same span.

Solution: Let $X_0, X_1, \dots$, be a sequence of embedded states and let $U_1, U_2, \dots$, be the corresponding durations in the semi-Markov process. Assume for now that the durations are all discrete rv's. Consider a joint sample path of $\{(X_i, U_i)\}$ in which $X_0 = j$ and $X_n = j$. Then $\sum_{\ell=1}^n U_\ell$ is the duration of one or more renewals of state j in the semi-Markov process. By hypothesis then, this is an arithmetic rv whose span is a multiple of d .

For some given state i , consider a subset of such joint sample paths such that $X_0 = j, X_m = i, X_k = i, X_n = j$ for $0 < m < k < n$ and such that $X_{m+1}, \dots, X_{k-1} \neq i$. Thus we are looking at a sample path in the embedded chain that goes from j to i to a renewal of i and then back to j . Considering the associated durations, $\sum_{\ell=1}^n U_\ell$ is again a multiple of d for all such sample paths. Also $\sum_{\ell=1}^m U_\ell + \sum_{\ell=k+1}^n U_\ell$ is a multiple of d since this also corresponds to one or more renewals of state j (with the cycle from i to i omitted). It follows that $\sum_{\ell=m+1}^k U_\ell$ is also a multiple of d .

We have thus shown that the inter-renewal intervals for state i are arithmetic with a span that is a multiple of d . This multiple must be 1, since we can reverse the roles of i and j above. Since i is arbitrary, the inter-renewal intervals for all states are arithmetic with span d .

Finally we must show that the rv's U_{ik} are all discrete. Consider any sample path from say j to j that includes a single transition from i to k . The duration of this path must be a multiple of d for all sample values of U_{ik} , and thus these sample values can differ only by multiples of d , and they are thus discrete. Note that it doesn't follow that the random variables U_{ik} are arithmetic since, for example, the embedded chain might have period 2 with odd numbered states transitioning only to even numbered states and vice versa. One set of transitions might have durations of the form $\alpha + md$ where $0 < \alpha < d$ is arbitrary and m is a variable positive integer. The other set might have durations equal to $md - \alpha$. Any renewal would contain an equal number of durations from each set and thus all renewal durations would be integer multiples of d .

A.8 Solutions for Chapter 8

Exercise 8.1: In this exercise, we evaluate $\Pr\{e_\eta \mid X = \mathbf{a}\}$ and $\Pr\{e_\eta \mid X = \mathbf{b}\}$ for binary detection from vector signals in Gaussian noise directly from (8.40) and (8.41).

a) By using (8.40) for each sample value $\mathbf{y}$ of $\mathbf{Y}$, show that

$$\mathbb{E}[\text{LLR}(\mathbf{Y}) \mid X = \mathbf{a}] = \frac{-(\mathbf{b} - \mathbf{a})^\top (\mathbf{b} - \mathbf{a})}{2\sigma^2}.$$

Hint: Note that, given $X = \mathbf{a}$, $\mathbf{Y} = \mathbf{a} + \mathbf{Z}$.

Solution: Taking the expectation of (8.40) conditional on $X = \mathbf{a}$,

$$\begin{aligned} \mathbb{E}[\text{LLR}(\mathbf{Y}) \mid X = \mathbf{a}] &= \frac{(\mathbf{b} - \mathbf{a})^\top}{\sigma^2} \mathbb{E} \left[\mathbf{Y} - \frac{\mathbf{b} + \mathbf{a}}{2} \right] \\ &= \frac{(\mathbf{b} - \mathbf{a})^\top}{\sigma^2} \left(\mathbf{a} - \frac{\mathbf{b} + \mathbf{a}}{2} \right), \end{aligned}$$

from which the desired result is obvious.

b) Defining $\gamma = \|\mathbf{b} - \mathbf{a}\|/(2\sigma)$, show that

$$\mathbb{E}[\text{LLR}(\mathbf{Y}) \mid X = \mathbf{a}] = -2\gamma^2.$$

Solution: The result in (a) can be expressed as $\mathbb{E}[\text{LLR}(\mathbf{Y})] = -\|\mathbf{b} - \mathbf{a}\|^2/2\sigma^2$, from which the result follows.

c) Show that

$$\text{VAR}[\text{LLR}(\mathbf{Y}) \mid X = \mathbf{a}] = 4\gamma^2.$$

Hint: Note that the fluctuation of $\text{LLR}(\mathbf{Y})$ conditional on $X = \mathbf{a}$ is $(1/\sigma^2)(\mathbf{b} - \mathbf{a})^\top \mathbf{Z}$.

Solution: Using the hint,

$$\begin{aligned} \text{VAR}[\text{LLR}(\mathbf{Y})] &= \frac{1}{\sigma^2} (\mathbf{b} - \mathbf{a})^\top \mathbb{E}[\mathbf{Z}\mathbf{Z}^\top] (\mathbf{b} - \mathbf{a}) \frac{1}{\sigma^2} \\ &= \frac{1}{\sigma^2} (\mathbf{b} - \mathbf{a})^\top [\mathbf{I}] (\mathbf{b} - \mathbf{a}) = \frac{1}{\sigma^2} \|\mathbf{b} - \mathbf{a}\|^2, \end{aligned}$$

from which the result follows.

d) Show that, conditional on $X = \mathbf{a}$, $\text{LLR}(\mathbf{Y}) \sim \mathcal{N}(-2\gamma^2, 4\gamma^2)$. Show that, conditional on $X = \mathbf{a}$, $\text{LLR}(\mathbf{Y})/2\gamma \sim \mathcal{N}(-\gamma, 1)$.

Solution: Conditional on $X = \mathbf{a}$, we see that $\mathbf{Y} = \mathbf{a} + \mathbf{Z}$ is Gaussian and thus $\text{LLR}(\mathbf{Y})$ is also Gaussian conditional on $X = \mathbf{a}$. Using the conditional mean and variance of $\text{LLR}(\mathbf{Y})$ found in (b) and (c), $\text{LLR}(\mathbf{Y}) \sim \mathcal{N}(-2\gamma^2, 4\gamma^2)$.

When the rv $\text{LLR}(\mathbf{Y})$ is divided by 2γ , the conditional mean is also divided by 2γ , and the variance is divided by $(2\gamma)^2$, leading to the desired result.

Note that $\text{LLR}(\mathbf{Y})/2\gamma$ is very different from $\text{LLR}(\mathbf{Y}/2\gamma)$. The first scales the LLR and the second scales the observation $\mathbf{Y}$. If the observation itself is scaled, the result is a sufficient statistic and the LLR is unchanged by the scaling.

e) Show that the first half of (8.44) is valid, *i.e.*, that

$$\Pr\{e_\eta \mid X=\mathbf{a}\} = \Pr\{\text{LLR}(\mathbf{Y}) \geq \ln \eta \mid X=\mathbf{a}\} = Q\left(\frac{\ln \eta}{2\gamma} + \gamma\right).$$

Solution: The first equality above is simply the result of a threshold test with the threshold η . The second uses the fact in (d) that $\text{LLR}(\mathbf{Y})/2\gamma$, conditional on $X = \mathbf{a}$, is $\mathcal{N}(-\gamma, 1)$. This is a unit variance Gaussian rv with mean $-\gamma$. The probability that it exceeds $\ln \eta/2\gamma$ is then $Q(\ln(\eta/2\gamma) + \gamma)$.

f) By essentially repeating (a) through (e), show that the second half of (8.44) is valid, *i.e.*, that

$$\Pr\{e_\eta \mid X=\mathbf{b}\} = Q\left(\frac{-\ln \eta}{2\gamma} + \gamma\right).$$

Solution: One can simply rewrite each equation above, but care is needed in observing that the likelihood ratio requires a convention for which hypothesis goes on top of the fraction. Thus, here the sign of the LLR is opposite to that in parts (a) to (e). This also means that the error event occurs on the opposite side of the threshold.

Exercise 8.2: (Generalization of Exercise 8.1) a) Let $U = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbf{Y}$. Find $\mathbb{E}[U \mid X=\mathbf{a}]$ and $\mathbb{E}[U \mid X=\mathbf{b}]$.

Solution: As explained in Example 8.2.5, U is a sufficient statistic. Conditional on $X = \mathbf{a}$, we have $\mathbf{Y} = \mathbf{a} + \mathbf{Z}$ and thus

$$\mathbb{E}[U \mid X=\mathbf{a}] = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbb{E}[\mathbf{a} + \mathbf{Z}] = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbf{a}.$$

Similarly

$$\mathbb{E}[U \mid X=\mathbf{b}] = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbb{E}[\mathbf{b} + \mathbf{Z}] = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbf{b}.$$

b) Find the conditional variance of U conditional on $X = \mathbf{a}$. Hint: see the hint in Exercise 8.1 (c)

Solution: The fluctuation of U given $X = \mathbf{a}$ is $(\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbf{Z}$. Thus

$$\begin{aligned} \text{VAR}[U \mid X=\mathbf{a}] &= (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} \mathbb{E}[\mathbf{Z} \mathbf{Z}^\top] K_Z^{-1} (\mathbf{b} - \mathbf{a}) \\ &= (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} (\mathbf{b} - \mathbf{a}). \end{aligned}$$

c) Give the threshold test in terms of the sample value u of U , and evaluate $\Pr\{e_\eta \mid X=\mathbf{a}\}$ and $\Pr\{e_\eta \mid X=\mathbf{b}\}$ from this and (b). Show that your answer agrees with (8.53).

Solution: As in (8.50), define $\gamma^2 = (\mathbf{b} - \mathbf{a})^\top K_Z^{-1} (\mathbf{b} - \mathbf{a})/4$. Then $\text{VAR}[U \mid X=\mathbf{a}] = 4\gamma^2$. The LLR for U is the same as that for $\mathbf{Y}$ and can be evaluated (somewhat tediously) from (a) and (b). It is easier to use the value in (8.49), $\mathbb{E}[\text{LLR}(U) \mid X = \mathbf{a}] = -2\gamma^2$. Similarly, $\text{VAR}[U \mid X=\mathbf{b}] = 4\gamma^2$ and $\mathbb{E}[\text{LLR}(U) \mid X = \mathbf{b}] = +2\gamma^2$. The error probabilities are then the same as those in (e) and (f) of Exercise 8.1, which is the same as (8.53).

d) Explain what happens if $[K_Z]$ is singular. Hint: you must look at two separate cases, depending on the vector $\mathbf{b} - \mathbf{a}$.

Solution: If $\mathbf{b} - \mathbf{a}$ is an eigenvector with eigenvalue 0 of $[K_Z]$, it means that $\mathbf{b} - \mathbf{a}$ is in a direction in which there is no noise. Thus the error probability will be zero if it is based on

a threshold rule on the component of $\mathbf{Y}$ in the direction of $\mathbf{b} - \mathbf{a}$. In fact, if $\mathbf{b} - \mathbf{a}$ has any component in the direction of an eigenvector of eigenvalue 0, the error probability can be made zero by looking only in that direction. This is called singular detection. Alternatively, if $\mathbf{b} - \mathbf{a}$ is orthogonal to all eigenvectors of eigenvalue 0, the error probability will be nonzero and can be found by using a coordinate system in which the directions of eigenvectors of eigenvalue 0 are omitted from the formulation.

Exercise 8.3: a) Let $\mathbf{Y}$ be the observation rv for a binary detection problem, let $\mathbf{y}$ be the observed sample value. Let $v = v(\mathbf{y})$ be a sufficient statistic and let V be the corresponding random variable. Show that $\Lambda(\mathbf{y})$ is equal to $\mathbf{p}_{V|X}(v(\mathbf{y}) | \mathbf{b}) / \mathbf{p}_{V|X}(v(\mathbf{y}) | \mathbf{a})$. In other words, show that the likelihood ratio of a sufficient statistic is the same as the likelihood ratio of the original observation.

Solution: This is the third statement of Theorem 8.2.8, and we don't see any way of improving on that proof. It relies on the second statement of the theorem, which is further investigated in Exercise 8.4

b) Show that this also holds for the ratio of probability densities if V is a continuous rv or random vector.

Solution: Repeating the proof of the third statement of Theorem 8.2.8 for the case in which $\mathbf{Y}$ and V have densities, we start with the second statement, *i.e.*, $\mathbf{p}_{X|Y}(x|\mathbf{y}) = \mathbf{p}_{X|V}(x|v(\mathbf{y}))$.

$$\begin{aligned} \frac{\mathbf{p}_{X|Y}(1|\mathbf{y})}{\mathbf{p}_{X|Y}(0|\mathbf{y})} &= \frac{\mathbf{p}_{X|V}(1|v(\mathbf{y}))}{\mathbf{p}_{X|V}(0|v(\mathbf{y}))} \\ \frac{f_{Y|X}(\mathbf{y}|1)\mathbf{p}_X(1)/f_Y(\mathbf{y})}{f_{Y|X}(\mathbf{y}|0)\mathbf{p}_X(0)/f_Y(\mathbf{y})} &= \frac{f_{V|X}(v(\mathbf{y})|1)\mathbf{p}_X(1)/f_V(v(\mathbf{y}))}{f_{V|X}(v(\mathbf{y})|0)\mathbf{p}_X(0)/f_V(v(\mathbf{y}))}, \end{aligned}$$

Where we have used Bayes' rule on each term. Cancelling terms, we get

$$\frac{f_{Y|X}(\mathbf{y}|1)}{f_{Y|X}(\mathbf{y}|0)} = \frac{f_{V|X}(v(\mathbf{y})|1)}{f_{V|X}(v(\mathbf{y})|0)}.$$

The left side of this is $\Lambda(\mathbf{y})$, so this is the desired result. We see that it is derived simply by replacing PMF's with PDF's, but there are some assumptions made about the densities being positive. Turning this into a mathematical theorem with precise conditions would require measure theory, and without that, it is better to rely on common sense applied to simple models.

Exercise 8.4:

a) Show that if $v(\mathbf{y})$ is a sufficient statistic according to condition 1 of Theorem 8.2.8, then

$$p_{X|YV}(x | \mathbf{y}, v(\mathbf{y})) = p_{X|Y}(x | \mathbf{y}). \quad (\text{A.160})$$

Solution: Let $V(\mathbf{Y})$ be the rv with sample values $v(\mathbf{y})$. Note that $\{\mathbf{Y} = \mathbf{y}\}$ is the same event as $\{\mathbf{Y} = \mathbf{y}\} \cap \{V = v(\mathbf{y})\}$. If $\mathbf{Y}$ is discrete and this event has positive probability, then (A.160) is obvious since the condition on both sides is the same and has positive probability. If Y is a rv with a positive density, then (A.160) is true if the condition $Y = y$ is replaced with $y - \delta < Y \leq y$. Then (A.160) holds if $\lim_{\delta \rightarrow 0} (1/\delta) \Pr\{y - \delta < Y \leq y\} > 0$, which is valid since Y has a positive density. This type of argument can be extended to the

case where Y is a random vector and holds whether or not $V(\mathbf{Y})$ is a sufficient statistic. This says that $X \rightarrow \mathbf{Y} \rightarrow V$ is Markov, which is not surprising since V is simply a function of $\mathbf{Y}$.

b) Consider the subspace of events conditional on $V(\mathbf{y}) = v$ for a given v . Show that for $\mathbf{y}$ such that $v(\mathbf{y}) = v$,

$$p_{X|YV}(x | \mathbf{y}, v(\mathbf{y})) = p_{X|V}(x | v). \quad (\text{A.161})$$

Solution: We must assume (as a natural extension of (a)), that $V(\mathbf{Y})$ is a sufficient statistic, *i.e.*, that there is a function u such that for each v , $u(v) = \Lambda(\mathbf{y})$ for all $\mathbf{y}$ such that $v(\mathbf{y}) = v$. We also assume that $\mathbf{Y} = \mathbf{y}$ has positive probability or probability density, since the conditional probabilities don't have much meaning otherwise. Then

$$\begin{aligned} \frac{p_{X|YV}(1 | \mathbf{y}, v(\mathbf{y}))}{p_{X|YV}(0 | \mathbf{y}, v(\mathbf{y}))} &= \frac{p_{X|Y}(1 | \mathbf{y})}{p_{X|Y}(0 | \mathbf{y})} = \frac{p_1}{p_0} \Lambda(\mathbf{y}) \\ &= \frac{p_1}{p_0} u(v(\mathbf{y})), \end{aligned}$$

where we first used (A.160), then Bayes' law, and then the assumption that $u(v)$ is a sufficient statistic. Since this ratio is the same for all $\mathbf{y}$ for which $v(\mathbf{y})$ has the same value, the ratio is a function of v alone,

$$\frac{p_{X|YV}(1 | \mathbf{y}, v(\mathbf{y}))}{p_{X|YV}(0 | \mathbf{y}, v(\mathbf{y}))} = \frac{p_{X|V}(1 | v)}{p_{X|V}(0 | v)}. \quad (\text{A.162})$$

Finally, since $p_{X|V}(0|v) = 1 - p_{X|V}(1|v)$ and $p_{X|YV}(0|\mathbf{y}, v(\mathbf{y})) = 1 - p_{X|YV}(1|\mathbf{y}, v(\mathbf{y}))$, we see that (A.162) implies (A.161). Note that this says that $X \rightarrow V \rightarrow \mathbf{Y}$ is Markov.

c) Explain why this argument is valid whether $\mathbf{Y}$ is a discrete or continuous random vector and whether V is discrete, continuous or part discrete and part continuous.

Solution: The argument above essentially makes no assumptions about either $\mathbf{Y}$ nor V being discrete or having a density. The argument does depend on the assumption that the given conditional probabilities are defined.

Exercise 8.5: a) Let $\mathbf{Y}$ be a discrete observation random vector and let $v(\mathbf{y})$ be a function of the sample values of $\mathbf{Y}$. Show that

$$p_{Y|VX}(\mathbf{y} | v(\mathbf{y}), x) = \frac{p_{Y|X}(\mathbf{y} | x)}{p_{V|X}(v(\mathbf{y}) | x)}. \quad (\text{A.163})$$

Solution: We must assume that $p_{V|X}(v(\mathbf{y}), x) > 0$ so that the expression on the left is defined. Then, using Bayes' law on $\mathbf{Y}$ and V for fixed x on the left side of (A.163),

$$p_{Y|VX}(\mathbf{y} | v(\mathbf{y}), x) = \frac{p_{V|YX}(v(\mathbf{y}) | \mathbf{y}x) p_{Y|X}(\mathbf{y} | x)}{p_{V|X}(v(\mathbf{y}) | x)}. \quad (\text{A.164})$$

Since V is a deterministic function of $\mathbf{Y}$, the first term in the numerator above is 1, so this is equivalent to (A.163),

b) Using Theorem 8.2.8, show that the above fraction is independent of X if and only if $v(\mathbf{y})$ is a sufficient statistic.

Solution: Using Bayes' law on the numerator and denominator of (A.163),

$$\frac{p_{Y|X}(\mathbf{y} | x)}{p_{V|X}(v(\mathbf{y}) | x)} = \frac{p_{X|Y}(x | \mathbf{y}) p_Y(\mathbf{y})}{p_{X|V}(x | v(\mathbf{y})) p_V(v(\mathbf{y}))}. \quad (\text{A.165})$$

If $v(\mathbf{y})$ is a sufficient statistic, then the second equivalent statement of Theorem 8.2.8, *i.e.*,

$$p_{X|Y}(x | \mathbf{y}) = p_{X|V}(x | v(\mathbf{y})),$$

shows that the first terms on the right side of (A.165) cancel, showing that the fraction is independent of x . Conversely, if the fraction is independent of x , then the ratio of $p_{X|Y}(x | \mathbf{y})$ to $p_{X|V}(x | v(\mathbf{y}))$, is a function only of $\mathbf{y}$. Since X is binary, this fraction must be 1, establishing the second statement of Theorem 8.2.8.

c) Now assume that $\mathbf{Y}$ is a continuous observation random vector, that $v(\mathbf{y})$ is a given function, and $V = v(\mathbf{Y})$ has a probability density. Define

$$f_{Y|VX}(\mathbf{y} | v(\mathbf{y}), x) = \frac{f_{Y|X}(\mathbf{y} | x)}{f_{V|X}(v(\mathbf{y}) | x)}. \quad (\text{A.166})$$

One can interpret this as a strange kind of probability density on a conditional sample space, but it is more straightforward to regard it simply as a fraction. Show that $v(\mathbf{y})$ is a sufficient statistic if and only if this fraction is independent of x . Hint: Model your derivation on that in (b), modifying (b) as necessary to do this.

Solution: The trouble with (A.164) can be seen by looking at (A.165). When this is converted to densities, the joint density of $\mathbf{Y}, V(\mathbf{Y})$ has the same problem as the densities in Example 8.2.10. The numerator and denominator of (A.163) are well defined as densities, however, and the argument in (b) carries through as before.

Exercise 8.6: Consider Example 8.2.5, and let $\mathbf{Z} = [A]\mathbf{W}$ where $\mathbf{W} \sim \mathcal{N}(0, I)$ is normalized IID Gaussian and $[A]$ is non-singular. The observation $\mathbf{r}$ is $\mathbf{a} + \mathbf{Z}$ given $X = \mathbf{a}$ and is $\mathbf{b} + \mathbf{Z}$ given $X = \mathbf{b}$. Suppose the observed sample value $\mathbf{y}$ is transformed into $\mathbf{v} = [A^{-1}]\mathbf{y}$. Explain why $\mathbf{v}$ is a sufficient statistic for this detection problem (and thus why MAP detection based on $\mathbf{v}$ must yield the same decision as that based on $\mathbf{y}$).

Solution: v is a sufficient statistic if there is a function u such that $u(v(\mathbf{y})) = \Lambda(\mathbf{y})$. Since $\mathbf{v}$ is an invertible function of $\mathbf{y}$, we can clearly map $\mathbf{v}$ back into $\mathbf{y}$ and map that into $\Lambda(\mathbf{y})$. In equations, using (8.64),

$$\Lambda(\mathbf{y}) = \frac{f_{V|X}(v(\mathbf{y}) | 1)}{f_{V|X}(v(\mathbf{y}) | 0)}.$$

b) Consider the detection problem where $\mathbf{V} = [A^{-1}]\mathbf{a} + \mathbf{W}$ given $X = \mathbf{a}$ and $[A^{-1}]\mathbf{b} + \mathbf{W}$ given $X = \mathbf{b}$. Find the log-likelihood ratio $\text{LLR}(\mathbf{v})$ for a sample value $\mathbf{v}$ of $\mathbf{V}$. Show that this is the same as the log-likelihood ratio for a sample value $\mathbf{y} = [A]\mathbf{v}$ of $\mathbf{Y}$.

Solution: The problem $\mathbf{V} = [A^{-1}]\mathbf{X} + \mathbf{W}$ is the same as that in Example 8.2.3 except that the received signal is $\mathbf{V}$ rather than $\mathbf{Y}$, the transmitted signals are $[A^{-1}]\mathbf{a}$ and $[A^{-1}]\mathbf{b}$,

and $\sigma^2 = 1$. The log likelihood ratio, from (8.40) is

$$\begin{aligned}\text{LLR}(\mathbf{v}) &= \left([A^{-1}]\mathbf{b} - [A^{-1}]\mathbf{a}\right)^{\top} \left(\mathbf{v} - \frac{1}{2}([A^{-1}]\mathbf{b} + [A^{-1}]\mathbf{a})\right) \\ \text{LLR}(\mathbf{v}(\mathbf{y})) &= (\mathbf{b} - \mathbf{a})^{\top} [A^{-1}]^{\top} \left([A^{-1}]\mathbf{y} - [A^{-1}]\frac{1}{2}(\mathbf{b} + \mathbf{a})\right) \\ &= \frac{1}{2}(\mathbf{b} - \mathbf{a})^{\top} [A^{-1}]^{\top} [A^{-1}] \left(\mathbf{y} - \frac{(\mathbf{b} + \mathbf{a})}{2}\right).\end{aligned}\quad (\text{A.167})$$

This is the same as $\text{LLR}(\mathbf{y})$ as given in (8.45) after making the observation that $[K_{\mathbf{Z}}] = [A^{\top}A]$.

c) Find $\Pr\{e \mid X=\mathbf{a}\}$ and $\Pr\{e \mid X=\mathbf{b}\}$ for the detection problem in (b) by using the results of Example 8.2.3. Show that your answer agrees with (8.53). Note: the methodology here is to transform the observed sample value to make the noise IID; this approach is often both useful and insightful.

Solution: The error probability using the results of Example 8.2.3 is given by (8.44) in terms of γ , which is defined as

$$\begin{aligned}\gamma &= \frac{1}{2}\|[A^{-1}]\mathbf{b} - [A^{-1}]\mathbf{a}\| = \frac{1}{2}\|[A^{-1}](\mathbf{b} - \mathbf{a})\| \\ &= \frac{1}{2}\sqrt{(\mathbf{b} - \mathbf{a})^{\top} [A^{-1}]^{\top} [A^{-1}](\mathbf{b} - \mathbf{a})}.\end{aligned}$$

The result in (8.53) is the same equation, except that γ is defined in (8.50) as

$$\gamma = \sqrt{\frac{(\mathbf{b} - \mathbf{a})^{\top}}{2} [K_{\mathbf{Z}}^{-1}] \frac{(\mathbf{b} - \mathbf{a})}{2}}.$$

Since $[K_{\mathbf{Z}}] = [A^{\top}A]$, these are the same.

Exercise 8.7: Binary frequency shift keying (FSK) with incoherent reception can be modeled in terms of a 4 dimensional observation vector $\mathbf{Y} = (Y_1, Y_2, Y_3, Y_4)^{\top}$. $\mathbf{Y} = \mathbf{U} + \mathbf{Z}$ where $\mathbf{Z} \sim \mathcal{N}(0, \sigma^2 I)$ and $\mathbf{Z}$ is independent of X . Under $X = 0$, $\mathbf{U} = (a \cos \phi, a \sin \phi, 0, 0)^{\top}$, whereas under $X = 1$, $\mathbf{U} = (0, 0, a \cos \phi, a \sin \phi)^{\top}$. The random variable ϕ is uniformly distributed between 0 and 2π and is independent of X and $\mathbf{Z}$. The *a priori* probabilities are $p_0 = p_1 = 1/2$.

a) Convince yourself from the circular symmetry of the situation that the ML receiver calculates the sample values v_0 and v_1 of $V_0 = Y_1^2 + Y_2^2$ and $V_1 = Y_3^2 + Y_4^2$ and chooses $\hat{x} = 0$ if $v_0 \geq v_1$ and chooses $\hat{x} = 1$ otherwise.

Solution: It is easier to visualize the here if we express the 4-rv's as pairs of complex rv's. That is, $U = (ae^{i\phi}, 0)$ under $X = 0$ and $U = (0, ae^{i\phi})$ under $X = 1$. Similarly, $\mathbf{Z} = (R_0 e^{i\theta_0}, R_1 e^{i\theta_1})$ where R_0 and R_1 are Rayleigh rv's and are independent of the other rv's as in the problem description. The received complex rv's are $(\sqrt{V_0} e^{i\psi_0}, \sqrt{V_1} e^{i\psi_1})$. The phases ϕ, θ_0 , and θ_1 are each uniformly distributed over $[0, 2\pi)$ and are independent of each other and of the other variables as described. The received phases, ψ_0 and ψ_1 are complicated functions of the other phases, but it is intuitively clear if one draws a picture of $\sqrt{V_0} e^{i\psi_0}$ that ψ_0 is uniformly distributed over $[0, 2\pi)$ and is independent of V_0, V_1 , and X . This is proven in a slightly more general context in the solution to Exercise 3.23. In the

same way, ψ_2 is uniform and independent of V_0, V_1 , and X . It follows that V_0 and V_1 form a sufficient statistic for X

b) Find $\Pr\{V_1 > v_1 \mid X=0\}$ as a function of $v_1 > 0$.

Solution: Conditional on $X = 0$, $V_1 = |Z_1|^2$. Since $|Z_1|^2$ is the sum of the squares of 2 IID real Gaussian rv's each of mean σ^2 ,

$$\Pr\{V_1 > v_1 \mid X=0\} = \exp(-v_1/2\sigma^2). \quad (\text{A.168})$$

This is derived in detail in the solution to Exercise 3.1.

c) Show that

$$f_{Y_1, Y_2 \mid X, \phi}(y_1, y_2 \mid 0, 0) = \frac{1}{2\pi\sigma^2} \exp\left[\frac{-y_1^2 - y_2^2 + 2y_1a - a^2}{2\sigma^2}\right]. \quad (\text{A.169})$$

Solution: Conditional on $X = 0$ and $\phi = 0$, Y_1 is $\mathcal{N}(a, \sigma^2)$ and Y_2 is $\mathcal{N}(0, \sigma^2)$ and independent of Y_1 . Thus their conditional joint density is as shown.

d) Show that

$$\Pr\{V_1 > V_0 \mid X=0, \phi=0\} = \int f_{Y_1, Y_2 \mid X, \phi}(y_1, y_2 \mid 0, 0) \Pr\{V_1 > y_1^2 + y_2^2\} dy_1 dy_2. \quad (\text{A.170})$$

Show that this is equal to $(1/2) \exp(-a^2/(4\sigma^2))$.

Solution: The right side of (A.170) is $\Pr\{V_1 > Y_1^2 + Y_2^2 \mid X = 0, \phi = 0\}$. Recalling that $V_0 = Y_1^2 + Y_2^2$, we see that (A.170) is satisfied. Substituting (A.168) and (A.169) into (A.170), we get

$$\begin{aligned} \Pr\{V_1 > V_0 \mid X=0, \phi=0\} &= \int \frac{1}{2\pi\sigma^2} \exp\left[\frac{-y_1^2 - y_2^2 + 2y_1a - a^2}{2\sigma^2}\right] \left[\frac{\exp(-y_1^2 - y_2^2)}{2\sigma^2}\right] dy_1 dy_2 \\ &= \int \frac{1}{2\pi\sigma^2} \exp\left[\frac{-2y_1^2 - 2y_2^2 + 2y_1a - a^2}{2\sigma^2}\right] dy_1 dy_2 \\ &= \frac{1}{2} \exp\left(\frac{-a^2}{4\sigma^2}\right). \end{aligned} \quad (\text{A.171})$$

e) Explain why this is the probability of error (i. e., why the event $V_1 > V_0$ is independent of ϕ), and why $\Pr\{e \mid X=0\} = \Pr\{e \mid X=1\}$.

Solution: If we use an arbitrary phase ϕ in (A.168), we can express Z_1, Z_2 in a coordinate system rotated by ϕ to see that (A.168) is still valid. Thus (A.171) is valid for all ϕ and thus valid with no conditioning on ϕ . The probability of error conditional on $X = 1$ is obviously the same because of the symmetry between $X = 0$ and $X = 1$.

Exercise 8.8: Binary frequency shift keying (FSK) on a Rayleigh fading channel can be modeled in terms of a 4 dimensional observation vector $\mathbf{Y} = (Y_1, Y_2, Y_3, Y_4)^T$. $\mathbf{Y} = \mathbf{U} + \mathbf{Z}$ where $\mathbf{Z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ and $\mathbf{Z}$ is independent of $\mathbf{U}$. Under $X = 0$, $\mathbf{X} = (U_1, U_2, 0, 0)^T$, whereas under $X = 1$, $\mathbf{U} = (0, 0, U_3, U_4)^T$. The random variables $U_i \sim \mathcal{N}(0, a^2)$ are IID. The *a priori* probabilities are $p_0 = p_1 = 1/2$.

a) Convince yourself from the circular symmetry of the situation that the ML receiver calculates sample values v_0 and v_1 for $V_0 = Y_1^2 + Y_2^2$ and $V_1 = Y_3^2 + Y_4^2$ and chooses $\hat{x} = 0$ if $v_0 > v_1$ and chooses $\hat{x} = 1$ otherwise.

Solution: See the solution to Exercise 8.7(a) for a proof that (V_0, V_1) is a sufficient statistic for this problem. From the symmetry between $X = 1$ and $X = 0$, it is then clear that the ML decision threshold is at $v_1 - v_0 = 0$.

b) Find $f_{V_0|X}(v_0 | 0)$ and find $f_{V_1|X}(v_1 | 0)$.

Solution: Conditional on $X = 0$, we see that $Y_1 = Z_1 + U_1$ and $Y_2 = Z_2 + U_2$. Thus, conditional on $X = 0$, $Y_1 \sim \mathcal{N}(0, \sigma^2 + a^2)$ and, independently, $Y_2 \sim \mathcal{N}(0, \sigma^2 + a^2)$. Since V_0 given $X = 0$ is the sum of the squares of two IID rv's, each $\mathcal{N}(0, \sigma^2 + a^2)$,

$$f_{V_0|X}(v_0 | 0) = \frac{1}{2(\sigma^2 + a^2)} \exp\left(\frac{-v_0}{2(\sigma^2 + a^2)}\right); \quad \text{for } v_0 \geq 0.$$

In the same way, conditional on $X = 0$, $Y_3 \sim \mathcal{N}(0, \sigma^2)$ and $Y_4 \sim \mathcal{N}(0, \sigma^2)$. Thus

$$f_{V_1|X}(v_1 | 0) = \frac{1}{2\sigma^2} \exp\left(\frac{-v_1}{2\sigma^2}\right); \quad \text{for } v_1 \geq 0.$$

c) Let $W = V_0 - V_1$ and find $f_W(w | X=0)$.

Solution: The joint density of V_0 and V_1 conditional on $X = 0$ is

$$f_{V_0 V_1|X}(v_0, v_1 | 0) = \frac{1}{4(\sigma^2 + a^2)\sigma^2} \exp\left(-\frac{v_0}{2(\sigma^2 + a^2)} - \frac{v_1}{2\sigma^2}\right); \quad \text{for } v_0, v_1 \geq 0.$$

Substitute $w = v_0 - v_1$ for v_0 in this and then integrate over v_1 . For $w > 0$, we integrate v_1 from 0 to ∞ , and for $w \leq 0$, we integrate from $-w$ to ∞ . The result is

$$f_W(w) = \begin{cases} \frac{1}{4\sigma^2 + 2a^2} \exp\left(\frac{-w}{2\sigma^2 + 2a^2}\right) & ; \quad \text{for } w > 0, \\ \frac{1}{4\sigma^2 + 2a^2} \exp\left(\frac{w}{2\sigma^2}\right) & ; \quad \text{for } w \leq 0. \end{cases}$$

d) Show that $\Pr\{e | X=0\} = [2 + a^2/\sigma^2]^{-1}$. Explain why this is also the unconditional probability of an incorrect decision.

Solution: An error occurs if $W \leq 0$. We have

$$\Pr\{e | X = 0\} = \int_{-\infty}^0 f_{W|X}(w|0) dw = \frac{2\sigma^2}{4\sigma^2 + 2a^2} = \frac{1}{2 + a^2/\sigma^2}.$$

The same probability of error clearly holds for $X = 1$ because of the symmetry between $X = 0$ and $X = 1$; thus this is the unconditional error probability.

Note the difference between error probability for the Rayleigh fading channel using FSK and that for incoherent reception in Exercise 8.7. In the Rayleigh fading case, $\Pr\{e\}$ decays essentially as $1/(a^2/\sigma^2)$, whereas it decays exponentially in a^2/σ^2 for the incoherent case. The reason is that, even with very large a^2/σ^2 , the signal has a significant probability of fading to a level smaller than the noise in the fading case.

Exercise 8.9: A disease has two strains, 0 and 1, which occur with *a priori* probabilities p_0 and $p_1 = 1 - p_0$ respectively.

a) Initially, a rather noisy test was developed to find which strain is present for patients with the disease. The output of the test is the sample value y_1 of a random variable Y_1 . Given strain 0 ($X=0$), $Y_1 = 5 + Z_1$, and given strain 1 ($X=1$), $Y_1 = 1 + Z_1$. The measurement noise Z_1 is independent of X and is Gaussian, $Z_1 \sim \mathcal{N}(0, \sigma^2)$. Give the MAP decision rule, *i.e.*, determine the set of observations y_1 for which the decision is $\hat{x}=1$. Give $\Pr\{e \mid X=0\}$ and $\Pr\{e \mid X=1\}$ in terms of the function $Q(x)$.

Solution: This is simply a case of binary detection with an additive Gaussian noise rv. To prevent simply copying the answer from Example 8.2.3, the signal a associated with $X = 0$ is 5 and the signal b associated with $X = 1$ is 1. Thus $b < a$, contrary to the assumption in Example 8.2.3. Looking at that example, we see that (8.27), repeated below, is still valid.

$$\text{LLR}(y) = \left[\left(\frac{b-a}{\sigma^2} \right) \left(y - \frac{b+a}{2} \right) \right] \begin{matrix} \geq & \hat{x}(y)=b \\ & \hat{x}(y)=a \\ < \end{matrix} \ln(\eta).$$

We can get a threshold test on y directly by first taking the negative of this expression and then dividing both sides by the positive term $(a-b)/\sigma^2$ to get

$$y \begin{matrix} \leq & \hat{x}(y)=b \\ & \hat{x}(y)=a \\ > \end{matrix} \frac{-\sigma^2 \ln(\eta)}{a-b} + \frac{b+a}{2}.$$

We get the same equation by switching the association of $X = 1$ and $X = 0$, which also changes the sign of the log threshold.

b) A budding medical researcher determines that the test is making too many errors. A new measurement procedure is devised with two observation random variables Y_1 and Y_2 . Y_1 is the same as in (a). Y_2 , under hypothesis 0, is given by $Y_2 = 5 + Z_1 + Z_2$, and, under hypothesis 1, is given by $Y_2 = 1 + Z_1 + Z_2$. Assume that Z_2 is independent of both Z_1 and X , and that $Z_2 \sim \mathcal{N}(0, \sigma^2)$. Find the MAP decision rule for $\hat{x}$ in terms of the joint observation (y_1, y_2) , and find $\Pr\{e \mid X=0\}$ and $\Pr\{e \mid X=1\}$. Hint: Find $f_{Y_2|Y_1,X}(y_2 \mid y_1, 0)$ and $f_{Y_2|Y_1,X}(y_2 \mid y_1, 1)$.

Solution: Note that Y_2 is simply Y_1 plus the noise term Z_2 , and that Z_2 is independent of X and Y_1 . Thus, Y_2 , conditional on Y_1 and X is simply $\mathcal{N}(Y_1, \sigma^2)$, which is independent of X . Thus Y_1 is a sufficient statistic and Y_2 is irrelevant. Including Y_2 does not change the probability of error.

c) Explain in laymen's terms why the medical researcher should learn more about probability.

Solution: It should have been clear intuitively that adding an additional observation that is only a noisy version of what has already been observed will not help in the decision, but knowledge of probability sharpens one's intuition so that something like this becomes self evident even without mathematical proof.

d) Now suppose that Z_2 , in (b), is uniformly distributed between 0 and 1 rather than being Gaussian. We are still given that Z_2 is independent of both Z_1 and X . Find the MAP decision rule for $\hat{x}$ in terms of the joint observation (y_1, y_2) and find $\Pr(e \mid X=0)$ and $\Pr(e \mid X=1)$.

Solution: The same argument as in (b) shows that Y_2 , conditional on Y_1 , is independent of X , and thus the decision rule and error probability do not change.

e) Finally, suppose that Z_1 is also uniformly distributed between 0 and 1. Again find the MAP decision rule and error probabilities.

Solution: By the same argument as before, Y_2 , conditional on Y_1 is independent of X , so Y_1 is a sufficient statistic and Y_2 is irrelevant. Since Z_1 is uniformly distributed between 0 and 1, then Y_1 lies between 5 and 6 for $X = 0$ and between 1 and 2 for $X = 1$. There is thus no possibility of error in this case.

Exercise 8.10: a) Consider a binary hypothesis testing problem, and denote the hypotheses as $X = 1$ and $X = -1$. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)^\top$ be an arbitrary real n -vector and let the observation be a sample value $\mathbf{y}$ of the random vector $\mathbf{Y} = X\mathbf{a} + \mathbf{Z}$ where $Z \sim \mathcal{N}(0, \sigma^2 I_n)$ and I_n is the $n \times n$ identity matrix. Assume that Z and X are independent. Find the maximum likelihood decision rule and find the probabilities of error $\Pr(e | X = -1)$ and $\Pr(e | X = 1)$ in terms of the function $Q(x)$.

Solution: This is a minor notational variation on Example 8.2.4. Since we are interested in maximum likelihood, $\ln \eta = 0$. The ML test, from (8.41), is then

$$\text{LLR}(\mathbf{y}) = \frac{2\mathbf{a}^\top \mathbf{y}}{\sigma^2} \underset{\hat{x}(\mathbf{y})=-1}{\overset{\hat{x}(\mathbf{y})=1}{\geq}} 0.$$

The error probabilities, from (8.44), are then

$$\Pr\{e | \mathbf{X}=1\} = Q(\gamma) \quad \Pr\{e | \mathbf{X}=-1\} = Q(\gamma),$$

where $\gamma = \|2\mathbf{a}\|/(2\sigma)$.

b) Now suppose a third hypothesis, $X = 0$, is added to the situation of (a). Again the observation random vector is $\mathbf{Y} = X\mathbf{a} + \mathbf{Z}$, but here X can take on values $-1, 0$, or $+1$. Find a one dimensional sufficient statistic for this problem (i.e., a one dimensional function of $\mathbf{y}$ from which the likelihood ratios

$$\Lambda_1(\mathbf{y}) = \frac{p_{Y|X}(\mathbf{y} | 1)}{p_{Y|X}(\mathbf{y} | 0)} \quad \text{and} \quad \Lambda_{-1}(\mathbf{y}) = \frac{p_{Y|X}(\mathbf{y} | -1)}{p_{Y|X}(\mathbf{y} | 0)}$$

can be calculated).

Solution: We have seen that the likelihood ratio for each of these binary decisions depends only on the noise in the direction of the difference between the vectors. Since each difference is $\mathbf{a}$, we conclude that $\mathbf{a}^\top \mathbf{y}$ is a sufficient statistic. One can verify this easily by calculating $\Lambda_1(\mathbf{y})$ and $\Lambda_{-1}(\mathbf{y})$.

c) Find the maximum likelihood decision rule for the situation in (b) and find the probabilities of error, $\Pr(e | X=x)$ for $x = -1, 0, +1$.

Solution: For $X = 1$, an error is made if $\Lambda_1(\mathbf{y})$ is less than 1. This occurs if $\mathbf{a}^\top \mathbf{y} < \|\mathbf{a}\|^2/2$ and has probability $\Pr\{e | X = 1\} = Q(\|\mathbf{a}\|/2\sigma)$. For $X = 0$, an error occurs if $\mathbf{a}^\top \mathbf{y} \geq \|\mathbf{a}\|^2/2$ or if $\mathbf{a}^\top \mathbf{y} < -\|\mathbf{a}\|^2/2$. Thus $\Pr\{e | X = 0\} = 2Q(\|\mathbf{a}\|/2\sigma)$. Finally, for $X = -1$, an error occurs if $\mathbf{a}^\top \mathbf{y} \geq -\|\mathbf{a}\|^2/2$. $\Pr\{e | X = -1\} = Q(\|\mathbf{a}\|/2\sigma)$.

d) Now suppose that $Z_1, \dots, Z_n$ in (a) are IID and each is uniformly distributed over the interval -2 to $+2$. Also assume that $\mathbf{a} = (1, 1, \dots, 1)^\top$. Find the maximum likelihood decision rule for this situation.

Solution: If $X = 1$, then each $Y_i, 1 \leq i \leq n$ lies between -1 and 3 and the conditional probability density of each such point is $(1/4)^n$. Similarly, for $X = -1$, each Y_i lies between -3 and 1 . If all Y_i are between -1 and $+1$, then the LLR is 0. If any are above 1, the LLR

is ∞ , and if any are below -1, the LLR is $-\infty$. Thus the ML rule is $\hat{x} = 1$ if any $Y_i > 1$ and $\hat{x} = -1$ if any $Y_i < 1$. Everything else (which has aggregate probability 2^{-n}) is ‘don’t care,’ which by convention is detected as $X = 1$.

Exercise 8.11: A sales executive hears that one of his salespeople is routing half of his incoming sales to a competitor. In particular, arriving sales are known to be Poisson at rate one per hour. According to the report (which we view as hypothesis $X=1$), each second arrival is routed to the competition; thus under hypothesis 1 the interarrival density for successful sales is $f(y|X=1) = ye^{-y}$; $y \geq 0$. The alternate hypothesis ($X=0$) is that the rumor is false and the interarrival density for successful sales is $f(y|X=0) = e^{-y}$; $y \geq 0$. Assume that, *a priori*, the hypotheses are equally likely. The executive, a recent student of stochastic processes, explores various alternatives for choosing between the hypotheses; he can only observe the times of successful sales however.

a) Starting with a successful sale at time 0, let S_i be the arrival time of the i^{th} subsequent successful sale. The executive observes $S_1, S_2, \dots, S_n$ ($n \geq 1$) and chooses the maximum a posteriori probability hypothesis given this data. Find the joint probability density $f(S_1, S_2, \dots, S_n|X=1)$ and $f(S_1, \dots, S_n|X=0)$ and give the decision rule.

Solution: The interarrival times are independent conditional each on $X = 1$ and $X = 0$. The density of an interarrival interval given $X = 1$ is Erlang of order 2, with density xe^{-x} , so

$$f_{S|X}(s_1, \dots, s_n|1) = \prod_{i=1}^n [(s_i - s_{i-1}) \exp -(s_i - s_{i-1})] = e^{-s_n} \prod_{i=1}^n (s_i - s_{i-1}).$$

The density of an interarrival interval given $X = 0$ is exponential, so

$$f_{S|X}(s_1, \dots, s_n|0) = e^{-s_n}.$$

The MAP rule, with $p_0 = p_1$ is then

$$\text{LLR}(\mathbf{y}) = \ln(s_0) + \sum_{i=2}^n \ln(s_i - s_{i-1}) \begin{matrix} \geq \\ < \end{matrix} \begin{matrix} \hat{x}(\mathbf{y})=1 \\ \hat{x}(\mathbf{y})=0 \end{matrix} 0. \quad (\text{A.172})$$

The executive might have based a decision only on the aggregate time for n sales to take place, but this would not have been a sufficient statistic for the sequence of sale times, so this would have yielded a higher probability of error. It is also interesting to note that a very short interarrival interval is weighed very heavily in (A.172), and this is not surprising since very short intervals are very improbable under $X = 1$. The sales person, if both fraudulent and a master of stochastic processes, would recognize that randomizing the sales to the competitor would make the fraud much more difficult to detect.

b) This is the same as (a) except that the system is in steady state at time 0 (rather than starting with a successful sale). Find the density of S_1 (the time of the first arrival after time 0) conditional on $X=0$ and on $X=1$. What is the decision rule now after observing $S_1, \dots, S_n$.

Solution: Under $X = 1$, the last arrival before 0 was successful with probability 1/2 and routed away with probability 1/2. Thus $f_{S_1|X}(s_1|1) = (1/2)e^{-s_1} + (1/2)s_1e^{-s_1}$. This could also be derived as a residual life probability. With this modification, the first term in the LLR of (A.172) would be changed from $\ln(s_1)$ to $\ln((s_1 + 1)/2)$.

c) This is the same as (b), except rather than observing n successful sales, the successful sales up to some given time t are observed. Find the probability, under each hypothesis, that the first successful sale occurs in $(s_1, s_1 + \Delta]$, the second in $(s_2, s_2 + \Delta]$, $\dots$, and the last in $(s_{N(t)}, s_{N(t)} + \Delta]$ (assume Δ very small). What is the decision rule now?

Solution: The somewhat artificial use of Δ here is to avoid dealing with the discrete rv $N(t)$ and the density of a random number of rv's. One can eliminate this artificiality after understanding the solution. Under $X = 0$, the probability that $N(t) = n$ and $S_i \in [s_i, s_i + \Delta)$ for $1 \leq i \leq n$ and $s_n \leq t$ (using an approximation for Δ very small) is

$$\Delta^n e^{-s_1} \left[\prod_{i=2}^n e^{-(s_i - s_{i-1})} \right] e^{-(t - s_n)} = \Delta^n e^{-t}. \quad (\text{A.173})$$

The term $e^{-(t - s_n)}$ on the left side is the probability of no arrivals in $(s_n, t]$, which along with the other arrival times specifies that $N(t) = n$.

Under $X = 1$, the term e^{-s_1} in (A.173) is changed to $(1/2)(s_1 + 1)e^{-s_1}$ as we saw in (b). The final term, $e^{-(t - s_n)}$ in (A.173) must be changed to the probability of no successful sales in $(s_n, t]$ under $X = 1$. This is $(t - s_n + 1)e^{-(t - s_n)}$. The term Δ^n cancels out in the likelihood ratio, so the decision rule is

$$\text{LLR}(\mathbf{y}) = \ln \left(\frac{1 + s_1}{2} \right) + \ln(t - s_n + 1) + \sum_{i=2}^n \ln(s_i - s_{i-1}) \underset{\hat{x}(\mathbf{y})=0}{\overset{\hat{x}(\mathbf{y})=1}{>}} 0.$$

Exercise 8.12: This exercise generalizes the MAP rule to cases where neither a PMF nor PDF exist. To view the problem in its simplest context, assume a binary decision, a one-dimensional observation ($y \in \mathbb{R}^1$), and fixed *a priori* probabilities p_0, p_1 . An arbitrary test, denoted test A , can be defined by the set A of observations that are mapped into decision $\hat{x} = 1$. Using such a test, the overall probability of correct decision is given by (8.6) as

$$\Pr\{\hat{X}_A(Y) = X\} = p_0 \Pr\{Y \in A^c \mid X = 0\} + p_1 \Pr\{Y \in A \mid X = 1\}. \quad (\text{A.174})$$

The maximum of this probability over all tests A is then

$$\sup_A \Pr\{\hat{X}_A(Y) = X\} = \sup_A [p_0 \Pr\{Y \in A^c \mid X = 0\} + p_1 \Pr\{Y \in A \mid X = 1\}]. \quad (\text{A.175})$$

We first consider this supremum over all A consisting of a finite union of disjoint intervals. Then (using measure theory), we show that the supremum over all measurable sets A is the same as that over finite unions of disjoint intervals.

a) If A is a union of k intervals, show that A^c is a union of at most $k + 1$ disjoint intervals. Intervals can be open or closed on each end and can be bounded by $\pm\infty$.

Solution: First suppose A is a union of k disjoint finite intervals, $(a_1, b_1), (a_2, b_2), \dots, (a_k, b_k)$, ordered so that $a_1 < a_2 < \dots < a_k$ and where the notation throughout omits distinguishing between the inclusion of end points. Then A^c is $(-\infty, a_1), (b_1, a_2), \dots, (b_k, \infty)$, which is $k + 1$ intervals. If $a_1 = -\infty$ or $b_k = \infty$, this simply reduces the number of intervals in A^c . If the intervals are not disjoint, then at most one interval in A^c can start at each b_i and at most one at $-\infty$, so the number is at most $k + 1$ in general.

b) Let $\mathcal{I}$ be the partition of $\mathbb{R}$ created by the intervals of both A and of A^c . Let $\mathcal{I}(j)$ be the j th interval in this partition. Show that

$$\Pr\{\hat{x}_A(Y) = X\} \leq \sum_j \max \left[p_0 \Pr\{Y \in \mathcal{I}(j) \mid X = 0\}, p_1 \Pr\{Y \in \mathcal{I}(j) \mid X = 1\} \right]. \quad (\text{A.176})$$

Hint: Break (A.174) into intervals and apply the MAP principle on an interval basis.

Solution: Express $\Pr\{Y \in A \mid X\}$ as $\sum_{i=1}^k \Pr\{Y \in (a_i, b_i) \mid X\}$. $\Pr\{Y \in A^c \mid X\}$ is expressed in the same way over the complementary intervals. Thus $\Pr\{\hat{x}_A(Y) = X\}$ is the sum in (A.176) without the max but rather a particular rule over each interval of $\mathcal{I}$.

c) The expression on the right in (A.176) is a function of the partition but is otherwise independent of A . It corresponds to a test where Y is first quantized to a finite set of intervals, and the MAP test is then applied to this discrete problem. We denote this test as $\text{MAP}(\mathcal{I})$. Let the partition $\mathcal{I}'$ be a refinement of $\mathcal{I}$ in the sense that each interval of $\mathcal{I}'$ is contained in some interval of $\mathcal{I}$. Show that

$$\Pr\{\hat{x}_{\text{MAP}(\mathcal{I})}(Y) = X\} \leq \Pr\{\hat{x}_{\text{MAP}(\mathcal{I}')} (Y) = X\}. \quad (\text{A.177})$$

Solution: To avoid a lot of notation, suppose $\mathcal{I}'$ is same as $\mathcal{I}$ except for one interval $(a, b) \in \mathcal{I}$ that is broken into $(a, c), (c, b)$ in $\mathcal{I}'$. Then $\text{MAP}(\mathcal{I})$ is the same as $\text{MAP}(\mathcal{I}')$ except that $\text{MAP}(\mathcal{I})$ is restricted to the same decision on (a, c) and (c, b) , and thus (A.177) is satisfied in this case. Any finite refinement can be expressed as a sequence of single refinements of this type.

d) Show that for any two finite partitions of $\mathbb{R}$ into intervals, there is a third finite partition that is a refinement of each of them.

Solution: If $\mathcal{I}$ is a finite partition of n disjoint intervals, it is specified by the $n - 1$ points separating these intervals. Similarly if $\mathcal{I}'$ has m disjoint intervals, it is specified by $m - 1$ points. Taking the union of these point sets, we specify another partition which is a refinement of both.

e) Consider a sequence $\{A_j, j \geq 1\}$ (each is a finite union of intervals) that approaches the sup (over finite unions of intervals) of (A.175). Demonstrate a corresponding sequence of successive refinements of partitions $\{\mathcal{I}_j, j \geq 1\}$ for which $\Pr\{\hat{x}_{\text{MAP}(\mathcal{I}_j)}(Y) = X\}$ approaches the same limit.

Note what this exercise has shown so far: if the likelihoods have no PDF or PMF, there is no basis for a MAP test on an individual observation y . However, by quantizing the observations sufficiently finely, the quantized MAP rule has an overall probability of being correct that is as close as desired to the optimum over all rules where A is a finite (but arbitrarily large) union of intervals. The arbitrarily fine quantization means that the decisions are arbitrarily close to pointwise decisions, and the use of quantization means that infinitely fine resolution is not required for the observations.

Solution: We demonstrated in (b) that each finite union of intervals A_j corresponds to a finite partition, say $\mathcal{I}_j$ for which (A.176) is satisfied. Each test $\text{MAP}(\mathcal{I}_j)$ also corresponds to a test A'_j where A'_j is the union of intervals in $\mathcal{I}_j$ that are mapped into decision 1. Thus, for each j ,

$$\Pr\{\hat{x}_{\text{MAP}(\mathcal{I}_j)}(Y) = X\} \leq \sup_A \Pr\{\hat{X}_A(Y) = X\}.$$

where the sup is over finite unions of intervals. Thus $\lim_{j \rightarrow \infty} \Pr\{\hat{x}_{\text{MAP}(\mathcal{I}_j)}(Y) = X\}$ must have this same limit.

f) Next suppose the supremum in (A.175) is taken over all measurable sets $A \in \mathbb{R}$. Show that, given any measurable set A and any $\epsilon > 0$, there is a finite union of intervals A' such that $\Pr\{A \neq A' \mid X = \ell\} \leq \epsilon$ both for $\ell = 0$ and $\ell = 1$. Show from this that the sup in (A.175) is the same whether taken over measurable sets or finite unions of intervals.

Solution: This is a standard result in measure theory.

Exercise 8.13: For the min-cost hypothesis testing problem of Section 8.3, assume that there is a cost C_0 of choosing $X=1$ when $X=0$ is correct, and a cost C_1 of choosing $X=0$ when $X=1$ is correct. Show that a threshold test minimizes the expected cost using the threshold $\eta = (C_0 p_0)/(C_1 p_1)$.

Solution: For a given sample vector $\mathbf{y}$ of the received rv $\mathbf{Y}$, the expected cost if the decision $\hat{x}(\mathbf{y}) = 0$ is made is the probability of an incorrect decision (*i.e.*, that $X = 1$) times the cost C_1 . Thus

$$E[\text{Cost} \mid \mathbf{Y} = \mathbf{y}, \hat{x}(\mathbf{y}) = 0] = C_1 \Pr\{X = 1 \mid \mathbf{Y} = \mathbf{y}\}.$$

Similarly, if the decision $\hat{x} = 1$ is made,

$$E[\text{Cost} \mid \mathbf{Y} = \mathbf{y}, \hat{x}(\mathbf{y}) = 1] = C_0 \Pr\{X = 0 \mid \mathbf{Y} = \mathbf{y}\}.$$

The min-cost decision, given $\mathbf{Y} = \mathbf{y}$, is the decision for which this conditional cost is minimum (with the usual convention of choosing $\hat{x}(\mathbf{y}) = 1$ in case of a tie. Thus the decision can be expressed as

$$C_1 \Pr\{X = 1 \mid \mathbf{Y} = \mathbf{y}\} \begin{matrix} \geq \\ < \end{matrix} \begin{matrix} \hat{x}(\mathbf{y})=1 \\ \hat{x}(\mathbf{y})=0 \end{matrix} C_0 \Pr\{X = 0 \mid \mathbf{Y} = \mathbf{y}\}.$$

Using Bayes' law on each probability, and rearranging terms,

$$\Lambda(\mathbf{y}) \begin{matrix} \geq \\ < \end{matrix} \begin{matrix} \hat{x}(\mathbf{y})=1 \\ \hat{x}(\mathbf{y})=0 \end{matrix} \frac{C_0 p_0}{C_1 p_1}.$$

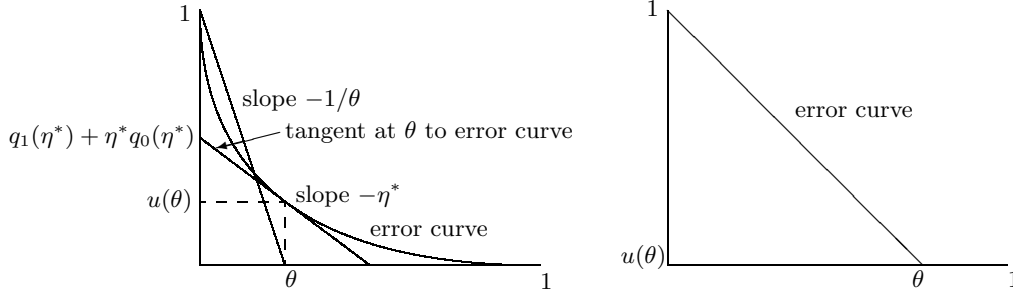
Exercise 8.14: a) For given θ , $0 < \theta \leq 1$, let η^* achieve the supremum $\sup_{0 \leq \eta < \infty} q_1(\eta) + \eta(q_0(\eta) - \theta)$. Show that $\eta^* \leq 1/\theta$. Hint: Think in terms of Lemma 8.4.1 applied to a very simple test.

Solution: Recall that $q_0(\eta)$ and $q_1(\eta)$ are the error probabilities, given $X = 0$ and $X = 1$ respectively, using a threshold test with threshold η . As shown in Lemma 8.4.1, for each η , there is a straight line of slope $-\eta$ passing through the point $(q_0(\eta), q_1(\eta))$ and for every test A , the point $(q_0(A), q_1(A))$ lies North East of that line.

Since this is true for all $\eta > 0$, it is also true for the error curve, defined as the supremum of all these straight lines, $\sup_{0 \leq \eta < \infty} q_1(\eta) + \eta(q_0(\eta) - \theta)$. Then (see Theorem 8.4.2), for all tests A , $(q_0(A), q_1(A))$ lies above or on the curve $(\theta, u(\theta))$; $0 \leq \theta \leq 1$.

From the first figure below, we can see that the slope $|1/\theta| \geq |\eta^*|$ where η^* is the slope of the error curve at θ . To see this analytically, $|1/\theta|$ is the slope of the hypotenuse of a right triangle of height 1 and width θ . Similarly, $|\eta^*|$ is the slope of the hypotenuse of a right

triangle of height $q_1(\eta^*) + \eta^* q_0(\eta^*) - u(\theta)$ and width θ . Now $u(\theta) \geq 0$ and (as explained more fully in the next paragraph), $q_1(\eta) + \eta q_0(\eta) \leq 1$ for all $\eta \geq 0$, showing that $|1/\theta| \geq |\eta^*|$. The second figure simply gives an example (similar to Example 8.4.3) in which this inequality is met with equality.



Finally, to verify that $q_1(\eta) + \eta q_0(\eta) \leq 1$, note from the proof of Lemma 8.4.1 that $q_1(\eta) + \eta q_0(\eta) \leq q_1(A) + \eta q_0(A)$ for all sets A . Choosing A to be the empty set (*i.e.*, always decide hypothesis 0), we have $q_1(A) = 1$ and $q_0(A) = 0$, so $q_1(\eta) + \eta q_0(\eta) \leq 1$.

b) Show that the magnitude of the slope of the error curve $u(\theta)$ at θ is at most $1/\theta$.

Solution: The slope of the error curve at θ is η^* as defined and shown in (a). It was also shown in (a) that $|\eta^*| \leq |1/\theta|$.

Exercise 8.15: Consider a binary hypothesis testing problem where X is 0 or 1 and a one dimensional observation Y is given by $Y = X + U$ where U is uniformly distributed over $[-1, 1]$ and is independent of X .

a) Find $f_{Y|X}(y | 0)$, $f_{Y|X}(y | 1)$ and the likelihood ratio $\Lambda(y)$.

Solution: Note that $f_{Y|X}$ is simply the density of U shifted by X , *i.e.*,

$$f_{Y|X}(y | 0) = \begin{cases} 1/2; & -1 \leq y \leq 1 \\ 0; & \text{elsewhere} \end{cases} \quad f_{Y|X}(y | 1) = \begin{cases} 1/2; & 0 \leq y \leq 2 \\ 0; & \text{elsewhere} \end{cases}.$$

The likelihood ratio $\Lambda(y)$ is defined only for $-1 \leq y \leq 2$ since neither conditional density is non-zero outside this range.

$$\Lambda(y) = \frac{f_{Y|X}(y | 1)}{f_{Y|X}(y | 0)} = \begin{cases} 0; & -1 \leq y < 0 \\ 1; & 0 \leq y \leq 1 \\ \infty; & 1 < y \leq 2 \end{cases}.$$

b) Find the threshold test at η for each η , $0 < \eta < \infty$ and evaluate the conditional error probabilities, $q_0(\eta)$ and $q_1(\eta)$.

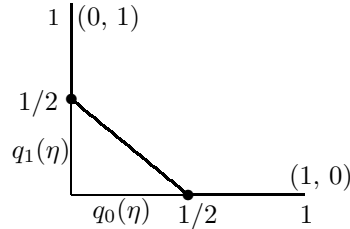
Solution: Since $\Lambda(y)$ has finitely many (3) possible values, all values of η between any adjacent pair lead to the same threshold test. Thus, for $\eta > 1$, $\Lambda(y) \geq \eta$, if and only if (iff) $\Lambda(y) = \infty$. Thus $\hat{x} = 1$ iff $1 < y \leq 2$. For $\eta = 1$, $\hat{x} = 1$ iff $\Lambda(y) \geq 1$, *i.e.*, iff $\Lambda(y)$ is 1 or ∞ . Thus $\hat{x} = 1$ iff $0 \leq y \leq 2$. For $\eta < 1$, $\Lambda(y) \geq \eta$ iff $\Lambda(y)$ is 1 or ∞ . Thus $\hat{x} = 1$ iff $0 \leq y \leq 2$. Note that the MAP test is the same for $\eta = 1$ and $\eta < 1$, in both cases choosing $\hat{x} = 1$ for $0 \leq y \leq 2$.

Consider $q_1(\eta)$ (the error probability using a threshold test at η conditional of $X = 1$). For $\eta > 1$, we have seen that $\hat{x} = 1$ (no error) for $1 < y \leq 2$. This occurs with probability $1/2$

given $X = 1$. Thus $q_1(\eta) = 1/2$ for $\eta > 1$. Also, for $\eta > 1$, $\hat{x} = 0$ for $-1 \leq y \leq 1$. Thus $q_0(\eta) = 0$. Reasoning in the same way for $\eta \leq 1$, we have $q_1(\eta) = 0$ and $q_0(\eta) = 1/2$.

c) Find the error curve $u(\alpha)$ and explain carefully how $u(0)$ and $u(1/2)$ are found (hint: $u(0) = 1/2$).

Solution: Each $\eta > 1$ maps into the pair of error probabilities $(q_0(\eta), q_1(\eta)) = (0, 1/2)$. Similarly, each $\eta \leq 1$ maps into the pair of error probabilities $(q_0(\eta), q_1(\eta)) = (1/2, 0)$. The error curve contains these points and also contains the supremum of the straight lines of each slope $-\eta$ around $(0, 1/2)$ for $\eta > 1$ and around $(1/2, 0)$ for $\eta \leq 1$. The resulting curve is given below.



Another approach (perhaps more insightful) is to repeat (a) and (b) for the alternative threshold tests that choose $\hat{x} = 0$ in the don't care cases, *i.e.*, the cases for $\eta = 1$ and $0 \leq y \leq 1$. It can be seen that Lemma 8.4.1 and Theorem 8.4.2 apply to these alternative threshold tests also. The points on the straight line between $(0, 1/2)$ and $(1/2, 0)$ can then be achieved by randomizing the choice between the threshold tests and the alternative threshold tests.

d) Find a *discrete* sufficient statistic $v(y)$ for this problem that has 3 sample values.

Solution: $v(y) = \lambda(y)$ is a discrete sufficient statistic with 3 sample values.

e) Describe a decision rule for which the error probability under each hypothesis is $1/4$. You need not use a randomized rule, but you need to handle the don't-care cases under the threshold test carefully.

Solution: The don't care cases arise for $0 \leq y \leq 1$ when $\eta = 1$. With the decision rule of (8.11), these don't care cases result in $\hat{x} = 1$. If half of those don't care cases are decided as $\hat{x} = 0$, then the error probability given $X = 1$ is increased to $1/4$ and that for $X = 0$ is decreased to $1/4$. This could be done by random choice, or more easily, by mapping $y > 1/2$ into $\hat{x} = 1$ and $y \leq 1/2$ into $\hat{x} = 0$.

A.9 Solutions for Chapter 9

Exercise 9.1: Consider the simple random walk $\{S_n; n \geq 1\}$ of Section 9.1.1 with $S_n = X_1 + \cdots + X_n$ and $\Pr\{X_i = 1\} = p$; $\Pr\{X_i = -1\} = 1 - p$; assume that $p \leq 1/2$.

a) Show that $\Pr\left\{\bigcup_{n \geq 1} \{S_n \geq k\}\right\} = \left[\Pr\left\{\bigcup_{n \geq 1} \{S_n \geq 1\}\right\}\right]^k$ for any positive integer k . Hint: Given that the random walk ever reaches the value 1, consider a new random walk starting at that time and explore the probability that the new walk ever reaches a value 1 greater than its starting point.

Solution: Since $\{S_n; n \geq 1\}$ changes only in increments of 1 or -1, the only way that the walk can reach a threshold at integer $k > 1$ (i.e., the only way that the event $\bigcup_{n \geq 1} \{S_n \geq k\}$ can occur), is if the walk first eventually reaches the value 1, and then starting from the first time that 1 is reached, goes on to eventually reach 2, and so forth onto k .

The probability of eventually reaching 2 given that 1 is reached is the same as the probability of eventually reaching 1 starting from 0; this is most clearly seen from the Markov chain depiction of the simple random walk given in Figure 6.1. Similarly, the probability of eventually reaching any j starting from $j - 1$ is again the same, so (using induction if one insists on being formal), we get the desired relationship. This relationship also holds for $p > 1/2$.

b) Find a quadratic equation for $y = \Pr\left\{\bigcup_{n \geq 1} \{S_n \geq 1\}\right\}$. Hint: explore each of the two possibilities immediately after the first trial.

Solution: As explained in Example 5.5.4, $y = p + (1 - p)y^2$.

c) For $p < 1/2$, show that the two roots of this quadratic equation are $p/(1 - p)$ and 1. Argue that $\Pr\left\{\bigcup_{n \geq 1} \{S_n \geq 1\}\right\}$ cannot be 1 and thus must be $p/(1 - p)$.

Solution: This is also explained in Example 5.5.4.

d) For $p = 1/2$, show that the quadratic equation in (c) has a double root at 1, and thus $\Pr\left\{\bigcup_{n \geq 1} \{S_n \geq 1\}\right\} = 1$. Note: this is the very peculiar case explained in Section 5.5.1.

Solution: As explained in Example 5.5.4, the fact that the roots are both 1 means that $\Pr\left\{\bigcup_{n \geq 1} \{S_n \geq 1\}\right\} = 1$.

e) For $p < 1/2$, show that $p/(1 - p) = \exp(-r^*)$ where r^* is the unique positive root of $g(r) = 1$ where $g(r) = \mathbb{E}[e^{rX}]$.

Solution: Note that $g(r) = \mathbb{E}[e^{rX}] = pe^r + (1 - p)e^{-r}$. The positive root of $g(r) = 1$ is the $r^* > 0$ for which

$$1 = pe^{r^*} + (1 - p)e^{-r^*}.$$

This is quadratic in e^{r^*} (and also in e^{-r^*}) and is the same equation as in (b) if we substitute y for e^{-r^*} . The two solutions are $e^{-r^*} = 1$ ($r^* = 0$) and $e^{-r^*} = p/(1 - p)$ ($r^* = \ln[(1 - p)/p]$). Thus the unique positive solution for r^* is $\ln[(1 - p)/p]$. Thus the optimized Chernoff bound for crossing a threshold is satisfied with equality by the simple combinatorial solution in (c).

Exercise 9.2: Consider a G/G/1 queue with IID arrivals $\{X_i; i \geq 1\}$, IID FCFS service times $\{Y_i; i \geq 0\}$, and an initial arrival to an empty system at time 0. Define $U_i = Y_{i-1} - X_i$ for $i \geq 1$. (Note:

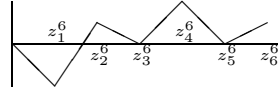
the problem statement in the text incorrectly defined this as $U_i = X_i - Y_{i-1}$, which is inconsistent with the treatment in Section 9.2) Consider a sample path where $(u_1, \dots, u_6) = (1, -2, 2, -1, 3, -2)$.

a) Let $Z_i^6 = U_6 + U_{6-1} + \dots + U_{6-i+1}$. Find the queueing delay for customer 6 as the maximum of the ‘backward’ random walk with elements $0, Z_1^6, Z_2^6, \dots, Z_6^6$; sketch this random walk.

Solution:

$$\begin{aligned} Z_1^6 &= U_6 &= -2 \\ Z_2^6 &= U_6 + U_5 &= 1 \\ Z_3^6 &= U_6 + U_5 + U_4 &= 0 \\ Z_4^6 &= U_6 + U_5 + U_4 + U_3 &= 2 \\ Z_5^6 &= U_6 + U_5 + U_4 + U_3 + U_2 &= 0 \\ Z_6^6 &= U_6 + U_5 + U_4 + U_3 + U_2 + U_1 &= 1 \end{aligned}$$

Note that the equation for Z_i^n is $\sum_{j=0}^{i-1} U_{n-j}$, rather than the slight typo given for Z_i^n in the equation between (9.5) and (9.6) in the text. The queueing delay W_6 , for customer 6 is the maximum of $(0, Z_1^6, \dots, Z_6^6)$, *i.e.*, 2.



b) Find the queueing delay for customers 1 to 5.

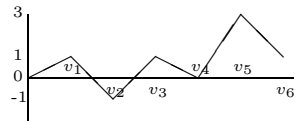
Solution: In the same way, $(W_1, \dots, W_5) = (1, 0, 2, 1, 4)$

c) Which customers start a busy period (*i.e.*, arrive when the queue and server are both empty)? Verify that if Z_i^6 maximizes the random walk in (a), then a busy period starts with arrival $6 - i$.

Solution: Customer 2 starts a busy period (*i.e.*, $W_2 = 0$) and Z_{6-i}^6 is maximized in the random walk in (a) by $i = 2$.

d) Now consider a forward random walk $V_n = U_1 + \dots + U_n$. Sketch this walk for the sample path above and show that the queueing delay for each customer is the difference between two appropriately chosen values of this walk.

Solution: $(v_1, \dots, v_6) = (1, -1, 1, 0, 3, 1)$. The corresponding sample path segment is then



Note that $Z_i^n = U_n + \dots + U_{n-i+1}$ and this can be rewritten as $V_n - V_{n-i}$. Thus $W_n = \max_{i \leq n} V_n - V_{n-i} = V_n - \min_{i \leq n} V_i$. Thus the queue delay at any n is the difference between the V random walk at n and the lowest previous point on that random walk. To understand this, consider the random walk $\{V_n; n \geq 1\}$. The first arrival $i > 0$ for which $v_i < 0$ is the first $i > 0$ to see an already empty system, and thus $W_i = 0$ rather than $W_i = V_i < 0$. Subsequent delays are then determined as if the random walk is reset to 0 at the i th arrival. The next arrival j that sees an already empty system is the first j for which $V_j < V_i$. and

so forth at each arrival to see an already empty system. Thus the smallest V_j for $j \leq n$ is the last $j \leq n$ to start a new busy period. This is somewhat more intuitive than the backward random walk, but the backward random walk is the one that connects G/G/1 queue-waiting with random walks.

Exercise 9.3: A G/G/1 queue has a deterministic service time of 2 and interarrival times that are 3 with probability $p < 1/2$ and 1 with probability $1 - p$.

a) Find the distribution of W_1 , the wait in queue of the first arrival after the beginning of a busy period.

Solution: The service time is always 2 units, so the first arrival after the beginning of a busy period must arrive either after 1 unit of the service is completed or 1 unit after the service is completed. Thus $\Pr\{W_1 = 0\} = p$ and $\Pr\{W_1 = 1\} = 1 - p$.

b) Find the distribution of W_∞ , the steady-state wait in queue.

Solution: The rv $U_i = Y_{i+1} - X_i$ is binary (either 2-1 or 2-3), so $\Pr\{U_i = 1\} = p$ and $\Pr\{U_i = -1\} = 1 - p$. The backward random walk is thus a simple random walk. From Theorem 9.2.1, $\Pr\{W \geq k\}$ is the probability that the random walk based on $\{U_i; i \geq 1\}$ ever crosses k . From (9.2), recognizing that the simple random walk is integer valued,

$$\Pr\{W \geq k\} = \left(\frac{p}{1-p} \right)^k.$$

c) Repeat (a) and (b) assuming the service times and interarrival times are exponentially distributed with rates μ and λ respectively.

Solution: This is an M/M/1 queue. With probability $\mu/(\lambda + \mu)$, the first arrival appears after the initial departure and thus has no wait in the queue. With probability $\lambda/(\lambda + \mu)$ the first arrival appears before the initial departure, and then waits in queue for an exponentially distributed time of rate μ . Thus $\Pr\{W_1 > w\} = \lambda/(\lambda + \mu) \exp(-\mu w)$ for $w \geq 0$.

In steady state, *i.e.*, to find $\Pr\{W > w\} = \lim_{n \rightarrow \infty} \Pr\{W_n > w\}$, we recall that the steady-state number N in the system is geometric with $\mathbf{p}_N(n) = (1 - \lambda/\mu)(\lambda/\mu)^n$ for $n \geq 0$. Given $N = 0$ (an event of probability $1 - \lambda/\mu$), the wait is 0. Given $N = n$, the wait in the queue is $\sum_{i=1}^n Y_i^*$ where each Y_i^* is exponential with rate μ . Thus for $N > 0$ (an event of probability λ/μ) the wait is a geometrically distributed sum of exponential rv's. From Example 2.3.3, this is equivalent to a sum of two Poisson processes, one of rate λ and one of rate $\mu - \lambda$ where the first arrival from the $\mu - \lambda$ process is geometrically distributed in the combined μ process. Thus,

$$\Pr\{W \geq w\} = \frac{\lambda}{\mu} \exp(-(\mu - \lambda)w); \quad \text{for } w \geq 0.$$

Exercise 9.4: Let $U = V_1 + \cdots + V_n$ where $V_1, \dots, V_n$ are IID rv's with the MGF $\mathbf{g}_V(s)$. Show that $g_U(s) = [g_V(s)]^n$. Hint: You should be able to do this simply in a couple of lines.

Solution:

$$g_U(s) = \mathbf{E}[\exp(sU)] = \mathbf{E}[\exp(sV_1 + sV_2 + \cdots + sV_n)] = [\mathbf{g}_V(s)]^n,$$

where we have used the fact that the expectation of a product of independent rv's is equal to the product of the expectations.

Exercise 9.5: Let $\{a_n; n \geq 1\}$ and $\{\alpha_n; n \geq 1\}$ be sequences of numbers. For some b , assume that $a_n \leq \alpha_n e^{-bn}$ for all $n \geq 1$. For each of the following choices for α_n and arbitrary b , determine whether the bound is exponentially tight. Note: you may assume $b \geq 0$ if you wish, but it really does not matter.

a) $\alpha_n = \sum_{j=1}^k c_j n^{-j}$.

b) $\alpha_n = \exp(-\sqrt{n})$.

c) $\alpha_n = e^{-n^2}$.

Solution to (a), (b), (c): The question is ill-posed, since we know nothing about the sequence $\{a_n; n \geq 1\}$ other than it is upper-bounded by $a_n \leq \alpha_n e^{-bn}$. For example, $a_n = 0$ for each $n \geq 1$ would satisfy the indicated bounds (assuming each $c_j > 0$ in part (a)). To make sense of the exercise, assume that $a_n = \alpha_n e^{-bn}$ for each $n \geq 1$ and assume that the intended bound is $a_n \leq e^{-bn}$ for each n . Thus the question has to do with whether the coefficients $\{\alpha_n; n \geq 1\}$ are 'significant enough' to destroy exponential tightness.

By definition of exponential tightness, the sequence of bounds $a_n \leq e^{-bn}$ for each $n \geq 1$ is exponentially tight if $\lim_{n \rightarrow \infty} (1/n) \ln[a_n] = \lim_{n \rightarrow \infty} \ln[e^{-bn}]$, i.e., if

$$\lim_{n \rightarrow \infty} (1/n) \ln[\alpha_n e^{-bn}] = \lim_{n \rightarrow \infty} (1/n) \ln[e^{-bn}].$$

Simplifying, the sequence of bounds is exponentially tight if $\lim_{n \rightarrow \infty} (1/n) \ln[\alpha_n] = 0$. Thus, assuming $c_j > 0$ for $1 \leq j \leq k$, we have $\lim_{n \rightarrow \infty} (1/n) \ln[\sum_{j=1}^k c_j n^{-j}] = 0$. Thus the bound is exponentially tight in part (a), and is similarly exponentially tight in (b). The bound is not exponentially tight in (c) since $\lim_{n \rightarrow \infty} (1/n) \ln[e^{-n^2}] = -\infty$. In other words, the 'coefficient' e^{-n^2} overwhelms the exponential effect in part (c).

Exercise 9.6: Define $\gamma(r)$ as $\ln[g(r)]$ where $g(r) = E[\exp(rX)]$. Assume that X is discrete with possible outcomes $\{a_i; i \geq 1\}$, let p_i denote $\Pr\{X = a_i\}$, and assume that $g(r)$ exists in some open interval (r_-, r_+) containing $r = 0$. For any given r , $r_- < r < r_+$, define a random variable X_r with the same set of possible outcomes $\{a_i; i \geq 1\}$ as X , but with a probability mass function $q_i = \Pr\{X_r = a_i\} = p_i \exp[a_i r - \gamma(r)]$. X_r is not a function of X , and is not even to be viewed as in the same probability space as X ; it is of interest simply because of the behavior of its defined probability mass function. It is called a tilted random variable relative to X , and this exercise, along with Exercise 9.11 will justify our interest in it.

a) Verify that $\sum_i q_i = 1$.

Solution: Note that these tilted probabilities are described in Section 9.3.2.

$$\sum_i q_i = \sum_i p_i \exp[a_i r - \gamma(r)] = \frac{1}{g(r)} \sum_i p_i \exp[a_i r] = \frac{g(r)}{g(r)} = 1.$$

b) Verify that $E[X_r] = \sum_i a_i q_i$ is equal to $\gamma'(r)$.

Solution:

$$\begin{aligned} E[X_r] &= \sum_i a_i p_i \exp[a_i r - \gamma(r)] = \frac{1}{g(r)} \sum_i a_i p_i \exp[a_i r] \\ &= \frac{1}{g(r)} \frac{d}{dr} \sum_i p_i e^{r a_i} = \frac{g'(r)}{g(r)} = \gamma'(r). \end{aligned} \tag{A.178}$$

c) Verify that $\text{VAR}[X_r] = \sum_i a_i^2 q_i - (\mathbb{E}[X_r])^2$ is equal to $\gamma''(r)$.

Solution: We first calculate the second moment, $\mathbb{E}[X_r^2]$.

$$\begin{aligned} \mathbb{E}[X_r^2] &= \sum_i a_i^2 p_i \exp[a_i r - \gamma(r)] = \frac{1}{g(r)} \sum_i a_i^2 p_i \exp[a_i r] \\ &= \frac{1}{g(r)} \frac{d^2}{dr^2} \sum_i p_i e^{ra_i} = \frac{g''(r)}{g(r)}. \end{aligned} \quad (\text{A.179})$$

Using (A.178) and (A.179),

$$\text{VAR}[X_r] = \frac{g''(r)}{g(r)} - \frac{[g'(r)]^2}{[g(r)]^2} = \frac{d^2}{dr^2} \ln(g(r)) = \gamma''(r).$$

d) Argue that $\gamma''(r) \geq 0$ for all r such that $g(r)$ exists, and that $\gamma''(r) > 0$ if $\gamma''(0) > 0$.

Solution: Since $\gamma''(r)$ is the variance of a rv, it is nonnegative. If $\text{VAR}[X] > 0$, then X is non-atomic, which shows that X_r is non-atomic and thus has a positive variance wherever it exists.

e) Give a similar definition of X_r for a random variable X with a density, and modify (a) to (d) accordingly.

Solution: If X has a density, $f_X(x)$, and also has an MGF over some region of r , then the tilted variable X_r is defined to have the density $f_{X_r}(x) = f_X(x) \exp(xr - \gamma(r))$. The derivations above follow as before except for the need of more care about convergence.

Exercise 9.7: a) Suppose Z is uniformly distributed over $[-b, +b]$. Find $g_Z(r)$, $g'_Z(r)$, and $\gamma'(r)$ as a function of r .

Solution:

$$\begin{aligned} g_Z(r) &= \mathbb{E}[e^{rZ}] = \int_{-b}^b \frac{1}{2b} e^{rz} dz = \frac{e^{br} - e^{-br}}{2br}. \\ g'_Z(r) &= \frac{d}{dr} \left(\frac{e^{br} - e^{-br}}{2br} \right) = \frac{(br - 1)e^{br} + (br + 1)e^{-br}}{2br^2}. \\ \gamma'_Z(r) &= \frac{g'_Z(r)}{g_Z(r)} = \frac{(br - 1)e^{br} + (br + 1)e^{-br}}{r(e^{br} - e^{-br})}. \end{aligned} \quad (\text{A.180})$$

b) Show that the interval over which $g_Z(r)$ exists is the real line, i.e., $r_+ = \infty$ and $r_- = -\infty$. Show that $\lim_{r \rightarrow \infty} \gamma'(r) = b$.

Solution: We know from the theoretical development that all these quantities exist over an interval r_- to r_+ , and it is clear from (A.180) that they are finite over all nonzero r , but it is not immediately clear from (A.180) itself that they are finite for $r = 0$. If one expands e^{br} and e^{-br} into power series around $r = 0$, however, it can be seen that $g_Z(0) = 1$ and

$g'_Z(0) = 0$. Finally, by dividing numerator and denominator of (A.180) by re^{br} , it is clear that $\lim_{r \rightarrow \infty} \gamma'_Z(r) = b$.

c) Show that if $a > b$, then $\inf_{r \in I(X)} [\gamma(r) - ra] = -\infty$, so that the optimized Chernoff bound in (9.10) says that $\Pr\{S_n > na\} = 0$. Explain without any mathematics why $\Pr\{S_n > na\} = 0$ must be valid. Explain (with as little mathematical verbiage as possible) why an infimum rather than a minimum must be used in (9.10).

Solution: We know that $\gamma''(r) \geq 0$ and thus, since $\lim_{r \rightarrow \infty} \gamma'_Z(r) = b$, we must have $\gamma'(r) \leq b$ for all r . Since $\gamma_Z(0) = 0$, it then follows that $\gamma(r) \leq br$. Thus $\gamma(r) - ar \leq (b-a)r$, which approaches $-\infty$ as $r \rightarrow \infty$. Thus the optimized Chernoff bound is 0. Since S_n is the sum of n rv's each upper bounded by b , it is obvious that $\Pr\{S_n > bn\} = 0$ and thus $\Pr\{S_n > an\} = 0$. Finally an infimum is needed in (9.10) because there is no minimum over finite r .

d) For an arbitrary rv X , show that that if $r_+ = \infty$ and $\lim_{r \rightarrow \infty} \gamma'(r) = b$, then (9.10) says that $\Pr\{S_n > na\} = 0$ for $a > b$.

Solution: The argument in (c) is in fact general for any rv under the above assumptions.

Exercise 9.8: Note that the MGF of the nonnegative exponential rv with density e^{-x} is $(1-r)^{-1}$ for $r < r_+ = 1$. It can be seen from this that $g(r_+)$ does not exist (*i.e.*, it is infinite) and both $\lim_{r \rightarrow r_+} g(r) = \infty$ and $\lim_{r \rightarrow r_+} g'(r) = \infty$, where the limit is over $r < r_+$. In this exercise, you are first to assume an arbitrary rv X for which $r_+ < \infty$ and $g(r_+) = \infty$ and show that both $\lim_{r \rightarrow r_+} g(r) = \infty$ and $\lim_{r \rightarrow r_+} g'(r) = \infty$. You will then use this to show that if $\sup_{r < r_+} g'(r) < \infty$, then $\gamma(r_+) < \infty$ and $\mu(a)$ in (9.10) is given by $\mu(a) = \gamma(r_+) - r_+a$ for $a > \sup_{r < r_+} \gamma'(r)$.

a) For X such that $r_+ < \infty$ and $g(r_+) = \infty$, explain why

$$\lim_{A \rightarrow \infty} \int_0^A e^{xr_+} dF(x) = \infty. \quad (\text{A.181})$$

Solution: $g(r_+) = E[re^{xr_+}] = \int_0^\infty e^{xr_+} dF(x)$. This is, by the definition of an integral with an infinite upper limit, the integral from 0 to ∞ is the limit on the left of (A.181).

b) Show that for any $\epsilon > 0$ and any $A > 0$,

$$g(r_+ - \epsilon) \geq e^{-\epsilon A} \int_0^A e^{xr_+} dF(x). \quad (\text{A.182})$$

Solution: Since $e^{x(r_+ - \epsilon)} \geq 0$,

$$g(r_+ - \epsilon) \geq \int_0^A e^{x(r_+ - \epsilon)} dF(x) = e^{-\epsilon A} \int_0^A e^{xr_+} dF(x).$$

c) Choose $A = 1/\epsilon$ and show that

$$\lim_{\epsilon \rightarrow 0} g(r_+ - \epsilon) = \infty.$$

Solution: Using $A = 1/\epsilon$ in (A.182),

$$g(r_+ - \epsilon) \geq e^{-1} \int_0^{1/\epsilon} e^{xr_+} dF(x).$$

From (A.181), the limit of this as $\epsilon \rightarrow 0$ is ∞ .

d) Show that $\lim_{\epsilon \rightarrow 0} g'(r_+ - \epsilon) = \infty$.

Solution: Using the same approach as in (A.182) and then lower-bounding the coefficient x by 1 for $x \geq 1$,

$$\begin{aligned} g'(r_+ - \epsilon) &\geq \int_0^A x e^{x(r_+ - \epsilon)} dF_X(x) \geq \int_1^A e^{x(r_+ - \epsilon)} dF_X(x) \\ &\geq \int_0^A e^{x(r_+ - \epsilon)} dF_X(x) - e^{r_+}. \end{aligned}$$

As shown in (c), if $1/\epsilon$ is substituted for A in the final integral, and then the limit $\epsilon \rightarrow 0$ is taken, we get $\lim_{\epsilon \rightarrow 0} g'(r_+ - \epsilon) = \infty$.

e) Use (a) to (d) to show that if $r_+ < \infty$ and $\sup_{r < r_+} g'(r) < \infty$, then $g(r_+) < \infty$.

Solution: Given that $r_+ < \infty$ and $\epsilon > 0$, we have shown that $g(r_+) = \infty$ implies that $\lim_{\epsilon \rightarrow 0} g'(r_+ - \epsilon) = \infty$. Thus if $\lim_{\epsilon \rightarrow 0} g'(r_+ - \epsilon) < \infty$, we must have $g(r_+) < \infty$. Since $g'(r)$ is increasing with r , $\sup_{r < r_+} g'(r) = \lim_{\epsilon \rightarrow 0} g'(r_+ - \epsilon)$, so $\sup_{r < r_+} g'(r) < \infty$ also implies that $g(r_+) < \infty$.

f) Show that if $r_+ < \infty$ and $\sup_{r < r_+} g'(r) < \infty$, then $\mu(a) = \gamma_X(r_+) - ar_+$ for $a > \sup_{r < r_+} \gamma'_X(r)$.

Solution: If $\sup_{r < r_+} g'(r) < \infty$ then $g(r_+) < \infty$ and thus $\sup_{r < r_+} \gamma'_X(r) < \infty$. If $a > \sup_{r < r_+} \gamma'_X(r)$, then $\frac{d}{dr}(\gamma(r) - ra) < 0$ for all $r < r_+$. Thus $\gamma(r) - ra$ is decreasing in r for $r < r_+$, so

$$\mu(a) = \inf_{r < r_+} \gamma_X(r) - ra = \gamma(r_+) - ar_+.$$

See Exercise 1.25, (c) and (d) for examples of this.

Exercise 9.9: [Details in proof of Theorem 9.3.3] a) Show that the two appearances of ϵ in (9.24) can be replaced with two independent arbitrary positive quantities ϵ_1 and ϵ_2 , getting

$$\Pr\{S_n \geq n(\gamma'(r) - \epsilon_1)\} \geq (1 - \delta) \exp[-n(r\gamma'(r) + r\epsilon_2 - \gamma(r))]. \quad (\text{A.183})$$

Show that if this equation is valid for ϵ_1 and ϵ_2 , then it is valid for all larger values of ϵ_1 and ϵ_2 . Hint: Note that the left side of (9.24) is increasing in ϵ and the right side is decreasing.

Solution: For an arbitrary ϵ_1 and ϵ_2 , let $\epsilon = \min(\epsilon_1, \epsilon_2)$. For that ϵ and any $\delta > 0$, there is an n_o such that (9.24) is satisfied for all $n \geq n_o$. Then, using the hint to replace ϵ by $\epsilon_1 \geq \epsilon$ on the left of (9.24),

$$\begin{aligned} \Pr\{S_n \geq n(\gamma'(r) - \epsilon_1)\} &\geq \Pr\{S_n \geq n(\gamma'(r) - \epsilon)\} \\ &\geq (1 - \delta) \exp[-n(r\gamma'(r) + r\epsilon - \gamma(r))] \\ &\geq (1 - \delta) \exp[-n(r\gamma'(r) + r\epsilon_2 - \gamma(r))]. \end{aligned}$$

This also shows that if (A.183) is satisfied for a given ϵ_1, ϵ_2 and n , then it is satisfied for all larger ϵ_1, ϵ_2 for that n .

b) Show that by increasing the required value of n_o , the factor of $(1 - \delta)$ can be eliminated in (A.183).

Solution: We can rewrite (A.183) as

$$\Pr\{S_n \geq n(\gamma'(r) - \epsilon_1)\} \geq (1 - \delta)e^{nr\epsilon_2} \exp[-n(r\gamma'(r) + 2r\epsilon_2 - \gamma(r))].$$

This applies for $n \geq n_o$ for the n_o in part (a). Now choose n'_o large enough so that both $(1 - \delta)e^{n_o r \epsilon'_2} \geq 1$ and (A.183) (for the given (ϵ_1, ϵ_2)) is satisfied. Then for $n \geq n'_o$,

$$\begin{aligned} \Pr\{S_n \geq n(\gamma'(r) - \epsilon_1)\} &\geq (1 - \delta)e^{nr\epsilon'_2} \exp[-n(r\gamma'(r) + 2r\epsilon_2 - \gamma(r))] \\ &\geq \exp[-n(r\gamma'(r) + 2r\epsilon_2 - \gamma(r))] \\ &= \exp[-n(r\gamma'(r) + r\epsilon'_2 - \gamma(r))], \end{aligned} \quad (\text{A.184})$$

where $\epsilon'_2 = 2\epsilon_2$. Since $\epsilon_2 > 0$ is arbitrary, $\epsilon'_2 > 0$ is also arbitrary, with the corresponding change to n'_o .

c) For any $r \in (0, r_+)$, let δ_1 be an arbitrary number in $(0, r_+ - r)$, let $r_1 = r + \delta_1$, and let $\epsilon_1 = \gamma'(r_1) - \gamma'(r)$. Show that there is an m such that for all $n \geq m$,

$$\Pr\{S_n \geq n\gamma'(r)\} \geq \exp\{-n[(r + \delta_1)\gamma'(r + \delta_1) + (r + \delta_1)\epsilon_2 - \gamma(r + \delta_1)]\}. \quad (\text{A.185})$$

Using the continuity of γ and its derivatives, show that for any $\epsilon > 0$, there is a $\delta_1 > 0$ so that the right side of (A.185) is greater than or equal to $\exp[-n(\gamma'(r) - r\gamma(r) + r\epsilon)]$.

Solution: Since $\gamma'(r)$ is increasing in r , $\epsilon_1 = \gamma'(r_1) - \gamma'(r) > 0$. Now (A.184) applies for arbitrary $r \in (0, r_+)$ and ϵ_1, ϵ_2 (using the appropriate n'_o for those parameters). Thus we can apply (A.184) to r_1 and ϵ_1 as defined above and with ϵ_2 replacing ϵ'_2 .

$$\Pr\{S_n \geq n(\gamma'(r_1) - \epsilon_1)\} \geq \exp[-n(r_1\gamma'(r_1) + r_1\epsilon_2 - \gamma(r_1))]$$

Replacing $\gamma'(r_1) - \epsilon_1$ on the left with $\gamma'(r)$ and r_1 on the right by $r + \delta_1$, we get (A.185). This is valid for $n \geq m$ where m is the value of n'_o for these modifications of r, ϵ_1 , and ϵ_2 .

Let $\epsilon > 0$ be arbitrary. Since $r\gamma'(r)$ is continuous in r , we see that for all small enough δ_1 , $(r + \delta_1)\gamma'(r + \delta_1) \leq r\gamma'(r) + r\epsilon/3$. Similarly, $-\gamma(r + \delta_1) \leq -\gamma(r) + r\epsilon/3$ for all small enough δ_1 . Finally, for any such small enough δ_1 , we can choose $\epsilon_2 > 0$ small enough that $(r + \delta_1)\epsilon_2 \leq r\epsilon/3$. Substituting these inequalities into (A.185), there is an m such that for $n \geq m$,

$$\Pr\{S_n \geq n\gamma'(r)\} \geq \exp\{-n[r\gamma'(r) + r\epsilon - \gamma(r)]\},$$

thus completing the final details of the proof of Theorem 9.3.3.

Exercise 9.10: In this exercise, we show that the optimized Chernoff bound is tight for the 3rd case in (9.12) as well as the first case. That is, we show that if $r_+ < \infty$ and $a > \sup_{r < r_+} \gamma'(r)$, then for any $\epsilon > 0$, $\Pr\{S_n \geq na\} \geq \exp\{n[\gamma(r_+) - r_+a - \epsilon]\}$ for all large enough n .

a) Let Y_i be the truncated version of X_i , truncated for some given b to $Y_i = X_i$ for $X_i \leq b$ and $Y_i = b$ otherwise. Let $W_n = Y_1 + \cdots + Y_n$. Show that $\Pr\{S_n \geq na\} \geq \Pr\{W_n \geq na\}$.

Solution: Since $X_i - Y_i$ is a nonnegative rv for each i , $S_n - W_n$ is also nonnegative. Thus all sample points satisfying $W_n \geq na$ also satisfy $S_n \geq na$ so $\Pr\{W_n \geq na\} \leq \Pr\{S_n \geq na\}$.

b) Let $g_b(r)$ be the MGF of Y . Show that $g_b(r) < \infty$ and that $g_b(r)$ is non-decreasing in b for all $r < \infty$.

Solution: Assuming a PMF for notational simplicity, we have

$$g_b(r) = \sum_{x \leq b} p_X(x)e^{rx} + \sum_{x > b} p_X(x)e^{rb} \leq \sum_x p_X(x)e^{rb} = e^{rb} < \infty.$$

If b is replaced by $b_1 > b$, the terms above for $x \leq b$ are unchanged and the terms with $x > b$ are increased from $p_X(x)e^{rb}$ to either $p_X(x)e^{rx}$ or e^{rb_1} , increasing the sum. For those who prefer calculus to logic, $\frac{\partial}{\partial b} g_b(r) = re^{br} \Pr\{X > b\}$, which is nonnegative.

c) Show that $\lim_{b \rightarrow \infty} g_b(r) = \infty$ for all $r > r_+$ and that $\lim_{b \rightarrow \infty} g_b(r) = g(r)$ for all $r \leq r_+$.

Solution: For $r > r_+$, we have $g(r) = \infty$ (by definition of r_+). Also $g_b(r) \geq \sum_{x \leq b} p_X(x)e^{rx}$, which goes to $g(r) = \infty$ in the limit $b \rightarrow \infty$. For $r \leq r_+$, we have $\sum_{x \leq b} p_X(x)e^{rx} \leq g_b(r) \leq g(r)$, so $\lim_{b \rightarrow \infty} g_b(r) = g(r)$.

d) Let $\gamma_b(r) = \ln g_b(r)$. Show that $\gamma_b(r) < \infty$ for all $r < \infty$. Also show that $\lim_{b \rightarrow \infty} \gamma_b(r) = \infty$ for $r > r_+$ and $\lim_{b \rightarrow \infty} \gamma_b(r) = \gamma(r)$ for $r \leq r_+$. Hint: Use b) and c).

Solution: From (b), $g_b(r) < \infty$, from which it follows that $\gamma_b(r) = \ln g_b(r) < \infty$. Also, from (c), $\lim_{b \rightarrow \infty} g_b(r) = \infty$ for $r > r_+$, from which it follows that $\lim_{b \rightarrow \infty} \gamma_b(r) = \infty$ for $r > r_+$. Finally, since $\lim_{b \rightarrow \infty} g_b(r) = g(r)$ for $r \leq r_+$, it follows that $\lim_{b \rightarrow \infty} \gamma_b(r) = \gamma(r)$ for $r \leq r_+$.

e) Let $\gamma'_b(r) = \frac{\partial}{\partial r} \gamma_b(r)$ and let $\delta > 0$ be arbitrary. Show that for all large enough b , $\gamma'_b(r_+ + \delta) > a$. Hint: First show that $\gamma'_b(r_+ + \delta) \geq [\gamma_b(r_+ + \delta) - \gamma(r_+)]/\delta$.

Solution: This is the first part of the exercise where we must consider the situation in the problem statement, where we are looking at the case $r_+ < \infty$ and $a > \sup_{r < r_+} \gamma'(r)$. This implies that $\sup_{r < r_+} \gamma'(r) < \infty$, and, as shown in Exercise 9.8e, this implies that $g(r_+) < \infty$ and thus $\gamma(r_+) < \infty$.

Since $\gamma'_b(r)$ is the slope of $\gamma_b(r)$ and is non-decreasing in r , we have

$$\begin{aligned} \gamma'_b(r_+ + \delta) &\geq \frac{\gamma_b(r_+ + \delta) - \gamma_b(r_+)}{\delta} \\ &\geq \frac{\gamma_b(r_+ + \delta) - \gamma(r_+)}{\delta}, \end{aligned}$$

where in the final step we used the fact that $\gamma_b(r_+) \leq \gamma(r_+)$. Since $\gamma(r_+) < \infty$ and $\lim_{b \rightarrow \infty} \gamma_b(r_+ + \delta) = \infty$, we see that $\gamma'_b(r_+ + \delta)$ is unbounded with increasing b , and thus exceeds a for large enough b .

f) Show that the optimized Chernoff bound for $\Pr\{W_n \geq na\}$ is exponentially tight for the values of b in (e). Show that the optimizing r is less than $r_+ + \delta$.

Solution: For each $b < \infty$, the MGF of W_n is finite for all r , $0 \leq r < \infty$. Thus, from Theorem 9.3.3, the result is exponentially tight if $a = \gamma'_b(r)$ for some $r < \infty$. For any b such that $\gamma'_b(r_+ + \delta) > a$, the continuity of $\gamma'_b(r)$ in r and the fact that $\gamma'_b(0) < \bar{X}$ shows that such an r must exist. Since $\gamma'_b(r_+ + \delta) > a$, the optimizing r is less than $r_+ + \delta$.

g) Show that for any $\epsilon > 0$ and all sufficiently large b ,

$$\gamma_b(r) - ra \geq \gamma(r) - ra - \epsilon \geq \gamma(r_+) - r_+a - \epsilon \quad \text{for } 0 < r \leq r_+.$$

Hint: Show that the convergence of $\gamma_b(r)$ to $\gamma(r)$ is uniform in r for $0 < r < r_+$.

Solution: Since $\mathbf{g}(r_+) < \infty$, it can be seen that $\mathbf{g}(r)$ is uniformly continuous in r over $0 \leq r \leq r_+$. It then follows that $\mathbf{g}_b(r)$ must converge to $\mathbf{g}(r)$ uniformly in r for $0 \leq r \leq r_+$ and the uniform convergence of $\gamma_b(r)$ follows from this. For any $\epsilon > 0$, and for large enough b , this means that $\gamma(r) - \gamma_b(r) \leq \epsilon$ for all $r \in (0, r_+)$. This is equivalent to the first part of the desired statement. The second part follows from the fact that $\gamma'(r)$ is increasing in r and thus $a > \sup_{r < r_+} \gamma'(r) \geq \gamma'(r)$ for $0 \leq r \leq r_+$. This implies that $\gamma(r) - ar$ is decreasing in r over $0 \leq r \leq r_+$ and is thus lower bounded, over $0 \leq r \leq r_+$, at r_+ .

h) Show that for arbitrary $\epsilon > 0$, there exists $\delta > 0$ and b_o such that

$$\gamma_b(r) - ra \geq \gamma(r_+) - r_+a - \epsilon \quad \text{for } r_+ < r < r_+ + \delta \text{ and } b \geq b_o.$$

Solution: For the given $\epsilon > 0$, choose b_o so that $\gamma_b(r_+) > \gamma(r_+) - \epsilon/3$ for $b \geq b_o$. We also choose $\delta = \epsilon/[3(a - \gamma'(r_+))]$. Finally, define $\gamma'(r_+)$ as $\sup_{r < r_+} \gamma'(r)$. Since $\gamma'_b(r_+)$ is non-decreasing in b and $\lim_{b \rightarrow \infty} \gamma'_b(r_+) = \gamma'(r_+)$, we can choose b_o large enough that $\gamma'_b(r_+) \geq \gamma'(r_+) - \epsilon/3\delta$ for $b \geq b_o$. We then have

$$\begin{aligned} \gamma_b(r) - ra &\geq \gamma_b(r_+) + (r - r_+)\gamma'_b(r_+) - ra \\ &\geq \gamma(r_+) - \epsilon/3 + (r - r_+)[\gamma'(r_+) - \epsilon/3\delta] - ra \\ &= \gamma(r_+) - r_+a - \epsilon/3 + (r - r_+)[\gamma'(r_+) - a - \epsilon/3\delta] \\ &= \gamma(r_+) - r_+a - \epsilon/3 + (r - r_+)[2\gamma'(r_+) - 2a] \\ &\geq \gamma(r_+) - r_+a - \epsilon/3 + 2\delta[\gamma'(r_+) - a] \\ &= \gamma(r_+) - r_+a - \epsilon. \end{aligned}$$

i) Note that if we put (g), (h), and d) together, and use the δ of (h) in (d), then we have shown that the optimized exponent in the Chernoff bound for $\Pr\{W_n \geq na\}$ satisfies $\mu_b(a) \geq \gamma(r_+) - r_+a - \epsilon$ for sufficiently large b . Show that this means that $\Pr\{S_n \geq na\} \geq \gamma(r_+) - r_+a - 2\epsilon$ for sufficiently large n .

Solution: Theorem 9.3.3 applies to $\{\Pr\{W_n \geq na\}; n \geq 1\}$, so for sufficiently large n and large enough b ,

$$\Pr\{W_n \geq na\} \geq \exp[n(\mu_b(a) - \epsilon)] \geq \exp[n(\gamma(r_+) - r_+a - 2\epsilon)].$$

Using the result of (a) completes the argument.

Exercise 9.11: Assume that X is discrete, with possible values $\{a_i; i \geq 1\}$ and probabilities $\Pr\{X = a_i\} = p_i$. Let X_r be the corresponding tilted random variable as defined in Section 9.3.2. Let $S_n = X_1 + \dots + X_n$ be the sum of n IID rv 's with the distribution of X , and let $S_{n,r} = X_{1,r} + \dots + X_{n,r}$ be the sum of n IID tilted rv 's with the distribution of X_r . Assume that $\bar{X} < 0$ and that $r > 0$ is such that $\gamma(r)$ exists.

a) Show that $\Pr\{S_{n,r} = s\} = \Pr\{S_n = s\} \exp[sr - n\gamma(r)]$. Hint: Read the text.

Solution: Since X_r is the tilted random variable of X , it satisfies

$$\Pr\{X_r = a_i\} = \Pr\{X = a_i\} \exp[a_i r - \gamma(r)].$$

Let $v_1, \dots, v_n$ be any set of values drawn from the set $\{a_i\}$ of possible values for X . Let $s = v_1 + v_2 + \dots + v_n$. For any such set, since $\{X_{i,r}\}$ is a set of IID variables, each with PMF $\mathbf{p}_{X_{i,r}}(v_i) = \mathbf{p}_X(v_i) \exp[v_i r - \gamma(r)]$,

$$\Pr\{X_{1,r}=v_1, \dots, X_{n,r}=v_n\} = \Pr\{X_1=v_1, \dots, X_n=v_n\} \exp[sr - n\gamma(r)].$$

Let $V(s)$ be the set of all vectors $\mathbf{v} = (v_1, \dots, v_n)$ such that $v_1 + v_2 + \dots + v_n = s$. Then

$$\begin{aligned} \Pr\{S_{n,r} = s\} &= \sum_{\mathbf{v} \in V(s)} \Pr\{X_{1,r}=v_1, \dots, X_{n,r}=v_n\} \\ &= \sum_{\mathbf{v} \in V(s)} \Pr\{X_1=v_1, \dots, X_n=v_n\} \exp[sr - n\gamma(r)] \\ &= \Pr\{S_n = s\} \exp[sr - n\gamma(r)]. \end{aligned}$$

b) Find the mean and variance of $S_{n,r}$ in terms of $\gamma(r)$.

Solution: $E[S_{n,r}] = nE[X_r] = n\gamma'(r)$ and $\text{VAR}[S_{n,r}] = n\text{VAR}[X_r] = n\gamma''(r)$, as shown in Exercise 9.6.

c) Define $a = \gamma'(r)$ and $\sigma_r^2 = \gamma''(r)$. Show that $\Pr\{|S_{n,r} - na| \leq \sqrt{2n}\sigma_r\} > 1/2$. Use this to show that

$$\Pr\{|S_n - na| \leq \sqrt{2n}\sigma_r\} > (1/2) \exp[-r(an + \sqrt{2n}\sigma_r) + n\gamma(r)]. \quad (\text{i})$$

Solution: Assuming $\sigma_r > 0$ and applying the Chebyshev inequality, $\Pr\{|Y - \bar{Y}| \geq \epsilon\} \leq \epsilon^{-2} \text{VAR}[Y]$, to $S_{n,r}$,

$$\Pr\{|S_{n,r} - na| \geq \sqrt{2n}\sigma_r\} \leq \frac{n\sigma_r^2}{2n\sigma_r^2} = \frac{1}{2}.$$

The complementary event then satisfies

$$\Pr\{|S_{n,r} - na| < \sqrt{2n}\sigma_r\} \geq \frac{1}{2}.$$

The strictness of the inequalities is not important here. We will use the result in this form, but if one traces out the case of equality in the Chebyshev bound, one also gets the form in the problem statement. Combining this with (a),

$$\begin{aligned} 1/2 &\leq \Pr\{|S_{n,r} - na| < \sqrt{2n}\sigma_r\} \\ &= \sum_{s: |s-na| < \sqrt{2n}\sigma_r} \Pr\{S_{n,r} = s\} \\ &= \sum_{s: |s-na| < \sqrt{2n}\sigma_r} \Pr\{S_n = s\} \exp[sr - n\gamma(r)] \\ &< \sum_{s: |s-na| < \sqrt{2n}\sigma_r} \Pr\{S_n = s\} \exp[nar + \sqrt{2n}r\sigma_r - n\gamma(r)] \\ &= \Pr\{|S_n - na| < \sqrt{2n}\sigma_r\} \exp[nar + \sqrt{2n}r\sigma_r - n\gamma(r)]. \end{aligned}$$

In the third step, we used part (a), in the fourth step, we recognized that $s < na + \sqrt{2n}\gamma_r$ over the terms in the sum, and in the fifth step, we rewrote the sum of terms as a single probability. Rearranging terms, we have (i).

d) Use this to show that for any ϵ and for all sufficiently large n ,

$$\Pr\{S_n \geq n(\gamma'(r) - \epsilon)\} > \frac{1}{2} \exp[-rn(\gamma'(r) + \epsilon) + n\gamma(r)].$$

Solution: Note that the event $\{|S_n - na| < \sqrt{2n}\sigma_r\}$ is contained in the event $\{S_n > na - \sqrt{2n}\sigma_r\}$. Also, for any $\epsilon > 0$, we can choose n_0 such that $n_0\epsilon > \sqrt{2n}\sigma_r$. For all $n \geq n_0$, the event $\{S_n > na - \sqrt{2n}\sigma_r\}$ is contained in the event $\{S_n > na - n\epsilon\} = \{S_n > n(\gamma'(r) - \epsilon)\}$. Thus

$$\Pr\{|S_n > n(\gamma'(r) - \epsilon)\} > \frac{1}{2} \exp[-r(na + \sqrt{2n}\sigma_r) + n\gamma(r)].$$

Also, for $n \geq n_0$, the right hand side of this equation can be further bounded by

$$\Pr\{|S_n > n(\gamma'(r) - \epsilon)\} > \frac{1}{2} \exp[-rn(\gamma'(r) + \epsilon) + n\gamma(r)].$$

Note that this exercise gives a slightly different proof from Theorem 9.3.3 that the Chernoff bound is exponentially tight.

Exercise 9.12: Let $\mathbf{p} = (p_1, \dots, p_M)$ and $\tilde{\mathbf{p}} = (\tilde{p}_1, \dots, \tilde{p}_M)$ be probability vectors. Show that the divergence satisfies $D(\tilde{\mathbf{p}}|\mathbf{p}) \geq 0$ and is 0 if and only if $\tilde{\mathbf{p}} = \mathbf{p}$.

Solution: Recall that

$$D(\tilde{\mathbf{p}}|\mathbf{p}) = \sum_j \tilde{p}_j \ln \left(\frac{\tilde{p}_j}{p_j} \right).$$

If $p_j = 0$ and $\tilde{p}_j > 0$ for one or more j , this is infinite and by convention greater than 0. Thus assume $\mathbf{p}$ and $\tilde{\mathbf{p}}$ to be strictly positive. Using the inequality $\ln x \leq x - 1$ for $x > 0$, we can upper bound $-D(\tilde{\mathbf{p}}|\mathbf{p})$ by

$$-D(\tilde{\mathbf{p}}|\mathbf{p}) = \sum_j \tilde{p}_j \ln \left(\frac{p_j}{\tilde{p}_j} \right) \leq \sum_j \tilde{p}_j \frac{p_j}{\tilde{p}_j} - \sum_j \tilde{p}_j = 0. \quad (\text{A.186})$$

Since $\ln x = x - 1$ if and only if $x = 1$, the inequality in (A.186) is satisfied with equality if and only if $\tilde{\mathbf{p}} = \mathbf{p}$.

b Let $\mathbf{u} = u_1, \dots, u_M$ satisfy $\sum_j u_j = 0$ and show that $\mathbf{p} + \epsilon \mathbf{u}$ is a probability vector over a sufficiently small region of ϵ around 0. Show that

$$\frac{\partial}{\partial \epsilon} D(\mathbf{p} + \epsilon \mathbf{u}|\mathbf{p})|_{\epsilon=0} = 0.$$

Solution: It is necessary to add the condition that $p_j > 0$ for all j , since if $p_j = 0$ then $p_j + \epsilon u_j < 0$ for all $\epsilon < 0$ when $u_j > 0$ and for all $\epsilon > 0$ when $u_j < 0$. If all components of $\mathbf{p}$ are positive, then $p_j + \epsilon u_j$ is positive for all j for small enough $|\epsilon|$. Note that $\sum_j p_j + \epsilon u_j = 1$, so $\mathbf{p} + \epsilon \mathbf{u}$ is a probability vector for small enough $|\epsilon|$.

Next, taking the derivative,

$$\begin{aligned} \frac{\partial}{\partial \epsilon} D(\mathbf{p} + \epsilon \mathbf{u}|\mathbf{p}) &= \frac{\partial}{\partial \epsilon} \sum_j (p_j + \epsilon u_j) \ln \left(\frac{p_j + \epsilon u_j}{p_j} \right) \\ &= \sum_j u_j \left[\ln \left(\frac{p_j + \epsilon u_j}{p_j} \right) + 1 \right], \end{aligned}$$

which is clearly 0 at $\epsilon = 0$.

Viewing $D(\mathbf{p} + \epsilon \mathbf{u} || \mathbf{p})$ geometrically as a function in $\mathbb{R}^M$ of $\mathbf{p} + \epsilon \mathbf{u}$, this says that the function is flat at $\mathbf{p}$ in each probability vector direction.

c Show that

$$\frac{\partial^2}{\partial \epsilon^2} D(\tilde{\mathbf{p}} + \epsilon \mathbf{u} || \mathbf{p}) \geq 0,$$

where $\tilde{\mathbf{p}} + \epsilon \mathbf{u}$ is a probability vector with non-zero components. Explain why this implies that D is convex in $\tilde{\mathbf{p}}$ and nonnegative over the region of probability vectors.

Solution: Taking the second derivative with respect to ϵ , and here allowing ϵ to be positive or negative,

$$\frac{\partial^2}{\partial \epsilon^2} D(\tilde{\mathbf{p}} + \epsilon \mathbf{u} || \mathbf{p}) = \sum_j \frac{u_j^2}{\tilde{p}_j + \epsilon u_j} \geq 0 \quad \text{for } |\epsilon| \text{ sufficiently small.}$$

This shows that for fixed positive probability vectors $\tilde{\mathbf{p}}, \mathbf{p}$, and arbitrary $\mathbf{u}$, $D(\tilde{\mathbf{p}} + \epsilon \mathbf{u} || \mathbf{p})$ is a convex function of ϵ . This says that the function D over any straight line probability vector direction from $\tilde{\mathbf{p}}$ is convex, and this implies that D is convex over the convex region where $\tilde{\mathbf{p}}$ is a probability vector.

Exercise 9.13: Consider a random walk $\{S_n; n \geq 1\}$ where $S_n = X_1 + \cdots + X_n$ and $\{X_i; i \geq 1\}$ is a sequence of IID exponential rv's with the PDF $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. In other words, the random walk is the sequence of arrival epochs in a Poisson process.

a) Show that for $\lambda a > 1$, the optimized Chernoff bound for $\Pr\{S_n \geq na\}$ is given by

$$\Pr\{S_n \geq na\} \leq (a\lambda)^n e^{-n(a\lambda-1)}.$$

Solution: The moment generating function is $\mathbf{g}(r) = \mathbf{E}[e^{Xr}] = \lambda/(\lambda - r)$ for $r < \lambda$. Thus $\gamma(r) = \ln \mathbf{g}(r) = \ln(\lambda/(\lambda - r))$ and $\gamma'(r) = 1/(\lambda - r)$. The optimizing r for the Chernoff bound is then the solution to $a = 1/(\lambda - r)$, which is $r = \lambda - 1/a$. Using this r in the Chernoff bound,

$$\Pr\{S_n \geq na\} \leq \exp \left[n \ln \left(\frac{\lambda}{\lambda - r} \right) - nra \right] = \exp[n \ln(a\lambda) - n(a\lambda - 1)],$$

which is equivalent to the desired expression.

b) Show that the exact value of $\Pr\{S_n \geq na\}$ is given by

$$\Pr\{S_n \geq na\} = \sum_{i=0}^{n-1} \frac{(na\lambda)^i e^{-na\lambda}}{i!}. \quad (\text{A.187})$$

Solution: For a Poisson counting process $\{N(t); t > 0\}$, the event $\{S_n > na\}$ is the same as $\{N(na) < n\} = \bigcup_{i=0}^{n-1} \{N(na) = i\}$. Since this is a union of disjoint events,

$$\Pr\{S_n > na\} = \sum_{i=0}^{n-1} \Pr\{N(na) = i\}.$$

Using the Poisson PMF, the right side of this is equal to the right side of (A.187). Since S_n is continuous, $\Pr\{S_n > na\} = \Pr\{S_n \geq na\}$.

c) By upper and lower bounding the quantity on the right of (A.187), show that

$$\frac{(na\lambda)^n e^{-na\lambda}}{n! a\lambda} \leq \Pr\{S_n \geq na\} \leq \frac{(na\lambda)^n e^{-na\lambda}}{n!(a\lambda - 1)}.$$

Hint: Use the term at $i = n - 1$ for the lower bound and note that the term on the right can be bounded by a geometric series starting at $i = n - 1$.

Solution: The lower bound on the left is the single term with $i = n - 1$ of the sum in (A.187). For the upper bound, rewrite the sum in (b) as

$$\begin{aligned} \sum_{i=0}^{n-1} \frac{(na\lambda)^i e^{-na\lambda}}{i!} &= \frac{(na\lambda)^n e^{-na\lambda}}{n!} \left[\frac{n}{na\lambda} + \frac{n(n-1)}{(na\lambda)^2} + \cdots \right] \\ &\leq \frac{(na\lambda)^n e^{-na\lambda}}{n!} \left[\frac{1}{a\lambda} + \frac{1}{(a\lambda)^2} + \cdots \right] = \frac{(na\lambda)^n e^{-na\lambda}}{n!(a\lambda - 1)}. \end{aligned}$$

d) Use the Stirling bounds on $n!$ to show that

$$\frac{(a\lambda)^n e^{-n(a\lambda-1)}}{\sqrt{2\pi n} a\lambda \exp(1/12n)} \leq \Pr\{S_n \geq na\} \leq \frac{(a\lambda)^n e^{-n(a\lambda-1)}}{\sqrt{2\pi n} (a\lambda - 1)}.$$

Solution: The Stirling bounds are

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/12n}.$$

Substituting these for $n!$ in (c) and cancelling terms gives the desired expression. Note that the Chernoff bound contains all the factors that vary exponentially with n . Note also that the Erlang expression for S_n and the Poisson expression for $N(t)$ are quite simple, but the corresponding CDF's are quite messy, and this makes the Chernoff bound more attractive in this case.

Exercise 9.14: Consider a random walk with thresholds $\alpha > 0$, $\beta < 0$. We wish to find $\Pr\{S_J \geq \alpha\}$ in the absence of a lower threshold. Use the upper bound in (9.46) for the probability that the random walk crosses α before β .

a) Given that the random walk crosses β first, find an upper bound to the probability that α is now crossed before a yet lower threshold at 2β is crossed.

Solution: Let J_1 be the stopping trial at which the walk first crosses either α or β . Let J_2 be the stopping trial at which the random walk first crosses either α or 2β (assuming the random walk continues forever rather than actually stopping at any stopping trial. Note that if $S_{J_1} \geq \alpha$, then $S_{J_2} = S_{J_1}$, but if $S_{J_1} \leq \beta$, then it is still possible to have $S_{J_2} \geq \alpha$. In order for this to happen, a random walk starting at trial J_1 must reach a threshold of $\alpha - S_{J_1}$ before reaching $2\beta - S_{J_1}$. Putting this into equations,

$$\begin{aligned} \Pr\{S_{J_2} \geq \alpha\} &= \Pr\{S_{J_1} \geq \alpha\} + \Pr\{S_{J_2} \geq \alpha \mid S_{J_1} \leq \beta\} \Pr\{S_{J_1} \leq \beta\} \\ &\leq \Pr\{S_{J_1} \geq \alpha\} + \Pr\{S_{J_2} \geq \alpha \mid S_{J_1} \leq \beta\}. \end{aligned}$$

The second term above upper bounds the probability that the RW starting at trial J_1 reaches $\alpha - S_{J_1}$ before $2\beta - S_{J_1}$, given that $S_{J_1} \leq \beta$. Since $\alpha - S_{J_1} \geq \alpha - \beta$,

$$\Pr\{S_{J_2} \geq \alpha \mid S_{J_1} \leq \beta\} \leq \exp[-r^*(\alpha - \beta)].$$

Thus,

$$\Pr\{S_{J_2} \geq \alpha\} \leq \exp(-r^*\alpha) + \exp[-r^*(\alpha - \beta)].$$

Note that it is conceivable that $S_{J_1} = S_{J_2}$, *i.e.*, that β and 2β are crossed at the same time. The argument above includes this case, and no further note about it is necessary.

b) Given that 2β is crossed before α , upper bound the probability that α is crossed before a threshold at 3β . Extending this argument to successively lower thresholds, find an upper bound to each successive term, and find an upper bound on the overall probability that α is crossed. By observing that β is arbitrary, show that (9.46) is valid with no lower threshold.

Solution: Let J_k for each $k \geq 1$ be the stopping trial at which α or $k\beta$ is crossed. By the same argument as above,

$$\begin{aligned} \Pr\{S_{J_{k+1}} \geq \alpha\} &= \Pr\{S_{J_k} \geq \alpha\} + \Pr\{S_{J_{k+1}} \geq \alpha \mid S_{J_k} \leq k\beta\} \Pr\{S_{J_k} \leq k\beta\} \\ &\leq \Pr\{S_{J_k} \geq \alpha\} + \exp[-r^*(\alpha - k\beta)], \end{aligned}$$

Finally, let J_∞ be the defective stopping time at which α is first crossed. We see from above that the event $S_{J_\infty} > \alpha$ is the union of the events $S_{J_k} \geq \alpha$ over all $k \geq 1$. We can upper bound this by

$$\begin{aligned} \Pr\{S_{J_\infty} \geq \alpha\} &\leq \Pr\{S_{J_1} \geq \alpha\} + \sum_{k=1}^{\infty} \Pr\{S_{J_{k+1}} \geq \alpha \mid S_{J_k} \leq k\beta\} \\ &\leq \exp[-r^*\alpha] \frac{1}{1 - \exp[-r^*\beta]}. \end{aligned}$$

Since this is true for all $\beta < 0$, it is valid in the limit $\beta \rightarrow -\infty$, yielding $e^{-r^*\alpha}$.

The reason why we did not simply take the limit $\beta \rightarrow -\infty$ in the first place is that such a limit would not define a non-defective stopping rule. The approach here is to define the limit as a union of non-defective stopping rules.

Exercise 9.15: This exercise verifies that Corollary 9.4.4 holds in the situation where $\gamma(r_+) < 0$ (see Figure 9.4). We have defined $r^* = r_+$ in this case.

a) Use the Wald identity at $r = r_+$ to show that

$$\Pr\{S_J \geq \alpha\} \mathbf{E}[\exp(r_+ S_J - J\gamma(r_+)) \mid S_J \geq \alpha] \leq 1. \quad (\text{A.188})$$

Hint: Look at the first part of the proof of Corollary 9.9.4.

Solution: Starting with the Wald identity expressed for two thresholds,

$$\begin{aligned} 1 &= \Pr\{S_J \geq \alpha\} \mathbf{E}[\exp(r_+ S_J - J\gamma(r_+)) \mid S_J \geq \alpha] + \Pr\{S_J \leq \beta\} \mathbf{E}[\exp(r_+ S_J - J\gamma(r_+)) \mid S_J \leq \beta] \\ &\geq \Pr\{S_J \geq \alpha\} \mathbf{E}[\exp(r_+ S_J - J\gamma(r_+)) \mid S_J \geq \alpha], \end{aligned}$$

which is the same as (A.188)

b) Show that

$$\Pr\{S_J \geq \alpha\} \exp[r_+ \alpha - \gamma(r_+)] \leq 1. \quad (\text{A.189})$$

Solution: Since the expectation is conditional on $S_J \geq \alpha$, we can lower bound S_J by α in the expectation. Also J can be lower bounded by 1. Thus

$$\begin{aligned} 1 &\geq \Pr\{S_J \geq \alpha\} \mathbf{E}[\exp(r_+ S_J - J\gamma(r_+)) | S_J \geq \alpha] \\ &\geq \Pr\{S_J \geq \alpha\} \mathbf{E}[\exp(r_+ \alpha - \gamma(r_+)) | S_J \geq \alpha] \\ &= \Pr\{S_J \geq \alpha\} \exp(r_+ \alpha - \gamma(r_+)). \end{aligned}$$

c) Show that

$$\Pr\{S_J \geq \alpha\} \leq \exp[-r_+ \alpha + \gamma(r_+)] \quad \text{and that } \Pr\{S_J \geq \alpha\} \leq \exp[-r^* \alpha].$$

Note that the first bound above is a little stronger than the second, and note that lower bounding J by 1 is not necessarily a very weak bound, since this is the case where if α is crossed, it tends to be crossed for small J .

Solution: The first bound is simply a rearrangement of (A.189) and the second follows from the first since $\gamma(r_+)$ is negative by assumption.

Exercise 9.16: a) Use Wald's equality to show that if $\bar{X} = 0$, then $\mathbf{E}[S_J] = 0$ where J is the time of threshold crossing with one threshold at $\alpha > 0$ and another at $\beta < 0$.

Solution: Wald's equality holds since $\mathbf{E}[|J|] < \infty$, which follows from Lemma 9.4.1, which says that J has an MGF in an interval around 0. Thus $\mathbf{E}[S_J] = \bar{X} \mathbf{E}[J]$. Since $\bar{X} = 0$, it follows that $\mathbf{E}[S_J] = 0$.

b) Obtain an expression for $\Pr\{S_J \geq \alpha\}$. Your expression should involve the expected value of S_J conditional on crossing the individual thresholds (you need not try to calculate these expected values).

Solution: Writing out $\mathbf{E}[S_J] = 0$ in terms of conditional expectations,

$$\begin{aligned} \mathbf{E}[S_J] &= \Pr\{S_J \geq \alpha\} \mathbf{E}[S_J | S_J \geq \alpha] + \Pr\{S_J \leq \beta\} \mathbf{E}[S_J | S_J \leq \beta] \\ &= \Pr\{S_J \geq \alpha\} \mathbf{E}[S_J | S_J \geq \alpha] + [1 - \Pr\{S_J \geq \alpha\}] \mathbf{E}[S_J | S_J \leq \beta]. \end{aligned}$$

Using $\mathbf{E}[S_J] = 0$, we can solve this for $\Pr\{S_J \geq \alpha\}$,

$$\Pr\{S_J \geq \alpha\} = \frac{\mathbf{E}[-S_J | S_J \leq \beta]}{\mathbf{E}[-S_J | S_J \leq \beta] + \mathbf{E}[S_J | S_J \geq \alpha]}.$$

c) Evaluate your expression for the case of a simple random walk.

Solution: A simple random walk moves up or down only by unit steps. Thus if α and β are integers, there can be no overshoot when a threshold is crossed. Thus $\mathbf{E}[S_J | S_J \geq \alpha] = \alpha$ and $\mathbf{E}[S_J | S_J \leq \beta] = \beta$. Thus $\Pr\{S_J \geq \alpha\} = \frac{|\beta|}{|\beta| + \alpha}$. If α is non-integer, then a positive threshold crossing occurs at $\lceil \alpha \rceil$ and a lower threshold crossing at $\lfloor \beta \rfloor$. Thus, in this general case $\Pr\{S_J \geq \alpha\} = \frac{\lfloor \beta \rfloor}{\lfloor \beta \rfloor + \lceil \alpha \rceil}$.

d) Evaluate your expression when X has an exponential density, $f_X(x) = a_1 e^{-\lambda x}$ for $x \geq 0$ and $f_X(x) = a_2 e^{\mu x}$ for $x < 0$ and where a_1 and a_2 are chosen so that $\bar{X} = 0$.

Solution: Let us condition on $J = n$, $S_n \geq \alpha$, and $S_{n-1} = s$, for $s < \alpha$. The overshoot, $V = S_J - \alpha$ is then $V = X_n + s - \alpha$. Because of the memoryless property of the exponential,

the density of V , conditioned as above, is exponential and $f_V(v) = \lambda e^{-\lambda v}$ for $v \geq 0$. This does not depend on n or s , and is thus the overshoot density conditioned only on $S_J \geq \alpha$. Thus $E[S_J | J \geq \alpha] = \alpha + 1/\lambda$. In the same way, $E[S_J | S_J \leq \beta] = \beta - 1/\mu$. Thus

$$\Pr\{S_J \geq \alpha\} = \frac{|\beta| + \mu^{-1}}{\alpha + \lambda^{-1} + |\beta| + \mu^{-1}}.$$

Note that it is not necessary to calculate a_1 or a_2 .

Exercise 9.17: A random walk $\{S_n; n \geq 1\}$, with $S_n = \sum_{i=1}^n X_i$, has the following probability density for X_i

$$f_X(x) = \begin{cases} \frac{e^{-x}}{e - e^{-1}}; & -1 \leq x \leq 1, \\ 0; & \text{elsewhere.} \end{cases}$$

a) Find the values of r for which $g(r) = E[\exp(rX)] = 1$.

Solution: Evaluating $g(r)$,

$$g(r) = \int_{-1}^1 e^{rx} \frac{e^{-x}}{e - e^{-1}} dx = \frac{1}{r - 1} \frac{e^{r-1} - e^{-(r-1)}}{e - e^{-1}}.$$

One solution to $g(r) = 1$ (for any rv) is $r = 0$, and this is easily verified for this special case. To find the other solution, one can either plot the function $g(r)$ numerically, or one can observe that the function is symmetric around $r = 1$. Either way, it can be verified that the other solution is $r = 2$.

b) Let P_α be the probability that the random walk ever crosses a threshold at α for some $\alpha > 0$. Find an upper bound to P_α of the form $P_\alpha \leq e^{-\alpha A}$ where A is a constant that does not depend on α ; evaluate A .

Solution: Using Corollary 9.4.4 (which is the major result for upper bounding the probability of a threshold crossing for a random walk), we see that $\Pr\{S_J \geq \alpha\} \leq \exp(-r^*\alpha)$. Here J is the stopping rule that stops on the first crossing of a positive threshold at α or a negative threshold at $-\beta$. The negative threshold can be set at $-\infty$, as can be seen by going to the limit, or by doing Exercise 9.14, or by using Theorem 9.8.5. The parameter r^* here is the positive root of $\ln g(r)$, which, from (a) is $r^* = 2$. Thus

$$P_\alpha \leq \exp(-2\alpha).$$

c) Find a lower bound to P_α of the form $P_\alpha \geq B e^{-\alpha A}$ where A is the same as in (b) and B is a constant that does not depend on α . Hint: keep it simple — you are not expected to find an elaborate bound. Also recall that $E[e^{r^* S_J}] = 1$ where J is a stopping trial for the random walk and $g(r^*) = 1$.

Solution: Wald's identity, with $r = r^*$, is $E[\exp(r^* S_J)] = 1$, which is evaluated as

$$\begin{aligned} 1 &= P_\alpha E[\exp(r^* S_J) | S_J \geq \alpha] + (1 - P_\alpha) E[\exp(r^* S_J) | S_J \leq \beta] \\ &\leq P_\alpha E[\exp(r^* S_J) | S_J \geq \alpha] + (1 - P_\alpha) e^{r^* \beta}. \end{aligned}$$

The second term approaches 0 as $\beta \rightarrow -\infty$, so the limiting value of P_α as the lower threshold approaches ∞ is

$$P_\alpha \geq \left(E[\exp(r^* S_J) | S_J \geq \alpha] \right)^{-1} \geq \left(\exp(2(\alpha + 1)) \right)^{-1} = (1/e)^2 e^{-2\alpha}.$$

The second inequality above follows since $|X| \leq 1$ and therefore the overshoot in crossing α is at most 1, *i.e.*, $r^*S_J \leq r^*(\alpha + 1)$ and $r^* = 2$.

Exercise 9.18: Let $\{X_n; n \geq 1\}$ be a sequence of IID integer-valued random variables with the probability mass function $p_X(k) = Q_k$. Assume that $Q_k > 0$ for $|k| \leq 10$ and $Q_k = 0$ for $|k| > 10$. Let $\{S_n; n \geq 1\}$ be a random walk with $S_n = X_1 + \cdots + X_n$. Let $\alpha > 0$ and $\beta < 0$ be integer-valued thresholds, let J be the smallest value of n for which either $S_n \geq \alpha$ or $S_n \leq \beta$. Let $\{S_n^*; n \geq 1\}$ be the stopped random walk; *i.e.*, $S_n^* = S_n$ for $n \leq J$ and $S_n^* = S_J$ for $n > J$. Let $\pi_i^* = \Pr\{S_J = i\}$.

a) Consider a Markov chain in which this stopped random walk is run repeatedly until the point of stopping. That is, the Markov chain transition probabilities are given by $P_{ij} = Q_{j-i}$ for $\beta < i < \alpha$ and $P_{i0} = 1$ for $i \leq \beta$ and $i \geq \alpha$. All other transition probabilities are 0 and the set of states is the set of integers $[-9 + \beta, 9 + \alpha]$. Show that this Markov chain is ergodic.

Solution: The number of states in this repeating Markov chain is finite, so we need only show that there is a single class of states and it is aperiodic. All states clearly communicate with each other, so there is a single class of states and it is recurrent. Also $P_{00} > 0$, so as shown in Exercise 4.1, the chain is aperiodic.

b) Let $\{\pi_i\}$ be the set of steady-state probabilities for this Markov chain. Find the set of probabilities $\{\pi_i^*\}$ for the stopping states of the stopped random walk in terms of $\{\pi_i\}$.

Solution: The steady-state probabilities $\{\pi_i\}$ are rather a mess, even for given values of $\{Q_i\}$. We specified X with so many (21) possible values to discourage you from spending much time on finding the π_i . Fortunately, finding $\{\pi_i^*\}$ from $\{\pi_i\}$ is fairly easy. The repeating Markov process can be viewed as a (delayed) renewal process, associating each visit to any of the stopping states as a renewal. The expected time between renewals is then

$$\mathbb{E}[T] = \frac{1}{\sum_{i=\beta-9}^{\beta} \pi_i + \sum_{i=\alpha}^{\alpha+9} \pi_i}.$$

The probabilities of the stopping values in the random walk are then $\pi_i^* = \pi_i/\bar{T}$ for $\beta - 9 \leq i \leq \beta$ and $\alpha \leq i \leq \alpha + 9$.

Note that we cannot use visits to state 0 as renewals here, since visits to state 0 occur both on transitions from the stopping states and on transitions from states neighboring state 0.

c) Find $\mathbb{E}[S_J]$ and $\mathbb{E}[J]$ in terms of $\{\pi_i\}$.

Solution: A renewal requires J steps to cross a threshold, and the time to the next renewal requires an additional step to return to 0. Thus $\mathbb{E}[J] = \mathbb{E}[T] - 1$. The expected stopping value, $\mathbb{E}[S_J]$ is

$$\mathbb{E}[S_J] = \sum_{i=\beta-9}^{\beta} i\pi_i^* + \sum_{i=\alpha}^{\alpha+9} i\pi_i^* = \sum_{i=\beta-9}^{\beta} \frac{i\pi_i}{\mathbb{E}[T]} + \sum_{i=\alpha}^{\alpha+9} \frac{i\pi_i}{\mathbb{E}[T]}.$$

The Wald equality could also be used to find $\mathbb{E}[S_J]$ as $\mathbb{E}[Q]\mathbb{E}[J]$, but then $\mathbb{E}[Q]$ would be required in the solution.

Exercise 9.19: a) Conditional on $X=0$ for the hypothesis testing problem, consider the random variables $Z_i = \ln[f_{Y_i|X}(y_i|1)/f_{Y_i|X}(y_i|0)]$. Show that r^* , the positive solution to $\mathbf{g}_0(r) = 1$, where $\mathbf{g}_0(r) =$

$E[\exp(rZ)|X=0]$, is given by $r^* = 1$.

Solution: First assume that Y and Z are one-dimensional. Then

$$\begin{aligned} g_0(r) &= E[\exp(rZ)|X=0] = \int_{-\infty}^{\infty} f_{Y|X}(y|0) \exp \left[r \ln \left(\frac{f_{Y|X}(y|1)}{f_{Y|X}(y|0)} \right) \right] dy \\ &= \int_{-\infty}^{\infty} f_{Y|X}^{1-r}(y|0) f_{Y|X}^r(y|1) dy \\ &= 1 \quad \text{for } r = 1. \end{aligned}$$

Since $g_0(r)$ is convex in r over the region where it exists, $g_0(r) = 1$ can have only two roots, one of them at $r = 0$, and as just shown, the other at $r = 1$. For the finite dimensional case, it is simply necessary to replace the one dimensional integral with a multidimensional integral; there is no requirement that $Y_1, \dots, Y_n$ are independent conditional on X .

b) Assuming that Y is a discrete random variable (under each hypothesis), show that the tilted random variable Z_r with $r = 1$ has the PMF $p_{Y|X}(y|1)$.

Solution: The assumption that Y and thus Z are conditional on $X = 0$ carries over from (a). Thus,

$$\begin{aligned} q_{Z_r|X}(z|0) &= p_{Z|X}(z|0) \exp[rz - \gamma_0(r)] \\ &= p_{Y|X}(y|0) \exp \left[r \ln \frac{p_{Y|X}(y|1)}{p_{Y|X}(y|0)} - \gamma_0(r) \right] \\ &= p_{Y|X}(y|1) \quad \text{for } r = 1. \end{aligned}$$

Exercise 9.20: (Proof of Stein's lemma) **a)** Use a power series expansion around $r = 0$ to show that $\gamma_0(r) - r\gamma'_0(r) = -r^2\gamma''_0(0)/2 + o(r^2)$.

Solution: Assuming that the MGF exists in an open interval around 0, $g(r)$ has derivatives of all orders at $r = 0$ and $\gamma_0(r) = \ln g(r)$ has the same. Then using a second order Taylor expansion at $r = 0$, $\gamma_0(r) = \gamma_0(0) + r\gamma'_0(0) + r^2\gamma''_0(0)/2 + o(r^2)$. Using a single order Taylor expansion for $\gamma'_0(r)$ around $r = 0$, we get $\gamma'_0(r) = \gamma'_0(0) + r\gamma''_0(0) + o(r)$. This shows that $r\gamma'_0(r) = r\gamma'_0(0) + r^2\gamma''_0(0) + o(r^2)$. Combining these expansions, $\gamma_0(r) - r\gamma'_0(r) = \gamma_0(0) - r^2\gamma''_0(0)/2 + o(r^2)$. Since $\gamma_0(0) = 0$, this is the desired result.

This can also be derived (with considerably more effort) by applying the power series expansion of $g(r)$ to $\gamma_0(r) = \ln(g(r))$ and $r\gamma'_0(r)$.

b) Choose $r = n^{-1/3}$ in (9.61) and (9.62). Show that (9.61) can then be rewritten as

$$\Pr\{e(n) | X=0\} \leq \exp \{n[-r^2\gamma''_0(0)/2 + o(r^2)]\} \quad (\text{A.190})$$

$$= \exp \left\{ -n^{1/3} \gamma''_0(0)/2 + o(n^{1/3}) \right\}. \quad (\text{A.191})$$

Thus $\Pr\{e(n) | X = 0\}$ approaches 0 with increasing n exponentially in $n^{1/3}$.

Solution: Eq. (A.190) is simply the result of substituting the result of (a) into (9.61). Then (A.191) follows by choosing $r = n^{-1/3}$. The term $o(n^{1/3})$ is a function of n that satisfies $\lim_{n \rightarrow \infty} o(n^{1/3})/n^{1/3} = 0$. In other words,

$$\lim_{n \rightarrow \infty} \frac{\ln \Pr\{e(n)|X=0\}}{n^{1/3}} \leq \frac{-\gamma''_0(0)}{2}.$$

c) Using the same power series expansion for r in (9.62), show that

$$\begin{aligned}\Pr\{e(n) \mid X=1\} &\leq \exp\{n[r^2\gamma_0''(0)/2 + \gamma_0'(0) + o(r)]\} \\ &= \exp\{n[\gamma_0'(0) + o(n)]\}.\end{aligned}$$

Thus $\lim_n (1/n) \ln \Pr\{e(n) \mid X=1\} = \gamma_0'(0) = \mathbf{E}[Z \mid X=0]$.

Solution: Using the expansion of (a) in (9.62), we get

$$\begin{aligned}\Pr\{e(n) \mid X=1\} &\leq \exp\{n[-r^2\gamma_0''(0)/2 + o(r^2) + \gamma_0'(r)]\} \\ &= \exp\{n[-r^2\gamma_0''(0)/2 + o(r^2) + \gamma_0'(0) + r\gamma_0''(0) + o(r)]\} \\ &= \exp\{n[\gamma_0'(0) + r\gamma_0''(0) + o(r)]\} \tag{A.192}\end{aligned}$$

$$= \exp\{n[\gamma_0'(0) + o(n)]\}, \tag{A.193}$$

where in (A.192) we recognized that $r^2\gamma_0''(0)/2$ can be included in the $o(r)$ term and in (A.193) we substituted $n^{-1/3}$ for r , as is necessary for the bounds on $\Pr\{e(n) \mid X=0\}$ and $\Pr\{e(n) \mid X=1\}$ to represent the same threshold rule and thus be jointly valid.

d) Explain why the bound in (9.62) is exponentially tight given that $\Pr\{e(n) \mid X=0\} \rightarrow 0$ as $n \rightarrow \infty$.

Solution: Part (d) is not stated precisely, but is intended to show that (9.62) at $r=0$ (*i.e.*, (A.193)) is exponentially tight for any sequence of tests such that $\lim_{n \rightarrow \infty} \Pr\{e(n) \mid X=0\} = 0$.

Assume that the bound in (A.193) is not exponentially tight, *i.e.*, that for some $\epsilon > 0$, $\Pr\{e(n) \mid X=1\} \leq \exp(n[\gamma_0'(0) - \epsilon])$ for some non-terminating sub-sequence of n . For each such n , a threshold test minimizes $\Pr\{e(n) \mid X=0\}$ given this bound on $\Pr\{e(n) \mid X=1\}$. Since the MGF exists in some open interval around 0, there is an $\epsilon > 0$ small enough that $\gamma_0'(0) - \epsilon = \gamma_0'(r)$ for some r less than 0. For the sequence of threshold tests at the threshold $\eta = e^{n\gamma_0'(r)}$, the Chernoff bound on $\Pr\{e(n) \mid X=0\}$ is an exponential bound on the probability of correct decision (since $r < 0$). Thus for this subsequence of n , $\Pr\{e(n) \mid X=0\}$ must approach 1 exponentially rather than 0. Thus $\lim_{n \rightarrow \infty} \Pr\{e(n) \mid X=0\}$ cannot exist and be equal to 0 if (A.193) is not exponentially tight.

This completes the proof of Stein's lemma.

Exercise 9.21: (The secretary problem or marriage problem) This illustrates a very different type of sequential decision problem from those of Section 9.5. Alice is looking for a spouse and dates a set of n suitors sequentially, one per week. For simplicity, assume that Alice must make her decision to accept a suitor for marriage immediately after dating that suitor; she can not come back later to accept a formerly rejected suitor. Her decision must be based only on whether the current suitor is more suitable than all previous suitors. Mathematically, we can view the dating as continuing for all n weeks, but the choice at week m is a stopping rule. Assume that each suitor's suitability is represented by a real number and that all n numbers are different. Alice does not observe the suitability numbers, but only observes at each time m whether suitor m has the highest suitability so far. The suitors are randomly permuted before Alice dates them.

a) A reasonable algorithm for Alice is to reject the first k suitors (for some k to be optimized later) and then to choose the first suitor $m > k$ that is more suitable than all the previous $m-1$ suitors. Find the probability q_k that Alice chooses the most suitable of all n when she uses this algorithm for given k . Hint 1: What is the probability (as a function of m) that the most suitable of all n is dated on week m . Hint 2: Given that the most suitable of all suitors is dated at week m , what is the probability that Alice chooses

that suitor? For $m > k$, show how this involves the location of the most suitable choice from 1 to $m - 1$, conditional on m 's overall optimality. Note: if no suitor is selected by the algorithm, it makes no difference whether suitor n is chosen at the end or no suitor is chosen.

Solution: Since the suitability numbers are randomly permuted (*i.e.*, permuted so that each permutation is equally likely), the largest appears at position m with probability $1/n$ for $1 \leq m \leq n$.

Conditional on the largest number appearing at some given position m , note that for $m \leq k$, $\Pr\{\text{choose } m \mid m \text{ best}\} = 0$ since the first k suitors are rejected out of hand. Next suppose that the position m of the largest number is $m > k$. Then Alice will choose m (and thus choose the best suitor) unless Alice has chosen someone earlier, *i.e.*, unless some suitor j , $k < j < m$, is more suitable than suitors 1 to k . To state this slightly more simply, Alice chooses someone earlier if the best suitor from 1 to $m - 1$ lies in a position from $k + 1$ to $m - 1$. Since the best suitor from 1 to $m - 1$ is in each position with equal probability,

$$\Pr\{\text{choose } m \mid m \text{ best}\} = \frac{k}{m-1} \quad k < j < m.$$

Averaging over m , we get

$$q_k = \sum_{m=k+1}^n \frac{k}{n(m-1)}.$$

b) Approximating $\sum_{i=1}^j 1/i$ by $\ln j$, show that for n and k large,

$$q_k \approx \frac{k}{n} \ln(n/k).$$

Ignoring the constraint that n and k are integers, show that the right hand side above is maximized at $k/n = e$ and that $\max_k q_k \approx 1/e$.

Solution: Using the solution for q_k in (a), and letting $i = m - 1$,

$$q_k = \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i} = \frac{k}{n} \left[\sum_{i=1}^{n-1} \frac{1}{i} - \sum_{i=1}^{k-1} \frac{1}{i} \right] \approx \frac{k}{n} [\ln(n-1) - \ln(k-1)] \approx \frac{k}{n} \left[\ln \frac{n}{k} \right].$$

The approximations here are pretty crude. The first one can be improved by using the Euler-Mascheroni constant, but q_k is still approximately optimized at $k = n/e$, and at that point $q_k \approx 1/e$. The exact value is too easy to calculate to scrutinize the approximations closely.

c) (Optional) Show that the algorithm of (a), optimized over k , is optimal over all algorithms (given the constraints of the problem). Hint: Let p_m be the maximum probability of choosing the optimal suitor given that no choice has been made before time m . Show that $p_m = \max \left[\frac{1}{n} + \frac{m-1}{m} p_{m+1}, p_{m+1} \right]$; part of the problem here is understanding exactly what p_m means.

Solution outline: We have looked at the class of algorithms for which the first k suitors are rejected and the first subsequent 'best-yet' suitor is chosen. Alice has many other choices, including choosing the suitor at time m using the past history of best-yet suitors. It turns

out, however, that if we start at time $m = n$ and work back inductively, we can find the optimal probability of success conditional on no suitor being chosen before time m . Denote this optimal conditional probability as p_m . As we shall see, the remarkable fact which makes this possible is that this conditional probability depends neither on the decisions made before time m nor history of best-yet suitors before m .

Given that no decision is made before time n , then, a selection at time n is correct only if suitor n is the best suitor, and this is an event of probability $1/n$, independent of past history and past decisions. Given no prior decision, the optimal choice is for Alice to choose suitor n if n is best-yet and thus best overall. Thus $p_n = 1/n$.

Next suppose no decision has been made before time $n - 1$. The suitor at $n - 1$ is optimal with probability $1/n$, and in this case must be a best-yet. Consider the following possible decisions at time $n - 1$: first, choose $n - 1$ if best-yet, or, second, wait until time n whether or not $n - 1$ is best-yet. We evaluate p_{n-1} under the first of these decisions and then show it is optimal. These decisions at $n - 1$ and n are successful overall if $n - 1$ is optimal (an event of probability $1/n$ independent of past history and choices) and also if $n - 1$ is not best-yet and n is best. The event that $n - 1$ is not best-yet has probability $(n - 2)/(n - 1)$ and is independent of the event that n is best, so

$$p_{n-1} = \frac{1}{n} + \frac{n - 2}{n(n - 1)}. \quad (\text{A.194})$$

If Alice waits until time n even if $n - 1$ is best-yet, the probability of success drops to $1/n$. Finally, if Alice bases her decision on the past history of best-yet suitors, the probability of success is a linear combination of p_{n-1} above and $1/n$, thus verifying (A.194) in general.

We next find p_m in terms of p_{m+1} for arbitrary m . We first assume that Alice's optimal strategy at time m , given that no prior decision has been made, is to choose m if it is best-yet, and if not best-yet to wait until time $m + 1$. As before, m is optimal with probability $1/n$ and if so, must be a best-yet choice. If m is not best-yet (an event of probability $(m - 1)/m$), then the event that success will occur at time $m + 1$ or later has probability p_{m+1} . This event is independent of the event that m is not best-yet, so if this is the best decision,

$$p_m = \frac{1}{n} + \frac{m - 1}{m} p_{m+1} \quad \text{tentatively.} \quad (\text{A.195})$$

Now recall that Alice can also reject suitor m even if best-yet, and in this case the probability of success becomes p_{m+1} . If the choice is based on past history, then the probability of success is a linear combination of p_{m+1} and the value in (A.195). We then have

$$p_m = \max \left[\frac{1}{n} + \frac{m - 1}{m} p_{m+1}, \quad p_{m+1} \right]. \quad (\text{A.196})$$

If we subtract p_{m+1} from both sides, above, we see that $p_m - p_{m+1} = \max [1/n - p_{m+1}/m, 0]$, from which we conclude that p_m increases as m decreases from n , and eventually becomes constant, with the strategy rejecting all suitors for small m . If we expand the solution in (A.196), we see that it is simply the strategy in (a), which is thus optimal.

This proves the desired result, but it is of interest to see what happens if Alice receives more information. If at each m , she receives the ordering of the first m numbers, then it is easy to see by an extension of the argument above, that the additional information is irrelevant and that the algorithm of (a), optimized over k , is still optimal.

On the other hand if Alice sees the actual suitability numbers, she can do better. For example, if the sequence of suitability numbers is a sequence of n IID rv's, then if Alice knows the distribution of those numbers, she can solve the problem in a very different way and increase the probability of best choice significantly. If she doesn't know the distribution, and n is very large, she can estimate the distribution and improve the algorithm. The case of an 'unknown' distribution is flaky, however, and the notion of optimality becomes undefined.

d) Explain why this is a poor model for choosing a spouse (or for making a best choice in a wide variety of similar problems). Caution: It is not enough to explain why this is not closely related to these real problems, You should also explain why this gives very little insight into such real problems.

Solution: The assumptions that there are a given number n of candidates, that they are randomly permuted, that they can be evaluated on a one-dimensional scale, that Alice is unaware of that evaluation, but only of a best-yet ranking, and finally that decisions must be made immediately are often individually appropriate. but, as discussed below, are very strange when taken together.

We often model problems where some given parameter (n in this case) is taken to be either large or infinite according to analytical convenience, and a stopping rule is needed to terminate an essentially infinite search. Here, however, the optimal stopping rule depends critically on n , so that n should be a meaningful parameter of the underlying problem. In such cases, the reason for stopping and making an early decision is usually related to a cost of making measurements, which does not exist here. Here, the need to make decisions sequentially is quite artificial.

We next ask whether this model can illustrate limited aspects about choosing a spouse (or similar problems of choice). The many artificial and arbitrary aspects of the model make this difficult. The worst part is that most real world problems of sequential choice involve a combination of learning, decision making, and avoidance of delay. This model, in rejecting the first k suitors, appears to be like this, but in fact no learning is involved and no criterion of time-savng is involved.

On the other hand, this model is clearly interesting from a mathematical standpoint — there is a large literature that illustrates its inherent interest. Problems that are inherently interesting are certainly worthy of attention. In addition, limiting the decision maker to best-yet data can serve as a lower bound to situations of immediate choice where the decision maker has more available data. What is objectionable is claiming applicability to a broad set of real-world decision problems where no such applicability exists.

Exercise 9.22: **a)** Suppose $\{Z_n; n \geq 1\}$ is a martingale. Verify (9.83); *i.e.*, $E[Z_n] = E[Z_1]$ for $n > 1$.

Solution: The special case of Lemma 9.6.5 with $i = 1$ is $E[Z_n|Z_1] = Z_1$. Since $E[|Z_n|] < \infty$ by definition of a martingale, (1.44) (*i.e.*, the total expectation theorem) shows that $E[Z_n] = E[Z_1]$

b) If $\{Z_n; n \geq 1\}$ is a submartingale, verify (9.88), and if a supermartingale, verify (9.89).

Solution: As explained in (9.86), $E[Z_n | Z_1] \geq Z_1$, so again (1.44) shows that $E[Z_n] \geq E[Z_1]$. The same argument with inequalities reversed applies to supermartingales.

Exercise 9.23: Suppose $\{Z_n; n \geq 1\}$ is a martingale. Show that

$$E[Z_m | Z_{n_i}, Z_{n_i-1}, \dots, Z_{n_1}] = Z_{n_i} \text{ for all } 0 < n_1 < n_2 < \dots < n_i < m.$$

Solution: First observe from Lemma 9.6.5 that for $n_i < m$,

$$E[Z_m | Z_{n_i}, Z_{n_i-1}, Z_{n_i-2}, \dots, Z_1] = Z_{n_i}.$$

This is valid for every sample value of every conditioning variable. Thus consider Z_{n_i-1} for example. Since this equation has the same value for each sample value of Z_{n_i-1} , the expected value of this conditional expectation over Z_{n_i-1} is $E[Z_m | Z_{n_i}, Z_{n_i-2}, \dots, Z_1] = Z_{n_i}$. In the same way, any subset of these conditioning rv's could be removed, leaving us with the desired form.

Exercise 9.24: a) Assume that $\{Z_n; n \geq 1\}$ is a submartingale. Show that

$$E[Z_m | Z_n, Z_{n-1}, \dots, Z_1] \geq Z_n \text{ for all } n < m.$$

Solution: The proof of Lemma 9.6.5 with inequalities used in place of equalities works here.

b) Show that

$$E[Z_m | Z_{n_i}, Z_{n_i-1}, \dots, Z_{n_1}] \geq Z_{n_i} \text{ for all } 0 < n_1 < n_2 < \dots < n_i < m.$$

Solution: This follows by using inequalities in place of equalities in the same set of steps used in the solution to Exercise (9.23).

c) Assume now that $\{Z_n; n \geq 1\}$ is a supermartingale. Show that (a) and (b) still hold with $\geq$ replaced by $\leq$.

Solution: One can repeat the steps of (a) and (b), blindly replacing each $\geq$ with $\leq$. Alternatively, note that if $\{Z_n; n \geq 1\}$ is a supermartingale, then $\{-Z_n; n \geq 1\}$ is a submartingale. This technique is frequently useful in converting results about submartingales into results about supermartingales.

Exercise 9.25: Let $\{Z_n = \exp[rS_n - n\gamma(r)]; n \geq 1\}$ be the generating function martingale of (9.77) where $S_n = X_1 + \dots + X_n$ and where $X_1, X_2, \dots$, are IID with mean $\bar{X} < 0$. Let J be the possibly-defective stopping trial for which the process stops after crossing a threshold at $\alpha > 0$ (there is no negative threshold). Show that $\exp[-r^*\alpha]$ is an upper bound to the probability of threshold crossing by considering the stopped process $\{Z_n^*; n \geq 1\}$.

The purpose of this exercise is to illustrate that the stopped process can yield useful upper bounds even when the stopping trial is defective.

Solution: Example 9.6.4 shows that if $g_X(r)$ exists for any given r , then $\{Z_n = \exp[rS_n - n\gamma(r)]; n \geq 1\}$ is a martingale and $E[Z_n] = 1$ for all n . From Theorem 9.8.3, the corresponding stopped process, $\{Z_n^*; n \geq 1\}$ is a martingale. From Theorem 9.8.5, $E[Z_n^*] = 1$ for all $n \geq 1$.

There is a confusing notational issue here, since we are using Z_n^* to refer to the stopped process for some given r , and we earlier defined r^* as the positive root of $\gamma_X(r)$. We now look at the generating function process $\{Z_n; n \geq 1\}$ for $r = r^*$ and the stopped process $\{Z_n^*; n \geq 1\}$ for $r = r^*$. Then $Z_n = e^{r^* S_n}$. If $S_J \geq \alpha$, then $Z_n^* \geq e^{r^* \alpha}$ for some sufficiently large n , and we have

$$\Pr\{S_J \geq \alpha\} = \lim_{n \rightarrow \infty} \Pr\{Z_n^* \geq e^{r^* \alpha}\}.$$

By the Markov inequality, $\Pr\{Z_n^* \geq e^{r^* \alpha}\} \leq e^{-r^* \alpha}$. Since this holds for all n , it holds for the limit, and $\Pr\{S_J \geq \alpha\} \leq e^{-r^* \alpha}$.

Exercise 9.26: This exercise uses a martingale to find the expected number of successive trials $\mathbf{E}[J]$ until some fixed pattern, $a_1, a_2, \dots, a_k$, of successive binary digits occurs within a sequence of IID binary random variables $X_1, X_2, \dots$ (see Example 4.5.1 and Exercise 5.35 for alternative approaches). We take the stopping time J to be the smallest n for which $(X_{n-k+1}, \dots, X_n) = (a_1, \dots, a_k)$. A mythical casino and sequence of gamblers who follow a prescribed strategy will be used to determine $\mathbf{E}[J]$. The outcomes of the plays (trials), $\{X_n; n \geq 1\}$ at the casino is a binary IID sequence for which $\Pr\{X_n = i\} = p_i$ for $i \in \{0, 1\}$.

If a gambler places a bet s on 1 at play n , the return is s/p_1 if $X_n = 1$ and 0 otherwise. With a bet s on 0, the return is s/p_0 if $X_n = 0$ and 0 otherwise; *i.e.*, the game is fair.

a) Assume an arbitrary choice of bets on 0 and 1 by the various gamblers on the various trials. Let Y_n be the net gain of the casino on trial n . Show that $\mathbf{E}[Y_n] = 0$. Let $Z_n = Y_1 + Y_2 + \dots + Y_n$ be the aggregate gain of the casino over n trials. Show that for any given pattern of bets, $\{Z_n; n \geq 1\}$ is a martingale.

Solution: The net gain of the casino on trial n is the sum of the gains on each gambler. If a gambler bets s on outcome 1, the expected gain for the casino is $s - p_1 s/p_1 = 0$. Similarly, it is 0 for a bet on outcome 0. Since the expected gain from each gambler is 0, independent of earlier gains, we have $\mathbf{E}[Y_n | Y_{n-1}, \dots, Y_1] = 0$. As seen in Example 9.6.3, this implies that $\{Z_n; n \geq 1\}$ is a martingale.

b) In order to determine $\mathbf{E}[J]$ for a given pattern $(a_1, a_2, \dots, a_k)$, we program our gamblers to bet as follows:

- i) Gambler 1 has an initial capital of 1 which is bet on a_1 at trial 1. If $X_1 = a_1$, the capital grows to $1/p_{a_1}$, all of which is bet on a_2 at trial 2. If $X_2 = a_2$, the capital grows to $1/(p_{a_1} p_{a_2})$, all of which is bet on a_3 at trial 3. Gambler 1 continues in this way until either losing at some trial (in which case he leaves with no money) or winning on k successive trials (in which case he leaves with $1/[p_{a_1} \dots p_{a_k}]$).
- ii) Gambler ℓ , for each $\ell > 1$, follows the same strategy, but starts at trial ℓ . Note that if the string $(a_1, \dots, a_k)$ appears for the first time at trials $n-k+1, n-k+2, \dots, n$, *i.e.*, if $J = n$, then gambler $n-k+1$ leaves at time n with capital $1/[p(a_1) \dots p(a_k)]$ and gamblers $j < n-k+1$ have all lost their capital. We will come back later to investigate the capital at time n for gamblers $n-k+2$ to n .

First consider the string $a_1=0, a_2=1$ with $k = 2$. Find the sample values of Z_1, Z_2, Z_3 for the sample sequence $X_1 = 1, X_2 = 0, X_3 = 1, \dots$. Note that gamblers 1 and 3 have lost their capital, but gambler 2 now has capital $1/p_0 p_1$. Show that the sample value of the stopping time for this case is $J = 3$. Given an arbitrary sample value $n \geq 2$ for J , show that $Z_n = n - 1/p_0 p_1$.

Solution: Since gambler 1 bets on 0 at the first trial and $X_1 = 1$, gambler 1 loses and $Z_1 = 1$. At trial 2, gambler 2 bets on 0 and $X_2 = 0$. Gambler 2's capital increases from 1 to $1/p_0$ so $Y_2 = 1 - 1/p_0$. Thus $Z_2 = 2 - 1/p_0$. On trial 3, gambler 1 is broke and doesn't bet, gambler 2's capital increases from $1/p_0$ to $1/p_0 p_1$ and gambler 3 loses. Thus $Y_3 = 1 + 1/p_0 - 1/p_0 p_1$ and $Z_3 = 3 - 1/p_0 p_1$. It is preferable here to look only at the casino's aggregate gain Z_3 and not the immediate gain Y_3 . In aggregate, the casino keeps all 3 initial bets, and pays out $1/p_0 p_1$.

$J = 3$ since $(X_2, X_3) = (a_1, a_2) = (0, 1)$ and this is the first time that the pattern $(0, 1)$ has appeared. For an arbitrary sample value n for J , note that each gambler before $n - 1$ loses unit capital, gambler $n - 1$ retires to Maui with capital increased from 1 to $1/p_0p_1$, and gambler n loses. Thus the casino has $n - 1/p_0p_1$ as its gain.

c) Find $E[Z_J]$ from (a). Use this plus (b) to find $E[J]$. Hint: This uses the special form of the solution in (b), not the Wald equality.

Solution: The casino's expected gain up to each time n is $E[Z_n] = 0$, so it follows that $E[Z_J] = 0$ (It is easy to verify that the condition in 9.104 is satisfied in this case). We saw in (b) that $E[Z_n | J = n] = n - 1/p_0p_1$, so $E[Z_J] = E[J] - 1/p_0p_1$. Thus $E[J] = 1/p_0p_1$. Note that this is the mean first passage time for the same problem in Exercise 4.28. The approach there was simpler than this for this short string. For long strings, the approach here will be simpler.

d) Repeat parts b) and c) using the string $(a_1, \dots, a_k) = (1, 1)$ and initially assuming $(X_1X_2X_3) = (011)$. Be careful about gambler 3 for $J = 3$. Show that $E[J] = 1/p_1^2 + 1/p_1$.

Solution: This is almost the same as (b) except that here gambler 3 wins at time 3. In other words, since $a_1 = a_2$, gamblers 2 and 3 both bet on 1 at time 3. As before, $J = 3$ for this sample outcome. We also see that for J equal to an arbitrary n , gamblers $n - 1$ and n both bet on 1 and since $X_n = 1$, both win. Thus $E[J] = 1/p_1^2 + 1/p_1$.

e) Repeat parts b) and c) for $(a_1, \dots, a_k) = (1, 1, 1, 0, 1, 1)$.

Solution: Given that $J = n$, we know that $(X_{n-5}, \dots, X_n) = (111011)$ so gambler $n - 5$ leaves with $1/p_1^5p_0$ and all previous gamblers lose their capital. For the gamblers after $n - 5$, note that gambler n makes a winning bet on 1 at time n and gambler $n - 1$ makes winning bets on $(1, 1)$ at times $(n - 1, n)$. Thus the casino wins $n - 1/p_1^5p_0 - 1/p_1 - 1/p_1^2$. Averaging over J , we see that $E[J] = 1/(p_1^5p_0) + 1/p_1 + 1/p_1^2$. In general, we see that, given $J = n$, gambler n wins if $a_1 = a_k$, gambler 2 wins if $(a_1, a_2) = (a_{k-1}, a_k)$ and so forth.

f) Consider an arbitrary binary string $a_1, \dots, a_k$ and condition on $J = n$ for some $n \geq k$. Show that the sample capital of gambler ℓ is then equal to

- 0 for $\ell < n - k$.
- $1/[p_{a_1}p_{a_2} \cdots p_{a_k}]$ for $\ell = n - k + 1$.
- $1/[p_{a_1}p_{a_2} \cdots p_{a_i}]$ for $\ell = n - i + 1$, $1 \leq i < k$ if $(a_1, \dots, a_i) = (a_{k-i+1}, \dots, a_k)$.
- 0 for $\ell = n - i + 1$, $1 \leq i < k$ if $(a_1, \dots, a_i) \neq (a_{k-i+1}, \dots, a_k)$.

Verify that this general formula agrees with parts (b), (d), and (e).

Solution: Gambler ℓ for $\ell \leq n - k$ bets (until losing a bet) on $a_1, a_2, \dots, a_k$. Since the first occurrence of $(a_1, \dots, a_k)$ occurs at n , we see that each of these gamblers loses at some point and thus is reduced to 0 capital at that point and stays there. Gambler $n - k + 1$ bets on $a_1, \dots, a_k$ at times $n - k + 1, \dots, n$ and thus wins each bet for $J = n$. Finally, gambler $\ell = n - i + 1$ bets (until losing) on $a_1, a_2, \dots, a_i$ at times $n - i + 1$ to n . Since $J = n$ implies that $X_{n-k+1}, \dots, X_n = a_1, \dots, a_k$, gambler $n - i + 1$ is successful on all i bets if and only if $(a_1, \dots, a_i) = (a_{k-i+1}, \dots, a_k)$.

For (b), gambler n is unsuccessful and in (d), gambler n is successful. In (e), gamblers $n - 1$ and n are each successful. It might be slightly mystifying at first that conditioning on J is

enough to specify what happens to each gambler after time $n - k + 1$, but the sample values of X_{n-k+1} to X_n are specified by $J = n$, and the bets of the gamblers are also specified.

g) For a given binary string $(a_1, \dots, a_k)$, and each j , $1 \leq j \leq k$ let $\mathbb{I}_j = 1$ if $(a_1, \dots, a_j) = (a_{k-j+1}, \dots, a_k)$ and let $\mathbb{I}_j = 0$ otherwise. Show that

$$\mathbb{E}[J] = \sum_{i=1}^k \frac{\mathbb{I}_i}{\prod_{\ell=1}^i p_{a_\ell}}.$$

Note that this is the same as the final result in Exercise 5.35. The argument is shorter here, but more motivated and insightful there. Both approaches are useful and lead to simple generalizations.

Solution: The i th term in the above expansion is the capital of gambler $n - i + 1$ at time n . The final term at $i = k$ corresponds to the gambler who retires to Maui and $\mathbb{I}_k = 1$ in all cases. How many other terms are non-zero depends on the choice of string. These other terms can all be set to zero by choosing a string for which no prefix is equal to the suffix of the same length.

Exercise 9.27: a) This exercise shows why the condition $\mathbb{E}[|Z_J|] < \infty$ is required in Lemma 9.8.4. Let $Z_1 = -2$ and, for $n \geq 1$, let $Z_{n+1} = Z_n[1 + X_n(3n+1)/(n+1)]$ where $X_1, X_2, \dots$ are IID and take on the values $+1$ and -1 with probability $1/2$ each. Show that $\{Z_n; n \geq 1\}$ is a martingale.

Solution: From the definition of Z_n above,

$$\mathbb{E}[Z_n \mid Z_{n-1}, Z_{n-2}, \dots, Z_1] = \mathbb{E}[Z_{n-1}[1 + X_{n-1}(3n-2)/n] \mid Z_{n-1}, \dots, Z_1].$$

Since the X_n are zero mean and IID, this is just $\mathbb{E}[Z_{n-1} \mid Z_{n-1}, \dots, Z_1]$, which is Z_{n-1} . We also must show that $\mathbb{E}[|Z_n|] < \infty$ for each n . Note that

$$|Z_n| \leq |Z_{n-1}|[1 + (3n-2)/n] \leq 4|Z_{n-1}|.$$

Thus for each n , $|Z_n|$ is bounded (although the bound is rapidly increasing in n), so $\mathbb{E}[|Z_n|] < \infty$ for each n . It follows that $\{Z_n; n \geq 1\}$ is a martingale.

b) Consider the stopping trial J such that J is the smallest value of $n > 1$ for which Z_n and Z_{n-1} have the same sign. Show that, conditional on $n < J$, $Z_n = (-2)^n/n$ and, conditional on $n = J$, $Z_J = -(-2)^n(n-2)/(n^2-n)$.

Solution: It can be seen from the iterative definition of Z_n that Z_n and Z_{n-1} have different signs if $X_{n-1} = -1$ and have the same sign if $X_{n-1} = 1$. Thus the stopping trial is the smallest trial $n \geq 2$ for which $X_{n-1} = 1$. Thus for $n < J$, we must have $X_i = -1$ for $1 \leq i < n$. For $n = 2 < J$, X_1 must be -1 , so from the formula above, $Z_2 = Z_1[1 - 4/2] = 2$. Thus $Z_n = (-2)^n/n$ for $n = 2 < J$. We can use induction now for arbitrary $n < J$. Thus for $X_n = -1$,

$$Z_{n+1} = Z_n \left[1 - \frac{3n+1}{n+1} \right] = \frac{(-2)^n}{n} \frac{-2n}{n+1} = \frac{(-2)^{n+1}}{n+1}.$$

The remaining task is to compute Z_n for $n = J$. Using the result just derived for $n = J - 1$ and using $X_{J-1} = 1$,

$$Z_J = Z_{J-1} \left[1 + \frac{3(J-1)+1}{J} \right] = \frac{(-2)^{J-1}}{J-1} \frac{4J-2}{J} = \frac{-(-2)^J(2J-1)}{J(J-1)}.$$

c) Show that $E[|Z_J|]$ is infinite, so that $E[Z_J]$ does not exist according to the definition of expectation, and show that $\lim_{n \rightarrow \infty} E[Z_n | J > n] \Pr\{J > n\} = 0$.

Solution: We have seen that $J = n$ if and only if $X_i = -1$ for $1 \leq i \leq n-2$ and $X_{J-1} = 1$. Thus $\Pr\{J = n\} = 2^{-n+1}$ so

$$E[|Z_J|] = \sum_{n=2}^{\infty} 2^{-n+1} \cdot \frac{2^n(2n-1)}{n(n-1)} = \sum_{n=2}^{\infty} \frac{2(2n-1)}{n(n-1)} \geq \sum_{n=2}^{\infty} \frac{4}{n} = \infty, \quad (i)$$

since the harmonic series diverges.

Finally, we see that $\Pr\{J > n\} = 2^{-n+1}$ (since this event occurs if and only if $X_i = -1$ for $1 \leq i < n$). Thus

$$E[Z_n | J > n] \Pr\{J > n\} = \frac{2^{-n+1}(-2)^n}{n} = \frac{2(-1)^n}{n} \rightarrow 0.$$

This martingale and stopping rule gives an example where the condition $E[|Y_J|]$ is needed in Lemma 9.8.4. Note that $E[Y_J]$, if it exists, is $\lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} E[Z_J=i] \Pr\{J=i\}$. For this example, this is the alternating version of the harmonic series in (i), which converges in the Cauchy sense but not in the usual sense. Exercise 1.8 provides a reminder of why this series is defined to be nonexistent, *i.e.*, of why the condition $E[|Y_J|] < \infty$ is needed.

Exercise 9.28: This exercise shows why the sup of a martingale can behave markedly differently than the maximum of an arbitrarily large number of the variables. More precisely, it shows that $\Pr\{\sup_{n \geq 1} Z_n \geq a\}$ can be unequal to $\Pr\{\bigcup_{n \geq 1} \{Z_n \geq a\}\}$.

a) Consider a martingale where Z_n can take on only the values 2^{-n-1} and $1 - 2^{-n-1}$, each with probability $1/2$. Given that Z_n , conditional on Z_{n-1} , is independent of $Z_1, \dots, Z_{n-2}$, find $\Pr\{Z_n | Z_{n-1}\}$ for each n so that the martingale condition is satisfied.

Solution: This is a positive martingale for which the sample values approach 0 or 1 with increasing n . We will see that as $n \rightarrow \infty$, the transitions between the larger and smaller value become less frequent. Let $p_0(n) = \Pr\{Z_n = 2^{-n-1} | Z_{n-1} = 2^{-n}\}$, *i.e.*, the probability that the sample value close to 0 at trial $n-1$ goes to the yet smaller value at trial n . For $n \geq 1$, with Z_0 taken to be $1/2$, we have

$$E[Z_n | Z_{n-1} = 2^{-n}] = 2^{-n-1}p_0(n) + (1 - 2^{-n-1})(1 - p_0(n)).$$

To satisfy the martingale condition, this must be equal to 2^{-n} , so we can solve for $p_0(n)$, getting

$$p_0(n) = \frac{2^n - 3/2}{2^n - 1}. \quad (A.197)$$

We can define $p_1(n) = \Pr\{Z_n = 1 - 2^{-n-1} | Z_{n-1} = 1 - 2^{-n}\}$ and solve for $p_1(n)$ in the same way, getting $p_1(n) = p_0(n)$ for $n \geq 1$. Thus a martingale exists with the transition probabilities for $p_0(n)$ (and $p_1(n) = p_0(n)$) in (A.197).

We should also show that $\Pr\{Z_n = 2^{-n-1}\} = 1/2$ as specified in the problem statement. For $n = 1$, it can be seen from (A.197) that $p_0(1) = 1/2$, and similarly $p_1(1) = 1/2$. It

follows that $E[Z_1] = 1/2$. From (9.83), then, $E[Z_n] = 1/2$ for all $n \geq 1$. It follows that $\Pr\{Z_n = 2^{-n-1}\} = \Pr\{Z_n = 1 - 2^{-n-1}\} = 1/2$.

b) Show that $\Pr\{\sup_{n \geq 1} Z_n \geq 1\} = 1/2$ and show that $\Pr\{\bigcup_n \{Z_n \geq 1\}\} = 0$.

Solution: First we show that $\Pr\{\bigcup_n \{Z_n \geq 1\}\} = 0$. Since Z_n has no sample values of 1 or more, $\Pr\{Z_n \geq 1\} = 0$ for all n . From the union bound, $\Pr\{\bigcup_n Z_n \geq 1\}$ is upper bounded by a sequence of 0's, so $\Pr\{\bigcup_n \{Z_n \geq 1\}\} = 0$.

Finding $\Pr\{\sup_{n \geq 1} Z_n \geq 1\}$ is a little harder, since a sample sequence of $\{Z_n; n \geq 1\}$ can jump back and forth between values less than $1/2$ and greater than $1/2$. This is an ideal opportunity to use the martingale convergence theorem, which says that $\{Z_n; n \geq 1\}$ converges WP1. Since the sample values greater than $1/2$ approach 1 and the sample values less than $1/2$ approach 0, Z_n must converge to 1 with some probability q and to 0 with probability $1 - q$. Either using the symmetry between 0 and 1 or using the fact that each sample value has probability $1/2$ for each n , $\Pr\{\sup_{n \geq 1} Z_n \geq 1\} = 1/2$.

c) Show that for every $\epsilon > 0$, $\Pr\{\sup_{n \geq 1} Z_n \geq a\} \leq \frac{\bar{Z}_1}{a - \epsilon}$. Hint: Use the relationship between $\Pr\{\sup_{n \geq 1} Z_n \geq a\}$ and $\Pr\{\bigcup_n \{Z_n \geq a\}\}$ while getting around the issue in (b).

Solution: For any $\epsilon > 0$, if $\sup_{n \geq 1} Z_n \geq a$, then for some large enough m , $\max_{1 \leq n \leq m} Z_n \geq a - \epsilon$. The probability of this latter event is non-decreasing with m and bounded by 1, so it has a limit, and

$$\Pr\left\{\sup_{n \geq 1} Z_n \geq a\right\} \leq \lim_{m \rightarrow \infty} \Pr\left\{\max_{1 \leq n \leq m} Z_n \geq a - \epsilon\right\} \leq \lim_{m \rightarrow \infty} \frac{E[Z_1]}{a - \epsilon} = \frac{E[Z_1]}{a - \epsilon},$$

where we used (9.110) in the middle step.

d) Use (c) to establish (9.111).

Solution: Since (c) is valid for all $\epsilon > 0$, it is valid in the limit as $\epsilon > 0$ approaches 0, yielding (9.111).

Exercise 9.29: Show that Theorem 9.7.4 is also valid for martingales relative to a joint process. That is, show that if h is a convex function of a real variable and if $\{Z_n; n \geq 1\}$ is a martingale relative to a joint process $\{Z_n, X_n; n \geq 1\}$, then $\{h(Z_n); n \geq 1\}$ is a submartingale relative to $\{h(Z_n), X_n; n \geq 1\}$.

Solution: This is simply a matter of adding the additional conditioning to the proof of Theorem 9.7.4. Specifically, using Jensen's inequality and then the martingale property,

$$E[h(Z_n) | Z_{n-1}, X_{n-1}, \dots, Z_1, X_1] \geq h(E[Z_n | Z_{n-1}, X_{n-1}, \dots, Z_1, X_1]) = h(Z_{n-1}).$$

This shows that $\{h(Z_n); n \geq 1\}$ is a submartingale relative to $\{h(Z_n), X_n; n \geq 1\}$.

Exercise 9.30: Show that if $\{Z_n; n \geq 1\}$ is a martingale (submartingale or supermartingale) relative to a joint process $\{Z_n, X_n; n \geq 1\}$ and if J is a stopping trial for $\{Z_n; n \geq 1\}$ relative to $\{Z_n, X_n; n \geq 1\}$, then the stopped process is a martingale (submartingale or supermartingale) respectively relative to the joint process.

Solution: This simply extends Theorem 9.8.3 to include a joint process. The proof that $E[|Z_n|] < \infty$ is unchanged by the inclusion of a joint process. Now suppose that $\{Z_n; n \geq 1\}$

is a submartingale relative to $\{Z_n, X_n; n \geq 1\}$ and let $\{Z_n^*; n \geq 1\}$ be the corresponding stopped process. For any $n \geq 1$, let $\{z_1, x_1, z_2, x_2, \dots, z_{n-1}, x_{n-1}\}$ be a sample sequence for $\{Z_1, X_1, Z_2, X_2, \dots, Z_{n-1}, X_{n-1}\}$. This sample sequence specifies whether or not $J \leq n-1$ and specifies the sample value of J if $J \leq n-1$. Thus, as in the proof of Theorem 9.8.3, if the sample sequence is such that $J \leq n-1$, then $z_n^* = z_{n-1}^*$ and

$$\mathbb{E} \left[Z_n^* | Z_1^{*(n-1)} = z_1^{*(n-1)}, X_1^{(n-1)} = x_1^{(n-1)} \right] = z_{n-1}^*.$$

Alternatively, if $J > n-1$, then $z_n^* = z_n$ and

$$\mathbb{E} \left[Z_n^* | Z_1^{*(n-1)} = z_1^{*(n-1)}, X_1^{(n-1)} = x_1^{(n-1)} \right] \geq z_{n-1}^*.$$

Thus $\{Z_n^*; n \geq 1\}$ is a submartingale relative to the joint process. The argument for a supermartingale is the same with a reversal of inequality, and that for a martingale follows since it is both a sub and supermartingale.

Exercise 9.31: Prove Corollaries 9.9.3 to 9.9.5, i.e., prove the following three statements:

a) Let $\{Z_n; n \geq 1\}$ be a martingale with $\mathbb{E}[Z_n^2] < \infty$ for all $n \geq 1$. Then

$$\Pr \left\{ \max_{1 \leq n \leq m} |Z_n| \geq b \right\} \leq \frac{\mathbb{E}[Z_m^2]}{b^2}; \text{ for all integer } m \geq 2, \text{ and all } b > 0. \quad (\text{A.198})$$

Hint: First show that $\{Z_n^2; n \geq 1\}$ is a submartingale.

Solution: Z_n^2 is a convex function of Z_n and $\mathbb{E}[Z_n^2] < \infty$ for all $n \geq 1$. Thus, from Theorem 9.7.4, $\{Z_n^2; n \geq 1\}$ is a submartingale. Applying the submartingale inequality to $\{Z_n^2; n \geq 1\}$,

$$\Pr \left\{ \max_{1 \leq n \leq m} Z_n^2 \geq a \right\} \leq \frac{\mathbb{E}[Z_m^2]}{a}.$$

Substituting b^2 for a , this becomes (A.198).

b) [Kolmogorov's random walk inequality] Let $\{S_n; n \geq 1\}$ be a random walk with $S_n = X_1 + \dots + X_n$ where $\{X_i; i \geq 1\}$ is a set of IID random variables with mean $\bar{X}$ and variance σ^2 . Then for any positive integer m and any $\epsilon > 0$,

$$\Pr \left\{ \max_{1 \leq n \leq m} |S_n - n\bar{X}| \geq m\epsilon \right\} \leq \frac{\sigma^2}{m\epsilon^2}. \quad (\text{A.199})$$

Hint: First show that $\{S_n - n\bar{X}; n \geq 1\}$ is a martingale.

Solution: Let $Z_n = S_n - n\bar{X}$. Then $\{Z_n; n \geq 1\}$ is a martingale as shown in Example 9.6.2. $\mathbb{E}[Z_m^2] = m\sigma^2$, so using (A.198) with $m\epsilon$ substituted for b , we get (A.199).

c) Let $\{S_n; n \geq 1\}$ be a random walk, $S_n = X_1 + \dots + X_n$, where each X_i has mean $\bar{X} < 0$ and semi-invariant moment generating function $\gamma(r)$. For any $r > 0$ such that $0 < \gamma(r) < \infty$, and for any $\alpha > 0$. show that

$$\Pr \left\{ \max_{1 \leq i \leq n} S_i \geq \alpha \right\} \leq \exp\{-r\alpha + n\gamma(r)\}. \quad (\text{A.200})$$

Hint: First show that $\{e^{rS_n}; n \geq 1\}$ is a submartingale.

Solution: Again let $Z_n = S_n - n\bar{X}$, so $\{Z_n, n \geq 1\}$ is a martingale. Then $e^{rS_n} = e^{rZ_n + rn\bar{X}}$ and we see by differentiating twice with respect to Z_n that e^{rS_n} is a convex function of Z_n . Thus by Theorem 9.7.4, $\{e^{rS_n}; n \geq 1\}$ is a submartingale. Thus by the Kolmogorov submartingale inequality,

$$\Pr \left\{ \max_{1 \leq i \leq n} e^{rS_i} \geq a \right\} \leq \frac{\mathbb{E}[e^{rS_n}]}{a}. \quad (\text{A.201})$$

Now $\mathbb{E}[e^{rS_n}] = [\mathbf{g}_X(r)]^n = e^{n\gamma(r)}$. Substituting $e^{r\alpha}$ for a , (A.201) becomes (A.200)

Exercise 9.32: a) Let $\{Z_n; n \geq 1\}$ be the scaled branching process of Section 9.6.2 and assume that Y , the number of offspring of an element, has finite mean $\bar{Y}$ and finite variance σ^2 . Assume that the population at time 0 is 1. Show that

$$\mathbb{E}[Z_n^2] = 1 + \sigma^2 [\bar{Y}^{-2} + \bar{Y}^{-3} + \cdots + \bar{Y}^{-n-1}]. \quad (\text{A.202})$$

Solution: Let $\{X_n; n \geq 1\}$ be the unscaled process as discussed in Section 6.7. Thus $Z_n = X_n/\bar{Y}^n$. We first find the second moment of X_n . Recall that $X_n = \sum_{i=1}^{X_{n-1}} Y_{i,n-1}$. Thus

$$\mathbb{E}[X_n | X_{n-1}] = X_{n-1} \bar{Y} \quad \text{and} \quad \mathbb{E}[X_n] = \bar{Y} \mathbb{E}[X_{n-1}].$$

Recurring over n , we see that $\mathbb{E}[X_n] = \bar{Y}^n$. We use the same approach to find $\mathbb{E}[X_n^2]$.

$$\begin{aligned} \mathbb{E}[X_n^2 | X_{n-1}] &= \mathbb{E} \left[\left(\sum_{i=1}^{X_{n-1}} Y_{i,n-1} \right)^2 \mid X_{n-1} \right] \\ &= \mathbb{E}[Y^2] X_{n-1} + \bar{Y}^2 (X_{n-1} - 1) X_{n-1} \\ &= \sigma^2 X_{n-1} + \bar{Y}^2 X_{n-1}^2 \\ \mathbb{E}[X_n^2] &= \sigma^2 \bar{Y}^{n-1} + \bar{Y}^2 \mathbb{E}[X_{n-1}^2]. \end{aligned}$$

In the third line, we used $\mathbb{E}[Y^2] = \sigma^2 + \bar{Y}^2$. Recursing on n down to $n = 0$, we get

$$\mathbb{E}[X_n^2] = \sigma^2 (\bar{Y}^{n-1} + \bar{Y}^n + \cdots + \bar{Y}^{2n-2}) + \bar{Y}^{2n}.$$

Since $Z_n^2 = X_n^2/\bar{Y}^{2n}$, this implies (A.202).

b) Assume that $\bar{Y} > 1$ and find $\lim_{n \rightarrow \infty} \mathbb{E}[Z_n^2]$. Show from this that the conditions for the martingale convergence theorem, Theorem 9.9.8, are satisfied.

Note: This condition is not satisfied if $\bar{Y} \leq 1$. The general martingale convergence theorem requires only a bounded first absolute moment, so it holds for $\bar{Y} \leq 1$.

Solution: From (A.202), $\mathbb{E}[Z_n^2] = 1 + (\sigma^2/\bar{Y}^2)(1 + \bar{Y}^{-1} + \cdots + \bar{Y}^{-n+1})$. Summing the geometric series,

$$\mathbb{E}[Z_n^2] = \frac{\sigma^2(1 - \bar{Y}^{-n})}{\bar{Y}^2(1 - \bar{Y}^{-1})}.$$

Since $\bar{Y} > 1$, the limit converges to $\sigma^2/(\bar{Y}^2(1 - \bar{Y}^{-1}))$. See Example 9.9.10 for an interpretation of this result.

Exercise 9.33: Show that if $\{Z_n; n \geq 1\}$ is a martingale (submartingale or supermartingale) relative to a joint process $\{Z_n, X_n; n \geq 1\}$ and if J is a stopping trial for $\{Z_n; n \geq 1\}$ relative to $\{Z_n, X_n; n \geq 1\}$, then the stopped process satisfies (9.98), (9.99), or (9.100) respectively.

Solution: This follows from Exercise 9.30 by the same argument as used in proving Theorem 9.8.3 from Theorem 9.8.2.

Exercise 9.34: Show that if $\{Z_n; n \geq 1\}$ is a martingale relative to a joint process $\{Z_n, X_n; n \geq 1\}$ and if J is a stopping trial for $\{Z_n; n \geq 1\}$ relative to $\{Z_n, X_n; n \geq 1\}$, then $E[Z_J] = E[Z_1]$ if and only if (9.104) is satisfied.

Solution: This follows from the martingale part of Exercise 9.30 in the same way as Lemma 9.8.4 follows from Theorem 9.8.2.

Exercise 9.35: (The double-or-quarter game) Consider an investment example similar to that of Example 9.10.1 in which there is only one investment other than cash. The ratio X of that investment value at the end of an epoch to that at the beginning is either 2 or 1/4, each with equal probability. Thus $\Pr\{X = 2\} = 1/2$ and $\Pr\{X = 1/4\} = 1/2$. Successive ratios, $X_1, X_2, \dots$, are IID.

a) In (a) to (c), assume a constant allocation strategy where a fraction λ is constantly placed in the double-or-quarter investment and the remaining $1 - \lambda$ is kept in cash. Find the expected wealth, $E[W_n(\lambda)]$ and the expected log-wealth $E[L_n(\lambda)]$ as a function of $\lambda \in [0, 1]$ and $n \geq 1$. Assume unit initial wealth.

Solution: With probability 1/2, the fraction λ that is invested during epoch 1 becomes 2λ and the $1 - \lambda$ in cash remains $1 - \lambda$, leading to wealth $W_1 = 1 + \lambda$. Similarly, with probability 1/2, the fraction λ that is invested becomes $\lambda/4$, yielding the final wealth $W_1 = 1 - 3\lambda/4$. The expected wealth at the end of the first epoch is then

$$E[W_1(\lambda)] = (1/2)(1 + \lambda) + (1/2)(1 - 3\lambda/4) = 1 + \lambda/8.$$

As in Example 9.10.1, this grows geometrically with n , so

$$E[W_n(\lambda)] = (1 + \lambda/8)^n.$$

The log-wealth $L_1(\lambda)$ is the log of $W_1(\lambda)$, which is $\ln(1 + \lambda)$ with probability 1/2 and $\ln(1 - 3\lambda/4)$ with probability 1/2. Thus $E[L_1(\lambda)] = (1/2) \ln((1 + \lambda)(1 - 3\lambda/4))$. Thus

$$E[L_n(\lambda)] = \frac{n}{2} \ln((1 + \lambda)(1 - 3\lambda/4)).$$

b) For $\lambda = 1$, find the PMF for $W_4(1)$ and give a brief explanation of why $E[W_n(1)]$ is growing exponentially with n but $E[L_n(1)]$ is decreasing linearly toward $-\infty$.

Solution: The probability of k successes out of n is given by the binomial $\binom{n}{k}2^{-n}$, which for $n = 4$ is, respectively, 1/16, 4/16, 6/16, 4/16, 1/16 for $k = 0, 1, 2, 3, 4$. Thus, using

logs to the base 2,

$$\begin{aligned}
 \Pr\{W_4(1) = 16\} &= \Pr\{L_4(1) = 4\} = \frac{1}{16} \\
 \Pr\{W_4(1) = 2\} &= \Pr\{L_4(1) = 1\} = \frac{4}{16} \\
 \Pr\left\{W_4(1) = \frac{1}{4}\right\} &= \Pr\{L_4(1) = -2\} = \frac{6}{16} \\
 \Pr\left\{W_4(1) = \frac{1}{32}\right\} &= \Pr\{L_4(1) = -5\} = \frac{4}{16} \\
 \Pr\left\{W_4(1) = \frac{1}{256}\right\} &= \Pr\{L_4(1) = -8\} = \frac{1}{16}.
 \end{aligned}$$

As seen above, the sample values of $W_4(1)$ are exponential in the number of successes, whereas those of $L_4(1)$ are linear in the number of successes. As n increases, the probabilities cluster around the mean of L_n (since L_n is the sum of IID rv's), but the very large low probability values of W_n lead to an exponentially increasing value of $E[W_n]$.

c) Using the same approach as in Example 9.10.1, find the value of λ^* that maximizes $E[L_n(\lambda)]$. Show that your solution satisfies the optimality conditions in (9.136).

Solution: The easiest way to do this is to use the fact that the log function is monotonic. Therefore we can maximize $L_n(\lambda)$ over λ by maximizing $(1 + \lambda)(1 - 3\lambda/4)$. The maximum occurs at $\lambda^* = 1/6$. We then see that $E[L_n(\lambda^*)] = (n/2) \log(49/48)$, which is slowly but steadily increasing in n .

The optimality conditions of (9.136) in this case involve only 2 investments; the double or quarter price-relative, denoted by X , and cash, denoted by 1. Letting λ^* and $1 - \lambda^*$ be the corresponding allocations, (9.136) for the double or quarter investment becomes

$$E\left[\frac{X}{\lambda^*X + (1 - \lambda^*)}\right] \begin{cases} = 1 & ; & \text{for } \lambda^* > 0 \\ \leq 1 & ; & \text{for } \lambda^* = 0. \end{cases}$$

To verify that $\lambda^* = 1/6$ satisfies this equation, note that $X/(\lambda^*X + (1 - \lambda^*))$ is $12/7$ for $X = 2$ and $2/7$ for $X = 1/4$. Since each value occurs with probability $1/2$, we have

$$E\left[\frac{X}{\lambda^*X + (1 - \lambda^*)}\right] = \frac{1}{2} \cdot \frac{12}{7} + \frac{1}{2} \cdot \frac{2}{7} = 1.$$

The same type of calculation works for the cash investment. The lesson to be learned from this is that the optimality conditions, while simple appearing and highly useful theoretically, are not trivial to interpret or work with.

d) Find the range of λ over which $E[L_n(\lambda)]$ is positive.

Solution: $E[L_n(\lambda)]$ is positive where $(1 + \lambda)(1 - 3\lambda/4)$ is greater than 1. Solving for $(1 + \lambda)(1 - 3\lambda/4) = 1$, we get $\lambda = 1/3$ and $\lambda = 0$. We then see that $E[L_n(\lambda)]$ is positive between those limits, *i.e.*, for $0 < \lambda < 1/3$.

e) Find the PMF of the rv Z_n/Z_{n-1} as given in (9.138) for any given λ_n .

Solution: Again let X_n be the price-relative of the double or quarter investment and 1 be that for cash. Letting λ and $1 - \lambda$ be the corresponding allocations and λ^* and $1 - \lambda^*$ be the optimal allocations, (9.138) becomes

$$Z_n/Z_{n-1} = \lambda_n \frac{X_n}{\lambda_n^* X_n + (1 - \lambda_n^*)} + (1 - \lambda_n) \frac{1}{\lambda_n^* X_n + (1 - \lambda_n^*)}.$$

For the two possible values of X_n , this becomes

$$\begin{aligned} Z_n/Z_{n-1} &= \lambda_n \frac{2}{7/6} + (1 - \lambda_n) \frac{1}{7/6} = \frac{6}{7} + \frac{6\lambda}{7} \quad \text{for } X = 2 \\ Z_n/Z_{n-1} &= \lambda_n \frac{1/4}{7/8} + (1 - \lambda_n) \frac{1}{7/8} = \frac{8}{7} - \frac{6\lambda}{7} \quad \text{for } X = 1/2. \end{aligned}$$

Thus, Z_n/Z_{n-1} is $6/7 + 6\lambda/7$ with probability $1/2$ and $8/7 - 6\lambda/7$ with probability $1/2$. As a check, $E[Z_n/Z_{n-1}] = 1$ for all λ , since as we have seen $\{Z_n; n \geq 1\}$ is a product martingale in this case.

Exercise 9.36: (Kelly's horse-racing game) An interesting special case of this simple theory of investment is the horse-racing game due to J. Kelly and described in [5]. There are $\ell - 1$ horses in a race and each horse j wins with some probability $p(j) > 0$. One and only one horse wins, and if j wins, the gambler receives $r(j) > 0$ for each dollar bet on j and nothing for the bets on other horses. In other words, the price-relative $X(j)$ for each j , $1 \leq j \leq \ell - 1$, is $r(j)$ with probability $p(j)$ and 0 otherwise. For cash, $X(\ell) = 1$.

The gambler's allocation of wealth on the race is denoted by $\lambda(j)$ on each horse j with $\lambda(\ell)$ kept in cash. As usual, $\sum_j \lambda(j) = 1$ and $\lambda(j) \geq 0$ for $1 \leq j \leq \ell$. Note that $X(1), \dots, X(\ell - 1)$ are highly dependent, since only one has a nonzero sample value in any race.

a) For any given allocation λ find the expected wealth and the expected log-wealth at the end of a race for unit initial wealth.

Solution: With probability $p(j)$, horse j wins and the resulting value of $W_1(\lambda)$ is $\lambda(j)r(j) + \lambda(\ell)$. Thus

$$\begin{aligned} E[W_1(\lambda)] &= \sum_{j=1}^{\ell-1} p(j) [\lambda(j)r(j) + \lambda(\ell)], \\ E[L_1(\lambda)] &= \sum_{j=1}^{\ell-1} p(j) \ln [\lambda(j)r(j) + \lambda(\ell)]. \end{aligned}$$

b) Assume that a statistically identical sequence of races are run, i.e., $\mathbf{X}_1, \mathbf{X}_2, \dots$, are IID where each $\mathbf{X}_n = (X_n(1), \dots, X_n(\ell))^T$. Assuming unit initial wealth and a constant allocation λ on each race, find the expected log-wealth $E[L_n(\lambda)]$ at the end of the n th race and express it as $nE[Y(\lambda)]$.

Solution: Using (9.128) to express $E[L_n(\lambda)]$ as $nE[Y(\lambda)]$, we have

$$E[Y(\lambda)] = \sum_{j=1}^{\ell-1} p(j) \ln [\lambda(j)r(j) + \lambda(\ell)]. \quad (\text{A.203})$$

c) Let λ^* maximize $E[Y(\lambda)]$. Use the necessary and sufficient condition (9.136) on λ^* for horse j , $1 \leq j < \ell$ to show that $\lambda^*(j)$ can be expressed in the following two equivalent ways; each uniquely specifies each $\lambda^*(j)$ in terms of $\lambda^*(\ell)$.

$$\lambda^*(j) \geq p(j) - \frac{\lambda^*(\ell)}{r(j)}; \quad \text{with equality if } \lambda^*(j) > 0 \quad (\text{A.204})$$

$$\lambda^*(j) = \max \left\{ p(j) - \frac{\lambda^*(\ell)}{r(j)}, 0 \right\}. \quad (\text{A.205})$$

Solution: The necessary and sufficient condition for λ^* in (9.136) for horse j is

$$E \left[\frac{X(j)}{\sum_k \lambda^*(k)X(k)} \right] \leq 1; \quad \text{with equality if } \lambda^*(j) > 0.$$

In the event that horse j wins, $X(j) = r(j)$ while $X(k) = 0$ for horses $k \neq j$. Also $X(\ell) = 1$. Thus in the event that horse j wins, $\frac{X(j)}{\sum_k \lambda^*(k)X(k)} = \frac{r(j)}{\lambda^*(j)r(j) + \lambda^*(\ell)}$. If any other horse wins, $\frac{X(j)}{\sum_k \lambda^*(k)X(k)} = 0$. Thus, since j wins with probability $p(j)$,

$$E \left[\frac{X(j)}{\sum_k \lambda^*(k)X(k)} \right] = \frac{p(j)r(j)}{\lambda^*(j)r(j) + \lambda^*(\ell)} \leq 1; \quad \text{with equality if } \lambda^*(j) > 0. \quad (\text{A.206})$$

Rearranging this inequality yields (A.204); (A.205) is then easily verified by looking separately at the cases $\lambda^*(j) > 0$ and $\lambda^*(j) = 0$.

Solving for $\lambda^*(\ell)$ (which in turn specifies the other components of λ^*) breaks into 3 special cases which are treated below in parts d), e), and f) respectively. The first case, in (d), shows that if $\sum_{j < \ell} 1/r(j) < 1$, then $\lambda^*(\ell) = 0$. The second case, in (e), shows that if $\sum_{j < \ell} 1/r(j) > 1$, then $\lambda^*(\ell) > \min_j (p(j)r(j))$, with the specific value specified by the unique solution to (A.208). The third case, in (f), shows that if $\sum_{j < \ell} 1/r(j) = 1$, then $\lambda^*(\ell)$ is nonunique, and its set of possible values occupy the range $[0, \min_j (p(j)r(j))]$.

d) Sum (A.204) over j to show that if $\lambda^*(\ell) > 0$, then $\sum_{j < \ell} 1/r(j) \geq 1$. Note that the logical obverse of this is that $\sum_{j < \ell} 1/r(j) < 1$ implies that $\lambda^*(\ell) = 0$ and thus that $\lambda^*(j) = p(j)$ for each horse j .

Solution: Summing (A.204) over $j < \ell$ and using the fact that $\sum_{j < \ell} \lambda^*(j) = 1 - \lambda^*(\ell)$, we get

$$1 - \lambda^*(\ell) \geq 1 - \lambda^*(\ell) \sum_{j < \ell} 1/r(j).$$

If $\lambda^*(\ell) > 0$, this shows that $\sum_j 1/r(j) \geq 1$. The logical obverse is that if $\sum_j 1/r(j) < 1$, then $\lambda^*(\ell) = 0$. This is the precise way of saying that if the returns on the horses are sufficiently large, then no money should be retained in cash.

When $\lambda^*(\ell) = 0$ is substituted into (A.204), we see that each $\lambda^*(j)$ must be positive and thus equal to $p(j)$. This is very surprising, since it says that the allocation of bets does not depend on the rewards $r(j)$ (so long as the rewards are large enough to satisfy $\sum_j 1/r(j) < 1$). This will be further explained by example in (g).

e) In (c), $\lambda^*(j)$ for each $j < \ell$ was specified in terms of $\lambda^*(\ell)$; here you are to use the necessary and sufficient condition (9.136) on cash to specify $\lambda^*(\ell)$. More specifically, you are to show that $\lambda^*(\ell)$ satisfies each of the

following two equivalent inequalities:

$$\sum_{j < \ell} \frac{p(j)}{\lambda^*(j)r(j) + \lambda^*(\ell)} \leq 1; \quad \text{with equality if } \lambda^*(\ell) > 0 \quad (\text{A.207})$$

$$\sum_{j < \ell} \frac{p(j)}{\max[p(j)r(j), \lambda^*(\ell)]} \leq 1; \quad \text{with equality if } \lambda^*(\ell) > 0. \quad (\text{A.208})$$

Show from (A.208) that if $\lambda^*(\ell) \leq p(j)r(j)$ for each j , then $\sum_j 1/r(j) \leq 1$. Point out that the logical obverse of this is that if $\sum_j 1/r(j) > 1$, then $\lambda^*(\ell) > \min_j(p(j)r(j))$. Explain why (A.208) has a unique solution for $\lambda^*(\ell)$ in this case. Note that $\lambda^*(j) = 0$ for each j such that $p(j)r(j) < \lambda^*(\ell)$.

Solution: The necessary and sufficient condition for cash (investment ℓ) is

$$\mathbb{E} \left[\frac{X(\ell)}{\sum_k \lambda^*(k)X(k)} \right] \leq 1; \quad \text{with equality if } \lambda^*(\ell) > 0. \quad (\text{A.209})$$

In the event that horse j wins, $X(\ell)$ has the sample value 1 and $\sum_k \lambda^*(k)X(k)$ has the sample value $\lambda^*(j)r(j) + \lambda^*(\ell)$. Taking the expectation by multiplying by $p(j)$ and summing over $j < \ell$, (A.209) reduces to (A.207). Now if we multiply both sides of (A.205) by $r(j)$ and then add $\lambda^*(\ell)$ to both sides, we get

$$\lambda^*(j)r(j) + \lambda^*(\ell) = \max[p(j)r(j), \lambda^*(\ell)],$$

which converts (A.207) into (A.208). Now assume that $\lambda^*(\ell) \leq p(j)r(j)$ for each j . Then the max in the denominator of the left side of (A.208) is simply $p(j)r(j)$ for each j and (A.208) becomes $\sum_{j < \ell} 1/r(j) \leq 1$. The logical obverse is that $\sum_{j < \ell} 1/r(j) > 1$ implies that $\lambda^*(\ell) > \min_j(p(j)r(j))$, as was to be shown.

Finally, we must show that if $\sum_{j < \ell} 1/r(j) > 1$, then (A.208) has a unique solution for $\lambda^*(\ell)$. The left side of (A.208), viewed as a function of $\lambda^*(\ell)$, is $\sum_{j < \ell} 1/r(j) > 1$ for $\lambda^*(\ell) = \min_j(p(j)r(j))$. This function is continuous and strictly decreasing with further increases in $\lambda^*(\ell)$ and is less than or equal to 1 at $\lambda^*(\ell) = 1$. Thus there must be a unique value of $\lambda^*(\ell)$ at which (A.208) is satisfied.

It is interesting to observe from (A.205) that $\lambda^*(j) = 0$ for each j such that $p(j)r(j) \leq \lambda^*(\ell)$. In other words, no bets are placed on any horse j for which $\mathbb{E}[X(j)] < \lambda^*(\ell)$. This is in marked contrast to the case in (d) where the allocation does not depend on $r(j)$ (within the assumed range).

f) Now consider the case in which $\sum_{j < \ell} 1/r(j) = 1$. Show that (A.208) is satisfied with equality for each choice of $\lambda^*(\ell)$, $0 \leq \lambda^*(\ell) \leq \min_{j < \ell} p(j)r(j)$.

Solution: Note that $\max[p(j)r(j), \lambda^*(\ell)] = p(j)r(j)$ over the given range of $\lambda^*(\ell)$, so the left side of (A.208) is $\sum_{j < \ell} 1/r(j) = 1$ over this range. Thus the inequality in (A.208) is satisfied for all $\lambda^*(\ell)$ in this range. From (A.204), $\lambda^*(j) = p(j) - \lambda^*(\ell)/r(j)$ can be used for each $j < \ell$, to see that all the necessary and sufficient conditions are satisfied for maximizing $\mathbb{E}[Y(\lambda)]$.

g) Consider the special case of a race with only two horses. Let $p(1) = p(2) = 1/2$. Assume that $r(1)$ and $r(2)$ are large enough to satisfy $1/r(1) + 1/r(2) < 1$; thus no cash allotment is used in maximizing $\mathbb{E}[Y(\lambda)]$. With $\lambda(3) = 0$, we have

$$\mathbb{E}[Y(\lambda)] = \frac{1}{2} \ln[\lambda(1)r(1)] + \frac{1}{2} \ln[\lambda(2)r(2)] = \frac{1}{2} \ln[\lambda(1)r(1)(1 - \lambda(1))r(2)]. \quad (\text{A.210})$$

Use this equation to give an intuitive explanation for why $\lambda^*(1) = 1/2$, independent of $r(1)$ and $r(2)$.

Solution: Suppose that $r(1) \gg r(2)$. Choosing $\lambda(1)$ to be large so as to enhance the profit when horse 1 wins is counter-productive, since (A.210) shows that there is a corresponding loss when horse 2 wins. This gain and loss cancel each other in the expected log wealth. To view this slightly differently, if each horse wins $n/2$ times, W_n is given by

$$W_n = (\lambda(1))^{n/2} (1 - \lambda(1))^{n/2} (r(1))^{n/2} (r(2))^{n/2},$$

which again makes it clear that $\lambda^*(1)$ does not depend on $r(1)$ and $r(2)$.

h) Again consider the special case of two horses with $p(1) = p(2) = 1/2$, but let $r(1) = 3$ and $r(2) = 3/2$. Show that λ^* is nonunique with $(1/2, 1/2, 0)$ and $(1/4, 0, 3/4)$ as possible values. Show that if $r(2) > 3/2$, then the first solution above is unique, and if $r(2) < 3/2$, then the second solution is unique, assuming $p(1) = 1/2$ and $r(1) = 3$ throughout. Note: When $3/2 < r(2) < 2$, this is an example where an investment is used to maximize log-wealth even though $E[X(2)] = p(2)r(2) < 1$, *i.e.*, horse 2 is a lousy investment, but is preferable to cash in this case as a hedge against horse 1 losing.

Solution: Approach 1: Substitute $\lambda^* = (1/2, 1/2, 0)^T$ and then $(1/4, 0, 3/4)^T$ into the necessary and sufficient conditions; each satisfies those conditions. Approach 2: Note that $1/r(1) + 1/r(2) = 1$. Thus, from (f), each of these values is satisfied.

Both choices of λ^* here lead to the same rv, *i.e.*, $Y(\lambda^*) = \ln[3/2]$ for the event that horse 1 wins and $Y(\lambda^*) = \ln[3/4]$ for the event that horse 2 wins. In other words, the maximizing rv $Y(\lambda^*)$ is uniquely specified, even though λ^* is not unique. All points on the straight line between these two choices of λ^* , *i.e.*, $(1/2 - \alpha, 1/2 - 2\alpha, 3\alpha)^T$ for $0 \leq \alpha \leq 1/4$ also lead to the same optimizing $Y(\lambda^*)$.

For $r(2) > 3/2$, we have $1/r(1) + 1/r(2) < 1$, so from (d), the solution $(1/2, 1/2, 0)$ is valid and in this case unique. This can also be seen by substituting this choice of λ^* into the necessary and sufficient conditions, first with $r(2) > 3/2$ and then with $r(2) < 3/2$.

From (e), the choice $\lambda^* = (1/4, 0, 3/4)$ is the unique solution for $1/r(1) + 1/r(2) > 0$, *i.e.*, for $r(2) < 3/2$. This can be recognized as the allocation that maximizes $E[Y(\lambda)]$ for the triple-or-nothing investment.

i) For the case where $\sum_{j < \ell} 1/r(j) = 1$, define $q(j) = 1/r(j)$ as a PMF on $\{1, \dots, \ell - 1\}$. Show that $E[Y(\lambda^*)] = D(\mathbf{p} \parallel \mathbf{q})$ for the conditions of (f). Note: To interpret this, we could view a horse race where each horse j has probability $q(j)$ of winning the reward $r(j) = 1/q(j)$ as a ‘fair game’. Our gambler, who knows that the true probabilities are $\{p(j); 1 \leq j < \ell\}$, then has ‘inside information’ if $p(j) \neq q(j)$, and can establish a positive rate of return equal to $D(\mathbf{p} \parallel \mathbf{q})$.

Solution: From (f), $(p(1), \dots, p(\ell - 1), 0)^T$ is one choice for the optimizing λ^* . Using this choice,

$$E[Y(\lambda^*)] = \sum_{j < \ell} p(j) \ln[p(j)r(j)] = D(\mathbf{p} \parallel \mathbf{q}).$$

To further clarify the notion of a fair game, put on rose-colored glasses to envision a race track that simply accumulates the bets from all gamblers and distributes all income to the

bets on the winning horse. In this sense, $q(j) = 1/r(j)$ is the ‘odds’ on horse j as established by the aggregate of the gamblers. Fairness is not a word that is always used the same way, and here, rather than meaning anything about probabilities and expectations, it simply refers to the unrealistic assumption that neither the track nor the horse owners receive any expected return from the betting.

Exercise 9.37: (Proof of Theorem 9.10.2) Let $Z_n = \frac{1}{n}L_n(\boldsymbol{\lambda}) - \mathbf{E}[Y(\boldsymbol{\lambda})]$ and note that $\lim_{n \rightarrow \infty} Z_n = 0$ WP1. Recall that this means that the set of sample points ω for which $\lim Z_n(\omega) = 0$ has probability 1. Let $\epsilon > 0$ be arbitrary but fixed throughout the exercise and let $n_o \geq 1$ be an integer. Let $A(n_o, \epsilon) = \{\omega : |Z_n(\omega)| \leq \epsilon \text{ for all } n \geq n_o\}$.

a) Consider an ω for which $\lim_{n \rightarrow \infty} Z_n(\omega) = 0$ and explain why $\omega \in A(n_o, \epsilon)$ for some n_o .

Solution: By the definition of the limit (to 0) of a sequence of real numbers, *i.e.*, of $\lim Z_n(\omega)$, there must be an n_o large enough that $|Z_n(\omega)| \leq \epsilon$ for all $n \geq n_o$. Clearly, $\omega \in A(n_o, \epsilon)$ for that n_o .

b) Show that $\Pr\left\{\bigcup_{n_o=1}^{\infty} A(n_o, \epsilon)\right\} = 1$.

Solution: The set of ω for which $\lim_{n \rightarrow \infty} Z_n(\omega) = 0$ has probability 1, and, from (a), each such ω is in $A(n_o, \epsilon)$ for some n_o . Thus each such ω is in the above union, so that the union has probability 1.

c) Show that for any $\delta > 0$, there is an n_o large enough that $\Pr\{A(n_o, \epsilon)\} > 1 - \delta$. Hint: Use (1.9).

Solution: The sequence $\{A(n_o, \epsilon)\}$ is nested in n_o , *i.e.*, $A(n_o, \epsilon) \subseteq A(n_o+1, \epsilon)$ for all $n_o \geq 1$. Thus, from (1.9), $\lim_{n_o \rightarrow \infty} \Pr\{A(n_o, \epsilon)\} = 1$. Using the definition of a limit of real numbers (plus the fact that probabilities are at most equal to 1), it follows that for any $\delta > 0$, there is an n_o large enough that $\Pr\{A(n_o, \epsilon)\} > 1 - \delta$.

d) Show that for any $\epsilon > 0$ and $\delta > 0$, there is an n_o such that

$$\Pr\left\{\exp\left[n(\mathbf{E}[Y(\boldsymbol{\lambda})] - \epsilon)\right] \leq W_n(\boldsymbol{\lambda}) \leq \exp\left[n(\mathbf{E}[Y(\boldsymbol{\lambda})] + \epsilon)\right] \text{ for all } n \geq n_o\right\} > 1 - \delta.$$

Solution: (c) applies to any given ϵ , and (from the definition of $A(n_o, \epsilon)$) says that, for any $\epsilon, \delta > 0$, there is an n_o such that $\Pr\{|Z_n| \leq \epsilon \text{ for all } n \geq n_o\} > 1 - \delta$. Substituting the definition of Z_n into this,

$$\Pr\{n\mathbf{E}[Y(\boldsymbol{\lambda})] - \epsilon \leq L_n(\boldsymbol{\lambda}) \leq n\mathbf{E}[Y(\boldsymbol{\lambda})] \text{ for all } n \geq n_o\} > 1 - \delta.$$

Finally, by exponentiating the terms above, we get the desired expression.

Exercise 9.38: In this exercise, you are to express the buy-and-hold strategy and the two-pile strategy of (9.137) as special cases of a time-varying allocation.

a) Suppose the buy-and-hold strategy starts with an allocation λ_1 . Let $W_n(k)$ be the investors wealth in investment k at time n . Show that

$$W_n(k) = \lambda_1(k) \prod_{m=1}^n X_m(k).$$

Solution: There is hardly anything to be shown. $\lambda_1(k)$ is the original investment, and each $X_m(k)$ is the ratio of wealth in k at the beginning and end of epoch m , so the product gives the final wealth at epoch n .

b) Show that $\lambda_n(k)$ can be expressed in each of the following ways.

$$\lambda_n(k) = \frac{W_{n-1}(k)}{\sum_j W_{n-1}(j)} = \frac{\lambda_{n-1}(k)X_{n-1}(k)}{\sum_j \lambda_{n-1}(j)X_{n-1}(j)}. \quad (\text{A.211})$$

Solution: Note that $\lambda_n(k)$ is the fraction of total wealth in investment k at the beginning of epoch n . Thus, as shown in the first equality above, it is the wealth in k at the end of epoch $n-1$ divided by the total wealth at the end of epoch $n-1$. For the second equality, we start by rewriting the terms in the first equality as

$$\begin{aligned} \lambda_n(k) &= \frac{W_{n-2}(k)X_{n-1}(k)}{\sum_j W_{n-2}(j)X_{n-1}(j)} \\ &= \frac{\frac{W_{n-2}(k)}{\sum_i W_{n-2}(i)}X_{n-1}(k)}{\sum_j \frac{W_{n-2}(j)}{\sum_i W_{n-2}(i)}X_{n-1}(j)} \\ &= \frac{\lambda_{n-1}(k)X_{n-1}(k)}{\sum_j \lambda_{n-1}(j)X_{n-1}(j)}, \end{aligned}$$

where in the last equality we have used the first equality in (A.211), applied to $n-2$.

Note that there is no real reason for wanting to know $\lambda_n(k)$ for any given investment strategy, other than simply convincing oneself that $\lambda_n(k)$ (which depends on $X_m(i)$ for all $m < n$ and investments i) exists for any strategy and in fact specifies any given strategy.

c) For the two-pile strategy of (9.137), verify (9.137) and find an expression for $\lambda_n(k)$.

Solution: We omit the parenthesis for k since there is only one investment (other than cash). For a constant allocation strategy, we see from (9.126) that $E[W_n(\lambda)] = (1 + \lambda/2)^n$. The two-pile strategy invests $1/2$ the initial capital in a constant allocation with $\lambda = 1$ and $1/2$ with $\lambda = 1/4$. Thus the resulting expected capital is $E[W_n] = \frac{1}{2}(1 + \frac{1}{2})^n + \frac{1}{2}(1 + \frac{1}{8})^n$ as desired.

To find $E[L_n]$, note that for the n -tuple $X_m = 3$ for $1 \leq m \leq n$, we have $L_n = \ln[\frac{1}{2}3^n + \frac{1}{2}(3/2)^n]$. This n -tuple occurs with probability 2^{-n} and thus this value of L_n can be neglected in approximating $E[L_n]$. For all other n -tuples, W_n is $1/2$ its value with the constant allocation $\lambda = 1/4$, so L_n is reduced by $\ln(2)$ from the pure strategy value, giving rise to (9.137).

To find λ_n , we look first at λ_1 and λ_2 . Since each pile initially has $1/2$ (of the initial investment), and $1/4$ of the first pile is bet, and all of the second pile is bet, $\lambda_1 = \frac{1}{2}(1/4) + \frac{1}{2} = 5/8$.

Now note that λ_2 is a rv depending on X_1 . If $X_1 = 3$, then the piles contain $3/4$ and $3/2$ respectively. Since $1/4$ of the first pile is bet and all the second pile is bet, we have $3/16 + 3/2$ bet overall out of $3/4$ and $3/2$. Thus, for $X_1 = 3$, we have $\lambda_2 = 3/4$. For $X_1 = 0$, only the first pile is non-empty and $\lambda_2 = 1/4$.

In general, $\lambda_n = 1/4$ except where $X_m = 3$ for $1 \leq m < n$ and then

$$\lambda_n = \frac{1 + 2^{n+1}}{4 + 2^{n+1}}.$$

Exercise 9.39: a) Consider the martingale $\{Z_n; n \geq 1\}$ where $Z_n = W_n/W_n^*$ and W_n is the wealth at time n for the pure triple-or-nothing strategy of Example 9.10.1 and W_n^* is the wealth using the constant allocation λ^* . Find the PMF of Z where $Z = \lim_{n \rightarrow \infty} Z_n$ WP1.

Solution: The martingale convergence theorem applies, so Z actually exists as a rv. Actually, the power of the martingale convergence theorem is wasted here, since it is obvious that $\lim_{n \rightarrow \infty} Z_n(\omega) = 0$ for all sample sequences ω except that for which $X_n = 3$ for all n . Thus Z is the constant rv $Z = 0$ WP1.

b) Now consider the two-pile strategy of (9.137) and find the PMF of the limiting rv Z for this strategy.

Solution: The martingale convergence theorem again applies, and again is unnecessary since $Z_n(\omega) = 1/2$ for all ω other than those for which $X_m = 3$ for all $m \leq n$. Thus $Z = \lim_{n \rightarrow \infty} Z_n = 1/2$ WP1.

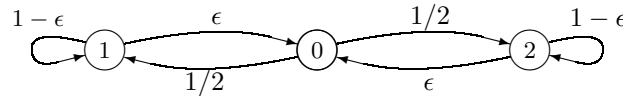
c) Now consider a ‘tempt-the-fates’ strategy where one uses the pure triple-or-nothing strategy for the first 3 epochs and uses the constant allocation λ^* thereafter. Again find the PMF of the limiting rv Z .

Solution: This is slightly less trivial since Z is now a non-trivial (*i.e.*, non-constant) rv. The martingale convergence theorem again applies, although the answer can be seen without using that theorem. If $X_1 = 3$, then $W_1 = 3$ and $W_1^* = 3/2$ so $Z_1 = 2$. If $X_1 = 0$, then $Z_n = 0$ for all $n \geq 1$.

If $X_1 = X_2 = 3$, then $W_2 = 9$ and $W_2^* = (3/2)^2$ so $Z_2 = 4$. If $X_2 = 0$, then $Z_n = 0$ for all $n \geq 2$. Similarly, if $X_m = 3$ for $1 \leq m \leq 3$, then $Z_n = 8$ and otherwise $Z_n = 0$ for all $n \geq m$.

Since the allocation λ^* is used for $m > 3$, we see that $Z_m = Z_3$ for $m \geq 3$. Thus $Z = \lim_{n \rightarrow \infty} Z_n = 8$ with probability $1/8$ (*i.e.*, if $X_1 = X_2 = X_3 = 3$) and $Z = 0$ with probability $7/8$.

Exercise 9.40 Consider the Markov modulated random walk in the figure below. The random variables Y_n in this example take on only a single value for each transition, that value being 1 for all transitions from state 1, 10 for all transitions from state 2, and 0 otherwise. $\epsilon > 0$ is a very small number, say $\epsilon < 10^{-6}$.



a) Show that the steady-state gain per transition is $5.5/(1 + \epsilon)$. Show that the relative-gain vector is

$$\mathbf{w} = \left(0, \quad \frac{\epsilon - 4.5}{\epsilon(1 + \epsilon)}, \quad \frac{10\epsilon + 4.5}{\epsilon(1 + \epsilon)} \right)^T. \quad (\text{A.212})$$

Solution: Observe that the Markov chain goes from state 0 to either 1 or 2 with equal probability, stays there for a very long, geometrically distributed time, then goes back to 0 for another equiprobable choice between 1 and 2. The chain is a birth-death chain, so its steady-state probabilities are easily calculated to be

$$\pi_0 = \frac{\epsilon}{1 + \epsilon}, \quad \pi_1 = \frac{1}{2(1 + \epsilon)}, \quad \pi_2 = \frac{1}{2(1 + \epsilon)}.$$

The steady-state gain per transition is then $g = 5.5/(1 + \epsilon)$. The relative gain equation is $\mathbf{w} + g\mathbf{e} = \bar{\mathbf{Y}} + [P]\mathbf{w}$. Solving this with $w_0 = 0$, we get (A.212)

b) Let $S_n = Y_0 + Y_1 + \dots + Y_{n-1}$ and take the starting state X_0 to be 0. Let J be the smallest value of n for which $S_n \geq 100$. Find $\Pr\{J = 11\}$ and $\Pr\{J = 101\}$. Find an estimate of $E[J]$ that is exact in the limit $\epsilon \rightarrow 0$.

Solution: If the first transition goes to state 2 and the state remains in state 2 for transitions 1 to 10, then $S_{11} = 100$. Since $Y_i \leq 10$ and $Y_0 = 0$, this is the smallest value of n for which $S_n \geq 100$ and $\Pr\{S_{11} = 100\} = (1/2)(1 - \epsilon^{10})$. As $\epsilon \rightarrow 0$, $\Pr\{S_{11} = 100\} = (1/2)$. If the first transition goes to state 1 and no transitions out of state 1 occur for the next 100 transitions, then the threshold is crossed at epoch 101 and $\Pr\{J = 101\} \approx (1/2)(1 - \epsilon)^{100}$, which is $1/2$ in the limit $\epsilon \rightarrow 0$. This is an approximation because there are other low probability ways of crossing the threshold at time 101. The most probable of these is to stay in state 1 for 98 transitions and then go to 0 and then 2. The probability of this is $(1/4)\epsilon(1 - \epsilon)^{98}$. In the limit $\epsilon \rightarrow 0$, $J = 11$ with probability $1/2$ and $J = 101$ with probability $1/2$. Thus $\lim_{\epsilon \rightarrow 0} E[J] = 11/2 + 101/2 = 56/2$.

c) Show that $\Pr\{X_J = 1\} = (1 - 45\epsilon + o(\epsilon))/2$ and that $\Pr\{X_J = 2\} = (1 + 45\epsilon + o(\epsilon))/2$. Verify, to first order in ϵ that (9.159) is satisfied.

Solution: We can consider only sample paths that include at most one ϵ transition since all other paths are $o(\epsilon)$. In order to stop in state 2 (*i.e.*, satisfy $\{X_J = 2\}$), there is one sample path of probability $(1/2)(1 - \epsilon)^{10} = 1/2 - 5\epsilon + o(\epsilon)$, and we see that $5\epsilon + o(\epsilon)$ is the probability of going to state 2 initially and then returning to 0 before stopping. Half of these sample segments then go from 0 to 2, a set of paths of probability $5\epsilon/2 + o(\epsilon)$. In the same way, there is a set of paths of probability $50\epsilon/2 + o(\epsilon)$ that go from 0 to 1 initially, return to 0 before stopping, and then go to state 2. Adding these terms together,

$$\Pr\{X_J = 2\} = 1/2 - 5\epsilon + 5\epsilon/2 + 50\epsilon/2 + o(\epsilon) = 1/2 + 45\epsilon/2 + o(\epsilon).$$

Then $\Pr\{X_J = 1\} = 1/2 - 45\epsilon/2 + o(\epsilon)$.

Exercise 9.42: Let $\{Z_n; n \geq 1\}$ be a martingale, and for some integer m , let $Y_n = Z_{n+m} - Z_m$.

a) Show that $E[Y_n | Z_{n+m-1} = z_{n+m-1}, Z_{n+m-2} = z_{n+m-2}, \dots, Z_m = z_m, \dots, Z_1 = z_1] = z_{n+m-1} - z_m$.

Solution: This is more straightforward if the desired result is written in the more abbreviated form

$$E[Y_n | Z_{n+m-1}, Z_{n+m-2}, \dots, Z_m, \dots, Z_1] = Z_{n+m-1} - Z_m.$$

Since $Y_n = Z_{n+m} - Z_m$, the left side above is

$$E[Z_{n+m} - Z_m | Z_{n+m-1}, \dots, Z_1] = Z_{n+m-1} - E[Z_m | Z_{n+m-1}, \dots, Z_m, \dots, Z_1].$$

Given sample values for each conditioning rv on the right of the above expression, and particularly given that $Z_m = z_m$, the expected value of Z_m is simply the conditioning value z_m for Z_m . This is one of those strange things that are completely obvious, and yet somehow obscure. We then have $E[Y_n | Z_{n+m-1}, \dots, Z_1] = Z_{n+m-1} - Z_m$.

b) Show that $E[Y_n | Y_{n-1} = y_{n-1}, \dots, Y_1 = y_1] = y_{n-1}$.

Solution: In abbreviated form, we want to show that $E[Y_n | Y_{n-1}, \dots, Y_1] = Y_{n-1}$. We showed in (a) that $E[Y_n | Z_{n+m-1}, \dots, Z_1] = Y_{n-1}$. For each sample point ω , $Y_{n-1}(\omega), \dots, Y_1(\omega)$ is a function of $Z_{n+m-1}(\omega), \dots, Z_1(\omega)$. Thus, the rv $E[Y_n | Z_{n+m-1}, \dots, Z_1]$ specifies the rv $E[Y_n | Y_{n-1}, \dots, Y_1]$, which then must be Y_{n-1} .

c) Show that $E[|Y_n|] < \infty$. Note that b) and c) show that $\{Y_n; n \geq 1\}$ is a martingale.

Solution: Since $Y_n = Z_{n+m} - Z_m$, we have $|Y_n| \leq |Z_{n+m}| + |Z_m|$. Since $\{Z_n; n \geq 1\}$ is a martingale, $E[|Z_n|] < \infty$ for each n so

$$E[|Y_n|] \leq E[|Z_{n+m}|] + E[|Z_m|] < \infty.$$

Exercise 9.43: This exercise views the continuous-time branching process of Exercise 7.15 as a stopped random walk. Recall that the process was specified there as a Markov process such that for each state j , $j \geq 0$, the transition rate to $j+1$ is $j\lambda$ and to $j-1$ is $j\mu$. There are no other transitions, and in particular, there are no transitions out of state 0, so that the Markov process is reducible. Recall that the embedded Markov chain is the same as the embedded chain of an M/M/1 queue except that there is no transition from state 0 to state 1.

a) To model the possible extinction of the population, convert the embedded Markov chain above to a stopped random walk, $\{S_n; n \geq 0\}$. The stopped random walk starts at $S_0 = 0$ and stops on reaching a threshold at -1 . Before stopping, it moves up by one with probability $\frac{\lambda}{\lambda+\mu}$ and downward by 1 with probability $\frac{\mu}{\lambda+\mu}$ at each step. Give the (very simple) relationship between the state X_n of the Markov chain and the state S_n of the stopped random walk for each $n \geq 0$.

Solution: Consider a random walk where $S_n = Y_1 + \dots + Y_n$ and the sequence $\{Y_n; n \geq 0\}$ is IID with the PMF

$$p_{Y_n}(1) = \frac{\lambda}{\lambda + \mu} \quad \text{and} \quad p_{Y_n}(-1) = \frac{\mu}{\lambda + \mu}.$$

We note that X_n in the Markov chain is the sum of 1 (for the initial organism) and ± 1 for each subsequent birth or death transition. Since each of those birth and death transitions in the Markov chain, is an IID random variable so long as the number of organisms is positive, we see that $S_n = X_n - 1$ so long as the number of organisms is positive. When the number of organisms goes to 0, the number of deaths exceeds the number of births by 1, so $S_n = -1$. The stopped process, stopping when $S_n = -1$, is then identical to the Markov chain, which stays in state 0 after it is once entered.

b) Find the probability that the population eventually dies out as a function of λ and μ . Be sure to consider all three cases $\lambda > \mu$, $\lambda < \mu$, and $\lambda = \mu$.

Solution: The probability P_d that the population dies out is the probability that the random walk in (a) crosses a threshold at -1 . If $\lambda < \mu$, the random walk has a negative drift (*i.e.*, the rv's Y_n of (a) have a negative mean), so the walk crosses the threshold eventually with probability 1 and $P_d = 1$.

For $\lambda > \mu$, the random walk has a positive drift and the problem is identical to that of the negative of the random walk (where the step $-Y$ has a negative drift) crossing a positive threshold at $+1$. The probability of threshold crossing, (using the Wald identity

and Corollary 9.4.4) is upper bounded by $P_d \leq e^{-r^*}$ where r^* is the r where $\gamma_{(-Y)}(r^*) = 0$. Solving for r^* , we get $r^* = \ln(\mu/\lambda)$, so $P_d \leq \mu/\lambda$. Since the random walk changes by ± 1 at each step, and the threshold is an integer, this inequality is satisfied with equality, so $P_d = \mu/\lambda$. This same analysis works for $\lambda = \mu$, so $P_d = 1$ in that case.

An alternative approach is to note that this negative random walk with a threshold at $+1$ is the same as Example 5.5.4 (Stop when you're ahead in coin tossing).

A.10 Solutions for Chapter 10

Exercise 10.1: a) Consider the joint probability density $f_{X,Z}(x, z) = e^{-z}$ for $0 \leq x \leq z$ and $f_{X,Z}(x, z) = 0$ otherwise. Find the pair x, z of values that maximize this density. Find the marginal density $f_Z(z)$ and find the value of z that maximizes this.

Solution: e^{-z} has value 1 when $x = z = 0$, and the joint density is smaller whenever $z > 0$, and is zero when $z < 0$, so $f_{X,Z}(x, z)$ is maximized by $x = z = 0$. The marginal density is found by integrating $f_{X,Z}(x, z) = e^{-z}$ over x in the range 0 to z , so $f_Z(z) = ze^{-z}$ for $z \geq 0$. This is maximized at $z = 1$.

b) Let $f_{X,Z,Y}(x, z, y)$ be $y^2 e^{-yz}$ for $0 \leq x \leq z$, $1 \leq y \leq 2$ and be 0 otherwise. Conditional on an observation $Y = y$, find the joint MAP estimate of X, Z . Find $f_{Z|Y}(z|y)$, the marginal density of Z conditional on $Y = y$, and find the MAP estimate of Z conditional on $Y = y$.

Solution: The joint MAP estimate is the value of x, z in the range $0 \leq x \leq z$, that maximizes $f_{X,Z|Y}(x, z|y) = f_{X,Z,Y}(x, z, y)/f_Y(y) = (y^2 e^{-yz})/f_Y(y)$. For any given y , $0 < y \leq 1$, this is maximized, as in (a), for $x = z = 0$. Next, integrating $f_{X,Z,Y}(x, z, y)$ over x from 0 to z , we get $f_{Z,Y}(z, y) = y^2 z e^{-yz}$. Integrating over z , we get $f_Y(y) = 1$ for $1 \leq y \leq 2$. Thus

$$f_{Z|Y}(z|y) = f_{Z,Y}(z, y) = y^2 z e^{-yz}.$$

This is maximized by $z = 1/y$, which is thus the MAP estimate for Z alone.

This shows that MAP estimation on joint rv's does not necessarily agree with the MAP estimates of the individual rv's. This illustrates that MAP estimates do not necessarily have the properties one might desire for estimation. More specifically, estimation, as distinguished from hypothesis testing, usually involves approximation of the desired quantity, and MAP makes no allowance for closeness in the approximation.

Exercise 10.2: Let $Y = X + Z$ where X and Z are IID and $\mathcal{N}(0, 1)$. Let $U = Z - X$.

a) Explain (using the results of Chapter 3) why Y and U are jointly Gaussian and why they are statistically independent.

Solution: Since Y and U are both linear combinations of X, Z , they are jointly Gaussian by definition 3.3.1. Since $E[YU] = 0$, i.e., Y and U are uncorrelated, and since they are jointly Gaussian, they are independent.

b) Without using any matrices, write down the joint probability density of Y and U . Verify your answer from (3.22).

Solution: Since Y and U are independent and are each $\mathcal{N}(0, \sqrt{2})$, the joint density is

$$f_{YU}(y, u) = \frac{1}{4\pi} \exp \left[\frac{-y^2 - u^2}{4} \right].$$

Since the covariance matrix of Y and U is $[K] = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, this is the same as (3.22).

c) Find the MMSE estimate $\hat{x}(y)$ of X conditional on a given sample value y of Y . You can derive this from first principles, or use (10.9) or Example 10.2.2.

Solution: From first principles, $\hat{x}(y) = \mathbb{E}[X|Y = y]$. To find this, we first find $f_{X|Y}(x|y)$.

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} = \frac{1}{\sqrt{\pi}} \exp\left(-\frac{(y-x)^2}{2} - \frac{x^2}{2} + \frac{y^2}{4}\right) \\ &= \frac{1}{\sqrt{\pi}} \exp(-(x - y/2)^2). \end{aligned}$$

Thus, given $Y = y$, $X \sim \mathcal{N}(y/2, 1/\sqrt{2})$. It follows that $\hat{x}(y) = \mathbb{E}[X|Y = y] = y/2$.

d) Show that the estimation error $\xi = X - \hat{X}(Y)$ is equal to $-U/2$.

Solution: From (c), $\hat{X}(Y) = Y/2$ so

$$\xi = X - \hat{X}(Y) = X - (X + Z)/2 = (X - Z)/2 = -U/2.$$

e) Write down the probability density of U conditional on $Y = y$ and that of ξ conditional on $Y = y$.

Solution: Since U and Y are statistically independent,

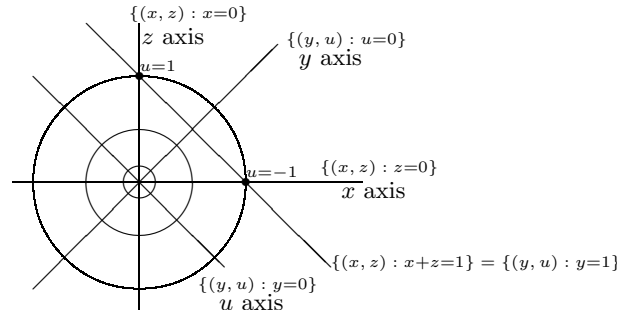
$$f_{U|Y}(u|y) = f_U(u) = \frac{1}{\sqrt{4\pi}} \exp(-u^2/4).$$

Since $\xi = -U/2$ (or since ξ , conditional on $Y = y$, is the fluctuation of X , conditional on $Y = y$),

$$f_{\xi|Y}(\xi|y) = f_{\xi}(\xi) = \frac{1}{\sqrt{\pi}} \exp(-\xi^2).$$

f) Draw a sketch, in the x, z plane of the equiprobability contours of X and Z . Explain why these are also equiprobability contours for Y and U . For some given sample value of Y , say $Y = 1$, illustrate the set of points for which $x + z = 1$. For a given point on this line, illustrate the sample value of U .

Solution: The circles below are equiprobability contours of X, Z . Since $y^2 + u^2 = 2(x^2 + z^2)$, they are also equiprobable contours of Y, U .



The point of the problem, in associating the estimation error with $-u/2$, is to give a graphical explanation of why the estimation error is independent of the estimate. Y and U are independent since the y axis and the u axis are at 45° rotations from the x and y axes, and thus 90° from each other.

Exercise 10.3: a) Let $X, Z_1, Z_2, \dots, Z_n$ be independent zero-mean Gaussian rv's with variances $\sigma_X^2, \sigma_{Z_1}^2, \dots, \sigma_{Z_n}^2$ respectively. Let $Y_j = h_j X + Z_j$ for $j \geq 1$ and let $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$. Use (10.9) to show that the MMSE estimate of X conditional on $\mathbf{Y} = \mathbf{y} = (y_1, \dots, y_n)^\top$, is given by

$$\hat{x}(\mathbf{y}) = \sum_{j=1}^n g_j y_j; \quad \text{where} \quad g_j = \frac{h_j / \sigma_{Z_j}^2}{(1/\sigma_X^2) + \sum_{i=1}^n h_i^2 / \sigma_{Z_i}^2}. \quad (\text{A.213})$$

Hint: Let the row vector $\mathbf{g}^\top$ be $[K_{X \cdot \mathbf{Y}}][K_{\mathbf{Y}}^{-1}]$ and multiply $\mathbf{g}^\top$ by $K_{\mathbf{Y}}$ to solve for $\mathbf{g}^\top$.

Solution: From (10.9), we see that $\hat{x}(\mathbf{y}) = \mathbf{g}^\top \mathbf{y}$ where $\mathbf{g}^\top = [K_{X \cdot \mathbf{Y}}][K_{\mathbf{Y}}^{-1}]$. Since $\mathbf{Y} = \mathbf{h}X + \mathbf{Z}$, we see that $[K_{\mathbf{Y}}] = \mathbf{h}\sigma_X^2\mathbf{h}^\top + [K_{\mathbf{Z}}]$ and $[K_{X \cdot \mathbf{Y}}] = \sigma_X^2\mathbf{h}^\top$. We can avoid inverting $[K_{\mathbf{Y}}]$ by taking the hint to solve the vector equation $\mathbf{g}^\top[K_{\mathbf{Y}}] = [K_{X \cdot \mathbf{Y}}]$, *i.e.*,

$$\mathbf{g}^\top \mathbf{h} \sigma_X^2 \mathbf{h}^\top + \mathbf{g}^\top [K_{\mathbf{Z}}] = \sigma_X^2 \mathbf{h}^\top.$$

Since $\mathbf{g}^\top \mathbf{h}$ is a scalar, we can rewrite this as $(1 - \mathbf{g}^\top \mathbf{h}) \sigma_X^2 \mathbf{h}^\top = \mathbf{g}^\top [K_{\mathbf{Z}}]$. The j th component of this equation is

$$g_j = \frac{(1 - \mathbf{g}^\top \mathbf{h}) \sigma_X^2 h_j}{\sigma_{Z_j}^2}. \quad (\text{A.214})$$

This shows that the weighting factors g_j in $\hat{x}(\mathbf{y})$ depend on j only through h_j/σ_{Z_j} (if we visualize scaling Y_j by scaling both h_j and Z_j , we see that the estimate should involve h_j and σ_j only in the ratio h_j/σ_j). We still must determine the unknown constant $1 - \mathbf{g}^\top \mathbf{h}$. To do this, multiply (A.214) by h_j and sum over j , getting

$$\mathbf{g}^\top \mathbf{h} = (1 - \mathbf{g}^\top \mathbf{h}) \sum_j \frac{\sigma_X^2 h_j^2}{\sigma_{Z_j}^2}.$$

Solving for $\mathbf{g}^\top \mathbf{h}$, from this,

$$\mathbf{g}^\top \mathbf{h} = \frac{\sum_j \sigma_X^2 h_j^2 / \sigma_{Z_j}^2}{1 + \sum_j \sigma_X^2 h_j^2 / \sigma_{Z_j}^2}; \quad 1 - \mathbf{g}^\top \mathbf{h} = \frac{1}{1 + \sum_j \sigma_X^2 h_j^2 / \sigma_{Z_j}^2}. \quad (\text{A.215})$$

Substituting the expression for $1 - \mathbf{g}^\top \mathbf{h}$ into (A.214) yields (A.213).

b) Let $\xi = X - \hat{X}(\mathbf{Y})$ and show that (10.29) is valid, *i.e.*, that

$$1/\sigma_\xi^2 = 1/\sigma_X^2 + \sum_{i=1}^n \frac{h_i^2}{\sigma_{Z_i}^2}.$$

Solution: Using (10.6) in one dimension, $\sigma_\xi^2 = \mathbf{E}[\xi X] = \sigma_X^2 - \mathbf{E}[\hat{X}(\mathbf{Y})X]$. Since $\hat{X}(\mathbf{Y}) = \sum_j g_j Y_j$ from (A.213), and since $\mathbf{E}[Y_i X] = \sigma_X^2 h_i$, we have

$$\begin{aligned} \sigma_\xi^2 &= \sigma_X^2 - \sum_{i=1}^n g_i \mathbf{E}[Y_i X] = \sigma_X^2 \left(1 - \sum_{i=1}^n g_i h_i \right) \\ &= \sigma_X^2 (1 - \mathbf{g}^\top \mathbf{h}) = \frac{\sigma_X^2}{1 + \sum_j \sigma_X^2 h_j^2 / \sigma_{Z_j}^2} = \frac{1}{1/\sigma_X^2 + \sum_j h_j^2 / \sigma_{Z_j}^2}, \end{aligned}$$

where we have used (A.215). This is equivalent to (10.29).

c) Show that (10.28), i.e., $\hat{x}(\mathbf{y}) = \sigma_\xi^2 \sum_{j=1}^n h_j y_j / \sigma_{Z_j}^2$, is valid.

Solution: From (b), the denominator of the expression for g_j in (A.213) is equal to $1/\sigma_\xi^2$. Thus $g_j = \sigma_\xi^2 h_j / \sigma_{Z_j}^2$ and $\hat{x}(\mathbf{y}) = \sum_j g_j y_j$ is the desired expression.

d) Show that the expression in (10.29) is equivalent to the iterative expression in (10.25).

Solution: First, we show that (10.29) implies (10.27). We use ξ_n to refer to the error for n observations and ξ_{n-1} for the error using the first $n-1$ of those observations. Using (10.29),

$$\begin{aligned} \frac{1}{\sigma_{\xi_n}^2} &= \frac{1}{\sigma_X^2} + \sum_{i=1}^n \frac{h_i^2}{\sigma_{Z_i}^2} = \frac{1}{\sigma_X^2} + \sum_{i=1}^{n-1} \frac{h_i^2}{\sigma_{Z_i}^2} + \frac{h_n^2}{\sigma_{Z_n}^2} \\ &= \frac{1}{\sigma_{\xi_{n-1}}^2} + \frac{h_n^2}{\sigma_{Z_n}^2}, \end{aligned} \quad (\text{A.216})$$

which is (10.27). This holds for all $n > 1$, so (10.27) for all n also implies (10.29).

e) Show that the expression in (10.28) is equivalent to the iterative expression in (10.25).

Solution: Breaking (10.28) into the first $n-1$ terms followed by the term for n , we get

$$\hat{x}(y_1^n) = \sigma_{\xi_n}^2 \sum_{j=1}^{n-1} \frac{h_j y_j}{\sigma_{Z_j}^2} + \sigma_{\xi_n}^2 \frac{h_n y_n}{\sigma_{Z_n}^2} = \frac{\sigma_{\xi_n}^2}{\sigma_{\xi_{n-1}}^2} \hat{x}(y_1^{n-1}) + \sigma_{\xi_n}^2 \frac{h_n y_n}{\sigma_{Z_n}^2}, \quad (\text{A.217})$$

where we used (10.28) for $\hat{x}(y_1^{n-1})$. We can solve for $\sigma_{\xi_n}^2 / \sigma_{\xi_{n-1}}^2$ by multiplying (A.216) by $\sigma_{\xi_n}^2$, getting

$$\frac{\sigma_{\xi_n}^2}{\sigma_{\xi_{n-1}}^2} = 1 - \frac{\sigma_{\xi_n}^2 h_n^2}{\sigma_{Z_n}^2}.$$

Substituting this into (A.217) yields

$$\hat{x}(y_1^n) = \hat{x}(y_1^{n-1}) + \sigma_{\xi_n}^2 \frac{h_n y_n - h_n^2 \hat{x}(y_1^{n-1})}{\sigma_{Z_n}^2}.$$

Finally, if we invert (A.216), we get

$$\sigma_{\xi_n}^2 = \frac{\sigma_{\xi_{n-1}}^2 \sigma_{Z_n}^2}{h_n^2 \sigma_{\xi_{n-1}}^2 + \sigma_{Z_n}^2}.$$

Substituting this into (A.217), we get (10.27).

Exercise 10.4: Let $X, Z_1, \dots, Z_n$ be independent, zero-mean, and Gaussian, satisfying $Y_i = h_i X + Z_i$ for $1 \leq i \leq n$. The point of this exercise is to point out that estimation of X from $\mathbf{Y}_1^n$ is virtually no more general here than the same problem with $h_i = 1$ for $1 \leq i \leq n$.

a) Write out the MMSE estimate and variance of the estimation error from (10.28) and (10.29) where each $h_i = 1$.

Solution:

$$\hat{x}(\mathbf{y}_1^n) = \sigma_{\xi_n}^2 \sum_{j=1}^n \frac{y_j}{\sigma_{Z_j}^2}; \quad \frac{1}{\sigma_{\xi_n}^2} = \frac{1}{\sigma_X^2} + \sum_{i=1}^n \frac{1}{\sigma_{Z_i}^2}.$$

b) For arbitrary numbers $h_1, \dots, h_n$, consider the equations $h_i Y_i = h_i X + h_i Z_i$. Let $U_i = h_i Y_i$. Write out the MMSE estimate of X as a function of $U_1, \dots, U_n$, still using the rv's $Z_1, \dots, Z_n$. Hint: Note that if you scale each observation U_i to U_i/h_i you have the same estimation problem as in (a).

Solution: Assume that $h_i \neq 0$ for $1 \leq i \leq n$. Letting $U_i = h_i Y_i$, the modified equations become $U_i = h_i X + h_i Z_i$ for $1 \leq i \leq n$. Scaling these equations, $U_i/h_i = X + Z_i$ for $1 \leq i \leq n$. The MMSE estimate of X from $U_1, \dots, U_n$ must be the same as the MMSE estimate from $U_1/h_1, \dots, U_n/h_n$, since either set of rv's specifies the other set and the MMSE estimate is a minimum over all types of processing of the observations. Since $U_i/h_i = X + Z_i$, the MMSE estimation problem is that solved in (a), with the observations now equal to u_i/h_i . Thus

$$\hat{x}(u_1, \dots, u_n) = \sigma_{\xi_n}^2 \sum_{j=1}^n \frac{u_j}{h_j \sigma_{Z_j}^2}; \quad \frac{1}{\sigma_{\xi_n}^2} = \frac{1}{\sigma_X^2} + \sum_{i=1}^n \frac{1}{\sigma_{Z_i}^2}. \quad (\text{A.218})$$

If the assumption that all $h_i \neq 0$ is not made, then for any i such that $h_i = 0$, there is no observation for that i and the equation above must be changed by omitting all such i .

c) Now let $W_i = h_i Z_i$. Write out the same MMSE estimate in terms of both $U_i, \dots, U_n$ and $W_1, \dots, W_n$ and show that you now have (10.28) with all the h_i 's back in the equation. Hint: you still have the basic equation in (a), but the variance of the Z_i 's must be expressed in terms of the W_i 's. Write out the error variance from (10.29) in the same way.

Solution: The equations in terms of $\{W_i; 1 \leq i \leq n\}$ are $U_i = h_i X + W_i$ and this set of variables is still jointly Gaussian with independence between X and $W_1, \dots, W_n$. The MMSE estimate given in (A.218) is still valid, since we are merely denoting Z_i as W_i/h_i . Noting that $\sigma_{W_i}^2 = h_i^2 \sigma_{Z_i}^2$, we have

$$\hat{x}(u_1, \dots, u_n) = \sigma_{\xi_n}^2 \sum_{j=1}^n \frac{h_j u_j}{\sigma_{W_j}^2}; \quad \frac{1}{\sigma_{\xi_n}^2} = \frac{1}{\sigma_X^2} + \sum_{i=1}^n \frac{h_i^2}{\sigma_{W_i}^2}.$$

This gives the MMSE estimate and error variance for the observations $U_i = h_i X + W_i$, and the expression is the same as the general equations in (10.28) and (10.29). In other words, the more general problem is the restricted problem with appropriate scaling.

Exercise 10.5: a) Assume that $X_1 \sim \mathcal{N}(\bar{X}_1, \sigma_{X_1}^2)$ and that for each $n \geq 1$, $X_{n+1} = \alpha X_n + W_n$ where $0 < \alpha < 1$, $W_n \sim \mathcal{N}(0, \sigma_W^2)$, and $X_1, W_1, W_2, \dots$, are independent. Show that for each $n \geq 1$,

$$\mathbb{E}[X_n] = \alpha^{n-1} \bar{X}_1; \quad \sigma_{X_n}^2 = \frac{(1 - \alpha^{2(n-1)})\sigma_W^2}{1 - \alpha^2} + \alpha^{2(n-1)}\sigma_{X_1}^2.$$

Solution: For each $n > 1$, $\mathbb{E}[X_n] = \alpha \mathbb{E}[X_{n-1}]$, so by iteration, $\mathbb{E}[X_n] = \alpha^{n-1} \bar{X}_1$. Simi-

larly,

$$\begin{aligned}
 \sigma_{X_n}^2 &= \alpha^2 \sigma_{X_{n-1}}^2 + \sigma_W^2 = \alpha^2 [\alpha^2 \sigma_{X_{n-2}}^2 + \sigma_W^2] + \sigma_W^2 \\
 &= \alpha^4 \sigma_{X_{n-2}}^2 + (1 + \alpha^2) \sigma_W^2 = \dots \\
 &= \alpha^{2(n-1)} \sigma_{X_1}^2 + (1 + \alpha^2 + \alpha^4 + \dots + \alpha^{2(n-2)}) \sigma_W^2 \\
 &= \frac{(1 - \alpha^{2(n-1)}) \sigma_W^2}{1 - \alpha^2} + \alpha^{2(n-1)} \sigma_{X_1}^2.
 \end{aligned} \tag{A.219}$$

b) Show directly, by comparing the equation $\sigma_{X_n}^2 = \alpha^2 \sigma_{X_{n-1}}^2 + \sigma_W^2$ for each two adjoining values of n that $\sigma_{X_n}^2$ moves monotonically from $\sigma_{X_1}^2$ to $\sigma_W^2/(1 - \alpha^2)$ as $n \rightarrow \infty$.

Solution: View

$$\sigma_{X_n}^2 = \alpha^2 \sigma_{X_{n-1}}^2 + \sigma_W^2 \tag{A.220}$$

as a function of $\sigma_{X_{n-1}}^2$ for fixed α^2 and σ_W^2 . Since $\alpha^2 > 0$, this is monotonic increasing in $\sigma_{X_{n-1}}^2$. Now consider two cases, first where $\sigma_{X_2}^2 > \sigma_{X_1}^2$ and second where $\sigma_{X_2}^2 < \sigma_{X_1}^2$.

Case 1: Consider (A.220) for $n = 1$ and $n = 2$. Since $\sigma_{X_2}^2 > \sigma_{X_1}^2$ and since (A.220) is increasing in $\sigma_{X_{n-1}}^2$, we see that $\sigma_{X_3}^2 > \sigma_{X_2}^2$. Iterating this argument, $\sigma_{X_n}^2 > \sigma_{X_{n-1}}^2$ for all $n > 1$, i.e., $\sigma_{X_n}^2$ is monotonic increasing in n . Next, substituting $\sigma_{X_n}^2 > \sigma_{X_{n-1}}^2$ into (A.220), we get $\sigma_{X_n}^2 < \alpha^2 \sigma_{X_n}^2 + \sigma_W^2$ for all $n \geq 1$, so that $\sigma_{X_n}^2 < \sigma_W^2/(1 - \alpha^2)$ for all $n \geq 1$. Since $\{\sigma_{X_n}^2; n > 1\}$ is a bounded increasing sequence, it has a limit. The limit satisfies $\lim_{n \rightarrow \infty} (\sigma_{X_n}^2) = \alpha^2 \lim_{n \rightarrow \infty} (\sigma_{X_n}^2) + \sigma_W^2$. Thus $\lim_{n \rightarrow \infty} \sigma_{X_n}^2 = \sigma_W^2/(1 - \alpha^2)$.

Case 2: This follows the same argument as case 1, except here $\sigma_{X_n}^2$ is decreasing in n from an initial value above $\sigma_W^2/(1 - \alpha^2)$. Finally, for $\sigma_{X_1}^2 = \sigma_{X_2}^2$, we see that $\sigma_{X_n}^2 = \sigma_W^2/(1 - \alpha^2)$ for all $n \geq 1$.

c) Assume that sample values of $Y_1, Y_2, \dots$, are observed, where $Y_n = hX_n + Z_n$ and where $Z_1, Z_2, \dots$, are IID zero-mean Gaussian rv's with variance σ_Z^2 . Assume that $Z_1, \dots, W_1, \dots, X_1$ are independent and assume $h \geq 0$. Rewrite the recursion for the variance of the estimation error in (10.41) for this special case. Show that if $h/\sigma_Z = 0$, then $\sigma_{\xi_n}^2 = \sigma_{X_n}^2$ for each $n \geq 1$. Hint: Compare the recursion in (b) to that for $\sigma_{\xi_n}^2$.

Solution: Rewriting (10.41) for this special case,

$$\frac{1}{\sigma_{\xi_n}^2} = \frac{1}{\alpha^2 \sigma_{\xi_{n-1}}^2 + \sigma_W^2} + \frac{h^2}{\sigma_Z^2}. \tag{A.221}$$

If $h/\sigma_Z = 0$, this simplifies to

$$\sigma_{\xi_n}^2 = \alpha^2 \sigma_{\xi_{n-1}}^2 + \sigma_W^2. \tag{A.222}$$

This is the same recursion as (A.219) with $\sigma_{X_n}^2$ replaced by $\sigma_{\xi_n}^2$. Now for $n = 1$, $\sigma_{\xi_1}^2$ is the variance of the error in the MMSE estimate of X_1 with no measurement, i.e., $\sigma_{X_1}^2 = \sigma_{\xi_1}^2$ (this is also clear from (10.33)). Thus from the recursion in (A.222), $\sigma_{\xi_n}^2 = \sigma_{X_n}^2$ for all $n \geq 1$.

d) Show from the recursion that $\sigma_{\xi_n}^2$ is a decreasing function of h/σ_Z for each $n \geq 2$. Use this to show that $\sigma_{\xi_n}^2 \leq \sigma_{X_n}^2$ for each n . Explain (without equations) why this result must be true.

Solution: From (10.33), $\sigma_{\xi_1}^2$ is decreasing in h . We use this as the basis for induction on (A.221), *i.e.*, as h/σ_Z increases, $\sigma_{\xi_{n-1}}^2$ decreases by the inductive hypothesis, and thus from (A.221), $\sigma_{\xi_n}^2$ also decreases. Since $\sigma_{\xi_n}^2 = \sigma_{X_n}^2$ for $h = 0$, we must then have $\sigma_{\xi_n}^2 \leq \sigma_{X_n}^2$ for $h > 0$. This can also be seen because one possible estimate for X_n is $E[X_n]$, *i.e.*, the mean of X_n in the absence of measurements. The error variance is then $\sigma_{X_n}^2$, which must be greater than or equal to the variance with a MMSE estimate using the measurements.

e) Show that the sequence $\{\sigma_{\xi_n}^2; n \geq 1\}$ is monotonic in n . Hint: Use the same technique as in (b). From this and (d), show that $\lambda = \lim_{n \rightarrow \infty} \sigma_{\xi_n}^2$ exists. Show that the limit satisfies (10.42) (note that (10.42) must have 2 roots, one positive and one negative, so the limit must be the positive root).

Solution: View (A.221) as a monotonic increasing function of $\sigma_{\xi_{n-1}}^2$, viewing all other quantities as constants. Thus if $\sigma_{\xi_{n-1}}^2$ is replaced by $\sigma_{\xi_n}^2$ and $\sigma_{\xi_{n-1}}^2 < \sigma_{\xi_n}^2$, then $\sigma_{\xi_n}^2 < \sigma_{\xi_{n+1}}^2$. By induction, then, $\sigma_{\xi_n}^2$ is increasing in n . Similarly, if $\sigma_{\xi_{n-1}}^2 > \sigma_{\xi_n}^2$, then $\sigma_{\xi_n}^2$ is decreasing in n , so either way $\sigma_{\xi_n}^2$ is monotonic in n . From (b), $\sigma_{X_n}^2$ is also bounded independent of n , so $\lambda = \lim_{n \rightarrow \infty} \sigma_{\xi_n}^2$ must exist. Eq. (10.42) is simply a rearrangement of (A.221) replacing $\sigma_{\xi_n}^2$ and $\sigma_{\xi_{n-1}}^2$ by λ , yielding

$$\alpha^2 h^2 \sigma_Z^{-2} \lambda^2 + [h^2 \sigma_Z^{-2}] \sigma_W^2 + 1 - \alpha^2 \lambda - \sigma_W^2 = 0. \quad (\text{A.223})$$

This must be satisfied by $\lambda = \lim_{n \rightarrow \infty} \sigma_{\xi_n}^2$. Since each term within the brackets above is positive, the coefficient of λ is positive. Since the constant term is negative, this quadratic equation must have two solutions, one positive and one negative. The limiting error variance is, of course, the positive term, and thus can be found uniquely simply by solving this equation.

f) Show that for each $n \geq 1$, $\sigma_{\xi_n}^2$ is increasing in σ_W^2 and increasing in α . Note: This increase with α is surprising, since when α is close to one, X_n changes slowly, so we would expect to be able to track X_n well. The problem is that $\lim_n \sigma_{X_n}^2 = \sigma_W^2/(1 - \alpha^2)$ so the variance of the untracked X_n is increasing without limit as α approaches 1. (g) is somewhat messy, but resolves this issue.

Solution: The argument that $\sigma_{\xi_n}^2$ is increasing in α and increasing in σ_W^2 is the same as the argument in (d).

g) Show that if the recursion is expressed in terms of $\beta = \sigma_W^2/(1 - \alpha^2)$ and α , then λ is decreasing in α for constant β .

Solution: If we make the suggested substitution in (A.223) and rearrange it, we get

$$\frac{\alpha^2 \lambda^2}{1 - \alpha^2} + \left(\beta + \frac{\sigma_Z^2}{h^2} \right) \lambda - \frac{\beta \sigma_Z^2}{h^2} = 0.$$

Since the quadratic term is increasing in α over the range $0 < \alpha < 1$, the positive root is decreasing.

Exercise 10.6: a) Assume that $\mathbf{X}$ and $\mathbf{Y}$ are zero-mean, jointly Gaussian, jointly non-singular, and related by $\mathbf{Y} = [\mathbf{H}]\mathbf{X} + \mathbf{Z}$. Show that the MMSE estimate can be expressed as

$$\hat{\mathbf{x}}(\mathbf{y}) = [\mathbf{K}_\xi][\mathbf{H}^\top][\mathbf{K}_Z^{-1}]\mathbf{y}. \quad (\text{A.224})$$

Hint: Recall from (10.9) that $\hat{\mathbf{x}}(\mathbf{y}) = [G]\mathbf{y}$. Relate $[G]$ and $[H]$ by using (3.115).

Solution: We assume that $\mathbf{X}$ and $\mathbf{Z}$ are independent (as required for $[H]$ and $[K_{\mathbf{Z}}]$ to specify the joint covariance of $\mathbf{X}$ and $\mathbf{Y}$ and as required for the development in Section 3.5 to make any sense). As shown in Section 3.5, the relation between $\mathbf{X}$ and $\mathbf{Y}$ can be expressed both as $\mathbf{Y} = [H]\mathbf{X} + \mathbf{Z}$ (with $\mathbf{X}$ and $\mathbf{Z}$ independent) and $\mathbf{X} = [G]\mathbf{Y} + \mathbf{V}$ (with $\mathbf{Y}$ and $\mathbf{V}$ independent). From (3.43) and (3.49), $[G]$ and $[H]$ are symmetrically given as

$$[G] = [K_{\mathbf{X} \cdot \mathbf{Y}} K_{\mathbf{Y}}^{-1}]; \quad [H] = [K_{\mathbf{X} \cdot \mathbf{Y}}^{\top} K_{\mathbf{X}}^{-1}]. \quad (\text{A.225})$$

From the relationship $\mathbf{X} = [G]\mathbf{y} + \mathbf{V}$, it is clear that the MMSE estimate of $\mathbf{X}$ for a sample observation $\mathbf{y}$ is $\hat{\mathbf{x}}(\mathbf{y}) = [G]\mathbf{y}$ and that the estimation error is $\mathbf{v}$ and thus that $[K_{\boldsymbol{\xi}}] = [K_{\mathbf{V}}]$. The matrices $[G]$ and $[H]$ are related in (3.115) as $[G^{\top} K_{\mathbf{V}}^{-1}] = [K_{\mathbf{Z}}^{-1} H]$ and thus

$$[G^{\top}] = [K_{\mathbf{Z}}^{-1} H K_{\mathbf{V}}] = [K_{\mathbf{Z}}^{-1} H K_{\boldsymbol{\xi}}]. \quad (\text{A.226})$$

Substituting the transpose of this into $\hat{\mathbf{x}}(\mathbf{y}) = [G]\mathbf{y}$, we get (A.224).

b) Show that

$$[K_{\boldsymbol{\xi}}^{-1}] = [K_{\mathbf{X}}^{-1}] + [H^{\top} K_{\mathbf{Z}}^{-1} H]. \quad (\text{A.227})$$

Hint: A reasonable approach is to start with (10.10), *i.e.*, $[K_{\boldsymbol{\xi}}] = [K_{\mathbf{X}}] - [K_{\mathbf{X} \cdot \mathbf{Y}} K_{\mathbf{Y}}^{-1} K_{\mathbf{X} \cdot \mathbf{Y}}^{\top}]$. Premultiply both sides by $[K_{\mathbf{X}}^{-1}]$ and post-multiply both sides by $[K_{\boldsymbol{\xi}}^{-1}]$. The final product of matrices here can be expressed as $[H^{\top} K_{\mathbf{Z}}^{-1} H]$ by using the hint in (a) on $[K_{\mathbf{Z}}^{-1} H]$.

Solution: Following the hint,

$$[K_{\mathbf{X}}^{-1}] = [K_{\boldsymbol{\xi}}^{-1}] - [K_{\mathbf{X}}^{-1} K_{\mathbf{X} \cdot \mathbf{Y}} K_{\mathbf{Y}}^{-1} K_{\mathbf{X} \cdot \mathbf{Y}}^{\top} K_{\boldsymbol{\xi}}^{-1}].$$

From (A.225), we can substitute $[H^{\top}]$ for $[K_{\mathbf{X}}^{-1} K_{\mathbf{X} \cdot \mathbf{Y}}]$ and $[G^{\top}]$ for $[K_{\mathbf{Y}}^{-1} K_{\mathbf{X} \cdot \mathbf{Y}}^{\top}]$ above, getting

$$[K_{\mathbf{X}}^{-1}] = [K_{\boldsymbol{\xi}}^{-1}] - [H^{\top} G^{\top} K_{\boldsymbol{\xi}}^{-1}].$$

Rearranging and substituting (A.226) into this, we get (A.227).

One minor detail we have omitted above is to verify that $[K_{\mathbf{Z}}]$ and $[K_{\mathbf{V}}]$ are non-singular. If $[K_{\mathbf{Z}}]$ is singular, then there is some vector $\mathbf{a}$ such that $\mathbf{a}^{\top} \mathbf{Z}$ is 0. This implies that $\mathbf{a}^{\top} \mathbf{X} = [G]\mathbf{Y}$, which means that $\mathbf{X}$ and $\mathbf{Y}$ can not be mutually non-singular. Since we have assumed that $\mathbf{X}$ and $\mathbf{Y}$ are mutually non-singular, this means that $\mathbf{Z}$ (and thus $[K_{\mathbf{Z}}]$) is non-singular. The same argument applies to $\mathbf{V}$.

Exercise 10.7: a) Write out $\mathbb{E}[(\mathbf{X} - \mathbf{g}^{\top} \mathbf{Y})^2] = \sigma_{\mathbf{X}}^2 - 2[K_{\mathbf{X} \cdot \mathbf{Y}}]\mathbf{g} + \mathbf{g}^{\top} [K_{\mathbf{Y}}]\mathbf{g}$ as a function of $\mathbf{g} = (g_1, g_2, \dots, g_n)^{\top}$ and take the partial derivative of the function with respect to g_i for each i , $1 \leq i \leq n$. Show that the vector of these partial derivatives is $-2[K_{\mathbf{X} \cdot \mathbf{Y}}] + 2\mathbf{g}^{\top} [K_{\mathbf{Y}}]$.

Solution: Note that $\frac{\partial}{\partial g_i} [K_{\mathbf{X} \cdot \mathbf{Y}}]\mathbf{g} = [K_{\mathbf{X} \cdot \mathbf{Y}}]_{\cdot i}$. For the second term, we take the partial derivative with respect to g_i for any given i , getting

$$\frac{\partial}{\partial g_i} \mathbf{g}^{\top} [K_{\mathbf{Y}}]\mathbf{g} = [K_{\mathbf{Y}}]_{i \cdot} \mathbf{g} + \mathbf{g}^{\top} [K_{\mathbf{Y}}]_{\cdot i} = 2\mathbf{g}^{\top} [K_{\mathbf{Y}}]_{\cdot i},$$

where $[K_{\mathbf{Y}}]_{\cdot i}$ denotes the i th column of $[K_{\mathbf{Y}}]$ and $[K_{\mathbf{Y}}]_i$ denotes the i th row. We have used the fact that $[K_{\mathbf{Y}}]$ is symmetric for the latter equality. Putting these equations into vector form,

$$\frac{\partial}{\partial \mathbf{g}} [\sigma_X^2 - 2[K_{X \cdot \mathbf{Y}}]\mathbf{g} + \mathbf{g}^\top [K_{\mathbf{Y}}]\mathbf{g}] = -2[K_{X \cdot \mathbf{Y}}] + 2\mathbf{g}^\top [K_{\mathbf{Y}}].$$

b) Explain why the stationary point here is actually a minimum.

Solution: The stationary point, say $\tilde{\mathbf{g}}$, *i.e.*, the point at which this vector partial derivative is 0, is $\tilde{\mathbf{g}}^\top = [K_{X \cdot \mathbf{Y}} K_{\mathbf{Y}}^{-1}]$. Taking the transpose, $\tilde{\mathbf{g}} = [K_{\mathbf{Y}}^{-1}]^\top [K_{X \cdot \mathbf{Y}}]^\top$. There are two ways of showing that $\tilde{\mathbf{g}}$ minimizes $\mathbb{E}[(X - \mathbf{g}^\top \mathbf{Y})^2]$ over $\mathbf{g}$. For the first of these, recall that for the Gaussian case, $\hat{x}_{\text{MMSE}}(\mathbf{y}) = \tilde{\mathbf{g}}^\top \mathbf{y}$. Since the MMSE estimate is then linear in $\mathbf{y}$, the LLSE estimate must be the same, and thus $\tilde{\mathbf{g}}$ must minimize $\mathbb{E}[(X - \mathbf{g}^\top \mathbf{Y})^2]$ over $\mathbf{g}$.

The second way to show that the stationary point $\tilde{\mathbf{g}}$ minimizes $\mathbb{E}[(X - \mathbf{g}^\top \mathbf{Y})^2]$ is to note that $\mathbb{E}[(X - \mathbf{g}^\top \mathbf{Y})^2]$ is convex in $\mathbf{g}$, and thus must be minimized by a stationary point if a stationary point exists. As we have just shown, a stationary point does exist in this case.

Exercise 10.8: For a real inner-product space, show that n vectors, $Y_1, \dots, Y_n$ are linearly dependent if and only if the matrix of inner products, $\{\langle Y_j, Y_k \rangle; 1 \leq j, k \leq n\}$, is singular.

Solution: Let $[K]$ be the matrix of inner products, $\{\langle Y_j, Y_i \rangle; 1 \leq i, j \leq n\}$, and let $[K]_i = (\langle Y_1, Y_i \rangle, \dots, \langle Y_n, Y_i \rangle)^\top$ be the i th column of $[K]$. Then $[K]$ is singular iff there is a non-zero vector $\mathbf{a} = (a_1, \dots, a_n)^\top$ such that $\sum_i a_i [K]_i = 0$. This vector equation, written out by components, is $\sum_i a_i \langle Y_j, Y_i \rangle = 0$ for $1 \leq j \leq n$. This can be rewritten as $\langle Y_j, \sum_i a_i Y_i \rangle = 0$ for $1 \leq j \leq n$. Thus, if $[K]$ is singular, there is an $\mathbf{a}$ for which this equation is satisfied for $1 \leq j \leq n$. It follows that $\langle \sum_j a_j Y_j, \sum_i a_i Y_i \rangle = 0$. This implies that $\sum_i a_i Y_i = 0$ and $\{Y_i; 1 \leq i \leq n\}$ is a linearly dependent set. Conversely, if $\{Y_i; 1 \leq i \leq n\}$ is a linearly dependent set, then there is a non-zero $\mathbf{a}$ such that $\sum_i a_i Y_i = 0$. Thus $\langle Y_j, \sum_i a_i Y_i \rangle = 0$ for $1 \leq j \leq n$ and $[K]$ is singular.

Exercise 10.9: Show that (10.89) - (10.91) agree with (10.11).

Solution: From (10.90),

$$\beta = \mathbb{E}[X] - \sum_j \alpha_j \mathbb{E}[Y_j]. \quad (\text{A.228})$$

Note that $\mathbb{E}[XY_i] = \mathbb{E}[X] \mathbb{E}[Y_i] + K_{XY_i}$ and $\mathbb{E}[Y_i Y_j] = \mathbb{E}[Y_i] \mathbb{E}[Y_j] + K_{Y_i Y_j}$. Substituting this in (10.91),

$$\mathbb{E}[X] \mathbb{E}[Y_i] + K_{XY_i} = \beta \mathbb{E}[Y_i] + \sum_j \alpha_j \mathbb{E}[Y_i] \mathbb{E}[Y_j] + \sum_j \alpha_j K_{Y_i Y_j}. \quad (\text{A.229})$$

Multiplying (A.228) by $\mathbb{E}[Y_i]$ and subtracting from (A.229),

$$K_{XY_i} = \sum_j \alpha_j K_{Y_i Y_j}.$$

This implies that $\mathbf{\alpha}^\top = K_{XY} K_Y^{-1}$ so that (10.89) becomes $\hat{X}(\mathbf{Y}) = \beta D + K_{XY} K_Y^{-1} \mathbf{Y}$. In terms of any given sample value $\mathbf{y}$, this is $\hat{X}(\mathbf{y}) = \beta + K_{XY} K_Y^{-1} \mathbf{y}$. Associating β with $\mathbf{b}$, we see that this is the one dimensional (in X) form of (10.12), which is equivalent to (10.11). Recall that (10.11) is the MMSE estimate for the Gaussian case and thus is also the LLSE estimate for the general case with the same covariances.

Exercise 10.10: Let $\mathbf{X} = (X_1, \dots, X_n)^\top$ be a zero-mean complex rv with real and imaginary components $X_{\text{re},j}, X_{\text{im},j}$, $1 \leq j \leq n$ respectively. Express $\mathbb{E}[X_{\text{re},j} X_{\text{re},k}]$, $\mathbb{E}[X_{\text{re},j} X_{\text{im},k}]$, $\mathbb{E}[X_{\text{im},j} X_{\text{im},k}]$, $\mathbb{E}[X_{\text{im},j} X_{\text{re},k}]$ as functions of the components of $[K_X]$ and $\mathbb{E}[\mathbf{X} \mathbf{X}^\top]$.

Solution: Note that $\mathbb{E}[\mathbf{X} \mathbf{X}^\top]$ is the pseudo-covariance matrix as treated in Section 3.7.2 and the desired covariances of real and imaginary parts are given in (3.100). We simply rewrite those results in the notation of the problem statement..

$$\begin{aligned} \mathbb{E}[X_{\text{re},j} X_{\text{re},k}] &= \frac{1}{2} \Re([K_X]_{jk} + [M_X]_{jk}), \\ \mathbb{E}[X_{\text{re},j} X_{\text{im},k}] &= \frac{1}{2} \Im(-[K_X]_{jk} + [M_X]_{jk}), \\ \mathbb{E}[X_{\text{im},j} X_{\text{im},k}] &= \frac{1}{2} \Re([K_X]_{jk} - [M_X]_{jk}), \\ \mathbb{E}[X_{\text{im},j} X_{\text{re},k}] &= \frac{1}{2} \Im([K_X]_{jk} + [M_X]_{jk}). \end{aligned}$$

Exercise 10.11: Let $Y = Y_{\text{re}} + iY_{\text{im}}$ be a complex random variable. For arbitrary real numbers a, b, c, d , find complex numbers α and β such that

$$\Re[\alpha Y + \beta Y^*] = aY_{\text{re}} + bY_{\text{im}},$$

$$\Im[\alpha Y + \beta Y^*] = cY_{\text{re}} + dY_{\text{im}}.$$

Solution: Writing out the first equation above in terms of the real and imaginary parts of α, β , and Y , we get

$$\Re[\alpha Y + \beta Y^*] = \alpha_{\text{re}} Y_{\text{re}} - \alpha_{\text{im}} Y_{\text{im}} + \beta_{\text{re}} Y_{\text{re}} + \beta_{\text{im}} Y_{\text{im}} = aY_{\text{re}} + bY_{\text{im}}.$$

This must be satisfied for all sample values of Y_{re} and Y_{im} , so it breaks into two equations,

$$\alpha_{\text{re}} + \beta_{\text{re}} = a; \quad -\alpha_{\text{im}} + \beta_{\text{im}} = d.$$

Treating the second equation in the same way, we get

$$\alpha_{\text{im}} + \beta_{\text{im}} = c; \quad \alpha_{\text{re}} - \beta_{\text{re}} = d.$$

We now have two equations in the real variables α_{re} and β_{re} and two equations in α_{im} and β_{im} . They can be trivially solved for $\alpha_{\text{re}}, \beta_{\text{re}}, \alpha_{\text{im}}$, and β_{im} , getting

$$\begin{aligned} \alpha &= \frac{a+d}{2} + i \frac{c-b}{2} \\ \beta &= \frac{a-d}{2} + i \frac{b+c}{2}. \end{aligned}$$

Note: The complex field is similar to the real field in so many ways that it is important to sometimes focus on essentially trivial exercises like this that force one to be aware of the differences, especially when going from a complex rv Y to the real rv's Y_{re} and Y_{im} .

Exercise 10.12: (Derivation of circularly symmetric Gaussian density) Let $\mathbf{X} = \mathbf{X}_{\text{re}} + i\mathbf{X}_{\text{im}}$ be a zero-mean, circularly-symmetric, n dimensional Gaussian complex rv. Let $\mathbf{U} = (\mathbf{X}_{\text{re}}^{\top}, \mathbf{X}_{\text{im}}^{\top})^{\top}$ be the corresponding $2n$ dimensional real rv. Let $[K_{\text{re}}] = \mathbb{E}[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{re}}^{\top}]$ and $[K_{\text{ri}}] = \mathbb{E}[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{im}}^{\top}]$.

a) Show that

$$[K_{\mathbf{U}}] = \begin{bmatrix} K_{\text{re}} & K_{\text{ri}} \\ -K_{\text{ri}} & K_{\text{re}} \end{bmatrix}.$$

Solution:

$$\begin{aligned} E[\mathbf{X}\mathbf{X}^{\top}] &= E[(\mathbf{X}_{\text{re}} + i\mathbf{X}_{\text{im}})(\mathbf{X}_{\text{re}} + i\mathbf{X}_{\text{im}})^{\top}] \\ &= E[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{re}}^{\top}] - E[\mathbf{X}_{\text{im}}\mathbf{X}_{\text{im}}^{\top}] + iE[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{im}}^{\top}] + iE[\mathbf{X}_{\text{im}}\mathbf{X}_{\text{re}}^{\top}]. \end{aligned}$$

This is the pseudo-covariance matrix, so it must be zero from the circular-symmetry constraint. Thus

$$\begin{aligned} K_{\text{re}} &= E[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{re}}^{\top}] = E[\mathbf{X}_{\text{im}}\mathbf{X}_{\text{im}}^{\top}] \\ K_{\text{ri}} &= E[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{im}}^{\top}] = -E[\mathbf{X}_{\text{im}}\mathbf{X}_{\text{re}}^{\top}]. \end{aligned}$$

Relating this to $[K_{\mathbf{U}}]$,

$$[K_{\mathbf{U}}] = \begin{bmatrix} \mathbf{X}_{\text{re}} \\ \mathbf{X}_{\text{im}} \end{bmatrix} [\mathbf{X}_{\text{re}}^{\top}, \mathbf{X}_{\text{im}}^{\top}] = \begin{bmatrix} \mathbf{X}_{\text{re}}\mathbf{X}_{\text{re}}^{\top} & \mathbf{X}_{\text{re}}\mathbf{X}_{\text{im}}^{\top} \\ \mathbf{X}_{\text{im}}\mathbf{X}_{\text{re}}^{\top} & \mathbf{X}_{\text{im}}\mathbf{X}_{\text{im}}^{\top} \end{bmatrix} = \begin{bmatrix} K_{\text{re}} & K_{\text{ri}} \\ -K_{\text{ri}} & K_{\text{re}} \end{bmatrix}.$$

This is the desired solution, but note in addition that since $[K_{\mathbf{U}}]$ is a covariance matrix of a (real) rv, it is symmetric, so that the lower left matrix is also equal to $K_{\text{ri}}^{\top}$. Thus $K_{\text{ri}}^{\top} = -K_{\text{ri}}$.

b) Show that

$$[K_{\mathbf{U}}^{-1}] = \begin{bmatrix} B & C \\ -C & B \end{bmatrix}.$$

and find the B, C for which this is true.

Solution: We assume throughout that $[K_{\mathbf{U}}]$ is non-singular, *i.e.*, that none of the $2n$ real and imaginary parts of $\mathbf{X}$ are linear combinations of the others. We see later that this is equivalent to $[K_{\mathbf{X}}]$ being non-singular. From (3.46), the upper left portion of $[K_{\mathbf{U}}^{-1}]$ is $[B] = [K_{\text{re}}] + [K_{\text{ri}}K_{\text{re}}^{-1}K_{\text{ri}}]$. The same formula applies to the lower right matrix, so it is $[B]$ also. Similarly, from (3.47), the upper right matrix is $[C] = -[BK_{\text{ri}}K_{\text{re}}^{-1}]$. The lower left matrix is then $-[C] = [BK_{\text{ri}}K_{\text{re}}^{-1}]$.

c) Show that $[K_{\mathbf{X}}] = 2[K_{\text{re}} - iK_{\text{ri}}]$.

Solution: Since $[K_{\mathbf{X}}] = \mathbb{E}[\mathbf{X}\mathbf{X}^{\dagger}]$,

$$\begin{aligned} [K_{\mathbf{X}}] &= E[(\mathbf{X}_{\text{re}} + i\mathbf{X}_{\text{im}})(\mathbf{X}_{\text{re}} - i\mathbf{X}_{\text{im}})^{\top}] \\ &= E[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{re}}^{\top}] + [\mathbf{X}_{\text{im}}\mathbf{X}_{\text{im}}^{\top}] + i[\mathbf{X}_{\text{im}}\mathbf{X}_{\text{re}}^{\top}] - i[\mathbf{X}_{\text{re}}\mathbf{X}_{\text{im}}^{\top}] \\ &= 2[K_{\text{re}}] - 2i[K_{\text{ri}}]. \end{aligned}$$

Note: After some notational changes, this agrees with 3.105. In 3.105, $[K_{\text{re}}]$ was defined as $\Re[K_{\mathbf{X}}]$, which is $(1/2)[K_{\text{re}}]$ as defined here. In 3.105, $[K_{\text{im}}]$ was defined as $\Im[K_{\mathbf{X}}]$, which is $(-1/2)[K_{\text{ri}}]$ as defined here.

d) Show that $[K_{\mathbf{X}}^{-1}] = \frac{1}{2}[B - iC]$.

Solution: We are to show that $[K_{\mathbf{X}}^{-1}] = \frac{1}{2}[B - iC]$. This is equivalent to showing that $[K_{\mathbf{X}}] \frac{1}{2}[B - iC] = [I_n]$, which is equivalent to showing that $[K_{\text{re}} - iK_{\text{ri}}][B - iC] = [I_n]$. To show this, consider

$$[I_{2n}] = [K_U][K_U^{-1}] = \begin{bmatrix} K_{\text{re}} & K_{\text{ri}} \\ -K_{\text{ri}} & K_{\text{re}} \end{bmatrix} \begin{bmatrix} B & C \\ -C & B \end{bmatrix}.$$

Expanding, we see that $[I_n] = [K_{\text{re}}B] - [K_{\text{ri}}C]$ and $0 = [K_{\text{re}}C] + [K_{\text{ri}}B]$. Since the matrices $[K_{\text{re}}]$, $[K_{\text{ri}}]$, $[B]$, and $[C]$ are real, this is equivalent to $[K_{\text{re}} - iK_{\text{ri}}][B - iC] = [I_n]$.

e) Define $\mathbf{f}_{\mathbf{X}}(\mathbf{x}) = \mathbf{f}_U(\mathbf{u})$ for $\mathbf{u} = (\mathbf{x}_{\text{re}}^T, \mathbf{x}_{\text{im}}^T)^T$ and show that

$$\mathbf{f}_{\mathbf{X}}(\mathbf{x}) = \frac{\exp -\mathbf{x}^* K_{\mathbf{X}}^{-1} \mathbf{x}^T}{(2\pi)^n \sqrt{\det[K_U]}}.$$

Solution: Since $\mathbf{U}$ is a non-singular, real, zero-mean Gaussian 2n-rv, its density is

$$\mathbf{f}_U(\mathbf{u}) = \frac{\exp -\frac{1}{2} [\mathbf{u}^T K_U^{-1} \mathbf{u}^T]}{(2\pi)^n \sqrt{\det[K_U]}}. \quad (\text{A.230})$$

Expanding the term in brackets, we get

$$\begin{aligned} \mathbf{u}^T K_U^{-1} \mathbf{u}^T &= (\mathbf{x}_{\text{re}}^T, \mathbf{x}_{\text{im}}^T) \begin{bmatrix} B & C \\ -C & B \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\text{re}} \\ \mathbf{x}_{\text{im}} \end{bmatrix} \\ &= \mathbf{x}_{\text{re}}^T B \mathbf{x}_{\text{re}} + \mathbf{x}_{\text{im}}^T B \mathbf{x}_{\text{im}} + \mathbf{x}_{\text{re}}^T C \mathbf{x}_{\text{im}} - \mathbf{x}_{\text{im}}^T C \mathbf{x}_{\text{re}}. \end{aligned} \quad (\text{A.231})$$

Since B is real, the first two terms in (A.231) can be rewritten as $\Re[(\mathbf{x}_{\text{re}} - i\mathbf{x}_{\text{im}})^T B (\mathbf{x}_{\text{re}} + i\mathbf{x}_{\text{im}})]$. Since $[B]$ is symmetric, $\Im[(\mathbf{x}_{\text{re}} - i\mathbf{x}_{\text{im}})^T B (\mathbf{x}_{\text{re}} + i\mathbf{x}_{\text{im}})] = 0$ and thus the first two terms of (A.231) are equal to $[(\mathbf{x}_{\text{re}} - i\mathbf{x}_{\text{im}})^T B (\mathbf{x}_{\text{re}} + i\mathbf{x}_{\text{im}})]$. Since $C = -C^T$, we have $\mathbf{x}^T C \mathbf{x} = 0$ for any real $\mathbf{x}$ (and thus for both $\mathbf{x}_{\text{re}}$ and $\mathbf{x}_{\text{im}}$). Thus the latter two terms in (A.231) can be rewritten as

$$(\mathbf{x}_{\text{re}} - i\mathbf{x}_{\text{im}})^T [-iC] (\mathbf{x}_{\text{re}} + i\mathbf{x}_{\text{im}}) = \mathbf{x}_{\text{re}}^T [-iC] (i\mathbf{x}_{\text{im}}) + (i\mathbf{x}_{\text{im}}^T) [iC] \mathbf{x}_{\text{re}} = \mathbf{x}_{\text{re}}^T C \mathbf{x}_{\text{im}} - \mathbf{x}_{\text{im}}^T C \mathbf{x}_{\text{re}}.$$

Thus, we have

$$\mathbf{u}^T K_U^{-1} \mathbf{u}^T = \mathbf{x}^\dagger (B - iC) \mathbf{x} = 2\mathbf{x}^\dagger K_{\mathbf{X}}^{-1} \mathbf{x}.$$

Substituting this back in (A.230), we have the desired expression.

f) Show that

$$\det[K_U] = \begin{vmatrix} K_{\text{re}} + iK_{\text{ri}} & K_{\text{ri}} - iK_{\text{re}} \\ -K_{\text{ri}} & K_{\text{re}} \end{vmatrix}.$$

Hint: Recall that elementary row operations do not change the value of a determinant.

Solution: Start with the $2n \times 2n$ matrix $[K_U]$ and add $-i$ times row $n+1$ to row 1. Similarly, for each j , $2 \leq j \leq n$, add $-i$ times row $n+j$ to row j . This yields

$$\det[K_U] = \det \begin{bmatrix} K_{\text{re}} + iK_{\text{ri}} & K_{\text{ri}} - iK_{\text{re}} \\ -K_{\text{ri}} & K_{\text{re}} \end{bmatrix}.$$

g) Show that

$$\det[K_U] = \begin{bmatrix} K_{\text{re}} + iK_{\text{ri}} & 0 \\ -K_{\text{ri}} & K_{\text{re}} - iK_{\text{ri}} \end{bmatrix}.$$

Hint: Recall that elementary column operations do not change the value of a determinant.

Solution: This is essentially the same as (f). Add i times each of the first n columns to the corresponding final n columns to yield

$$\det[K_U] = \det \begin{bmatrix} K_{\text{re}} + iK_{\text{ri}} & 0 \\ -K_{\text{ri}} & K_{\text{re}} - iK_{\text{ri}} \end{bmatrix}.$$

h) Show that

$$\det[K_U] = 2^{-2n} (\det[K_X])^2,$$

and from this conclude that (3.108) is valid.

Solution: The determinant in (g) is just the determinant of the upper left block times the determinant of the lower right block. From (c), the upper left block is $(1/2)[K_X^*]$ and the lower right block is $(1/2)[K_X]$. Since multiplying each element of an $n \times n$ matrix by $1/2$ multiplies the determinant by $(1/2)^n$, we have

$$\det[K_U] = 2^{-2n} (\det[K_X])^2.$$

This also shows that $[K_U]$ is non-singular if and only if $[K_X]$ is.

Exercise 10.13: (Alternate derivation of circularly-symmetric Gaussian density)

a) Let X be a circularly-symmetric, zero-mean, complex Gaussian rv with variance 1. Show that

$$f_X(x) = \frac{\exp -x^* x}{\pi}.$$

Recall that the real part and imaginary part each have variance $1/2$.

Solution: Since X_{re} and X_{im} are independent and $\sigma_{X_{\text{re}}}^2 = \sigma_{X_{\text{im}}}^2 = \frac{1}{2}$, we have

$$f_X(x) = \frac{1}{2\pi\sigma_{X_{\text{re}}}\sigma_{X_{\text{im}}}} \exp \left[\frac{-x_{\text{re}}^2}{2\sigma_{X_{\text{re}}}^2} - \frac{x_{\text{im}}^2}{2\sigma_{X_{\text{im}}}^2} \right] = \frac{\exp -x^* x}{\pi}.$$

b) Let $\mathbf{X}$ be an n -dimensional, circularly-symmetric, complex Gaussian zero-mean random vector with $K_X = [I_n]$. Show that

$$f_X(\mathbf{x}) = \frac{\exp -\mathbf{x}^\dagger \mathbf{x}}{\pi^n}.$$

Solution: Since all real and imaginary components are independent, the joint probability over all n complex components is the product of n terms with the form in (a).

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\exp - \sum_{i=1}^n x_i^* x_i}{\pi^n} = \frac{\exp - \mathbf{x}^{*T} \mathbf{x}}{\pi^n}.$$

c) Let $\mathbf{Y} = H\mathbf{X}$ where H is $n \times n$ and invertible. Show that

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{\exp [-\mathbf{y}^\dagger (H^{-1})^\dagger H^{-1} \mathbf{y}]}{v\pi^n},$$

where v is $d\mathbf{y}/d\mathbf{x}$, the ratio of an incremental $2n$ dimensional volume element after being transformed by H to that before being transformed.

Solution: The argument here is exactly the same as that in Section 3.3.4. Since $\mathbf{Y} = H\mathbf{X}$, we have $\mathbf{X} = H^{-1}\mathbf{Y}$, so for any $\mathbf{y}$ and $\mathbf{x} = H^{-1}\mathbf{y}$,

$$f_{\mathbf{Y}}(\mathbf{y})|d\mathbf{y}| = f_{\mathbf{X}}(\mathbf{x})|d\mathbf{x}|.$$

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{f_{\mathbf{X}}(H^{-1}\mathbf{y})}{|d\mathbf{y}|/|d\mathbf{x}|}.$$

Substituting the result of b) into this,

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{\exp [-\mathbf{y}^\dagger [H^\dagger]^{-1} [H]^{-1} \mathbf{y}]}{|d\mathbf{y}|/|d\mathbf{x}|}.$$

d) Use this to show that that (3.108) is valid.

Solution: First note that

$$[K_{\mathbf{Y}}] = \mathbb{E} [\mathbf{Y} \mathbf{Y}^\dagger] = \mathbb{E} [H \mathbf{X} \mathbf{X}^\dagger H^\dagger] = [H H^\dagger].$$

Thus $[K_{\mathbf{Y}}]^{-1} = [H^\dagger]^{-1} [H]^{-1}$ and (b) can be rewritten as

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{\exp [-\mathbf{y}^\dagger [K_{\mathbf{Y}}]^{-1} \mathbf{y}]}{\pi^n |d\mathbf{y}|/|d\mathbf{x}|}.$$

Next, view $|d\mathbf{x}|$ as an incremental volume in $2n$ dimensional space (n real and n imaginary components) and view $|d\mathbf{y}|$ as the corresponding incremental volume for $\mathbf{y} = H\mathbf{x}$, i.e.,

$$\begin{bmatrix} \mathbf{y}_{\text{re}} \\ \mathbf{y}_{\text{im}} \end{bmatrix} = \begin{bmatrix} H_{\text{re}} & -H_{\text{im}} \\ H_{\text{im}} & H_{\text{re}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\text{re}} \\ \mathbf{x}_{\text{im}} \end{bmatrix}.$$

Using the same manipulations on the determinant as in Exercise 10.13, parts (f) and (g), we then have

$$\begin{aligned} \frac{|d\mathbf{y}|}{|d\mathbf{x}|} &= \left| \det \begin{bmatrix} H_{\text{re}} & -H_{\text{im}} \\ H_{\text{im}} & H_{\text{re}} \end{bmatrix} \right| = \left| \det \begin{bmatrix} H_{\text{re}} + iH_{\text{im}} & iH_{\text{re}} - H_{\text{im}} \\ H_{\text{im}} & H_{\text{re}} \end{bmatrix} \right| \\ &= \left| \det \begin{bmatrix} H_{\text{re}} + iH_{\text{im}} & 0 \\ H_{\text{im}} & H_{\text{re}} - iH_{\text{im}} \end{bmatrix} \right| = |\det[H] \det[H^*]| = \det[K_{\mathbf{Y}}]. \end{aligned}$$

Substituting this into the expression for $f_Y(y)$ above, we have 3.108.

Exercise 10.14: a) Let $Y = X^2 + Z$ where Z is a zero-mean unit variance Gaussian random variable. Show that no unbiased estimate of X exists from observation of Y . Hint. Consider any $x > 0$ and compare with $-x$.

Solution: Let $x > 0$ be arbitrary. Then $f_{Y|X}(y|x) = f_{Z|X}(y - x^2)$. Similarly, $f_{Y|X}(y|-x) = f_{Z|X}(y - x^2)$. Thus for all choices of parameter x , $f_{Y|X}(y|x) = f_{Y|X}(y|-x)$. It follows that

$$\begin{aligned} \mathbb{E} [\hat{X}(Y)|X=x] &= \int \hat{x}(y) f_{Y|X}(y|x) dy \\ &= \int \hat{x}(y) f_{Y|X}(y|-x) dy = \mathbb{E} [\hat{X}(Y)|X=-x]. \end{aligned}$$

As a consequence, the bias is given by

$$\begin{aligned} b_{\hat{x}}(x) &= \mathbb{E} [\hat{X}(Y) - X | X=x] = \mathbb{E} [\hat{X}(Y) | X=x] - x \\ b_{\hat{x}}(-x) &= \mathbb{E} [\hat{X}(Y) - X | X=-x] = \mathbb{E} [\hat{X}(Y) | X=x] + x. \end{aligned}$$

Thus $b_{\hat{x}}(-x) = b_{\hat{x}}(x) + 2x$. This shows that for each $x \neq 0$, if $\hat{x}$ is unbiased at $X = x$, then it must be biased at $X = -x$ and vice-versa.

This is not a bizarre type of example; it is unusual only in its extreme simplicity. The underlying problem here occurs whether or not one assumes an *a priori* distribution for X , but it is easiest to understand if X has a symmetric distribution around 0. In this case, Y is independent of the sign of X , so one should not expect any success in estimating the sign of X from Y . The point of interest here, however, is that bias is not a very fundamental quantity. Estimation usually requires some sort of tradeoff between successful estimation of different values of x , and focussing on bias hides this tradeoff.

b) Let $Y = X + Z$ where Z is uniform over $(-1, 1)$ and X is a parameter lying in $(-1, 1)$. Show that $\hat{x}(y) = y$ is an unbiased estimate of x . Find a biased estimate $\hat{x}_1(y)$ for which $|\hat{x}_1(y) - x| \leq |\hat{x}(y) - x|$ for all x and y with strict inequality with positive probability for all $x \in (-1, 1)$.

Solution: Choosing $\hat{x}(y) = y$ implies that $\mathbb{E} [\hat{X}(Y)|X=x] = \mathbb{E} [X + Z|X=x] = x$ since Z is independent of X and satisfies $\bar{Z} = 0$. Thus $b_{\hat{x}}(x) = 0$ for all x , $-1 < x < 1$ and $\hat{x}(y)$ is unbiased. This is clearly a stupid estimation rule, however, since whenever $y > 1$, the rule chooses $\hat{x}(y)$ to be larger than any possible x . Reducing the estimate to 1 whenever $y > 1$ clearly reduces the error $|\hat{x}(y) - x|$ for all x when $y > 1$. Increasing the estimate to -1 when $y < -1$ has the same advantage. Thus a more sensible estimate is

$$\hat{x}_1(y) = \begin{cases} 1; & y \geq 1 \\ y; & -1 < y < 1 \\ -1; & y \leq -1 \end{cases}.$$

For all $y > 1$, $|\hat{x}_1(y) - x| = 1 - x < y - x = |\hat{x}(y) - x|$. Similarly, for $y < -1$, $|\hat{x}_1(y) - x| < |\hat{x}(y) - x|$. For $-1 \leq y \leq 1$, $|\hat{x}_1(y) - x| = |\hat{x}(y) - x|$. In summary, $|\hat{x}_1(y) - x| \leq |\hat{x}(y) - x|$

with strict inequality for $|y| > 1$. For each $x \neq 0$, there is a positive probability that $|Y| > 0$, so the expected magnitude of error is smaller for $\hat{x}_1$ than for $\hat{x}$ for all $x \neq 0$.

To be more explicit, it is straightforward to calculate that $b_{\hat{x}_1}(x) = -x^2/4$ for $x \geq 0$ and $b_{\hat{x}_1}(x) = x^2/4$ for $x < 0$, so the new estimate is biased for all x except $x = 0$. The expected error magnitude for $\hat{x}_1$ is $E[\hat{x}_1(Y) - x] = 1/2 - x^2/4$ whereas for the unbiased estimate, $E[\hat{x}(Y) - x] = 1/2$. Thus the expected error magnitude using $\hat{x}_1$ is smaller than that for $\hat{x}$ for all $x \neq 0$.

Neither this example nor the one in (a) are exotic. Together they give the picture that unbiased estimators can neither be counted on to exist nor to provide sensibly small estimation errors without checking for additional specialized structure.

Exercise 10.15: a) Assume that for each parameter value x , Y is Gaussian, $\mathcal{N}(x, \sigma^2)$. Show that $v_x(y)$ as defined in (10.103) is equal to $(y - x)/\sigma^2$ and show that the Fisher information is equal to $1/\sigma^2$.

Solution: Note that we can view Y as $X + Z$ where Z is $N(0, \sigma^2)$. We have $f(y|x) = (2\pi\sigma^2)^{-1/2} \exp(-(y-x)^2/(2\sigma^2))$. Thus

$$\partial f(y|x)/\partial x = [(y - x)/\sigma^2](2\pi\sigma^2)^{-1/2} \exp(-(y - x)^2/(2\sigma^2)).$$

Dividing by $f(y|x)$, we see that $v_x(y) = (y - x)/\sigma^2$. Then the random variable $V_x(Y)$, conditional on x , is $(Y - x)/\sigma^2$. Since $Y \sim N(x, \sigma^2)$, the variance of $(Y - x)/\sigma^2$, conditional on x , is $\sigma^2/\sigma^4 = \sigma^{-2}$. By definition (see (10.107)), this is the Fisher information, *i.e.*, $J(x) = 1/\sigma^2$.

b) Show that for ML estimation, the bias is 0 for all x . Show that the Cramer-Rao bound is satisfied with equality for all x in this case.

Solution: For the ML estimate, $\hat{x}_{\text{ML}}(y) = y$, and for any given x ,

$$E[\hat{X}_{\text{ML}}(Y) | X = x] = E[Y | X = x] = x.$$

Thus $b_{\hat{x}_{\text{ML}}}(x) = 0$ and the estimate is unbiased. The estimation error is just Z , so the mean-square estimation error, $E[(\hat{X}_{\text{ML}}(Y) - x)^2 | X = x]$, is σ^2 for each x . Combining this with (a),

$$E[(\hat{X}_{\text{ML}}(Y) - x)^2 | X = x] = 1/J(x) \quad \text{for each } x.$$

Comparing with (10.113), the Cramer-Rao bound for an unbiased estimator is met with equality for all x by using the ML estimate.

c) Consider using the MMSE estimator for the *a priori* distribution $X \sim \mathcal{N}(0, \sigma_X^2)$. Show that the bias satisfies $b_{\hat{x}_{\text{MMSE}}}(x) = -x\sigma^2/(\sigma^2 + \sigma_X^2)$.

Solution: For the MMSE estimator, $\hat{x}_{\text{MMSE}}(y) = \frac{y\sigma_X^2}{\sigma^2 + \sigma_X^2}$. Thus, since $E[Y|X=x] = x$, we have $E[\hat{X}(Y) | X=x] = \frac{x\sigma_X^2}{\sigma^2 + \sigma_X^2}$ and

$$b_{\hat{x}_{\text{MMSE}}}(x) = \frac{-x\sigma^2}{\sigma^2 + \sigma_X^2}.$$

d) Show that the MMSE estimator in (c) satisfies the Cramer-Rao bound with equality for each x . Note that the mean-squared error, as a function of x is smaller than that for the ML case for small x and larger for large x .

Solution: From (10.109), the Cramer-Rao bound for a biased estimate is

$$\begin{aligned} \text{VAR} [\hat{X}(Y)|X=x] &\geq \frac{\left[1 + \frac{\partial b_{\hat{X}}(x)}{\partial x}\right]^2}{J(x)} = \frac{[1 - \sigma^2/(\sigma^2 + \sigma_X^2)]^2}{1/\sigma^2} \\ &= \frac{[\sigma_X^2/(\sigma^2 + \sigma_X^2)]^2}{1/\sigma^2} = \frac{\sigma^2 \sigma_X^4}{(\sigma^2 + \sigma_X^2)^2}. \end{aligned} \quad (\text{A.232})$$

Calculating $\text{VAR} [\hat{X}(Y)|X=x]$ directly,

$$\begin{aligned} \text{VAR} [\hat{X}(Y)|X=x] &= \text{VAR} \left[\frac{Y \sigma_X^2}{\sigma^2 + \sigma_X^2} \mid X=x \right] \\ &= \text{VAR} \left[\frac{(x + Z) \sigma_X^2}{\sigma^2 + \sigma_X^2} \mid X=x \right] \\ &= \text{VAR} \left[\frac{Z \sigma_X^2}{\sigma^2 + \sigma_X^2} \right] = \frac{\sigma^2 \sigma_X^4}{(\sigma^2 + \sigma_X^2)^2}. \end{aligned} \quad (\text{A.233})$$

Comparing (A.232) and (A.233), we see that the Cramer-Rao bound is met with equality for all x . Note also that the variance of the estimation error is smaller for all σ_X^2 than that for the ML estimate, and that this variance shrinks to 0 as $\sigma_X^2 \rightarrow 0$. This seems surprising until we recognize that the estimate is also shrinking to 0 as $\sigma_X^2 \rightarrow 0$.

If we use the Cramer-Rao bound to look at the mean-square error given x rather than the variance given x (see (10.112)), we get

$$\mathbb{E} [(\hat{X}(Y) - x)^2 | X=x] \geq \frac{\sigma^2 \sigma_X^4}{(\sigma^2 + \sigma_X^2)^2} + \frac{\sigma^2 x^4}{(\sigma^2 + \sigma_X^2)^2}.$$

We see then that the mean-square error using MMSE estimation is small when $|x|$ is small and large when $|x|$ is large. This is not surprising, since the MMSE estimate chooses $\hat{x}(y)$ to be a fraction smaller than 1 of the observation y .

Exercise 10.16: Assume that Y is $\mathcal{N}(0, x)$. Show that $V_x(y)$ as defined in (10.103) is $v_x(y) = [y^2/x - 1]/(2x)$. Verify that $V_x(Y)$ is zero mean for each x . Find the Fisher information, $J(x)$.

Solution: given x , Y is simply a zero-mean Gaussian rv, so

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{1}{\sqrt{2\pi x}} \exp\left(\frac{-y^2}{2x}\right) \\ \frac{\partial f_{Y|X}(y|x)}{\partial x} &= \frac{1}{\sqrt{2\pi x}} \left[-\frac{1}{2x} + \frac{y^2}{2x^2}\right] \exp\left(\frac{-y^2}{2x}\right) \\ V_x(y) &= -\frac{1}{2x} + \frac{y^2}{2x^2} = \frac{(y^2/x) - 1}{2x}. \end{aligned}$$

Since the variance of Y , conditional on $X = x$ is x , the expected value of $V_x(Y)$ conditional on $X = x$ is 0 for all x .

Since $E[V_x(Y)|X = x] = 0$, we have

$$\begin{aligned} J(x) &= \text{VAR}[V_x(Y)|X=x] = E[V_x^2(Y)|X=x] \\ &= \frac{1}{4x^2} E[(Y^2/x - 1)^2|X=x] = \frac{1}{4x^2} E\left[\frac{Y^4}{x^2} - \frac{2Y^2}{x} + 1 \mid X = x\right] \\ &= \frac{1}{4x^2} \left[\frac{3x^2}{x^2} - \frac{2x}{x} + 1\right] = \frac{1}{2x^2}. \end{aligned}$$

It is slightly easier to use (10.108) to calculate $J(x)$, recognizing that the second derivative in (10.108) is the first derivative of the score $V_x(Y)$.